REGULARITY OF p-HARMONIC FUNCTIONS ON THE PLANE: BLUEPRINT OF THE LEAN FORMALIZATION

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Abstract. This document collects the natural-language statements and detailed proofs that accompany the Lean 4 formalization of the theorems, corollaries and lemmas of T. Iwaniec and J. J. Manfredi, *Regularity of p-harmonic functions on the plane*, Rev. Mat. Iberoamericana **5** (1989), 1–19. Three groups of results that the article quotes from the literature (regularity of the complex gradient of a p-harmonic function; the basic theory of quasiregular and quasiconformal mappings; smoothness off the zero set) are not proved; they are stated precisely and carried as explicit hypotheses (`CGradPackage`, `QCPackage`, `SmoothOffZeros`). Everything else is proved. Each subsection corresponds to the Lean declaration of the same name in the directory `IwaniecManfredi/`; cross-references point to the results that a proof uses. Section 3 contains the formalized results of the article. Within each section, items appear in dependency order.

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1. Definitions

1.1. Preamble.

1.2. MemLpLocOn.

Definition 1.1. Let $\Omega \subset \mathbb{C}$, let $1 \le s \le \infty$ and let $u\colon \mathbb{C} \to \mathbb{C}$. We say that u belongs to $L^s_{\mathrm{loc}}(\Omega)$ if for every compact $K \subset \Omega$ the function $\mathbf{1}_K\, u$, equal to u on K and to 0 off K, lies in $L^s(\mathbb{C})$ with respect to two-dimensional Lebesgue measure; that is, $\mathbf{1}_K u$ is measurable and

$$\int_K |u|^s \, d\sigma < \infty \qquad (s < \infty), \qquad \operatorname*{ess\,sup}_K |u| < \infty \qquad (s = \infty).$$

This is the space $L^s_{\mathrm{loc}}(\Omega)$ used in Section 1 of the paper.

1.3. `IsTestFn`.

Definition 1.2. Let $\Omega \subset \mathbb{C}$. A function $\phi\colon \mathbb{C} \to \mathbb{C}$ is a *test function for* Ω if

(a) ϕ is C^∞-smooth as a function of the two real variables (x, y) on all of $\mathbb{C}$,
(b) the support of ϕ is compact, and
(c) that support is contained in Ω.

Here the support is the closure of $\{z : \phi(z) \neq 0\}$. This is the class $C_c^\infty(\Omega)$ of smooth functions with compact support in Ω, used below to express distributional derivatives and the weak form of the p-harmonic equation (1).

1.4. `IsWeakPartialOn`.

Definition 1.3. Let $\Omega \subset \mathbb{C}$, let $j \in \{0, 1\}$ and let $u, v\colon \mathbb{C} \to \mathbb{C}$. Identify $\mathbb{C}$ with $\mathbb{R}^2$ by $z = x+iy$ and write $\partial_0 = \partial/\partial x$, $\partial_1 = \partial/\partial y$. We say that v is the *distributional ∂_j-derivative of u on Ω* if

$$\int_{\mathbb{C}} u(z)\, \partial_j \phi(z)\, d\sigma(z) = -\int_{\mathbb{C}} v(z)\, \phi(z)\, d\sigma(z)$$

for every test function ϕ for Ω in the sense of `IsTestFn`. (Both integrands vanish off the compact support of ϕ, so both integrals are taken over a compact subset of Ω.)

This is the usual meaning of the phrase "distributional derivative $D^\nu u$" in Section 1 of the paper, for a multi-index ν of order one.

1.5. `PHarmonicOn`.

Definition 1.4. Let $\Omega \subset \mathbb{C} = \mathbb{R}^2$ be open and let $1 < p < \infty$. A real-valued function $u\colon \mathbb{C} \to \mathbb{R}$ is *p-harmonic on Ω* if there is a function $g\colon \mathbb{C} \to \mathbb{C}$ such that

(a) u and g both lie in $L^p_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`;
(b) $\operatorname{Re} g$ is the distributional $\partial/\partial x$-derivative of u on Ω and $\operatorname{Im} g$ is its distributional $\partial/\partial y$-derivative, both in the sense of `IsWeakPartialOn`; and
(c) for every real-valued test function ϕ for Ω in the sense of `IsTestFn`,

$$\int_{\mathbb{C}} |\nabla u|^{p-2}\, \langle \nabla u, \nabla \phi \rangle\, d\sigma = 0.$$

Conditions (a) and (b) say precisely that u belongs to the Sobolev class $W^{1,p}_{\mathrm{loc}}(\Omega)$ with weak gradient $\nabla u = (u_x, u_y)$ recorded as the single complex-valued function $g = u_x + iu_y$; under this encoding $|\nabla u| = |g|$ and, writing $\nabla\phi = \phi_x + i\phi_y$ in the same way, $\langle \nabla u, \nabla \phi\rangle = \operatorname{Re}\big(g\,\overline{\nabla\phi}\big) = u_x\phi_x + u_y\phi_y$, so that the integrand of (c) is $|g|^{p-2}\operatorname{Re}(g\,\overline{\nabla\phi})$. Both integrands vanish off the compact support of ϕ, so the integral is effectively taken over a compact subset of Ω, as in Section 1.

This is the definition of Section 1 of the paper: "a real-valued function u in the Sobolev class $W^{1,p}_{\mathrm{loc}}(\Omega)$, $1 < p < \infty$, is said to be p-harmonic if it is a weak solution of the p-harmonic equation $\operatorname{div}(|\nabla u|^{p-2}\nabla u) = 0$, or equivalently ... $\int_\Omega |\nabla u|^{p-2}\langle \nabla u(x), \nabla\phi(x)\rangle\, dx = 0$". The paper offers test functions $\phi \in W^{1,p}(\Omega)$ with compact support in Ω; we use the smooth compactly supported test functions, which is the primary reading of "weak solution". Where ∇u vanishes the factor $|\nabla u|^{p-2}$ is read as 0 for $p \neq 2$, which is the standard convention and the one under which the integrand $|\nabla u|^{p-2}\langle \nabla u, \nabla\phi\rangle$, bounded in modulus by $|\nabla u|^{p-1}|\nabla\phi|$, extends continuously by 0.

1.6. `IsWeakWirtingerOn`.

Definition 1.5. Let $\Omega \subseteq \mathbb{C}$ and let $f, a, b\colon \mathbb{C} \to \mathbb{C}$. Identify $\mathbb{C}$ with $\mathbb{R}^2$ by $z = x + iy$ and write $\partial_0 = \partial/\partial x$, $\partial_1 = \partial/\partial y$. We say that a and b are the *weak Wirtinger derivatives of f on Ω* if there are functions $f_x, f_y\colon \mathbb{C} \to \mathbb{C}$ such that

(a) f, f_x and f_y belong to $L^1_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`;
(b) f_x is the distributional ∂_0-derivative of f on Ω and f_y is the distributional ∂_1-derivative of f on Ω, both in the sense of `IsWeakPartialOn`;
(c) for every $z \in \mathbb{C}$,

$$a(z) = \tfrac{1}{2}\big(f_x(z) - i\, f_y(z)\big), \qquad b(z) = \tfrac{1}{2}\big(f_x(z) + i\, f_y(z)\big).$$

We then write $a = \partial f/\partial z$ and $b = \partial f/\partial \bar{z}$ in the weak sense, and also $a = f_z$, $b = f_{\bar{z}}$.

This is the pair of operators

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big), \qquad \frac{\partial}{\partial \bar{z}} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big)$$

of Section 2 of the paper, applied in the distributional sense. It is the reading the paper's text forces where those operators are first used. A mapping is called K-quasiregular at paper.tex:129–133 if $f \in W^{1,2}_{\mathrm{loc}}(\Omega)$ *and* the inequality (11) between $\partial f/\partial \bar{z}$ and $\partial f/\partial z$ holds almost everywhere; at that point in the paper f carries no hypothesis beyond Sobolev membership, so the derivatives appearing in (11) — and in (12), which is about the same f one sentence later — are necessarily the distributional ones whose existence that membership asserts. The same is true of the system (18), whose solutions the paper assumes only to lie in $W^{1,2}$.

Clause (a) is part of the data for two reasons. First, it is what makes clause (b) say what it should: the integration-by-parts identity of `IsWeakPartialOn` is an identity between integrals, and a divergent integral carries no information, so without local integrability that identity could be met by functions which are not derivatives of f in any sense. Second, (a) and (b) are exactly the hypotheses under which the weak partial derivatives of an almost everywhere differentiable function are known to agree almost everywhere with its classical ones. Consequently, when Ω is open and f is in addition differentiable at almost every point of Ω, one has $f_x = \partial_0 f$ and $f_y = \partial_1 f$ almost everywhere, hence $a = \partial f/\partial z$ and $b = \partial f/\partial \bar{z}$ almost everywhere with the right-hand sides read through the real Fréchet derivative of f. So for an almost everywhere differentiable function the weak and the pointwise readings of the two operators agree almost everywhere, which is what lets the pointwise computations of Sections 4–6 proceed. For a function known only up to a null set, by contrast, only the weak reading has content: the Fréchet derivative is taken to be 0 wherever a function fails to be differentiable, so both pointwise operators vanish identically on a nowhere differentiable representative of a class, and any inequality or identity between them is satisfied vacuously there.

Finally, a and b are determined by f and Ω up to null sets when Ω is open: if f_x and f'_x are both distributional ∂_0-derivatives of f on Ω lying in $L^1_{\mathrm{loc}}(\Omega)$, then $\int (f_x - f'_x)\phi \, d\sigma = 0$ for every test function ϕ for Ω, whence $f_x = f'_x$ almost everywhere on Ω, and likewise for f_y. So the choice of the family in (b) does not matter, and a statement of the form "there are weak Wirtinger derivatives a, b of f on Ω such that $P(a, b)$ holds almost everywhere" is unambiguous for every property P which is insensitive to changes of its arguments on null sets.

1.7. `SolvesEq12`.

Definition 1.6. Let $\Omega \subset \mathbb{C}$ be open and let $1 < p < \infty$. A mapping $f\colon \mathbb{C} \to \mathbb{C}$ is said to *satisfy the system (12) on* Ω if there are weak Wirtinger derivatives $a = \partial f/\partial z$ and $b = \partial f/\partial \bar{z}$ of f on Ω, in the sense of `IsWeakWirtingerOn`, such that

$$b(z) = \Big(\frac{1}{p} - \frac{1}{2}\Big)\Big[\frac{\overline{f(z)}}{f(z)}\, a(z) + \frac{f(z)}{\overline{f(z)}}\,\overline{a(z)}\Big]$$

at almost every $z \in \Omega$ with $f(z) \neq 0$, "almost every" referring to two-dimensional Lebesgue measure.

This is equation (12) of the paper,

$$\frac{\partial f}{\partial \bar{z}} = \Big(\frac{1}{p} - \frac{1}{2}\Big)\Big[\frac{\bar{f}}{f}\frac{\partial f}{\partial z} + \frac{f}{\bar{f}}\,\overline{\frac{\partial f}{\partial z}}\Big], \tag{12}$$

the one complex equation into which Section 2 recasts the p-harmonic equation (1) for the complex gradient $f = \partial u/\partial z$. The right-hand side involves the quotients $\bar{f}/f$ and $f/\bar{f}$ and is therefore defined only where f does not vanish, which is why the condition is imposed off the zero set of f; the paper asserts (12) "almost everywhere", and in its situation the zero set of f is discrete, hence of measure zero, so the two readings agree.

The derivatives are the weak ones. Equation (12) is introduced at paper.tex:140–143 for the same f, and in the same breath, as the distortion inequality (11) of `QuasiregularOn`, and the only hypothesis that f carries there is $f \in W^{1,2}_{\mathrm{loc}}(\Omega)$; so $\partial f/\partial z$ and $\partial f/\partial \bar{z}$ in (12) are the

distributional derivatives, and satisfying (12) is a property of the class of f. Read through the real Fréchet derivative of a chosen representative, (12) would be satisfied vacuously by every nowhere differentiable representative of every L^2-class, since both of its sides would vanish identically. By the next-to-last paragraph of `IsWeakWirtingerOn`, whenever f is in addition differentiable at almost every point of Ω the pair (a, b) agrees almost everywhere on Ω with the pointwise pair $(f_z, f_{\bar{z}})$ read through $\mathrm{d}f$, which is how the pointwise form of (12) — the form consumed by the elliptic-regularity step and by the hodograph transformation — is recovered. By the last paragraph of `IsWeakWirtingerOn` the choice of the witnessing pair is immaterial.

1.8. `MemWLocOn`.

Definition 1.7. Let $\Omega \subset \mathbb{C}$, let $k \geq 0$ be an integer, let $1 \leq s \leq \infty$ and let $u\colon \mathbb{C} \to \mathbb{C}$. We say that u belongs to $W^{k,s}_{\mathrm{loc}}(\Omega)$ if there is a family of functions $\big(u_w\big)_w$, indexed by the finite words w with entries in $\{0, 1\}$, such that

(a) $u_{()} = u$ for the empty word;
(b) $u_w \in L^s_{\mathrm{loc}}(\Omega)$, in the sense of `MemLpLocOn`, for every word w with $|w| \leq k$;
(c) for every word w with $|w| < k$ and every $j \in \{0, 1\}$, the function u_{wj} obtained by appending j to w is the distributional ∂_j-derivative of u_w on Ω, in the sense of `IsWeakPartialOn`.

This is the space $W^{k,s}_{\mathrm{loc}}(\Omega)$ of Section 1 of the paper: the space of functions $u\colon \Omega \to \mathbb{C}$ whose distributional derivatives $D^\nu u$, $|\nu| \leq k$, belong to $L^s_{\mathrm{loc}}(\Omega)$. Conditions (a) and (c) say that u_w is the iterated distributional derivative of u along the word w, so that (b) is exactly the requirement that $D^\nu u \in L^s_{\mathrm{loc}}(\Omega)$ for $|\nu| \leq k$; indexing by words rather than by multi-indices records the same collection of derivatives, since a word of length m determines a multi-index of order m and every multi-index of order $m \leq k$ arises from some word.

1.9. `QuasiregularOn`.

Definition 1.8. Let $\Omega \subset \mathbb{C}$ be open and let $1 \leq K < \infty$. A mapping $f\colon \mathbb{C} \to \mathbb{C}$ is *K-quasiregular on Ω* if

(a) $f \in W^{1,2}_{\mathrm{loc}}(\Omega)$, in the sense of `MemWLocOn`; and
(b) there are weak Wirtinger derivatives $a = \partial f/\partial z$ and $b = \partial f/\partial \bar{z}$ of f on Ω, in the sense of `IsWeakWirtingerOn`, such that

$$\big|b(z)\big| \leq \frac{K-1}{K+1}\big|a(z)\big|$$

for almost every $z \in \Omega$, "almost every" referring to two-dimensional Lebesgue measure.

This is the definition of K-quasiregularity given at paper.tex:129–133, together with its differential inequality

$$\Big|\frac{\partial f(z)}{\partial \bar{z}}\Big| \leq \frac{K-1}{K+1}\Big|\frac{\partial f(z)}{\partial z}\Big| \tag{11}$$

for almost every $z \in \Omega$.

Two remarks on the reading of (11).

First, the derivatives in (b) are the weak ones. This is what the paper's sentence says: the only hypothesis f carries where (11) is stated is $f \in W^{1,2}_{\mathrm{loc}}(\Omega)$, and the two Wirtinger operators have just been introduced, two lines earlier, as the combinations $\frac{1}{2}(\partial_x \mp i\partial_y)$; so the derivatives in (11) are the distributional ones whose existence the Sobolev membership asserts. Reading them instead through the real Fréchet derivative of the given representative would not formalize (11) at all. Clause (a) constrains f only up to null sets — by `MemWLocOn` it is a family of integral identities and local L^2-conditions — so a class in $W^{1,2}_{\mathrm{loc}}(\Omega)$ has representatives that are differentiable at no point whatever; for such a representative every pointwise Wirtinger derivative is 0, by the convention that the differential is 0 where it does not exist, and the pointwise form of (11) would hold vacuously however large the class's genuine distortion. With (b) as stated this cannot happen: the weak derivatives depend only on the class of f, so (11) is a condition on the class, as the paper intends.

Second, the witnessing pair in (b) is immaterial. By the last paragraph of `IsWeakWirtingerOn`, two pairs of weak Wirtinger derivatives of f on an open Ω agree almost everywhere there, so the validity of (b) does not depend on which pair is chosen; and (b) for one pair gives (b) for every pair.

1.10. `CGradPackage`.

Definition 1.9. A *complex-gradient package* is a datum consisting of proofs of the following two statements.

First statement. Let $\Omega \subset \mathbb{C}$ be open, let $1 < p < \infty$, let $u\colon \mathbb{C} \to \mathbb{R}$ be p-harmonic on Ω in the sense of `PHarmonicOn`, and let $g\colon \mathbb{C} \to \mathbb{C}$ be such that $\operatorname{Re} g$ and $\operatorname{Im} g$ are the distributional $\partial/\partial x$- and $\partial/\partial y$-derivatives of u on Ω, in the sense of `IsWeakPartialOn`; that is, $g = u_x + iu_y$ records the weak gradient ∇u. Then the complex gradient

$$f = \frac{\partial u}{\partial z} = \tfrac{1}{2}\big(u_x - iu_y\big) = \tfrac{1}{2}\,\overline{g}$$

satisfies

(a) $f \in W^{1,2}_{\mathrm{loc}}(\Omega)$ in the sense of `MemWLocOn`;
(b) f satisfies equation (12) on Ω in the sense of `SolvesEq12`; and
(c) f is K-quasiregular on Ω in the sense of `QuasiregularOn`, with

$$K = \max\Big\{p - 1, \frac{1}{p-1}\Big\},$$

the exponent of equation (13).

Second statement. Let Ω, p, u and g be as above, and let $f_c\colon \mathbb{C} \to \mathbb{C}$ satisfy $f_c = f$ almost everywhere on Ω and be continuous on Ω. Then

(d) f_c is real-differentiable at almost every point of Ω; and
(e) f_c satisfies equation (12) on Ω in the sense of `SolvesEq12`.

This is item (C1) of the paper's note for the formalization, which quotes it from [BI] and [Ma]: "If $u \in W^{1,p}_{\mathrm{loc}}(\Omega)$, $1 < p < \infty$, is p-harmonic on a plane domain Ω, then its complex gradient $f = \partial u/\partial z$ belongs to $W^{1,2}_{\mathrm{loc}}(\Omega)$ and satisfies (12) almost everywhere; in particular f is K-quasiregular with K as in (13)." The note directs that such quoted results be carried as explicit hypotheses, bundled into a structure or passed as arguments, and that they not be introduced as axioms; this definition is that bundle, so that a theorem needing (C1) takes a complex-gradient package as an argument. The hypothesis $u \in W^{1,p}_{\mathrm{loc}}(\Omega)$ of the quoted statement is part of p-harmonicity as `PHarmonicOn` defines it.

The second statement reads the same quoted sentence for a continuous representative of the complex gradient. It is not a strengthening of the first in its clause (e): equation (12) as `SolvesEq12` states it is an identity between the *weak* Wirtinger derivatives of its argument, and by `IsWeakWirtingerOn` those depend only on the almost-everywhere class of that argument on Ω — the local integrability and the integration-by-parts clauses see the argument only through integrals, and the pointwise values of f_c occur in (12) only inside an "almost every z" quantifier. So (e) for f_c and (b) for $\frac{1}{2}\overline{g}$ are equivalent, given $f_c = f$ almost everywhere, and (e) is recorded only to spare the consumer that unfolding. This is the right behaviour: the paper's complex gradient is the continuous object — paper.tex:156 states that $f = \partial u/\partial z$ is continuous and writes its zero set as $f^{-1}(0) = \{z \in \Omega : \nabla u(z) = 0\}$ — and the quoted assertion that f "satisfies (12) almost everywhere" should hold of that representative as well as of the one attached to a chosen weak gradient g, which with the weak reading of (12) it does.

The one item of the second statement that the quoted sentence does not contain is clause (d), the almost-everywhere real differentiability of the continuous representative. It is quoted from the same sources and in the same spirit as the rest: a quasiregular mapping is differentiable almost everywhere, by the $W^{1,2}_{\mathrm{loc}}$ membership of clause (a) together with the standard regularity theory of [Ah], [LV], and clause (c) makes f quasiregular. It is carried here rather than proved because it is an *input* to the elliptic-regularity bundle (C3), which assumes it of the representative whose smoothness it concludes, so it cannot be obtained from (C3); and it is not among

the three conclusions that (C2) quotes for the inverse of a quasiconformal homeomorphism. The representative itself is not produced here: the consumer supplies it, obtaining it from the continuity clause of (C2).

1.11. `DZ`.

Definition 1.10. Identify $\mathbb{C}$ with $\mathbb{R}^2$ by $z = x+iy$, $(x,y) \in \mathbb{R}^2$. For a map $f\colon \mathbb{C} \to \mathbb{C}$ let $Df(z)$ denote the real Fréchet derivative of f at z, a real-linear map $\mathbb{C} \to \mathbb{C}$, so that $Df(z)[1] = f_x(z)$ and $Df(z)[i] = f_y(z)$. The operator of complex differentiation

$$\frac{\partial}{\partial z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\Big)$$

of Section 2 of the paper is then given by

$$\frac{\partial f}{\partial z}(z) = \frac{1}{2}\big(Df(z)[1] - i\,Df(z)[i]\big).$$

We write f_z for this function of z, and H_ξ when the variable is called ξ.

At a point where f is not real-differentiable, $Df(z)$ is taken to be the zero map, so the expression above is defined at every point of $\mathbb{C}$.

1.12. `DZbar`.

Definition 1.11. With the identification $z = x+iy$ of $\mathbb{C}$ with $\mathbb{R}^2$ and the real Fréchet derivative $Df(z)$ of a map f at z (a real-linear map $\mathbb{C} \to \mathbb{C}$, so that $Df(z)[1] = f_x(z)$ and $Df(z)[i] = f_y(z)$), the second operator of complex differentiation

$$\frac{\partial}{\partial \bar z} = \frac{1}{2}\Big(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\Big)$$

of Section 2 of the paper is given, for $f\colon \mathbb{C} \to \mathbb{C}$, by

$$\frac{\partial f}{\partial \bar z}(z) = \frac{1}{2}\big(Df(z)[1] + i\,Df(z)[i]\big).$$

We write $f_{\bar z}$ for this function of z, and $H_{\bar\xi}$ when the variable is called ξ.

At a point where f is not real-differentiable, $Df(z)$ is taken to be the zero map, so the expression is defined at every point of $\mathbb{C}$.

1.13. `Jac`.

Definition 1.12. For a map $H\colon \mathbb{C} \to \mathbb{C}$ put

$$J(\xi) = |H_\xi(\xi)|^2 - |H_{\bar\xi}(\xi)|^2,$$

with H_ξ as in `DZ` and $H_{\bar\xi}$ as in `DZbar`. This is the Jacobian determinant of H, viewed as a map of two real variables; it is the quantity $J(\xi)$ appearing in equation (17) of the paper and again in equation (42).

1.14. `SolvesEq18`.

Definition 1.13. Let $\Omega \subset \mathbb{C}$ be open, let $n \geq 1$ be an integer and let $1 < p < \infty$. A mapping $H\colon \mathbb{C} \to \mathbb{C}$ is said to *satisfy the system (18) on* Ω if there are weak Wirtinger derivatives $a = \partial H/\partial\xi$ and $b = \partial H/\partial\bar\xi$ of H on Ω, in the sense of `IsWeakWirtingerOn`, such that

$$b(\xi) = \Big(\frac{1}{2} - \frac{1}{p}\Big)\Big[\frac{\xi}{\bar\xi}\,a(\xi) + \frac{\bar\xi^{\,n}}{\xi^{\,n}}\,\overline{a(\xi)}\Big]$$

at almost every $\xi \in \Omega$ with $\xi \neq 0$, "almost every" referring to two-dimensional Lebesgue measure.

This is equation (18) of the paper, the linear first order system that the hodograph transformation of Section 3 produces for the inverse H of the quasiconformal factor χ, written

$$H_{\bar\xi} = \Big(\frac{1}{2} - \frac{1}{p}\Big)\Big[\frac{\xi}{\bar\xi}H_\xi + \frac{\bar\xi^{\,n}}{\xi^{\,n}}\,\overline{H_\xi}\Big]. \tag{18}$$

Its coefficients $\xi/\bar\xi$ and $\bar\xi^{\,n}/\xi^{\,n}$ are defined only for $\xi \neq 0$, which is why the origin is excluded; the single point $\xi = 0$ has measure zero.

The derivatives are the weak ones, for the same reason as in `QuasiregularOn`: the only hypothesis the paper's solutions of (18) carry is membership in $W^{1,2}$, so H_ξ and $H_{\bar\xi}$ are the distributional derivatives, and a solution in this sense is a condition on the class of H rather than on a chosen representative. Read through the real Fréchet derivative of the given representative the system would be satisfied by every nowhere differentiable representative of every L^2-class, both sides vanishing identically, and the paper's assertion that a $W^{1,2}$ solution of (18) is K-quasiregular with $K = \max\{p-1, 1/(p-1)\}$ would be false. By the next-to-last paragraph of `IsWeakWirtingerOn`, whenever H is in addition differentiable at almost every point of Ω the pair (a, b) agrees almost everywhere with the pointwise pair $(H_\xi, H_{\bar\xi})$ read through $\mathrm{d}H$; this is how Sections 4–6, which work with almost everywhere differentiable maps, recover the pointwise form of (18). By the last paragraph of `IsWeakWirtingerOn` the choice of the witnessing pair is immaterial.

1.15. `QuasiconformalOn`.

Definition 1.14. Let $\Omega \subset \mathbb{C}$ be open and let $1 \le K < \infty$. A mapping $\chi\colon \mathbb{C} \to \mathbb{C}$ is a *K-quasiconformal homeomorphism of Ω* if

(a) χ is K-quasiregular on Ω in the sense of `QuasiregularOn`;
(b) χ is injective on Ω and continuous on Ω;
(c) the image $\chi(\Omega)$ is open; and
(d) there is a map ψ with $\psi(\chi(z)) = z$ for every $z \in \Omega$ which is continuous on $\chi(\Omega)$.

Conditions (b)–(d) say that χ restricted to Ω is a homeomorphism of Ω onto the open set $\chi(\Omega)$, with inverse the restriction of ψ: injectivity and (d) give $\psi \circ \chi = \mathrm{id}$ on Ω, and for $w = \chi(z) \in \chi(\Omega)$ one gets $\chi(\psi(w)) = \chi(z) = w$, so ψ restricted to $\chi(\Omega)$ is a two-sided inverse; both maps are continuous on the relevant sets.

This is the notion of quasiconformal homeomorphism used throughout the paper, in particular for the factor χ of the factorization (15) and in the standing assumptions of Sections 3 and 5. A quasiconformal homeomorphism is by definition a quasiregular mapping which is in addition a homeomorphism onto its image.

1.16. `PartialWord`.

Definition 1.15. Identify $\mathbb{C}$ with $\mathbb{R}^2$ by $z = x + iy$ and write $\partial_0 = \partial/\partial x$, $\partial_1 = \partial/\partial y$, each evaluated through the real Fréchet derivative $Dg(z)$, so that $\partial_0 g(z) = Dg(z)[1]$ and $\partial_1 g(z) = Dg(z)[i]$.

For a finite word $w = (j_1, j_2, \dots, j_m)$ with entries $j_t \in \{0, 1\}$ and a map $f\colon \mathbb{C} \to \mathbb{C}$ put

$$D^w f = \partial_{j_1} \partial_{j_2} \cdots \partial_{j_m} f,$$

with $D^{()} f = f$ for the empty word.

A word w of length m determines a partial derivative operator of order m, and as w ranges over the words of length m these operators are exactly the operators D^ν with $|\nu| = m$ that the paper writes with multi-index notation; words that differ by a permutation of their entries give the same operator on functions whose partials of order m are continuous. A sum $\sum_{|\nu|=m}$ over multi-indices is therefore rendered below as a sum over the words of length m (up to a combinatorial factor depending only on m, which is harmless in every estimate of the paper, all of which carry an unspecified constant C_m depending on m).

At a point where the relevant function fails to be real-differentiable its derivative is taken to be zero, so $D^w f$ is defined at every point of $\mathbb{C}$.

1.17. `LambdaCoef`.

Definition 1.16. For integers $n \ge 0$ and $k \ge 0$ and a real number p put

$$\lambda_k(n, p) = \frac{1}{2}\Big(-np + \sqrt{4k^2(p-1) + n^2(p-2)^2}\Big).$$

This is the quantity $\lambda_k = \lambda_k(n, p)$ of equation (20) of the paper (equation (37) repeats the same formula).

Here $\sqrt{\cdot}$ denotes the real square root, extended by $\sqrt{t} = 0$ for $t < 0$, so that the formula defines a real number for every real p. The paper uses $\lambda_k(n,p)$ only in the range $1 < p < \infty$, $k \geq n \geq 1$, where the radicand $4k^2(p-1) + n^2(p-2)^2$ is a sum of two nonnegative terms and the extension is never invoked.

1.18. `FourierCoefOfH`.

Definition 1.17. Let $n \geq 0$ be an integer, let $H\colon \mathbb{C} \to \mathbb{C}$, let $k \in \mathbb{Z}$ and let $r \in \mathbb{R}$. Put

$$a_k(r) = \frac{1}{2\pi} \int_0^{2\pi} H\big(re^{i\theta}\big)\, e^{i(n-k)\theta}\, d\theta.$$

This is the k-th Fourier coefficient (30) of the paper, of the function $\theta \mapsto H(r,\theta) = H(re^{i\theta})$ relative to the system $\big(e^{i(k-n)\theta}\big)_{k\in\mathbb{Z}}$ used in the expansion (29). The scan of the article prints the upper limit of integration in (30) as "$2k$"; the transcription records that it is 2π, and that is the reading used here.

The integral is the Lebesgue integral over the interval $[0, 2\pi]$ and is read as 0 when the integrand is not integrable there; the paper uses $a_k(r)$ only for $0 \leq r < 1$ and for H continuous on the unit disk, where the integrand is continuous in θ and the integral is an ordinary one. The shift by n in the exponent is the paper's: it is what makes a_n the coefficient of the constant mode, so that (23) $\lambda_n = \epsilon_n = 0$ corresponds to the $k = n$ term of (19) being constant.

1.19. `HRad`.

Definition 1.18. For a map $H\colon \mathbb{C} \to \mathbb{C}$ and a point $\xi \in \mathbb{C}$ put

$$H_r(\xi) = DH(\xi)\Big[\frac{\xi}{|\xi|}\Big],$$

the value of the real Fréchet derivative $DH(\xi)$ on the vector $\xi/|\xi|$.

This is the radial derivative H_r of Section 4 of the paper, written as a function of ξ rather than of the polar coordinates. Indeed, Section 4 writes $\xi = re^{i\theta}$ with $0 \leq r < 1$ and $H(r,\theta) = H(re^{i\theta})$; for $r > 0$ the curve $r \mapsto re^{i\theta}$ has velocity $\partial(re^{i\theta})/\partial r = e^{i\theta} = \xi/|\xi|$, so the chain rule gives

$$\frac{\partial}{\partial r} H(r,\theta) = DH(re^{i\theta})\big[e^{i\theta}\big] = H_r\big(re^{i\theta}\big),$$

whenever H is real-differentiable at $re^{i\theta}$. Recording H_r as a function of ξ rather than of (r,θ) avoids carrying the polar chart and is what makes the formulas of Section 4 usable at a point.

At $\xi = 0$ the quotient $\xi/|\xi|$ is read as 0 (the convention $w/0 = 0$), and at a point where H is not real-differentiable $DH(\xi)$ is taken to be the zero map; so the expression is defined at every point of $\mathbb{C}$. Section 4 uses it only for $0 < |\xi| < 1$.

1.20. `HAng`.

Definition 1.19. For a map $H\colon \mathbb{C} \to \mathbb{C}$ and a point $\xi \in \mathbb{C}$ put

$$H_\theta(\xi) = DH(\xi)\big[i\xi\big],$$

the value of the real Fréchet derivative $DH(\xi)$ on the vector $i\xi$.

This is the angular derivative H_θ of Section 4 of the paper, written as a function of ξ rather than of the polar coordinates. With $\xi = re^{i\theta}$ and $H(r,\theta) = H(re^{i\theta})$ as in Section 4, the curve $\theta \mapsto re^{i\theta}$ has velocity $\partial(re^{i\theta})/\partial\theta = ire^{i\theta} = i\xi$, so the chain rule gives

$$\frac{\partial}{\partial\theta} H(r,\theta) = DH(re^{i\theta})\big[ire^{i\theta}\big] = H_\theta\big(re^{i\theta}\big),$$

whenever H is real-differentiable at $re^{i\theta}$.

At a point where H is not real-differentiable, $DH(\xi)$ is taken to be the zero map, so the expression is defined at every point of $\mathbb{C}$; note that unlike H_r this one needs no convention at $\xi = 0$, where it is $DH(0)[0] = 0$.

1.21. `EpsCoef`.

Definition 1.20. For integers $n \geq 0$ and $k \geq 0$ and a real number p put

$$\epsilon_k(n,p) = \frac{\lambda_k(n,p) + n - k}{\lambda_k(n,p) + n + k},$$

where $\lambda_k(n,p)$ is the exponent of `LambdaCoef`. This is equation (21) of the paper.

The quotient is interpreted as 0 whenever its denominator vanishes. The paper uses $\epsilon_k(n,p)$ only in the range $1 < p < \infty$, $k \geq n \geq 1$, where $\lambda_k(n,p) \geq 0$ and hence $\lambda_k(n,p)+n+k \geq 2n > 0$, so that convention is never invoked.

1.22. `Thm2Term`.

Definition 1.21. Let $n \geq 0$ and $k \geq 0$ be integers, let p be real, let $A\colon \mathbb{Z} \to \mathbb{C}$, and let $\xi \in \mathbb{C}$. Put

$$T_k(\xi) = \left(A_k\,\xi^k + \epsilon_k\,\overline{A_k}\,\bar{\xi}^{\,k}\right)|\xi|^{\lambda_k+n-k}\,\xi^{-n},$$

where $A_k = A(k)$, $\lambda_k = \lambda_k(n,p)$ is the exponent of `LambdaCoef` and $\epsilon_k = \epsilon_k(n,p)$ is the coefficient of `EpsCoef`.

This is the k-th summand of the series (19) of Theorem 2 of the paper,

$$H(\xi) = \sum_{k=n}^{\infty} \big(A_k\xi^k + \epsilon_k\bar{A}_k\bar{\xi}^k\big)|\xi|^{\lambda_k+n-k}\xi^{-n},$$

so that (19) reads $H(\xi) = \sum_{k\geq n} T_k(\xi)$. The coefficient family is indexed by all of $\mathbb{Z}$ because the compatibility condition (39) of the paper determines A_{-k} from A_k; only the values A_k with $k \geq n$ enter T_k.

Two conventions make the formula total. The factor $|\xi|^{\lambda_k+n-k}$ is a real power of a nonnegative real, read with $0^a = 0$ for $a > 0$ and $0^0 = 1$; and ξ^{-n} is an integer power, read as 0 at $\xi = 0$ when $n > 0$ (Lean's convention $w/0 = 0$). The paper uses (19) for $0 < |\xi| < 1$, where neither convention is invoked.

1.23. `GammaExp`.

Definition 1.22. For an integer $n \geq 0$ and a real number p put

$$\gamma_n = \gamma_n(p) = \lambda_{n+1}(n,p),$$

with $\lambda_k(n,p)$ as in `LambdaCoef`. This is the exponent γ_n of equation (44) of the paper.

Equation (44) displays γ_n in the expanded form

$$\gamma_n = \frac{n}{2}\Big\{-p + \sqrt{4\big(1+\tfrac{1}{n}\big)^2(p-1) + (p-2)^2}\Big\},$$

which for $n \geq 1$ is the same number, because $4(n+1)^2(p-1) + n^2(p-2)^2 = n^2\big(4(1+\frac{1}{n})^2(p-1) + (p-2)^2\big)$ and $n > 0$.

1.24. `WindingNumber`.

Definition 1.23. Let $\gamma\colon [0,1] \to \mathbb{C}$ be continuous with $\gamma(t) \neq 0$ for every $t \in [0,1]$. The *index of γ about* 0 is defined as follows.

Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and let $q\colon \mathbb{R} \to \mathbb{T}$ be the quotient map, which is a covering map. The principal argument $\arg\colon \mathbb{C} - \{0\} \to (-\pi, \pi]$ is discontinuous on the negative reals, but its reduction $\xi \mapsto q(\arg\xi)$ is continuous at every $\xi \neq 0$; hence

$$\alpha = q \circ \arg \circ \gamma\colon [0,1] \to \mathbb{T}$$

is continuous. Because q is a covering map and $[0,1]$ is an interval, α lifts along q: there is a unique continuous $\vartheta\colon [0,1] \to \mathbb{R}$ with $q \circ \vartheta = \alpha$ and $\vartheta(0) = \arg\gamma(0)$. We set

$$\operatorname{ind}(\gamma) = \frac{\vartheta(1) - \vartheta(0)}{2\pi} = \frac{\vartheta(1) - \arg\gamma(0)}{2\pi}.$$

Thus ϑ is a continuous branch of the argument along γ, normalised at $t = 0$, and $\operatorname{ind}(\gamma)$ is the normalised increment of that branch. This is the quantity behind the paper's step "Applying argument principle, we find that $A_{n+1} \neq 0$" in the proof of Corollary 1.

Note that *no loop condition is imposed*: γ is an arbitrary non-vanishing path, and $\operatorname{ind}(\gamma)$ is defined for it. For a general path the value is a real number, measuring the total turning of γ in units of a full turn; it is an integer precisely when $\gamma(0) = \gamma(1)$, since only then do the two endpoints of the lift project to the same point of $\mathbb{T}$. Several of the lemmas built on this definition are applied to genuine paths rather than loops – for instance to a branch traversed a fractional number of times – which is why the definition is stated at this generality.

Two remarks on the formal rendering. The loop is taken as a continuous map on the unit interval $[0, 1]$, so the parametrisation $\theta \in [0, 2\pi]$ used in the paper corresponds to $t = \theta/2\pi$. And the lift ϑ is the one produced by the path-lifting property of the covering q with the specified initial value, so no choice is involved: the normalisation $\vartheta(0) = \arg \gamma(0)$ determines it.

1.25. `WirtingerIter`.

Definition 1.24. For non-negative integers i, j and a map $H \colon \mathbb{C} \to \mathbb{C}$ put

$$\frac{\partial^{i+j} H}{\partial \xi^i \partial \bar{\xi}^j} = \Big(\frac{\partial}{\partial \xi}\Big)^i \Big(\frac{\partial}{\partial \bar{\xi}}\Big)^j H,$$

that is, the function obtained from H by applying the operator $\partial/\partial\bar{\xi}$ of `DZbar` j times and then the operator $\partial/\partial\xi$ of `DZ` i times. For $i = j = 0$ this is H itself.

These are the iterated partials $\partial^s H/\partial\xi^i\partial\bar{\xi}^j$ with $s = i + j$ appearing in the convention of Section 5 of the paper. The definition fixes one order of application, so that the expression is defined for every H without any smoothness assumption, using the convention of `DZ` and `DZbar` that a point of non-differentiability contributes a zero derivative. On an open set where H is C^∞ the two operators commute, so the choice of order is immaterial there; that commutation is proved separately in the tablet and is not part of this definition.

1.26. `DerivSymbol`.

Definition 1.25. A *derivative symbol* is a triple $d = (i, j, \beta)$ consisting of two non-negative integers i and j and a Boolean value β, subject to the requirement

$$i + j > 0.$$

The positive integer $|d| = i + j$ is called the *order* of d.

A derivative symbol is the formal name of one of the factors H^s in the convention of Section 5 of the paper, which reads: "Given a positive integer s let H^s denote one of the partials $\partial^s H/\partial\xi^i\partial\bar{\xi}^j$ or $\overline{\partial^s H/\partial\xi^i\partial\bar{\xi}^j}$, where i and j run over non-negative integers such that $i + j = s$." The symbol (i, j, β) names the partial $\partial^{i+j} H/\partial\xi^i\partial\bar{\xi}^j$ when β is false and its complex conjugate when β is true; its order $i + j$ is the superscript s, and the requirement $i + j > 0$ is the paper's restriction to positive s.

The paper leaves the particular i, j and the presence or absence of the conjugation unspecified ("we do not specify which one of them"), recording only the order s. Here they are recorded as data. This is a presentational choice with no change of mathematical content: a statement about all monomials of a given type is a statement about all choices of these data.

1.27. `SymbolValue`.

Definition 1.26. Let $H \colon \mathbb{C} \to \mathbb{C}$ and let $d = (i, j, \beta)$ be a derivative symbol in the sense of `DerivSymbol`. The *value of d on H* is the function $H^d \colon \mathbb{C} \to \mathbb{C}$ given by

$$H^d = \begin{cases} \dfrac{\partial^{i+j} H}{\partial \xi^i \partial \bar{\xi}^j} & \text{if } \beta \text{ is false,} \\[2ex] \overline{\dfrac{\partial^{i+j} H}{\partial \xi^i \partial \bar{\xi}^j}} & \text{if } \beta \text{ is true,} \end{cases}$$

where $\partial^{i+j}H/\partial\xi^i\partial\bar{\xi}^j$ is the iterated Wirtinger derivative of `WirtingerIter` and the bar denotes complex conjugation of the value.

Thus H^d is precisely one of the partials denoted H^s in Section 5 of the paper, with $s = i+j = |d|$.

1.28. `MonomialValue`.

Definition 1.27. Let $H\colon \mathbb{C} \to \mathbb{C}$ and let $M = (d_1, d_2, \dots, d_k)$ be a finite list of derivative symbols in the sense of `DerivSymbol`. The *monomial determined by* M is the function

$$H^M = H^{d_1} H^{d_2} \cdots H^{d_k},$$

the pointwise product of the symbol values of `SymbolValue`. For the empty list, H^M is the constant function 1.

The list M carries two numerical invariants, both computed from M:

$k(M) = k$ (the length of M), $\qquad s(M) = |d_1|+|d_2|+\cdots+|d_k|$ (the sum of the orders of its entries).

We say that M, and also the function H^M, is *of type* $(s(M), k(M))$.

With this terminology H^M is exactly an expression $H^{s_1}H^{s_2}\cdots H^{s_k}$ with $s_1 + s_2 + \cdots + s_k = s(M)$, which is what Section 5 of the paper calls a monomial of type (s, k): "Any expression of the form $H^{s_1}H^{s_2}\cdots H^{s_k}$, where $s_1 + s_2 + \cdots + s_k = s$ will be called a monomial of type (s, k)." Because the type is read off from the list rather than imposed on it, the order $|d_t|$ of each individual factor stays visible in M.

1.29. `PClass`.

Definition 1.28. Let $H\colon \mathbb{C} \to \mathbb{C}$ and let s, k be non-negative integers. A function $P\colon \mathbb{C} \to \mathbb{C}$ is said to belong to $\mathbb{P}(s, k)$ *relative to* H if there are an integer $r \geq 0$, complex numbers $c_1, \dots, c_r$, and lists $M_1, \dots, M_r$ of derivative symbols, each of type (s, k) in the sense of `MonomialValue`, i.e.

$$s(M_\alpha) = s \quad \text{and} \quad k(M_\alpha) = k \qquad (\alpha = 1, \dots, r),$$

such that

$$P(\xi) = \sum_{\alpha=1}^{r} c_\alpha \, H^{M_\alpha}(\xi) \qquad \text{for every } \xi \in \mathbb{C},$$

with H^M the monomial of `MonomialValue`. The case $r = 0$ is allowed, so the zero function belongs to every $\mathbb{P}(s, k)$.

This is the set $\mathbb{P}(s, k)$ of Section 5 of the paper: "The set of all linear combinations of monomials of type (s, k) is denoted by $\mathbb{P}(s, k)$." The paper uses it for $s \geq 1$ and $k \geq 1$. Note that the condition is an equality of functions on all of $\mathbb{C}$, and that it is existential in the presentation: to know $P \in \mathbb{P}(s, k)$ is to have some expression of P as a linear combination of monomials of type (s, k).

1.30. `QCPackage`.

Definition 1.29. A *quasiregular-theory package* is a datum consisting of proofs of the following four statements.

(a) (Continuity.) For every K, every open $\Omega \subset \mathbb{C}$ and every f which is K-quasiregular on Ω in the sense of `QuasiregularOn`, there is $g\colon \mathbb{C} \to \mathbb{C}$ with $g = f$ almost everywhere on Ω and g continuous on Ω.

(b) (Discreteness of the zero set.) For every K, every open connected Ω, every f which is K-quasiregular on Ω, and every g with $g = f$ almost everywhere on Ω and g continuous on Ω such that g does not vanish identically on Ω, every point $z_0 \in \Omega$ has a punctured neighbourhood in Ω on which g does not vanish: there is $r > 0$ such that $g(z) \neq 0$ for all $z \in \Omega$ with $z \neq z_0$ and $|z - z_0| < r$.

(c) (Factorization near a zero.) For every K, every open connected Ω, every f which is K-quasiregular on Ω, every g with $g = f$ almost everywhere on Ω and g continuous on Ω such that g does not vanish identically on Ω, and every $z_0 \in \Omega$ with $g(z_0) = 0$, there are an integer $n \geq 1$, a constant K', an open set V with $z_0 \in V \subset \Omega$ and a map χ which

is a K'-quasiconformal homeomorphism of V in the sense of `QuasiconformalOn`, such that

$$\chi(z_0) = 0 \qquad \text{and} \qquad g(z) = \big[\chi(z)\big]^n \quad \text{for } z \in V.$$

(d) (The inverse of a quasiconformal homeomorphism.) For every K, every open V, every χ which is a K-quasiconformal homeomorphism of V, and every H with $H(\chi(z)) = z$ for all $z \in V$, the map H belongs to $W^{1,2}_{\mathrm{loc}}(\chi(V))$ in the sense of `MemWLocOn`, is real-differentiable at almost every point of $\chi(V)$, and satisfies $J(\xi) > 0$ for almost every $\xi \in \chi(V)$, where J is the Jacobian of `Jac`.

This is item (C2) of the paper's note for the formalization, quoted there from [Ah] and [LV]: "A K-quasiregular mapping is continuous; if it is not identically zero its zero set is discrete. Near a zero z_0 it factors as $f = \chi^n$ with $n \geq 1$ and χ a quasiconformal homeomorphism of a neighborhood of z_0 with $\chi(z_0) = 0$; the inverse H of a quasiconformal homeomorphism belongs to $W^{1,2}_{\mathrm{loc}}$, has Jacobian $J > 0$ almost everywhere, and satisfies (17)." Clause (c) is the factorization (15) of Section 3. The note directs that such quoted results be carried as explicit hypotheses rather than axioms; this definition is that bundle, so that a theorem needing (C2) takes a quasiregular-theory package as an argument.

Three points of reading.

First, why the first three clauses speak of a representative g rather than of f. Quasiregularity as `QuasiregularOn` defines it is a property of the almost-everywhere class of f and of nothing more. Its two clauses are membership in $W^{1,2}_{\mathrm{loc}}$, which by `MemWLocOn` is a family of integral identities and indicator L^2-conditions, and an almost-everywhere inequality between the weak Wirtinger derivatives of f, which by `IsWeakWirtingerOn` depend only on that class as well. Both are unchanged when f is altered on a set of measure zero, so neither can force f to be continuous at any point. The function f equal to 1 at the origin and to 0 elsewhere is K-quasiregular on $\mathbb{C}$ for $K = 1$: it vanishes almost everywhere, so the integral identities and the indicator conditions are those of the zero function, and the zero function is a distributional ∂_0- and ∂_1-derivative of f on $\mathbb{C}$, since $\int f\,\partial_j\phi\,d\sigma = 0 = -\int 0 \cdot \phi\,d\sigma$ for every test function; so both weak Wirtinger derivatives of f may be taken to be 0 and the inequality reads $0 \leq 0$. Yet f is discontinuous at the origin, has $f(1) = 0$ with f vanishing on a whole punctured neighbourhood of 1, and cannot factor as χ^n near 1 with χ injective and continuous. So the three clauses stated of f would be false, the package would be uninhabited, and every theorem taking one would be vacuously true; 'paper.tex:107' forbids exactly that, requiring the quoted results not to be vacuous. Stated of a continuous representative they are the literature's statements: a quasiregular mapping *is* a $W^{1,2}_{\mathrm{loc}}$ map with a distortion bound on its weak derivatives, and "it is continuous" means that it has a continuous representative. The representative is unique: two continuous functions on an open set agreeing almost everywhere agree everywhere on it.

It matters here that the distortion inequality of `QuasiregularOn` is read for the weak derivatives. Were it read for the pointwise Wirtinger derivatives of the given representative, which are built from $\mathrm{d}f$ and vanish wherever f is not differentiable, the hypothesis K-quasiregular would be met by representatives with no continuous representative at all — for instance by $f_0(z) = \log\log(1/|z|)$ on $\{|z| < 1/2\}$, which lies in $W^{1,2}$ there but is essentially unbounded at the origin, altered by 1 on the countable dense null set $\mathbb{Q} + i\mathbb{Q}$, so that it becomes nowhere continuous and hence nowhere differentiable while both pointwise derivatives vanish identically — and clause (a) would then be false, making this bundle uninhabited. With the weak reading, (a) asserts no more and no less than the literature's continuity theorem.

Second, the zero set in (b) and (c) is the representative's. The set $f^{-1}(0)$ is not determined by f modulo null sets, whereas $g^{-1}(0)$ is; and the paper itself writes the exceptional set as $f^{-1}(0) = \{z \in \Omega : \nabla u(z) = 0\}$ at paper.tex:156, that is, as the zero set of the continuous complex gradient. Only clause (a) produces a representative, so a consumer of the package introduces a single g and runs (b), (c) and the bundle (C3) on that same g.

Third, clause (d) needs no such reading, and the final clause "and satisfies (17)" of the quoted (C2) is deliberately *not* among the assumptions above. The hypotheses of (d) constrain χ through `QuasiconformalOn`, which carries genuine pointwise content — injectivity, continuity,

openness of the image and a continuous inverse — and its three conclusions are themselves almost-everywhere statements, so it is inhabited as it stands. Equation (17) is derived in the tablet from the chain rule for the Wirtinger operators, wherever H and χ are differentiable at matched points and $J \neq 0$, and the almost-everywhere differentiability and positivity of J which make that derivation applicable almost everywhere are assumed in (d). Omitting a clause that is proved rather than quoted leaves the package no stronger than the literature statement, which is what the note requires.

1.31. `DExp`.

Definition 1.30. For a real number p put

$$d = d(p) = \frac{1}{6}\left(1 + \frac{1}{p-1} + \sqrt{1 + \frac{14}{p-1} + \frac{1}{(p-1)^2}}\right).$$

This is the exponent $d = d(p)$ of equation (50) of the paper, which Section 6 introduces as the smallest of the numbers n/γ_n, $n = 1, 2, \dots$

Two conventions make the formula total: $\sqrt{t} = 0$ for $t < 0$, and $1/0 = 0$. The paper uses $d(p)$ only for $1 < p < \infty$, where $p - 1 > 0$ and the radicand is positive, so neither convention is invoked.

The scan of the article prints the last denominator as $(p-2)^2$; the transcription records that the correct denominator is $(p-1)^2$, and that is the reading used here. It is the reading for which $d = 1/\gamma_1$ and for which the right-hand side of equation (9) equals $1 + d$.

1.32. `ExtremalH`.

Definition 1.31. Let $1 < p < \infty$, let $d = d(p)$ be the exponent of `DExp` and let $\epsilon = \epsilon_2(1, p)$ be the coefficient of `EpsCoef` taken with $n = 1$ and $k = 2$. Define $H \colon \mathbb{C} \to \mathbb{C}$ by

$$H(\xi) = \left(\frac{\xi}{|\xi|} + \epsilon\,\frac{|\xi|^3}{\xi^3}\right)|\xi|^{1/d}.$$

This is the extremal function (54) of Section 7 of the paper, which Section 7 obtains from the series (19) by taking $n = 1$ in the factorization (15) and $A_1 = 0$, $A_2 = 1$, $A_3 = A_4 = \cdots = 0$ in (19); there $\lambda_2 = \gamma_1 = 1/d$ and $\epsilon = \epsilon_2 = (1 - d)/(1 + 3d)$, which is the number $\epsilon_2(1, p)$ used here.

At $\xi = 0$ both quotients are read as 0, under the convention $w/0 = 0$, so the first factor vanishes and the formula gives $H(0) = 0$; this is the value the paper's H takes at the origin, and it is what makes H continuous there, since $1/d > 0$.

1.33. `KAlphaExp`.

Definition 1.32. For a real number p put

$$\kappa(p) = \frac{1}{6}\left(7 + \frac{1}{p-1} + \sqrt{1 + \frac{14}{p-1} + \frac{1}{(p-1)^2}}\right).$$

This is exactly the right-hand side of equation (9) of the paper, that is, the number which the integer $k \geq 1$ and the exponent $\alpha \in (0, 1]$ of Theorem 1 are required to satisfy $k + \alpha = \kappa(p)$.

As for the exponent $d(p)$ of equation (50), the conventions $\sqrt{t} = 0$ for $t < 0$ and $1/0 = 0$ make the formula total; the paper uses it only for $1 < p < \infty$, where $p - 1 > 0$ and the radicand is positive.

1.34. `MemHolderLocOn`.

Definition 1.33. Let $\Omega \subset \mathbb{C}$ be open, let $k \geq 0$ be an integer and let $0 < \alpha \leq 1$. A function $u \colon \mathbb{C} \to \mathbb{C}$ belongs to the Hölder space $C^{k,\alpha}_{\mathrm{loc}}(\Omega)$ if

(a) u is of class C^k on Ω, and
(b) for every compact $F \subset \Omega$ there is a constant $C \geq 0$ with

$$\left\|D^k u(x) - D^k u(y)\right\| \leq C\,|x - y|^{\alpha} \qquad \text{for all } x, y \in F.$$

Here $D^k u(x)$ denotes the k-th total derivative of u at x, a symmetric k-linear map on $\mathbb{R}^2$, and $\|\cdot\|$ its operator norm.

This is the space $C^{k,\alpha}_{\mathrm{loc}}(\Omega)$ of Section 1 of the paper, which is described there as the space of complex-valued $u \in C^k(\Omega)$ whose k-th order partial derivatives $D^\nu u$, $|\nu| = k$, are locally Hölder continuous with exponent α; condition (b) is the finiteness, for each compact F, of the second seminorm in the displayed norm $\|u\|_{C^{k,\alpha}(F)}$ of Section 1. Using the operator norm of the k-th total derivative in place of the sum $\sum_{|\nu|=k} |D^\nu u(x) - D^\nu u(y)|$ defines the same space: the two quantities bound one another up to factors depending only on k, because the partial derivatives $D^\nu u(x)$ with $|\nu| = k$ are precisely the values of $D^k u(x)$ on the k-fold tuples drawn from the basis $\{1, i\}$ of $\mathbb{R}^2$, and condition (b) is unchanged when C is multiplied by such a factor.

1.35. `MemHolderSmallOn`.

Definition 1.34. Let $\Omega \subset \mathbb{C}$ be open, let $k \geq 0$ be an integer and let $0 < \alpha \leq 1$. A function $u\colon \mathbb{C} \to \mathbb{C}$ belongs to $C^{k+\alpha}_{\mathrm{loc}}(\Omega)$ if one of the following holds.

(a) $\alpha = 1$ and u is of class C^{k+1} on Ω.

(b) $\alpha < 1$, u is of class C^k on Ω, and

$$\big\|D^k u(x) - D^k u(y)\big\| = o\big(|x-y|^\alpha\big)$$

uniformly on compact subsets of $\Omega \times \Omega$; explicitly, for every compact $F \subset \Omega$ and every $\delta > 0$ there is $\rho > 0$ such that

$$\big\|D^k u(x) - D^k u(y)\big\| \leq \delta\, |x-y|^\alpha \qquad \text{whenever } x, y \in F \text{ and } |x-y| < \rho.$$

Here $D^k u(x)$ and $\|\cdot\|$ are as in `MemHolderLocOn`.

This is the space $C^{k+\alpha}_{\mathrm{loc}}(\Omega)$ of Section 1 of the paper. Section 1 introduces it as the completion of $C^\infty(\Omega)$ in the topology of $C^{k,\alpha}_{\mathrm{loc}}(\Omega)$ and then states exactly the two descriptions used above: "for $\alpha = 1$ the space $C^{k+\alpha}_{\mathrm{loc}}(\Omega)$ coincides with $C^{k+1}(\Omega)$ whereas functions in $C^{k+\alpha}_{\mathrm{loc}}(\Omega)$ with $0 < \alpha < 1$ are characterized by the condition $\sum_{|\nu|=k} |D^\nu u(x) - D^\nu u(y)| = o(|x-y|^\alpha)$ uniformly on compact subsets of $\Omega \times \Omega$". We take these two descriptions as the definition. As in `MemHolderLocOn`, replacing the sum $\sum_{|\nu|=k}$ by the operator norm of the k-th total derivative changes the two sides only by factors depending on k alone, which leaves a little-o condition unchanged.

1.36. `SmoothOffZeros`.

Definition 1.35. A *smoothness-off-zeros package* is a single datum: a proof of the following statement.

Let $\Omega \subset \mathbb{C}$ be open, let $1 < p < \infty$, let $f\colon \mathbb{C} \to \mathbb{C}$ belong to $W^{1,2}_{\mathrm{loc}}(\Omega)$ in the sense of `MemWLocOn`, and let $g\colon \mathbb{C} \to \mathbb{C}$ satisfy

(i) $g = f$ almost everywhere on Ω;

(ii) g is continuous on Ω;

(iii) g is real-differentiable at almost every point of Ω;

(iv) g satisfies equation (12) on Ω in the sense of `SolvesEq12`.

Then g is C^∞ on $\Omega \setminus g^{-1}(0)$.

This is item (C3) of the paper's note for the formalization: "(General elliptic regularity, as used in Section 2.) A $W^{1,2}_{\mathrm{loc}}$ solution of (12) is C^∞ outside its zero set." The note directs that such quoted results be carried as explicit hypotheses rather than axioms; this definition is that bundle, so that a theorem needing (C3) takes a smoothness-off-zeros package as an argument.

The conclusion is asserted of the representative g, and the exceptional set is $g^{-1}(0)$. It has to be. Both $W^{1,2}_{\mathrm{loc}}$ membership and equation (12) see their argument only almost everywhere: the first by `MemWLocOn`, which is a family of integral identities and indicator L^2-conditions; the second because `SolvesEq12` is an identity between the weak Wirtinger derivatives of its argument, and by `IsWeakWirtingerOn` those are determined by the argument's almost-everywhere class, the pointwise values of the argument entering (12) only inside an "almost every z" quantifier. Consequently nothing about the smoothness of f at any individual point follows from a

hypothesis on f alone, the set $f^{-1}(0)$ is not determined modulo null sets, and the statement with f in place of g throughout would be false, making the package uninhabited; paper.tex:107 forbids exactly that. The paper writes the exceptional set as $f^{-1}(0) = \{z \in \Omega : \nabla u(z) = 0\}$ at paper.tex:156, that is, as the zero set of the continuous complex gradient, which is $g^{-1}(0)$ here. Note that, (12) being a condition on the class, hypothesis (iv) for g and the same condition for f are equivalent given (i); it is stated for g because g is the function the conclusion is about.

Hypotheses (ii) and (iii) only weaken the statement, so the package remains no stronger than the quoted result; they are what pin g down. Continuity makes g the unique continuous representative of f, two continuous functions agreeing almost everywhere on an open set being equal there; and almost-everywhere differentiability identifies the weak derivatives of g with its classical ones almost everywhere, so that the classical Wirtinger derivatives of g — the objects the elliptic theory differentiates — do satisfy (12) almost everywhere, which is the form of the equation the quoted result is about.

2. Lemmas

2.1. `WirtingerChainRule`.

Lemma 2.1. *Let $F, G\colon \mathbb{C} \to \mathbb{C}$, let $\xi \in \mathbb{C}$, and suppose that G is real-differentiable at ξ and that F is real-differentiable at $G(\xi)$. Then*

$$\frac{\partial(F\circ G)}{\partial\xi}(\xi) = \frac{\partial F}{\partial z}\big(G(\xi)\big)\frac{\partial G}{\partial\xi}(\xi) + \frac{\partial F}{\partial\bar z}\big(G(\xi)\big)\,\overline{\frac{\partial G}{\partial\bar\xi}(\xi)},$$

$$\frac{\partial(F\circ G)}{\partial\bar\xi}(\xi) = \frac{\partial F}{\partial z}\big(G(\xi)\big)\frac{\partial G}{\partial\bar\xi}(\xi) + \frac{\partial F}{\partial\bar z}\big(G(\xi)\big)\,\overline{\frac{\partial G}{\partial\xi}(\xi)},$$

where $\partial/\partial\xi$, $\partial/\partial z$ are as in `DZ` and $\partial/\partial\bar\xi$, $\partial/\partial\bar z$ are as in `DZbar`.

Proof. For a map h real-differentiable at a point w write $Dh(w)$ for its real Fréchet derivative there, a real-linear map $\mathbb{C} \to \mathbb{C}$. By `DZ` and `DZbar`,

$$\frac{\partial h}{\partial z}(w) = \tfrac12\big(Dh(w)[1] - i\,Dh(w)[i]\big), \qquad \frac{\partial h}{\partial\bar z}(w) = \tfrac12\big(Dh(w)[1] + i\,Dh(w)[i]\big). \tag{0}$$

Step 1: the real derivative in Wirtinger form. Let h be real-differentiable at w. We claim that

$$Dh(w)[v] = \frac{\partial h}{\partial z}(w)\,v + \frac{\partial h}{\partial\bar z}(w)\,\bar v \qquad \text{for every } v \in \mathbb{C}. \tag{1}$$

Both sides of (1) are $\mathbb{R}$-linear functions of v: the left side because $Dh(w)$ is $\mathbb{R}$-linear, the right side because $v \mapsto v$ and $v \mapsto \bar v$ are $\mathbb{R}$-linear and the two coefficients are fixed complex numbers. Since 1 and i form a basis of $\mathbb{C}$ over $\mathbb{R}$, it suffices to verify (1) for $v = 1$ and $v = i$.

For $v = 1$ we have $\bar v = 1$ and, by (0),

$$\frac{\partial h}{\partial z}(w) + \frac{\partial h}{\partial\bar z}(w) = \tfrac12\big(Dh(w)[1] - i\,Dh(w)[i]\big) + \tfrac12\big(Dh(w)[1] + i\,Dh(w)[i]\big) = Dh(w)[1].$$

For $v = i$ we have $\bar v = -i$ and, by (0),

$$\frac{\partial h}{\partial z}(w) - \frac{\partial h}{\partial\bar z}(w) = \tfrac12\big(-2i\,Dh(w)[i]\big) = -i\,Dh(w)[i],$$

so

$$\frac{\partial h}{\partial z}(w)\,i + \frac{\partial h}{\partial\bar z}(w)(-i) = i\Big(\frac{\partial h}{\partial z}(w) - \frac{\partial h}{\partial\bar z}(w)\Big) = i\big(-i\,Dh(w)[i]\big) = Dh(w)[i],$$

using $i(-i) = 1$. This proves (1).

Step 2: the real chain rule. Since G is real-differentiable at ξ and F is real-differentiable at $G(\xi)$, the composite $F\circ G$ is real-differentiable at ξ and $D(F\circ G)(\xi) = DF\big(G(\xi)\big)\circ DG(\xi)$, that is,

$$D(F\circ G)(\xi)[v] = DF\big(G(\xi)\big)\big[DG(\xi)[v]\big] \qquad (v \in \mathbb{C}).$$

Applying (1) to $h = F$, $w = G(\xi)$ and the vector $DG(\xi)[v]$ gives

$$D(F\circ G)(\xi)[v] = \frac{\partial F}{\partial z}\big(G(\xi)\big)DG(\xi)[v] + \frac{\partial F}{\partial \bar z}\big(G(\xi)\big)\overline{DG(\xi)[v]}. \tag{2}$$

Step 3: taking the two Wirtinger combinations. Abbreviate $A = \frac{\partial F}{\partial z}(G(\xi))$ and $B = \frac{\partial F}{\partial \bar z}(G(\xi))$. By (0) applied to $F\circ G$ at ξ, and then (2) with $v = 1$ and $v = i$,

$$\frac{\partial (F\circ G)}{\partial \xi}(\xi) = \tfrac12\Big(D(F\circ G)(\xi)[1] - i\,D(F\circ G)(\xi)[i]\Big) = A\cdot\tfrac12\big(DG(\xi)[1] - i\,DG(\xi)[i]\big) + B\cdot\tfrac12\Big(\overline{DG(\xi)[1]} - i\,\overline{DG(\xi)[i]}\Big),$$

after regrouping the four terms (legitimate since A and B are fixed complex numbers). By (0) the first bracket is $\frac{\partial G}{\partial \xi}(\xi)$. For the second, $\overline{i\,w} = -i\bar w$ for every $w \in \mathbb{C}$, so $-i\,\overline{DG(\xi)[i]} = \overline{i\,DG(\xi)[i]}$ and, conjugation being additive and $\frac12$ real,

$$\tfrac12\Big(\overline{DG(\xi)[1]} - i\,\overline{DG(\xi)[i]}\Big) = \overline{\tfrac12\big(DG(\xi)[1] + i\,DG(\xi)[i]\big)} = \overline{\frac{\partial G}{\partial \bar\xi}(\xi)}.$$

This gives the first identity of the lemma.

For the second, (0) gives

$$\frac{\partial (F\circ G)}{\partial \bar\xi}(\xi) = \tfrac12\Big(D(F\circ G)(\xi)[1] + i\,D(F\circ G)(\xi)[i]\Big) = A\cdot\tfrac12\big(DG(\xi)[1] + i\,DG(\xi)[i]\big) + B\cdot\tfrac12\Big(\overline{DG(\xi)[1]} + i\,\overline{DG(\xi)[i]}\Big).$$

The first bracket is $\frac{\partial G}{\partial \bar\xi}(\xi)$ by (0). For the second, $\overline{-i\,w} = i\bar w$, so $i\,\overline{DG(\xi)[i]} = \overline{-i\,DG(\xi)[i]}$ and

$$\tfrac12\Big(\overline{DG(\xi)[1]} + i\,\overline{DG(\xi)[i]}\Big) = \overline{\tfrac12\big(DG(\xi)[1] - i\,DG(\xi)[i]\big)} = \overline{\frac{\partial G}{\partial \xi}(\xi)}.$$

This gives the second identity. □

2.2. `ChiFirstDerivs`.

Lemma 2.2. *Let $U \subset \mathbb{C}$ be open, let $\chi, H\colon \mathbb{C}\to\mathbb{C}$ and let $\xi \in U$ be such that*

$$\chi\big(H(\zeta)\big) = \zeta \qquad \text{for every } \zeta \in U.$$

Assume that H is real-differentiable at ξ, that χ is real-differentiable at $H(\xi)$, and that $J(\xi) \neq 0$, where $J = |H_\xi|^2 - |H_{\bar\xi}|^2$ is the Jacobian of `Jac`. Then

$$\frac{\partial \chi}{\partial z}\big(H(\xi)\big) = \frac{\overline{H_\xi(\xi)}}{J(\xi)}, \qquad \frac{\partial \chi}{\partial \bar z}\big(H(\xi)\big) = -\frac{H_{\bar\xi}(\xi)}{J(\xi)},$$

with $\partial/\partial z$, $\partial/\partial\xi$ as in `DZ` and $\partial/\partial\bar z$, $\partial/\partial\bar\xi$ as in `DZbar`.

These are the two formulas (17) of Section 3 of the paper, repeated as (46) in Section 5. The hypotheses involve the pair (χ, H) only through the relation $\chi\circ H = \mathrm{id}$ on U together with differentiability at the two matched points, so the statement may equally be applied with the roles of χ and H exchanged; that is the direction in which Section 7 runs the hodograph correspondence.

Proof. Abbreviate

$$a = \frac{\partial \chi}{\partial z}\big(H(\xi)\big), \quad b = \frac{\partial \chi}{\partial \bar z}\big(H(\xi)\big), \quad A = \frac{\partial H}{\partial \xi}(\xi), \quad B = \frac{\partial H}{\partial \bar\xi}(\xi).$$

Step 1: the composite has the derivatives of the identity. By hypothesis $\chi\circ H$ agrees with the identity map $\zeta \mapsto \zeta$ at every point of U. Since U is open and $\xi \in U$, the two functions agree on a neighbourhood of ξ, hence have the same real Fréchet derivative at ξ, and therefore, by `DZ` and `DZbar`, the same $\partial/\partial\xi$ and $\partial/\partial\bar\xi$ at ξ. The identity map is real-differentiable with $D\,\mathrm{id}(\xi)[v] = v$ for every v, so by `DZ` and `DZbar`

$$\frac{\partial\,\mathrm{id}}{\partial\xi}(\xi) = \tfrac12(1 - i\cdot i) = 1, \qquad \frac{\partial\,\mathrm{id}}{\partial\bar\xi}(\xi) = \tfrac12(1 + i\cdot i) = 0.$$

Consequently

$$\frac{\partial(\chi\circ H)}{\partial\xi}(\xi)=1,\qquad \frac{\partial(\chi\circ H)}{\partial\bar\xi}(\xi)=0. \tag{1}$$

Step 2: the chain rule gives a linear system for a and b. The hypotheses of `WirtingerChainRule` hold with $F=\chi$ and $G=H$ at the point ξ: H is real-differentiable at ξ and χ is real-differentiable at $H(\xi)$. That lemma therefore gives

$$\frac{\partial(\chi\circ H)}{\partial\xi}(\xi)=a\,A+b\,\overline{B},\qquad \frac{\partial(\chi\circ H)}{\partial\bar\xi}(\xi)=a\,B+b\,\overline{A}.$$

Combining with (1),

$$a\,A+b\,\overline{B}=1, \tag{2}$$

$$a\,B+b\,\overline{A}=0. \tag{3}$$

Step 3: the determinant of the system is $J(\xi)$. For any $w\in\mathbb{C}$ one has $|w|^2=w\bar w$ as complex numbers, so by `Jac`

$$J(\xi)=|A|^2-|B|^2=A\overline{A}-B\overline{B}. \tag{4}$$

By hypothesis $J(\xi)\neq 0$.

Step 4: solving for a. Multiply (2) by $\overline{A}$ and (3) by $\overline{B}$:

$$a\,A\overline{A}+b\,\overline{B}\,\overline{A}=\overline{A},\qquad a\,B\overline{B}+b\,\overline{A}\,\overline{B}=0.$$

Subtracting the second from the first, the terms in b cancel because $\overline{B}\,\overline{A}=\overline{A}\,\overline{B}$, and we obtain

$$a\big(A\overline{A}-B\overline{B}\big)=\overline{A},$$

that is, $a\,J(\xi)=\overline{A}$ by (4). Since $J(\xi)\neq 0$,

$$a=\frac{\overline{A}}{J(\xi)}=\frac{\overline{H_\xi(\xi)}}{J(\xi)},$$

which is the first assertion.

Step 5: solving for b. Multiply (2) by B and (3) by A:

$$a\,AB+b\,\overline{B}B=B,\qquad a\,BA+b\,\overline{A}A=0.$$

Subtracting the second from the first, the terms in a cancel because $AB=BA$, and we obtain

$$b\big(\overline{B}B-\overline{A}A\big)=B,$$

that is, $-b\,J(\xi)=B$ by (4). Since $J(\xi)\neq 0$,

$$b=-\frac{B}{J(\xi)}=-\frac{H_{\bar\xi}(\xi)}{J(\xi)},$$

which is the second assertion. □

2.3. `MollificationL1Convergence`.

Lemma 2.3. Let $d\sigma$ denote Lebesgue measure on $\mathbb{C}\cong\mathbb{R}^2$ and let $g\in L^1(\mathbb{C})$, i.e. $g\colon\mathbb{C}\to\mathbb{C}$ is integrable for $d\sigma$. For each $m\in\mathbb{N}$ let φ_m be a smooth bump function centred at 0: a C^∞ function $\varphi_m\colon\mathbb{C}\to\mathbb{R}$ with radii $0<r_m^{\mathrm{in}}<r_m^{\mathrm{out}}$ such that $0\le\varphi_m\le 1$, $\varphi_m\equiv 1$ on $\{|y|\le r_m^{\mathrm{in}}\}$ and $\varphi_m(y)=0$ for $|y|\ge r_m^{\mathrm{out}}$. Write

$$\varrho_m=\frac{\varphi_m}{\int\varphi_m\,d\sigma}$$

for its normalisation, so that $\varrho_m\in C^\infty(\mathbb{C})$, $\varrho_m\ge 0$, $\varrho_m(y)=0$ for $|y|\ge r_m^{\mathrm{out}}$, and $\int\varrho_m\,d\sigma=1$. Assume

$$r_m^{\mathrm{out}}\le 1\quad\text{for every } m,\qquad\text{and}\qquad r_m^{\mathrm{out}}\longrightarrow 0.$$

Then, with $(\varrho_m*g)(x)=\int\varrho_m(t)\,g(x-t)\,d\sigma(t)$,

$$\int_{\mathbb{C}}\big|(\varrho_m*g)(x)-g(x)\big|\,d\sigma(x)\longrightarrow 0\qquad(m\to\infty).$$

This is the L^1-norm form of the convergence of mollifications. Only the pointwise form (at a point of continuity) and the almost-everywhere form are standard background; the L^1 form is what is needed whenever the limit $m \to \infty$ has to be taken inside a further integration, for there almost-everywhere convergence of the $\varrho_m * g$ would have to be accompanied by an m-uniform dominating function, which mere local integrability of g does not supply. The normalisation $\int \varrho_m \, d\sigma = 1$ is what makes the statement true, and the bound $r_m^{\text{out}} \le 1$ only serves to confine all the supports involved to one fixed ball.

Proof. Throughout, all integrals without an indicated domain are over $\mathbb{C}$ against $d\sigma$. Note first that each ϱ_m is well defined, since $\varphi_m \ge 0$ is continuous, is ≥ 1 on the ball $\{|y| \le r_m^{\text{in}}\}$ of positive measure and vanishes off a bounded set, so $0 < \int \varphi_m \, d\sigma < \infty$; and that $\varrho_m \in L^1(\mathbb{C})$ with $\int |\varrho_m| \, d\sigma = \int \varrho_m \, d\sigma = 1$, because $\varrho_m \ge 0$.

Step 1: convolution with an L^1 kernel is an L^1 contraction (Young's inequality at $p = 1$). We claim that for every real-valued $\rho \in L^1(\mathbb{C})$ and every $f \in L^1(\mathbb{C})$,

$$\int \big|(\rho * f)(x)\big| \, d\sigma(x) \le \Big(\int |\rho(t)| \, d\sigma(t)\Big)\Big(\int |f(x)| \, d\sigma(x)\Big). \tag{1}$$

Indeed, the map $(x,t) \mapsto \rho(t) f(x-t)$ is measurable on $\mathbb{C} \times \mathbb{C}$, being the product of a measurable function of t with the composition of the measurable f with the continuous map $(x,t) \mapsto x - t$; and it is integrable for $d\sigma \otimes d\sigma$, since by Tonelli's theorem and the translation invariance of $d\sigma$

$$\iint |\rho(t)| \, |f(x-t)| \, d\sigma(x) \, d\sigma(t) = \int |\rho(t)| \Big(\int |f(x-t)| \, d\sigma(x)\Big) d\sigma(t) = \Big(\int |\rho|\Big)\Big(\int |f|\Big) < \infty.$$

In particular $(\rho * f)(x) = \int \rho(t) f(x-t) \, d\sigma(t)$ is an absolutely convergent integral for almost every x, the function $\rho * f$ is in $L^1(\mathbb{C})$, and

$$\int \big|(\rho * f)(x)\big| \, d\sigma(x) \le \int \Big(\int |\rho(t)| \, |f(x-t)| \, d\sigma(t)\Big) d\sigma(x) = \Big(\int |\rho|\Big)\Big(\int |f|\Big),$$

where the inequality is the triangle inequality for the inner integral and the equality is the computation just displayed, read with the order of integration exchanged by Fubini's theorem. This proves (1). Applied to $\rho = \varrho_m$, for which $\int |\varrho_m| = 1$, it gives

$$\int \big|(\varrho_m * f)(x)\big| \, d\sigma(x) \le \int |f| \, d\sigma \qquad \big(f \in L^1(\mathbb{C}),\ m \in \mathbb{N}\big). \tag{2}$$

Step 2: the assertion for a continuous compactly supported function. Let $h \colon \mathbb{C} \to \mathbb{C}$ be continuous with compact support. We claim

$$\int \big|(\varrho_m * h)(x) - h(x)\big| \, d\sigma(x) \longrightarrow 0. \tag{3}$$

Being continuous with compact support, h is bounded, say $|h| \le C$, and integrable; choose $R > 0$ with $\operatorname{supp} h \subseteq \{|y| \le R\}$ and put

$$S = \{x \in \mathbb{C} : |x| \le 1 + R\},$$

a closed ball, hence measurable with $\sigma(S) < \infty$.

First, both h and $\varrho_m * h$ vanish off S. For h this is clear from $R \le 1 + R$. For $\varrho_m * h$, the support of a convolution is contained in the closure of the sum of the supports, so if $(\varrho_m * h)(x) \ne 0$ then $x = a + b$ with $|a| < r_m^{\text{out}}$ and $|b| \le R$, whence $|x| \le |a| + |b| < r_m^{\text{out}} + R \le 1 + R$ by the hypothesis $r_m^{\text{out}} \le 1$, i.e. $x \in S$.

Second, on all of $\mathbb{C}$ we have the m-independent bound

$$\big|(\varrho_m * h)(x) - h(x)\big| \le 2C + 1. \tag{4}$$

Indeed, since $\int \varrho_m \, d\sigma = 1$,

$$(\varrho_m * h)(x) - h(x) = \int \varrho_m(t) \big[h(x-t) - h(x)\big] \, d\sigma(t),$$

and $\varrho_m \ge 0$ vanishes for $|t| \ge r_m^{\text{out}}$, so the modulus of the left-hand side is at most $\sup_{|y-x| < r_m^{\text{out}}} |h(y) - h(x)| \le 2C \le 2C + 1$.

Third, for each fixed x we have $(\varrho_m * h)(x) \to h(x)$. This is the pointwise form of the convergence of mollifications: given $\epsilon > 0$, continuity of h at x provides $\tau > 0$ with $|h(y)-h(x)| \le \epsilon$ whenever $|y - x| \le \tau$; since $r_m^{\mathrm{out}} \to 0$ we have $r_m^{\mathrm{out}} \le \tau$ for all large m, and then the display above bounds $|(\varrho_m * h)(x) - h(x)|$ by $\int \varrho_m(t)\,\epsilon\,d\sigma(t) = \epsilon$.

Now apply the dominated convergence theorem to the functions $x \mapsto |(\varrho_m * h)(x) - h(x)|$, which are measurable (each $\varrho_m * h$ is in L^1 by Step 1, and h is measurable). By the three observations they converge pointwise to 0 and are dominated by the fixed integrable function $(2C + 1)\mathbf{1}_S$: on S this is (4), and off S the function is identically 0. Hence their integrals tend to 0, which is (3).

Step 3: the general case, by density. Let $\eta > 0$. The continuous compactly supported functions are dense in $L^1(\mathbb{C})$, so there is a continuous h with compact support such that

$$\int |g - h|\,d\sigma \le \frac{\eta}{3}; \tag{5}$$

such an h is in particular integrable. Fix m. Since $\varrho_m \in L^1$ and $g, h \in L^1$, Step 1 shows that for almost every x both defining integrals of $(\varrho_m * g)(x)$ and $(\varrho_m * h)(x)$ converge absolutely, so for almost every x

$$(\varrho_m * g)(x) - (\varrho_m * h)(x) = \int \varrho_m(t)\big[g(x - t) - h(x - t)\big]\,d\sigma(t) = \big(\varrho_m * (g - h)\big)(x).$$

Combining this with (2) applied to $f = g - h \in L^1$ and with (5),

$$\int \big|(\varrho_m * g)(x) - (\varrho_m * h)(x)\big|\,d\sigma(x) = \int \big|\big(\varrho_m * (g - h)\big)(x)\big|\,d\sigma(x) \le \int |g - h|\,d\sigma \le \frac{\eta}{3}. \tag{6}$$

The pointwise triangle inequality

$$\big|(\varrho_m * g)(x) - g(x)\big| \le \big|(\varrho_m * g)(x) - (\varrho_m * h)(x)\big| + \big|(\varrho_m * h)(x) - h(x)\big| + \big|h(x) - g(x)\big|$$

holds for every x, and each of the three functions on the right is integrable: the first and second because $\varrho_m * g$, $\varrho_m * h$, g and h all lie in L^1 by Step 1, the third likewise. Integrating and using additivity of the integral, then (6) and (5), we get for every m

$$\int \big|(\varrho_m * g)(x) - g(x)\big|\,d\sigma(x) \le \frac{\eta}{3} + \int \big|(\varrho_m * h)(x) - h(x)\big|\,d\sigma(x) + \frac{\eta}{3}. \tag{7}$$

By Step 2 applied to this h, there is M such that the middle term is $< \eta/3$ for every $m \ge M$. Hence

$$\int \big|(\varrho_m * g)(x) - g(x)\big|\,d\sigma(x) < \eta \qquad (m \ge M).$$

Since $\eta > 0$ was arbitrary and the quantity on the left is nonnegative, it tends to 0 as $m \to \infty$, which is the assertion. □

2.4. `WeakPartialLineIntegral`.

Lemma 2.4. *Let $\Omega \subseteq \mathbb{C}$ be open, let $j \in \{0, 1\}$, write $e_0 = 1$ and $e_1 = i$, and let $u, D\colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *$u \in L^1_{\mathrm{loc}}(\Omega)$ and $D \in L^1_{\mathrm{loc}}(\Omega)$, in the sense of `MemLpLocOn`;*

(H2) *D is the distributional j-th partial derivative of u on Ω, in the sense of `IsWeakPartialOn`.*

Let $K \subseteq \Omega$ be compact. Then there is $\delta > 0$ such that for every real t with $|t| \le \delta$,

$$u(z + te_j) - u(z) = \int_0^t D(z + se_j)\,ds \qquad \text{for almost every } z \in K.$$

This is the absolute continuity of u along almost every line parallel to e_j, localised to a compact subset of Ω. It is the only route from (H2), which sees u exclusively through integrals against test functions, to pointwise information about the increments of u itself, and it is what the paper relies on implicitly wherever Sections 4 to 6 treat the $W^{1,2}$ derivatives of H, and of the inverse χ, as classical derivatives: the integration-by-parts identity alone says nothing about any particular representative of the L^1-class of u, whereas the displayed identity pins down every increment of u along e_j from D. The restriction to a compact K and to small $|t|$ is not a weakening: the

translate $z + te_j$ must stay inside Ω for the right-hand side to be meaningful, and no uniform bound on $|t|$ is available on Ω itself.

Proof. Write $d\sigma$ for plane Lebesgue measure. Throughout, $\partial_j v(z) = \mathrm{d}v(z)[e_j]$ denotes the classical partial derivative of a differentiable v along e_j.

Step 0: the scale δ. Since K is compact, Ω is open and $K \subseteq \Omega$, there is $\delta > 0$ with

$$K^{3\delta} := \{z : \operatorname{dist}(z, K) \le 3\delta\} \subseteq \Omega;$$

indeed either $\Omega = \mathbb{C}$, where any $\delta > 0$ serves, or $\operatorname{dist}(K, \mathbb{C} - \Omega) > 0$ because K is compact and $\mathbb{C} - \Omega$ is closed, nonempty and disjoint from K, and then any δ below a third of that distance serves. The set $K^{3\delta}$ is closed and bounded, hence compact, and $K \subseteq K^{\delta} \subseteq K^{2\delta} \subseteq K^{3\delta}$. We prove the assertion with this δ. For $t = 0$ both sides vanish, so we may assume $0 < |t| \le \delta$.

Step 1: the mollifier. Fix a function $\varrho\colon \mathbb{C} \to \mathbb{R}$ of class C^∞ with $\varrho \ge 0$, $\varrho(y) = 0$ for $|y| \ge 1$ and $\int \varrho \, d\sigma = 1$, and for $0 < \varepsilon \le \delta$ put $\varrho_\varepsilon(y) = \varepsilon^{-2}\varrho(y/\varepsilon)$, which is C^∞, vanishes for $|y| \ge \varepsilon$ and has $\int \varrho_\varepsilon \, d\sigma = 1$. Let

$$U = \mathbf{1}_{K^{3\delta}}\, u, \qquad V = \mathbf{1}_{K^{3\delta}}\, D.$$

By (H1) applied to the compact set $K^{3\delta} \subseteq \Omega$ and `MemLpLocOn`, U and V belong to $L^1(\mathbb{C})$. Put $u_\varepsilon = U * \varrho_\varepsilon$ and $v_\varepsilon = V * \varrho_\varepsilon$.

Step 2: u_ε is smooth and $\partial_j u_\varepsilon = v_\varepsilon$ near K. The convolution of an L^1 function with a compactly supported C^∞ function is C^∞, with derivatives obtained by differentiating the smooth factor; so u_ε is C^∞ on $\mathbb{C}$ and

$$\partial_j u_\varepsilon(z) = \big(U * \partial_j \varrho_\varepsilon\big)(z) = \int U(w)\, (\partial_j \varrho_\varepsilon)(z - w)\, d\sigma(w). \tag{1}$$

Fix z with $\operatorname{dist}(z, K) \le 2\delta$ and $0 < \varepsilon \le \delta$, and define $\varphi(w) = \varrho_\varepsilon(z - w)$, read as a complex-valued function. Then φ is C^∞ and vanishes off the closed ball $\{w : |w - z| \le \varepsilon\}$, which is compact and contained in $K^{3\delta} \subseteq \Omega$ because $\operatorname{dist}(z, K) \le 2\delta$ and $\varepsilon \le \delta$. Hence $\operatorname{tsupp}\varphi \subseteq \Omega$ and φ is a test function for Ω in the sense of `IsTestFn`. The map $w \mapsto z - w$ is affine with derivative $-\mathrm{id}$, so by the chain rule

$$\mathrm{d}\varphi(w)[e_j] = -(\partial_j \varrho_\varepsilon)(z - w).$$

Therefore (H2), applied to φ, reads

$$-\int u(w)\, (\partial_j \varrho_\varepsilon)(z - w)\, d\sigma(w) = -\int D(w)\, \varrho_\varepsilon(z - w)\, d\sigma(w),$$

that is

$$\int u(w)\, (\partial_j \varrho_\varepsilon)(z - w)\, d\sigma(w) = \int D(w)\, \varrho_\varepsilon(z - w)\, d\sigma(w). \tag{2}$$

Both integrands vanish off $\{w : |w - z| \le \varepsilon\} \subseteq K^{3\delta}$, where $u = U$ and $D = V$; so by (1) the left side of (2) is $\partial_j u_\varepsilon(z)$ and the right side is $v_\varepsilon(z)$. Hence

$$\partial_j u_\varepsilon(z) = v_\varepsilon(z) \qquad \big(\operatorname{dist}(z, K) \le 2\delta,\ 0 < \varepsilon \le \delta\big). \tag{3}$$

Step 3: the finite-difference identity for u_ε. Let $z \in K$ and let s lie between 0 and t. Then $\operatorname{dist}(z + se_j, K) \le |s| \le |t| \le \delta \le 2\delta$, so (3) applies at $z + se_j$. The function $s \mapsto u_\varepsilon(z + se_j)$ is C^∞ on $\mathbb{R}$ with derivative $\partial_j u_\varepsilon(z + se_j)$, by the chain rule, so the fundamental theorem of calculus and (3) give

$$u_\varepsilon(z + te_j) - u_\varepsilon(z) = \int_0^t \partial_j u_\varepsilon(z + se_j)\, ds = \int_0^t v_\varepsilon(z + se_j)\, ds. \tag{4}$$

Step 4: letting $\varepsilon \to 0$ in (4). For $g \in L^1(\mathbb{C})$ the mollifications converge in L^1, that is $\|g * \varrho_\varepsilon - g\|_{L^1(\mathbb{C})} \to 0$ as $\varepsilon \to 0$, by `MollificationL1Convergence`; applied to U and V,

$$\|u_\varepsilon - U\|_{L^1(\mathbb{C})} \to 0, \qquad \|v_\varepsilon - V\|_{L^1(\mathbb{C})} \to 0. \tag{5}$$

Regard both sides of (4) as functions of $z \in K$, with t fixed. For the left side, translation invariance of $d\sigma$ gives

$$\int_K \bigl|u_\varepsilon(z+te_j) - U(z+te_j)\bigr|\, d\sigma(z) \le \|u_\varepsilon - U\|_{L^1(\mathbb{C})},$$

and the same bound without the translation, so $z \mapsto u_\varepsilon(z+te_j) - u_\varepsilon(z)$ converges in $L^1(K)$ to $z \mapsto U(z+te_j) - U(z)$, which equals $u(z+te_j) - u(z)$ for $z \in K$ because z and $z+te_j$ both lie in $K^{3\delta}$. For the right side, write I_t for the interval between 0 and t; the function $(z,s) \mapsto \bigl(v_\varepsilon - V\bigr)(z+se_j)$ is measurable on $K \times I_t$, being the composition of a measurable function with the continuous map $(z,s) \mapsto z+se_j$, so by Tonelli's theorem and translation invariance

$$\int_K \Bigl|\int_0^t \bigl(v_\varepsilon - V\bigr)(z+se_j)\, ds\Bigr| d\sigma(z) \le \int_{I_t}\Bigl(\int_K \bigl|v_\varepsilon - V\bigr|(z+se_j)\, d\sigma(z)\Bigr) ds \le |t|\, \|v_\varepsilon - V\|_{L^1(\mathbb{C})},$$

which tends to 0 by (5); and $V(z+se_j) = D(z+se_j)$ for $z \in K$ and $|s| \le \delta$, because then $z+se_j \in K^\delta \subseteq K^{3\delta}$. Thus the left sides of (4) converge in $L^1(K)$ to $z \mapsto u(z+te_j) - u(z)$ and the right sides converge in $L^1(K)$ to $z \mapsto \int_0^t D(z+se_j)\, ds$. Two $L^1(K)$ limits of one and the same family agree almost everywhere, so

$$u(z+te_j) - u(z) = \int_0^t D(z+se_j)\, ds \qquad \text{for almost every } z \in K,$$

which is the assertion for this t. As t with $0 < |t| \le \delta$ was arbitrary, and the case $t = 0$ is trivial, the proof is complete. □

2.5. `WeakPartialDiffQuotientAe`.

Lemma 2.5. *Let $\Omega \subseteq \mathbb{C}$ be open, let $j \in \{0,1\}$, write $e_0 = 1$ and $e_1 = i$, and let $u, D\colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *$u \in L^1_{\mathrm{loc}}(\Omega)$ and $D \in L^1_{\mathrm{loc}}(\Omega)$, in the sense of `MemLpLocOn`;*
(H2) *D is the distributional j-th partial derivative of u on Ω, in the sense of `IsWeakPartialOn`.*

Let $K \subseteq \Omega$ be compact. Then there is a sequence $(t_l)_{l\ge1}$ of nonzero real numbers with $t_l \to 0$ such that

$$\frac{u(z+t_le_j) - u(z)}{t_l} \longrightarrow D(z) \qquad (l \to \infty)$$

for almost every $z \in K$.

This is the statement that the distributional derivative is a genuine limit of difference quotients of u, almost everywhere, and it is what makes the distributional derivative comparable with a classical one. The paper uses it implicitly throughout Sections 4 to 6, where the $W^{1,2}$ derivatives of H and of the inverse χ are manipulated as classical derivatives.

Two features of the statement are deliberate and are exactly what the argument supplies. First, the increments form a sequence rather than a continuous parameter: the difference quotients converge to D in $L^1(K)$ as $t \to 0$, and almost-everywhere convergence is obtained from that only along a subsequence. A full statement, with $t \to 0$ continuously and one exceptional null set, is false in general, because the exceptional set depends on the increment. Second, the conclusion is confined to a compact K, since the translate $z+te_j$ must remain in Ω. Neither restriction costs anything where the statement is used: at a point where u is differentiable, every null sequence of increments gives the classical derivative, and Ω is a countable union of compact sets.

Proof. Write $d\sigma$ for plane Lebesgue measure.

Step 1: a compact collar and the truncation of D. Since K is compact, Ω is open and $K \subseteq \Omega$, there is $\delta_1 > 0$ with

$$K^{\delta_1} := \{z : \operatorname{dist}(z,K) \le \delta_1\} \subseteq \Omega;$$

either $\Omega = \mathbb{C}$, where any $\delta_1 > 0$ serves, or $\operatorname{dist}(K, \mathbb{C}-\Omega) > 0$ because K is compact and $\mathbb{C}-\Omega$ is closed, nonempty and disjoint from K, and then any δ_1 below that distance serves. The set K^{δ_1} is closed and bounded, hence compact. Put $V = \mathbf{1}_{K^{\delta_1}} D$. By (H1) applied to the compact set $K^{\delta_1} \subseteq \Omega$ and `MemLpLocOn`, $V \in L^1(\mathbb{C})$; by the same hypothesis applied to K^{δ_1}, u is almost

everywhere strongly measurable on K^{δ_1}, so for each fixed t the difference quotient below is almost everywhere strongly measurable on K.

Step 2: the line-integral identity. By `WeakPartialLineIntegral` applied to Ω, j, u, D and K there is $\delta_2 > 0$ such that for every real t with $|t| \le \delta_2$,

$$u(z + te_j) - u(z) = \int_0^t D(z + se_j)\, ds \qquad \text{for almost every } z \in K. \tag{1}$$

Put $\delta = \min\{\delta_1, \delta_2\} > 0$.

Step 3: the difference quotients converge in $L^1(K)$. For $0 < |t| \le \delta$ put

$$\Delta_t(z) = \frac{u(z + te_j) - u(z)}{t} \qquad (z \in \mathbb{C}).$$

Since $\frac{1}{t}\int_0^t D(z)\, ds = D(z)$ for every z, (1) gives, for almost every $z \in K$,

$$\Delta_t(z) - D(z) = \frac{1}{t}\int_0^t \big[D(z + se_j) - D(z)\big]\, ds. \tag{2}$$

Let $z \in K$ and $|s| \le \delta$. Then $\operatorname{dist}(z + se_j, K) \le |s| \le \delta_1$, so $z + se_j \in K^{\delta_1}$ and hence $D(z + se_j) = V(z + se_j)$; and $z \in K \subseteq K^{\delta_1}$, so $D(z) = V(z)$. Therefore the bracket in (2) may be replaced by $V(z + se_j) - V(z)$. Writing I_t for the interval between 0 and t, whose length is $|t|$, and noting that $(z, s) \mapsto V(z + se_j) - V(z)$ is measurable on $\mathbb{C} \times I_t$, being built from the measurable V and the continuous map $(z, s) \mapsto z + se_j$, we get from (2), the triangle inequality for integrals, Tonelli's theorem and $K \subseteq \mathbb{C}$ that

$$\int_K \big|\Delta_t - D\big|\, d\sigma \le \frac{1}{|t|}\int_{I_t}\Big(\int_{\mathbb{C}} \big|V(z + se_j) - V(z)\big|\, d\sigma(z)\Big) ds \le \sup_{|s|\le|t|} \big\|V(\cdot + se_j) - V\big\|_{L^1(\mathbb{C})},$$

the last step because the inner integral is at most that supremum for every $s \in I_t$ and $|I_t| = |t|$. Translation is continuous in $L^1(\mathbb{C})$, so $\|V(\cdot + se_j) - V\|_{L^1(\mathbb{C})} \to 0$ as $s \to 0$, and therefore the right-hand side tends to 0 as $t \to 0$. Hence

$$\int_K \big|\Delta_t - D\big|\, d\sigma \longrightarrow 0 \qquad \text{as } t \to 0, \tag{3}$$

that is, $\Delta_t \to D$ in $L^1(K)$.

Step 4: extraction of an almost everywhere convergent sequence. Put $\tau_l = \delta/l$ for $l \ge 1$. Then $\tau_l \ne 0$, $|\tau_l| \le \delta$ and $\tau_l \to 0$, so by (3) we have $\int_K |\Delta_{\tau_l} - D|\, d\sigma \to 0$, i.e. $\Delta_{\tau_l} \to D$ in $L^1(K)$. Convergence in $L^1(K)$ implies convergence in measure on K: for $\eta > 0$,

$$\sigma\big(\{z \in K : |\Delta_{\tau_l}(z) - D(z)| \ge \eta\}\big) \le \frac{1}{\eta}\int_K \big|\Delta_{\tau_l} - D\big|\, d\sigma \longrightarrow 0.$$

A sequence of almost everywhere strongly measurable functions converging in measure has a subsequence converging almost everywhere; let $l_1 < l_2 < \cdots$ index such a subsequence and put $t_i = \tau_{l_i}$. Then each t_i is nonzero, $t_i \to 0$ because $l_i \to \infty$, and $\Delta_{t_i} \to D$ almost everywhere on K, which is precisely the assertion. □

2.6. `WeakPartialEqFDerivAE`.

Lemma 2.6. *Let $\Omega \subseteq \mathbb{C}$ be open, let $u\colon \mathbb{C} \to \mathbb{C}$ and let $D_0, D_1\colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *$u \in L^1_{\mathrm{loc}}(\Omega)$ and $D_j \in L^1_{\mathrm{loc}}(\Omega)$ for $j = 0, 1$, in the sense of `MemLpLocOn`;*

(H2) *D_j is the distributional j-th partial derivative of u on Ω for $j = 0, 1$, in the sense of `IsWeakPartialOn`;*

(H3) *u is differentiable at almost every point of Ω, in the real Fréchet sense.*

Then for $j = 0, 1$,

$$D_j(z) = \partial_j u(z) \qquad \text{for almost every } z \in \Omega,$$

where ∂_0 and ∂_1 are the classical partial derivatives along 1 and i, that is $\partial_j u = D^{[j]}u$ in the sense of `PartialWord`.

The tablet carries two unrelated notions of derivative: $W^{k,s}_{\mathrm{loc}}$ and quasiregularity are written with the integration-by-parts identity of `IsWeakPartialOn`, while the Wirtinger operators, the

polar derivatives of Section 4 and the iterated partial derivatives D^ν of Sections 5 and 6 are all built from the real Fréchet derivative. This lemma is the bridge. The hypotheses are exactly what the data of a $W^{1,1}_{\mathrm{loc}}$ function provides, with D_j the member of the coherent family indexed by the one-letter word $[j]$, together with the almost-everywhere differentiability that item (C2) of the paper's note supplies for the inverse of a quasiconformal homeomorphism. Neither (H2) nor (H3) is dispensable. Without (H3) the classical derivative need not exist at any point, so there is nothing to compare D_j with. And (H3) alone does not make the classical derivative a distributional one: the Cantor function on $[0,1]$, extended to the plane so as not to depend on the second coordinate, is continuous and differentiable almost everywhere with classical derivative 0 almost everywhere, yet 0 is not its distributional derivative. So the proof below must consume (H2), and it does, at Step 2.

Proof. Write $d\sigma$ for plane Lebesgue measure, $e_0 = 1$ and $e_1 = i$, so that $\partial_j v(z) = \mathrm{d}v(z)[e_j]$ and, by the recursion of `PartialWord`, $D^{[j]}v = \partial_j v$. Put $w_j := \partial_j u$, defined at every point where u is differentiable. Fix $j \in \{0,1\}$.

Step 0: reduction to a compact set. Every open subset of the plane is a countable union of compact sets, say $\Omega = \bigcup_{m\ge1} K^{(m)}$ with each $K^{(m)} \subseteq \Omega$ compact. A countable union of null sets is null, so it suffices to prove that $D_j = w_j$ almost everywhere on each $K^{(m)}$. Fix therefore a compact $K \subseteq \Omega$. Since K is compact, Ω is open and $K \subseteq \Omega$, there is $\delta > 0$ with

$$K^{3\delta} := \{z : \operatorname{dist}(z,K) \le 3\delta\} \subseteq \Omega;$$

indeed either $\Omega = \mathbb{C}$, where any δ serves, or $\operatorname{dist}(K, \mathbb{C}-\Omega) > 0$ because K is compact and $\mathbb{C}-\Omega$ is closed and disjoint from K, and then any δ below a third of that distance serves. The set $K^{3\delta}$ is closed and bounded, hence compact, and $K \subseteq K^{\delta} \subseteq K^{2\delta} \subseteq K^{3\delta}$.

Step 1: the mollifier. Fix once and for all a function $\varrho\colon \mathbb{C} \to \mathbb{R}$ of class C^∞ with $\varrho \ge 0$, $\varrho(y) = 0$ for $|y| \ge 1$ and $\int \varrho\, d\sigma = 1$, and for $0 < \varepsilon \le \delta$ put $\varrho_\varepsilon(y) = \varepsilon^{-2}\varrho(y/\varepsilon)$, which is C^∞, vanishes for $|y| \ge \varepsilon$ and has $\int \varrho_\varepsilon\, d\sigma = 1$. Let

$$U = \mathbf{1}_{K^{3\delta}}\, u, \qquad V_j = \mathbf{1}_{K^{3\delta}}\, D_j.$$

By (H1) applied to the compact set $K^{3\delta} \subseteq \Omega$ and `MemLpLocOn`, U and V_j belong to $L^1(\mathbb{C})$. Put $u_\varepsilon = U * \varrho_\varepsilon$ and $v_{j,\varepsilon} = V_j * \varrho_\varepsilon$.

Step 2: u_ε is smooth and $\partial_j u_\varepsilon = v_{j,\varepsilon}$ near K. The convolution of an L^1 function with a compactly supported C^∞ function is C^∞, with derivatives obtained by differentiating the smooth factor; so u_ε is C^∞ on $\mathbb{C}$ and

$$\partial_j u_\varepsilon(z) = \big(U * \partial_j \varrho_\varepsilon\big)(z) = \int U(w)\,(\partial_j \varrho_\varepsilon)(z-w)\, d\sigma(w). \tag{1}$$

Fix z with $\operatorname{dist}(z,K) \le 2\delta$ and $0 < \varepsilon \le \delta$, and define $\varphi(w) = \varrho_\varepsilon(z-w)$, read as a complex-valued function. Then φ is C^∞ and vanishes off the closed ball $\{w : |w-z| \le \varepsilon\}$, which is compact and contained in $K^{3\delta} \subseteq \Omega$ because $\operatorname{dist}(z,K) \le 2\delta$ and $\varepsilon \le \delta$. Hence $\operatorname{tsupp}\varphi \subseteq \Omega$ and φ is a test function for Ω in the sense of `IsTestFn`. The map $w \mapsto z - w$ is affine with derivative $-\mathrm{id}$, so by the chain rule

$$\mathrm{d}\varphi(w)[e_j] = -(\partial_j \varrho_\varepsilon)(z-w).$$

Therefore (H2) for this j, applied to φ, reads

$$-\int u(w)\,(\partial_j \varrho_\varepsilon)(z-w)\, d\sigma(w) = -\int D_j(w)\,\varrho_\varepsilon(z-w)\, d\sigma(w),$$

that is

$$\int u(w)\,(\partial_j \varrho_\varepsilon)(z-w)\, d\sigma(w) = \int D_j(w)\,\varrho_\varepsilon(z-w)\, d\sigma(w). \tag{2}$$

Both integrands vanish off $\{w : |w-z| \le \varepsilon\} \subseteq K^{3\delta}$, where $u = U$ and $D_j = V_j$; so by (1) the left side of (2) is $\partial_j u_\varepsilon(z)$ and the right side is $v_{j,\varepsilon}(z)$. Hence

$$\partial_j u_\varepsilon(z) = v_{j,\varepsilon}(z) \qquad \big(\operatorname{dist}(z,K) \le 2\delta,\ 0 < \varepsilon \le \delta\big). \tag{3}$$

Step 3: the finite-difference identity for u_ε. Let $0 < |t| \le \delta$, let $z \in K$ and let s lie between 0 and t. Then $\operatorname{dist}(z + se_j, K) \le |s| \le \delta \le 2\delta$, so (3) applies at $z + se_j$. The function $s \mapsto u_\varepsilon(z + se_j)$ is C^∞ on $\mathbb{R}$ with derivative $\partial_j u_\varepsilon(z + se_j)$, by the chain rule, so the fundamental theorem of calculus and (3) give

$$u_\varepsilon(z + te_j) - u_\varepsilon(z) = \int_0^t \partial_j u_\varepsilon(z + se_j)\, ds = \int_0^t v_{j,\varepsilon}(z + se_j)\, ds. \tag{4}$$

Step 4: letting $\varepsilon \to 0$ in (4). For $g \in L^1(\mathbb{C})$ one has $\|g * \varrho_\varepsilon - g\|_{L^1(\mathbb{C})} \to 0$ as $\varepsilon \to 0$; applied to U and V_j,

$$\|u_\varepsilon - U\|_{L^1(\mathbb{C})} \to 0, \qquad \|v_{j,\varepsilon} - V_j\|_{L^1(\mathbb{C})} \to 0. \tag{5}$$

Fix t with $0 < |t| \le \delta$ and regard both sides of (4) as functions of $z \in K$. For the left side, translation invariance of $d\sigma$ gives

$$\int_K \big|u_\varepsilon(z + te_j) - U(z + te_j)\big|\, d\sigma(z) \le \|u_\varepsilon - U\|_{L^1(\mathbb{C})},$$

and the same bound without the translation, so $z \mapsto u_\varepsilon(z + te_j) - u_\varepsilon(z)$ converges in $L^1(K)$ to $z \mapsto U(z + te_j) - U(z)$, which equals $u(z + te_j) - u(z)$ for $z \in K$ because z and $z + te_j$ both lie in $K^{3\delta}$. For the right side, write I_t for the interval between 0 and t; the integrand is measurable on $K \times I_t$, and by Tonelli's theorem and translation invariance

$$\int_K \Big|\int_0^t \big(v_{j,\varepsilon} - V_j\big)(z + se_j)\, ds\Big| d\sigma(z) \le \int_{I_t}\Big(\int_K \big|v_{j,\varepsilon} - V_j\big|(z + se_j)\, d\sigma(z)\Big) ds \le |t|\, \|v_{j,\varepsilon} - V_j\|_{L^1(\mathbb{C})},$$

which tends to 0 by (5); and $V_j(z + se_j) = D_j(z + se_j)$ for $z \in K$ and $|s| \le \delta$. Two $L^1(K)$ limits of one and the same family agree almost everywhere, so

$$u(z + te_j) - u(z) = \int_0^t D_j(z + se_j)\, ds \qquad \text{for a.e. } z \in K, \tag{6}$$

for each fixed t with $0 < |t| \le \delta$.

Step 5: the difference quotients converge in $L^1(K)$. For $0 < |t| \le \delta$ put $\Delta_t(z) = \big(u(z + te_j) - u(z)\big)/t$. By (6) and $\frac{1}{t}\int_0^t D_j(z)\, ds = D_j(z)$,

$$\Delta_t(z) - D_j(z) = \frac{1}{t}\int_0^t \big[D_j(z + se_j) - D_j(z)\big] ds \qquad \text{for a.e. } z \in K,$$

so, by Tonelli's theorem again, by $D_j = V_j$ on $K^{3\delta}$ and by $K \subseteq K^{3\delta}$, $K + se_j \subseteq K^{3\delta}$ for $|s| \le \delta$,

$$\int_K \big|\Delta_t - D_j\big|\, d\sigma \le \frac{1}{|t|}\int_{I_t}\Big(\int_{\mathbb{C}} \big|V_j(z + se_j) - V_j(z)\big|\, d\sigma(z)\Big) ds \le \sup_{|s| \le |t|} \big\|V_j(\cdot + se_j) - V_j\big\|_{L^1(\mathbb{C})},$$

the last step because the inner integral is at most the supremum and $|I_t| = |t|$. Translation is continuous in $L^1(\mathbb{C})$, so the right-hand side tends to 0 as $t \to 0$. Hence

$$\Delta_t \to D_j \quad \text{in } L^1(K) \text{ as } t \to 0. \tag{7}$$

Step 6: the pointwise limit. By (H3) there is a null set N such that u is differentiable at every $z \in \Omega - N$. For such a z the definition of the real Fréchet derivative gives, for real $t \to 0$,

$$\Delta_t(z) = \frac{u(z + te_j) - u(z)}{t} \longrightarrow \mathrm{d}u(z)[e_j] = w_j(z).$$

Step 7: conclusion on K. Put $t_l = \delta/l$ for $l \ge 1$. By (7), $\Delta_{t_l} \to D_j$ in $L^1(K)$, so some subsequence $\Delta_{t_{l_i}}$ converges to D_j almost everywhere on K. By Step 6 the same subsequence converges to $w_j(z)$ at every $z \in K - N$. A sequence of complex numbers has at most one limit, so $D_j = w_j$ almost everywhere on K.

Step 8: conclusion on Ω. By Step 0 the set $\{z \in \Omega : D_j(z) \ne w_j(z)\}$ is contained in a countable union of null sets, one for each $K^{(m)}$, hence is null. As $j \in \{0, 1\}$ was arbitrary, this is the assertion. □

2.7. `Eq18Rescale`.

Lemma 2.7. *Let $n \geq 0$ be an integer, let $1 < p < \infty$, let $K \geq 1$ and $t > 0$, let $V \subseteq \mathbb{C}$ be open with $\overline{\mathbb{B}(0,t)} = \{z : |z| \leq t\} \subseteq V$, and let $H \colon \mathbb{C} \to \mathbb{C}$ be continuous on V, differentiable at almost every point of V, K-quasiconformal on V in the sense of `QuasiconformalOn`, and a solution of equation (18) on V in the sense of `SolvesEq18`. Write $\mathbb{B} = \{\xi : |\xi| < 1\}$, $d\sigma$ for plane Lebesgue measure, and*

$$G(\zeta) = H(t\zeta).$$

Then

(a) *G is continuous on $\mathbb{B}$;*
(b) *G is differentiable at almost every point of $\mathbb{B}$;*
(c) *G solves (18) on $\mathbb{B}$ in the sense of `SolvesEq18`;*
(d) *$G \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`;*
(e) *$|G_\zeta|^2 + |G_{\bar\zeta}|^2$ is integrable on $\mathbb{B}$, where G_ζ, $G_{\bar\zeta}$ are the Wirtinger derivatives of `DZ` and `DZbar`;*
(f) *G is K-quasiconformal on $\mathbb{B}$ in the sense of `QuasiconformalOn`;*
(g) *if G is of class C^∞ on $\mathbb{B} - \{0\}$, then H is of class C^∞ on $\{z : 0 < |z| < t\}$ and*

$$\big(D^w H\big)(t\zeta) = t^{-|w|} D^w G(\zeta) \qquad \text{for every word } w \text{ and every } \zeta \text{ with } 0 < |\zeta| < 1,$$

where D^w is the iterated partial derivative of `PartialWord` and $|w|$ its length.

Clauses (a)–(f) are the content of the paper's opening sentence in the proof of Corollary 1 (paper.tex:376), "By rescaling the variable ξ one may assume without loss of generality that H is a quasiconformal solution of (18) on the unit disk", licensed at paper.tex:183; they are exactly the hypotheses under which Theorem 2 applies, so they are what makes that "without loss of generality" legitimate rather than a change of problem. No smoothness of H is assumed: rescaling a variable cannot manufacture it, and in the paper's order it is not yet available here — the smoothness off the origin is produced later, by term-by-term differentiation of the expansion (19) at paper.tex:387–388. That is why clause (g), which is what transports the conclusions (41), (42), (43) — stated for H — back from G, is a conditional whose hypothesis is the smoothness of G: at the point of use that is exactly what has just been produced for G, and (g) hands it on to H together with the derivative dictionary.

Proof. Write $s(\zeta) = t\zeta$, so that $G = H \circ s$, and put $c = \frac{1}{2} - \frac{1}{p}$.

Step 0: the scaling map. Since $t > 0$, s is an $\mathbb{R}$-linear bijection of $\mathbb{C}$ with inverse $z \mapsto z/t$, hence a C^∞ diffeomorphism, and its Fréchet derivative at every point is s itself. We record three consequences used repeatedly.

First, $s(\mathbb{B}) = \mathbb{B}(0,t) \subseteq \overline{\mathbb{B}(0,t)} \subseteq V$; more precisely $|s(\zeta)| = t|\zeta| < t$ for $|\zeta| < 1$. Second, $s(\zeta) = 0$ if and only if $\zeta = 0$, so s maps $\mathbb{B} - \{0\}$ into $V - \{0\}$. Third, identifying $\mathbb{C}$ with $\mathbb{R}^2$, s is multiplication by the matrix $t\,\mathrm{Id}$, whose determinant is t^2; so for every measurable E,

$$\sigma\big(s(E)\big) = t^2 \sigma(E), \qquad \int_E f \circ s \, d\sigma = t^{-2} \int_{s(E)} f \, d\sigma \tag{1}$$

for every nonnegative measurable f, and in particular E is null if and only if $s(E)$ is null.

Step 1: proof of (a). s is continuous and maps $\mathbb{B}$ into V by Step 0, and H is continuous on V; so $G = H \circ s$ is continuous on $\mathbb{B}$.

Step 2: proof of (b). Let $N \subseteq \mathbb{C}$ be the null set outside which every point of V is a point of differentiability of H, as the hypothesis provides. By (1) the set $s^{-1}(N)$ is null, and for $\zeta \in \mathbb{B}$ outside it we have $s(\zeta) \in V$ by Step 0 and H differentiable at $s(\zeta)$; since s is C^∞, the chain rule makes $G = H \circ s$ differentiable at ζ, with

$$\mathrm{d}G(\zeta)[v] = \mathrm{d}H(t\zeta)\big[s(v)\big] = t\,\mathrm{d}H(t\zeta)[v] \qquad (v \in \mathbb{C}), \tag{1′}$$

using $\mathrm{d}s = s$, $s(v) = tv$ and the $\mathbb{R}$-linearity of $\mathrm{d}H(t\zeta)$. So G is differentiable at almost every point of $\mathbb{B}$, which is (b).

Step 3: the first-order partials scale by t. By `PartialWord`, $D^{[j]}f = \partial_j f$ is the value of the real Fréchet derivative of f on the vector $e_0 = 1$ or $e_1 = i$. So $(1')$ of Step 2 says: at every $\zeta \in \mathbb{B}$ outside the null set $s^{-1}(N)$,

$$D^{[j]}G(\zeta) = t\,\big(D^{[j]}H\big)(t\zeta) \qquad (j = 0, 1). \tag{1''}$$

Step 4: the Wirtinger derivatives scale by t. By `DZ` and `DZbar`,

$$f_\zeta = \tfrac{1}{2}\big(\partial_x f - i\,\partial_y f\big), \qquad f_{\bar\zeta} = \tfrac{1}{2}\big(\partial_x f + i\,\partial_y f\big),$$

the two partial derivatives being the values of the real Fréchet derivative on 1 and i, i.e. $D^{[0]}f$ and $D^{[1]}f$. So $(1'')$ of Step 3 gives, for every $\zeta \in \mathbb{B}$ outside the null set $s^{-1}(N)$,

$$G_\zeta(\zeta) = t\,H_\xi(t\zeta), \qquad G_{\bar\zeta}(\zeta) = t\,H_{\bar\xi}(t\zeta). \tag{2}$$

Step 4′: weak and classical Wirtinger derivatives agree. We record once, in general form, the passage between the two readings of the Wirtinger operators, and use it twice below. *Let $\Omega \subseteq \mathbb{C}$ be open, let $u\colon \mathbb{C} \to \mathbb{C}$ be differentiable at almost every point of Ω, and let a, b be weak Wirtinger derivatives of u on Ω in the sense of `IsWeakWirtingerOn`. Then $a = u_z$ and $b = u_{\bar z}$ almost everywhere on Ω, with u_z, $u_{\bar z}$ the pointwise operators of `DZ` and `DZbar`.* Indeed, by `IsWeakWirtingerOn` there are u_x, u_y with $u, u_x, u_y \in L^1_{\mathrm{loc}}(\Omega)$, with u_x and u_y the distributional ∂_0- and ∂_1-derivatives of u on Ω, and with $a = \frac{1}{2}(u_x - iu_y)$, $b = \frac{1}{2}(u_x + iu_y)$ at every point. The first two facts are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for Ω, u and the pair $D_0 = u_x$, $D_1 = u_y$, and the almost-everywhere differentiability is its (H3); so $u_x = \partial_0 u$ and $u_y = \partial_1 u$ almost everywhere on Ω, where $\partial_j = D^{[j]}$ is the classical partial derivative of `PartialWord`, that is $\partial_0 u = \mathrm{d}u[1]$ and $\partial_1 u = \mathrm{d}u[i]$. By `DZ` and `DZbar`, u_z and $u_{\bar z}$ are the same two combinations $\frac{1}{2}\big(\mathrm{d}u[1] \mp i\,\mathrm{d}u[i]\big)$, which gives the claim. The two applications are: $u = H$ on $\Omega = V$, where V is open and almost-everywhere differentiability is a hypothesis; and $u = G$ on $\Omega = \mathbb{B}$, where $\mathbb{B}$ is open and almost-everywhere differentiability is (b) of Step 2.

Step 5: the pointwise form of (18) for G. By `SolvesEq18` there are weak Wirtinger derivatives a_H, b_H of H on V and a null set Z_0 such that

$$b_H(z) = c\Big[\frac{z}{\bar z}a_H(z) + \frac{\bar z^{\,n}}{z^n}\overline{a_H(z)}\Big] \qquad (z \in V - Z_0,\ z \neq 0),$$

and by Step 4′ applied to H on V there is a null set Z_2 with $a_H = H_\xi$ and $b_H = H_{\bar\xi}$ on $V - Z_2$. So, with $Z = Z_0 \cup Z_2$, a null set, every $z \notin Z$ with $z \in V$ and $z \neq 0$ satisfies

$$H_{\bar\xi}(z) = c\Big[\frac{z}{\bar z}H_\xi(z) + \frac{\bar z^{\,n}}{z^n}\overline{H_\xi(z)}\Big]. \tag{3}$$

Put $Z' = s^{-1}(Z) \cup s^{-1}(N) \cup \{0\}$, a null set by (1) and because a singleton is null; the term $s^{-1}(N)$ is there so that (2) is available at the points considered. Let $\zeta \notin Z'$ with $\zeta \in \mathbb{B}$ and $\zeta \neq 0$, and set $z = t\zeta$. Then $z \in V$ and $z \neq 0$ by Step 0, and $z \notin Z$. Since t is a positive real, $\bar z = t\bar\zeta$ and therefore

$$\frac{z}{\bar z} = \frac{t\zeta}{t\bar\zeta} = \frac{\zeta}{\bar\zeta}, \qquad \frac{\bar z^{\,n}}{z^n} = \frac{t^n\bar\zeta^{\,n}}{t^n\zeta^n} = \frac{\bar\zeta^{\,n}}{\zeta^n}.$$

Multiplying (3) by t and using (2), together with $t\,\overline{x} = \overline{t\,x}$ for real t,

$$G_{\bar\zeta}(\zeta) = t\,H_{\bar\xi}(z) = c\Big[\frac{\zeta}{\bar\zeta}\,t\,H_\xi(z) + \frac{\bar\zeta^{\,n}}{\zeta^n}\,\overline{t\,H_\xi(z)}\Big] = c\Big[\frac{\zeta}{\bar\zeta}G_\zeta(\zeta) + \frac{\bar\zeta^{\,n}}{\zeta^n}\overline{G_\zeta(\zeta)}\Big].$$

As Z' is null, the pointwise identity

$$G_{\bar\zeta}(\zeta) = c\Big[\frac{\zeta}{\bar\zeta}G_\zeta(\zeta) + \frac{\bar\zeta^{\,n}}{\zeta^n}\overline{G_\zeta(\zeta)}\Big] \tag{3'}$$

holds at almost every $\zeta \in \mathbb{B}$ with $\zeta \neq 0$. Step 7′ below turns this into (c).

The same argument gives the pointwise distortion inequality for H, which Step 9 uses: H being K-quasiregular on V, `QuasiregularOn` provides weak Wirtinger derivatives a', b' of H on

V and a null set with $|b'| \le \kappa_0 |a'|$ off it, where $\kappa_0 = (K-1)/(K+1)$, and Step 4′ applied to that pair gives $a' = H_\xi$, $b' = H_{\bar\xi}$ almost everywhere on V; hence

$$|H_{\bar\xi}(z)| \le \kappa_0 \, |H_\xi(z)| \qquad \text{for almost every } z \in V. \tag{3''}$$

Step 6: proof of (d). By `QuasiconformalOn` and `QuasiregularOn`, $H \in W^{1,2}_{\mathrm{loc}}(V)$, so by `MemWLocOn` there is a family D of functions indexed by words with $D[\,] = H$, with $D(w) \in L^2_{\mathrm{loc}}(V)$ for $|w| \le 1$ in the sense of `MemLpLocOn`, and with $D[j]$ the distributional j-th partial derivative of H on V in the sense of `IsWeakPartialOn`. Define

$$D'[\,] = G, \qquad D'[j] = t\,\big(D[j] \circ s\big) \ (j = 0, 1), \qquad D'(w) = 0 \ \ (|w| \ge 2).$$

We check the three clauses of `MemWLocOn` for D' with $k = 1$, $s = 2$.

$D'[\,] = G$ holds by definition.

Local square integrability: let $\kappa \subseteq \mathbb{B}$ be compact. Then $s(\kappa)$ is compact, and $s(\kappa) \subseteq \mathbb{B}(0,t) \subseteq V$ by Step 0. By (1), for $j = 0, 1$,

$$\int_\kappa \big|D'[j]\big|^2 d\sigma = t^2 \int_\kappa \big|D[j] \circ s\big|^2 d\sigma = t^2 \cdot t^{-2} \int_{s(\kappa)} \big|D[j]\big|^2 d\sigma < \infty,$$

the last integral being finite because $D[j] \in L^2_{\mathrm{loc}}(V)$ and $s(\kappa)$ is a compact subset of V; and likewise $\int_\kappa |G|^2 d\sigma = t^{-2} \int_{s(\kappa)} |H|^2 d\sigma < \infty$. The functions $D'(w)$ with $|w| \ge 2$ are 0, trivially square integrable.

The weak-derivative clause: the only word of length < 1 is the empty word, so we must show that $D'[j]$ is the distributional j-th partial derivative of G on $\mathbb{B}$. Let φ be a test function for $\mathbb{B}$ in the sense of `IsTestFn` and put $\tilde\varphi(z) = \varphi(z/t)$. Then $\tilde\varphi$ is C^∞ on $\mathbb{C}$, with compact support $s(\operatorname{supp}\varphi) \subseteq \mathbb{B}(0,t) \subseteq V$, so $\tilde\varphi$ is a test function for V. Differentiating $\tilde\varphi = \varphi \circ s^{-1}$ gives $\partial_j \tilde\varphi(z) = t^{-1} (\partial_j \varphi)(z/t)$, i.e. $(\partial_j \varphi)(z/t) = t\, \partial_j \tilde\varphi(z)$. Hence, substituting $z = t\zeta$ by (1),

$$\int G\, \partial_j \varphi \, d\sigma = t^{-2} \int H(z)\, (\partial_j \varphi)(z/t)\, d\sigma(z) = t^{-1} \int H\, \partial_j \tilde\varphi \, d\sigma = -t^{-1} \int D[j]\, \tilde\varphi \, d\sigma,$$

the last step by `IsWeakPartialOn` for H on V applied to the test function $\tilde\varphi$. Substituting back by (1),

$$-t^{-1} \int D[j]\, \tilde\varphi \, d\sigma = -t^{-1} \cdot t^2 \int \big(D[j] \circ s\big) \varphi \, d\sigma = -\int D'[j]\, \varphi \, d\sigma.$$

As φ was arbitrary, $D'[j]$ is the distributional j-th partial of G on $\mathbb{B}$. This proves (d).

Step 7: the classical and weak first derivatives agree almost everywhere. The hypotheses of `WeakPartialEqFDerivAE` hold with $\Omega = V$, which is open, the function H and the family $D_j = D[j]$ of Step 6: each of H, $D[0]$, $D[1]$ lies in $L^1_{\mathrm{loc}}(V)$, because it lies in $L^2_{\mathrm{loc}}(V)$ by Step 6 and a compact set has finite measure, so the Cauchy–Schwarz inequality gives $\int_\kappa |g| \le (\int_\kappa |g|^2)^{1/2} \sigma(\kappa)^{1/2} < \infty$ for compact $\kappa \subseteq V$; $D[j]$ is the distributional j-th partial derivative of H on V by Step 6; and H is differentiable at almost every point of V by hypothesis. Hence

$$D[j] = \partial_j H \qquad \text{almost everywhere on } V \quad (j = 0, 1). \tag{4}$$

Step 7′: proof of (c). Define

$$a_G = \tfrac{1}{2}\big(D'[0] - i\, D'[1]\big), \qquad b_G = \tfrac{1}{2}\big(D'[0] + i\, D'[1]\big)$$

at every point of $\mathbb{C}$, with D' the family of Step 6. Then a_G and b_G are weak Wirtinger derivatives of G on $\mathbb{B}$ in the sense of `IsWeakWirtingerOn`: its clause (b) is the weak-derivative clause proved in Step 6, its clause (c) is the definition just made, and its clause (a) holds because G, $D'[0]$ and $D'[1]$ lie in $L^2_{\mathrm{loc}}(\mathbb{B})$ by Step 6 and hence in $L^1_{\mathrm{loc}}(\mathbb{B})$, by the Cauchy–Schwarz inequality on compact subsets of $\mathbb{B}$ exactly as in Step 7.

By Step 4′ applied to G on $\mathbb{B}$ — legitimate by (b) — we have $a_G = G_\zeta$ and $b_G = G_{\bar\zeta}$ almost everywhere on $\mathbb{B}$. Combining this with (3′), the identity

$$b_G(\zeta) = c\Big[\frac{\zeta}{\bar\zeta} a_G(\zeta) + \frac{\bar\zeta^{\,n}}{\zeta^n} \overline{a_G(\zeta)}\Big]$$

holds at almost every $\zeta \in \mathbb{B}$ with $\zeta \neq 0$, the union of the two exceptional null sets being null. With the pair a_G, b_G this is exactly the assertion that G solves (18) on $\mathbb{B}$ in the sense of `SolvesEq18`, which is (c).

Step 8: proof of (e). Put $E = |H_\xi|^2 + |H_{\bar\xi}|^2$ and $E_G = |G_\zeta|^2 + |G_{\bar\zeta}|^2$. By (2),

$$E_G(\zeta) = t^2 E(t\zeta) \qquad (0 < |\zeta| < 1). \tag{5}$$

By Step 4 and the parallelogram law in $\mathbb{C}$, namely $|a-ib|^2+|a+ib|^2 = 2|a|^2+2|b|^2$ for $a, b \in \mathbb{C}$, we have on $V - \{0\}$

$$E = \tfrac{1}{4}\big(|\partial_x H - i\partial_y H|^2 + |\partial_x H + i\partial_y H|^2\big) = \tfrac{1}{2}\big(|\partial_x H|^2 + |\partial_y H|^2\big),$$

so by (4), $E = \frac{1}{2}\big(|D[0]|^2 + |D[1]|^2\big)$ almost everywhere on V. Since $\overline{\mathbb{B}(0,t)}$ is a compact subset of V and $D[j] \in L^2_{\mathrm{loc}}(V)$,

$$\int_{\mathbb{B}(0,t)} E\, d\sigma \le \tfrac{1}{2}\int_{\overline{\mathbb{B}(0,t)}} \big(|D[0]|^2 + |D[1]|^2\big) d\sigma < \infty.$$

Therefore, by (5) and (1) applied to E_G on $\mathbb{B} - \{0\}$, whose complement in $\mathbb{B}$ is the null set $\{0\}$,

$$\int_{\mathbb{B}} E_G\, d\sigma = t^2 \int_{\mathbb{B}-\{0\}} E \circ s\, d\sigma = t^2 \cdot t^{-2} \int_{\mathbb{B}(0,t)-\{0\}} E\, d\sigma \le \int_{\mathbb{B}(0,t)} E\, d\sigma < \infty.$$

Finally E_G is measurable: by (2) it agrees almost everywhere on $\mathbb{B}$ with $t^2(E \circ s)$, and E agrees almost everywhere on V with $\frac{1}{2}(|D[0]|^2 + |D[1]|^2)$ by (4), a measurable function, so $E \circ s$ is measurable by (1); a function agreeing almost everywhere with a measurable function is measurable for the completed Lebesgue measure. A nonnegative measurable function with finite integral is integrable, which is (e).

Step 9: proof of (f). We verify the five clauses of `QuasiconformalOn` for G on $\mathbb{B}$ with the same K. Write $\kappa_0 = (K-1)/(K+1)$.

K-quasiregularity. By `QuasiregularOn` we need $1 \le K$, which is a hypothesis; $G \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$, which is (d); and a pair of weak Wirtinger derivatives of G on $\mathbb{B}$ satisfying the distortion inequality almost everywhere. We take the pair a_G, b_G of Step 7′, for which $a_G = G_\zeta$ and $b_G = G_{\bar\zeta}$ almost everywhere on $\mathbb{B}$, also by Step 7′; so it suffices to prove $|G_{\bar\zeta}| \le \kappa_0 |G_\zeta|$ almost everywhere on $\mathbb{B}$. By (3″) of Step 5 there is a null set Z_1 with $|H_{\bar\xi}(z)| \le \kappa_0 |H_\xi(z)|$ for every $z \in V$ outside Z_1. Let $\zeta \in \mathbb{B}$ lie outside the null set $s^{-1}(Z_1) \cup s^{-1}(N) \cup \{0\}$, which is null by (1); then $t\zeta \in V - \{0\}$ lies outside Z_1, and (2) is available at ζ, so

$$|G_{\bar\zeta}(\zeta)| = t\,|H_{\bar\xi}(t\zeta)| \le t\,\kappa_0 |H_\xi(t\zeta)| = \kappa_0 |G_\zeta(\zeta)|.$$

Injectivity. If $\zeta_1, \zeta_2 \in \mathbb{B}$ and $G(\zeta_1) = G(\zeta_2)$ then $H(t\zeta_1) = H(t\zeta_2)$ with $t\zeta_1, t\zeta_2 \in V$ by Step 0, so $t\zeta_1 = t\zeta_2$ because H is injective on V, whence $\zeta_1 = \zeta_2$ as $t \neq 0$.

Continuity is (a).

Openness of the image. By `QuasiconformalOn` applied to H there is ψ with $\psi(H(z)) = z$ for every $z \in V$ and ψ continuous on $H(V)$, and $H(V)$ is open. We claim

$$G(\mathbb{B}) = H\big(\mathbb{B}(0,t)\big) = \big\{w \in H(V) : \psi(w) \in \mathbb{B}(0,t)\big\}. \tag{6}$$

The first equality holds because $G(\mathbb{B}) = \{H(t\zeta) : |\zeta| < 1\} = \{H(z) : |z| < t\}$. For the second: if $w = H(z)$ with $|z| < t$ then $z \in V$ by Step 0, so $w \in H(V)$ and $\psi(w) = z \in \mathbb{B}(0,t)$; conversely if $w \in H(V)$ with $\psi(w) \in \mathbb{B}(0,t)$, write $w = H(z)$ with $z \in V$, so that $\psi(w) = z$, whence $|z| < t$ and $w \in H(\mathbb{B}(0,t))$. Now the right side of (6) is $\psi^{-1}\big(\mathbb{B}(0,t)\big) \cap H(V)$, which by continuity of ψ on $H(V)$ is open in the subspace $H(V)$, that is, of the form $U \cap H(V)$ with $U \subseteq \mathbb{C}$ open; and $H(V)$ is open in $\mathbb{C}$. So $G(\mathbb{B})$ is open in $\mathbb{C}$.

The inverse. Put $\psi_G(w) = \psi(w)/t$. For $\zeta \in \mathbb{B}$ we have $\psi_G(G(\zeta)) = \psi(H(t\zeta))/t = t\zeta/t = \zeta$. And $G(\mathbb{B}) \subseteq H(V)$ by (6), where ψ is continuous, so ψ_G is continuous on $G(\mathbb{B})$, division by the nonzero constant t being continuous.

All five clauses hold, which is (f), and the proof is complete.

Step 10: proof of (g). Assume G is of class C^∞ on $\mathbb{B} - \{0\}$ and put $W = \{z : 0 < |z| < t\}$, an open set. By Step 0 the map $\sigma_0(z) = z/t$, the inverse of s, is C^∞ and carries W onto $\mathbb{B} - \{0\}$:

indeed $0 < |z| < t$ if and only if $0 < |z/t| < 1$. For $z \in W$ we have $H(z) = H\big(s(z/t)\big) = G(z/t)$, so

$$H = G \circ \sigma_0 \qquad \text{on } W,$$

a composition of C^∞ maps between open sets; since smoothness on an open set is a local property and H agrees with that composition on all of W, H is C^∞ on W.

For the derivative dictionary we induct on w, proving

$$\big(D^w H\big)(t\zeta) = t^{-|w|} D^w G(\zeta) \qquad (0 < |\zeta| < 1). \tag{6}$$

If w is empty, D^w is the identity by `PartialWord` and both sides are $H(t\zeta) = G(\zeta)$, since $t^0 = 1$.

Let $w = j :: w'$, so that $D^w f = \partial_j (D^{w'} f)$ by `PartialWord`, where ∂_j is the value of the real Fréchet derivative on $e_0 = 1$ or $e_1 = i$. By the inductive hypothesis the two functions

$$\zeta \mapsto \big(D^{w'} H\big)(t\zeta) \qquad \text{and} \qquad \zeta \mapsto t^{-|w'|} D^{w'} G(\zeta)$$

agree at every point of the open set $\mathbb{B} - \{0\}$, so their Fréchet derivatives agree there. Both are differentiable: $D^{w'} G$ is C^∞ on $\mathbb{B} - \{0\}$ because G is, and $D^{w'} H$ is C^∞ on W because H is, by the first half of this step, while $\zeta \mapsto t\zeta$ carries $\mathbb{B} - \{0\}$ into W. Differentiating the left-hand function at $\zeta \in \mathbb{B} - \{0\}$ by the chain rule, with $\mathrm{d}s = s$ and $s(v) = tv$,

$$\mathrm{d}\big(\big(D^{w'} H\big) \circ s\big)(\zeta)[v] = t\, \mathrm{d}\big(D^{w'} H\big)(t\zeta)[v],$$

so taking $v = e_j$ and equating with the derivative of the right-hand function,

$$t\, \big(D^w H\big)(t\zeta) = t^{-|w'|} D^w G(\zeta),$$

that is $\big(D^w H\big)(t\zeta) = t^{-|w'|-1} D^w G(\zeta) = t^{-|w|} D^w G(\zeta)$, since $|w| = |w'| + 1$. This completes the induction and proves (g). □

2.8. `EulerOdeSolution`.

Lemma 2.8. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, and let $k \in \mathbb{Z}$ satisfy the non-degeneracy condition*

$$k \neq 0 \quad \text{or} \quad p \neq 2.$$

Put

$$\lambda^+ = \lambda_{|k|}(n, p), \qquad \lambda^- = -pn - \lambda^+,$$

where $\lambda_{|k|}(n,p)$ is the exponent of `LambdaCoef`. Let $y \colon \mathbb{R} \to \mathbb{C}$ be C^∞ on $(0,1)$ and satisfy the Euler equation

$$r\big(r\, y'\big)'(r) + pn\, r\, y'(r) + (n^2 - k^2)(p-1)\, y(r) = 0 \qquad \text{for every } r \in (0,1).$$

Then there are $A^+, A^- \in \mathbb{C}$ with

$$y(r) = A^+ r^{\lambda^+} + A^- r^{\lambda^-} \qquad \text{for every } r \in (0,1).$$

Explicitly, with $S = \sqrt{4k^2(p-1) + n^2(p-2)^2}$ one has $2\lambda^\pm = -pn \pm S$, so λ^+ and λ^- are the two roots displayed at paper.tex:318–321 of the indicial equation $\lambda^2 + pn\lambda + (n^2 - k^2)(p-1) = 0$ of paper.tex:315. Indeed $\lambda_{|k|}(n,p) = \frac{1}{2}(-pn + \sqrt{4|k|^2(p-1) + n^2(p-2)^2})$ and $|k|^2 = k^2$; in particular $\lambda_k = \lambda_{-k}$, as the paper notes at line 331.

The non-degeneracy condition is exactly the positivity of the indicial discriminant $4k^2(p-1) + n^2(p-2)^2$, which is what the paper's assertion at line 313 that the two roots are distinct *requires: for $n \geq 1$ and $p > 1$ the discriminant vanishes precisely when $k = 0$ and $p = 2$. In that single degenerate case the indicial polynomial is $(\lambda + n)^2$, the two roots coincide at $-n$, and the conclusion above is false, since $(\log r)\, r^{-n}$ then solves the same equation without lying in the span of r^{λ^+} and r^{λ^-}; that case is treated separately.*

Proof. Write $S = \sqrt{4k^2(p-1) + n^2(p-2)^2}$, so that by `LambdaCoef` and $|k|^2 = k^2$,

$$\lambda^+ = \frac{-pn + S}{2}, \qquad \lambda^- = -pn - \lambda^+ = \frac{-pn - S}{2}. \tag{0}$$

Step 0: the roots are real, distinct, and satisfy Vieta's relations. Since $p > 1$ we have $4k^2(p-1) \geq 0$, with $4k^2(p-1) > 0$ when $k \neq 0$; and $n^2(p-2)^2 \geq 0$, with $n^2(p-2)^2 > 0$ when $p \neq 2$,

because $n \geq 1$. By the non-degeneracy hypothesis at least one of $k \neq 0$ and $p \neq 2$ holds, so in either case the radicand $4k^2(p-1)+n^2(p-2)^2$ is a sum of two nonnegative terms of which one is positive. Hence the radicand is positive and

$$S > 0, \qquad S^2 = 4k^2(p-1)+n^2(p-2)^2. \tag{1}$$

From (0), $\lambda^+ - \lambda^- = S > 0$, so the two roots are distinct. Also from (0),

$$\lambda^+ + \lambda^- = -pn, \tag{2}$$

and, using (1) and the identity $p^2-(p-2)^2 = 4(p-1)$,

(3)

$$\lambda^+\lambda^- = \frac{(-pn)^2 - S^2}{4} = \frac{p^2n^2 - 4k^2(p-1) - n^2(p-2)^2}{4} = \frac{n^2\big(p^2-(p-2)^2\big) - 4k^2(p-1)}{4} = (n^2-k^2)(p-1).$$

Step 1: the substitution $r = e^t$. Put $J = (-\infty, 0)$ and

$$Y(t) = y\big(e^t\big) \qquad (t \in J).$$

For $t < 0$ we have $e^t \in (0,1)$, so Y is well defined on J, and Y is C^∞ on J as the composite of the C^∞ map $t \mapsto e^t$ (with values in $(0,1)$) and the map y, which is C^∞ on $(0,1)$. By the chain rule, writing $r = e^t$,

$$Y'(t) = e^t y'(e^t) = r\, y'(r),$$

and differentiating once more,

$$Y''(t) = e^t y'(e^t) + e^{2t} y''(e^t) = r\, y'(r) + r^2 y''(r).$$

On the other hand $r\big(r\, y'\big)'(r) = r\big(y'(r) + r\, y''(r)\big) = r\, y'(r) + r^2 y''(r)$, so

$$Y''(t) = r\big(r\, y'\big)'(r) \quad \text{and} \quad Y'(t) = r\, y'(r) \qquad \big(r = e^t\big). \tag{4}$$

Every $r \in (0,1)$ is e^t for exactly one $t \in J$, namely $t = \log r$, so the Euler equation together with (4) says precisely that

$$Y''(t) + pn\, Y'(t) + (n^2-k^2)(p-1)\, Y(t) = 0 \qquad \text{for every } t \in J,$$

and by (2) and (3) this is

$$Y''(t) - \big(\lambda^+ + \lambda^-\big)Y'(t) + \lambda^+\lambda^- Y(t) = 0 \qquad (t \in J). \tag{5}$$

Step 2: a function with vanishing derivative on J is constant. We isolate the step used twice below, because the standard statement is for a closed interval while J is open. Let $g\colon J \to \mathbb{C}$ be differentiable with $g'(t) = 0$ for every $t \in J$. Let $t_0, t_1 \in J$ with $t_0 \leq t_1$. Since J is an interval, $[t_0, t_1] \subseteq J$; g is continuous on the compact interval $[t_0, t_1]$, and for every $x \in [t_0, t_1)$ the derivative of g at x within $[x, \infty)$ exists and equals 0, being the restriction of the two-sided derivative $g'(x) = 0$. The theorem that a function continuous on $[t_0, t_1]$ whose derivative within $[x, \infty)$ vanishes at every $x \in [t_0, t_1)$ is constant on $[t_0, t_1]$ therefore gives $g(t_1) = g(t_0)$. As any two points of J are comparable, g takes the same value at every point of J: g is constant on J.

Step 3: first integrating factor. Put

$$W(t) = Y'(t) - \lambda^- Y(t) \qquad (t \in J),$$

which is C^∞ on J. By (5),

$$W'(t) = Y''(t) - \lambda^- Y'(t) = \big(\lambda^+ + \lambda^-\big)Y'(t) - \lambda^+\lambda^- Y(t) - \lambda^- Y'(t) = \lambda^+\big(Y'(t) - \lambda^- Y(t)\big) = \lambda^+ W(t).$$

Hence the function $g(t) = e^{-\lambda^+ t} W(t)$, which is differentiable on J, satisfies

$$g'(t) = e^{-\lambda^+ t}\big(W'(t) - \lambda^+ W(t)\big) = 0 \qquad (t \in J).$$

By Step 2, g is constant on J; call its value $C_1 \in \mathbb{C}$. Thus

$$Y'(t) - \lambda^- Y(t) = W(t) = C_1 e^{\lambda^+ t} \qquad (t \in J). \tag{6}$$

Step 4: second integrating factor. Since $\lambda^+ \neq \lambda^-$ by Step 0, we may define

$$h(t) = e^{-\lambda^- t} Y(t) - \frac{C_1}{\lambda^+ - \lambda^-} e^{(\lambda^+ - \lambda^-)t} \qquad (t \in J),$$

which is differentiable on J. Using (6),

$$\frac{d}{dt}\Big(e^{-\lambda^- t}Y(t)\Big) = e^{-\lambda^- t}\big(Y'(t) - \lambda^- Y(t)\big) = C_1 e^{(\lambda^+ - \lambda^-)t},$$

so

$$h'(t) = C_1 e^{(\lambda^+ - \lambda^-)t} - \frac{C_1}{\lambda^+ - \lambda^-}\big(\lambda^+ - \lambda^-\big)e^{(\lambda^+ - \lambda^-)t} = 0 \qquad (t \in J).$$

By Step 2 again, h is constant on J; call its value $C_2 \in \mathbb{C}$. Multiplying $h \equiv C_2$ by $e^{\lambda^- t}$ and rearranging,

$$Y(t) = \frac{C_1}{\lambda^+ - \lambda^-}e^{\lambda^+ t} + C_2 e^{\lambda^- t} \qquad (t \in J). \tag{7}$$

Step 5: back to the variable r. Put

$$A^+ = \frac{C_1}{\lambda^+ - \lambda^-}, \qquad A^- = C_2.$$

Let $r \in (0,1)$ and $t = \log r$. Then $t < 0$, so $t \in J$, and $e^t = r$. For a real exponent λ and $r > 0$ one has $e^{\lambda \log r} = r^\lambda$. Hence by (7)

$$y(r) = Y(\log r) = A^+ e^{\lambda^+ \log r} + A^- e^{\lambda^- \log r} = A^+ r^{\lambda^+} + A^- r^{\lambda^-},$$

which is the assertion. □

2.9. `EulerOdeDegenerate`.

Lemma 2.9. *Let $n \geq 1$ be an integer and let $y\colon \mathbb{R} \to \mathbb{C}$ be C^∞ on $(0,1)$ and satisfy*

$$r\big(r\,y'\big)'(r) + 2n\,r\,y'(r) + n^2\,y(r) = 0 \qquad \text{for every } r \in (0,1).$$

Then there are $A, B \in \mathbb{C}$ with

$$y(r) = \big(A + B\log r\big)r^{-n} \qquad \text{for every } r \in (0,1).$$

This is the equation (35) of the paper at the single pair $(k,p) = (0,2)$, where $pn = 2n$ and $(n^2 - k^2)(p-1) = n^2$. It is exactly the case in which the indicial polynomial $\lambda^2 + pn\lambda + (n^2 - k^2)(p-1)$ degenerates to $(\lambda+n)^2$, so that the two roots coincide at $-n$ and the paper's assertion at line 313 that they are distinct fails; the solution $(\log r)\,r^{-n}$ obtained here for $A = 0$, $B = 1$ is not a combination of powers r^λ at all.

Proof. First note the degeneration of the indicial polynomial recorded in the statement: at $k = 0$ and $p = 2$,

$$\lambda^2 + pn\lambda + (n^2 - k^2)(p-1) = \lambda^2 + 2n\lambda + n^2 = (\lambda + n)^2,$$

so $-n$ is a double root. This is the algebraic reason a logarithm appears below, and the reason the two-power conclusion is unavailable.

Step 1: the substitution $r = e^t$. Put $J = (-\infty, 0)$ and

$$Y(t) = y\big(e^t\big) \qquad (t \in J).$$

For $t < 0$ one has $e^t \in (0,1)$, so Y is defined on J, and Y is C^∞ on J, being the composite of the C^∞ map $t \mapsto e^t$ (with values in $(0,1)$) with y, which is C^∞ on $(0,1)$. By the chain rule, with $r = e^t$,

$$Y'(t) = e^t y'(e^t) = r\,y'(r),$$

and differentiating once more,

$$Y''(t) = e^t y'(e^t) + e^{2t} y''(e^t) = r\,y'(r) + r^2 y''(r).$$

Since $r\big(r\,y'\big)'(r) = r\big(y'(r) + r\,y''(r)\big) = r\,y'(r) + r^2 y''(r)$,

$$Y''(t) = r\big(r\,y'\big)'(r), \qquad Y'(t) = r\,y'(r) \qquad \big(r = e^t\big). \tag{1}$$

Every $r \in (0,1)$ equals e^t for exactly one $t \in J$, namely $t = \log r$, so the hypothesis together with (1) says precisely that

$$Y''(t) + 2n\,Y'(t) + n^2\,Y(t) = 0 \qquad \text{for every } t \in J. \tag{2}$$

Step 2: a function with vanishing derivative on J is constant. We isolate this step, used twice below, because the standard statement is for a closed interval while J is open. Let $g\colon J \to \mathbb{C}$ be differentiable with $g'(t) = 0$ for every $t \in J$, and let $t_0, t_1 \in J$ with $t_0 \le t_1$. Since J is an interval, $[t_0, t_1] \subseteq J$; g is continuous on the compact interval $[t_0, t_1]$, and for every $x \in [t_0, t_1)$ the derivative of g at x within $[x, \infty)$ exists and equals 0, being a restriction of the two-sided derivative $g'(x) = 0$. The theorem that a function continuous on $[t_0, t_1]$ whose derivative within $[x, \infty)$ vanishes at every $x \in [t_0, t_1)$ is constant on $[t_0, t_1]$ gives $g(t_1) = g(t_0)$. As any two points of J are comparable, g is constant on J.

Step 3: first integrating factor. Put

$$W(t) = Y'(t) + n\,Y(t) \qquad (t \in J),$$

which is C^∞ on J. By (2),

$$W'(t) = Y''(t) + n\,Y'(t) = -2n\,Y'(t) - n^2 Y(t) + n\,Y'(t) = -n\big(Y'(t) + n\,Y(t)\big) = -n\,W(t).$$

Hence $g(t) = e^{nt}W(t)$ is differentiable on J with

$$g'(t) = e^{nt}\big(W'(t) + n\,W(t)\big) = 0 \qquad (t \in J),$$

so by Step 2 it is constant; call its value $B \in \mathbb{C}$. Thus

$$Y'(t) + n\,Y(t) = W(t) = B\,e^{-nt} \qquad (t \in J). \tag{3}$$

Step 4: second integrating factor — this is where the logarithm enters. Let

$$h(t) = e^{nt}Y(t) - B\,t \qquad (t \in J),$$

which is differentiable on J. Using (3),

$$\frac{d}{dt}\Big(e^{nt}Y(t)\Big) = e^{nt}\big(Y'(t) + n\,Y(t)\big) = e^{nt}\cdot B\,e^{-nt} = B,$$

so $h'(t) = B - B = 0$ for every $t \in J$. By Step 2, h is constant on J; call its value $A \in \mathbb{C}$. (Note what has happened: because the two roots coincide, the right-hand side of (3) after multiplication by e^{nt} is the *constant* B rather than a nonconstant exponential, and its primitive is the linear function $B\,t$. This is the single point at which the argument differs from the non-degenerate one, and it is what produces the logarithm.) Hence

$$e^{nt}Y(t) = A + B\,t, \qquad \text{i.e.} \qquad Y(t) = \big(A + B\,t\big)e^{-nt} \qquad (t \in J). \tag{4}$$

Step 5: back to the variable r. Let $r \in (0,1)$ and put $t = \log r$. Then $t < 0$, so $t \in J$, and $e^t = r$. For a real exponent λ and $r > 0$ one has $e^{\lambda \log r} = r^\lambda$, so $e^{-nt} = r^{-n}$. Therefore by (4)

$$y(r) = Y(\log r) = \big(A + B\log r\big)r^{-n},$$

which is the assertion, with the constants A and B produced in Steps 4 and 3. □

2.10. `FourierCoefForm36.`

Lemma 2.10. *Let $n \ge 1$ be an integer, let $1 < p < \infty$, and let $k \in \mathbb{Z}$. Let $a\colon \mathbb{R} \to \mathbb{C}$ be continuous on $[0,1)$, C^∞ on $(0,1)$, and a solution of the Euler equation*

$$r\big(r\,a'\big)'(r) + pn\,r\,a'(r) + (n^2 - k^2)(p-1)\,a(r) = 0 \qquad \textit{for every } r \in (0,1).$$

Then

$$a(r) = \begin{cases} 0 & \text{if } |k| < n, \\ A\,r^{\lambda_{|k|}(n,p)} & \text{if } |k| \ge n, \end{cases} \qquad \textit{for every } r \in [0,1),$$

where in the second case $A \in \mathbb{C}$ is a constant depending on a, and $\lambda_{|k|}(n,p)$ is the exponent of `LambdaCoef`. Real powers are taken with the conventions $0^\lambda = 0$ for $\lambda > 0$ and $0^0 = 1$.

This is formula (36) of the paper, with $\lambda_k = \lambda_{|k|}(n,p)$ its (37); the two agree because only k^2 enters the formula, which is the paper's remark $\lambda_k = \lambda_{-k}$ at line 331. No restriction $p \ne 2$ is imposed: the conclusion holds for every $1 < p < \infty$, including the single pair $(k,p) = (0,2)$ at which the indicial roots coincide.

Proof. Write $S = \sqrt{4k^2(p-1) + n^2(p-2)^2}$ and

$$\lambda^+ = \lambda_{|k|}(n,p) = \frac{-pn+S}{2}, \qquad \lambda^- = -pn - \lambda^+ = \frac{-pn-S}{2},$$

as in `LambdaCoef` (using $|k|^2 = k^2$).

Step 0a: the two cases. Since $p > 1$ we have $4k^2(p-1) \geq 0$, with strict inequality when $k \neq 0$; and $n^2(p-2)^2 \geq 0$, with strict inequality when $p \neq 2$, because $n \geq 1$. Hence the radicand $4k^2(p-1) + n^2(p-2)^2$ is positive unless $k = 0$ and $p = 2$, in which case it is 0. We therefore distinguish the *non-degenerate case* ($k \neq 0$ or $p \neq 2$), treated in Steps 0–5, and the *degenerate case* ($k = 0$ and $p = 2$), treated in Step 6. The degenerate case is the one in which the paper's assertion at line 313 that the roots are distinct fails. Only the bound (4) of Step 0b is used in both cases.

Step 0b: a is bounded near 0; this is common to both cases. The interval $[0, \frac{1}{2}]$ is a compact subset of $[0,1)$ and a is continuous on $[0,1)$, so

$$M = \sup_{r \in [0,1/2]} |a(r)| < \infty. \tag{4}$$

Steps 0–5 assume $k \neq 0$ or $p \neq 2$. Then the radicand is positive by Step 0a, so

$$S > 0, \qquad S^2 = 4k^2(p-1) + n^2(p-2)^2, \qquad \lambda^+ - \lambda^- = S > 0. \tag{1}$$

Step 0: $\lambda^- < 0$ always, and the sign of λ^+ in three regimes. Since $n \geq 1$ and $p > 1$ we have $pn > 0$, and $S \geq 0$, so

$$\lambda^- = \frac{-pn-S}{2} < 0. \tag{2}$$

For λ^+, note that λ^+ and 0 compare exactly as S and pn do, by (1); and both $S \geq 0$ and $pn > 0$, so S and pn compare exactly as S^2 and $(pn)^2$ do. Now, using the identity $p^2 - (p-2)^2 = 4(p-1)$,

$$S^2 - (pn)^2 = 4k^2(p-1) + n^2(p-2)^2 - n^2p^2 = 4k^2(p-1) - n^2\big(p^2 - (p-2)^2\big) = 4(p-1)\big(k^2 - n^2\big),$$

and $4(p-1) > 0$. Therefore

$$\lambda^+ < 0 \iff |k| < n, \qquad \lambda^+ = 0 \iff |k| = n, \qquad \lambda^+ > 0 \iff |k| > n, \tag{3}$$

since $k^2 - n^2$ is negative, zero or positive exactly as $|k|$ is less than, equal to or greater than n. The middle regime is the one the paper's sentence at line 323 passes over: at $|k| = n$ the exponent is exactly 0, so r^{λ^+} is the constant 1, which is bounded, and the corresponding constant need *not* vanish. Note also that the roots remain distinct there, since $\lambda^- < 0 = \lambda^+$ by (2).

Step 2: the two constants. The hypotheses of `EulerOdeSolution` are met: $n \geq 1$, $1 < p < \infty$, its non-degeneracy hypothesis "$k \neq 0$ or $p \neq 2$" is the case assumption in force, a is C^∞ on $(0,1)$, and a satisfies the Euler equation there. It therefore provides $A^+, A^- \in \mathbb{C}$ with

$$a(r) = A^+ r^{\lambda^+} + A^- r^{\lambda^-} \qquad \text{for every } r \in (0,1). \tag{5}$$

Step 3: $A^- = 0$. Multiply (5) by $r^{-\lambda^-}$, which is legitimate for $r > 0$, and use $r^{-\lambda^-} r^{\lambda^+} = r^{\lambda^+ - \lambda^-} = r^S$ and $r^{-\lambda^-} r^{\lambda^-} = 1$:

$$r^{-\lambda^-} a(r) = A^+ r^S + A^- \qquad \text{for every } r \in (0,1). \tag{6}$$

Let $r \to 0^+$. By (2), $-\lambda^- > 0$, so $r^{-\lambda^-} \to 0$, and by (4), $|r^{-\lambda^-} a(r)| \leq M\, r^{-\lambda^-} \to 0$ for $r \in (0, \frac{1}{2}]$; thus the left-hand side of (6) tends to 0. By (1), $S > 0$, so $r^S \to 0$ and the first term on the right tends to 0. Hence $A^- = 0$, and (5) becomes

$$a(r) = A^+ r^{\lambda^+} \qquad \text{for every } r \in (0,1). \tag{7}$$

Step 4: the case $|k| < n$. Here $\lambda^+ < 0$ by (3). Suppose $A^+ \neq 0$. Then by (7), for $r \in (0, \frac{1}{2}]$, $|a(r)| = |A^+| r^{\lambda^+} \to \infty$ as $r \to 0^+$, because $-\lambda^+ > 0$; this contradicts the bound (4). Hence $A^+ = 0$, and by (7) $a(r) = 0$ for every $r \in (0,1)$. Finally a is continuous at 0 from the right, so $a(0) = \lim_{r \to 0^+} a(r) = 0$ as well. Therefore $a(r) = 0$ for every $r \in [0,1)$, which is the first case of the assertion.

Step 5: the case $|k| \geq n$. Here $\lambda^+ \geq 0$ by (3). Put $A = A^+$. By (7), $a(r) = A\, r^{\lambda^+}$ for every $r \in (0,1)$, so it remains only to check the equality at $r = 0$, i.e. that $a(0) = A \cdot 0^{\lambda^+}$. We distinguish the two sub-regimes of (3).

If $|k| > n$ then $\lambda^+ > 0$, so $0^{\lambda^+} = 0$ by the convention for real powers, and $r^{\lambda^+} \to 0$ as $r \to 0^+$; by continuity of a at 0 and (7), $a(0) = \lim_{r\to 0^+} A\, r^{\lambda^+} = 0 = A \cdot 0^{\lambda^+}$.

If $|k| = n$ then $\lambda^+ = 0$, so $r^{\lambda^+} = 1$ for every $r > 0$ and $0^{\lambda^+} = 0^0 = 1$. Thus (7) says $a(r) = A$ for every $r \in (0,1)$, and by continuity of a at 0, $a(0) = \lim_{r\to 0^+} A = A = A \cdot 0^{\lambda^+}$. This is the regime in which the coefficient is a nonzero constant; it is exactly the $k = \pm n$ branch of (36), and it is consistent with the paper's remark $\lambda_n = 0$ at line 331.

In both sub-regimes $a(r) = A\, r^{\lambda^+}$ for every $r \in [0,1)$, which is the second case of the assertion. This completes the non-degenerate case.

Step 6: the degenerate case $k = 0$, $p = 2$. Here only the first branch of the assertion has to be proved, and the second is vacuous: $|k| = 0$, and $0 < n$ because $n \geq 1$, so $|k| < n$ and the hypothesis $|k| \geq n$ of the second branch is never met.

With $k = 0$ and $p = 2$ the Euler equation in the hypothesis reads

$$r\big(r\, a'\big)'(r) + 2n\, r\, a'(r) + n^2\, a(r) = 0 \qquad \text{for every } r \in (0,1),$$

because $pn = 2n$ and $(n^2 - k^2)(p-1) = n^2 \cdot 1 = n^2$. Together with $n \geq 1$ and the C^∞-smoothness of a on $(0,1)$, these are exactly the hypotheses of `EulerOdeDegenerate`, which therefore provides $A, B \in \mathbb{C}$ with

$$a(r) = \big(A + B \log r\big) r^{-n} \qquad \text{for every } r \in (0,1). \tag{8}$$

We show $B = 0$. Suppose $B \neq 0$. For $0 < r \leq \frac{1}{2}$ we have $r^{-n} \geq 1$, since $n \geq 1$ and $0 < r \leq 1$; and $\log r < 0$, so $|\log r| = -\log r$. Hence by (8) and the triangle inequality, for $0 < r \leq \frac{1}{2}$,

$$|a(r)| = \big|A + B \log r\big|\, r^{-n} \geq \big(|B|\, |\log r| - |A|\big) r^{-n} \geq |B|\, |\log r| - |A|$$

whenever the middle quantity is nonnegative. As $r \to 0^+$ we have $|\log r| \to \infty$, so $|B||\log r| - |A| \to \infty$, contradicting the bound $|a(r)| \leq M$ of (4). Therefore $B = 0$.

Now (8) reads $a(r) = A\, r^{-n}$ on $(0,1)$. If $A \neq 0$ then $|a(r)| = |A| r^{-n} \to \infty$ as $r \to 0^+$, because $n \geq 1$, again contradicting (4). Therefore $A = 0$ and $a(r) = 0$ for every $r \in (0,1)$. Finally a is continuous at 0 from the right, so $a(0) = \lim_{r\to 0^+} a(r) = 0$ as well, and $a(r) = 0$ for every $r \in [0,1)$. This is the first case of the assertion, and it is the only one that can arise in the degenerate case. □

2.11. `FourierCoefAnnulusRegularity`.

Lemma 2.11. Let $n \geq 0$ be an integer, let $k \in \mathbb{Z}$ and let c', d' be real with $c' > 0$. Write

$$\mathcal{A} = \{\zeta \in \mathbb{C} : c' \leq |\zeta| \leq d'\}$$

for the compact annulus between the two radii, and for $F \colon \mathbb{C} \to \mathbb{C}$ write $\widehat{F}_k(s) = \frac{1}{2\pi} \int_0^{2\pi} F(se^{i\theta}) e^{i(n-k)\theta} d\theta$ for its k-th Fourier coefficient at radius s, as in `FourierCoefOfH`. Then the operator $F \mapsto \widehat{F}_k$ has the following two properties.

(a) If $W \colon \mathbb{C} \to \mathbb{C}$ is measurable and integrable on $\mathcal{A}$, then $s \mapsto \widehat{W}_k(s)$ is integrable on (c', d'), and for almost every $s \in (c', d')$ the circle restriction $\theta \mapsto W(se^{i\theta})$ is interval-integrable over $[0, 2\pi]$.

(b) The coefficient depends continuously on the function in L^1: if $F_l, F \colon \mathbb{C} \to \mathbb{C}$ are measurable and integrable on $\mathbb{C}$ and $\int_{\mathbb{C}} |F_l - F|\, d\sigma \to 0$, then

$$\int_{(c',d')} \big\| \widehat{(F_l)}_k(s) - \widehat{F}_k(s) \big\|\, ds \longrightarrow 0,$$

the integral being the $[0,\infty]$-valued Lebesgue integral.

Both are properties of the one operator $F \mapsto \widehat{F}_k$, which is why they are stated together: (b) is proved from the same estimate that proves (a), and (a) is what makes the inner integral defining $\widehat{F}_k$ meaningful in the first place.

This is the measure-theoretic foundation under the circle-average argument. The paper reads $a_k \in C[0,1)\cap C^\infty(0,1)$ off its standing smoothness hypothesis at `paper.tex:265-266`; the tablet carries no such hypothesis and replaces that step by `CircleAvgLocAbsCont`, which needs exactly these two facts before it can begin — (a) to know that $\widehat{W}_k$ is a genuine integrable function of the radius, and (b) to pass to the limit in a mollification, where the approximating functions converge to W only in L^1.

The hypothesis $c' > 0$ is essential and is used twice: it keeps the Jacobian s of the polar map bounded below on $\mathcal{A}$, which is what converts an L^1 bound on the plane into an L^1 bound in the polar variables, and it bounds the factor $1/s$ implicit in the pointwise estimate behind (b). The upper radius d' plays no such role; it only makes the annulus bounded.

Proof. Throughout write

$$\mathrm{ci}(s,\theta) = se^{i\theta}, \qquad \mathrm{wt}(\theta) = e^{i(n-k)\theta},$$

and $d\sigma$ for plane Lebesgue measure. By `FourierCoefOfH`, $\widehat{F}_k(s) = \frac{1}{2\pi}\int_0^{2\pi} F(\mathrm{ci}(s,\theta))\,\mathrm{wt}(\theta)\,d\theta$. Note $|\mathrm{wt}(\theta)| = 1$ and $\mathrm{wt}(\theta) \neq 0$ for every θ, since $n-k$ is an integer and θ is real, so the exponent is purely imaginary; and $|\mathrm{ci}(s,\theta)| = |s|$. Both ci and wt are continuous. The set $\mathcal{A}$ is closed and bounded, hence compact, hence measurable.

Step 0: two standard transfers. First, the polar change of variables: for every measurable $G\colon \mathbb{C} \to [0,\infty]$,

$$\int_{(0,\infty)\times(-\pi,\pi)} s\,G\big(\mathrm{ci}(s,\theta)\big)\,d(s,\theta) = \int_{\mathbb{C}} G\,d\sigma, \tag{1}$$

the map $(s,\theta) \mapsto se^{i\theta}$ being a diffeomorphism of $(0,\infty)\times(-\pi,\pi)$ onto the plane minus a null ray, with Jacobian s. Second, the window change: for fixed s the function $\theta \mapsto F(\mathrm{ci}(s,\theta))\mathrm{wt}(\theta)$ is 2π-periodic, because ci and wt are, so

$$\int_0^{2\pi} F\big(\mathrm{ci}(s,\theta)\big)\mathrm{wt}(\theta)\,d\theta = \int_{(-\pi,\pi)} F\big(\mathrm{ci}(s,\theta)\big)\mathrm{wt}(\theta)\,d\theta, \tag{2}$$

a period's worth of integral being independent of where the period starts.

Step 1: the key lower bound, where $c' > 0$ enters. For every $F\colon \mathbb{C} \to \mathbb{C}$,

$$c'\int_{(c',d')\times(-\pi,\pi)} \big\|F\big(\mathrm{ci}(s,\theta)\big)\big\|\,d(s,\theta) \le \int_{\mathbb{C}} \|F\|\,d\sigma, \tag{3}$$

all integrals being $[0,\infty]$-valued. Indeed on $(c',d')\times(-\pi,\pi)$ we have $s > c'$, so pointwise $c'\|F(\mathrm{ci}(s,\theta))\| \le s\,\|F(\mathrm{ci}(s,\theta))\|$; hence the left side is at most $\int_{(c',d')\times(-\pi,\pi)} s\|F(\mathrm{ci}(s,\theta))\|$, which is at most the same integral over the larger set $(0,\infty)\times(-\pi,\pi)$ because $(c',d') \subseteq (0,\infty)$ and the integrand is nonnegative, and that equals $\int_{\mathbb{C}}\|F\|$ by (1) applied to $G = \|F\|$.

Step 2: the product is integrable on the rectangle. Let F be measurable and integrable on $\mathbb{C}$. Then $(s,\theta) \mapsto F(\mathrm{ci}(s,\theta))\mathrm{wt}(\theta)$ is measurable on $(c',d')\times(-\pi,\pi)$, being the product of the composition of the measurable F with the continuous ci and of the continuous wt; and it is integrable there, because $\|F(\mathrm{ci}(s,\theta))\mathrm{wt}(\theta)\| = \|F(\mathrm{ci}(s,\theta))\|$ by $|\mathrm{wt}| = 1$, and by (3)

$$\int_{(c',d')\times(-\pi,\pi)} \big\|F\big(\mathrm{ci}(s,\theta)\big)\big\|\,d(s,\theta) \le \frac{1}{c'}\int_{\mathbb{C}} \|F\|\,d\sigma < \infty, \tag{4}$$

where dividing by c' is legitimate because $c' > 0$.

Step 3: proof of (a). Let W be measurable and integrable on $\mathcal{A}$, and put $F = \mathbf{1}_{\mathcal{A}}W$. Then F is measurable, $\mathcal{A}$ being measurable, and F is integrable on all of $\mathbb{C}$, since it vanishes off $\mathcal{A}$ and agrees with W there. So Step 2 applies to F, and Fubini's theorem for integrable functions on the product $(c',d')\times(-\pi,\pi)$ gives both

$$\text{for a.e. } s \in (c',d'): \quad \theta \mapsto F\big(\mathrm{ci}(s,\theta)\big)\mathrm{wt}(\theta) \text{ is integrable on } (-\pi,\pi), \tag{5}$$

and

$$s \mapsto \int_{(-\pi,\pi)} F\big(\mathrm{ci}(s,\theta)\big)\mathrm{wt}(\theta)\,d\theta \text{ is integrable on } (c',d'). \tag{6}$$

Now let $s \in (c', d')$. For every θ we have $|\mathrm{ci}(s,\theta)| = |s| = s \in (c', d') \subseteq [c', d']$, so $\mathrm{ci}(s,\theta) \in \mathcal{A}$ and therefore $F(\mathrm{ci}(s,\theta)) = W(\mathrm{ci}(s,\theta))$. Hence, by the definition of the coefficient and by (2),

$$\widehat{W}_k(s) = \frac{1}{2\pi}\int_0^{2\pi} W\big(\mathrm{ci}(s,\theta)\big)\mathrm{wt}(\theta)\,d\theta = \frac{1}{2\pi}\int_{(-\pi,\pi)} F\big(\mathrm{ci}(s,\theta)\big)\mathrm{wt}(\theta)\,d\theta \qquad \big(s \in (c', d')\big),$$

so on (c', d') the function $s \mapsto \widehat{W}_k(s)$ is the constant $\frac{1}{2\pi}$ times the function of (6), and is therefore integrable on (c', d'). That is the first half of (a).

For the second half, fix an $s \in (c', d')$ for which (5) holds. Dividing by wt, which is continuous and never zero, the function $\theta \mapsto F(\mathrm{ci}(s,\theta))$ is integrable on $(-\pi, \pi)$; it equals $\theta \mapsto W(\mathrm{ci}(s,\theta))$, as just observed; and that function is 2π-periodic, so by the same period-shift argument as in (2) it is interval-integrable over $[0, 2\pi]$. Since the exceptional set of s is null, this holds for almost every $s \in (c', d')$, which is the second half of (a).

Step 4: the pointwise estimate behind (b). Let G_1, G_2 be measurable and integrable on $\mathbb{C}$ and let s be real. The coefficient is linear in the function, so $\widehat{(G_1)}_k(s) - \widehat{(G_2)}_k(s) = \frac{1}{2\pi}\int_0^{2\pi}(G_1 - G_2)(\mathrm{ci}(s,\theta))\mathrm{wt}(\theta)\,d\theta$, and hence, by the triangle inequality for the integral, by $|\mathrm{wt}| = 1$, and by (2),

$$\big\|\widehat{(G_1)}_k(s) - \widehat{(G_2)}_k(s)\big\| \le \frac{1}{2\pi}\int_{(-\pi,\pi)} \big\|(G_1 - G_2)\big(\mathrm{ci}(s,\theta)\big)\big\|\,d\theta. \tag{7}$$

Step 5: proof of (b). Let F_l, F be as in (b) and put $\eta_l = \int_{\mathbb{C}} \|F_l - F\|\,d\sigma$, so $\eta_l \to 0$ by hypothesis. Integrating (7) with $G_1 = F_l$, $G_2 = F$ over $s \in (c', d')$, using Tonelli's theorem to read the iterated integral of a nonnegative measurable function as an integral over the rectangle, and then applying (3) to $F_l - F$,

$$\int_{(c',d')} \big\|\widehat{(F_l)}_k(s) - \widehat{F}_k(s)\big\|\,ds \le \frac{1}{2\pi}\int_{(c',d')\times(-\pi,\pi)} \big\|(F_l - F)\big(\mathrm{ci}(s,\theta)\big)\big\|\,d(s,\theta) \le \frac{1}{2\pi}\cdot\frac{\eta_l}{c'}.$$

Both constants $\frac{1}{2\pi}$ and $\frac{1}{c'}$ are finite and independent of l, the first because $\pi > 0$ and the second because $c' > 0$, so the right-hand side tends to 0 as $l \to \infty$. The left-hand side is nonnegative, so it is squeezed between 0 and a sequence tending to 0 and therefore tends to 0 itself. That is (b), and the proof is complete. □

2.12. `CircleAvgLocAbsCont`.

Lemma 2.12. *Let $n \ge 0$ be an integer, let $0 \le a < b$, let $k \in \mathbb{Z}$, put*

$$A = \{\xi \in \mathbb{C} : a < |\xi| < b\},$$

and let $H, D_0, D_1 \colon \mathbb{C} \to \mathbb{C}$ satisfy

(H1) *$H \in L^1_{\mathrm{loc}}(A)$ and $D_j \in L^1_{\mathrm{loc}}(A)$ for $j = 0, 1$, in the sense of `MemLpLocOn`;*
(H2) *D_j is the distributional j-th partial derivative of H on A for $j = 0, 1$, in the sense of `IsWeakPartialOn`;*
(H3) *H is continuous on A;*
(H4) *H is differentiable at almost every point of A.*

Write

$$a_k(r) = \frac{1}{2\pi}\int_0^{2\pi} H(re^{i\theta})e^{i(n-k)\theta}\,d\theta, \qquad A_k(r) = \frac{1}{2\pi}\int_0^{2\pi} H_r(re^{i\theta})e^{i(n-k)\theta}\,d\theta, \qquad B_k(r) = \frac{1}{2\pi}\int_0^{2\pi} H_\theta(re^{i\theta})e^{i(n-k)\theta}\,d\theta,$$

where the three displayed expressions are the Fourier coefficient (30) of `FourierCoefOfH` formed from H, from the polar radial derivative H_r of `HRad` and from the polar angular derivative H_θ of `HAng` respectively. Then

(a) *for all $c, d \in (a, b)$, A_k is integrable on the closed interval with endpoints c and d;*
(b) *for all $c, d \in (a, b)$, $a_k(d) - a_k(c) = \int_c^d A_k(s)\,ds$;*
(c) *for almost every $r \in (a, b)$, a_k is differentiable at r with $a_k'(r) = A_k(r)$;*
(d) *for almost every $r \in (a, b)$, $B_k(r) = i(k - n)\,a_k(r)$;*

(e) *for almost every* $r \in (a,b)$*, the functions* $\theta \mapsto H_r(re^{i\theta})$ *and* $\theta \mapsto H_\theta(re^{i\theta})$ *are integrable on* $[0,2\pi]$*.*

This is the one-variable content of the paper's (31) and of the two identities that the term-by-term differentiation (32) amounts to. It is what replaces Section 3's standing assumption $H \in C^\infty(\mathbb{B}-\{0\})$*: the paper reads that assumption at exactly one place, paper.tex:265–266, to obtain* $a_k \in C[0,1)\cap C^\infty(0,1)$*, while paper.tex:269 justifies (32) from* $H \in W^{1,2}(\mathbb{B})$ *alone; and the* $C[0,1)$ *half of (31) is continuity, which the inverse of a quasiconformal homeomorphism has for nothing. (a), (b) and (c) say that* a_k *is locally absolutely continuous on* (a,b) *in the strongest usable sense – it is the indefinite integral of* A_k *– and (d) is the integration by parts in* θ *that the paper performs at paper.tex:281–286 to turn (28) into the system (33). The hypotheses are exactly the data of a* $W^{1,1}_{\mathrm{loc}}(A)$ *function, with* D_j *the member of the coherent family indexed by the one-letter word* $[j]$*, together with the continuity and almost-everywhere differentiability that item (C2) of the paper's note supplies for the inverse of a quasiconformal homeomorphism. That the paper means its circle averages to be differentiated in exactly this way is explicit at paper.tex:268–277, where* $H \in W^{1,2}(\mathbb{B})$ *alone is what justifies (32) and its two series are said to converge in* $L^2(\mathbb{B})$*, the derivatives* H_r *and* H_θ *appearing there being the classical ones.*

The conclusions are no stronger than that: the derivative identity can only hold almost everywhere, since H_r *and* H_θ *are defined through the Fréchet derivative of* H*, which (H4) provides only almost everywhere. They are also deliberately short of (31) itself. Nothing here concerns the centre of the disk, or the family of all* k *at once, or the* L^2 *expansion (29), or* C^∞ *regularity in* r*: the* C^∞ *half of (31) is a consequence of the system (33) satisfied by the coefficients, not of this lemma, and the* $C[0,1)$ *half is continuity.*

Proof. Write $d\sigma$ for plane Lebesgue measure, $e_0 = 1$, $e_1 = i$, and $\partial_j v(\zeta) = \mathrm{d}v(\zeta)[e_j]$. The set A is open, and $0 \notin A$ because $|\xi| > a \geq 0$ for $\xi \in A$. For $\zeta \in \mathbb{C}$ write $\zeta = \zeta_1 + i\zeta_2$ with ζ_1, ζ_2 real.

Step 1: the radial and angular combinations of the weak partials. Define, for $\zeta \neq 0$,

$$R(\zeta) = \frac{\zeta_1}{|\zeta|}D_0(\zeta) + \frac{\zeta_2}{|\zeta|}D_1(\zeta), \qquad S(\zeta) = -\zeta_2 D_0(\zeta) + \zeta_1 D_1(\zeta).$$

The hypotheses of `WeakPartialEqFDerivAE` hold with $\Omega = A$, which is open, the function H, the family D, and (H1), (H2), (H4). Its conclusion gives, for each of $j = 0$ and $j = 1$, a null set outside which $D_j = \partial_j H$ on A; let N_1 be the union of those two null sets and of the null set outside which (H4) makes H differentiable on A, so that N_1 is null and, for every $\zeta \in A - N_1$, H is differentiable at ζ and $D_j(\zeta) = \partial_j H(\zeta)$ for $j = 0, 1$. Fix $\zeta \in A - N_1$ and put $r = |\zeta| > 0$. Since $\zeta/|\zeta| = (\zeta_1/r)e_0 + (\zeta_2/r)e_1$ and $i\zeta = -\zeta_2 e_0 + \zeta_1 e_1$, and $\mathrm{d}H(\zeta)$ is $\mathbb{R}$-linear, `HRad` and `HAng` give

$$H_r(\zeta) = \mathrm{d}H(\zeta)\Big[\frac{\zeta}{|\zeta|}\Big] = R(\zeta), \qquad H_\theta(\zeta) = \mathrm{d}H(\zeta)[i\zeta] = S(\zeta) \qquad (\zeta \in A - N_1). \tag{1}$$

Step 2: polar consequences of (1). Let $[c', d'] \subseteq (a,b)$ be a closed interval and put $A' = \{\zeta : c' \leq |\zeta| \leq d'\}$, a compact subset of A. On A' the coefficients of R and S are bounded, by 1 and by d' respectively, so by (H1) applied to the compact set A' both R and S are integrable on A'. Polar coordinates, that is the change of variables $(s,\theta) \mapsto se^{i\theta}$ under which $d\sigma = s\,ds\,d\theta$, together with Tonelli's theorem and $s \geq c' > 0$ on A', give

$$\int_{c'}^{d'}\int_0^{2\pi} |R(se^{i\theta})|\,d\theta\,ds \leq \frac{1}{c'}\int_{A'} |R|\,d\sigma < \infty, \tag{2}$$

and the same bound for S. This estimate, together with the two consequences drawn from it below — that the coefficient of a function integrable on the annulus is integrable in the radius, and that the coefficient depends continuously on the function in L^1 — is `FourierCoefAnnulusRegularity`, stated there for an arbitrary measurable function on a compact annulus of positive inner radius, and it is what Step 4's mollification limit is applied through. By Fubini's theorem the function $s \mapsto \frac{1}{2\pi}\int_0^{2\pi} R(se^{i\theta})e^{i(n-k)\theta}\,d\theta$ is therefore defined for almost every $s \in [c', d']$, measurable, and integrable there; write $\widetilde{A}_k$ for it, and $\widetilde{B}_k$ for the same expression with S in place of R. Moreover the null set N_1 of Step 1 has $\int_{c'}^{d'}\int_0^{2\pi} \mathbf{1}_{N_1}(se^{i\theta})\,s\,d\theta\,ds = 0$, so for almost every $s \in [c', d']$ the

circle $\{se^{i\theta} : 0 \le \theta < 2\pi\}$ meets N_1 in a set of θ-measure zero; for such s the integrands of $A_k(s)$ and $\widetilde{A}_k(s)$ agree for almost every θ by (1), as do those of $B_k(s)$ and $\widetilde{B}_k(s)$. For almost every $s \in [c', d']$ the integrand of $\widetilde{A}_k(s)$ is moreover integrable in θ, by the Fubini theorem just invoked, hence so is the integrand of $A_k(s)$, two functions agreeing almost everywhere being integrable together; and then the two integrals coincide. The same applies to B_k and $\widetilde{B}_k$. Hence

$$A_k(s) = \widetilde{A}_k(s) \quad \text{and} \quad B_k(s) = \widetilde{B}_k(s) \qquad \text{for almost every } s \in [c', d']. \tag{3}$$

In particular A_k agrees almost everywhere on $[c', d']$ with an integrable function, so A_k is integrable on $[c', d']$. Since every $c, d \in (a, b)$ lie in such an interval $[c', d'] \subseteq (a, b)$, this proves (a).

Step 3: the mollification. Fix $c, d \in (a, b)$ and assume first that $c \le d$; the opposite case is reduced to this one at the end of Step 6. Choose $\delta > 0$ with $[c - 3\delta, d + 3\delta] \subseteq (a, b)$, which is possible because (a, b) is open and $[c, d] \subseteq (a, b)$ is compact; note $c - 3\delta > a \ge 0$. Put

$$A'' = \{\zeta : c - 3\delta \le |\zeta| \le d + 3\delta\}, \qquad A^\flat = \{\zeta : c - \delta \le |\zeta| \le d + \delta\},$$

both compact subsets of A, with $A^\flat \subseteq A''$. Fix $\varrho\colon \mathbb{C} \to \mathbb{R}$ of class C^∞ with $\varrho \ge 0$, $\varrho(y) = 0$ for $|y| \ge 1$ and $\int \varrho \, d\sigma = 1$, and for $0 < \varepsilon \le \delta$ put $\varrho_\varepsilon(y) = \varepsilon^{-2}\varrho(y/\varepsilon)$, so that ϱ_ε is C^∞, vanishes for $|y| \ge \varepsilon$ and has integral 1. By (H1) applied to the compact set A'' the functions $\mathbf{1}_{A''}H$ and $\mathbf{1}_{A''}D_j$ lie in $L^1(\mathbb{C})$; put

$$H_\varepsilon = (\mathbf{1}_{A''}H) * \varrho_\varepsilon, \qquad W_{j,\varepsilon} = (\mathbf{1}_{A''}D_j) * \varrho_\varepsilon.$$

These are C^∞ on $\mathbb{C}$, being convolutions of L^1 functions with a compactly supported C^∞ function, and $\partial_j H_\varepsilon = (\mathbf{1}_{A''}H) * \partial_j\varrho_\varepsilon$.

Step 4: $\partial_j H_\varepsilon = W_{j,\varepsilon}$ on $A^\flat$. Let $\zeta \in A^\flat$ and $0 < \varepsilon \le \delta$, and put $\varphi(w) = \varrho_\varepsilon(\zeta - w)$, read as a complex-valued function. Then φ is C^∞ and vanishes off $\{w : |w - \zeta| \le \varepsilon\}$, a compact set contained in $A'' \subseteq A$, because $|\,|w| - |\zeta|\,| \le \varepsilon \le \delta$ there and $c - \delta \le |\zeta| \le d + \delta$. So $\operatorname{tsupp}\varphi \subseteq A$ and φ is a test function for A in the sense of `IsTestFn`. The map $w \mapsto \zeta - w$ is affine with derivative $-\mathrm{id}$, so $\mathrm{d}\varphi(w)[e_j] = -(\partial_j\varrho_\varepsilon)(\zeta - w)$, and (H2) applied to φ reads

$$-\int H(w)(\partial_j\varrho_\varepsilon)(\zeta - w)\, d\sigma(w) = -\int D_j(w)\varrho_\varepsilon(\zeta - w)\, d\sigma(w).$$

Both integrands vanish off $\{w : |w - \zeta| \le \varepsilon\} \subseteq A''$, where $H = \mathbf{1}_{A''}H$ and $D_j = \mathbf{1}_{A''}D_j$, so the displayed identity says precisely

$$\partial_j H_\varepsilon(\zeta) = W_{j,\varepsilon}(\zeta) \qquad (\zeta \in A^\flat,\ 0 < \varepsilon \le \delta). \tag{4}$$

Consequently, with $R_\varepsilon(\zeta) = \frac{\zeta_1}{|\zeta|}W_{0,\varepsilon}(\zeta) + \frac{\zeta_2}{|\zeta|}W_{1,\varepsilon}(\zeta)$ and $S_\varepsilon(\zeta) = -\zeta_2 W_{0,\varepsilon}(\zeta) + \zeta_1 W_{1,\varepsilon}(\zeta)$, the computation of Step 1 applied to the everywhere differentiable H_ε gives

$$(H_\varepsilon)_r = R_\varepsilon \quad \text{and} \quad (H_\varepsilon)_\theta = S_\varepsilon \qquad \text{on } A^\flat. \tag{5}$$

Step 5: the radial identity for H_ε. Put

$$g_\varepsilon(r) = \frac{1}{2\pi}\int_0^{2\pi} H_\varepsilon(re^{i\theta})e^{i(n-k)\theta}\, d\theta.$$

The function $(r, \theta) \mapsto H_\varepsilon(re^{i\theta})e^{i(n-k)\theta}$ is C^∞ on $\mathbb{R}^2$ and the θ-range is a compact interval, so differentiation under the integral sign applies: g_ε is C^1 with

$$g_\varepsilon'(r) = \frac{1}{2\pi}\int_0^{2\pi} \frac{\partial}{\partial r}\big[H_\varepsilon(re^{i\theta})\big]e^{i(n-k)\theta}\, d\theta.$$

The curve $r \mapsto re^{i\theta}$ has velocity $e^{i\theta}$, so by the chain rule $\frac{\partial}{\partial r}[H_\varepsilon(re^{i\theta})] = \mathrm{d}H_\varepsilon(re^{i\theta})[e^{i\theta}]$, and for $r > 0$ the vector $e^{i\theta}$ is $\xi/|\xi|$ at $\xi = re^{i\theta}$, so this is $(H_\varepsilon)_r(re^{i\theta})$ by `HRad`. Hence, for $r > 0$, $g_\varepsilon'(r) = \frac{1}{2\pi}\int_0^{2\pi}(H_\varepsilon)_r(re^{i\theta})e^{i(n-k)\theta}\, d\theta$, and the fundamental theorem of calculus for the C^1 function g_ε on $[c, d] \subseteq (0, \infty)$ gives, using (5) and $\{se^{i\theta} : c \le s \le d\} \subseteq A^\flat$,

$$g_\varepsilon(d) - g_\varepsilon(c) = \int_c^d \Big(\frac{1}{2\pi}\int_0^{2\pi} R_\varepsilon(se^{i\theta})e^{i(n-k)\theta}\, d\theta\Big) ds. \tag{6}$$

Step 6: letting $\varepsilon \to 0$ in (6); proof of (b). First, $H_\varepsilon \to H$ uniformly on $A^\flat$. Indeed H is continuous on A by (H3) and $A'' \subseteq A$ is compact, so H is uniformly continuous on A''; and for $\zeta \in A^\flat$,

$$H_\varepsilon(\zeta) - H(\zeta) = \int \big[H(\zeta - y) - H(\zeta)\big]\, \varrho_\varepsilon(y)\, d\sigma(y),$$

because $\int \varrho_\varepsilon = 1$ and $\mathbf{1}_{A''} H = H$ on $\{\zeta - y : |y| \le \varepsilon\} \subseteq A''$; so $|H_\varepsilon(\zeta) - H(\zeta)| \le \sup\{|H(\zeta') - H(\zeta'')| : \zeta', \zeta'' \in A'',\ |\zeta' - \zeta''| \le \varepsilon\}$, which tends to 0 with ε by uniform continuity. Since $\{re^{i\theta} : r \in [c,d]\} \subseteq A^\flat$ and $|e^{i(n-k)\theta}| = 1$, it follows that $g_\varepsilon \to a_k$ uniformly on $[c,d]$; in particular $g_\varepsilon(c) \to a_k(c)$ and $g_\varepsilon(d) \to a_k(d)$.

Next, $\|W_{j,\varepsilon} - \mathbf{1}_{A''} D_j\|_{L^1(\mathbb{C})} \to 0$, since $g * \varrho_\varepsilon \to g$ in $L^1(\mathbb{C})$ for every $g \in L^1(\mathbb{C})$ by `MollificationL1Convergenc`. On $A^\flat$ the coefficients $\zeta_1/|\zeta|, \zeta_2/|\zeta|$ are bounded by 1 and $\mathbf{1}_{A''} D_j = D_j$, so

$$\int_{A^\flat} |R_\varepsilon - R|\, d\sigma \le \sum_{j=0}^{1} \|W_{j,\varepsilon} - \mathbf{1}_{A''} D_j\|_{L^1(\mathbb{C})} \longrightarrow 0.$$

Passing to polar coordinates exactly as in (2), with $s \ge c - \delta > 0$,

$$\Big|\int_c^d \Big(\frac{1}{2\pi}\int_0^{2\pi} (R_\varepsilon - R)(se^{i\theta}) e^{i(n-k)\theta} d\theta\Big) ds\Big| \le \frac{1}{2\pi(c-\delta)} \int_{A^\flat} |R_\varepsilon - R|\, d\sigma \longrightarrow 0.$$

So the right-hand side of (6) converges to $\int_c^d \widetilde{A}_k(s)\, ds$, which by (3) equals $\int_c^d A_k(s)\, ds$. Letting $\varepsilon \to 0$ in (6) therefore gives $a_k(d) - a_k(c) = \int_c^d A_k(s)\, ds$ whenever $c \le d$. For $d < c$ apply this to the pair (d,c) and use $\int_c^d = -\int_d^c$; for $c = d$ both sides vanish. This is (b).

Step 7: proof of (c). Fix $c_0 \in (a,b)$. By (a) the function A_k is integrable on every closed subinterval of (a,b), hence locally integrable there, and by (b) $a_k(r) = a_k(c_0) + \int_{c_0}^r A_k(s)\, ds$ for every $r \in (a,b)$. The indefinite integral of a locally integrable function on an interval is differentiable at almost every point of that interval, with derivative the integrand (the Lebesgue differentiation theorem in one variable). Hence for almost every $r \in (a,b)$ the function a_k is differentiable at r with $a_k'(r) = A_k(r)$, which is (c).

Step 8: proof of (d). Fix $c, d \in (a,b)$ with $c \le d$ and keep the notation of Steps 3–6. For $r \in [c,d]$ the function $\theta \mapsto H_\varepsilon(re^{i\theta})$ is C^∞ on $\mathbb{R}$, and the curve $\theta \mapsto re^{i\theta}$ has velocity $ire^{i\theta} = i\xi$ at $\xi = re^{i\theta}$, so by the chain rule and `HAng`

$$\frac{\partial}{\partial\theta}\big[H_\varepsilon(re^{i\theta})\big] = \mathrm{d}H_\varepsilon(re^{i\theta})[i\, re^{i\theta}] = (H_\varepsilon)_\theta(re^{i\theta}).$$

Both $\theta \mapsto H_\varepsilon(re^{i\theta})$ and $\theta \mapsto e^{i(n-k)\theta}$ are C^1 on $[0,2\pi]$ and take the same values at 0 and at 2π, the first because $e^{i\theta}$ is 2π-periodic and the second because $n-k$ is an integer. So integration by parts on $[0,2\pi]$ has vanishing boundary term and gives

$$\frac{1}{2\pi}\int_0^{2\pi} (H_\varepsilon)_\theta(re^{i\theta}) e^{i(n-k)\theta}\, d\theta = -i(n-k)\frac{1}{2\pi}\int_0^{2\pi} H_\varepsilon(re^{i\theta}) e^{i(n-k)\theta}\, d\theta = i(k-n)\, g_\varepsilon(r).$$

By (5) the left-hand side is $\frac{1}{2\pi}\int_0^{2\pi} S_\varepsilon(re^{i\theta}) e^{i(n-k)\theta} d\theta$, so

$$\frac{1}{2\pi}\int_0^{2\pi} S_\varepsilon(re^{i\theta}) e^{i(n-k)\theta}\, d\theta = i(k-n) g_\varepsilon(r) \qquad (r \in [c,d],\ 0 < \varepsilon \le \delta). \tag{7}$$

Now let $\varepsilon \to 0$ in (7), comparing the two sides in $L^1[c,d]$. On $A^\flat$ the coefficients of S and S_ε are bounded by $d + \delta$, so as in Step 6

$$\int_c^d \Big|\frac{1}{2\pi}\int_0^{2\pi} (S_\varepsilon - S)(se^{i\theta}) e^{i(n-k)\theta} d\theta\Big| ds \le \frac{d+\delta}{2\pi(c-\delta)} \sum_{j=0}^{1} \|W_{j,\varepsilon} - \mathbf{1}_{A''} D_j\|_{L^1(\mathbb{C})} \longrightarrow 0,$$

so the left-hand side of (7) converges in $L^1[c,d]$ to $\widetilde{B}_k$, which equals B_k almost everywhere on $[c,d]$ by (3). The right-hand side converges to $i(k-n)a_k$ uniformly on $[c,d]$ by Step 6, hence in $L^1[c,d]$. Two $L^1[c,d]$ limits of one family agree almost everywhere, so $B_k(r) = i(k-n)a_k(r)$ for almost every $r \in [c,d]$.

Finally, $(a,b)=\bigcup_{m\ge1}[c_m,d_m]$ with $c_m=a+\frac{b-a}{m+2}$ and $d_m=b-\frac{b-a}{m+2}$, each contained in (a,b); a countable union of null sets is null, so the exceptional sets assemble and (d) holds for almost every $r\in(a,b)$.

Step 9: proof of (e). Step 2, applied on the interval $[c_m,d_m]$, established that for almost every $s\in[c_m,d_m]$ the integrand of $\widetilde{A}_k(s)$ is integrable in θ and that the integrand of $A_k(s)$ is integrable with it, and likewise for $\widetilde{B}_k$ and B_k; those integrands are $\theta\mapsto R(se^{i\theta})e^{i(n-k)\theta}$ and $\theta\mapsto H_r(se^{i\theta})e^{i(n-k)\theta}$, and the two angular ones. Since $|e^{i(n-k)\theta}|=1$ and $\theta\mapsto e^{i(n-k)\theta}$ is continuous, a function is integrable on $[0,2\pi]$ if and only if its product with that factor is. So for almost every $s\in[c_m,d_m]$ the functions $\theta\mapsto H_r(se^{i\theta})$ and $\theta\mapsto H_\theta(se^{i\theta})$ are integrable on $[0,2\pi]$, and assembling the countably many exceptional sets over the exhaustion above gives (e). □

2.13. `FourierCoefRegularity`.

Lemma 2.13. *Let $n\ge0$ be an integer, write $\mathbb{B}=\{\xi:|\xi|<1\}$, and let $H\colon\mathbb{C}\to\mathbb{C}$ satisfy*

(H1) *H is continuous on $\mathbb{B}$;*
(H2) *$H\in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`;*
(H3) *H is differentiable at almost every point of $\mathbb{B}$.*

Let a_k, $k\in\mathbb{Z}$, be the Fourier coefficients of `FourierCoefOfH` of H, and write A_k for the Fourier coefficient of the same index of H_r, the polar radial derivative of `HRad`, and B_k for that of H_θ, the polar angular derivative of `HAng`. Then

(a) *each a_k is continuous on $[0,1)$;*
(b) *for every k and all $c,d\in(0,1)$, the function A_k is integrable on the closed interval with endpoints c and d, and $a_k(d)-a_k(c)=\displaystyle\int_c^d A_k(s)\,ds$;*
(c) *for every k and almost every $r\in(0,1)$, a_k is differentiable at r with $a_k'(r)=A_k(r)$;*
(d) *for every k and almost every $r\in(0,1)$, $B_k(r)=i(k-n)a_k(r)$;*
(e) *for almost every $r\in(0,1)$ the functions $\theta\mapsto H_r(re^{i\theta})$ and $\theta\mapsto H_\theta(re^{i\theta})$ are integrable on $[0,2\pi]$;*
(f) *for every $r\in[0,1)$ the expansion (29) holds in $L^2(0,2\pi)$:*

$$\int_0^{2\pi}\Big|H\big(re^{i\theta}\big)-\sum_{k=-N}^{N}a_k(r)e^{i(k-n)\theta}\Big|^2\,d\theta\ \longrightarrow\ 0\qquad(N\to\infty).$$

This is the paper's (31) and (32) with Section 3's C^∞ standing assumption removed. The paper reads that assumption at exactly one place, paper.tex:265–266, to obtain $a_k\in C[0,1)\cap C^\infty(0,1)$, whereas paper.tex:269 justifies the term-by-term differentiation (32) from $H\in W^{1,2}(\mathbb{B})$ alone. Accordingly the $C[0,1)$ half of (31) is (a), which needs only continuity – free for the inverse of a quasiconformal homeomorphism – while the $C^\infty(0,1)$ half is deliberately not *claimed here: it is regained downstream from the system (33) that the coefficients satisfy, which is the paper's own reason for (31) in the first place. Parts (c) and (d) are what (32) asserts, namely that the Fourier coefficients of H_r and H_θ are a_k' and $i(k-n)a_k$; (b) is the integrated form of (c), which is what survives when a_k is not yet known to be differentiable everywhere; (e) is the integrability on circles that the passage from (28) to (33) needs in order to integrate in θ; and (f) is the expansion (29).*

Proof. Fix $k\in\mathbb{Z}$ and write, for $r\in[0,1)$ and $\theta\in\mathbb{R}$,

$$\gamma_r(\theta)=re^{i\theta},\qquad\varphi_k(\theta)=e^{i(n-k)\theta},$$

so that, by `FourierCoefOfH`,

$$a_k(r)=\frac{1}{2\pi}\int_0^{2\pi}H\big(\gamma_r(\theta)\big)\varphi_k(\theta)\,d\theta.\tag{0}$$

Since $|e^{i\theta}|=1$ we have $|\gamma_r(\theta)|=r<1$, so $\gamma_r(\theta)\in\mathbb{B}$ for every θ; and $|\varphi_k(\theta)|=1$.

Step 1: joint continuity and local bounds. The map $(r,\theta)\mapsto\gamma_r(\theta)$ is continuous from $[0,1)\times\mathbb{R}$ into $\mathbb{B}$, so $(r,\theta)\mapsto H(\gamma_r(\theta))\varphi_k(\theta)$ is continuous on $[0,1)\times\mathbb{R}$, by (H1) and by continuity of φ_k. For $0\le\rho<1$ the set $\{\xi:|\xi|\le\rho\}$ is a compact subset of $\mathbb{B}$, so

$$M_\rho=\sup\big\{|H(\xi)|:|\xi|\le\rho\big\}<\infty,$$

and $|H(\gamma_r(\theta))\varphi_k(\theta)|\le M_\rho$ for all $r\in[0,\rho]$ and all θ.

Step 2: proof of (a). Let $r_0\in[0,1)$ and choose ρ with $r_0<\rho<1$. For $r\in[0,\rho]$ the integrand in (0) is dominated by the constant M_ρ, which is integrable on $[0,2\pi]$, and for each fixed θ it is continuous in r by Step 1. Hence for any sequence $r_j\to r_0$ in $[0,\rho]$ the dominated convergence theorem gives

$$\int_0^{2\pi}H\big(\gamma_{r_j}(\theta)\big)\varphi_k(\theta)\,d\theta\longrightarrow\int_0^{2\pi}H\big(\gamma_{r_0}(\theta)\big)\varphi_k(\theta)\,d\theta,$$

so a_k is continuous at r_0. As $r_0\in[0,1)$ was arbitrary, a_k is continuous on $[0,1)$, and k was arbitrary; this is (a).

Step 3: the punctured disk as an annulus, and the Sobolev data there. Put

$$A=\{\xi:0<|\xi|<1\}=\mathbb{B}-\{0\},$$

the annulus with radii $a=0$ and $b=1$, so that $0\le a<b$. By (H2) and `MemWLocOn` there is a family D of functions indexed by words in $\{0,1\}$ with $D[\,]=H$, with $D(w)\in L^2_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemLpLocOn` for every word w of length at most 1, and with $D([j])$ the distributional j-th partial derivative of H on $\mathbb{B}$ in the sense of `IsWeakPartialOn` for $j=0,1$ — the latter because the only word w with $|w|<1$ is the empty word, and $[\,]{+}{+}[j]=[j]$. Put $D_j=D([j])$. We check the four hypotheses of `CircleAvgLocAbsCont` for n, $a=0$, $b=1$, the index k, the function H and the family D_0,D_1.

For its (H1): let $\kappa\subseteq A$ be compact. Then κ is a compact subset of $\mathbb{B}$, so by the above $\mathbf{1}_\kappa H$ and $\mathbf{1}_\kappa D_j$ lie in L^2; and κ has finite Lebesgue measure, so by the Cauchy–Schwarz inequality $\int_\kappa|g|\le\big(\int_\kappa|g|^2\big)^{1/2}\sigma(\kappa)^{1/2}<\infty$ for $g\in\{H,D_0,D_1\}$. Hence H and the D_j lie in $L^1_{\mathrm{loc}}(A)$ in the sense of `MemLpLocOn`.

For its (H2): a test function for A is a test function for $\mathbb{B}$, since by `IsTestFn` the only clause mentioning the set is $\operatorname{tsupp}\varphi\subseteq A\subseteq\mathbb{B}$, and the integration-by-parts identity of `IsWeakPartialOn` does not mention the set; so D_j is the distributional j-th partial derivative of H on A as well.

For its (H3): H is continuous on $A\subseteq\mathbb{B}$ by (H1) of the present lemma.

For its (H4): by (H3) of the present lemma H is differentiable at almost every point of $\mathbb{B}$, hence at almost every point of the subset A.

Step 4: proof of (b), (c), (d), (e). `CircleAvgLocAbsCont`, applied as in Step 3, yields for this k: its (a), that A_k is integrable on the closed interval with endpoints c and d for all $c,d\in(0,1)$; its (b), that $a_k(d)-a_k(c)=\int_c^d A_k(s)\,ds$ for all such c,d; its (c), that for almost every $r\in(0,1)$ the function a_k is differentiable at r with $a_k'(r)=A_k(r)$; its (d), that $B_k(r)=i(k-n)a_k(r)$ for almost every $r\in(0,1)$; and its (e), that for almost every $r\in(0,1)$ the functions $\theta\mapsto H_r(re^{i\theta})$ and $\theta\mapsto H_\theta(re^{i\theta})$ are integrable on $[0,2\pi]$. The first two are (b), the third is (c), the fourth is (d), all for this k and hence, k being arbitrary, for every k; and the fifth is (e), which does not depend on k.

Step 5: proof of (f). Fix $r\in[0,1)$. By Step 1 the function $\theta\mapsto H(\gamma_r(\theta))$ is continuous on the compact interval $[0,2\pi]$, hence bounded and measurable there, so it belongs to $L^2(0,2\pi)$. The family $(e_m)_{m\in\mathbb{Z}}$, $e_m(\theta)=(2\pi)^{-1/2}e^{im\theta}$, is an orthonormal basis of $L^2(0,2\pi)$. Its m-th coefficient for our function is

$$\big\langle H\circ\gamma_r,\,e_m\big\rangle=(2\pi)^{-1/2}\int_0^{2\pi}H\big(\gamma_r(\theta)\big)e^{-im\theta}\,d\theta.$$

Taking $m=k-n$ and using $e^{-i(k-n)\theta}=e^{i(n-k)\theta}=\varphi_k(\theta)$, this equals $(2\pi)^{1/2}a_k(r)$ by (0); consequently

$$\big\langle H\circ\gamma_r,\,e_{k-n}\big\rangle\,e_{k-n}(\theta)=(2\pi)^{1/2}a_k(r)\cdot(2\pi)^{-1/2}e^{i(k-n)\theta}=a_k(r)\,e^{i(k-n)\theta}.$$

Because $(e_m)_{m\in\mathbb{Z}}$ is an orthonormal basis, the expansion of $H\circ\gamma_r$ converges unconditionally in $L^2(0,2\pi)$: the net of partial sums over finite subsets of the index set converges to $H\circ\gamma_r$. In

particular, along any sequence of finite index sets increasing to the whole index set the partial sums converge in $L^2(0,2\pi)$. The sets $\{k \in \mathbb{Z} : |k| \le N\}$ increase to $\mathbb{Z}$ as $N \to \infty$, and the corresponding partial sum is, by the display above, $\sum_{k=-N}^{N} a_k(r)e^{i(k-n)\theta}$. Hence

$$\int_0^{2\pi} \Big| H\big(\gamma_r(\theta)\big) - \sum_{k=-N}^{N} a_k(r)e^{i(k-n)\theta} \Big|^2 \, d\theta \longrightarrow 0$$

as $N \to \infty$, which is (f). □

2.14. `WirtingerPolarForm`.

Lemma 2.14. *Let $H\colon \mathbb{C} \to \mathbb{C}$ be real-differentiable at $\xi \in \mathbb{C}$ with $\xi \neq 0$, and write $r = |\xi| > 0$ and $e^{i\theta} = \xi/r$, so that $e^{-i\theta} = \bar{\xi}/r$. Then*

$$H_{\bar{\xi}}(\xi) = \frac{1}{2}\Big(H_r(\xi) + \frac{i}{r}H_\theta(\xi)\Big)e^{i\theta}, \qquad H_{\xi}(\xi) = \frac{1}{2}\Big(H_r(\xi) - \frac{i}{r}H_\theta(\xi)\Big)e^{-i\theta},$$

and equivalently

$$H_r(\xi) = H_\xi(\xi)\, e^{i\theta} + H_{\bar{\xi}}(\xi)\, e^{-i\theta},$$

where H_ξ is as in `DZ`, $H_{\bar{\xi}}$ as in `DZbar`, H_r as in `HRad` and H_θ as in `HAng`.

The first two identities are the transformation formulas displayed at the start of the proof of Theorem 2 in Section 4 of the paper. The third is the resolution of that pair for H_r; it is recorded here because the distortion estimates (26), (27) and the Fourier computation of Section 4 use H_r expressed through H_ξ and $H_{\bar{\xi}}$ rather than the other way round.

Proof. Put $r = |\xi|$, which is > 0 since $\xi \neq 0$, and

$$e = \frac{\xi}{r},$$

so that $|e| = 1$, that is $e\bar{e} = 1$, and $\bar{e} = \bar{\xi}/r$. Write $L = DH(\xi)$ for the real Fréchet derivative of H at ξ, a real-linear map $\mathbb{C} \to \mathbb{C}$, and put

$$A = H_\xi(\xi), \qquad B = H_{\bar{\xi}}(\xi).$$

By `DZ` and `DZbar`,

$$A = \tfrac{1}{2}\big(L[1] - i\,L[i]\big), \qquad B = \tfrac{1}{2}\big(L[1] + i\,L[i]\big). \tag{1}$$

Adding and subtracting the two formulas in (1) gives

$$L[1] = A + B, \qquad L[i] = i(A - B), \tag{2}$$

the second because $A - B = -i\,L[i]$ and $i(-i\,L[i]) = L[i]$.

Step 1: the two directional derivatives. By `HRad`, $H_r(\xi) = L[\xi/|\xi|] = L[e]$, and by `HAng`, $H_\theta(\xi) = L[i\xi] = L[i\,re] = r\,L[ie]$, the last step by real-linearity of L, r being a real scalar.

Write $e = c + is$ with $c, s \in \mathbb{R}$, so that $ie = -s + ic$ and $\bar{e} = c - is$. By real-linearity of L and (2),

$$H_r(\xi) = L[c\cdot 1 + s\cdot i] = c\,L[1] + s\,L[i] = c(A+B) + s\,i(A-B) = A(c+is) + B(c-is),$$

that is

$$H_r(\xi) = A\,e + B\,\bar{e}. \tag{3}$$

Likewise

$$L[ie] = L[(-s)\cdot 1 + c\cdot i] = -s\,L[1] + c\,L[i] = -s(A+B) + c\,i(A-B) = A(-s+ic) + B(-s-ic),$$

and $-s + ic = ie$ while $-s - ic = -i(c - is) = -i\bar{e}$, so $L[ie] = A(ie) - B(i\bar{e})$ and therefore

$$H_\theta(\xi) = r\,L[ie] = i\,r\big(A\,e - B\,\bar{e}\big). \tag{4}$$

Step 2: the two Wirtinger formulas. Since $r \neq 0$, (4) gives

$$\frac{i}{r}H_\theta(\xi) = \frac{i}{r}\cdot i\,r\big(A\,e - B\,\bar{e}\big) = i^2\big(A\,e - B\,\bar{e}\big) = -A\,e + B\,\bar{e}.$$

Adding and subtracting this and (3),

$$H_r(\xi) + \frac{i}{r}H_\theta(\xi) = 2B\,\bar{e}, \qquad H_r(\xi) - \frac{i}{r}H_\theta(\xi) = 2A\,e.$$

Multiplying the first by $\frac{1}{2}e$ and the second by $\frac{1}{2}\bar{e}$, and using $e\bar{e} = 1$,

$$\frac{1}{2}\Big(H_r(\xi) + \frac{i}{r}H_\theta(\xi)\Big)e = B\,\bar{e}\,e = B = H_{\bar{\xi}}(\xi),$$
$$\frac{1}{2}\Big(H_r(\xi) - \frac{i}{r}H_\theta(\xi)\Big)\bar{e} = A\,e\,\bar{e} = A = H_{\xi}(\xi).$$

Since $e = \xi/r = e^{i\theta}$ and $\bar{e} = \bar{\xi}/r = e^{-i\theta}$, these are the first two identities of the statement, and (3) is the third. □

2.15. `PolarEq28`.

Lemma 2.15. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, let $\Omega \subset \mathbb{C}$ be open, and let $H\colon \mathbb{C} \to \mathbb{C}$ be differentiable at almost every point of Ω and satisfy equation (18) on Ω in the sense of `SolvesEq18`. Then for almost every $\xi \in \Omega$ with $\xi \neq 0$,*

$$2r\,H_r(\xi) = -ip\,H_\theta(\xi) + (p-2)\,i\,e^{-2in\theta}\,\overline{H_\theta(\xi)},$$

where $r = |\xi|$, $e^{-2in\theta} = (\bar{\xi}/\xi)^n$, and H_r, H_θ are as in `HRad` and `HAng`.

This is equation (28) of the paper. Both sides are built from the Fréchet derivative of H, so almost-everywhere differentiability is exactly the right hypothesis: it is what item (C2) of the paper's note supplies for the inverse of a quasiconformal homeomorphism, and the conclusion, like (18) itself, is an almost-everywhere identity.

Proof. Put

$$c = \frac{1}{2} - \frac{1}{p}.$$

Since $p > 1$ we have $1/p > 0$, hence

$$1 - c = \frac{1}{2} + \frac{1}{p} > 0, \qquad \text{in particular } 1 - c \neq 0. \tag{0}$$

Step 0: (18) for the pointwise derivatives. By `SolvesEq18` there are weak Wirtinger derivatives a, b of H on Ω, in the sense of `IsWeakWirtingerOn`, and a null set Z_0 such that

$$b(\xi) = c\Big[\frac{\xi}{\bar{\xi}}a(\xi) + \frac{\bar{\xi}^n}{\xi^n}\overline{a(\xi)}\Big] \qquad (\xi \in \Omega - Z_0,\ \xi \neq 0). \tag{18′}$$

By `IsWeakWirtingerOn` there are in turn H_x, H_y such that $H, H_x, H_y \in L^1_{\mathrm{loc}}(\Omega)$, such that H_x and H_y are the distributional ∂_0- and ∂_1-derivatives of H on Ω, and such that $a = \frac{1}{2}(H_x - iH_y)$, $b = \frac{1}{2}(H_x + iH_y)$ at every point. These are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set Ω, the function H and the pair $D_0 = H_x$, $D_1 = H_y$, and its (H3) is the hypothesis that H is differentiable at almost every point of Ω; so that lemma provides a null set Z_1 with

$$H_x(\xi) = \partial_0 H(\xi), \qquad H_y(\xi) = \partial_1 H(\xi) \qquad (\xi \in \Omega - Z_1),$$

where $\partial_j = D^{[j]}$ is the classical partial derivative of `PartialWord`, that is $\partial_0 H(\xi) = \mathrm{d}H(\xi)[1]$ and $\partial_1 H(\xi) = \mathrm{d}H(\xi)[i]$. By `DZ` and `DZbar`, $H_\xi(\xi)$ and $H_{\bar{\xi}}(\xi)$ are the same two combinations $\frac{1}{2}\big(\mathrm{d}H(\xi)[1] \mp i\,\mathrm{d}H(\xi)[i]\big)$, so

$$a(\xi) = H_\xi(\xi), \qquad b(\xi) = H_{\bar{\xi}}(\xi) \qquad (\xi \in \Omega - Z_1). \tag{18″}$$

Substituting (18″) into (18′), the pointwise form

$$H_{\bar{\xi}}(\xi) = c\Big[\frac{\xi}{\bar{\xi}}H_\xi(\xi) + \frac{\bar{\xi}^n}{\xi^n}\overline{H_\xi(\xi)}\Big] \tag{18}$$

holds at every $\xi \in \Omega - (Z_0 \cup Z_1)$ with $\xi \neq 0$.

Discard the union of $Z_0 \cup Z_1$ and the null set where H fails to be differentiable, which is again null, and fix a remaining $\xi \in \Omega$ with $\xi \neq 0$. Then (18) holds at ξ and H is real-differentiable there, so `WirtingerPolarForm` applies at ξ.

Notation. Put

$$r = |\xi| > 0, \qquad e = \frac{\xi}{r}, \qquad u = H_r(\xi), \qquad v = \frac{i}{r} H_\theta(\xi), \qquad E = \Big(\frac{\bar\xi}{\xi}\Big)^n.$$

Then $|e| = 1$, i.e. $e\bar e = 1$, and $\bar e = \bar\xi / r$. Consequently

$$\frac{\xi}{\bar\xi} = \frac{re}{r\bar e} = \frac{e}{\bar e} = e^2, \qquad \frac{\bar\xi^{\,n}}{\xi^{\,n}} = \Big(\frac{\bar\xi}{\xi}\Big)^n = E,$$

the middle step because $1/\bar e = e$. Also $\overline{E} = (\xi/\bar\xi)^n$ and

$$E\,\overline{E} = 1. \tag{1}$$

By `WirtingerPolarForm`,

$$H_{\bar\xi}(\xi) = \tfrac{1}{2}(u+v)\,e, \qquad H_\xi(\xi) = \tfrac{1}{2}(u-v)\,\bar e, \tag{2}$$

and therefore, conjugation being a ring homomorphism with $\bar{\bar e} = e$,

$$\overline{H_\xi(\xi)} = \tfrac{1}{2}\,\overline{(u-v)}\,e. \tag{3}$$

Step 1: (18) in polar form. Substituting (2) and (3) into (18), and using $\xi/\bar\xi = e^2$ together with $e^2\bar e = e$ (which follows from $e\bar e = 1$),

$$\tfrac{1}{2}(u+v)e = c\Big[e^2\cdot\tfrac{1}{2}(u-v)\bar e + E\cdot\tfrac{1}{2}\overline{(u-v)}\,e\Big] = \tfrac{1}{2}\,c\,e\Big[(u-v) + E\,\overline{(u-v)}\Big].$$

Since $e \neq 0$, dividing by $\frac{1}{2}e$ gives

$$u + v = c\Big[(u-v) + E\,\overline{(u-v)}\Big]. \tag{4}$$

Step 2: conjugate and eliminate $\bar u$. Expanding (4) and collecting the terms containing $\bar u$ on the right,

$$(1-c)u + (1+c)v + cE\,\bar v = cE\,\bar u. \tag{5}$$

Conjugating (4) — the number c is real, so $\bar c = c$ — gives $\bar u + \bar v = c\big[\overline{(u-v)} + \overline{E}(u-v)\big]$, that is, collecting the $\bar u$ terms,

$$(1-c)\bar u = -(1+c)\bar v + c\,\overline{E}\,(u-v). \tag{6}$$

Multiply (5) by $1-c$ and then replace $(1-c)\bar u$ using (6):

$$(1-c)^2u + (1-c)(1+c)v + c(1-c)E\,\bar v = cE\cdot(1-c)\bar u = cE\Big[-(1+c)\bar v + c\,\overline{E}\,(u-v)\Big],$$

and the right side equals $-c(1+c)E\,\bar v + c^2(u-v)$ by (1). Bringing everything to one side and using $(1-c)(1+c) = 1-c^2$,

$$\big[(1-c)^2 - c^2\big]u + \big[(1-c^2) + c^2\big]v + \big[c(1-c) + c(1+c)\big]E\,\bar v = 0.$$

Now $(1-c)^2 - c^2 = 1 - 2c$, $(1-c^2) + c^2 = 1$ and $c(1-c) + c(1+c) = 2c$, so

$$(1-2c)\,u + v + 2c\,E\,\bar v = 0. \tag{7}$$

(Note that this elimination multiplied by $1-c$, which is nonzero by (0); it did not divide by c, so (7) is valid also when $c = 0$, i.e. when $p = 2$.)

Step 3: back to H_r and H_θ. From $c = \frac{1}{2} - \frac{1}{p}$,

$$1 - 2c = 1 - 1 + \frac{2}{p} = \frac{2}{p}, \qquad 2c = 1 - \frac{2}{p} = \frac{p-2}{p}.$$

Multiplying (7) by $p > 0$,

$$2u + p\,v + (p-2)\,E\,\bar v = 0. \tag{8}$$

Finally $v = \frac{i}{r}H_\theta(\xi)$ with r real, so

$$r\,v = i\,H_\theta(\xi), \qquad r\,\bar v = r\,\overline{\Big(\frac{i}{r}H_\theta(\xi)\Big)} = r\cdot\Big(-\frac{i}{r}\Big)\overline{H_\theta(\xi)} = -i\,\overline{H_\theta(\xi)},$$

using that r is real so $\overline{i/r} = -i/r$. Multiplying (8) by $r > 0$ and substituting,

$$2r\,H_r(\xi) + i\,p\,H_\theta(\xi) - (p-2)\,i\,E\,\overline{H_\theta(\xi)} = 0,$$

that is

$$2r\,H_r(\xi) = -i\,p\,H_\theta(\xi) + (p-2)\,i\,E\,\overline{H_\theta(\xi)},$$

which is the assertion, since $E = (\bar{\xi}/\xi)^n = e^{-2in\theta}$. □

2.16. `LinearOdeSystemSmooth`.

Lemma 2.16. *Let $a < b$ be real, write $I = (a,b)$, and let $u, v, U, V \colon \mathbb{R} \to \mathbb{C}$ and $M_{ij} \colon \mathbb{R} \to \mathbb{C}$, $i, j \in \{0,1\}$, satisfy*

(H1) *each M_{ij} is C^∞ on I;*
(H2) *U and V are integrable on the closed interval with endpoints c and d, for all $c, d \in I$;*
(H3) *$u(d) - u(c) = \int_c^d U(s)\,ds$ and $v(d) - v(c) = \int_c^d V(s)\,ds$ for all $c, d \in I$;*
(H4) *for almost every $r \in I$,*

$$U(r) = M_{00}(r)u(r) + M_{01}(r)v(r), \qquad V(r) = M_{10}(r)u(r) + M_{11}(r)v(r).$$

Then u and v are C^∞ on I.

This is the one-dimensional bootstrapping step that regains the paper's (31) once the Fourier coefficients are known to satisfy the system (33). At paper.tex:265–266 the paper obtains $a_k \in C[0,1) \cap C^\infty(0,1)$ from the standing assumption that H is continuous on $\mathbb{B}$ and C^∞ on $\mathbb{B} - \{0\}$; the content here is that the C^∞ half follows instead from the system the coefficients satisfy, so that only $W^{1,2}$ data is needed. Hypotheses (H2) and (H3) are local absolute continuity in the form in which an indefinite integral provides it, which is what the circle averages of a Sobolev function supply, and (H4) is the system holding only almost everywhere, which is all that an almost-everywhere derivative identity gives. No existence theorem for ordinary differential equations is involved: this is bootstrapping, not solving. In particular no nondegeneracy of M is assumed, and none is used – the conclusion holds even when a row of the system decouples, which happens in the intended application at $|k| = n$.

Proof. Step 1: u and v are continuous on I. Fix $c \in I$. For $r \in I$, (H3) gives $u(r) = u(c) + \int_c^r U(s)\,ds$. Let $r_0 \in I$ and choose $c_1, d_1 \in I$ with $c_1 < r_0 < d_1$ and $c \in [c_1, d_1]$; by (H2), U is integrable on $[c_1, d_1]$. The indefinite integral of a function integrable on a compact interval is continuous there: if $r_j \to r_0$ in $[c_1, d_1]$ then $\left|\int_c^{r_j} U - \int_c^{r_0} U\right| = \left|\int_{r_0}^{r_j} U\right| \le \int_{[r_0 \wedge r_j,\, r_0 \vee r_j]} |U|$, which tends to 0 by absolute continuity of the measure $|U|\,ds$ with respect to Lebesgue measure. Hence u is continuous at r_0, and as $r_0 \in I$ was arbitrary, u is continuous on I. The same argument applies to v with V.

Step 2: the hinge – u and v are C^1 on I with derivatives given by the system at every point. Put

$$W(r) = M_{00}(r)u(r) + M_{01}(r)v(r), \qquad Z(r) = M_{10}(r)u(r) + M_{11}(r)v(r) \qquad (r \in I).$$

By (H1) each M_{ij} is continuous on I and by Step 1 so are u and v; hence W and Z are continuous on I. By (H4), $U = W$ almost everywhere on I and $V = Z$ almost everywhere on I. Two functions that agree almost everywhere have the same integral over every interval on which one of them is integrable, so by (H2) and (H3), for all $c, d \in I$,

$$u(d) - u(c) = \int_c^d W(s)\,ds, \qquad v(d) - v(c) = \int_c^d Z(s)\,ds. \tag{1}$$

Fix $c \in I$ and $r \in I$. Choose $c_1, d_1 \in I$ with $c_1 < r < d_1$ and $c \in [c_1, d_1]$. On $[c_1, d_1]$ the function W is continuous, so by the fundamental theorem of calculus the map $t \mapsto \int_c^t W(s)\,ds$ is differentiable at r with derivative $W(r)$. By (1) that map is $t \mapsto u(t) - u(c)$, so u is differentiable at r with

$$u'(r) = W(r) = M_{00}(r)u(r) + M_{01}(r)v(r), \tag{2}$$

and likewise

$$v'(r) = Z(r) = M_{10}(r)u(r) + M_{11}(r)v(r). \tag{3}$$

These hold at *every* $r \in I$, not merely almost everywhere: that is the point of the step. Since W and Z are continuous on I, u and v are C^1 on I.

Step 3: induction. We show by induction on $m \geq 0$ that u and v are C^m on I. For $m = 0$ this is Step 1. Assume u and v are C^m on I. Each M_{ij} is C^∞ on I by (H1), hence C^m there, and the pointwise product and sum of C^m functions on I are C^m; so W and Z are C^m on I. By (2) and (3) the functions u and v are differentiable on the open set I with derivatives W and Z of class C^m, hence u and v are C^{m+1} on I. This completes the induction, and a function that is C^m on I for every m is C^∞ on I. □

2.17. `FourierCoefOde`.

Lemma 2.17. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, write $\mathbb{B} = \{\xi \in \mathbb{C} : |\xi| < 1\}$, and let $H \colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *H is continuous on $\mathbb{B}$;*
(H2) *$H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`;*
(H3) *H is differentiable at almost every point of $\mathbb{B}$;*
(H4) *H solves equation (18) on $\mathbb{B}$ in the sense of `SolvesEq18`.*

Let a_k, $k \in \mathbb{Z}$, be the Fourier coefficients of H in the sense of `FourierCoefOfH`. Then for every $k \in \mathbb{Z}$ and every $r \in (0,1)$,

$$2r\, a_k'(r) = -p(n-k)\, a_k(r) - (p-2)(n+k)\, \overline{a_{-k}(r)}, \tag{33}$$

and

$$r\big(r\, a_k'\big)'(r) + pn\, r\, a_k'(r) + (n^2 - k^2)(p-1)\, a_k(r) = 0, \tag{35}$$

and moreover each a_k is C^∞ on $(0,1)$.

The hypothesis surface is Theorem 2's as printed, $H \in W^{1,2}(\mathbb{B})$ solving (18), together with the continuity of H, which the inverse of a quasiconformal homeomorphism has for nothing, and the almost-everywhere differentiability that item (C2) of the paper's note supplies for it. Section 3's C^∞ standing assumption is absent, and the smoothness of the coefficients – the paper's (31), read at paper.tex:265–266 from that assumption – appears here as a conclusion*: it is what the system (33) itself yields, the pair $(a_k, \overline{a_{-k}})$ being closed under (33). That is also what lets (33) be asserted at every r rather than almost every r.*

Proof. Fix $k \in \mathbb{Z}$. For $r \in [0,1)$ and $\theta \in \mathbb{R}$ write $\gamma_r(\theta) = re^{i\theta}$, so that by `FourierCoefOfH`, for every $j \in \mathbb{Z}$,

$$a_j(r) = \frac{1}{2\pi} \int_0^{2\pi} H\big(\gamma_r(\theta)\big) e^{i(n-j)\theta}\, d\theta. \tag{0}$$

Write A_j and B_j for the Fourier coefficients of the same index of H_r and H_θ, the polar derivatives of `HRad` and `HAng`:

$$A_j(r) = \frac{1}{2\pi} \int_0^{2\pi} H_r\big(\gamma_r(\theta)\big) e^{i(n-j)\theta} d\theta, \qquad B_j(r) = \frac{1}{2\pi} \int_0^{2\pi} H_\theta\big(\gamma_r(\theta)\big) e^{i(n-j)\theta} d\theta.$$

Note $|\gamma_r(\theta)| = r$, so $\gamma_r(\theta) \in \mathbb{B}$ for $r \in [0,1)$ and $\gamma_r(\theta) \in \mathbb{B} - \{0\}$ for $r \in (0,1)$.

Step 1: the coefficient identities. The hypotheses (H1), (H2), (H3) are exactly those of `FourierCoefRegularity`, which therefore gives, for every $j \in \mathbb{Z}$:

(1b) for all $c, d \in (0,1)$, A_j is integrable on the closed interval with endpoints c and d, and $a_j(d) - a_j(c) = \int_c^d A_j(s)\, ds$;
(1c) for almost every $r \in (0,1)$, a_j is differentiable at r with $a_j'(r) = A_j(r)$;
(1d) for almost every $r \in (0,1)$, $B_j(r) = i(j-n)\, a_j(r)$;
(1e) for almost every $r \in (0,1)$, the functions $\theta \mapsto H_r(\gamma_r(\theta))$ and $\theta \mapsto H_\theta(\gamma_r(\theta))$ are integrable on $[0, 2\pi]$.

Step 2: equation (28) for almost every r, for almost every θ. The set $\mathbb{B}$ is open and, by (H3), H is differentiable almost everywhere on it, so `PolarEq28` applies with $\Omega = \mathbb{B}$ and yields a set $N_0 \subset \mathbb{C}$ of plane Lebesgue measure zero such that for every $\xi \in \mathbb{B}$ with $\xi \neq 0$ and $\xi \notin N_0$,

$$2|\xi|\, H_r(\xi) = -ip\, H_\theta(\xi) + (p-2)\, i \left(\frac{\bar\xi}{\xi}\right)^n \overline{H_\theta(\xi)}. \tag{2}$$

In polar coordinates plane Lebesgue measure is $r\, dr\, d\theta$, so by Fubini's theorem applied to the indicator of N_0 there is a set $N_1 \subset (0,1)$ of linear measure zero such that for every $r \in (0,1) \setminus N_1$ one has $\gamma_r(\theta) \notin N_0$ for almost every $\theta \in [0, 2\pi]$. For such an r and such a θ, (2) applies at $\xi = \gamma_r(\theta)$; since $|\gamma_r(\theta)| = r$ and

$$\Big(\frac{\overline{\gamma_r(\theta)}}{\gamma_r(\theta)}\Big)^n = \Big(\frac{re^{-i\theta}}{re^{i\theta}}\Big)^n = e^{-2in\theta},$$

this reads

$$2r\, H_r\big(\gamma_r(\theta)\big) = -ip\, H_\theta\big(\gamma_r(\theta)\big) + (p-2)\, i\, e^{-2in\theta}\, \overline{H_\theta\big(\gamma_r(\theta)\big)}. \tag{3}$$

Step 3: the coefficient identity off a null set. Let $N_2 \subset (0,1)$ be the union of N_1, of the two null sets outside which (1d) holds for the indices k and $-k$, and of the null set outside which (1e) holds; N_2 is null. Fix $r \in (0,1) \setminus N_2$. By (1e) the functions $\theta \mapsto H_r(\gamma_r(\theta))$ and $\theta \mapsto H_\theta(\gamma_r(\theta))$ are integrable on $[0, 2\pi]$, and multiplying an integrable function by the continuous unimodular factor $e^{i(n-j)\theta}$ or by $e^{-2in\theta}$ preserves integrability, so every integral written below is absolutely convergent. Multiplying (3) by $\frac{1}{2\pi} e^{i(n-k)\theta}$ and integrating over $\theta \in [0, 2\pi]$ — legitimate because (3) holds for almost every θ, and a null set of θ does not affect the integrals — gives

$$2r\, A_k(r) = -ip \cdot \frac{1}{2\pi}\int_0^{2\pi} H_\theta(\gamma_r(\theta)) e^{i(n-k)\theta} d\theta + (p-2)i \cdot \frac{1}{2\pi}\int_0^{2\pi} \overline{H_\theta(\gamma_r(\theta))} e^{i(n-k)\theta} e^{-2in\theta} d\theta.$$

For the first term on the right, (1d) with $j = k$ gives $B_k(r) = i(k-n)a_k(r)$, so that term is

$$-ip \cdot i(k-n)a_k(r) = p(k-n)a_k(r) = -p(n-k)a_k(r),$$

using $i \cdot i = -1$. For the second, $e^{i(n-k)\theta} e^{-2in\theta} = e^{-i(n+k)\theta} = \overline{e^{i(n+k)\theta}}$, and conjugation commutes with the integral, so

$$\frac{1}{2\pi}\int_0^{2\pi} \overline{H_\theta(\gamma_r(\theta))}\, \overline{e^{i(n+k)\theta}}\, d\theta = \overline{\frac{1}{2\pi}\int_0^{2\pi} H_\theta(\gamma_r(\theta)) e^{i(n+k)\theta}\, d\theta} = \overline{B_{-k}(r)},$$

because $n + k = n - (-k)$. By (1d) with $j = -k$, $B_{-k}(r) = i(-k-n)a_{-k}(r)$, so, using $\bar i = -i$, that $k + n$ is a real integer, and $i \cdot (-i) = 1$, the second term equals

$$(p-2)i \cdot \overline{i(-k-n)a_{-k}(r)} = (p-2)\, i\, (-i)(-k-n)\overline{a_{-k}(r)} = -(p-2)(n+k)\overline{a_{-k}(r)}.$$

Altogether

$$2r\, A_k(r) = -p(n-k)a_k(r) - (p-2)(n+k)\overline{a_{-k}(r)} \qquad \big(r \in (0,1) \setminus N_2\big), \tag{4}$$

and, $k \in \mathbb{Z}$ having been arbitrary in Steps 1–3, the same identity with k replaced by $-k$ holds off a null set N_3:

$$2r\, A_{-k}(r) = -p(n+k)a_{-k}(r) - (p-2)(n-k)\overline{a_k(r)} \qquad \big(r \in (0,1) \setminus N_3\big). \tag{5}$$

Step 4: each a_k is C^∞ on $(0,1)$. Put

$$u = a_k, \quad v = \overline{a_{-k}}, \quad U = A_k, \quad V = \overline{A_{-k}},$$

and, for $r \in (0,1)$,

$$M_{00}(r) = \frac{-p(n-k)}{2r}, \quad M_{01}(r) = \frac{-(p-2)(n+k)}{2r}, \quad M_{10}(r) = \frac{-(p-2)(n-k)}{2r}, \quad M_{11}(r) = \frac{-p(n+k)}{2r}.$$

We verify the hypotheses of `LinearOdeSystemSmooth` with $a = 0$, $b = 1$.

(H1): each M_{ij} is a real constant times $r \mapsto 1/(2r)$, which is C^∞ on $(0,1)$ because $r \neq 0$ there.

(H2): by (1b) with $j = k$, $U = A_k$ is integrable on the closed interval with endpoints c and d for all $c, d \in (0,1)$; by (1b) with $j = -k$ the same holds for A_{-k}, hence for $V = \overline{A_{-k}}$, since $|\overline{w}| = |w|$.

(H3): by (1b) with $j = k$, $u(d) - u(c) = \int_c^d U(s)\,ds$. By (1b) with $j = -k$, $a_{-k}(d) - a_{-k}(c) = \int_c^d A_{-k}(s)\,ds$; conjugating and using that conjugation is a continuous $\mathbb{R}$-linear map, hence commutes with the integral of an integrable function,

$$v(d) - v(c) = \overline{a_{-k}(d) - a_{-k}(c)} = \overline{\int_c^d A_{-k}(s)\,ds} = \int_c^d \overline{A_{-k}(s)}\,ds = \int_c^d V(s)\,ds.$$

(H4): let $r \in (0,1)$ lie outside the null set $N_2 \cup N_3$. Dividing (4) by $2r \neq 0$ gives $U(r) = M_{00}(r)u(r) + M_{01}(r)v(r)$. Conjugating (5), and using that p, n, k, r are real and that conjugation is a ring homomorphism with $\overline{\overline{w}} = w$,

$$2r\,\overline{A_{-k}(r)} = -p(n+k)\overline{a_{-k}(r)} - (p-2)(n-k)a_k(r),$$

and dividing by $2r$ gives $V(r) = M_{10}(r)u(r) + M_{11}(r)v(r)$.

`LinearOdeSystemSmooth` therefore applies and gives that $u = a_k$ and $v = \overline{a_{-k}}$ are C^∞ on $(0,1)$. Since $k \in \mathbb{Z}$ was arbitrary, every a_k is C^∞ on $(0,1)$; this is the third assertion.

Step 5: (33) for every $r \in (0,1)$. By Step 4 and (1c) with $j = k$ there is a null set off which $a_k'(r) = A_k(r)$, so by (4), for every $r \in (0,1)$ outside a null set,

$$2r\,a_k'(r) = -p(n-k)a_k(r) - (p-2)(n+k)\overline{a_{-k}(r)}. \tag{6}$$

Both sides of (6) are continuous on $(0,1)$: by Step 4 the functions a_k and a_{-k} are C^∞ on $(0,1)$, hence so is a_k', and $r \mapsto \overline{a_{-k}(r)}$ is continuous because complex conjugation is continuous. The complement of a null set is dense in $(0,1)$: if some nonempty open subinterval missed it, that subinterval would be null, which is false. Two continuous functions on $(0,1)$ agreeing on a dense subset agree everywhere. Hence (6) holds for every $r \in (0,1)$; this is (33).

Step 6: the elimination (34). Fix $r \in (0,1)$. Since k was arbitrary in Steps 1–5, (33) holds with k replaced by $-k$:

$$2r\,a_{-k}'(r) = -p(n+k)a_{-k}(r) - (p-2)(n-k)\overline{a_k(r)}.$$

Conjugating, and using that p, n, k, r are real and that conjugation is a ring homomorphism with $\overline{\overline{w}} = w$,

$$2r\,\overline{a_{-k}'(r)} = -p(n+k)\overline{a_{-k}(r)} - (p-2)(n-k)a_k(r).$$

Multiplying by $(p-2)$,

$$2(p-2)r\,\overline{a_{-k}'(r)} = -p\cdot(p-2)(n+k)\overline{a_{-k}(r)} - (p-2)^2(n-k)a_k(r).$$

By (33), $(p-2)(n+k)\overline{a_{-k}(r)} = -2r\,a_k'(r) - p(n-k)a_k(r)$. Substituting,

$$2(p-2)r\,\overline{a_{-k}'(r)} = p\big[2r\,a_k'(r) + p(n-k)a_k(r)\big] - (p-2)^2(n-k)a_k(r) = 2pr\,a_k'(r) + 4(p-1)(n-k)a_k(r), \tag{7}$$

where the last step uses the identity $p^2 - (p-2)^2 = 4(p-1)$. This is (34).

Step 7: (35). Differentiate (33) with respect to r on $(0,1)$. Every term is differentiable there: a_k and a_{-k} are C^∞ on $(0,1)$ by Step 4, and $\overline{a_{-k}}$ is too, with $\big(\overline{a_{-k}}\big)'(r) = \overline{a_{-k}'(r)}$, because conjugation is an $\mathbb{R}$-linear continuous map and therefore commutes with differentiation. We obtain, for $r \in (0,1)$,

$$\big(2r\,a_k'\big)'(r) = -p(n-k)a_k'(r) - (p-2)(n+k)\overline{a_{-k}'(r)}.$$

Multiply by $2r$ and use (7) for the quantity $2(p-2)r\,\overline{a_{-k}'(r)}$:

$$2r\big(2r\,a_k'\big)'(r) = -2pr(n-k)a_k'(r) - (n+k)\big[2pr\,a_k'(r) + 4(p-1)(n-k)a_k(r)\big].$$

Now expand both sides. On the left, $\big(2ra_k'\big)'(r) = 2a_k'(r) + 2r\,a_k''(r)$, so the left side is $4r\,a_k'(r) + 4r^2a_k''(r)$. On the right,

$$-2pr\,a_k'(r)\big[(n-k)+(n+k)\big] - 4(p-1)(n+k)(n-k)a_k(r) = -4pnr\,a_k'(r) - 4(p-1)(n^2-k^2)a_k(r).$$

Dividing by 4,

$$r\,a_k'(r) + r^2 a_k''(r) + pn\,r\,a_k'(r) + (n^2 - k^2)(p-1)a_k(r) = 0.$$

Finally $r\big(r\,a_k'\big)'(r) = r\big(a_k'(r) + r\,a_k''(r)\big) = r\,a_k'(r) + r^2 a_k''(r)$, so the last display is exactly

$$r\big(r\,a_k'\big)'(r) + pn\,r\,a_k'(r) + (n^2 - k^2)(p-1)a_k(r) = 0,$$

which is (35). □

2.18. `LambdaEpsEstimates`.

Lemma 2.18. *Let $n \geq 1$ be an integer and let $1 < p < \infty$. Then*

(23) $\lambda_n(n,p) = 0$ *and* $\epsilon_n(n,p) = 0$;
(24) $|\epsilon_k(n,p)| < 1$ *for* $k = n+1, n+2, \ldots$;
(25) $\dfrac{p-1}{np} \leq \lambda_k(n,p) \leq k\sqrt{p-1}$ *for* $k = n+1, n+2, \ldots$

Here $\lambda_k(n,p)$ is as in `LambdaCoef` and $\epsilon_k(n,p)$ is as in `EpsCoef`.

Proof. Throughout, $n \geq 1$ and $p > 1$ are fixed. Write $\lambda_k = \lambda_k(n,p)$ and

$$R_k = 4k^2(p-1) + n^2(p-2)^2,$$

so that, by `LambdaCoef`,

$$2\lambda_k = -np + \sqrt{R_k}.$$

Note $np > 0$ because $n \geq 1$ and $p > 1$, and $p - 1 > 0$.

Step 0 (the radicand). Each of $4k^2(p-1)$ and $n^2(p-2)^2$ is nonnegative, so $R_k \geq 0$ for every $k \geq 0$ and $\sqrt{R_k}$ is the ordinary square root of a nonnegative number. Moreover, for $k \geq j \geq 0$,

$$R_k - R_j = 4(p-1)(k^2 - j^2) \geq 0,$$

with strict inequality when $k > j$, since $p-1 > 0$. Hence $k \mapsto R_k$ is nondecreasing, and $k \mapsto \sqrt{R_k}$ and $k \mapsto \lambda_k$ are nondecreasing as well, strictly increasing where R strictly increases.

Step 1 ($\lambda_n = 0$). Taking $k = n$,

$$R_n = 4n^2(p-1) + n^2(p-2)^2 = n^2\big(4p - 4 + p^2 - 4p + 4\big) = n^2p^2.$$

Since $np > 0$ we get $\sqrt{R_n} = np$, and therefore $2\lambda_n = -np + np = 0$, i.e. $\lambda_n = 0$. This is the first half of (23).

Step 2 ($\epsilon_n = 0$). By `EpsCoef` and Step 1,

$$\epsilon_n(n,p) = \frac{\lambda_n + n - n}{\lambda_n + n + n} = \frac{0}{2n} = 0,$$

the denominator $2n$ being nonzero because $n \geq 1$. This completes (23).

Step 3 ($\lambda_k > 0$ for $k > n$). Let $k \geq n+1$. By Step 0 and Step 1,

$$R_k - n^2p^2 = R_k - R_n = 4(p-1)(k^2 - n^2) > 0,$$

since $p - 1 > 0$ and $k > n \geq 1$ gives $k^2 > n^2$. Hence $R_k > n^2p^2 \geq 0$, so $\sqrt{R_k} > \sqrt{n^2p^2} = np$, and therefore $2\lambda_k = -np + \sqrt{R_k} > 0$, i.e. $\lambda_k > 0$.

Step 4 (proof of (24)). Let $k \geq n+1$ and put $A = \lambda_k + n$ and $B = k$. By Step 3, $\lambda_k > 0$, and $n \geq 1$, so $A > 0$; also $B = k \geq n + 1 > 0$. Consequently $A + B > 0$, and

$$A - B < A + B \quad (\text{because } -B < B, \text{ as } B > 0), \qquad -(A+B) < A - B \quad (\text{because } 0 < 2A).$$

Thus $|A - B| < A + B$, and dividing by $A + B > 0$,

$$|\epsilon_k(n,p)| = \left|\frac{\lambda_k + n - k}{\lambda_k + n + k}\right| = \frac{|A - B|}{A + B} < 1.$$

Step 5 (upper bound in (25)). Let $k \geq n+1$. Since $p - 1 > 0$ we have $\sqrt{p-1} > 0$, so $np + 2k\sqrt{p-1} > 0$; and $\sqrt{R_k} \geq 0$. To conclude $\sqrt{R_k} \leq np + 2k\sqrt{p-1}$ it therefore suffices to compare the squares of these two nonnegative numbers. Using $(\sqrt{p-1})^2 = p - 1$,

$$\big(np + 2k\sqrt{p-1}\big)^2 - R_k = n^2p^2 + 4npk\sqrt{p-1} + 4k^2(p-1) - 4k^2(p-1) - n^2(p-2)^2$$

$$= n^2\big(p^2-(p-2)^2\big) + 4npk\sqrt{p-1} = 4n^2(p-1) + 4npk\sqrt{p-1} > 0,$$

where $p^2-(p-2)^2 = 4(p-1)$ and both displayed terms are positive because $p-1>0$, $n\ge 1$, $k>0$ and $p>0$. Hence $\sqrt{R_k} \le np + 2k\sqrt{p-1}$, so

$$2\lambda_k = -np + \sqrt{R_k} \le 2k\sqrt{p-1},$$

that is, $\lambda_k \le k\sqrt{p-1}$.

Step 6 (a lower bound for λ_{n+1}). By Step 1, $R_n = n^2p^2$, and $\sqrt{R_{n+1}} + np > 0$. Hence

$$2\lambda_{n+1} = \sqrt{R_{n+1}} - np = \frac{R_{n+1}-n^2p^2}{\sqrt{R_{n+1}}+np} = \frac{R_{n+1}-R_n}{\sqrt{R_{n+1}}+np} = \frac{4(p-1)(2n+1)}{\sqrt{R_{n+1}}+np},$$

using $(n+1)^2-n^2 = 2n+1$ and Step 0.

We next bound the denominator by $2np(2n+1)$, i.e. we show $\sqrt{R_{n+1}} \le np(4n+1)$. Both sides are nonnegative ($np>0$), so it is enough to show $R_{n+1} \le n^2p^2(4n+1)^2$. Now

$$n^2p^2(4n+1)^2 - R_{n+1} = n^2\big[\big(p(4n+1)\big)^2 - (p-2)^2\big] - 4(n+1)^2(p-1)$$
$$= n^2\big[p(4n+1)-(p-2)\big]\big[p(4n+1)+(p-2)\big] - 4(n+1)^2(p-1)$$
$$= n^2(4np+2)\big((4n+2)p-2\big) - 4(n+1)^2(p-1) = 4\Big[n^2(2np+1)\big((2n+1)p-1\big) - (n+1)^2(p-1)\Big],$$

so it suffices to prove

$$n^2(2np+1)\big((2n+1)p-1\big) \ge (n+1)^2(p-1).$$

Since $p>1$ we have $2np+1 > 2n+1 > 0$, and

$$(2n+1)p - 1 = (2n+1)(p-1) + 2n \ge (2n+1)(p-1) > 0,$$

both factors on the left being positive. Multiplying the two lower bounds,

$$n^2(2np+1)\big((2n+1)p-1\big) \ge n^2(2n+1)\cdot(2n+1)(p-1) = n^2(2n+1)^2(p-1).$$

Finally $n(2n+1) = 2n^2+n \ge n+1$ for $n\ge 1$, because $2n^2 \ge 2 > 1$; hence $n^2(2n+1)^2 \ge (n+1)^2$, and as $p-1>0$,

$$n^2(2n+1)^2(p-1) \ge (n+1)^2(p-1).$$

This gives the required inequality, hence $\sqrt{R_{n+1}} + np \le 2np(2n+1)$.

Substituting into the displayed formula for $2\lambda_{n+1}$, and using that its numerator $4(p-1)(2n+1)$ is positive,

$$2\lambda_{n+1} = \frac{4(p-1)(2n+1)}{\sqrt{R_{n+1}}+np} \ge \frac{4(p-1)(2n+1)}{2np(2n+1)} = \frac{2(p-1)}{np},$$

that is, $\lambda_{n+1} \ge \dfrac{p-1}{np}$.

Step 7 (lower bound in (25)). Let $k \ge n+1$. By Step 0, $\lambda_k \ge \lambda_{n+1}$, and by Step 6, $\lambda_{n+1} \ge (p-1)/(np)$. Hence $\lambda_k \ge (p-1)/(np)$.

Steps 1, 2, 4, 5 and 7 are exactly the four assertions (23), (24) and (25). □

2.19. `CoefCompatAndSeries`.

Lemma 2.19. *Let $n\ge 1$ be an integer, let $1<p<\infty$, write $\mathbb{B} = \{\xi : |\xi|<1\}$, and let $H\colon \mathbb{C}\to\mathbb{C}$ be continuous on $\mathbb{B}$, a member of $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, differentiable at almost every point of $\mathbb{B}$, and a solution of (18) on $\mathbb{B}$ in the sense of `SolvesEq18`. Let a_j, $j\in\mathbb{Z}$, be the Fourier coefficients of `FourierCoefOfH`, let $\lambda_j = \lambda_{|j|}(n,p)$ be the exponent of `LambdaCoef` and $\epsilon_k = \epsilon_k(n,p)$ the coefficient of `EpsCoef`. Suppose given $A\colon\mathbb{Z}\to\mathbb{C}$ with*

$$a_k(r) = A_k\, r^{\lambda_k} \qquad \text{for all } k\in\mathbb{Z} \text{ with } |k|\ge n \text{ and all } r\in(0,1).$$

Then:

(38) *for every $k\in\mathbb{Z}$ with $|k|\ge n$,*

$$\big[2\lambda_k + p(n-k)\big]A_k = (2-p)(n+k)\overline{A_{-k}};$$

(a) $A_{-n} = 0$;

(39) *for every integer $k>n$, $A_{-k} = \epsilon_k\overline{A_k}$;*

(b) *for every integer $k \geq n$, every $r > 0$ and every $\theta \in \mathbb{R}$, the k-th summand T_k of (19), in the sense of* `Thm2Term`*, satisfies*

$$T_k\big(re^{i\theta}\big) = \big(A_k e^{ik\theta} + \epsilon_k \overline{A_k} e^{-ik\theta}\big) r^{\lambda_k} e^{-in\theta};$$

(40) *for every $r \in (0,1)$,*

$$\int_0^{2\pi} \Big| H\big(re^{i\theta}\big) - \sum_{k=n}^{n+N} T_k\big(re^{i\theta}\big) \Big|^2 \, d\theta \;\longrightarrow\; 0 \qquad (N \to \infty),$$

that is, the series (40) converges to $H(r,\cdot)$ in $L^2(0,2\pi)$.

A coefficient family A as in the hypothesis exists by `FourierCoefForm36`*, which supplies for each k with $|k| \geq n$ a constant with this property; the family is taken as given here so that no choice is made inside the statement.*

Proof. Throughout, $\lambda_j = \lambda_{|j|}(n,p)$ depends on j only through j^2, so $\lambda_{-j} = \lambda_j$; this is the paper's remark at line 331.

Step 0: the indicial equation, without unfolding the square root. By `LambdaCoef`, for every $j \in \mathbb{Z}$,

$$2\lambda_j + pn = \sqrt{4j^2(p-1) + n^2(p-2)^2},$$

the right-hand side being nonnegative. Squaring, $(2\lambda_j + pn)^2 = 4j^2(p-1) + n^2(p-2)^2$, i.e.

$$4\lambda_j^2 + 4pn\lambda_j + n^2p^2 = 4j^2(p-1) + n^2(p-2)^2.$$

Hence, using $(p-2)^2 - p^2 = -4(p-1)$,

$$4\lambda_j^2 + 4pn\lambda_j = 4j^2(p-1) + n^2\big[(p-2)^2 - p^2\big] = 4j^2(p-1) - 4n^2(p-1) = 4(j^2-n^2)(p-1),$$

and dividing by 4,

$$\lambda_j^2 + pn\lambda_j + (n^2 - j^2)(p-1) = 0. \tag{I}$$

This is the indicial equation of paper.tex:315. It is the only property of λ_j used below, so no square root is ever unfolded.

Step 0: the hypotheses of the Section-4 nodes. Both `FourierCoefRegularity` and `FourierCoefOde` ask of H that it be continuous on $\mathbb{B}$, that it lie in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, and that it be differentiable at almost every point of $\mathbb{B}$; all three are hypotheses of the present lemma. `FourierCoefOde` asks in addition that H solve (18) on $\mathbb{B}$, which is a hypothesis as well.

Step 1: proof of (38). Let $k \in \mathbb{Z}$ with $|k| \geq n$ and let $r \in (0,1)$. By the first clause of `FourierCoefOde`, whose hypotheses hold by Step 0,

$$2r\, a_k'(r) = -p(n-k)a_k(r) - (p-2)(n+k)\overline{a_{-k}(r)}. \tag{33}$$

Since $|-k| = |k| \geq n$, the hypothesis applies to both indices and gives, on $(0,1)$,

$$a_k(r) = A_k r^{\lambda_k}, \qquad a_{-k}(r) = A_{-k} r^{\lambda_{-k}} = A_{-k} r^{\lambda_k},$$

using $\lambda_{-k} = \lambda_k$. Differentiating the first identity on the open interval $(0,1)$, $a_k'(r) = A_k \lambda_k r^{\lambda_k - 1}$, so $2r\, a_k'(r) = 2\lambda_k A_k r^{\lambda_k}$; and r^{λ_k} is a positive real, so $\overline{a_{-k}(r)} = \overline{A_{-k}}\, r^{\lambda_k}$. Substituting into (33),

$$2\lambda_k A_k r^{\lambda_k} = \Big[-p(n-k)A_k - (p-2)(n+k)\overline{A_{-k}}\Big] r^{\lambda_k},$$

and dividing by $r^{\lambda_k} \neq 0$,

$$2\lambda_k A_k = -p(n-k)A_k - (p-2)(n+k)\overline{A_{-k}},$$

that is $\big[2\lambda_k + p(n-k)\big]A_k = -(p-2)(n+k)\overline{A_{-k}} = (2-p)(n+k)\overline{A_{-k}}$, which is (38).

Step 2: proof of (a). The paper's sentence "For $k = \pm n$, in view of $\lambda_{-n} = \lambda_n = 0$, we obtain $A_{-n} = 0$" compresses two distinct derivations, since (38) is indexed over all k with $|k| \geq n$ and both $k = n$ and $k = -n$ are available. We use the corner $k = -n$, because it is the one that needs no restriction on p. By part (23) of `LambdaEpsEstimates`, $\lambda_n(n,p) = 0$, hence also $\lambda_{-n} = \lambda_n = 0$.

Corner $k=-n$. Take (38) at the index $k=-n$, which is permitted since $|-n|=n\geq n$. Its right-hand factor is $n+k=n-n=0$, and its left-hand coefficient is $2\lambda_{-n}+p\big(n-(-n)\big)=0+2pn$, so (38) reads

$$2pn\,A_{-n}=(2-p)\cdot 0\cdot\overline{A_n}=0.$$

Since $p>1>0$ and $n\geq 1$ we have $2pn>0$, in particular $2pn\neq 0$, so $A_{-n}=0$.

For comparison, the other corner $k=n$ gives the same conclusion but only for $p\neq 2$: there $2\lambda_n+p(n-n)=0$, so the left side of (38) vanishes and (38) reads $0=(2-p)(2n)\overline{A_{-n}}$, from which $\overline{A_{-n}}=0$ follows only after dividing by $(2-p)(2n)$, which requires $p\neq 2$. Since Theorem 2 as printed does not exclude $p=2$, we do not use that corner.

Note that there is no constraint on A_n: at $k=n$ the left side of (38) vanishes identically, and at $k=-n$ the right side does.

Step 3: proof of (39), and the two forms of ϵ_k. Let $k\in\mathbb{Z}$ with $k>n$. We apply (38) at the index $-k$, which is permitted since $|-k|=k\geq n$. As in Step 2, this is the instance that avoids any restriction on p: it has the *coefficient of the unknown* on the left, so the division below is by a quantity that is positive for every $1<p<\infty$.

First, three facts about signs. By part (25) of `LambdaEpsEstimates`, $\lambda_k\geq(p-1)/(np)>0$ for $k\geq n+1$, so

$$\lambda_k>0,\qquad \lambda_k+n+k>0,\qquad 2\lambda_k+p(n+k)>0, \tag{1}$$

the second because $n,k>0$ and the third because $\lambda_k>0$, $p>0$ and $n+k>0$. No claim like $(2-p)(n+k)\neq 0$ is needed anywhere below.

Now take (38) at the index $-k$. Using $\lambda_{-k}=\lambda_k$ and $-(-k)=k$, it reads

$$\big[2\lambda_k+p(n+k)\big]A_{-k}=(2-p)(n-k)\overline{A_k}. \tag{2}$$

By the third inequality of (1) we may divide by $2\lambda_k+p(n+k)$:

$$A_{-k}=\frac{(2-p)(n-k)}{2\lambda_k+p(n+k)}\,\overline{A_k}. \tag{3}$$

It remains to identify the real quotient in (3) with ϵ_k. This is the same polynomial identity as the one relating the paper's two displayed forms of ϵ_k at paper.tex:342, taken at $-k$ in place of k. We claim

$$(2-p)(n-k)\big(\lambda_k+n+k\big)=\big(\lambda_k+n-k\big)\big(2\lambda_k+p(n+k)\big). \tag{4}$$

Indeed, expanding the right-hand side,

$$2\lambda_k^2+p(n+k)\lambda_k+2(n-k)\lambda_k+p(n^2-k^2),$$

and substituting $\lambda_k^2=-pn\lambda_k-(n^2-k^2)(p-1)$ from (I),

$$\begin{aligned}
&=-2pn\lambda_k-2(n^2-k^2)(p-1)+p(n+k)\lambda_k+2(n-k)\lambda_k+p(n^2-k^2)\\
&=\lambda_k\big[-2pn+pn+pk+2n-2k\big]+(n^2-k^2)\big[-2(p-1)+p\big]\\
&=\lambda_k\big[-pn+pk+2n-2k\big]+(n^2-k^2)(2-p)\\
&=\lambda_k(2-p)(n-k)+(n-k)(n+k)(2-p)=(2-p)(n-k)\big(\lambda_k+n+k\big),
\end{aligned}$$

where the penultimate step used $-pn+2n=n(2-p)$ and $pk-2k=-k(2-p)$. This is (4). Dividing (4) by the positive number $\big(\lambda_k+n+k\big)\big(2\lambda_k+p(n+k)\big)$, legitimate by (1), gives

$$\frac{(2-p)(n-k)}{2\lambda_k+p(n+k)}=\frac{\lambda_k+n-k}{\lambda_k+n+k}=\epsilon_k, \tag{5}$$

the last equality by `EpsCoef`. So the quotient appearing in (3) is exactly ϵ_k, and the computation used only the indicial equation (I), never formula (37)'s square root.

Substituting (5) into (3),

$$A_{-k}=\epsilon_k\overline{A_k},$$

which is (39).

Two remarks. First, the second form of ϵ_k has denominator λ_k+n+k, which vanishes at $k=-n$ (there $\lambda_{-n}=0$ and $n+k=0$); that is exactly why the paper restricts (39) to $|k|>n$. Second,

the case $p = 2$, which the derivation above includes, is the one the paper computes explicitly at paper.tex:226–229. There the radicand of `LambdaCoef` is $4k^2(p-1) + n^2(p-2)^2 = 4k^2$, so $\lambda_k = \frac{1}{2}(-2n + 2|k|) = |k| - n$ for every k, and consequently $\epsilon_k = (\lambda_k + n - k)/(\lambda_k + n + k) = 0$ for $k \geq 0$ by `EpsCoef`. Then (39) gives $A_{-k} = 0$ for every $k \geq n$, and by Step 4 below the k-th summand of (19) is $T_k(\xi) = A_k \xi^k |\xi|^0 \xi^{-n} = A_k \xi^{k-n}$, so that (40) becomes $H(\xi) = \sum_{k \geq n} A_k \xi^{k-n}$ — the paper's displayed formula for $p = 2$.

Step 4: proof of (b), the identification of (19) with (40). Let $k \geq n$, $r > 0$, $\theta \in \mathbb{R}$ and $\xi = re^{i\theta}$. Since $|e^{i\theta}| = 1$ and $r > 0$ we have $|\xi| = r$, and

$$\xi^k = r^k e^{ik\theta}, \qquad \bar{\xi}^k = r^k e^{-ik\theta}, \qquad \xi^{-n} = r^{-n} e^{-in\theta}.$$

Hence, by `Thm2Term`,

$$T_k(\xi) = \big(A_k r^k e^{ik\theta} + \epsilon_k \overline{A_k} r^k e^{-ik\theta}\big) r^{\lambda_k + n - k}\, r^{-n} e^{-in\theta} = \big(A_k e^{ik\theta} + \epsilon_k \overline{A_k} e^{-ik\theta}\big) r^{k + (\lambda_k + n - k) - n} e^{-in\theta},$$

where the powers of the positive real r combine by $r^a r^b = r^{a+b}$. The exponent is $k + \lambda_k + n - k - n = \lambda_k$, which gives (b). This is the sense in which (40) is "the polar form of (19)", as the paper says at line 351.

Step 5: proof of (40). Fix $r \in (0, 1)$.

First, $a_j(r) = 0$ whenever $|j| < n$. Indeed, for each $j \in \mathbb{Z}$ the function a_j is continuous on $[0, 1)$ by part (a) of `FourierCoefRegularity` and C^∞ on $(0, 1)$ by the third clause of `FourierCoefOde`, and satisfies the Euler equation (35) on $(0, 1)$ by the second clause of `FourierCoefOde`; so `FourierCoefForm36` applies to a_j and its first case gives $a_j = 0$ on $[0, 1)$ when $|j| < n$.

Now let $N \geq 0$ and put $M = n + N$. By part (f) of `FourierCoefRegularity`, the expansion (29) converges in $L^2(0, 2\pi)$:

$$\int_0^{2\pi} \Big| H\big(re^{i\theta}\big) - \sum_{|j| \leq M} a_j(r) e^{i(j-n)\theta} \Big|^2 d\theta \;\longrightarrow\; 0 \qquad (M \to \infty).$$

We identify the partial sum. The index set $\{j : |j| \leq M\}$ splits into $\{j : |j| < n\}$, where $a_j(r) = 0$, and the two blocks $n \leq j \leq M$ and $-M \leq j \leq -n$, which correspond under $j \mapsto -j$; the blocks are disjoint because $n \geq 1$. Hence

$$\sum_{|j| \leq M} a_j(r) e^{i(j-n)\theta} = \sum_{j=n}^{M} \Big[a_j(r) e^{i(j-n)\theta} + a_{-j}(r) e^{i(-j-n)\theta} \Big].$$

By hypothesis $a_j(r) = A_j r^{\lambda_j}$ and $a_{-j}(r) = A_{-j} r^{\lambda_j}$ for $j \geq n$, so the bracket equals

$$r^{\lambda_j} \big[A_j e^{ij\theta} + A_{-j} e^{-ij\theta} \big] e^{-in\theta}.$$

Finally $A_{-j} = \epsilon_j \overline{A_j}$ for every $j \geq n$: for $j > n$ this is (39) from Step 3, and for $j = n$ both sides vanish, since $A_{-n} = 0$ by Step 2 and $\epsilon_n = 0$ by part (23) of `LambdaEpsEstimates`. Therefore, using Step 4,

$$\sum_{|j| \leq M} a_j(r) e^{i(j-n)\theta} = \sum_{j=n}^{M} \big(A_j e^{ij\theta} + \epsilon_j \overline{A_j} e^{-ij\theta} \big) r^{\lambda_j} e^{-in\theta} = \sum_{j=n}^{n+N} T_j\big(re^{i\theta}\big).$$

As $N \to \infty$ we have $M = n + N \to \infty$, so the convergence just quoted gives

$$\int_0^{2\pi} \Big| H\big(re^{i\theta}\big) - \sum_{j=n}^{n+N} T_j\big(re^{i\theta}\big) \Big|^2 d\theta \longrightarrow 0,$$

which is (40). □

2.20. `QCDistortionPolar`.

Lemma 2.20. *Let $\Omega \subset \mathbb{C}$ be open, let $1 \le K < \infty$, and let $H\colon \mathbb{C} \to \mathbb{C}$ be K-quasiregular on Ω in the sense of `QuasiregularOn` and differentiable at almost every point of Ω. Then for almost every $\xi \in \Omega$ with $\xi \neq 0$,*

$$K^{-1}|H_r(\xi)|^2 \;\le\; J(\xi) \;\le\; K|H_r(\xi)|^2, \tag{26}$$

and

$$\big(1+K^{-2}\big)|H_r(\xi)|^2 \;\le\; 2|H_\xi(\xi)|^2 + 2|H_{\bar\xi}(\xi)|^2 \;\le\; \big(1+K^{2}\big)|H_r(\xi)|^2, \tag{27}$$

where H_r is as in `HRad`, H_ξ as in `DZ`, $H_{\bar\xi}$ as in `DZbar`, and $J = |H_\xi|^2 - |H_{\bar\xi}|^2$ as in `Jac`.

These are the estimates (26) and (27) of Section 4 of the paper, there stated for the K-quasiregular H with $K = \max\{p-1, 1/(p-1)\}$ of (13). Every quantity in them is built from the Fréchet derivative of H, so almost-everywhere differentiability is the right hypothesis: it is what item (C2) of the paper's note supplies for the inverse of a quasiconformal homeomorphism, and the conclusion is an almost-everywhere statement like the distortion inequality it rearranges.

Proof. Put

$$\kappa = \frac{K-1}{K+1}.$$

Since $K \ge 1$ we have $K > 0$, $K+1 > 0$ and $0 \le \kappa < 1$.

Step 0: the distortion inequality for the pointwise derivatives. By `QuasiregularOn` there are weak Wirtinger derivatives a, b of H on Ω, in the sense of `IsWeakWirtingerOn`, and a null set Z_0 with

$$\big|b(\xi)\big| \le \kappa\,\big|a(\xi)\big| \qquad (\xi \in \Omega - Z_0). \tag{1$'$}$$

By `IsWeakWirtingerOn` there are in turn H_x, H_y such that $H, H_x, H_y \in L^1_{\mathrm{loc}}(\Omega)$, such that H_x and H_y are the distributional ∂_0- and ∂_1-derivatives of H on Ω, and such that $a = \frac{1}{2}(H_x - iH_y)$ and $b = \frac{1}{2}(H_x + iH_y)$ at every point. These are exactly hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set Ω, the function H and the pair $D_0 = H_x$, $D_1 = H_y$, and its (H3) is the hypothesis that H is differentiable at almost every point of Ω. That lemma therefore provides a null set Z_1 with

$$H_x(\xi) = \partial_0 H(\xi), \qquad H_y(\xi) = \partial_1 H(\xi) \qquad (\xi \in \Omega - Z_1),$$

$\partial_j = D^{[j]}$ being the classical partial derivative of `PartialWord`, that is $\partial_0 H(\xi) = \mathrm{d}H(\xi)[1]$ and $\partial_1 H(\xi) = \mathrm{d}H(\xi)[i]$. By `DZ` and `DZbar`, H_ξ and $H_{\bar\xi}$ are the same two combinations $\frac{1}{2}\big(\mathrm{d}H(\xi)[1] \mp i\,\mathrm{d}H(\xi)[i]\big)$ of those two numbers, so

$$a(\xi) = H_\xi(\xi), \qquad b(\xi) = H_{\bar\xi}(\xi) \qquad (\xi \in \Omega - Z_1). \tag{1$''$}$$

Combining (1′) and (1″), the distortion inequality

$$\big|H_{\bar\xi}(\xi)\big| \le \kappa\,\big|H_\xi(\xi)\big| \tag{1}$$

holds at every $\xi \in \Omega - (Z_0 \cup Z_1)$, hence for almost every $\xi \in \Omega$.

Discard the union of $Z_0 \cup Z_1$ and the null set on which H fails to be differentiable, which is again null, and fix a remaining $\xi \in \Omega$ with $\xi \neq 0$. Then H is real-differentiable at ξ, so `WirtingerPolarForm` applies there. Put

$$a = \big\|H_\xi(\xi)\big\| \ge 0, \qquad b = \big\|H_{\bar\xi}(\xi)\big\| \ge 0,$$

so that $b \le \kappa a$ by (1).

Step 1: the key inequality. We claim

$$a + b \le K\,(a - b), \tag{2}$$

and that $a - b \ge 0$. For (2): since $K+1 > 0$,

$$a+b \le K(a-b) \iff b(K+1) \le a(K-1) \iff b \le a\,\frac{K-1}{K+1} = \kappa a,$$

which holds. For $a-b\ge 0$: the left side of (2) is $a+b\ge 0$, so $K(a-b)\ge 0$, and $K>0$ gives $a-b\ge 0$.

Step 2: a two-sided bound on $|H_r(\xi)|$. By the third identity of `WirtingerPolarForm`,

$$H_r(\xi)=H_\xi(\xi)\,\frac{\xi}{|\xi|}+H_{\bar\xi}(\xi)\,\frac{\bar\xi}{|\xi|},$$

and $\big|\xi/|\xi|\big|=\big|\bar\xi/|\xi|\big|=1$, so the two summands have moduli exactly a and b. By the triangle inequality and its reverse form,

$$a-b\ \le\ \big|H_r(\xi)\big|\ \le\ a+b. \tag{3}$$

Step 3: proof of (26). By `Jac`,

$$J(\xi)=a^2-b^2=(a-b)(a+b).$$

For the upper bound, multiply (2) by $a-b\ge 0$:

$$J(\xi)=(a-b)(a+b)\le K\,(a-b)^2\le K\big|H_r(\xi)\big|^2,$$

the last step because $0\le a-b\le|H_r(\xi)|$ by (3), so $(a-b)^2\le|H_r(\xi)|^2$, and $K>0$.

For the lower bound, (3) and $a+b\ge 0$ give $|H_r(\xi)|^2\le(a+b)^2$, and multiplying (2) by $a+b\ge 0$ gives $(a+b)^2\le K(a-b)(a+b)=K\,J(\xi)$. Hence $|H_r(\xi)|^2\le K\,J(\xi)$, i.e. $K^{-1}|H_r(\xi)|^2\le J(\xi)$, dividing by $K>0$.

Step 4: proof of (27). We use the elementary identity

$$(a+b)^2+(a-b)^2=2\big(a^2+b^2\big)=2\big|H_\xi(\xi)\big|^2+2\big|H_{\bar\xi}(\xi)\big|^2. \tag{4}$$

For the upper bound: both sides of (2) are nonnegative (the left by $a,b\ge 0$), so squaring (2) gives $(a+b)^2\le K^2(a-b)^2$. With (4),

$$2\big(a^2+b^2\big)=(a+b)^2+(a-b)^2\le K^2(a-b)^2+(a-b)^2=(1+K^2)(a-b)^2\le(1+K^2)\big|H_r(\xi)\big|^2,$$

the last step by (3) with $a-b\ge 0$ and $1+K^2>0$.

For the lower bound: dividing (2) by $K>0$ gives $K^{-1}(a+b)\le a-b$, and both sides are nonnegative, so squaring gives $K^{-2}(a+b)^2\le(a-b)^2$. Hence, by (4),

$$\big(1+K^{-2}\big)(a+b)^2=(a+b)^2+K^{-2}(a+b)^2\le(a+b)^2+(a-b)^2=2\big(a^2+b^2\big).$$

Finally $|H_r(\xi)|^2\le(a+b)^2$ by (3), and $1+K^{-2}>0$, so

$$\big(1+K^{-2}\big)\big|H_r(\xi)\big|^2\le\big(1+K^{-2}\big)(a+b)^2\le 2\big(a^2+b^2\big),$$

which completes (27). □

2.21. `LambdaLinearLowerBound`.

Lemma 2.21. *Let $n\ge 1$ be an integer and let $1<p<\infty$. Then*

$$\lambda_k(n,p)\ \ge\ \frac{\gamma_n}{n+1}\,k\qquad\text{for } k=n+1,n+2,\dots,$$

where $\lambda_k(n,p)$ is as in `LambdaCoef` and $\gamma_n=\gamma_n(p)=\lambda_{n+1}(n,p)$ is as in `GammaExp`. Equality holds at $k=n+1$.

Proof. Fix $n\ge 1$ and $p>1$, and put

$$A=4(p-1),\qquad B=n^2(p-2)^2.$$

Then $A>0$ and $B\ge 0$, and $np>0$. For $k\ge 1$ write, as in `LambdaCoef`,

$$2\lambda_k=-np+\sqrt{4k^2(p-1)+n^2(p-2)^2}=-np+\sqrt{Ak^2+B}.$$

Define

$$g(t)=-np\,t+\sqrt{A+Bt^2}\qquad(t\ge 0).$$

Step 1 (a pointwise inequality). For every $t\ge 0$,

$$B\,t\ \le\ np\,\sqrt{A+Bt^2}.$$

Indeed, both sides are nonnegative: $B \geq 0$ and $t \geq 0$ on the left, and $np > 0$ with $A+Bt^2 \geq A > 0$ on the right. It therefore suffices to compare their squares. Using $\big(\sqrt{A+Bt^2}\big)^2 = A + Bt^2$ (legitimate since $A + Bt^2 \geq 0$) and substituting $A = 4(p-1)$, $B = n^2(p-2)^2$,

$$(np)^2\big(A + Bt^2\big) - (Bt)^2 = 4n^2p^2(p-1) + n^4p^2(p-2)^2t^2 - n^4(p-2)^4t^2$$

$$= 4n^2p^2(p-1) + n^4(p-2)^2t^2\big(p^2 - (p-2)^2\big) = 4(p-1)\Big(n^2p^2 + n^4(p-2)^2t^2\Big),$$

where the last step uses the identity $p^2 - (p-2)^2 = 4(p-1)$. Since $p - 1 > 0$ and $n^2p^2 + n^4(p-2)^2t^2 \geq 0$, this quantity is nonnegative, i.e. $(Bt)^2 \leq (np)^2(A + Bt^2)$. As both Bt and $np\sqrt{A+Bt^2}$ are nonnegative, the inequality of Step 1 follows.

Step 2 (g is nonincreasing on $[0,\infty)$). Let $0 \leq t_1 \leq t_2$. Then

$$g(t_2) - g(t_1) = -np\,(t_2 - t_1) + \Big(\sqrt{A+Bt_2^2} - \sqrt{A+Bt_1^2}\Big).$$

Both radicands are $\geq A > 0$, so both square roots are positive and their sum $S = \sqrt{A+Bt_2^2} + \sqrt{A+Bt_1^2}$ is positive. Multiplying and dividing by S,

$$\sqrt{A+Bt_2^2} - \sqrt{A+Bt_1^2} = \frac{\big(A+Bt_2^2\big) - \big(A+Bt_1^2\big)}{S} = \frac{B\,(t_2^2 - t_1^2)}{S} = \frac{B\,(t_2+t_1)(t_2-t_1)}{S}.$$

Hence

$$g(t_2) - g(t_1) = (t_2 - t_1)\Big[\frac{B\,(t_1+t_2)}{S} - np\Big].$$

By Step 1 applied with $t = t_1$ and with $t = t_2$,

$$B\,t_1 \leq np\sqrt{A+Bt_1^2}, \qquad B\,t_2 \leq np\sqrt{A+Bt_2^2},$$

and adding these gives $B(t_1+t_2) \leq np\,S$. Dividing by $S > 0$, the bracket is ≤ 0; and $t_2 - t_1 \geq 0$. Therefore $g(t_2) - g(t_1) \leq 0$, i.e. $g(t_2) \leq g(t_1)$.

Step 3 (the rescaling identity). For every integer $k \geq 1$,

$$g\Big(\frac{1}{k}\Big) = \frac{2\lambda_k}{k}.$$

Indeed, since $k > 0$,

$$\sqrt{A + \frac{B}{k^2}} = \sqrt{\frac{Ak^2+B}{k^2}} = \frac{\sqrt{Ak^2+B}}{k},$$

so

$$g\Big(\frac{1}{k}\Big) = -\frac{np}{k} + \frac{\sqrt{Ak^2+B}}{k} = \frac{1}{k}\Big(-np + \sqrt{Ak^2+B}\Big) = \frac{2\lambda_k}{k}.$$

Step 4 (conclusion). Let $k \geq n+1$. Then $k \geq n+1 \geq 2 > 0$, so

$$0 < \frac{1}{k} \leq \frac{1}{n+1},$$

so that Step 2 applies with $t_1 = 1/k$ and $t_2 = 1/(n+1)$ and yields

$$g\Big(\frac{1}{n+1}\Big) \leq g\Big(\frac{1}{k}\Big).$$

By Step 3 this reads

$$\frac{2\lambda_{n+1}}{n+1} \leq \frac{2\lambda_k}{k},$$

and $\lambda_{n+1} = \gamma_n$ by GammaExp. Multiplying by $k/2 > 0$,

$$\frac{\gamma_n}{n+1}\,k \leq \lambda_k,$$

as claimed. For $k = n+1$ both sides equal γ_n, so the inequality is an equality there. □

2.22. `GammaRatioMinAtOne`.

Lemma 2.22. *Let $1 < p < \infty$ and let $\gamma_n = \gamma_n(p)$ be as in `GammaExp`. Then*

(a) *$\gamma_n > 0$ for every integer $n \geq 1$, and*

(b) *$\dfrac{1}{\gamma_1} \leq \dfrac{n}{\gamma_n}$ for every integer $n \geq 1$.*

That is, the smallest of the numbers n/γ_n, $n = 1, 2, \ldots$, occurs for $n = 1$, as asserted in Section 6 of the paper.

Proof. Fix $p > 1$, so $p - 1 > 0$. By `GammaExp` and `LambdaCoef`, for every integer $n \geq 1$,

$$2\gamma_n = 2\lambda_{n+1}(n, p) = -np + \sqrt{4(n+1)^2(p-1) + n^2(p-2)^2},$$

and $np > 0$. Write $R(n) = 4(n+1)^2(p-1) + n^2(p-2)^2$, a sum of two nonnegative terms, so $R(n) \geq 0$.

Step 1 (proof of (a)). Fix $n \geq 1$. Then

$$R(n) - n^2p^2 = 4(n+1)^2(p-1) + n^2\big[(p-2)^2 - p^2\big] = 4(n+1)^2(p-1) - 4n^2(p-1) = 4(p-1)(2n+1),$$

where we used the identity $(p-2)^2 - p^2 = -4(p-1)$ and $(n+1)^2 - n^2 = 2n+1$. Since $p - 1 > 0$ and $n \geq 1$, this is positive. Hence $R(n) > n^2p^2 \geq 0$, and therefore

$$\sqrt{R(n)} > \sqrt{n^2p^2} = np$$

(the square root being strictly increasing on $[0, \infty)$, and $np \geq 0$). Consequently $2\gamma_n = -np + \sqrt{R(n)} > 0$, i.e. $\gamma_n > 0$.

Step 2 (a scale-free formula for γ_n/n). Fix $n \geq 1$. Since $n > 0$,

$$\frac{\sqrt{R(n)}}{n} = \sqrt{\frac{R(n)}{n^2}} = \sqrt{4\Big(\frac{n+1}{n}\Big)^2(p-1) + (p-2)^2} = \sqrt{4\Big(1 + \frac{1}{n}\Big)^2(p-1) + (p-2)^2},$$

and therefore

$$\frac{2\gamma_n}{n} = -p + \sqrt{4\Big(1 + \frac{1}{n}\Big)^2(p-1) + (p-2)^2}.$$

Step 3 ($\gamma_n/n \leq \gamma_1$). Fix $n \geq 1$. Then $1/n \leq 1$, so $0 < 1 + \frac{1}{n} \leq 2$ and hence $\big(1 + \frac{1}{n}\big)^2 \leq 4$. Since $p - 1 > 0$,

$$4\Big(1 + \frac{1}{n}\Big)^2(p-1) + (p-2)^2 \;\leq\; 16(p-1) + (p-2)^2,$$

and the square root is nondecreasing on $[0, \infty)$, so

$$\frac{2\gamma_n}{n} = -p + \sqrt{4\Big(1 + \tfrac{1}{n}\Big)^2(p-1) + (p-2)^2} \;\leq\; -p + \sqrt{16(p-1) + (p-2)^2} = \frac{2\gamma_1}{1} = 2\gamma_1,$$

the right-hand equality being Step 2 at $n = 1$, where $4\big(1 + \frac{1}{1}\big)^2 = 16$. Dividing by 2 gives $\gamma_n/n \leq \gamma_1$.

Step 4 (proof of (b)). Fix $n \geq 1$. By Step 1 both γ_n and γ_1 are positive, and $n > 0$, so $\gamma_n/n > 0$. Step 3 gives $0 < \gamma_n/n \leq \gamma_1$, and taking reciprocals of positive numbers reverses the inequality:

$$\frac{1}{\gamma_1} \leq \frac{1}{\gamma_n/n} = \frac{n}{\gamma_n}.$$

□

2.23. `ParsevalOnCircle`.

Lemma 2.23. *Let $f\colon \mathbb{R} \to \mathbb{C}$ be Lebesgue measurable and suppose that $|f|^2$ is integrable on $(0, 2\pi]$. For $m \in \mathbb{Z}$ set*

$$c_m = \frac{1}{2\pi}\int_0^{2\pi} f(\theta)e^{-im\theta}\,d\theta.$$

Then the family $\big(2\pi|c_m|^2\big)_{m\in\mathbb{Z}}$ *is summable and*

$$\sum_{m\in\mathbb{Z}} 2\pi\,|c_m|^2 = \int_0^{2\pi} \big|f(\theta)\big|^2\,d\theta.$$

This is Parseval's identity for the exponential system $\{e^{im\theta}\}_{m\in\mathbb{Z}}$ *on a period, i.e. the completeness of that system in* $L^2(0,2\pi)$*, stated for an arbitrary square-integrable* f*. The paper performs exactly this evaluation at paper.tex:353–357, where* $\int_0^{2\pi}|H_r(re^{i\theta})|^2d\theta$ *is replaced by* $2\pi\sum_j|a_j'(r)|^2$ *in the course of the energy identity; that substitution is the only step of Section 4 that needs completeness rather than mere orthogonality, and the energy identity cannot be obtained without it. Both directions of the identity are used downstream: the inequality* $\sum\le\int$ *gives the forward coefficient bound (22a) and the inequality* $\int\le\sum$ *gives the converse bound (22b).*

The hypotheses are exactly what the application supplies: measurability, which the radial derivative has because it is read off a Fréchet derivative, and square-integrability on a single period, which is what Fubini's theorem yields for almost every radius from the finiteness of the Dirichlet energy. No continuity and no periodicity of f *are assumed, since neither is available there; correspondingly the sum runs over all of* $\mathbb{Z}$ *and the coefficients are those of* f *itself, not of a smoothed version.*

Proof. Write $e_m(\theta)=e^{im\theta}$ and, for a finite set $s\subset\mathbb{Z}$ and a family $(b_m)_{m\in\mathbb{Z}}$ of complex numbers, $T_b^s(\theta)=\sum_{m\in s}b_me_m(\theta)$. Put $S_s=T_c^s$, the Fourier partial sum of f over the window s. All integrals below are over $[0,2\pi]$ unless stated otherwise.

Step 0: integrability. Since $|f|^2$ is integrable on $(0,2\pi]$ and $|f|\le\frac12\big(1+|f|^2\big)$ pointwise — because $(|f|-1)^2\ge0$ — and $(0,2\pi]$ has finite measure, f itself is integrable on $(0,2\pi]$. Each e_m is continuous with $|e_m|=1$, so every product fe_m and every $T_b^se_m$ is integrable on $[0,2\pi]$, and every integral written below converges absolutely.

Step 1: orthogonality. For $m\neq0$ the function $\theta\mapsto e^{im\theta}$ has the antiderivative $e^{im\theta}/(im)$, so $\int_0^{2\pi}e_m=\big(e^{2\pi im}-1\big)/(im)=0$, while $\int_0^{2\pi}e_0=2\pi$. Hence

$$\int_0^{2\pi} e_m(\theta)\,d\theta=\begin{cases}2\pi, & m=0,\\ 0, & m\neq0.\end{cases}\tag{1}$$

Consequently, for a finite $s\subset\mathbb{Z}$, any (b_m) and any $i\in s$, using $e_je_{-i}=e_{j-i}$ and (1),

$$\int_0^{2\pi} T_b^s(\theta)e_{-i}(\theta)\,d\theta=\sum_{j\in s}b_j\int_0^{2\pi}e_{j-i}(\theta)\,d\theta=2\pi b_i,\tag{2}$$

and, by the definition of c_i, $\int_0^{2\pi}f(\theta)e_{-i}(\theta)\,d\theta=2\pi c_i$; subtracting,

$$\int_0^{2\pi}\big(f(\theta)-T_b^s(\theta)\big)e_{-i}(\theta)\,d\theta=2\pi\big(c_i-b_i\big)\qquad(i\in s).\tag{3}$$

Step 2: a Pythagoras identity. Let R be integrable on $[0,2\pi]$ with $|R|^2$ integrable there, let $s\subset\mathbb{Z}$ be finite, let (d_m) be complex numbers, and suppose $\int_0^{2\pi}R(\theta)e_{-j}(\theta)\,d\theta=0$ for every $j\in s$. Write $D=T_d^s$. Using $|z|^2=z\bar z$ and expanding,

$$|R+D|^2=R\bar R+D\bar D+R\bar D+D\bar R.$$

All four products are integrable: $R\bar R=|R|^2$ by hypothesis; $D\bar D$ is continuous; and $R\bar D$, $D\bar R$ are products of an integrable function with a bounded continuous one. Now $\bar D=\sum_{j\in s}\bar d_je_{-j}$, because $\overline{e_j}=e_{-j}$, so $\int R\bar D=\sum_{j\in s}\bar d_j\int Re_{-j}=0$ by hypothesis; and $\int D\bar R=\overline{\int\bar DR}=\overline{\int R\bar D}=0$, since complex conjugation commutes with the integral. Therefore

$$\int_0^{2\pi}|R+D|^2=\int_0^{2\pi}|R|^2+\int_0^{2\pi}|D|^2.\tag{4}$$

Step 3: the squared norm of a partial sum, and Bessel's identity. Taking $R = 0$ is not enough, so we compute $\int |S_s|^2$ directly. By $|z|^2 = z\bar{z}$, $\bar{S}_s = \sum_{j\in s} \bar{c}_j e_{-j}$ and (2) with $b = c$,

$$\int_0^{2\pi} |S_s|^2 = \sum_{j\in s} \bar{c}_j \int_0^{2\pi} S_s(\theta) e_{-j}(\theta)\, d\theta = \sum_{j\in s} \bar{c}_j \cdot 2\pi c_j = 2\pi \sum_{j\in s} |c_j|^2. \tag{5}$$

Apply (4) with $R = f - S_s$ and $D = S_s$: the hypothesis of Step 2 holds for this R by (3) with $b = c$, which gives $\int (f - S_s) e_{-j} = 2\pi(c_j - c_j) = 0$ for $j \in s$; and $f - S_s$ is integrable with $|f - S_s|^2$ integrable, since $|f - S_s|^2 \le 2|f|^2 + 2|S_s|^2$. Since $(f - S_s) + S_s = f$, (4) and (5) give

$$\int_0^{2\pi} |f|^2 = \int_0^{2\pi} \big|f - S_s\big|^2 + 2\pi \sum_{j\in s} |c_j|^2. \tag{6}$$

Step 4: the partial sum is the best approximation from its window. Let (b_m) be arbitrary. Then $f - T^s_b = (f - S_s) + T^s_{c-b}$, and the hypothesis of Step 2 again holds for $R = f - S_s$, so (4) gives

$$\int_0^{2\pi} \big|f - T^s_b\big|^2 = \int_0^{2\pi} \big|f - S_s\big|^2 + \int_0^{2\pi} \big|T^s_{c-b}\big|^2 \;\ge\; \int_0^{2\pi} \big|f - S_s\big|^2,$$

the last step because the omitted integrand is nonnegative. Hence

$$\int_0^{2\pi} \big|f - S_s\big|^2 \le \int_0^{2\pi} \big|f - T^s_b\big|^2. \tag{7}$$

Step 5: trigonometric polynomials are dense in $L^2(0, 2\pi)$. Let $\varepsilon > 0$. Let $F = \mathbf{1}_{(0,2\pi]} f$, so that F is measurable on $\mathbb{R}$, agrees with f on $(0, 2\pi]$ and vanishes elsewhere, and $|F|^2 = \mathbf{1}_{(0,2\pi]}|f|^2$ is integrable on $\mathbb{R}$; thus $F \in L^2(\mathbb{R})$. Since continuous compactly supported functions are dense in $L^2(\mathbb{R})$, there is a continuous g with compact support and

$$\int_{\mathbb{R}} |F - g|^2 \le \varepsilon/9. \tag{8}$$

Being continuous with compact support, g is bounded: $|g| \le C_g$ on $\mathbb{R}$ for some $C_g \ge 0$.

Choose $\delta = \min\big(\pi/2,\ \varepsilon/(18(C_g^2 + 1))\big) > 0$; then $\delta < \pi$ and

$$C_g^2 \cdot 2\delta \le \varepsilon/9, \tag{9}$$

because $C_g^2 \cdot 2\delta \le (C_g^2 + 1) \cdot 2\delta \le \varepsilon/9$. By Urysohn's lemma applied to the disjoint closed sets $\mathbb{R} \setminus (\delta/2, 2\pi - \delta/2)$ and $[\delta, 2\pi - \delta]$ there is a continuous $\chi\colon \mathbb{R} \to [0, 1]$ with $\chi = 1$ on $[\delta, 2\pi - \delta]$ and $\chi = 0$ off $(\delta/2, 2\pi - \delta/2)$.

Put $u = \chi g$. Then u is continuous and vanishes off $(\delta/2, 2\pi - \delta/2)$; in particular u vanishes on a neighbourhood of each of 0 and 2π. Therefore u descends to the unit circle: the function

$$G(w) = u\big(\arg(-w) + \pi\big)$$

is defined for all w, and for $|w| = 1$ it satisfies $G\big(e^{i\theta}\big) = u(\theta)$ for every $\theta \in [0, 2\pi]$. Indeed for $\theta \in (0, 2\pi)$ one has $-e^{i\theta} = e^{i(\theta-\pi)}$ with $\theta - \pi \in (-\pi, \pi)$, so $\arg(-e^{i\theta}) = \theta - \pi$; and for $\theta \in \{0, 2\pi\}$ one has $e^{i\theta} = 1$ and $G(1) = u(\arg(-1) + \pi) = u(2\pi) = 0 = u(\theta)$. Moreover G is continuous on the unit circle: at $w \ne 1$ with $|w| = 1$ the point $-w$ avoids the closed negative real axis, so arg is continuous there; and at $w = 1$ the function G vanishes identically on the neighbourhood $\{|w| = 1 :\ \cos\delta/2 < \operatorname{Re} w\}$ of 1, because $\cos\big(\arg(-w)\big) = -\operatorname{Re} w < -\cos(\delta/2) = \cos(\pi - \delta/2)$ forces $|\arg(-w)| > \pi - \delta/2$, hence $\arg(-w) + \pi \notin (\delta/2, 2\pi - \delta/2)$.

The Laurent monomials $w \mapsto w^j$, $j \in \mathbb{Z}$, span a $*$-subalgebra of the continuous functions on the unit circle: it contains the constants ($j = 0$), is closed under multiplication ($w^j w^{j'} = w^{j+j'}$) and under conjugation ($\overline{w^j} = w^{-j}$ for $|w| = 1$), and it separates points, since $w \mapsto w^1$ is injective. By the Stone–Weierstrass theorem it is therefore uniformly dense in the continuous functions on the circle. So, with $\eta = \sqrt{\varepsilon/(18\pi)} > 0$, there are a finite $s \subset \mathbb{Z}$ and coefficients (b_j) with

$$\Big|G(w) - \sum_{j\in s} b_j w^j\Big| \le \eta \qquad \text{whenever } |w| = 1. \tag{10}$$

Evaluating at $w = e^{i\theta}$ and using $\big(e^{i\theta}\big)^j = e_j(\theta)$ together with $G(e^{i\theta}) = u(\theta)$,

$$\big|u(\theta) - T_b^s(\theta)\big| \le \eta \qquad (\theta \in [0, 2\pi]). \tag{11}$$

Now estimate pointwise on $[0, 2\pi]$. Since $f - T_b^s = (f-g) + (1-\chi)g + \big(u - T_b^s\big)$ and $0 \le \chi \le 1$, the triangle inequality with $|g| \le C_g$ and (11) gives

$$\big|f(\theta) - T_b^s(\theta)\big| \le \big|f(\theta) - g(\theta)\big| + \big(1 - \chi(\theta)\big)C_g + \eta,$$

hence, by $(x+y+z)^2 \le 3(x^2+y^2+z^2)$ and $\big(1-\chi\big)^2 \le 1 - \chi$ (valid since $0 \le 1 - \chi \le 1$),

$$\big|f - T_b^s\big|^2 \le 3\Big(\big|f-g\big|^2 + C_g^2\big(1-\chi\big) + \eta^2\Big).$$

Integrate over $[0, 2\pi]$. For the first term, $f = F$ on $(0, 2\pi]$ and the integrand is nonnegative, so by (8) $\int_0^{2\pi} |f-g|^2 \le \int_{\mathbb{R}} |F-g|^2 \le \varepsilon/9$. For the second, $1 - \chi$ vanishes on $[\delta, 2\pi - \delta]$ and is at most 1, so splitting $[0, 2\pi]$ into $[0, \delta]$, $[\delta, 2\pi - \delta]$ and $[2\pi - \delta, 2\pi]$ gives $\int_0^{2\pi}(1-\chi) \le \delta + 0 + \delta = 2\delta$, whence $C_g^2 \int_0^{2\pi}(1-\chi) \le \varepsilon/9$ by (9). For the third, $\int_0^{2\pi} \eta^2 = 2\pi\eta^2 = \varepsilon/9$. Altogether

$$\int_0^{2\pi} \big|f - T_b^s\big|^2 \le 3 \cdot 3 \cdot \frac{\varepsilon}{9} = \varepsilon. \tag{12}$$

Step 6: conclusion. The numbers $2\pi|c_m|^2$ are nonnegative, so by the characterisation of unordered summability for nonnegative families it suffices to show that $\int_0^{2\pi} |f|^2$ is the least upper bound of the finite partial sums $\sigma_s = \sum_{m \in s} 2\pi|c_m|^2$, $s \subset \mathbb{Z}$ finite.

It is an upper bound: by (6), $\sigma_s = \int_0^{2\pi} |f|^2 - \int_0^{2\pi} |f - S_s|^2 \le \int_0^{2\pi} |f|^2$, the omitted integral being nonnegative.

It is the least one: let y be an upper bound and suppose $y < \int_0^{2\pi} |f|^2$. Apply Step 5 with $\varepsilon = \frac{1}{2}\big(\int_0^{2\pi} |f|^2 - y\big) > 0$ to obtain s and (b_j) satisfying (12); then (7) gives $\int_0^{2\pi} |f - S_s|^2 \le \varepsilon$, so by (6)

$$\sigma_s = \int_0^{2\pi} |f|^2 - \int_0^{2\pi} \big|f - S_s\big|^2 \ge \int_0^{2\pi} |f|^2 - \varepsilon = y + \varepsilon > y,$$

contradicting that y is an upper bound. Hence $\int_0^{2\pi} |f|^2 \le y$ for every upper bound y, and the least upper bound is $\int_0^{2\pi} |f|^2$. This is the assertion. □

2.24. `EnergyIdentity`.

Lemma 2.24. *Let $n \ge 1$ be an integer, let $1 < p < \infty$, and let $K \ge 1$. Write $\mathbb{B} = \{\xi : |\xi| < 1\}$, $\gamma_n = \lambda_{n+1}(n,p)$, and $d\sigma$ for planar Lebesgue measure. Then there is a constant $C = C(n,p,K) > 0$ such that the following holds for every $H\colon \mathbb{C} \to \mathbb{C}$ and every $A\colon \mathbb{Z} \to \mathbb{C}$ satisfying*

- (H0) *H solves equation (18) on $\mathbb{B}$, in the sense of `SolvesEq18`;*
- (H1) *H is continuous on $\mathbb{B}$ and differentiable at almost every point of $\mathbb{B}$;*
- (H2) *H is K-quasiregular on $\mathbb{B}$, in the sense of `QuasiregularOn`;*
- (H3) *$|H_\xi|^2 + |H_{\bar\xi}|^2$ is integrable on $\mathbb{B}$ (this is the standing assumption $H \in W^{1,2}(\mathbb{B})$ of Section 3);*
- (H4) *$a_k(r) = A_k r^{\lambda_k}$ for all k with $|k| \ge n$ and all $r \in (0,1)$, where a_k is as in `FourierCoefOfH` and $\lambda_k = \lambda_{|k|}(n,p)$ as in `LambdaCoef`;*
- (H5) *$A_{-k} = \epsilon_k \overline{A_k}$ for every integer $k > n$, with ϵ_k as in `EpsCoef`.*

Namely, writing H_r for the radial derivative of `HRad`:

- (E) the energy identity*: the series $\sum_{k \ge n+1} \lambda_k(1+\epsilon_k^2)|A_k|^2$ converges, and*

$$\int_0^1 \int_0^{2\pi} \big|H_r\big(re^{i\theta}\big)\big|^2 \, r \, d\theta \, dr = \pi \sum_{k=n+1}^{\infty} \lambda_k\big(1 + \epsilon_k^2\big)|A_k|^2;$$

(22a) *the series* $\sum_{k\ge n+1} k|A_k|^2$ *converges, and*

$$\sum_{k=n+1}^{\infty} k\,|A_k|^2 \le C \iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big)\,d\sigma;$$

(22b) $$\iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big)\,d\sigma \le C \sum_{k=n+1}^{\infty} k\,|A_k|^2.$$

Together, (22a) and (22b) are the assertion (22) of Theorem 2 that $\sum_{k>n} k|A_k|^2$ *and the Dirichlet energy of* H *are comparable, with a constant depending only on* n, p *and* K. *The two convergence assertions are the paper's own "*$<\infty$*" in (22).*

Hypothesis (H0) cannot be omitted, and it is not a consequence of (H4)/(H5): those pin a_j *only for* $|j|\ge n$, *whereas the Parseval sum of Step 2 below runs over all of* $\mathbb{Z}$. *Without (H0) a free low mode is unconstrained, and* $a_0(r) = r$ *— which is continuous on* $[0,1)$ *and contributes* $2\pi\int_0^1 r\,dr = \pi$ *to the left-hand side of (E) while changing no* A_k *— already defeats (E). What (H0) supplies is exactly the Euler equation (35) for* every *index, through* `FourierCoefOde`, *and with it the vanishing of the modes* $|j|<n$ *by* `FourierCoefForm36`.

No hypothesis $p\ne 2$ *is imposed, since Theorem 2 as printed carries none; at* $p=2$ *one has* $K=1$, $\lambda_k = k-n$ *and* $\epsilon_k = 0$, *and the conclusions hold with the constants constructed below.*

Proof. The paper's line 353 says only "a standard computation shows". We write the computation out.

The constants. By part (a) of `GammaRatioMinAtOne`, $\gamma_n = \lambda_{n+1}(n,p) > 0$. Put

$$C_1 = \frac{2(n+1)}{\pi\,\gamma_n\,(1+K^{-2})}, \qquad C_2 = \big(1+K^2\big)\pi\sqrt{p-1}, \qquad C = \max(C_1, C_2).$$

Both are positive: $\gamma_n > 0$, $1+K^{-2} > 0$, and $p>1$ so $\sqrt{p-1}>0$; hence $C>0$. Note C depends only on n, p, K, which is why the quantifier on C precedes those on H and A.

Throughout, $\lambda_{-k} = \lambda_k$ and $\epsilon_k \in \mathbb{R}$, and for $k \ge n+1$ we have, by parts (23), (24), (25) of `LambdaEpsEstimates`,

$$(*) \qquad \lambda_n = 0, \quad \epsilon_n = 0, \quad |\epsilon_k| < 1, \quad 0 < \frac{p-1}{np} \le \lambda_k \le k\sqrt{p-1}.$$

Step 0: the hypotheses of the Section-4 nodes. Both `FourierCoefRegularity` and `FourierCoefOde` require of H that it be continuous on $\mathbb{B}$, that it lie in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, and that it be differentiable at almost every point of $\mathbb{B}$. All three hold here: the first and the third are (H1); and the second is contained in (H2), since by `QuasiregularOn` a K-quasiregular map on $\mathbb{B}$ lies in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$. `FourierCoefOde` needs in addition that H solve (18) on $\mathbb{B}$, which is (H0).

We also record here the integrability of the radial derivative, which Steps 2 and 3 both use. By the lower half of (27), that is by `QCDistortionPolar` applied to $\Omega = \mathbb{B}$ (open) with K, (H1)'s almost-everywhere differentiability and (H2),

$$(\mathrm{F0}) \qquad \big(1+K^{-2}\big)\big|H_r(\xi)\big|^2 \le 2|H_\xi(\xi)|^2 + 2|H_{\bar\xi}(\xi)|^2 \qquad \text{for a.e. } \xi \in \mathbb{B}-\{0\},$$

and the punctured disk differs from $\mathbb{B}$ by a null set; since $1+K^{-2}>0$ and, by (H3), $|H_\xi|^2+|H_{\bar\xi}|^2$ is integrable on $\mathbb{B}$, the function $|H_r|^2$ is integrable on $\mathbb{B}$ and the polar change of variables $\xi = re^{i\theta}$, $d\sigma = r\,dr\,d\theta$, gives

$$(\mathrm{F0}') \qquad \int_0^1\!\!\int_0^{2\pi} |H_r|^2 r\,d\theta\,dr = \iint_{\mathbb{B}} |H_r|^2 d\sigma \le \frac{2}{1+K^{-2}} \iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big)\,d\sigma < \infty.$$

By Tonelli's theorem the inner integral $r \mapsto r\int_0^{2\pi} |H_r(re^{i\theta})|^2 d\theta$ is therefore finite for almost every $r\in(0,1)$, hence, $r>0$ there,

$$(\mathrm{F0}'') \qquad \theta \mapsto H_r\big(re^{i\theta}\big) \text{ lies in } L^2(0,2\pi) \qquad \text{for almost every } r\in(0,1).$$

Step 1: the Fourier coefficients of H_r. By part (c) of `FourierCoefRegularity`, applied with the data of Step 0, for each $j \in \mathbb{Z}$ there is a null set of radii outside which a_j is differentiable with $a_j'(r)$ equal to the j-th Fourier coefficient of $H_r(r,\cdot)$ in the sense of `FourierCoefOfH`. The union of those null sets over $j \in \mathbb{Z}$ is a countable union of null sets, hence null, so for almost every $r \in (0,1)$ the identity holds for every $j \in \mathbb{Z}$ simultaneously. Fix such an r, chosen also to satisfy (F0″) — again a full-measure condition; every use of r below is inside an integral over $r \in (0,1)$, so this costs nothing. We evaluate the coefficient in three regimes.

If $|j| < n$: this is where (H0) is used, and it is the only place. The hypotheses of `FourierCoefOde` are met, by Step 0 together with $n \geq 1$ and $1 < p < \infty$, so its second clause gives, for this j and every $r \in (0,1)$,

$$r\big(r\,a_j'\big)'(r) + pn\,r\,a_j'(r) + (n^2 - j^2)(p-1)\,a_j(r) = 0,$$

which is the Euler equation (35). Moreover a_j is continuous on $[0,1)$ by part (a) of `FourierCoefRegularity` and C^∞ on $(0,1)$ by the third clause of `FourierCoefOde`. These are exactly the hypotheses of `FourierCoefForm36` at the index j (which imposes no restriction on p), and since $|j| < n$ its first case gives $a_j \equiv 0$ on $[0,1)$; hence $a_j'(r) = 0$.

If $|j| \geq n$: by (H4), $a_j(r) = A_j r^{\lambda_j}$ on $(0,1)$, so differentiating,

$$a_j'(r) = A_j \lambda_j r^{\lambda_j - 1}.$$

In particular, if $|j| = n$ then $\lambda_j = \lambda_n = 0$ by $(*)$, so $a_j'(r) = 0$ — *whatever* $A_{\pm n}$ *are.* This is the precise reason the sums below start at $n+1$ and not at n: the two terms $j = \pm n$ carry no energy, because $r^{\lambda_n} = r^0$ is constant in r.

So the nonzero coefficients of $H_r(r,\cdot)$ are exactly those with $|j| \geq n+1$, and there $a_j'(r) = A_j \lambda_j r^{\lambda_j - 1}$.

Step 2: Parseval on each circle. Still with the r fixed in Step 1, $H_r(r,\cdot) \in L^2(0,2\pi)$ by (F0″), so Parseval's identity for the orthogonal system $\{e^{i(j-n)\theta}\}_{j\in\mathbb{Z}}$, of squared norm 2π each, gives

$$\int_0^{2\pi} \big|H_r\big(re^{i\theta}\big)\big|^2 d\theta = 2\pi \sum_{j\in\mathbb{Z}} |a_j'(r)|^2 = 2\pi \sum_{|j|\geq n+1} \lambda_j^2 |A_j|^2 r^{2\lambda_j - 2},$$

by Step 1. We now fold j and $-j$. The two index blocks $\{j \geq n+1\}$ and $\{-j : j \geq n+1\}$ are disjoint, since $n+1 \geq 2 > 0$, and their union is $\{|j| \geq n+1\}$. Since $\lambda_{-j} = \lambda_j$, and by (H5) $A_{-j} = \epsilon_j \overline{A_j}$ with ϵ_j real, so that $|A_{-j}|^2 = \epsilon_j^2 |A_j|^2$ and therefore

$$|A_j|^2 + |A_{-j}|^2 = \big(1 + \epsilon_j^2\big)|A_j|^2,$$

we obtain

$$\int_0^{2\pi} \big|H_r\big(re^{i\theta}\big)\big|^2 d\theta = 2\pi \sum_{k=n+1}^{\infty} \lambda_k^2 \big(1 + \epsilon_k^2\big)|A_k|^2 r^{2\lambda_k - 2}. \tag{P}$$

Step 3: integration in r, and the energy identity (E). That the left-hand side of (E) is finite is (F0′) of Step 0.

Now multiply (P) by $r > 0$ and integrate over $r \in (0,1)$. Every term of the series in (P) is nonnegative, so Tonelli's theorem permits the interchange of $\int_0^1 dr$ with the sum, with both sides possibly $+\infty$ a priori:

$$\int_0^1 \int_0^{2\pi} \big|H_r\big|^2 r\, d\theta\, dr = 2\pi \sum_{k=n+1}^{\infty} \lambda_k^2 \big(1 + \epsilon_k^2\big)|A_k|^2 \int_0^1 r^{2\lambda_k - 1}\, dr.$$

For $k \geq n+1$ we have $\lambda_k > 0$ by $(*)$, hence $2\lambda_k - 1 > -1$, so the inner integral converges and

$$\int_0^1 r^{2\lambda_k - 1} dr = \frac{1}{2\lambda_k}.$$

(This is exactly where $\lambda_k > 0$ is needed, and exactly why the terms $k = n$, for which $\lambda_n = 0$ and the integral $\int_0^1 r^{-1}dr$ diverges, must be — and by Step 1 are — absent.) Therefore

$$\int_0^1\int_0^{2\pi} |H_r|^2 r\, d\theta\, dr = 2\pi \sum_{k=n+1}^{\infty} \frac{\lambda_k^2}{2\lambda_k}\big(1+\epsilon_k^2\big)|A_k|^2 = \pi \sum_{k=n+1}^{\infty} \lambda_k\big(1+\epsilon_k^2\big)|A_k|^2,$$

Combining with (F0′), the right-hand side is finite; being a series of nonnegative terms with finite sum, $\sum_{k\ge n+1}\lambda_k(1+\epsilon_k^2)|A_k|^2$ converges. This, together with the displayed identity, is (E).

Step 4: the forward bound (22a). Let $k \ge n+1$. By `LambdaLinearLowerBound`, $\lambda_k \ge \frac{\gamma_n}{n+1}k$, i.e.

$$k \le \frac{n+1}{\gamma_n}\lambda_k.$$

Also $1+\epsilon_k^2 \ge 1$, simply because $\epsilon_k^2 \ge 0$ — no use is made here of the sharper estimate $|\epsilon_k| < 1$. Hence, termwise,

$$k\,|A_k|^2 \le \frac{n+1}{\gamma_n}\lambda_k\big(1+\epsilon_k^2\big)|A_k|^2,$$

and since the series $\sum_{k\ge n+1}\lambda_k(1+\epsilon_k^2)|A_k|^2$ converges by (E), the comparison test gives the convergence of $\sum_{k\ge n+1} k|A_k|^2$, which is the first assertion of (22a); summing the termwise inequality and applying (E),

$$\sum_{k=n+1}^{\infty} k|A_k|^2 \le \frac{n+1}{\gamma_n}\cdot\frac{1}{\pi}\int_0^1\int_0^{2\pi} |H_r|^2 r\, d\theta\, dr. \tag{F1}$$

Combining with (F0′) of Step 0,

$$\sum_{k=n+1}^{\infty} k|A_k|^2 \le \frac{2(n+1)}{\pi\gamma_n\big(1+K^{-2}\big)}\iint_{\mathbb{B}}\big(|H_\xi|^2+|H_{\bar\xi}|^2\big)d\sigma = C_1\iint_{\mathbb{B}}\big(|H_\xi|^2+|H_{\bar\xi}|^2\big)d\sigma.$$

Since the right-hand integrand is nonnegative, so is the integral, and $C_1 \le C$ gives (22a).

Step 5: the converse bound (22b). Here both sharper estimates of $(*)$ enter. For $k \ge n+1$, $|\epsilon_k| < 1$ gives $1+\epsilon_k^2 < 2$, and $\lambda_k \le k\sqrt{p-1}$ gives

$$\lambda_k\big(1+\epsilon_k^2\big) \le 2k\sqrt{p-1}.$$

Hence, by (E),

$$\iint_{\mathbb{B}} |H_r|^2 d\sigma = \pi\sum_{k=n+1}^{\infty}\lambda_k\big(1+\epsilon_k^2\big)|A_k|^2 \le 2\pi\sqrt{p-1}\sum_{k=n+1}^{\infty} k|A_k|^2.$$

By the upper half of (27) from `QCDistortionPolar`,

$$2|H_\xi|^2 + 2|H_{\bar\xi}|^2 \le \big(1+K^2\big)|H_r|^2 \qquad \text{a.e. on } \mathbb{B},$$

so integrating and combining,

$$\iint_{\mathbb{B}}\big(|H_\xi|^2+|H_{\bar\xi}|^2\big)d\sigma \le \frac{1+K^2}{2}\iint_{\mathbb{B}}|H_r|^2 d\sigma \le \big(1+K^2\big)\pi\sqrt{p-1}\sum_{k=n+1}^{\infty} k|A_k|^2 = C_2\sum_{k=n+1}^{\infty} k|A_k|^2.$$

Since $\sum_{k>n} k|A_k|^2 \ge 0$ and $C_2 \le C$, this gives (22b), and the proof is complete. □

2.25. `Eq18ImpliesQuasiregular`.

Lemma 2.25. *Let $1 < p < \infty$, let $n \ge 0$ be an integer, let $\Omega \subseteq \mathbb{C}$ and let $H\colon \mathbb{C}\to\mathbb{C}$ belong to $W^{1,2}_{\mathrm{loc}}(\Omega)$ in the sense of `MemWLocOn` and solve equation (18) on Ω in the sense of `SolvesEq18`. Put*

$$K = \max\Big\{p-1,\ \frac{1}{p-1}\Big\}.$$

Then H is K-quasiregular on Ω in the sense of `QuasiregularOn`.

This is the paper's assertion at line 239, "Since H is K-quasiregular, $K = \max\{p-1, 1/(p-1)\}$", which the paper states without proof and which the estimates (26) and (27) then use.

Because K here is determined by p alone, a constant of the form $C(n,p,K)$ obtained from it is a constant $C(n,p)$; this is what allows Theorem 2 to carry no quasiregularity hypothesis.

Proof. By `QuasiregularOn` we must verify three things: $1 \le K$, that $H \in W^{1,2}_{\mathrm{loc}}(\Omega)$, and that there are weak Wirtinger derivatives $a = \partial H/\partial\xi$, $b = \partial H/\partial\bar{\xi}$ of H on Ω, in the sense of `IsWeakWirtingerOn`, with

$$|b(\xi)| \le \frac{K-1}{K+1}\,|a(\xi)| \qquad \text{for almost every } \xi \in \Omega. \tag{11}$$

The second is the hypothesis. We prove the first and third. Note that the derivatives in the hypothesis and in the conclusion are the same objects: by `SolvesEq18` the system (18) is an identity between weak Wirtinger derivatives of H on Ω, and by `QuasiregularOn` so is the distortion inequality, so no passage between two notions of derivative is needed here.

Step 1: $p-1>0$, and $1 \le K$. Since $p>1$ we have $p-1>0$, so $1/(p-1)$ is defined and positive. If $p \ge 2$ then $p-1 \ge 1$, so $K \ge p-1 \ge 1$. If $p<2$ then $0<p-1<1$, so $1/(p-1)>1$ and $K \ge 1/(p-1) > 1$. In both cases $1 \le K$.

Step 2: $\dfrac{K-1}{K+1} = \dfrac{|p-2|}{p}$. We compute the two cases separately.

If $p \ge 2$ then $p-1 \ge 1$, hence $1/(p-1) \le 1 \le p-1$, so $K = p-1$. Therefore $K-1 = p-2$ and $K+1 = p$, and since $p-2 \ge 0$,

$$\frac{K-1}{K+1} = \frac{p-2}{p} = \frac{|p-2|}{p}.$$

If $1<p<2$ then $0<p-1<1$, hence $p-1<1<1/(p-1)$, so $K = 1/(p-1)$. Multiplying numerator and denominator by the positive number $p-1$,

$$\frac{K-1}{K+1} = \frac{\frac{1}{p-1}-1}{\frac{1}{p-1}+1} = \frac{1-(p-1)}{1+(p-1)} = \frac{2-p}{p} = \frac{|p-2|}{p},$$

the last equality because $p-2<0$. In both cases

$$\frac{K-1}{K+1} = \frac{|p-2|}{p}. \tag{1}$$

Step 3: the estimate off the origin. By `SolvesEq18` there are weak Wirtinger derivatives a, b of H on Ω, in the sense of `IsWeakWirtingerOn`, and a Lebesgue null set Z such that

$$b(\xi) = \Big(\frac{1}{2} - \frac{1}{p}\Big)\Big[\frac{\xi}{\bar{\xi}}a(\xi) + \frac{\bar{\xi}^n}{\xi^n}\overline{a(\xi)}\Big] \tag{18}$$

for every $\xi \notin Z$ with $\xi \in \Omega$ and $\xi \ne 0$. Fix this pair a, b, and fix such a ξ. Since $\xi \ne 0$ we have $\bar{\xi} \ne 0$ and $|\bar{\xi}| = |\xi| \ne 0$, so the two bracketed factors are defined and

$$\Big|\frac{\xi}{\bar{\xi}}\Big| = \frac{|\xi|}{|\bar{\xi}|} = 1, \qquad \Big|\frac{\bar{\xi}^n}{\xi^n}\Big| = \frac{|\bar{\xi}|^n}{|\xi|^n} = 1.$$

Also $\big|\overline{a(\xi)}\big| = |a(\xi)|$. Hence by (18), the triangle inequality and multiplicativity of the modulus,

$$|b(\xi)| \le \Big|\frac{1}{2} - \frac{1}{p}\Big|\Big(\Big|\frac{\xi}{\bar{\xi}}\Big||a(\xi)| + \Big|\frac{\bar{\xi}^n}{\xi^n}\Big|\big|\overline{a(\xi)}\big|\Big) = 2\Big|\frac{1}{2} - \frac{1}{p}\Big|\,|a(\xi)|.$$

Finally, since $p>1>0$,

$$\Big|\frac{1}{2} - \frac{1}{p}\Big| = \Big|\frac{p-2}{2p}\Big| = \frac{|p-2|}{2p},$$

so $2\big|\frac{1}{2} - \frac{1}{p}\big| = |p-2|/p$ and, by (1),

$$|b(\xi)| \le \frac{|p-2|}{p}\,|a(\xi)| = \frac{K-1}{K+1}\,|a(\xi)|. \tag{2}$$

Step 4: removing the restriction $\xi \ne 0$. Step 3 establishes (2) at every $\xi \in \Omega - Z$ with $\xi \ne 0$. The singleton $\{0\}$ is a Lebesgue null set in $\mathbb{C} \cong \mathbb{R}^2$, and the union of two null sets is null; outside

$Z \cup \{0\}$ every $\xi \in \Omega$ satisfies $\xi \neq 0$ and hence (2). So (2) holds for almost every $\xi \in \Omega$, which is (11) for the pair a, b of Step 3.

Together with Step 1 and the hypothesis $H \in W^{1,2}_{\mathrm{loc}}(\Omega)$, the three clauses of `QuasiregularOn` hold for the constant K, the pair exhibited in its clause (b) being the pair a, b of Step 3; this is the assertion. Note that no property of n was used beyond $n \geq 0$: the factor $\bar{\xi}^n/\xi^n$ has modulus 1 for every such n. □

2.26. `WirtingerModulusPowerRule`.

Lemma 2.26. *Let $\mu \in \mathbb{R}$ and let $\xi \in \mathbb{C}$ with $\xi \neq 0$, and let $g(z) = |z|^\mu$, regarded as a complex-valued function of z. Then*

(a) *g is C^∞ at ξ, and*

(b) $\dfrac{\partial g}{\partial \xi}(\xi) = \dfrac{\mu}{2}\,\dfrac{|\xi|^\mu}{\xi}, \qquad \dfrac{\partial g}{\partial \bar{\xi}}(\xi) = \dfrac{\mu}{2}\,\dfrac{|\xi|^\mu}{\bar{\xi}},$

where $\partial/\partial\xi$ and $\partial/\partial\bar{\xi}$ are the Wirtinger operators of `DZ` and `DZbar`, and real powers are taken in the sense $t^\mu = \exp(\mu \log t)$ for $t > 0$.

This is the rule that differentiates the factor $|\xi|^{\lambda_k+n-k}$ of the series (19). The exponent is a real number, so the integer power rules do not apply to it.

Proof. Write $z = x + iy$ with $x, y \in \mathbb{R}$, so that $|z|^2 = x^2 + y^2$, and put $\xi = x_0 + iy_0$. Since $\xi \neq 0$ we have $x_0^2 + y_0^2 > 0$.

Step 1: proof of (a), smoothness at ξ. The map $(x, y) \mapsto x^2 + y^2$ is a polynomial, hence C^∞ on $\mathbb{R}^2$, and it takes the value $x_0^2 + y_0^2 > 0$ at (x_0, y_0); by continuity it is positive on an open neighbourhood U of (x_0, y_0). The map $t \mapsto t^{\mu/2} = \exp((\mu/2)\log t)$ is C^∞ on $(0, \infty)$, being a composite of log and exp, which are C^∞ there. Hence

$$g(z) = |z|^\mu = \bigl(x^2 + y^2\bigr)^{\mu/2}$$

is C^∞ on U, and so is its inclusion into $\mathbb{C}$, which is $\mathbb{R}$-linear. In particular g is C^∞ at ξ, which is (a), and all the derivatives used below exist.

Step 2: the two real partial derivatives. By the chain rule, on U,

$$\frac{\partial g}{\partial x} = \frac{\mu}{2}\bigl(x^2 + y^2\bigr)^{\mu/2-1} \cdot 2x = \mu\, x\, \bigl(x^2 + y^2\bigr)^{\mu/2-1},$$

and symmetrically

$$\frac{\partial g}{\partial y} = \mu\, y\, \bigl(x^2 + y^2\bigr)^{\mu/2-1}.$$

Writing $|z|^{\mu-2} = (x^2 + y^2)^{\mu/2-1}$, which is legitimate because $|z| > 0$ on U and $t \mapsto t^\mu$ obeys $t^a t^b = t^{a+b}$ for $t > 0$, these read

$$\frac{\partial g}{\partial x}(\xi) = \mu\, x_0\, |\xi|^{\mu-2}, \qquad \frac{\partial g}{\partial y}(\xi) = \mu\, y_0\, |\xi|^{\mu-2}. \tag{1}$$

Step 3: proof of (b), the Wirtinger derivatives. By `DZ` and `DZbar`,

$$\frac{\partial g}{\partial \xi}(\xi) = \tfrac{1}{2}\Bigl(\frac{\partial g}{\partial x}(\xi) - i\,\frac{\partial g}{\partial y}(\xi)\Bigr), \qquad \frac{\partial g}{\partial \bar{\xi}}(\xi) = \tfrac{1}{2}\Bigl(\frac{\partial g}{\partial x}(\xi) + i\,\frac{\partial g}{\partial y}(\xi)\Bigr),$$

the two partial derivatives being the values of the real Fréchet derivative of g at ξ on the vectors 1 and i. Substituting (1),

$$\frac{\partial g}{\partial \xi}(\xi) = \tfrac{1}{2}\mu\, |\xi|^{\mu-2}\bigl(x_0 - iy_0\bigr) = \frac{\mu}{2}\, |\xi|^{\mu-2}\, \bar{\xi}, \qquad \frac{\partial g}{\partial \bar{\xi}}(\xi) = \frac{\mu}{2}\, |\xi|^{\mu-2}\, \xi.$$

Finally $|\xi|^2 = \xi\bar{\xi}$, so $|\xi|^{\mu-2}\bar{\xi} = |\xi|^\mu \bar{\xi}/|\xi|^2 = |\xi|^\mu \bar{\xi}/(\xi\bar{\xi}) = |\xi|^\mu/\xi$, and likewise $|\xi|^{\mu-2}\xi = |\xi|^\mu/\bar{\xi}$; both divisions are legitimate because $\xi \neq 0$ and hence $\bar{\xi} \neq 0$. Therefore

$$\frac{\partial g}{\partial \xi}(\xi) = \frac{\mu}{2}\,\frac{|\xi|^\mu}{\xi}, \qquad \frac{\partial g}{\partial \bar{\xi}}(\xi) = \frac{\mu}{2}\,\frac{|\xi|^\mu}{\bar{\xi}},$$

which is (b). □

2.27. `WirtingerProductRule`.

Lemma 2.27. *Let $f, g\colon \mathbb{C} \to \mathbb{C}$ be real-differentiable at $\xi \in \mathbb{C}$. Then*

$$\frac{\partial(fg)}{\partial\xi}(\xi) = \frac{\partial f}{\partial\xi}(\xi)\, g(\xi) + f(\xi)\, \frac{\partial g}{\partial\xi}(\xi),$$

$$\frac{\partial(fg)}{\partial\bar{\xi}}(\xi) = \frac{\partial f}{\partial\bar{\xi}}(\xi)\, g(\xi) + f(\xi)\, \frac{\partial g}{\partial\bar{\xi}}(\xi),$$

where fg denotes the pointwise product $z \mapsto f(z)g(z)$, $\partial/\partial\xi$ is as in `DZ` and $\partial/\partial\bar{\xi}$ is as in `DZbar`.

Proof. For a map $h\colon \mathbb{C} \to \mathbb{C}$ real-differentiable at ξ write $Dh(\xi)$ for its real Fréchet derivative at ξ, a real-linear map $\mathbb{C} \to \mathbb{C}$. By `DZ` and `DZbar`,

$$\frac{\partial h}{\partial\xi}(\xi) = \tfrac{1}{2}\big(Dh(\xi)[1] - i\, Dh(\xi)[i]\big), \qquad \frac{\partial h}{\partial\bar{\xi}}(\xi) = \tfrac{1}{2}\big(Dh(\xi)[1] + i\, Dh(\xi)[i]\big).$$

Multiplication $\mathbb{C} \times \mathbb{C} \to \mathbb{C}$ is a continuous $\mathbb{R}$-bilinear map, so since f and g are real-differentiable at ξ the pointwise product fg is real-differentiable at ξ and its derivative is given by the bilinear product rule

$$(*) \qquad D(fg)(\xi)[v] = \big(Df(\xi)[v]\big)g(\xi) + f(\xi)\big(Dg(\xi)[v]\big) \qquad \text{for every } v \in \mathbb{C}.$$

Insert $(*)$ with $v = 1$ and with $v = i$ into the formula for $\partial/\partial\xi$:

$$\begin{aligned}
\frac{\partial(fg)}{\partial\xi}(\xi) &= \tfrac{1}{2}\Big(D(fg)(\xi)[1] - i\, D(fg)(\xi)[i]\Big) \\
&= \tfrac{1}{2}\Big(\big(Df(\xi)[1]\big)g(\xi) + f(\xi)\big(Dg(\xi)[1]\big) - i\big(Df(\xi)[i]\big)g(\xi) - i\, f(\xi)\big(Dg(\xi)[i]\big)\Big) \\
&= \tfrac{1}{2}\big(Df(\xi)[1] - i\, Df(\xi)[i]\big)\, g(\xi) + f(\xi)\, \tfrac{1}{2}\big(Dg(\xi)[1] - i\, Dg(\xi)[i]\big) \\
&= \frac{\partial f}{\partial\xi}(\xi)\, g(\xi) + f(\xi)\, \frac{\partial g}{\partial\xi}(\xi),
\end{aligned}$$

where the third equality only regroups the four terms, using that $g(\xi)$ and $f(\xi)$ are fixed complex numbers and that multiplication in $\mathbb{C}$ is commutative and distributive.

The second identity is the same computation with $+i$ in place of $-i$ throughout:

$$\begin{aligned}
\frac{\partial(fg)}{\partial\bar{\xi}}(\xi) &= \tfrac{1}{2}\Big(D(fg)(\xi)[1] + i\, D(fg)(\xi)[i]\Big) \\
&= \tfrac{1}{2}\big(Df(\xi)[1] + i\, Df(\xi)[i]\big)\, g(\xi) + f(\xi)\, \tfrac{1}{2}\big(Dg(\xi)[1] + i\, Dg(\xi)[i]\big) \\
&= \frac{\partial f}{\partial\bar{\xi}}(\xi)\, g(\xi) + f(\xi)\, \frac{\partial g}{\partial\bar{\xi}}(\xi).
\end{aligned}$$

□

2.28. `WirtingerPowerRule`.

Lemma 2.28. *Let $\Omega \subset \mathbb{C}$ be open, let $g\colon \mathbb{C} \to \mathbb{C}$ be C^∞ on Ω with $g(\zeta) \neq 0$ for every $\zeta \in \Omega$, and let $a \in \mathbb{Z}$. Write g^a for the function $z \mapsto g(z)^a$, the integer power. Then for every $\xi \in \Omega$*

$$\frac{\partial(g^a)}{\partial\xi}(\xi) = a\, g(\xi)^{a-1}\, \frac{\partial g}{\partial\xi}(\xi), \qquad \frac{\partial(g^a)}{\partial\bar{\xi}}(\xi) = a\, g(\xi)^{a-1}\, \frac{\partial g}{\partial\bar{\xi}}(\xi),$$

with $\partial/\partial\xi$ as in `DZ` and $\partial/\partial\bar{\xi}$ as in `DZbar`. (Negative powers of g are defined on Ω because g does not vanish there; at points where g vanishes a negative power is read as 0, which is irrelevant since the assertion concerns only $\xi \in \Omega$ and derivatives at such a ξ depend only on the values of the function on the open set Ω.)

Proof. Note first that for every integer b the function g^b is C^∞ on Ω: for $b \geq 0$ it is a finite product of copies of g, and for $b < 0$ it is a finite product of copies of $1/g$, which is C^∞ on Ω because g is C^∞ and non-vanishing there. In particular every function appearing below is real-differentiable at every point of Ω, so `WirtingerProductRule` is applicable at such points.

We give the argument for $\partial/\partial\xi$. The argument for $\partial/\partial\bar{\xi}$ is word for word the same, using at each step the second of the two identities of `WirtingerProductRule` in place of the first, and the formula of `DZbar` in place of that of `DZ`.

Step 1: non-negative exponents. We prove by induction on the integer $n \geq 0$ that

$$\frac{\partial(g^n)}{\partial\xi}(\xi) = n\, g(\xi)^{n-1}\, \frac{\partial g}{\partial\xi}(\xi) \qquad \text{for every } \xi \in \Omega. \tag{1}$$

For $n = 0$ the function g^0 is the constant function 1. A constant function is real-differentiable with zero derivative at every point, so by `DZ` the left-hand side of (1) is 0; the right-hand side is $0 \cdot g(\xi)^{-1}\, \partial g/\partial\xi(\xi) = 0$ as well.

Assume (1) for some $n \geq 0$ and let $\xi \in \Omega$. On all of Ω we have $g^{n+1} = g^n \cdot g$ pointwise, and both factors are C^∞ on Ω, hence real-differentiable at ξ. By `WirtingerProductRule`,

$$\frac{\partial(g^{n+1})}{\partial\xi}(\xi) = \frac{\partial(g^n)}{\partial\xi}(\xi)\, g(\xi) + g(\xi)^n\, \frac{\partial g}{\partial\xi}(\xi).$$

By the induction hypothesis the first term is $n\, g(\xi)^{n-1} g(\xi)\, \partial g/\partial\xi(\xi) = n\, g(\xi)^n\, \partial g/\partial\xi(\xi)$, where $g(\xi)^{n-1} g(\xi) = g(\xi)^n$ holds because $g(\xi) \neq 0$ (for $n = 0$ both sides of that identity are interpreted using $g(\xi)^{-1} = 1/g(\xi)$, and the term is 0 in any case). Adding the second term gives

$$\frac{\partial(g^{n+1})}{\partial\xi}(\xi) = (n+1)\, g(\xi)^n\, \frac{\partial g}{\partial\xi}(\xi),$$

which is (1) for $n + 1$.

Step 2: the exponent -1. On Ω we have $g \cdot g^{-1} = 1$, the constant function, and both factors are C^∞ on Ω. Let $\xi \in \Omega$. Differentiating the constant function 1 gives 0 as in Step 1, so `WirtingerProductRule` yields

$$0 = \frac{\partial g}{\partial\xi}(\xi)\, g(\xi)^{-1} + g(\xi)\, \frac{\partial(g^{-1})}{\partial\xi}(\xi).$$

Since $g(\xi) \neq 0$ we may divide by $g(\xi)$ and rearrange:

$$\frac{\partial(g^{-1})}{\partial\xi}(\xi) = -g(\xi)^{-2}\, \frac{\partial g}{\partial\xi}(\xi) = (-1)\, g(\xi)^{-1-1}\, \frac{\partial g}{\partial\xi}(\xi), \tag{2}$$

which is the assertion for $a = -1$.

Step 3: negative exponents. Let $a < 0$ and write $a = -n$ with $n \geq 1$ an integer. On Ω we have $g^a = (g^{-1})^n$ pointwise. The function g^{-1} is C^∞ on Ω and non-vanishing there, so Step 1 applies with g^{-1} in place of g and gives, for $\xi \in \Omega$,

$$\frac{\partial(g^a)}{\partial\xi}(\xi) = n\, \big(g(\xi)^{-1}\big)^{n-1}\, \frac{\partial(g^{-1})}{\partial\xi}(\xi) = n\, g(\xi)^{1-n} \cdot \Big(-g(\xi)^{-2}\, \frac{\partial g}{\partial\xi}(\xi)\Big),$$

using (2) at the second step. Since $g(\xi) \neq 0$, the powers combine: $g(\xi)^{1-n} g(\xi)^{-2} = g(\xi)^{-n-1}$. Hence

$$\frac{\partial(g^a)}{\partial\xi}(\xi) = -n\, g(\xi)^{-n-1}\, \frac{\partial g}{\partial\xi}(\xi) = a\, g(\xi)^{a-1}\, \frac{\partial g}{\partial\xi}(\xi),$$

because $-n = a$ and $-n - 1 = a - 1$.

Steps 1 and 3 together cover every $a \in \mathbb{Z}$. □

2.29. `WirtingerConjRule`.

Lemma 2.29. *Let $f\colon \mathbb{C} \to \mathbb{C}$ be real-differentiable at $\xi \in \mathbb{C}$, and let $\bar{f}$ denote the function $z \mapsto \overline{f(z)}$. Then*

$$\frac{\partial \bar{f}}{\partial\xi}(\xi) = \overline{\frac{\partial f}{\partial\bar{\xi}}(\xi)}, \qquad \frac{\partial \bar{f}}{\partial\bar{\xi}}(\xi) = \overline{\frac{\partial f}{\partial\xi}(\xi)},$$

with $\partial/\partial\xi$ as in `DZ` and $\partial/\partial\bar{\xi}$ as in `DZbar`.

Proof. Let $c\colon \mathbb{C} \to \mathbb{C}$, $c(w) = \bar{w}$, denote complex conjugation, so that $\bar{f} = c \circ f$. As a map of the two real variables, c is $\mathbb{R}$-linear and continuous, hence real-differentiable at every point with $Dc(w) = c$ for all w.

Since f is real-differentiable at ξ, the chain rule for the real Fréchet derivative shows that $c \circ f$ is real-differentiable at ξ with $D(c \circ f)(\xi) = Dc(f(\xi)) \circ Df(\xi) = c \circ Df(\xi)$, that is,

$$(*) \qquad D(\bar{f})(\xi)[v] = \overline{Df(\xi)[v]} \qquad \text{for every } v \in \mathbb{C}.$$

We use two elementary identities of complex arithmetic: for every $w \in \mathbb{C}$,

$$(**) \qquad \overline{i\,w} = \bar{i}\,\bar{w} = -i\,\bar{w}, \qquad \overline{-i\,w} = i\,\bar{w},$$

and $\overline{\frac{1}{2}w} = \frac{1}{2}\bar{w}$ because $\frac{1}{2}$ is real.

Now by DZ and $(*)$,

$$\frac{\partial \bar{f}}{\partial \xi}(\xi) = \tfrac{1}{2}\Big(D(\bar{f})(\xi)[1] - i\,D(\bar{f})(\xi)[i]\Big) = \tfrac{1}{2}\Big(\overline{Df(\xi)[1]} - i\,\overline{Df(\xi)[i]}\Big).$$

By the first identity of $(**)$ with $w = Df(\xi)[i]$ we have $-i\,\overline{Df(\xi)[i]} = \overline{i\,Df(\xi)[i]}$, so the right-hand side equals

$$\tfrac{1}{2}\Big(\overline{Df(\xi)[1]} + \overline{i\,Df(\xi)[i]}\Big) = \tfrac{1}{2}\,\overline{Df(\xi)[1] + i\,Df(\xi)[i]} = \overline{\tfrac{1}{2}\big(Df(\xi)[1] + i\,Df(\xi)[i]\big)} = \overline{\frac{\partial f}{\partial \xi}(\xi)},$$

using additivity of conjugation, then $\overline{\frac{1}{2}w} = \frac{1}{2}\bar{w}$, and finally the formula of DZbar. This is the first identity.

Symmetrically, by DZbar and $(*)$,

$$\frac{\partial \bar{f}}{\partial \bar{\xi}}(\xi) = \tfrac{1}{2}\Big(\overline{Df(\xi)[1]} + i\,\overline{Df(\xi)[i]}\Big),$$

and by the second identity of $(**)$ with $w = Df(\xi)[i]$ we have $i\,\overline{Df(\xi)[i]} = \overline{-i\,Df(\xi)[i]}$, so this equals

$$\tfrac{1}{2}\,\overline{Df(\xi)[1] - i\,Df(\xi)[i]} = \overline{\tfrac{1}{2}\big(Df(\xi)[1] - i\,Df(\xi)[i]\big)} = \overline{\frac{\partial f}{\partial \xi}(\xi)}$$

by the formula of DZ. □

2.30. `Thm2TermWirtingerDerivs`.

Lemma 2.30. Let $n, k \geq 0$ be integers, let p be real, let $A\colon \mathbb{Z} \to \mathbb{C}$ and let $\xi \in \mathbb{C}$ with $\xi \neq 0$. Write $A_k = A(k)$, $\lambda_k = \lambda_k(n,p)$ for the exponent of LambdaCoef, $\epsilon_k = \epsilon_k(n,p)$ for the coefficient of EpsCoef,

$$s = \lambda_k + n - k,$$

and let $T_k = T_k(n,p,A,\cdot)$ be the k-th summand of the series (19) as in Thm2Term, so that

$$T_k(z) = \Big(A_k z^k + \epsilon_k \overline{A_k}\,\bar{z}^{\,k}\Big)\,|z|^s\,z^{-n}.$$

Then, with $\partial/\partial\xi$ as in DZ and $\partial/\partial\bar{\xi}$ as in DZbar,

$$\frac{\partial T_k}{\partial \xi}(\xi) = \frac{|\xi|^s\,\xi^{-n}}{\xi}\Big(k\,A_k\xi^k + \Big(\frac{s}{2} - n\Big)\big(A_k\xi^k + \epsilon_k\overline{A_k}\,\bar{\xi}^{\,k}\big)\Big)$$

and

$$\frac{\partial T_k}{\partial \bar{\xi}}(\xi) = \frac{|\xi|^s\,\xi^{-n}}{\bar{\xi}}\Big(k\,\epsilon_k\overline{A_k}\,\bar{\xi}^{\,k} + \frac{s}{2}\big(A_k\xi^k + \epsilon_k\overline{A_k}\,\bar{\xi}^{\,k}\big)\Big).$$

These are the Cartesian form of the paper's polar transformation formulas $H_{\bar{\xi}} = \frac{1}{2}\big(H_r + \frac{i}{r}H_\theta\big)e^{i\theta}$ and $H_\xi = \frac{1}{2}\big(H_r - \frac{i}{r}H_\theta\big)e^{-i\theta}$ of `paper.tex:234-237`, specialised to a single summand of (19); every later use of equation (18) on one term reads them.

No constraint whatever is imposed on n, p or k: the identity is a formal differentiation of a product of integer and real powers, valid off the origin for every exponent. In particular neither $n \geq 1$, nor $p > 1$, nor $k \geq n$ is needed. That matters, because one consumer of this identity quantifies over every integer $k \geq 0$ with no lower bound at all.

Proof. Throughout, $\xi \neq 0$ is fixed, $s = \lambda_k + n - k$ is a fixed real number, and $|z|^s$ means $\exp(s \log |z|)$ for $z \neq 0$, read as a complex-valued function of z. Put

$$F(z) = A_k z^k + \epsilon_k \overline{A_k}\, \bar{z}^{\,k}, \qquad G(z) = |z|^s, \qquad H(z) = z^{-n},$$

so that $T_k = F \cdot G \cdot H$ pointwise, by `Thm2Term`.

Step 0: two derivatives of the elementary functions. First, for the identity function $\mathrm{id}(z) = z$ one has $D\,\mathrm{id}(z) = \mathrm{id}$, so $D\,\mathrm{id}(z)[1] = 1$ and $D\,\mathrm{id}(z)[i] = i$, whence by `DZ` and `DZbar`

$$\frac{\partial\, \mathrm{id}}{\partial \xi} = \frac{1}{2}(1 - i \cdot i) = 1, \qquad \frac{\partial\, \mathrm{id}}{\partial \bar{\xi}} = \frac{1}{2}(1 + i \cdot i) = 0. \tag{1}$$

Second, both operators are additive and commute with multiplication by a complex constant on functions real-differentiable at the point in question, because each is a fixed complex-linear expression in the values $Df(\xi)[1]$ and $Df(\xi)[i]$ of the real Fréchet derivative, and $D(f+g)(\xi) = Df(\xi) + Dg(\xi)$, $D(cf)(\xi) = c\,Df(\xi)$.

Step 1: smoothness of the three factors at ξ. The map $z \mapsto \bar{z}$ is $\mathbb{R}$-linear, hence C^∞ on $\mathbb{C}$; so $z \mapsto z^k$ and $z \mapsto \bar{z}^{\,k}$ are C^∞ on $\mathbb{C}$, being integer powers of C^∞ maps with $k \geq 0$, and therefore F is C^∞ on all of $\mathbb{C}$, in particular real-differentiable at ξ. G is C^∞ at ξ by part (a) of `WirtingerModulusPowerRule`, since $\xi \neq 0$. And H is C^∞ at ξ: on the open set $\Omega = \mathbb{C} - \{0\}$ the function $z \mapsto z^n$ is C^∞ and nonvanishing, so its reciprocal $z \mapsto z^{-n}$ is C^∞ there, and $\xi \in \Omega$.

Step 2: the derivatives of F. Since $\overline{\cdot}$ is a ring homomorphism, $\bar{z}^{\,k} = \overline{z^k}$. Applying `WirtingerPowerRule` with $\Omega = \mathbb{C} - \{0\}$, $g = \mathrm{id}$ and $a = k$, and then (1),

$$\frac{\partial (z^k)}{\partial \xi}(\xi) = k\, \xi^{k-1}, \qquad \frac{\partial (z^k)}{\partial \bar{\xi}}(\xi) = k\, \xi^{k-1} \cdot 0 = 0. \tag{2}$$

Hence, by `WirtingerConjRule` applied to $f(z) = z^k$,

$$\frac{\partial (\bar{z}^{\,k})}{\partial \xi}(\xi) = \overline{\frac{\partial (z^k)}{\partial \bar{\xi}}(\xi)} = \bar{0} = 0, \qquad \frac{\partial (\bar{z}^{\,k})}{\partial \bar{\xi}}(\xi) = \overline{\frac{\partial (z^k)}{\partial \xi}(\xi)} = \overline{k\, \xi^{k-1}} = k\, \bar{\xi}^{\,k-1}. \tag{3}$$

By the additivity and constant-multiplicativity of Step 0, together with (2) and (3),

$$\frac{\partial F}{\partial \xi}(\xi) = k\, A_k\, \xi^{k-1}, \qquad \frac{\partial F}{\partial \bar{\xi}}(\xi) = k\, \epsilon_k \overline{A_k}\, \bar{\xi}^{\,k-1}. \tag{4}$$

Step 3: the derivatives of G and H. Part (b) of `WirtingerModulusPowerRule` with $\mu = s$ gives

$$\frac{\partial G}{\partial \xi}(\xi) = \frac{s}{2} \cdot \frac{|\xi|^s}{\xi}, \qquad \frac{\partial G}{\partial \bar{\xi}}(\xi) = \frac{s}{2} \cdot \frac{|\xi|^s}{\bar{\xi}}. \tag{5}$$

`WirtingerPowerRule` with $\Omega = \mathbb{C} - \{0\}$, $g = \mathrm{id}$ and $a = -n$, together with (1), gives

$$\frac{\partial H}{\partial \xi}(\xi) = -n\, \xi^{-n-1}, \qquad \frac{\partial H}{\partial \bar{\xi}}(\xi) = -n\, \xi^{-n-1} \cdot 0 = 0. \tag{6}$$

Step 4: the product rule on the triple. All three factors are real-differentiable at ξ by Step 1, so two applications of `WirtingerProductRule` — first to $(FG) \cdot H$, then to $F \cdot G$ — give, for each of the two operators $\partial \in \{\partial/\partial\xi, \partial/\partial\bar{\xi}\}$,

$$\partial(FGH)(\xi) = \big(\partial F(\xi)\, G(\xi) + F(\xi)\, \partial G(\xi)\big) H(\xi) + F(\xi)\, G(\xi)\, \partial H(\xi). \tag{7}$$

(The product FG is real-differentiable at ξ, being a product of two such functions, so the outer application is legitimate.)

Step 5: collecting, for $\partial/\partial\xi$. Substituting (4), (5) and (6) into (7),

$$\frac{\partial T_k}{\partial \xi}(\xi) = \Big(k\, A_k \xi^{k-1}\, |\xi|^s + F(\xi)\, \frac{s}{2} \frac{|\xi|^s}{\xi}\Big) \xi^{-n} + F(\xi)\, |\xi|^s \left(-n\, \xi^{-n-1}\right).$$

Since $\xi \neq 0$ we may write $\xi^{k-1} = \xi^k/\xi$ and $\xi^{-n-1} = \xi^{-n}/\xi$, so every term carries the common factor $|\xi|^s \xi^{-n}/\xi$, and factoring it out leaves

$$\frac{\partial T_k}{\partial \xi}(\xi) = \frac{|\xi|^s \xi^{-n}}{\xi} \Big(k\, A_k \xi^k + \frac{s}{2} F(\xi) - n\, F(\xi)\Big) = \frac{|\xi|^s \xi^{-n}}{\xi} \Big(k\, A_k \xi^k + \Big(\frac{s}{2} - n\Big) F(\xi)\Big),$$

which, on substituting the definition of F, is the first asserted identity.

Step 6: collecting, for $\partial/\partial\bar{\xi}$. Substituting the second halves of (4), (5) and (6) into (7), the last summand of (7) vanishes because $\partial H/\partial\bar{\xi}(\xi) = 0$, and

$$\frac{\partial T_k}{\partial\bar{\xi}}(\xi) = \Big(k\,\epsilon_k\overline{A_k}\,\bar{\xi}^{\,k-1}\,|\xi|^s + F(\xi)\,\frac{s}{2}\,\frac{|\xi|^s}{\bar{\xi}}\Big)\xi^{-n}.$$

Since $\xi \neq 0$ we also have $\bar{\xi} \neq 0$, so $\bar{\xi}^{\,k-1} = \bar{\xi}^{\,k}/\bar{\xi}$, and both terms carry the common factor $|\xi|^s\xi^{-n}/\bar{\xi}$. Factoring it out leaves

$$\frac{\partial T_k}{\partial\bar{\xi}}(\xi) = \frac{|\xi|^s\xi^{-n}}{\bar{\xi}}\Big(k\,\epsilon_k\overline{A_k}\,\bar{\xi}^{\,k} + \frac{s}{2}F(\xi)\Big),$$

which, on substituting the definition of F, is the second asserted identity.

No step used any inequality between n, k or p: the only hypothesis consumed is $\xi \neq 0$, needed for the smoothness of G and H at ξ, for the three divisions by ξ and $\bar{\xi}$, and for the negative powers ξ^{-n} to be the genuine reciprocals rather than the value 0 assigned by convention at the origin. □

2.31. `WeakPartialAcrossPoint`.

Lemma 2.31. *Let $\Omega \subseteq \mathbb{C}$ be open, let $S \subseteq \Omega$ meet every compact subset of Ω in a finite set, let $j \in \{0,1\}$, let $g, v\colon \mathbb{C} \to \mathbb{C}$, and let β satisfy $0 \le \beta < 1$. Assume*

(H1) *g is of class C^1 on $\Omega - S$;*
(H2) *$\partial_j g = v$ on $\Omega - S$, where ∂_0 and ∂_1 are the partial derivatives of `PartialWord` along the words [0] and [1];*
(H3) *$g \in L^1_{\mathrm{loc}}(\Omega)$ and $v \in L^1_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`;*
(H4) *for every $z_0 \in S$ there are M and $\rho > 0$ with $\|g(z)\| \le M\,|z - z_0|^{-\beta}$ whenever $0 < |z - z_0| < \rho$.*

Then v is the distributional j-th partial derivative of g on Ω, in the sense of `IsWeakPartialOn`.

This is the step hidden in Section 6's "Hence $u \in W^{k+2,q}_{\mathrm{loc}}(\Omega)$" at paper.tex:534: the derivatives of the regular representative are classical only off the singular set, and near a singular point they grow like a negative power of the distance. The same statement carries Section 7's classical p-harmonic identity off the origin to the weak form, with $S = \{0\}$ and $\beta = 0$ there.

Proof. By `IsWeakPartialOn` we must show

$$\int_{\mathbb{C}} g\,\partial_j\varphi\,d\sigma = -\int_{\mathbb{C}} v\,\varphi\,d\sigma \tag{G}$$

for every test function φ for Ω in the sense of `IsTestFn`, that is, every φ smooth on $\mathbb{C}$ with compact support $T = \operatorname{supp}\varphi \subseteq \Omega$. Fix such a φ; write $\|\varphi\|_\infty$ and $\|\partial_j\varphi\|_\infty$ for the suprema of $|\varphi|$ and $|\partial_j\varphi|$, finite because both are continuous with compact support. Throughout $d\sigma$ is plane Lebesgue measure.

Step 0: the finitely many exceptional points in the support, and integrability. By hypothesis $S \cap T$ is finite; write $S \cap T = \{\zeta_1, \dots, \zeta_N\}$ (possibly $N = 0$). Both g and v are measurable on Ω: they are continuous on the open set $\Omega - S$ — g by (H1), and v by (H1) together with (H2), being there the continuous function $\partial_j g$ — and $S \cap T$ is finite, hence null. By (H3) and compactness of $T \subseteq \Omega$,

$$\int_T \|g\|\,d\sigma < \infty, \qquad \int_T \|v\|\,d\sigma < \infty, \tag{1}$$

so both integrals in (G) converge absolutely, their integrands vanishing off T.

Step 1: the cutoffs. Fix once and for all a smooth $\chi\colon \mathbb{R} \to \mathbb{R}$ with

$$0 \le \chi \le 1, \qquad \chi(t) = 0\ (t \le 1), \qquad \chi(t) = 1\ (t \ge 2), \qquad |\chi'| \le C_0$$

for an absolute constant C_0; such a χ exists by the standard construction of smooth bump functions (take $\chi(t) = \int_{-\infty}^{t}\zeta$ with $\zeta \ge 0$ smooth, supported in $(1,2)$, of total integral 1). By

(H4) choose for each $i \le N$ numbers M_i and $\rho_i > 0$ with

$$\|g(z)\| \le M_i\,|z-\zeta_i|^{-\beta} \qquad (0<|z-\zeta_i|<\rho_i), \tag{2}$$

and put $\rho_* = \min_i \rho_i > 0$ (with $\rho_* = 1$ if $N = 0$) and $M_* = \max_i M_i$. Fix also

$$0 < \varepsilon < \tfrac{1}{4}\min\Big\{\rho_*,\ \min_{i\neq i'}|\zeta_i-\zeta_{i'}|\Big\},$$

the second minimum being over distinct pairs and omitted if $N \le 1$, and set

$$\eta_\varepsilon(z) = \prod_{i=1}^{N} \chi\Big(\frac{|z-\zeta_i|}{\varepsilon}\Big)$$

(the empty product being 1 when $N = 0$). Each factor is smooth on $\mathbb{C}$: it vanishes identically on the ball $|z-\zeta_i|<\varepsilon$, and on the open set $z \neq \zeta_i$ the map $z \mapsto |z-\zeta_i|$ is smooth, so the factor is smooth on each of two open sets covering $\mathbb{C}$. Hence η_ε is smooth, and

$$\eta_\varepsilon = 0 \text{ near each } \zeta_i, \qquad \eta_\varepsilon = 1 \text{ off } \textstyle\bigcup_i\{|z-\zeta_i|<2\varepsilon\}, \qquad 0 \le \eta_\varepsilon \le 1. \tag{3}$$

Because 4ε is less than the distance between distinct ζ_i, the annuli $\varepsilon < |z-\zeta_i| < 2\varepsilon$ are pairwise disjoint and at most one factor of η_ε differs from 1 at any given point; so by the product rule, the chain rule and $|\nabla|z-\zeta_i|| = 1$,

$$\big|\partial_j\eta_\varepsilon(z)\big| \le \frac{C_0}{\varepsilon}, \qquad \partial_j\eta_\varepsilon = 0 \ \text{ unless } \varepsilon<|z-\zeta_i|<2\varepsilon \ \text{ for some } i. \tag{4}$$

Step 2: the product has vanishing total derivative. Put

$$\psi_\varepsilon = g\,\varphi\,\eta_\varepsilon.$$

We claim $\psi_\varepsilon \in C^1_c(\mathbb{C})$. Its support is contained in the compact T. For smoothness, the open sets

$$\{z : |z-\zeta_i|<\varepsilon\}\ (i \le N), \qquad \Omega - S, \qquad \mathbb{C} - T$$

cover $\mathbb{C}$: a point outside all the first ones and outside the third lies in $T \subseteq \Omega$ and is distinct from every ζ_i, hence lies outside $S \cap T$; and since it lies in T, lying in S would put it in $S \cap T$, so it lies in $\Omega - S$. On each of these sets ψ_ε is C^1: on the first ones it is 0 by (3), on the second it is a product of the C^1 function g with the smooth φ and η_ε, and on the third it is 0 because φ is. Hence ψ_ε is C^1 on $\mathbb{C}$ with compact support, and therefore

$$\int_{\mathbb{C}} \partial_j\psi_\varepsilon\,d\sigma = 0. \tag{5}$$

Indeed, choose R with $\operatorname{supp}\psi_\varepsilon \subseteq [-R,R]^2$; for $j = 0$, by Fubini and the fundamental theorem of calculus in the first variable,

$$\int_{\mathbb{C}} \partial_x\psi_\varepsilon\,d\sigma = \int_{-R}^{R}\Big(\int_{-R}^{R}\partial_x\psi_\varepsilon(x,y)\,dx\Big)dy = \int_{-R}^{R}\big(\psi_\varepsilon(R,y)-\psi_\varepsilon(-R,y)\big)dy = 0,$$

both boundary values vanishing; the case $j = 1$ is the same in the second variable. (This is the plane divergence theorem for a C^1 field of compact support, applied to the field whose only nonzero component is the j-th.)

Step 3: expanding (5). At every $z \in \Omega - S$ the product rule and (H2) give

$$\partial_j\psi_\varepsilon = v\,\varphi\,\eta_\varepsilon + g\,(\partial_j\varphi)\,\eta_\varepsilon + g\,\varphi\,(\partial_j\eta_\varepsilon).$$

At a point ζ_i both sides vanish, since ψ_ε is identically 0 near ζ_i and since η_ε and $\partial_j\eta_\varepsilon$ vanish there by (3) and (4); at a point of S outside T, and at any point outside Ω, all terms vanish because φ and $\partial_j\varphi$ are supported in $T \subseteq \Omega$. So the identity holds everywhere, and all three terms are integrable by (1), (3) and (4). Integrating and using (5),

$$0 = \int v\varphi\,\eta_\varepsilon\,d\sigma + \int g\,(\partial_j\varphi)\,\eta_\varepsilon\,d\sigma + \int g\,\varphi\,(\partial_j\eta_\varepsilon)\,d\sigma. \tag{6}$$

Step 4: the first two terms converge. By (3), for every z outside the finite set $\{\zeta_1,\dots,\zeta_N\}$ we have $\eta_\varepsilon(z) = 1$ once ε is small enough, so $\eta_\varepsilon \to 1$ pointwise off a null set. Since $|v\varphi\eta_\varepsilon| \le$

$\|\varphi\|_\infty \|v\| \mathbf{1}_T$ and $|g(\partial_j\varphi)\eta_\varepsilon| \le \|\partial_j\varphi\|_\infty \|g\|\mathbf{1}_T$, both integrable by (1), dominated convergence gives

$$\int v\varphi\,\eta_\varepsilon\,d\sigma \to \int v\varphi\,d\sigma, \qquad \int g\,(\partial_j\varphi)\,\eta_\varepsilon\,d\sigma \to \int g\,\partial_j\varphi\,d\sigma.$$

Step 5: the cutoff term vanishes — this is where $\beta < 1$ is used. By (4) the integrand of the third term of (6) is supported in $\bigcup_{i\le N} A_{i,\varepsilon}$ with $A_{i,\varepsilon} = \{z : \varepsilon < |z-\zeta_i| < 2\varepsilon\}$, and $2\varepsilon < \rho_i$, so (2) applies on $A_{i,\varepsilon}$. Hence, by (4),

$$\Big|\int g\,\varphi\,(\partial_j\eta_\varepsilon)\,d\sigma\Big| \le \|\varphi\|_\infty \frac{C_0}{\varepsilon}\sum_{i=1}^{N}\int_{A_{i,\varepsilon}} \|g\|\,d\sigma \le \|\varphi\|_\infty \frac{C_0 M_*}{\varepsilon}\sum_{i=1}^{N}\int_{A_{i,\varepsilon}} |z-\zeta_i|^{-\beta} d\sigma.$$

In polar coordinates about ζ_i,

$$\int_{A_{i,\varepsilon}} |z-\zeta_i|^{-\beta} d\sigma = 2\pi\int_\varepsilon^{2\varepsilon} r^{1-\beta}\,dr \le 2\pi(2\varepsilon)^{1-\beta}\cdot\varepsilon = 2^{2-\beta}\pi\,\varepsilon^{2-\beta},$$

using that $r \mapsto r^{1-\beta}$ is nondecreasing, as $1-\beta > 0$, so its maximum on $[\varepsilon, 2\varepsilon]$ is $(2\varepsilon)^{1-\beta}$, and that the interval has length ε. Therefore

$$\Big|\int g\,\varphi\,(\partial_j\eta_\varepsilon)\,d\sigma\Big| \le 2^{2-\beta}\pi C_0 M_* N\|\varphi\|_\infty\,\varepsilon^{1-\beta} \;\longrightarrow\; 0 \qquad (\varepsilon \to 0^+),$$

because $1-\beta > 0$. This is the only place the hypothesis $\beta < 1$ enters, and it is sharp: at $\beta = 1$ the bound is a constant. Note that N is fixed once φ is, which is why the local finiteness of S is needed and a merely discrete S would not do.

Step 6: conclusion. Letting $\varepsilon \to 0^+$ in (6) and using Steps 4 and 5,

$$0 = \int v\varphi\,d\sigma + \int g\,\partial_j\varphi\,d\sigma,$$

that is $\int g\,\partial_j\varphi\,d\sigma = -\int v\varphi\,d\sigma$, which is (G). As φ was an arbitrary test function for Ω, this is the assertion. □

2.32. `PowerIntegrableNearZero`.

Lemma 2.32. *Let α satisfy $0 < \alpha \le 1$, let $\delta > 0$ and let q satisfy*

$$1 \le q < \frac{2}{2-\alpha}.$$

Then the function $z \mapsto |z|^{(\alpha-2)q}$ is integrable on the disk $\{z : |z| < \delta\}$ for plane Lebesgue measure; equivalently, $|z|^{\alpha-2}$ belongs to L^q of that disk.

This is the assertion the paper makes in one line at paper.tex:534, "The function $|z|^{\alpha-2}$ is integrable over the disk $0 \le |z| < \delta$ with every exponent q such that $1 \le q < 2/(2-\alpha)$", and it is what turns the estimate (52) at $m = k+1$ into $u \in W^{k+2,q}_{\mathrm{loc}}(\Omega)$. The range of q is exactly (10).

Proof. Write $s = (\alpha-2)q$ and $d\sigma$ for plane Lebesgue measure.

Step 1: the exponent. Since $\alpha \le 1$ we have $2-\alpha \ge 1 > 0$, so the inequality $q < 2/(2-\alpha)$ may be multiplied by $2-\alpha > 0$ to give $(2-\alpha)q < 2$, that is

$$s = (\alpha-2)q = -(2-\alpha)q > -2. \tag{1}$$

Equivalently $s + 2 > 0$. Note also $s < 0$, since $\alpha - 2 < 0$ and $q \ge 1 > 0$.

Step 2: measurability. The function $z \mapsto |z|^s$ is continuous on $\mathbb{C} - \{0\}$, being a composition of the continuous $z \mapsto |z|$, which is positive there, with the continuous $t \mapsto t^s$ on $(0,\infty)$; and $\{0\}$ is a Lebesgue null set. Hence the function is measurable, and it is nonnegative, so integrability on the disk is equivalent to the finiteness of its integral there.

Step 3: the polar computation. Passing to polar coordinates about the origin, $d\sigma = r\,dr\,d\theta$,

$$\int_{|z|<\delta} |z|^s\,d\sigma = \int_0^{2\pi}\int_0^\delta r^s\cdot r\,dr\,d\theta = 2\pi\int_0^\delta r^{s+1}\,dr.$$

By (1) we have $s+1>-1$, so the one-dimensional integral converges:

$$\int_0^{\delta} r^{s+1}\,dr=\frac{\delta^{s+2}}{s+2}<\infty,$$

the denominator being positive by (1). Therefore

$$\int_{|z|<\delta}|z|^{(\alpha-2)q}\,d\sigma=\frac{2\pi\,\delta^{(\alpha-2)q+2}}{(\alpha-2)q+2}<\infty,$$

and the function is integrable on the disk, which is the assertion.

Conversely — and this is why the range of q in (10) is what it is, though the statement does not assert it — for $q\geq 2/(2-\alpha)$ one has $s+1\leq -1$ and the same computation gives a divergent integral; so the condition $q<2/(2-\alpha)$ cannot be relaxed. □

2.33. `Thm2TermSolvesEq18`.

Lemma 2.33. *Let $n\geq 1$ be an integer, let $1<p<\infty$, let $k\geq n$ be an integer and let $A\colon\mathbb{Z}\to\mathbb{C}$. Write $\lambda_k=\lambda_k(n,p)$ for the exponent of `LambdaCoef`, $\epsilon_k=\epsilon_k(n,p)$ for the coefficient of `EpsCoef`, and T_k for the k-th summand of (19) in the sense of `Thm2Term`, so that*

$$T_k(\xi)=\big(A_k\xi^k+\epsilon_k\overline{A_k}\bar{\xi}^k\big)|\xi|^{\lambda_k+n-k}\xi^{-n}.$$

Then:

(a) *T_k is C^∞ at every $\xi\neq 0$;*

(b) *for every $r>0$ and every $\theta\in\mathbb{R}$,*

$$T_k\big(re^{i\theta}\big)=r^{\lambda_k}\Big(A_ke^{i(k-n)\theta}+\epsilon_k\overline{A_k}e^{i(-k-n)\theta}\Big),$$

so T_k is homogeneous of degree λ_k with angular profile a trigonometric polynomial of frequencies $k-n$ and $-k-n$;

(c) *T_k solves equation (18) on the unit disk $\mathbb{B}=\{\xi:|\xi|<1\}$, in the sense of `SolvesEq18`.*

Part (c) is the paper's assertion at line 357 that "our arguments can be turned back", and it is the whole content of the converse half of Theorem 2. No restriction on p is imposed; in particular (c) holds at $p=2$, where $\lambda_k=k-n$, $\epsilon_k=0$ and $T_k(\xi)=A_k\xi^{k-n}$ is holomorphic.

Proof. Throughout, write $L=\lambda_k$, $e=\epsilon_k$, $\mu=L+n-k$ and

$$F(\xi)=A_k\xi^k+e\,\overline{A_k}\,\bar{\xi}^k,$$

so that by `Thm2Term`

$$T_k(\xi)=F(\xi)\,|\xi|^{\mu}\,\xi^{-n}\qquad(\xi\neq 0).\tag{0}$$

Note $e\in\mathbb{R}$ by `EpsCoef`.

Step 0: $L\geq 0$, $L+n+k>0$, and the indicial equation. By part (23) of `LambdaEpsEstimates`, $\lambda_n(n,p)=0$, and by part (25) of the same node $\lambda_k\geq(p-1)/(np)>0$ for $k\geq n+1$. Since $k\geq n$, in either case

$$L\geq 0,\qquad\text{hence}\qquad L+n+k\geq 0+1+1>0,\tag{1}$$

using $n\geq 1$ and $k\geq n\geq 1$.

Next the indicial equation. By `LambdaCoef`, with $|k|=k$,

$$2L+pn=\sqrt{4k^2(p-1)+n^2(p-2)^2},$$

the right-hand side being a nonnegative real number. Squaring, $(2L+pn)^2=4k^2(p-1)+n^2(p-2)^2$, that is $4L^2+4pnL+n^2p^2=4k^2(p-1)+n^2(p-2)^2$; hence, using $(p-2)^2-p^2=-4(p-1)$,

$$4L^2+4pnL=4k^2(p-1)+n^2\big[(p-2)^2-p^2\big]=4(k^2-n^2)(p-1),$$

and dividing by 4,

$$L^2+pnL+(n^2-k^2)(p-1)=0.\tag{I}$$

This is the indicial equation at paper.tex:315. It is the only property of L used below, so formula (37)'s square root is never unfolded again.

Step 1: proof of (b). Let $r>0$, $\theta\in\mathbb{R}$ and $\xi = re^{i\theta}$. Since $|e^{i\theta}|=1$ we have $|\xi|=r>0$, and

$$\xi^k = r^k e^{ik\theta}, \qquad \bar{\xi}^k = r^k e^{-ik\theta}, \qquad \xi^{-n} = r^{-n}e^{-in\theta},$$

the last because $\xi\neq 0$. Substituting into (0) and using $r^a r^b = r^{a+b}$ for $r>0$,

$$T_k(\xi) = \big(A_k r^k e^{ik\theta} + e\overline{A_k} r^k e^{-ik\theta}\big) r^\mu r^{-n} e^{-in\theta} = r^{k+\mu-n}\big(A_k e^{i(k-n)\theta} + e\overline{A_k} e^{i(-k-n)\theta}\big).$$

The exponent is $k+\mu-n = k+(L+n-k)-n = L$, which is (b).

Step 2: proof of (a). Fix $\xi\neq 0$. The four factors in (0) are C^∞ at ξ:

- $z\mapsto z^k$ is a polynomial, hence C^∞ on $\mathbb{C}$;
- $z\mapsto \bar{z}$ is $\mathbb{R}$-linear, hence C^∞, so $z\mapsto \bar{z}^{\,k}$ is C^∞ on $\mathbb{C}$;
- $z\mapsto |z|^\mu$ is C^∞ at ξ by part (a) of `WirtingerModulusPowerRule`;
- $z\mapsto z^{-n}$ is C^∞ on $\mathbb{C}-\{0\}$, being the reciprocal of the nonvanishing polynomial z^n there.

A product of finitely many functions C^∞ at ξ is C^∞ at ξ, so T_k is C^∞ at ξ. This is (a); in particular all the first derivatives used in Steps 3–5 exist, and each factor is in particular real-differentiable at ξ, which is the hypothesis the Wirtinger rules require.

Step 3: the Wirtinger derivatives of the four factors. Let $\Omega=\mathbb{C}-\{0\}$, which is open, and fix $\xi\in\Omega$.

The identity map $z\mapsto z$ has real Fréchet derivative the identity, so by `DZ` and `DZbar`

$$\frac{\partial z}{\partial \xi} = \tfrac{1}{2}\big(1 - i\cdot i\big) = 1, \qquad \frac{\partial z}{\partial \bar{\xi}} = \tfrac{1}{2}\big(1 + i\cdot i\big) = 0.$$

The identity map is C^∞ and nonvanishing on Ω, so `WirtingerPowerRule` applies to it with the integer exponents k and $-n$ and gives

$$\frac{\partial\, \xi^k}{\partial \xi} = k\,\xi^{k-1}, \qquad \frac{\partial\, \xi^k}{\partial \bar{\xi}} = 0, \qquad \frac{\partial\, \xi^{-n}}{\partial \xi} = -n\,\xi^{-n-1}, \qquad \frac{\partial\, \xi^{-n}}{\partial \bar{\xi}} = 0. \tag{2}$$

By `WirtingerConjRule` applied to the identity map, the map $c(z)=\bar{z}$ satisfies $\partial c/\partial\xi = \overline{\partial z/\partial\bar{\xi}} = 0$ and $\partial c/\partial\bar{\xi} = \overline{\partial z/\partial\xi} = 1$; and c is C^∞ and nonvanishing on Ω, so `WirtingerPowerRule` applied to c with exponent k gives

$$\frac{\partial\, \bar{\xi}^k}{\partial \xi} = k\,\bar{\xi}^{k-1}\cdot 0 = 0, \qquad \frac{\partial\, \bar{\xi}^k}{\partial \bar{\xi}} = k\,\bar{\xi}^{k-1}. \tag{3}$$

By `WirtingerModulusPowerRule`,

$$\frac{\partial\, |\xi|^\mu}{\partial \xi} = \frac{\mu}{2}\frac{|\xi|^\mu}{\xi}, \qquad \frac{\partial\, |\xi|^\mu}{\partial \bar{\xi}} = \frac{\mu}{2}\frac{|\xi|^\mu}{\bar{\xi}}. \tag{4}$$

From (2) and (3), by $\mathbb{C}$-linearity of each Wirtinger operator in the function (immediate from `DZ`, `DZbar` and linearity of the Fréchet derivative),

$$\frac{\partial F}{\partial \xi}(\xi) = k\,A_k\xi^{k-1}, \qquad \frac{\partial F}{\partial \bar{\xi}}(\xi) = k\,e\,\overline{A_k}\,\bar{\xi}^{k-1}. \tag{5}$$

Step 4: the Wirtinger derivatives of T_k. Apply `WirtingerProductRule` twice to the product (0) — legitimate by Step 2, since each factor is real-differentiable at ξ — and substitute (2), (4), (5):

$$\begin{aligned}\frac{\partial T_k}{\partial \xi}(\xi) &= \frac{\partial F}{\partial \xi}|\xi|^\mu\xi^{-n} + F\cdot\frac{\mu}{2}\frac{|\xi|^\mu}{\xi}\cdot\xi^{-n} + F|\xi|^\mu\cdot\big(-n\xi^{-n-1}\big)\\ &= |\xi|^\mu\xi^{-n}\Big[k\,A_k\xi^{k-1} + \Big(\frac{\mu}{2}-n\Big)\frac{F(\xi)}{\xi}\Big],\end{aligned}$$

$$\frac{\partial T_k}{\partial \bar{\xi}}(\xi) = \frac{\partial F}{\partial \bar{\xi}}|\xi|^\mu\xi^{-n} + F\cdot\frac{\mu}{2}\frac{|\xi|^\mu}{\bar{\xi}}\cdot\xi^{-n} + 0 = |\xi|^\mu\xi^{-n}\Big[k\,e\,\overline{A_k}\,\bar{\xi}^{k-1} + \frac{\mu}{2}\frac{F(\xi)}{\bar{\xi}}\Big].$$

Both displays used $\xi^{-n}/\xi = \xi^{-n-1}$ and $\xi\neq 0$. These two closed forms, with the factor $|\xi|^\mu\xi^{-n}/\xi$ and $|\xi|^\mu\xi^{-n}/\bar{\xi}$ pulled out, are `Thm2TermWirtingerDerivs`, which proves them with no constraint on n, p or k.

Step 5: reduction of (18) to two scalar identities. Fix $\xi \neq 0$ and put $c = \frac{1}{2} - \frac{1}{p}$. Equation (18) at ξ reads

$$\frac{\partial T_k}{\partial \bar{\xi}} = c\Big[\frac{\xi}{\bar{\xi}}\frac{\partial T_k}{\partial \xi} + \frac{\bar{\xi}^{\,n}}{\xi^n}\overline{\Big(\frac{\partial T_k}{\partial \xi}\Big)}\Big]. \tag{18}$$

Conjugating the first display of Step 4, and using that $|\xi|^\mu$ is real and e is real,

$$\overline{\frac{\partial T_k}{\partial \xi}}(\xi) = |\xi|^\mu \bar{\xi}^{-n}\Big[k\,\overline{A_k}\,\bar{\xi}^{k-1} + \Big(\frac{\mu}{2} - n\Big)\frac{\overline{F(\xi)}}{\bar{\xi}}\Big], \qquad \overline{F(\xi)} = \overline{A_k}\bar{\xi}^k + e\,A_k\xi^k,$$

so, since $\bar{\xi}^{\,n}\bar{\xi}^{-n} = 1$,

$$\frac{\bar{\xi}^{\,n}}{\xi^n}\overline{\frac{\partial T_k}{\partial \xi}}(\xi) = |\xi|^\mu \xi^{-n}\Big[k\,\overline{A_k}\,\bar{\xi}^{k-1} + \Big(\frac{\mu}{2} - n\Big)\frac{\overline{F(\xi)}}{\bar{\xi}}\Big].$$

The common factor $|\xi|^\mu\xi^{-n}$ is nonzero, since $|\xi|^\mu > 0$ and $\xi \neq 0$. Dividing (18) by it and then multiplying by $\bar{\xi} \neq 0$, and writing

$$u = A_k\xi^k, \qquad v = \overline{A_k}\,\bar{\xi}^k,$$

so that $F = u + e\,v$, $\overline{F} = v + e\,u$ and $F + \overline{F} = (1+e)(u+v)$, equation (18) becomes

$$k\,e\,v + \frac{\mu}{2}(u + e\,v) = c\Big[k\,u + k\,v + \Big(\frac{\mu}{2} - n\Big)(1+e)(u+v)\Big],$$

where on the left $\bar{\xi}\cdot\overline{A_k}\bar{\xi}^{k-1} = v$ and $\bar{\xi}\cdot F/\bar{\xi} = F$, and on the right $\bar{\xi}\cdot(\xi/\bar{\xi})\,A_k k\xi^{k-1} = k\,u$ and $\bar{\xi}\cdot(\xi/\bar{\xi})(\mu/2-n)F/\xi = (\mu/2-n)F$, while $\bar{\xi}\cdot k\overline{A_k}\bar{\xi}^{k-1} = k\,v$ and $\bar{\xi}\cdot(\mu/2-n)\overline{F}/\bar{\xi} = (\mu/2-n)\overline{F}$. Grouping the coefficients of u and of v, (18) holds at ξ as soon as the two scalar identities

$$\frac{\mu}{2} = c\Big[k + \Big(\frac{\mu}{2} - n\Big)(1+e)\Big] \tag{α}$$

and

$$e\Big(k + \frac{\mu}{2}\Big) = c\Big[k + \Big(\frac{\mu}{2} - n\Big)(1+e)\Big] \tag{β}$$

hold, because then the left side is $\frac{\mu}{2}u + e(k+\frac{\mu}{2})v$ and the right side is the same bracket times $u+v$. Note that no property of ξ, A_k or n beyond $\xi \neq 0$ entered this reduction: the remaining work is pure algebra in L, n, k and p.

Step 6: proof of (β) from (α). By `EpsCoef` and (1),

$$e = \frac{L+n-k}{L+n+k} = \frac{\mu}{L+n+k},$$

the denominator being positive by (1). Also $k + \frac{\mu}{2} = k + \frac{L+n-k}{2} = \frac{L+n+k}{2}$. Hence

$$e\Big(k + \frac{\mu}{2}\Big) = \frac{\mu}{L+n+k}\cdot\frac{L+n+k}{2} = \frac{\mu}{2}, \tag{6}$$

so (β) is exactly (α). (This identity is an algebraic triviality: it holds for every value of L, and uses neither (I) nor the sign of $p-2$.)

Step 7: proof of (α). First simplify the bracket. From $e = \mu/(L+n+k)$,

$$1 + e = \frac{(L+n+k)+(L+n-k)}{L+n+k} = \frac{2(L+n)}{L+n+k}, \qquad \frac{\mu}{2} - n = \frac{L+n-k}{2} - n = \frac{L-n-k}{2},$$

so

$$\Big(\frac{\mu}{2} - n\Big)(1+e) = \frac{L-n-k}{2}\cdot\frac{2(L+n)}{L+n+k} = \frac{(L-n-k)(L+n)}{L+n+k},$$

and therefore

$$k + \Big(\frac{\mu}{2} - n\Big)(1+e) = \frac{k(L+n+k)+(L-n-k)(L+n)}{L+n+k} = \frac{L^2+k^2-n^2}{L+n+k},$$

the numerator because $k(L+n+k)+(L-n-k)(L+n) = kL+kn+k^2+L^2+Ln-nL-n^2-kL-kn = L^2+k^2-n^2$. By (I), $L^2 = -pnL - (n^2-k^2)(p-1)$, so

$$L^2+k^2-n^2 = -pnL-(n^2-k^2)(p-1)+k^2-n^2 = -pnL+(k^2-n^2)\big[(p-1)+1\big] = p\big(k^2-n^2-nL\big).$$

Since $p > 1 > 0$ we have $c = \frac{1}{2} - \frac{1}{p} = \dfrac{p-2}{2p}$, and hence

$$c\Big[k + \Big(\frac{\mu}{2} - n\Big)(1+e)\Big] = \frac{p-2}{2p} \cdot \frac{p\big(k^2 - n^2 - nL\big)}{L+n+k} = \frac{(p-2)\big(k^2-n^2-nL\big)}{2(L+n+k)}. \tag{7}$$

So (α), which reads $\mu/2 = (L+n-k)/2$ on the left, is equivalent — after multiplying by $2(L+n+k) > 0$ — to

$$(p-2)\big(k^2 - n^2 - nL\big) = (L+n-k)(L+n+k) = (L+n)^2 - k^2. \tag{8}$$

We verify (8). Using (I) once more,

$$\begin{aligned}(L+n)^2 - k^2 &= L^2 + 2nL + n^2 - k^2 \\ &= -pnL - (n^2-k^2)(p-1) + 2nL + n^2 - k^2 \\ &= nL(2-p) + (k^2-n^2)(p-1) + (n^2-k^2) \\ &= nL(2-p) + (k^2-n^2)\big[(p-1)-1\big] = nL(2-p) + (k^2-n^2)(p-2),\end{aligned}$$

whereas the left-hand side of (8) is

$$(p-2)(k^2-n^2) - (p-2)nL = (k^2-n^2)(p-2) + nL(2-p).$$

The two agree, so (8) holds, and with it (α) and, by Step 6, (β).

Step 8: the pointwise form of (18). By Steps 5–7, equation (18) holds at every $\xi \neq 0$; that is,

$$\frac{\partial T_k}{\partial \bar\xi}(\xi) = c\Big[\frac{\xi}{\bar\xi}\frac{\partial T_k}{\partial \xi}(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{\frac{\partial T_k}{\partial \xi}(\xi)}\Big] \qquad (\xi \neq 0), \tag{9}$$

the derivatives being the pointwise operators of `DZ` and `DZbar`. In particular (9) holds at every $\xi \in \mathbb{B}$ with $\xi \neq 0$.

Step 9: growth bounds for T_k and its first derivatives. Put $M = (1+|e|)|A_k|$ and let $\xi \neq 0$, $r = |\xi|$. Since $|\bar\xi| = |\xi| = r$ and e is real,

$$|F(\xi)| \le |A_k| r^k + |e|\,|A_k| r^k = M\, r^k,$$

so by (0), $\big||\xi|^\mu\big| = r^\mu$, $|\xi^{-n}| = r^{-n}$ and $k + \mu - n = L$ (Step 1),

$$|T_k(\xi)| \le M\, r^{k+\mu-n} = M\, r^L. \tag{10}$$

By the two displays of Step 4, with $|\xi^{k-1}| = |\bar\xi^{k-1}| = r^{k-1}$, $|1/\xi| = |1/\bar\xi| = r^{-1}$ and the bound on $|F|$,

$$\Big|\frac{\partial T_k}{\partial \xi}(\xi)\Big| \le r^{\mu-n}\Big[k|A_k| r^{k-1} + \Big|\frac{\mu}{2} - n\Big| M\, r^{k-1}\Big], \qquad \Big|\frac{\partial T_k}{\partial \bar\xi}(\xi)\Big| \le r^{\mu-n}\Big[k|e||A_k| r^{k-1} + \frac{|\mu|}{2} M\, r^{k-1}\Big],$$

so, with $M_1 = M\big(k + \big|\frac{\mu}{2} - n\big| + \frac{|\mu|}{2}\big)$ and $\mu - n + k - 1 = L - 1$,

$$\Big|\frac{\partial T_k}{\partial \xi}(\xi)\Big| \le M_1 r^{L-1}, \qquad \Big|\frac{\partial T_k}{\partial \bar\xi}(\xi)\Big| \le M_1 r^{L-1} \qquad (\xi \neq 0), \tag{11}$$

using $|A_k| \le M$ and $|e|\,|A_k| \le M$.

Write $\partial_0 = D^{[0]}$ and $\partial_1 = D^{[1]}$ for the classical partial derivatives of `PartialWord`, so that $\partial_0 f(\xi) = \mathrm{d}f(\xi)[1]$ and $\partial_1 f(\xi) = \mathrm{d}f(\xi)[i]$. By `DZ` and `DZbar`, $\partial T_k/\partial\xi = \frac{1}{2}(\partial_0 T_k - i\,\partial_1 T_k)$ and $\partial T_k/\partial\bar\xi = \frac{1}{2}(\partial_0 T_k + i\,\partial_1 T_k)$ at every point, whence

$$\partial_0 T_k = \frac{\partial T_k}{\partial \xi} + \frac{\partial T_k}{\partial \bar\xi}, \qquad \partial_1 T_k = i\Big(\frac{\partial T_k}{\partial \xi} - \frac{\partial T_k}{\partial \bar\xi}\Big) \tag{12}$$

at every point. With (11), and with $L \ge 0$ from (1) so that $r^{L-1} = r^L r^{-1} \le r^{-1}$ for $0 < r \le 1$,

$$|T_k(\xi)| \le M, \qquad |\partial_j T_k(\xi)| \le 2M_1\,|\xi|^{-1} \qquad (0 < |\xi| < 1,\ j = 0, 1), \tag{13}$$

the first by (10) and $r^L \le 1$.

Step 10: the weak first partial derivatives of T_k on $\mathbb{B}$. Fix $j \in \{0, 1\}$. We verify the hypotheses of `WeakPartialAcrossPoint` with $\Omega = \mathbb{B}$, $S = \{0\}$, this j, $g = T_k$, $v = \partial_j T_k$ and $\beta = 0$.

The set $\mathbb{B}$ is open and $S = \{0\} \subseteq \mathbb{B}$; $S \cap \kappa$ is finite for every compact κ, S being a singleton; and $0 \le \beta = 0 < 1$.

g is of class C^1 on $\mathbb{B} - S$: the set $\mathbb{B} - \{0\}$ is open and T_k is C^∞ at each of its points by (a), and a function of class C^∞ on an open set is of class C^1 there. And $D^{[j]}g = v$ on $\mathbb{B} - S$ holds by the definition of v.

$g \in L^1_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemLpLocOn`: T_k is continuous on the open set $\mathbb{B}-\{0\}$ by (a), hence measurable there, and $\{0\}$ is a null set, so T_k is measurable on $\mathbb{B}$ for the completed Lebesgue measure; and by (13), for every compact $\kappa \subseteq \mathbb{B}$, $\int_\kappa |T_k|\, d\sigma \le M\, \sigma(\kappa) < \infty$, $\sigma(\kappa)$ being finite because κ is bounded.

$v \in L^1_{\mathrm{loc}}(\mathbb{B})$: measurability is the same argument, $\partial_j T_k$ being continuous on $\mathbb{B} - \{0\}$ by (a). For integrability, apply `PowerIntegrableNearZero` with $\alpha = 1$, $\delta = 1$ and $q = 1$; its hypotheses $0 < \alpha$, $\alpha \le 1$, $0 < \delta$, $1 \le q$ and $q < 2/(2-\alpha) = 2$ all hold, and its conclusion is that $\xi \mapsto |\xi|^{(\alpha-2)q} = |\xi|^{-1}$ is integrable on $\mathbb{B} = \{|\xi| < 1\}$. By (13), $|v| \le 2M_1|\xi|^{-1}$ on $\mathbb{B} - \{0\}$, hence almost everywhere on $\mathbb{B}$, so $\int_{\mathbb{B}} |v|\, d\sigma < \infty$ and a fortiori $\int_\kappa |v|\, d\sigma < \infty$ for every compact $\kappa \subseteq \mathbb{B}$.

The growth hypothesis: at the only point 0 of S, take the constants M and $\rho = 1$; by (13), $\|T_k(\xi)\| \le M = M\,|\xi - 0|^{-0}$ whenever $0 < |\xi - 0| < 1$.

Therefore $\partial_j T_k$ is the distributional ∂_j-derivative of T_k on $\mathbb{B}$, in the sense of `IsWeakPartialOn`, for $j = 0, 1$.

Step 11: conclusion. Put $a = \partial T_k/\partial\xi$ and $b = \partial T_k/\partial\bar\xi$, and $f_x = \partial_0 T_k$, $f_y = \partial_1 T_k$. The three clauses of `IsWeakWirtingerOn` hold for T_k, a, b on $\mathbb{B}$ with this f_x, f_y: clause (a) is the local integrability of T_k, f_x and f_y established in Step 10; clause (b) is the conclusion of Step 10; and clause (c) is the identity $a = \frac{1}{2}(f_x - i f_y)$, $b = \frac{1}{2}(f_x + i f_y)$, valid at every point of $\mathbb{C}$ by `DZ` and `DZbar` as recorded before (12). So a and b are weak Wirtinger derivatives of T_k on $\mathbb{B}$.

By Step 8 the identity (9), which is exactly

$$b(\xi) = c\Big[\frac{\xi}{\bar\xi} a(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{a(\xi)}\Big],$$

holds at every $\xi \in \mathbb{B}$ with $\xi \ne 0$, hence at almost every such ξ. With the pair a, b this is precisely the assertion that T_k solves (18) on $\mathbb{B}$ in the sense of `SolvesEq18`. This is (c), and the proof is complete. □

2.34. `Thm2TermGeometricSummability`.

Lemma 2.34. Let $n \ge 1$ be an integer, let $1 < p < \infty$ and let $A\colon \mathbb{Z} \to \mathbb{C}$. Write $A_k = A(k)$, let $\lambda_k = \lambda_k(n,p)$ be as in `LambdaCoef`, let $\epsilon_k = \epsilon_k(n,p)$ be as in `EpsCoef`, and let $T_k = T_k(n,p,A,\cdot)$ be the k-th summand of the series (19) as in `Thm2Term`. Assume the energy hypothesis

$$\sum_{m \ge 0} (n+m)\, |A_{n+m}|^2 < \infty.$$

Then there are real numbers b and C_A such that

(i) $0 < b$ and $b \le \frac{1}{2}$;
(ii) $b\, m \le \lambda_{n+m}$ for every integer $m \ge 0$;
(iii) $0 \le C_A$ and $|A_{n+m}| \le C_A$ for every integer $m \ge 0$;
(iv) $\rho^{\lambda_{n+m}} \le \big(\rho^b\big)^m$ for every ρ with $0 < \rho < 1$ and every $m \ge 0$;
(v) for every integer $M \ge 0$ and every ρ with $0 < \rho < 1$ the series

$$\sum_{m \ge 0} (n+m)^M\, |A_{n+m}|\, \rho^{\lambda_{n+m}}$$

converges;
(vi) $T_n(\xi) = A_n$ for every $\xi \ne 0$;
(vii) $|T_{n+m}(\xi)| \le 2|A_{n+m}|\, |\xi|^{\lambda_{n+m}}$ for every integer $m \ge 0$ and every $\xi \ne 0$;
(viii) for every ρ with $0 < \rho < 1$ the series $\sum_{m \ge 0} 2|A_{n+m}|\, \rho^{\lambda_{n+m}}$ converges.

This is the summability engine behind the series (19), and it is the paper's "we note that $\lim \lambda_k = \infty$. Therefore, we can perform any finite number of term-by-term differentiations in formula (19)" (`paper.tex:387`) made quantitative. The point of (ii) is that the rate is *linear*

in m and not merely positive: by (iv) it converts $\rho^{\lambda_{n+m}}$ into the geometric $(\rho^b)^m$, which is what makes (v) hold simultaneously for every polynomial weight M and so permits differentiating the series any finite number of times. The engine is `LambdaLinearLowerBound` alone; the constant lower bound $\lambda_k \geq (p-1)/(np)$ of (25) contributes nothing here, being constant in k, so it yields neither $\lambda_k \to \infty$ nor any decay.

The numbers b and C_A are produced existentially because no property of the particular witnesses $\min\big(\gamma_n/(n+1), \frac{1}{2}\big)$ and $\big(\sum_m (n+m)|A_{n+m}|^2\big)^{1/2}$ is used downstream — only the inequalities listed above.

Proof. Step 0: the four inputs. Write $\gamma_n = \gamma_n(p)$ for the exponent of `GammaExp`. We use:

$$\gamma_n > 0, \tag{1}$$

which is `GammaRatioMinAtOne` applied at the index $n \geq 1$;

$$\lambda_n = 0, \qquad \epsilon_n = 0, \tag{2}$$

which is (23) of `LambdaEpsEstimates`;

$$|\epsilon_{n+m}| \leq 1 \qquad (m \geq 0), \tag{3}$$

which for $m = 0$ is (2) and for $m \geq 1$ is (24) of `LambdaEpsEstimates`, since then $n+1 \leq n+m$; and

$$\lambda_k \geq \frac{\gamma_n}{n+1}\, k \qquad (k \geq n+1), \tag{4}$$

which is `LambdaLinearLowerBound`.

Step 1: the rate b. Put

$$b = \min\Big(\frac{\gamma_n}{n+1},\ \frac{1}{2}\Big).$$

Then $b \leq \frac{1}{2}$ by definition, and $b > 0$ because $\gamma_n/(n+1) > 0$ by (1) and $n+1 > 0$, and because $\frac{1}{2} > 0$; this is (i).

For (ii), fix $m \geq 0$. Since $b \leq \gamma_n/(n+1)$ and $m \geq 0$,

$$b\, m \leq \frac{\gamma_n}{n+1}\, m,$$

so it suffices to bound the right-hand side by λ_{n+m}. If $m = 0$ this right-hand side is 0 and $\lambda_{n+0} = \lambda_n = 0$ by (2), so the two are equal. If $m \geq 1$ then $n+m \geq n+1$, so (4) applies at $k = n+m$ and gives $\lambda_{n+m} \geq \frac{\gamma_n}{n+1}(n+m)$; and since $\frac{\gamma_n}{n+1} \geq 0$ and $m \leq n+m$ (as $n \geq 0$) we have $\frac{\gamma_n}{n+1} m \leq \frac{\gamma_n}{n+1}(n+m) \leq \lambda_{n+m}$. In both cases $b\, m \leq \lambda_{n+m}$.

Step 2: the coefficient bound C_A. Let

$$S = \sum_{m \geq 0} (n+m)|A_{n+m}|^2,$$

which is a convergent sum of nonnegative terms by hypothesis, so $S \geq 0$; put $C_A = \sqrt{S} \geq 0$. Fix $m \geq 0$. Since every term of the sum is nonnegative and the sum converges, the single term at index m is at most the whole sum:

$$(n+m)|A_{n+m}|^2 \leq S.$$

Also $n+m \geq 1$, because $n \geq 1$ and $m \geq 0$, so $|A_{n+m}|^2 \leq (n+m)|A_{n+m}|^2 \leq S$. Taking square roots of the nonnegative numbers $|A_{n+m}|^2 \leq S$ gives $|A_{n+m}| \leq \sqrt{S} = C_A$. This is (iii). (No Cauchy–Schwarz inequality is needed: a single term of a nonnegative convergent series is bounded by its sum.)

Step 3: geometric domination, (iv). Let $0 < \rho < 1$ and $m \geq 0$. Since $\rho > 0$, the real-power laws give $\big(\rho^b\big)^m = \rho^{bm}$. Since $0 < \rho \leq 1$, the function $t \mapsto \rho^t$ is nonincreasing on $\mathbb{R}$, and $bm \leq \lambda_{n+m}$ by (ii), so

$$\rho^{\lambda_{n+m}} \leq \rho^{bm} = \big(\rho^b\big)^m.$$

Step 4: summability at every polynomial weight, (v). Fix an integer $M \geq 0$ and ρ with $0 < \rho < 1$, and put $q = \rho^b$. Then $q > 0$, being a real power of a positive number, and $q < 1$, because $0 < \rho < 1$ and $b > 0$. By Step 3,

$$\rho^{\lambda_{n+m}} \leq q^m \qquad (m \geq 0). \tag{5}$$

Next, an elementary inequality: for every real $x \geq 0$,

$$(x+1)^M \leq 2^M\big(x^M + 1\big). \tag{6}$$

Indeed if $x \leq 1$ then $x + 1 \leq 2$, so $(x+1)^M \leq 2^M \leq 2^M(x^M+1)$ because $x^M \geq 0$; and if $x > 1$ then $x + 1 \leq 2x$, so $(x+1)^M \leq (2x)^M = 2^M x^M \leq 2^M(x^M+1)$. Consequently, for $m \geq 0$,

$$(n+m)^M \leq \big((n+1)(m+1)\big)^M = (n+1)^M(m+1)^M \leq (n+1)^M 2^M\big(m^M+1\big), \tag{7}$$

where the first inequality holds because $(n+1)(m+1) = nm + n + m + 1 \geq n + m$ and both bases are nonnegative, and the last step is (6) at $x = m$ multiplied by the nonnegative factor $(n+1)^M$.

Combining (7), $|A_{n+m}| \leq C_A$ from Step 2, and (5) — all three factors being nonnegative — we get, for every $m \geq 0$,

$$(n+m)^M |A_{n+m}|\, \rho^{\lambda_{n+m}} \leq (n+1)^M 2^M\big(m^M+1\big)\, C_A\, q^m = \big((n+1)^M 2^M C_A\big)\, m^M q^m + \big((n+1)^M 2^M C_A\big)\, q^m.$$

The majorant on the right is summable in m: since $0 \leq q < 1$, the series $\sum_m m^M q^m$ converges (the standard polynomial-times-geometric series) and $\sum_m q^m$ converges (geometric series), and a convergent series stays convergent after multiplication by a constant and addition of another convergent series. The terms of the series in (v) are nonnegative, so by comparison with this majorant that series converges.

Step 5: the constant summand, (vi). Let $\xi \neq 0$. By (2), $\lambda_n = 0$ and $\epsilon_n = 0$, so the exponent of the modulus factor is $\lambda_n + n - n = 0$ and the second term of the bracket vanishes. Hence, by `Thm2Term`,

$$T_n(\xi) = \big(A_n \xi^n + 0 \cdot \overline{A_n}\, \bar{\xi}^n\big)\, |\xi|^0\, \xi^{-n} = A_n\, \xi^n\, \xi^{-n} = A_n,$$

using $|\xi|^0 = 1$ and $\xi^n \xi^{-n} = 1$, legitimate because $\xi \neq 0$ and so $\xi^n \neq 0$.

Step 6: the pointwise bound, (vii). First a bound for a general index. Let $k \geq 0$ be an integer and $\xi \neq 0$, and write $r = |\xi| > 0$. Since $|\overline{A_k}| = |A_k|$, $|\bar{\xi}| = |\xi| = r$ and ϵ_k is real,

$$\big|A_k \xi^k + \epsilon_k \overline{A_k}\, \bar{\xi}^k\big| \leq |A_k|\, r^k + |\epsilon_k|\, |A_k|\, r^k = \big(1 + |\epsilon_k|\big)|A_k|\, r^k \tag{8}$$

by the triangle inequality and multiplicativity of the modulus. Taking moduli in the definition of T_k and using $\big||\xi|^a\big| = r^a$ for real a (the modulus factor is a positive real) and $|\xi^{-n}| = r^{-n}$,

$$|T_k(\xi)| = \big|A_k \xi^k + \epsilon_k \overline{A_k} \bar{\xi}^k\big| \cdot r^{\lambda_k + n - k} \cdot r^{-n} \leq \big(1 + |\epsilon_k|\big)|A_k|\, r^k\, r^{\lambda_k + n - k}\, r^{-n} = \big(1 + |\epsilon_k|\big)|A_k|\, r^{\lambda_k},$$

where the inequality is (8) multiplied by the nonnegative factors $r^{\lambda_k + n - k}$ and r^{-n}, and the last step adds the exponents of the positive base r: $k + (\lambda_k + n - k) + (-n) = \lambda_k$. Now take $k = n + m$. By (3) we have $1 + |\epsilon_{n+m}| \leq 2$, and $|A_{n+m}| \geq 0$ and $r^{\lambda_{n+m}} \geq 0$, so

$$|T_{n+m}(\xi)| \leq \big(1 + |\epsilon_{n+m}|\big)|A_{n+m}|\, r^{\lambda_{n+m}} \leq 2|A_{n+m}|\, |\xi|^{\lambda_{n+m}},$$

which is (vii).

Step 7: (viii). Apply (v) with $M = 0$: since $(n+m)^0 = 1$, the series $\sum_m |A_{n+m}| \rho^{\lambda_{n+m}}$ converges for every $\rho \in (0, 1)$. Multiplying a convergent series of reals by the constant 2 leaves it convergent, and $2|A_{n+m}|\rho^{\lambda_{n+m}} = 2\big(|A_{n+m}|\rho^{\lambda_{n+m}}\big)$ termwise, which gives (viii).

With b from Step 1 and C_A from Step 2, all eight assertions have been established, so the existential statement holds. □

2.35. `Thm2TermWordDerivBound`.

Lemma 2.35. Let $n \geq 1$ be an integer, let $1 < p < \infty$ and let $A\colon \mathbb{Z} \to \mathbb{C}$. Write $T_k = T_k(n,p,A,\cdot)$ for the k-th summand of the series (19) as in `Thm2Term`, $\lambda_k = \lambda_k(n,p)$ as in `LambdaCoef` and $\epsilon_k = \epsilon_k(n,p)$ as in `EpsCoef`.

For a word $w = (v_1,\dots,v_q)$ of complex letters define the iterated directional derivative D_w on functions $f\colon \mathbb{C} \to \mathbb{C}$ by

$$D_{()}f = f, \qquad D_{(v,v_1,\dots,v_q)}f = \big(z \mapsto \mathrm{d}\big(D_{(v_1,\dots,v_q)}f\big)(z)[v]\big),$$

so that each letter contributes one real Fréchet differentiation in its own direction, the leading letter being the outermost one; and put

$$V_w = \prod_{i=1}^{q} |v_i|.$$

Then such a pair (D,V) exists, satisfies the two recursions above together with $V_{()} = 1$, $V_{(v,v_1,\dots,v_q)} = |v|\,V_{(v_1,\dots,v_q)}$ and $V_w \geq 0$, and has the following two properties. For every word w of length $q = |w|$, every integer $m \geq 0$ and every $\xi \neq 0$,

$$\big|D_w T_{n+m}(\xi)\big| \leq \Big(2\big((n+m)\big(3+\sqrt{p-1}\big)+q\big)\Big)^q\, V_w\,\big(2|A_{n+m}|\big)\,|\xi|^{\lambda_{n+m}-q},$$

and $D_w T_{n+m}$ is of class C^∞ at every $\xi \neq 0$.

This is the paper's "we note that $\lim \lambda_k = \infty$. Therefore, we can perform any finite number of term-by-term differentiations in formula (19)" (`paper.tex:387`) made quantitative on a single summand. Each differentiation lowers the common exponent of the summand by exactly one and inflates its coefficient by a factor at most $2\big((n+m)(3+\sqrt{p-1})+q\big)$, which is polynomial in $n+m$; against the geometric decay of $\rho^{\lambda_{n+m}}$ supplied by the linear lower bound for λ_k, such a factor is harmless at every order. The operators D and V are produced existentially rather than as separate definition nodes because only the stated recursions are ever used.

Proof. Throughout, $n \geq 1$ and $1 < p < \infty$ are fixed and $k = n+m$. All assertions about ξ are for $\xi \neq 0$, where $|\xi|^a$ is an ordinary real power of a positive real and ξ^j, $\bar\xi^{\,l}$ are ordinary integer powers of a nonzero complex number; in particular all the functions below are defined and C^∞ on the open set $\mathbb{C} - \{0\}$.

Step 0: the operators, and the estimates for λ_k and ϵ_k. Define

$$D_w f = \text{the right fold of } w \text{ under } (v,g) \mapsto \big(z \mapsto \mathrm{d}g(z)[v]\big) \text{ starting from } f, \qquad V_w = \prod_i |v_i|,$$

which satisfy $D_{()}f = f$ and $D_{(v)\frown w}f = (z \mapsto \mathrm{d}(D_w f)(z)[v])$ by definition of a fold, and likewise $V_{()} = 1$ and $V_{(v)\frown w} = |v|\,V_w$; and $V_w \geq 0$ because it is a product of moduli, by induction on w.

From `LambdaEpsEstimates` we record, for every $m \geq 0$,

$$\lambda_{n+m} \geq 0, \qquad |\epsilon_{n+m}| \leq 1, \qquad \lambda_{n+m} \leq \big((n:\mathbb{R})+m\big)\sqrt{p-1}. \tag{1}$$

For $m = 0$ these are (23): $\lambda_n = 0$ and $\epsilon_n = 0$, and the third reads $0 \leq n\sqrt{p-1}$, which holds since $n \geq 0$ and $p - 1 > 0$. For $m \geq 1$ we have $n+1 \leq n+m$, so (24) gives $|\epsilon_{n+m}| < 1$ and (25) gives both $\lambda_{n+m} \geq (p-1)/(np) > 0$ and $\lambda_{n+m} \leq (n+m)\sqrt{p-1}$.

Step 1: the monomial class and its first derivatives. For $s \in \mathbb{R}$ and $j,l \in \mathbb{Z}$ put

$$E_{s,j,l}(\xi) = |\xi|^s\,\xi^j\,\bar\xi^{\,l} \qquad (\xi \neq 0).$$

Each factor is C^∞ on $\mathbb{C} - \{0\}$ — $|\xi|^s$ because $\xi \mapsto |\xi|$ is C^∞ and positive there and $t \mapsto t^s$ is C^∞ on $(0,\infty)$; ξ^j and $\bar\xi^{\,l}$ because they are quotients of polynomials in ξ, $\bar\xi$ with nonvanishing denominators — so $E_{s,j,l}$ is C^∞ on $\mathbb{C} - \{0\}$. Its Wirtinger derivatives are

$$\partial_\xi E_{s,j,l} = \Big(\frac{s}{2} + j\Big)E_{s,j-1,l}, \qquad \partial_{\bar\xi} E_{s,j,l} = \Big(\frac{s}{2} + l\Big)E_{s,j,l-1}. \tag{2}$$

Indeed $|\xi|^s = (\xi\bar\xi)^{s/2}$, so by `WirtingerModulusPowerRule` $\partial_\xi|\xi|^s = \frac{s}{2}|\xi|^s\xi^{-1}$ and $\partial_{\bar\xi}|\xi|^s = \frac{s}{2}|\xi|^s\bar\xi^{-1}$; by `WirtingerPowerRule` and `WirtingerConjRule` $\partial_\xi\xi^j = j\xi^{j-1}$, $\partial_{\bar\xi}\xi^j = 0$, $\partial_\xi\bar\xi^{\,l} = 0$

and $\partial_{\bar\xi}\bar\xi^{\,l} = l\bar\xi^{\,l-1}$. Applying `WirtingerProductRule` twice and collecting terms gives (2), using $\xi^j\xi^{-1} = \xi^{j-1}$ and $\bar\xi^{\,l}\bar\xi^{\,-1} = \bar\xi^{\,l-1}$.

Step 2: finite combinations, and the three bookkeeping quantities. By a *configuration* we mean a finite list $L = \big((c_1, j_1, l_1), \dots, (c_N, j_N, l_N)\big)$ with $c_i \in \mathbb{C}$ and $j_i, l_i \in \mathbb{Z}$. For $s \in \mathbb{R}$ put

$$S_{s,L}(\xi) = \sum_{i=1}^{N} c_i\, E_{s,j_i,l_i}(\xi), \qquad \|L\| = \sum_{i=1}^{N} |c_i|.$$

Say that L *has exponent total* e (relative to s) if $s + j_i + l_i = e$ for every i, and that L *has frequency bound* K if $\frac{1}{2}|s| + |j_i| + |l_i| \le K$ for every i. Two elementary consequences:

(2a) if L has exponent total e then

$$\big|S_{s,L}(\xi)\big| \le \|L\|\,|\xi|^e \qquad (\xi \neq 0), \tag{3}$$

because $|E_{s,j_i,l_i}(\xi)| = |\xi|^s|\xi|^{j_i}|\xi|^{l_i} = |\xi|^{s+j_i+l_i} = |\xi|^e$ and the triangle inequality applies to the finite sum;

(2b) $S_{s,L}$ is C^∞ on $\mathbb{C} - \{0\}$, being a finite linear combination of the $E_{s,j,l}$ of Step 1.

Step 3: one differentiation of a configuration. Fix s, a letter $v \in \mathbb{C}$ and a configuration L, and define a new configuration of length $2N$,

$$\mathcal{D}_{s,v}L = \Big(\big(c_i(\tfrac{s}{2}+j_i)v,\ j_i - 1,\ l_i\big)\Big)_{i\le N} \frown \Big(\big(c_i(\tfrac{s}{2}+l_i)\bar v,\ j_i,\ l_i - 1\big)\Big)_{i\le N}.$$

For a function F that is real-differentiable at ξ one has $\mathrm{d}F(\xi)[v] = v\,\partial_\xi F(\xi) + \bar v\,\partial_{\bar\xi}F(\xi)$, which is just the definition of the Wirtinger derivatives $\partial_\xi = \frac{1}{2}(\partial_1 - i\partial_2)$, $\partial_{\bar\xi} = \frac{1}{2}(\partial_1 + i\partial_2)$ of `DZ` and `DZbar` read backwards, together with the $\mathbb{R}$-linearity of $v \mapsto \mathrm{d}F(\xi)[v]$ and $v = v_1 + iv_2$. Applying this to $F = S_{s,L}$, which is differentiable at every $\xi \neq 0$ by (2b), and using (2) termwise,

$$\mathrm{d}S_{s,L}(\xi)[v] = S_{s,\mathcal{D}_{s,v}L}(\xi) \qquad (\xi \neq 0). \tag{4}$$

Moreover:

(i) if L has exponent total e then $\mathcal{D}_{s,v}L$ has exponent total $e-1$: the two families of exponent pairs are $(j_i - 1, l_i)$ and $(j_i, l_i - 1)$, and in either case the sum $s + j + l$ drops by exactly 1;

(ii) if L has frequency bound K then $\mathcal{D}_{s,v}L$ has frequency bound $K+1$, since $|j_i - 1| \le |j_i| + 1$ and $|l_i - 1| \le |l_i| + 1$;

(iii) if L has frequency bound K with $K \ge 0$ then

$$\big\|\mathcal{D}_{s,v}L\big\| \le 2K\,|v|\,\|L\|, \tag{5}$$

because $\big|c_i(\frac{s}{2}+j_i)v\big| = |c_i|\,\big|\frac{s}{2}+j_i\big|\,|v| \le |c_i|\,K\,|v|$ by $\big|\frac{s}{2}+j_i\big| \le \frac{1}{2}|s|+|j_i| \le K$, and likewise $\big|c_i(\frac{s}{2}+l_i)\bar v\big| \le |c_i|\,K\,|v|$ using $|\bar v| = |v|$; summing the two families gives $2K|v|\|L\|$.

Step 4: the induction on the length of the word. We claim: for every word w, every $s, e, K \in \mathbb{R}$ with $K \ge 0$ and every configuration L with exponent total e and frequency bound K, there is a configuration L' with

$$\begin{aligned} &D_w\big(S_{s,L}\big)(\xi) = S_{s,L'}(\xi) \quad (\xi \neq 0),\\ &L' \text{ has exponent total } e - |w| \text{ and frequency bound } K + |w|,\\ &\|L'\| \le \big(2(K+|w|)\big)^{|w|}\,V_w\,\|L\|. \end{aligned} \tag{6}$$

We induct on w. If w is empty, take $L' = L$: then $D_{()}S_{s,L} = S_{s,L}$, the exponent total and frequency bound are $e - 0$ and $K + 0$, and $\|L\| \le 1 \cdot 1 \cdot \|L\|$ since $(2K)^0 = 1$ and $V_{()} = 1$.

Let $w = (v) \frown w_0$ and let L_0 be a configuration provided by the inductive hypothesis for w_0, so that $D_{w_0}S_{s,L} = S_{s,L_0}$ off the origin, L_0 has exponent total $e - |w_0|$ and frequency bound $K + |w_0|$, and $\|L_0\| \le (2(K+|w_0|))^{|w_0|}V_{w_0}\|L\|$. Put $L' = \mathcal{D}_{s,v}L_0$. Fix $\xi \neq 0$. The two functions $D_{w_0}S_{s,L}$ and S_{s,L_0} agree on the open set $\mathbb{C} - \{0\}$, which is a neighbourhood of ξ, so they have the same Fréchet derivative at ξ; hence, by the defining recursion of D and by (4),

$$D_wS_{s,L}(\xi) = \mathrm{d}\big(D_{w_0}S_{s,L}\big)(\xi)[v] = \mathrm{d}S_{s,L_0}(\xi)[v] = S_{s,\mathcal{D}_{s,v}L_0}(\xi) = S_{s,L'}(\xi).$$

By (i) and (ii) of Step 3, L' has exponent total $(e-|w_0|)-1 = e-|w|$ and frequency bound $(K+|w_0|)+1 = K+|w|$. Finally, by (5) applied with the frequency bound $K+|w_0| \geq 0$ of L_0, and then by the inductive bound on $\|L_0\|$,

$$\|L'\| \leq 2\big(K+|w_0|\big)|v|\,\|L_0\| \leq 2\big(K+|w_0|\big)|v|\,\big(2(K+|w_0|)\big)^{|w_0|}V_{w_0}\|L\| = \big(2(K+|w_0|)\big)^{|w|}V_w\|L\|,$$

using $|v|\,V_{w_0} = V_w$ and $|w| = |w_0|+1$. Since $0 \leq 2(K+|w_0|) \leq 2(K+|w|)$, raising to the $|w|$-th power preserves the inequality, and $V_w\|L\| \geq 0$, so $\|L'\| \leq (2(K+|w|))^{|w|}V_w\|L\|$. This proves (6).

Step 5: the summand as a configuration. Fix $m \geq 0$ and put $k = n+m$,

$$s = \lambda_k+n-k = \lambda_{n+m}-m, \qquad K = (n+m)\big(3+\sqrt{p-1}\big), \qquad L_k = \big((A_k,\ m,\ 0),\ (\epsilon_k\overline{A_k},\ -n,\ k)\big).$$

By `Thm2Term`, for $\xi \neq 0$,

$$T_k(\xi) = \big(A_k\xi^k + \epsilon_k\overline{A_k}\bar{\xi}^k\big)|\xi|^s\xi^{-n} = A_k|\xi|^s\xi^{k-n} + \epsilon_k\overline{A_k}\,|\xi|^s\xi^{-n}\bar{\xi}^k,$$

and $k-n = m$, so

$$T_k(\xi) = S_{s,L_k}(\xi) \qquad (\xi \neq 0). \tag{7}$$

We check the three bookkeeping quantities of L_k.

Exponent total λ_k. For the first entry $s+m+0 = \lambda_k+n-k+m = \lambda_k$ since $k = n+m$; for the second $s+(-n)+k = \lambda_k+n-k-n+k = \lambda_k$.

Frequency bound K. Write $N = (n:\mathbb{R})+m$, so $K = 3N+N\sqrt{p-1}$ and, by (1), $0 \leq \lambda_k \leq N\sqrt{p-1}$. Since $\lambda_k \geq 0$ and $m \geq 0$ we get $-(\lambda_k+m) \leq s = \lambda_k-m \leq \lambda_k+m$, i.e.

$$|s| \leq \lambda_k+m \leq N\sqrt{p-1}+m. \tag{8}$$

For the first entry, $\frac{1}{2}|s|+|m|+|0| \leq \frac{1}{2}N\sqrt{p-1}+\frac{1}{2}m+m \leq N\sqrt{p-1}+3N = K$, because $\frac{1}{2}m+m = \frac{3}{2}m \leq 3N$ (as $m \leq N$) and $\frac{1}{2}N\sqrt{p-1} \leq N\sqrt{p-1}$. For the second entry, $\frac{1}{2}|s|+|-n|+|k| \leq \frac{1}{2}N\sqrt{p-1}+\frac{1}{2}m+n+N$, and $\frac{1}{2}m+n+N \leq \frac{1}{2}N+N+N \leq 3N$ using $m \leq N$ and $n \leq N$; so again the total is at most $N\sqrt{p-1}+3N = K$. Note $K \geq 0$.

Mass at most $2|A_k|$. By (1), $|\epsilon_k| \leq 1$, so

$$\|L_k\| = |A_k| + \big|\epsilon_k\overline{A_k}\big| = |A_k| + |\epsilon_k|\,|A_k| \leq |A_k| + |A_k| = 2|A_k|.$$

Step 6: conclusion. Fix a word w with $|w| = q$, an integer $m \geq 0$ and $\xi \neq 0$, and keep k, s, K, L_k as in Step 5. The two functions T_k and S_{s,L_k} agree on the open set $\mathbb{C}-\{0\}$ by (7), so by the same local-agreement argument used in Step 4 (applied q times, i.e. by the evident induction on w) one has $D_wT_k = D_wS_{s,L_k}$ at every point of $\mathbb{C}-\{0\}$. Let L' be the configuration produced by Step 4 for the word w and the data (s, λ_k, K, L_k), legitimate because $K \geq 0$ and L_k has exponent total λ_k and frequency bound K. Then $D_wT_k(\xi) = S_{s,L'}(\xi)$, L' has exponent total λ_k-q and

$$\|L'\| \leq \big(2(K+q)\big)^qV_w\|L_k\| \leq \big(2(K+q)\big)^qV_w\cdot 2|A_k|,$$

the last step because $(2(K+q))^qV_w \geq 0$. Hence by (3),

$$\big|D_wT_k(\xi)\big| = \big|S_{s,L'}(\xi)\big| \leq \|L'\|\,|\xi|^{\lambda_k-q} \leq \Big(2\big((n+m)(3+\sqrt{p-1})+q\big)\Big)^qV_w\big(2|A_{n+m}|\big)|\xi|^{\lambda_{n+m}-q},$$

which is the asserted bound.

For the smoothness, $S_{s,L'}$ is C^∞ on $\mathbb{C}-\{0\}$ by (2b), in particular C^∞ at ξ; and D_wT_k agrees with $S_{s,L'}$ on the open set $\mathbb{C}-\{0\}$, which is a neighbourhood of ξ. Being C^∞ at a point depends only on the germ of the function there, so D_wT_{n+m} is C^∞ at ξ. As w, m and $\xi \neq 0$ were arbitrary, both conclusions hold, and the pair (D,V) constructed in Step 0 witnesses the existential statement. $\square$

2.36. `DiscEnergyPolarCoord`.

Lemma 2.36. Write $\mathbb{B} = \{\xi \in \mathbb{C} : |\xi| < 1\}$ and let $d\sigma$ denote plane Lebesgue measure. Let $f\colon \mathbb{C} \to \mathbb{C}$ be Lebesgue measurable. Then, as an identity between Lebesgue integrals with values in $[0, \infty]$,

$$\int_0^1 r\Big(\int_0^{2\pi} \big|f\big(re^{i\theta}\big)\big|^2 \, d\theta\Big) dr = \int_{\mathbb{B}} |f(\xi)|^2 \, d\sigma(\xi),$$

the integrals on the left being over $r \in (0,1)$ and $\theta \in (0, 2\pi)$.

This is the polar evaluation of the Dirichlet-type energy of a function on the unit disc, and it is the first move of the energy computation of Section 4 (`paper.tex:353-357`), where the area integral of $|H_r|^2$ over $\mathbb{B}$ is rewritten as $\int_0^1 r \int_0^{2\pi} |H_r|^2 d\theta \, dr$ before Parseval is applied to the inner integral.

Two features of the statement are what make it reusable. First, the integrals are the $[0,\infty]$-valued Lebesgue integrals of nonnegative measurable functions, so no integrability hypothesis is needed or asserted: the two sides are infinite together when the energy is infinite, and measurability of f is the only assumption. That is what lets the identity be applied to each of H, H_ξ and $H_{\bar\xi}$, and to partial sums of the series (19), without first knowing any of them to be square-integrable. Second, the angular window is $(0, 2\pi)$, which is the window in which the Fourier computation of Section 4 is carried out; the standard polar parametrisation uses $(-\pi, \pi)$ instead, and moving between the two is part of the proof.

Proof. Throughout, all integrals are Lebesgue integrals of nonnegative measurable functions with values in $[0,\infty]$, for which Tonelli's theorem and countable additivity hold with no finiteness hypothesis. Write $\mathbb{B} = \{\xi : |\xi| < 1\}$.

Step 0: two elementary facts. The set $\mathbb{B}$ is open, being the preimage of $(-\infty, 1)$ under the continuous map $\xi \mapsto |\xi|$, hence measurable. And for every $r > 0$ and every θ,

$$\big|re^{i\theta}\big| = |r| \cdot \big|e^{i\theta}\big| = r, \tag{1}$$

since $|e^{i\theta}| = 1$ and $r > 0$. In particular $re^{i\theta} \in \mathbb{B}$ if and only if $r < 1$, for every θ.

Step 1: pass to the indicator. Define $F\colon \mathbb{C} \to [0,\infty]$ by

$$F(\zeta) = \mathbf{1}_{\mathbb{B}}(\zeta)\, |f(\zeta)|^2,$$

i.e. $F = |f|^2$ on $\mathbb{B}$ and $F = 0$ off $\mathbb{B}$. Then F is measurable: $\zeta \mapsto |f(\zeta)|^2$ is measurable because f is and both the modulus and squaring are continuous, the passage to $[0,\infty]$ is measurable, and $\mathbb{B}$ is measurable by Step 0. By the defining property of the integral of an indicator,

$$\int_{\mathbb{B}} |f(\xi)|^2 \, d\sigma(\xi) = \int_{\mathbb{C}} F \, d\sigma. \tag{2}$$

Step 2: the polar change of variables. The polar parametrisation $(r,\theta) \mapsto re^{i\theta}$ is a diffeomorphism from $(0,\infty) \times (-\pi, \pi)$ onto $\mathbb{C}$ minus the closed ray $(-\infty, 0]$, with Jacobian r; the removed ray is a Lebesgue-null subset of the plane. Hence for every measurable $G\colon \mathbb{C} \to [0,\infty]$,

$$\iint_{(0,\infty)\times(-\pi,\pi)} r\, G\big(re^{i\theta}\big)\, d(r,\theta) = \int_{\mathbb{C}} G \, d\sigma. \tag{3}$$

Apply (3) to $G(\zeta) = F(-\zeta)$, which is measurable because $\zeta \mapsto -\zeta$ is continuous. The map $\zeta \mapsto -\zeta$ is a measure-preserving bijection of $\mathbb{C}$, so $\int_{\mathbb{C}} F(-\zeta)\, d\sigma(\zeta) = \int_{\mathbb{C}} F \, d\sigma$, and (3) becomes

$$\iint_{(0,\infty)\times(-\pi,\pi)} r\, F\big(-re^{i\theta}\big)\, d(r,\theta) = \int_{\mathbb{C}} F \, d\sigma. \tag{4}$$

The detour through $-\zeta$ is deliberate: it is exactly what converts the angular window $(-\pi,\pi)$ into $(0, 2\pi)$ in Step 4, because $-re^{i\theta} = re^{i(\theta+\pi)}$.

Step 3: Tonelli. The function $(r,\theta) \mapsto r\, F\big(-re^{i\theta}\big)$ is measurable on $\mathbb{R} \times \mathbb{R}$: it is the product of the measurable $(r,\theta) \mapsto r$ with the composition of the measurable F with negation and with

the continuous map $(r,\theta)\mapsto re^{i\theta}$. Plane Lebesgue measure on $\mathbb{R}^2$ is the product of the one-dimensional Lebesgue measures, and its restriction to a product of sets is the product of the restrictions, so Tonelli's theorem applies and gives

$$\iint_{(0,\infty)\times(-\pi,\pi)} r\,F\big(-re^{i\theta}\big)d(r,\theta)=\int_0^\infty r\Big(\int_{-\pi}^{\pi}F\big(-re^{i\theta}\big)\,d\theta\Big)dr, \tag{5}$$

the factor r being a finite constant in the inner integral and so coming out of it.

Step 4: the angular shift. Fix r. Since $e^{i\pi}=-1$ we have

$$-re^{i\theta}=re^{i\theta}e^{i\pi}=re^{i(\theta+\pi)}\qquad(\theta\in\mathbb{R}), \tag{6}$$

and plainly $\theta\in(-\pi,\pi)$ if and only if $\theta+\pi\in(0,2\pi)$. Therefore, for every $\theta\in\mathbb{R}$,

$$\mathbf{1}_{(-\pi,\pi)}(\theta)\,F\big(-re^{i\theta}\big)=\mathbf{1}_{(0,2\pi)}(\theta+\pi)\,F\big(re^{i(\theta+\pi)}\big)\,:$$

when $\theta\in(-\pi,\pi)$ both sides equal $F(-re^{i\theta})$ by (6), and otherwise both indicators vanish and both sides are 0. Writing each angular integral as the integral over $\mathbb{R}$ of the corresponding indicator and using the invariance of Lebesgue measure on $\mathbb{R}$ under the translation $\theta\mapsto\theta+\pi$,

$$\int_{-\pi}^{\pi}F\big(-re^{i\theta}\big)d\theta=\int_0^{2\pi}F\big(re^{i\theta}\big)d\theta. \tag{7}$$

Step 5: restricting the radius, and removing the indicator. Combining (4), (5), (7) and (2),

$$\int_0^\infty r\Big(\int_0^{2\pi}F\big(re^{i\theta}\big)d\theta\Big)dr=\int_{\mathbb{C}}F\,d\sigma=\int_{\mathbb{B}}|f(\xi)|^2d\sigma(\xi). \tag{8}$$

It remains to see that the left-hand side of (8) equals the left-hand side of the assertion, i.e. that replacing the radial window $(0,\infty)$ by $(0,1)$ and F by $|f|^2$ changes nothing. Write both radial integrals as integrals over $\mathbb{R}$ of the corresponding indicators and compare the two integrands at each $r\in\mathbb{R}$.

If $0<r<1$, then by (1) $|re^{i\theta}|=r<1$, so $re^{i\theta}\in\mathbb{B}$ and $F\big(re^{i\theta}\big)=\big|f\big(re^{i\theta}\big)\big|^2$ for every θ; both indicators are 1 and the two inner integrals coincide termwise, so the integrands agree.

If $r\geq 1$, the indicator of $(0,1)$ vanishes, so the integrand of the asserted left-hand side is 0 at r. On the other side, by (1) $|re^{i\theta}|=r\geq 1$, so $re^{i\theta}\notin\mathbb{B}$ and hence $F\big(re^{i\theta}\big)=0$ for every θ; the inner integral is the integral of the zero function, which is 0, and $r\cdot 0=0$. So both integrands are 0 at r.

If $r\leq 0$ both indicators vanish and both integrands are 0.

Thus the two radial integrands agree at every r, so the two radial integrals agree, and with (8) this is exactly the asserted identity. □

2.37. `TwoFamilyTrigParseval`.

Lemma 2.37. Let $F\subset\mathbb{N}$ be finite, let $g_1,g_2\colon\mathbb{N}\to\mathbb{Z}$ and $c_1,c_2\colon\mathbb{N}\to\mathbb{C}$, and assume

(i) g_1 is injective on F: for $k,k'\in F$, $g_1(k)=g_1(k')$ implies $k=k'$;
(ii) g_2 is injective on F: for $k,k'\in F$, $g_2(k)=g_2(k')$ implies $k=k'$;
(iii) the two frequency families never collide: $g_1(k)\neq g_2(k')$ for all $k,k'\in F$.

Then

$$\int_0^{2\pi}\Big|\sum_{k\in F}\big(c_1(k)\,e^{ig_1(k)\theta}+c_2(k)\,e^{ig_2(k)\theta}\big)\Big|^2d\theta=2\pi\sum_{k\in F}\big(|c_1(k)|^2+|c_2(k)|^2\big).$$

This is the Parseval step of the energy computation of Section 4 (`paper.tex:353-357`), in exactly the generality that computation needs. On the circle of radius r the summand T_k of the series (19) carries precisely the two frequencies $k-n$ and $-k-n$; the family $k\mapsto k-n$ is injective, the family $k\mapsto -k-n$ is injective, and the two never meet because $(k-n)-(-k'-n)=k+k'>0$ for $k,k'\geq n\geq 1$. Applying the identity to a partial sum of (19) and then integrating in r against the Jacobian is what produces the $\pi\sum\lambda_k(1+\epsilon_k^2)|A_k|^2$ of the energy identity.

Hypotheses (i)–(iii) are exactly what makes the diagonal the whole story: the two injectivity assumptions kill the off-diagonal pairs within each family and (iii) kills every cross pair. Nothing

is assumed about the sign or the size of the frequencies, and F is an arbitrary finite index set, so the identity applies verbatim to any partial sum.

Proof. For $l \in \mathbb{Z}$ write $E_l(\theta) = e^{il\theta}$, and abbreviate

$$S(\theta) = \sum_{k \in F} \big(c_1(k) E_{g_1(k)}(\theta) + c_2(k) E_{g_2(k)}(\theta)\big).$$

Step 0: orthogonality of the characters. For every $l \in \mathbb{Z}$,

$$\int_0^{2\pi} E_l(\theta)\, d\theta = \begin{cases} 2\pi, & l = 0, \\ 0, & l \neq 0. \end{cases} \tag{1}$$

Indeed, if $l = 0$ the integrand is the constant 1 and the integral is 2π. If $l \neq 0$, put $a = il \in \mathbb{C}$, which is nonzero because $l \neq 0$ and $i \neq 0$; writing $E_l(\theta) = e^{a\theta}$ and using the antiderivative $\theta \mapsto e^{a\theta}/a$ of $\theta \mapsto e^{a\theta}$,

$$\int_0^{2\pi} e^{a\theta} d\theta = \frac{e^{2\pi a} - 1}{a} = \frac{e^{2\pi i l} - 1}{il} = 0,$$

since $e^{2\pi i l} = 1$ for every integer l.

Step 1: integrability. For each l and each $c \in \mathbb{C}$ the function $\theta \mapsto c E_l(\theta)$ is continuous on $\mathbb{R}$, hence interval-integrable on $[0, 2\pi]$. Consequently S, a finite sum of such functions, is continuous and interval-integrable there, and so are $\theta \mapsto \overline{S(\theta)}$ and $\theta \mapsto |S(\theta)|^2$. Every integral written below is therefore defined, and the integral of a finite sum of such functions is the sum of their integrals.

Step 2: expanding the square. For every θ we have $|S(\theta)|^2 = S(\theta)\overline{S(\theta)}$, an identity between a real number and a complex one, read through the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$. Also

$$\overline{E_l(\theta)} = \overline{e^{il\theta}} = e^{-il\theta} = E_{-l}(\theta), \tag{2}$$

because conjugation is a continuous field automorphism fixing $\mathbb{R}$ and sending i to $-i$. Expanding the product of the two finite sums and using (2) termwise,

$$\begin{aligned} S(\theta)\overline{S(\theta)} = \sum_{k \in F} \sum_{k' \in F} \Big[& c_1(k)\overline{c_1(k')}\, E_{g_1(k) - g_1(k')}(\theta) + c_1(k)\overline{c_2(k')}\, E_{g_1(k) - g_2(k')}(\theta) \\ & + c_2(k)\overline{c_1(k')}\, E_{g_2(k) - g_1(k')}(\theta) + c_2(k)\overline{c_2(k')}\, E_{g_2(k) - g_2(k')}(\theta) \Big], \end{aligned} \tag{3}$$

using $E_a(\theta) E_b(\theta) = E_{a+b}(\theta)$, which is the additivity of the exponential.

Step 3: integrating, and discarding the off-diagonal terms. Integrate (3) over $[0, 2\pi]$. The right-hand side is a finite sum of functions of the form $c E_l$, so by Step 1 the integral is the corresponding finite sum of integrals, and each is $c \int_0^{2\pi} E_l$, evaluated by (1). We examine the four families of terms.

The first family contributes only when $g_1(k) - g_1(k') = 0$, i.e. $g_1(k) = g_1(k')$, which by (i) happens exactly when $k = k'$; for such a pair the contribution is $2\pi\, c_1(k)\overline{c_1(k)} = 2\pi |c_1(k)|^2$.

The fourth family likewise contributes only when $g_2(k) = g_2(k')$, which by (ii) happens exactly when $k = k'$, with contribution $2\pi\, c_2(k)\overline{c_2(k)} = 2\pi |c_2(k)|^2$.

The second family would need $g_1(k) = g_2(k')$ for some $k, k' \in F$, which (iii) forbids, so every one of its terms integrates to 0. The third family would need $g_2(k) = g_1(k')$ for some $k, k' \in F$; applying (iii) with the two indices interchanged gives $g_1(k') \neq g_2(k)$, so this too is impossible and every term integrates to 0.

Hence only the diagonal pairs $k = k'$ of the first and fourth families survive, and

$$\int_0^{2\pi} S(\theta)\overline{S(\theta)}\, d\theta = \sum_{k \in F} 2\pi \big(|c_1(k)|^2 + |c_2(k)|^2\big) = 2\pi \sum_{k \in F} \big(|c_1(k)|^2 + |c_2(k)|^2\big). \tag{4}$$

Step 4: descending to the real identity. By Step 2 the integrand on the left of (4) is the image of the real-valued function $\theta \mapsto |S(\theta)|^2$ under the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$, and that inclusion commutes with the integral of an interval-integrable real function. So the left side of (4) is the image of $\int_0^{2\pi} |S(\theta)|^2 d\theta$, while the right side is the image of the real number $2\pi \sum_{k \in F} (|c_1(k)|^2 + |c_2(k)|^2)$. The inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ is injective, so the two real numbers are equal, which is the assertion. $\square$

2.38. `PolarSeparatedSumEnergy`.

Lemma 2.38. Write $\mathbb{B} = \{\xi \in \mathbb{C} : |\xi| < 1\}$ and $d\sigma$ for plane Lebesgue measure. Let $F \subset \mathbb{N}$ be finite, let $f_k \colon \mathbb{C} \to \mathbb{C}$ be measurable for every k, let $g_1, g_2 \colon \mathbb{N} \to \mathbb{Z}$, let $\gamma, \delta \colon \mathbb{N} \to \mathbb{C}$ and let $e \colon \mathbb{N} \to \mathbb{R}$. Assume, for all $k, k' \in F$,

(i) $e(k) > -1$;
(ii) $g_1(k) = g_1(k')$ implies $k = k'$, and $g_2(k) = g_2(k')$ implies $k = k'$;
(iii) $g_1(k) \neq g_2(k')$;
(iv) f_k is polar-separated with radial exponent $e(k)$ and the two angular frequencies $g_1(k), g_2(k)$: for every $r > 0$ and every θ,

$$f_k\big(re^{i\theta}\big) = r^{e(k)}\Big(\gamma(k)\, e^{ig_1(k)\theta} + \delta(k)\, e^{ig_2(k)\theta}\Big).$$

Then, as an identity between Lebesgue integrals with values in $[0, \infty]$,

$$\int_{\mathbb{B}} \Big|\sum_{k\in F} f_k(\xi)\Big|^2 d\sigma(\xi) = \sum_{k\in F} \frac{2\pi\big(|\gamma(k)|^2 + |\delta(k)|^2\big)}{2e(k)+2}.$$

This is the finite-sum engine for the display the paper calls "a standard computation" at `paper.tex:351-353`,

$$\int_0^1 r\Big(\int_0^{2\pi} |H_r(r,\theta)|^2 d\theta\Big) dr = \pi \sum_{k=n+1}^{\infty} \lambda_k\big(1+\epsilon_k^2\big)|A_k|^2,$$

which together with (24), (25) and (27) is what yields Theorem 2's bound (22). Each summand of the polar form (40) of the series (19) has exactly the shape (iv): differentiating $\big(A_k e^{ik\theta} + \epsilon_k \overline{A_k} e^{-ik\theta}\big) r^{\lambda_k} e^{-in\theta}$ in r gives radial exponent $e(k) = \lambda_k - 1$ and the two frequencies $k - n$ and $-k - n$, with $\gamma(k) = \lambda_k A_k$ and $\delta(k) = \lambda_k \epsilon_k \overline{A_k}$; and then $2e(k) + 2 = 2\lambda_k$, so the k-th term of the conclusion is

$$\frac{2\pi\lambda_k^2\big(|A_k|^2 + \epsilon_k^2|A_k|^2\big)}{2\lambda_k} = \pi\lambda_k\big(1+\epsilon_k^2\big)|A_k|^2,$$

the paper's summand exactly.

Hypotheses (ii) and (iii) are what make the right-hand side a bare sum over F rather than a double sum: every cross term between two distinct indices, and every cross term between the two frequency families, integrates to zero over a full period. Hypothesis (i) is precisely the convergence condition for the radial integral $\int_0^1 r^{2e(k)+1} dr = 1/(2e(k)+2)$, and it is where the denominator comes from. Both sides are read in $[0, \infty]$, so no integrability hypothesis is needed; (i) already makes the right-hand side finite.

Proof. Write $S(\xi) = \sum_{k\in F} f_k(\xi)$. All integrals below are Lebesgue integrals of nonnegative measurable functions with values in $[0, \infty]$.

Step 1: to polar coordinates. Each f_k is measurable, and F is finite, so S is measurable. Hence `DiscEnergyPolarCoord` applies to S and gives

$$\int_{\mathbb{B}} |S(\xi)|^2 d\sigma(\xi) = \int_0^1 r\Big(\int_0^{2\pi} \big|S\big(re^{i\theta}\big)\big|^2 d\theta\Big) dr. \tag{1}$$

Step 2: the angular integral at a fixed radius. Fix r with $0 < r < 1$. Since $r > 0$, hypothesis (iv) applies at every $k \in F$ and every θ, so

$$S\big(re^{i\theta}\big) = \sum_{k\in F}\Big(\big(r^{e(k)}\gamma(k)\big)e^{ig_1(k)\theta} + \big(r^{e(k)}\delta(k)\big)e^{ig_2(k)\theta}\Big), \tag{2}$$

which is a two-family trigonometric sum over F with coefficients $c_1(k) = r^{e(k)}\gamma(k)$ and $c_2(k) = r^{e(k)}\delta(k)$. Hypotheses (ii) and (iii) are verbatim the three side conditions of `TwoFamilyTrigParseval`, so that node applies and yields

$$\int_0^{2\pi} \big|S\big(re^{i\theta}\big)\big|^2 d\theta = 2\pi \sum_{k\in F}\Big(\big|r^{e(k)}\gamma(k)\big|^2 + \big|r^{e(k)}\delta(k)\big|^2\Big).$$

Now $r^{e(k)}$ is a positive real, because $r>0$, so $\left|r^{e(k)}\gamma(k)\right| = r^{e(k)}|\gamma(k)|$ and likewise for δ; squaring,

$$\int_0^{2\pi} \left|S\big(re^{i\theta}\big)\right|^2 d\theta = 2\pi \sum_{k\in F} r^{2e(k)}\big(|\gamma(k)|^2+|\delta(k)|^2\big) \qquad (0<r<1). \tag{3}$$

(The integrand is continuous in θ, by (2) and the continuity of the exponentials, so its interval integral over $[0,2\pi]$ and its Lebesgue integral over $(0,2\pi)$ agree, and both are finite; this is what lets (3) be substituted into the $[0,\infty]$-valued inner integral of (1).)

Step 3: the radial integral. Substituting (3) into (1) and using $r\cdot r^{2e(k)} = r^{2e(k)+1}$ for $r>0$,

$$\int_{\mathbb{B}} |S|^2 d\sigma = \int_0^1 \sum_{k\in F} 2\pi\big(|\gamma(k)|^2+|\delta(k)|^2\big) r^{2e(k)+1}\, dr = \sum_{k\in F} 2\pi\big(|\gamma(k)|^2+|\delta(k)|^2\big)\int_0^1 r^{2e(k)+1} dr,$$

where the finite sum and the integral are interchanged, legitimate because there are finitely many summands and each is a nonnegative measurable function of r, and the constant factors are pulled out of the integrals.

It remains to evaluate the radial integral. For a real $s>-1$ the function $r\mapsto r^s$ is integrable on $(0,1)$ with

$$\int_0^1 r^s\, dr = \frac{1}{s+1}, \tag{4}$$

since $r\mapsto r^{s+1}/(s+1)$ is an antiderivative on $(0,1)$ whose limits at the endpoints are 0 and $1/(s+1)$, both finite because $s+1>0$. Apply (4) with $s=2e(k)+1$: by hypothesis (i) we have $e(k)>-1$, hence $s+1=2e(k)+2>0$, i.e. $s>-1$, and

$$\int_0^1 r^{2e(k)+1} dr = \frac{1}{2e(k)+2}.$$

Therefore

$$\int_{\mathbb{B}} \Big|\sum_{k\in F} f_k\Big|^2 d\sigma = \sum_{k\in F} \frac{2\pi\big(|\gamma(k)|^2+|\delta(k)|^2\big)}{2e(k)+2},$$

which is the assertion. Each term on the right is a finite nonnegative real, again because $2e(k)+2>0$, so the right-hand side is a finite sum of finite terms. □

2.39. `Thm2TermDerivBound`.

Lemma 2.39. Let $n\ge 1$ be an integer, let $1<p<\infty$, let $A\colon \mathbb{Z}\to\mathbb{C}$, let $m\ge 0$ be an integer and let $\xi\in\mathbb{C}$ with $\xi\neq 0$. Write $k=n+m$, $A_k = A(k)$, $\lambda_k = \lambda_k(n,p)$ as in `LambdaCoef`, $\epsilon_k = \epsilon_k(n,p)$ as in `EpsCoef`, and let $T_k = T_k(n,p,A,\cdot)$ be the k-th summand of the series (19) as in `Thm2Term`. Then

$$\left|\frac{\partial T_k}{\partial \xi}(\xi)\right| \le \big(4+\sqrt{p-1}\big)(n+m)\,|A_{n+m}|\,|\xi|^{\lambda_{n+m}-1}$$

and

$$\left|\frac{\partial T_k}{\partial \bar\xi}(\xi)\right| \le \big(4+\sqrt{p-1}\big)(n+m)\,|A_{n+m}|\,|\xi|^{\lambda_{n+m}-1},$$

with $\partial/\partial\xi$ as in `DZ` and $\partial/\partial\bar\xi$ as in `DZbar`.

The content of the statement is that the constant $4+\sqrt{p-1}$ does not depend on m: the only m-dependence left on the right is the explicit factor $n+m$ and the explicit power $|\xi|^{\lambda_{n+m}-1}$. That uniformity comes entirely from the Remark's estimates (23)–(25), carried by `LambdaEpsEstimates`: $0\le\lambda_{n+m}\le(n+m)\sqrt{p-1}$ and $|\epsilon_{n+m}|\le 1$. Without the *upper* bound on λ_k the coefficients of the closed form would grow in k at an unknown rate and no uniform constant would exist.

This is the pointwise estimate the $W^{1,2}$ convergence argument needs before it can sum over k: on a disc of radius $\rho<1$ the factor $|\xi|^{\lambda_{n+m}-1}$ decays geometrically in m, so this bound makes the tail of the differentiated series dominated by a convergent one, which is what allows (19) to be differentiated term by term in the $W^{1,2}$ norm.

The family is indexed by m rather than by $k \geq n$ because that is the shape in which `LambdaEpsEstimates` is used throughout: all three of its estimates are read as statements about λ_{n+m} and ϵ_{n+m}. The constant 4 is not sharp; the route below gives 3.

Proof. Throughout, $k = n + m$, $r = |\xi| > 0$, and

$$s = \lambda_k + n - k = \lambda_{n+m} - m, \qquad F = A_k \xi^k + \epsilon_k \overline{A_k}\, \bar{\xi}^k,$$

so that $T_k(\xi) = F\,|\xi|^s \xi^{-n}$ by `Thm2Term`.

Step 0: the three estimates used. From `LambdaEpsEstimates` we record, for every $m \geq 0$,

$$0 \leq \lambda_{n+m}, \qquad \lambda_{n+m} \leq \big((n:\mathbb{R}) + m\big)\sqrt{p-1}, \qquad |\epsilon_{n+m}| \leq 1. \tag{1}$$

For $m = 0$ these are (23), $\lambda_n = 0$ and $\epsilon_n = 0$, together with $0 \leq n\sqrt{p-1}$; for $m \geq 1$ we have $n + 1 \leq n + m$, so the first and second are (25) and the third is (24).

Step 1: the closed form. By `Thm2TermWirtingerDerivs`, applied at the index $k = n + m$ and the point $\xi \neq 0$,

$$\frac{\partial T_k}{\partial \xi}(\xi) = \frac{|\xi|^s \xi^{-n}}{\xi}\Big(k\, A_k \xi^k + \Big(\frac{s}{2} - n\Big)F\Big), \qquad \frac{\partial T_k}{\partial \bar{\xi}}(\xi) = \frac{|\xi|^s \xi^{-n}}{\bar{\xi}}\Big(k\, \epsilon_k \overline{A_k}\, \bar{\xi}^k + \frac{s}{2}F\Big). \tag{2}$$

That node requires no constraint on n, p or k, only $\xi \neq 0$.

Step 2: the modulus of the common prefactor. Since $r > 0$, real and integer powers of r multiply by adding exponents, and $\big||\xi|^s\big| = r^s$, $|\xi^{-n}| = r^{-n}$, $|\xi| = |\bar{\xi}| = r$. Hence

$$\Big|\frac{|\xi|^s \xi^{-n}}{\xi}\Big| = \Big|\frac{|\xi|^s \xi^{-n}}{\bar{\xi}}\Big| = r^{s-n-1} = r^{\lambda_k - k - 1}, \tag{3}$$

using $s - n - 1 = (\lambda_k + n - k) - n - 1 = \lambda_k - k - 1$.

Step 3: the two brackets. Since $|\overline{A_k}| = |A_k|$, $|\bar{\xi}| = r$ and ϵ_k is real, the two terms of F have moduli $|A_k| r^k$ and $|\epsilon_k||A_k| r^k$, so by the triangle inequality and (1),

$$|F| \leq \big(1 + |\epsilon_k|\big)|A_k|\, r^k \leq 2|A_k|\, r^k. \tag{4}$$

Therefore, again by the triangle inequality and (4),

$$\Big|k\, A_k \xi^k + \Big(\frac{s}{2} - n\Big)F\Big| \leq k\,|A_k| r^k + \Big|\frac{s}{2} - n\Big| \cdot 2|A_k| r^k = \big(k + |s - 2n|\big)|A_k|\, r^k,$$

$$\Big|k\, \epsilon_k \overline{A_k} \bar{\xi}^k + \frac{s}{2}F\Big| \leq k\,|\epsilon_k||A_k| r^k + \Big|\frac{s}{2}\Big| \cdot 2|A_k| r^k \leq \big(k + |s|\big)|A_k|\, r^k,$$

where $2\big|\frac{s}{2} - n\big| = |s - 2n|$, $2\big|\frac{s}{2}\big| = |s|$, and $|\epsilon_k| \leq 1$ was used in the second display.

Step 4: the two coefficient sizes are at most $(4 + \sqrt{p-1})(n+m)$. Write $N = (n:\mathbb{R}) + m$, so $k = N$ as a real number and, by (1), $0 \leq \lambda_k \leq N\sqrt{p-1}$. Since $\lambda_k \geq 0$ and $m \geq 0$, the number $s = \lambda_k - m$ satisfies $-(\lambda_k + m) \leq s \leq \lambda_k + m$, i.e.

$$|s| \leq \lambda_k + m \leq N\sqrt{p-1} + m. \tag{5}$$

Hence, using $m \leq N$ and $n \leq N$,

$$k + |s| \leq N + N\sqrt{p-1} + m \leq 2N + N\sqrt{p-1} = N\big(2 + \sqrt{p-1}\big) \leq N\big(4 + \sqrt{p-1}\big),$$

and, since $|s - 2n| \leq |s| + 2n \leq N\sqrt{p-1} + m + 2n \leq N\sqrt{p-1} + 2N$,

$$k + |s - 2n| \leq N + N\sqrt{p-1} + 2N = N\big(3 + \sqrt{p-1}\big) \leq N\big(4 + \sqrt{p-1}\big).$$

(The second chain is where the constant is largest, and it is the reason 3 would already do.)

Step 5: collecting. Multiply the bound of Step 3 by the modulus (3) of the prefactor, which is legitimate because both factors are nonnegative, and use $r^k\, r^{\lambda_k - k - 1} = r^{\lambda_k - 1}$ (again $r > 0$). For the $\partial/\partial\xi$ derivative this gives, by (2),

$$\Big|\frac{\partial T_k}{\partial \xi}(\xi)\Big| \leq \big(k + |s - 2n|\big)|A_k|\, r^{\lambda_k - 1} \leq \big(4 + \sqrt{p-1}\big)N\,|A_k|\, r^{\lambda_k - 1},$$

the last step by Step 4 together with $|A_k| r^{\lambda_k - 1} \geq 0$; and for the $\partial/\partial\bar{\xi}$ derivative, identically,

$$\Big|\frac{\partial T_k}{\partial \bar{\xi}}(\xi)\Big| \leq \big(k + |s|\big)|A_k|\, r^{\lambda_k - 1} \leq \big(4 + \sqrt{p-1}\big)N\,|A_k|\, r^{\lambda_k - 1}.$$

Since $N = (n : \mathbb{R}) + m$, $k = n + m$ and $A_k = A_{n+m}$, these are the two asserted inequalities. □

2.40. `Thm2SeriesSmoothOffZero`.

Lemma 2.40. Let $n \geq 1$ be an integer, let $1 < p < \infty$ and let $A\colon \mathbb{Z} \to \mathbb{C}$ satisfy the energy hypothesis

$$\sum_{m \geq 0} (n+m)\,|A_{n+m}|^2 < \infty.$$

Write $T_k = T_k(n, p, A, \cdot)$ for the k-th summand of the series (19) as in `Thm2Term`, $\lambda_k = \lambda_k(n, p)$ as in `LambdaCoef`, and $\mathbb{B} = \{\xi \in \mathbb{C} : |\xi| < 1\}$. Then the pointwise sum of the series is of class C^∞ on the punctured disc:

$$\xi \longmapsto \sum_{m \geq 0} T_{n+m}(\xi) \quad \text{is } C^\infty \text{ on } \mathbb{B} - \{0\}.$$

This is the paper's "we note that $\lim \lambda_k = \infty$. Therefore, we can perform any finite number of term-by-term differentiations in formula (19)" (`paper.tex:387`), carried out in full. Both ingredients are already nodes and the content here is the induction that combines them: every summand is C^∞ off the origin and obeys the all-order word-derivative bound of `Thm2TermWordDerivBound`, and that bound is summable in m at every order against the geometric majorant of `Thm2TermGeometricSummabil`. So on a small ball about any punctured point the term-by-term differentiated series converges uniformly at every order, and a locally uniform limit of C^∞ functions whose derived series of every order converge locally uniformly is itself C^∞.

The statement is about the series itself and carries no hypothesis tying it to any particular function: a consumer holding $H = \sum_m T_{n+m}$ off the origin transfers the conclusion to H by congruence, which is how `Thm2SeriesW12Convergence` uses it to obtain its conjunct (b).

Proof. Throughout, ρ and radii are real, $\mathbb{B} - \{0\}$ is open, and for a word w of complex letters D_w denotes the iterated directional derivative and V_w the product of the moduli of its letters, both as supplied by `Thm2TermWordDerivBound`.

Step 0: the two inputs. From `Thm2TermWordDerivBound` we take, for every word w with $|w| = q$, every $m \geq 0$ and every $\xi \neq 0$,

$$\big\|D_w T_{n+m}(\xi)\big\| \leq \Big(2\big((n+m)(3+\sqrt{p-1})+q\big)\Big)^q V_w\big(2|A_{n+m}|\big)\,|\xi|^{\lambda_{n+m}-q}, \tag{1}$$

together with the fact that each $D_w T_{n+m}$ is C^∞ at every $\xi \neq 0$. From `Thm2TermGeometricSummability` we take that for every integer $M \geq 0$ and every $\rho \in (0,1)$

$$\sum_{m \geq 0} (n+m)^M |A_{n+m}|\,\rho^{\lambda_{n+m}} < \infty, \tag{2}$$

and from `LambdaEpsEstimates` that $\lambda_{n+m} \geq 0$ for every $m \geq 0$ — for $m = 0$ this is (23) and for $m \geq 1$ it is the lower bound of (25).

Step 1: reduction to a finite order on a small ball. Being C^∞ on a set means being C^N there for every finite N, so fix N. Being C^N on a set is a local property: it suffices to produce, for each $x \in \mathbb{B} - \{0\}$, an open neighbourhood of x on which the sum is C^N. So fix x with $0 < |x| < 1$ and put

$$\rho_x = \min\Big(\frac{|x|}{2},\ \frac{1-|x|}{2}\Big) > 0, \qquad \rho = \frac{1+|x|}{2} \in (0,1).$$

For z in the ball $B(x, \rho_x)$ we have, by the triangle inequality,

$$\frac{|x|}{2} \leq |x| - \rho_x \leq |z| \leq |x| + \rho_x \leq \rho < 1,$$

so in particular $z \neq 0$ and $B(x, \rho_x) \subseteq \mathbb{B} - \{0\}$.

Step 2: a uniform bound on the ball, for each word. Fix a word w with $|w| = q$ and let $z \in B(x, \rho_x)$. Since $\lambda_{n+m} \geq 0$ and $|z| \leq \rho$ we have $|z|^{\lambda_{n+m}} \leq \rho^{\lambda_{n+m}}$; and since $|z| \geq |x|/2 > 0$ we have $|z|^{-q} \leq (|x|/2)^{-q}$. Writing $|z|^{\lambda_{n+m}-q} = |z|^{\lambda_{n+m}}|z|^{-q}$ and substituting into (1),

$$\sup_{z \in B(x,\rho_x)} \big\|D_w T_{n+m}(z)\big\| \leq C(w,x)\,(n+m)^q\,|A_{n+m}|\,\rho^{\lambda_{n+m}}, \tag{3}$$

where $C(w,x)$ collects the factors independent of m: the constant $2^q(3+\sqrt{p-1})^q$-type expansion of $\big(2((n+m)(3+\sqrt{p-1})+q)\big)^q$ into a multiple of $(n+m)^q$, which is legitimate because $n+m\ge 1$ and q is fixed, together with V_w, the factor 2, and $(|x|/2)^{-q}$.

Step 3: uniform convergence of every derived series. By (2) with $M=q$ and this $\rho\in(0,1)$, the right-hand side of (3) is the general term of a convergent series of nonnegative reals. Hence, by the Weierstrass M-test, the series

$$\sum_{m\ge 0} D_w T_{n+m}$$

converges uniformly on $B(x,\rho_x)$, for every word w. Taking w empty, and using $D_{()}f=f$, the series $\sum_m T_{n+m}$ itself converges uniformly there.

Step 4: differentiating the sum, by induction on the word. We show by induction on $|w|$, downwards from N, that for every word w with $|w|\le N$ the function $z\mapsto\sum_m D_w T_{n+m}(z)$ is $C^{N-|w|}$ on $B(x,\rho_x)$.

For the base case $|w|=N$ the claim is continuity, which holds because by Step 3 the series converges uniformly on the ball and each term is continuous there, being C^∞ at every point $z\neq 0$ by Step 0.

For the inductive step, let $|w|<N$ and consider the two words $1::w$ and $\mathrm{i}::w$ obtained by prefixing the letters 1 and i. Each $D_w T_{n+m}$ is differentiable on the ball, again by Step 0, with $\mathrm{d}(D_w T_{n+m})(z)[v]$ equal to $D_{v::w}T_{n+m}(z)$ for $v\in\{1,\mathrm{i}\}$ by the defining recursion of D. By Step 3 the series $\sum_m D_w T_{n+m}$ converges uniformly on the ball and so do the two derived series $\sum_m D_{1::w}T_{n+m}$ and $\sum_m D_{\mathrm{i}::w}T_{n+m}$. By the standard theorem on differentiating a uniformly convergent series — if a series of differentiable functions on an open set converges pointwise and its derived series converges uniformly, the sum is differentiable and its derivative is the sum of the derived series — the function $z\mapsto\sum_m D_w T_{n+m}(z)$ is differentiable on the ball, and its Fréchet derivative at z is the real-linear map sending v to the corresponding limit.

Here the two directional limits determine the whole derivative. A real-linear map $L\colon\mathbb{C}\to\mathbb{C}$ satisfies

$$Lv=(\operatorname{Re}v)\,L(1)+(\operatorname{Im}v)\,L(\mathrm{i})\qquad(v\in\mathbb{C}),$$

because $v=(\operatorname{Re}v)\cdot 1+(\operatorname{Im}v)\cdot\mathrm{i}$ and L is additive and commutes with real scalars. So the assignment

$$(q_1,q_2)\longmapsto\big(v\mapsto(\operatorname{Re}v)q_1+(\operatorname{Im}v)q_2\big)$$

is itself a continuous linear map $\mathbb{C}\times\mathbb{C}\to(\mathbb{C}\to_{\mathbb{R}}\mathbb{C})$, and the Fréchet derivative of $\sum_m D_w T_{n+m}$ is its composition with the pair of sums $\big(\sum_m D_{1::w}T_{n+m},\ \sum_m D_{\mathrm{i}::w}T_{n+m}\big)$. By the inductive hypothesis applied to the two longer words, each component of that pair is $C^{N-|w|-1}$ on the ball; composing with a continuous linear map preserves that class, so the derivative is $C^{N-|w|-1}$ and therefore $\sum_m D_w T_{n+m}$ is $C^{N-|w|}$, as claimed.

Step 5: conclusion. Taking w empty in Step 4 gives that $z\mapsto\sum_m T_{n+m}(z)$ is C^N on $B(x,\rho_x)$, an open neighbourhood of x contained in $\mathbb{B}-\{0\}$. Since $x\in\mathbb{B}-\{0\}$ was arbitrary and being C^N is local, the sum is C^N on $\mathbb{B}-\{0\}$; and since N was arbitrary it is C^∞ there. □

2.41. `Thm2TermPolarWirtingerDerivs`.

Lemma 2.41. Let $n\ge 0$ and $k\ge 0$ be integers, let p be real, let $A\colon\mathbb{Z}\to\mathbb{C}$, let $r>0$ be real and let θ be real. Write $A_k=A(k)$, $\lambda_k=\lambda_k(n,p)$ for the exponent of `LambdaCoef`, $\epsilon_k=\epsilon_k(n,p)$ for the coefficient of `EpsCoef`, and let $T_k=T_k(n,p,A,\cdot)$ be the k-th summand of the series (19) as in `Thm2Term`. Then, with $\partial/\partial\xi$ as in `DZ` and $\partial/\partial\bar\xi$ as in `DZbar`, evaluated at the point $\xi=re^{i\theta}$,

$$\frac{\partial T_k}{\partial\xi}\big(re^{i\theta}\big)=r^{\lambda_k-1}\Big(\frac{\lambda_k+k-n}{2}\,A_k\,e^{i(k-n-1)\theta}+\frac{\lambda_k-k-n}{2}\,\epsilon_k\overline{A_k}\,e^{i(-k-n-1)\theta}\Big)$$

and

$$\frac{\partial T_k}{\partial\bar\xi}\big(re^{i\theta}\big)=r^{\lambda_k-1}\Big(\frac{\lambda_k+n-k}{2}\,A_k\,e^{i(k-n+1)\theta}+\frac{\lambda_k+n+k}{2}\,\epsilon_k\overline{A_k}\,e^{i(-k-n+1)\theta}\Big).$$

This is the summand-level instance of the polar transformation formulas that open Section 4 of the paper at `paper.tex:234-237`,

$$H_{\bar\xi} = \tfrac{1}{2}\Big(H_r + \tfrac{i}{r}H_\theta\Big)e^{i\theta}, \qquad H_\xi = \tfrac{1}{2}\Big(H_r - \tfrac{i}{r}H_\theta\Big)e^{-i\theta},$$

in the one case where both sides can be written out: for $H = T_k$ the radial and angular dependences separate completely, the modulus contributes the single power r^{λ_k-1}, and the angle contributes exactly two pure frequencies, $k-n\mp1$ and $-k-n\mp1$. It is the form in which the two derivatives of a summand enter the circlewise Parseval computation of Section 4, where the two frequencies are what makes distinct summands orthogonal, and the radial integral, where r^{λ_k-1} is what is integrated against $r\,dr$.

No inequality among n, p and k is imposed; the only hypothesis is $r>0$, which keeps the point off the origin, where T_k need not be differentiable and where the negative powers of the statement are not defined.

Proof. Throughout put $s=\lambda_k+n-k$, fix $r>0$ and θ, and write $\xi = re^{i\theta}$, so that $\xi\neq0$ because $r\neq0$ and $e^{i\theta}\neq0$. Two elementary identities are used repeatedly:

$$|\xi| = \big|re^{i\theta}\big| = r, \qquad \overline{e^{i\theta}} = e^{-i\theta}, \tag{1}$$

the first because $|e^{i\theta}|=1$ and $r>0$, the second because conjugation is a ring homomorphism fixing the reals and sending i to $-i$, so that $\overline{\exp(i\theta)} = \exp(\overline{i\theta}) = \exp(-i\theta)$. From the second, $\bar\xi = r\,e^{-i\theta}$.

Step 1: the Cartesian closed form. Since $\xi\neq0$, `Thm2TermWirtingerDerivs` applies verbatim and gives

$$\frac{\partial T_k}{\partial\xi}(\xi) = \frac{|\xi|^s\,\xi^{-n}}{\xi}\Big(k\,A_k\xi^k + \Big(\frac{s}{2}-n\Big)\big(A_k\xi^k + \epsilon_k\overline{A_k}\,\bar\xi^{\,k}\big)\Big) \tag{2}$$

and

$$\frac{\partial T_k}{\partial\bar\xi}(\xi) = \frac{|\xi|^s\,\xi^{-n}}{\bar\xi}\Big(k\,\epsilon_k\overline{A_k}\,\bar\xi^{\,k} + \frac{s}{2}\big(A_k\xi^k + \epsilon_k\overline{A_k}\,\bar\xi^{\,k}\big)\Big). \tag{3}$$

Nothing beyond $\xi\neq0$ is needed for (2) and (3); in particular no relation between n, k and p is assumed.

Step 2: the four powers at $\xi = re^{i\theta}$. Because $r>0$ and $e^{i\theta}\neq0$, each of the following is a product of powers of the two nonzero factors separately:

$$\xi^k = r^k e^{ik\theta}, \qquad \bar\xi^{\,k} = r^k e^{-ik\theta}, \qquad \xi^{-n} = r^{-n}e^{-in\theta}, \qquad |\xi|^s = r^s, \tag{4}$$

the second by $\bar\xi = re^{-i\theta}$ and the last by the first identity of (1).

Step 3: the common prefactor, for $\partial/\partial\xi$. Substituting (4) into the prefactor of (2),

$$\frac{|\xi|^s\,\xi^{-n}}{\xi} = \frac{r^s\,r^{-n}e^{-in\theta}}{r\,e^{i\theta}} = r^{s-n-1}\,e^{-i(n+1)\theta} = r^{\lambda_k-k-1}\,e^{-i(n+1)\theta},$$

using $s-n-1=\lambda_k-k-1$. Here the three separate real powers of r collapse into one because $r>0$, so $r^ar^b=r^{a+b}$ and $r^a/r=r^{a-1}$ hold for all real exponents; this is the only place where $r>0$ rather than $r\neq0$ is used.

Step 4: the bracket, for $\partial/\partial\xi$. Substituting (4) into the bracket of (2) and taking out the common factor r^k,

$$k\,A_k\xi^k + \Big(\frac{s}{2}-n\Big)\big(A_k\xi^k + \epsilon_k\overline{A_k}\,\bar\xi^{\,k}\big) = r^k\Big(\Big(k+\frac{s}{2}-n\Big)A_k\,e^{ik\theta} + \Big(\frac{s}{2}-n\Big)\epsilon_k\overline{A_k}\,e^{-ik\theta}\Big).$$

The two scalars simplify, by $s=\lambda_k+n-k$:

$$k+\frac{s}{2}-n = \frac{2k+\lambda_k+n-k-2n}{2} = \frac{\lambda_k+k-n}{2}, \qquad \frac{s}{2}-n = \frac{\lambda_k+n-k-2n}{2} = \frac{\lambda_k-k-n}{2}. \tag{5}$$

Step 5: the first identity. Multiplying Step 3 by Step 4, the powers of r combine as $r^{\lambda_k-k-1}r^k = r^{\lambda_k-1}$, and the exponentials combine as

$$e^{-i(n+1)\theta}e^{ik\theta} = e^{i(k-n-1)\theta}, \qquad e^{-i(n+1)\theta}e^{-ik\theta} = e^{i(-k-n-1)\theta},$$

since $e^{ia\theta}e^{ib\theta} = e^{i(a+b)\theta}$. With (5) this is exactly the first asserted identity.

Step 6: the second identity. The prefactor of (3) differs from that of (2) only in carrying $\bar{\xi} = re^{-i\theta}$ in place of $\xi = re^{i\theta}$ in the denominator, so by (4)

$$\frac{|\xi|^s\,\xi^{-n}}{\bar{\xi}} = \frac{r^s\,r^{-n}e^{-in\theta}}{r\,e^{-i\theta}} = r^{\lambda_k-k-1}\,e^{-i(n-1)\theta}.$$

The bracket of (3), again after taking out r^k, is

$$r^k\Big(\frac{s}{2}\,A_k\,e^{ik\theta} + \Big(k+\frac{s}{2}\Big)\epsilon_k\overline{A_k}\,e^{-ik\theta}\Big),$$

and the two scalars are

$$\frac{s}{2} = \frac{\lambda_k+n-k}{2}, \qquad k+\frac{s}{2} = \frac{2k+\lambda_k+n-k}{2} = \frac{\lambda_k+n+k}{2}. \tag{6}$$

Multiplying, the powers of r again give r^{λ_k-1}, while

$$e^{-i(n-1)\theta}e^{ik\theta} = e^{i(k-n+1)\theta}, \qquad e^{-i(n-1)\theta}e^{-ik\theta} = e^{i(-k-n+1)\theta},$$

which with (6) is the second asserted identity. □

2.42. `Thm2SeriesCircleFourierCoef.`

Lemma 2.42. Let $n \geq 1$ be an integer, let $p > 1$ be real, let $A\colon \mathbb{Z} \to \mathbb{C}$ satisfy the energy hypothesis

$$\sum_{m=0}^{\infty}(n+m)\,|A_{n+m}|^2 < \infty,$$

write $A_k = A(k)$, $\lambda_k = \lambda_k(n,p)$ for the exponent of `LambdaCoef`, $\epsilon_k = \epsilon_k(n,p)$ for the coefficient of `EpsCoef`, and let $T_k = T_k(n,p,A,\cdot)$ be the k-th summand of the series (19) as in `Thm2Term`. Let

$$H(\xi) = \begin{cases} A_n, & \xi = 0, \\ \displaystyle\sum_{m=0}^{\infty} T_{n+m}(\xi), & \xi \neq 0, \end{cases}$$

be the sum of the series (19), completed at the origin by the value A_n of its constant summand T_n, and let $a_j(r)$ denote the Fourier coefficient (30) of `FourierCoefOfH` formed from n and H. Then both of the following hold.

(a) For every integer $m \geq 0$, every $j \in \mathbb{Z}$, every $r > 0$ and every real θ,

$$T_{n+m}\big(re^{i\theta}\big)\,e^{i(n-j)\theta} = r^{\lambda_{n+m}}A_{n+m}\,e^{i((n+m)-j)\theta} + r^{\lambda_{n+m}}\epsilon_{n+m}\overline{A_{n+m}}\,e^{i(-(n+m)-j)\theta}.$$

(b) For every $j \in \mathbb{Z}$ and every r with $0 < r < 1$,

$$a_j(r) = \frac{1}{2\pi}\sum_{m=0}^{\infty} 2\pi\,r^{\lambda_{n+m}}\Big(\big[\,n+m=j\,\big]A_{n+m} + \big[\,n+m=-j\,\big]\epsilon_{n+m}\overline{A_{n+m}}\Big),$$

where $[\,P\,]$ denotes 1 if P holds and 0 otherwise, so that each bracket selects at most one index m.

These are two statements about one computation, the Fourier expansion (29)–(30) of the paper evaluated on the series (19) rather than on an abstract solution of (18). Part (a) is the per-summand algebra: tested against the frequency $e^{i(n-j)\theta}$ of (30), one summand of (19) on the circle $|\xi| = r$ is a sum of exactly two pure angular exponentials, of frequencies $(n+m)-j$ and $-(n+m)-j$, both carried by the single radial factor $r^{\lambda_{n+m}}$; this is the polar form (40) of the paper for a single term. Part (b) is its consequence for the coefficient: integration over the circle annihilates every nonzero frequency, so the j-th coefficient of the series sees only those m with $n+m=j$ or $n+m=-j$. In particular $a_j(r) = A_j r^{\lambda_j}$ when $j \geq n$, $a_j(r) = \epsilon_{-j}\overline{A_{-j}}r^{\lambda_{-j}}$ when $j \leq -n$, and $a_j(r) = 0$ when $|j| < n$, since then no index $n+m$ meets either selection.

The factor 2π is left standing inside the sum, uncancelled against the $1/(2\pi)$ of (30), so that (b) is exactly what term-by-term integration produces and nothing is folded in by hand. The value of H at the origin is irrelevant to both parts, which evaluate only at points $re^{i\theta}$ with $r > 0$.

Proof. Step 1: part (a). Fix $m \geq 0$, $j \in \mathbb{Z}$, $r > 0$ and θ. Since $n + m \geq n$, part (b) of `Thm2TermSolvesEq18` applies to the index $k = n+m$ and gives the polar form of that summand on the circle of radius r,

$$T_{n+m}\big(re^{i\theta}\big) = r^{\lambda_{n+m}}\Big(A_{n+m}e^{i((n+m)-n)\theta} + \epsilon_{n+m}\overline{A_{n+m}}\,e^{i(-(n+m)-n)\theta}\Big), \tag{1}$$

which is the paper's (40) for one term. Multiplying (1) by $e^{i(n-j)\theta}$ and using $e^{ia\theta}e^{ib\theta} = e^{i(a+b)\theta}$ on each of the two summands, the two frequencies become

$$\big((n+m)-n\big) + (n-j) = (n+m) - j, \qquad \big(-(n+m)-n\big) + (n-j) = -(n+m) - j,$$

and distributing $r^{\lambda_{n+m}}$ over the two terms gives precisely the asserted identity. This is (a).

Step 2: two consequences of (a) used below. First, each function

$$\theta \mapsto T_{n+m}\big(re^{i\theta}\big)e^{i(n-j)\theta}$$

is continuous on $\mathbb{R}$: by (a) it equals a fixed linear combination of $\theta \mapsto e^{il\theta}$ for two integers l, and those are continuous. Second, for every integer l,

$$\int_0^{2\pi} e^{il\theta}\,d\theta = \begin{cases} 2\pi, & l = 0, \\ 0, & l \neq 0, \end{cases} \tag{2}$$

by the elementary evaluation of $\int_0^{2\pi} e^{c\theta}d\theta$ at $c = il \neq 0$, whose value is $(e^{2\pi il} - 1)/(il) = 0$ because $e^{2\pi il} = 1$ for integer l, and by direct computation when $l = 0$.

Step 3: the integral of one summand. Fix m, j and $r > 0$. Integrating the identity of (a) over $[0, 2\pi]$ — legitimate term by term because the right-hand side is a sum of two continuous functions — and applying (2) to each of the two frequencies,

$$\int_0^{2\pi} T_{n+m}\big(re^{i\theta}\big)e^{i(n-j)\theta}\,d\theta = 2\pi\, r^{\lambda_{n+m}}\Big(\big[\,n+m=j\,\big]A_{n+m} + \big[\,n+m=-j\,\big]\epsilon_{n+m}\overline{A_{n+m}}\Big). \tag{3}$$

Indeed the first frequency vanishes exactly when $(n+m) - j = 0$ and the second exactly when $-(n+m) - j = 0$, i.e. when $n+m = j$ and when $n+m = -j$ respectively; the three cases ($n+m = j$; $n+m = -j$; neither) are disjoint because $n + m \geq n \geq 1$ is positive, so $n+m = j$ and $n+m = -j$ cannot both hold.

Step 4: term-by-term integration. Fix j and $0 < r < 1$. By definition (30),

$$a_j(r) = \frac{1}{2\pi}\int_0^{2\pi} H\big(re^{i\theta}\big)e^{i(n-j)\theta}\,d\theta,$$

so it suffices to evaluate the integral. For each θ the point $re^{i\theta}$ is nonzero, because $r \neq 0$ and $e^{i\theta} \neq 0$; hence $H(re^{i\theta}) = \sum_m T_{n+m}(re^{i\theta})$ by the definition of H, and multiplying a convergent series by the constant $e^{i(n-j)\theta}$,

$$H\big(re^{i\theta}\big)e^{i(n-j)\theta} = \sum_{m=0}^{\infty} T_{n+m}\big(re^{i\theta}\big)e^{i(n-j)\theta}. \tag{4}$$

To interchange this sum with the integral we check the hypothesis of the dominated-convergence form of term-by-term integration: each summand is measurable on $[0, 2\pi]$ by Step 2, and

$$\sum_{m=0}^{\infty}\int_0^{2\pi}\big\|T_{n+m}\big(re^{i\theta}\big)e^{i(n-j)\theta}\big\|\,d\theta < \infty. \tag{5}$$

For the pointwise bound, $\big|e^{i(n-j)\theta}\big| = 1$ and $\big|re^{i\theta}\big| = r$, so the summand bound of `Thm2TermGeometricSummabili` gives

$$\big\|T_{n+m}\big(re^{i\theta}\big)e^{i(n-j)\theta}\big\| = \big\|T_{n+m}\big(re^{i\theta}\big)\big\| \leq 2\,|A_{n+m}|\,r^{\lambda_{n+m}}$$

for every θ; the bound is independent of θ, so the m-th integral in (5) is at most $2\pi\cdot 2|A_{n+m}|r^{\lambda_{n+m}}$. The series $\sum_m 2|A_{n+m}|r^{\lambda_{n+m}}$ converges, by the final clause of `Thm2TermGeometricSummability` applied at $\rho = r$ with $0 < r < 1$ — this is where the energy hypothesis on A and the geometric domination $r^{\lambda_{n+m}} \leq (r^b)^m$ are consumed — so (5) holds.

Therefore the integral of (4) may be taken term by term, and by (3)

$$\int_0^{2\pi} H\big(re^{i\theta}\big)e^{i(n-j)\theta}\,d\theta = \sum_{m=0}^{\infty} 2\pi\, r^{\lambda_{n+m}}\Big(\big[\,n+m=j\,\big]A_{n+m} + \big[\,n+m=-j\,\big]\epsilon_{n+m}\overline{A_{n+m}}\Big).$$

Dividing by 2π gives (b). □

2.43. `Thm2SeriesContinuousOnDisc`.

Lemma 2.43. Let $n \geq 1$ be an integer, let $p > 1$ be real, let $A\colon \mathbb{Z} \to \mathbb{C}$ satisfy

$$\sum_{m=0}^{\infty} (n+m)\,|A_{n+m}|^2 < \infty,$$

write $A_k = A(k)$, $\lambda_k = \lambda_k(n,p)$ for the exponent of `LambdaCoef`, and let $T_k = T_k(n,p,A,\cdot)$ be the k-th summand of the series (19) as in `Thm2Term`. Then the function

$$H(\xi) = \begin{cases} A_n, & \xi = 0, \\ \displaystyle\sum_{m=0}^{\infty} T_{n+m}(\xi), & \xi \neq 0, \end{cases}$$

is continuous on the whole unit disc $\mathbb{B} = \{\xi : |\xi| < 1\}$, the origin included.

This is conjunct (a) of Theorem 2 for the constructed solution, the statement $H \in C(\mathbb{B})$ of `paper.tex:264`. Off the origin it is local uniform convergence of a series of continuous summands. At the origin it is the real content, and it is what forces the completing value: the $m = 0$ summand of (19) is the constant A_n, every later summand carries a positive power of $|\xi|$, and so the sum tends to A_n as $\xi \to 0$. No other value at the origin would make the function continuous there.

The sum is written out as a case distinction rather than named, so that the statement depends on nothing but n, p and A.

Proof. By `Thm2TermGeometricSummability`, applied with the hypothesis on A, fix once and for all a rate b with $0 < b \leq \frac{1}{2}$ and a bound $CA \geq 0$ such that

$$(1) \qquad |A_{n+m}| \leq CA \quad \text{for all } m, \qquad |\xi|^{\lambda_{n+m}} \leq \big(|\xi|^b\big)^m \quad \text{for } 0 < |\xi| < 1 \text{ and all } m,$$

and such that, for $0 < \rho < 1$,

$$(2) \qquad \|T_{n+m}(\xi)\| \leq 2\,|A_{n+m}|\,|\xi|^{\lambda_{n+m}} \ (\xi \neq 0), \qquad \sum_{m=0}^{\infty} 2\,|A_{n+m}|\,\rho^{\lambda_{n+m}} < \infty.$$

The same node gives $T_n(\xi) = A_n$ for $\xi \neq 0$, the $m = 0$ summand being constant. Finally $\lambda_{n+m} \geq 0$ for every m, by `LambdaEpsEstimates`: $\lambda_n = 0$, and for $m \geq 1$ one has $\lambda_{n+m} \geq (p-1)/(np) > 0$. Each summand T_{n+m} is continuous away from the origin, by part (a) of `Thm2TermSolvesEq18` applied to the index $n + m \geq n$.

Two preliminary remarks. First, for $\xi \neq 0$ with $|\xi| < 1$ the series $\sum_m \|T_{n+m}(\xi)\|$ converges: by (2) its terms are at most $2|A_{n+m}|\,|\xi|^{\lambda_{n+m}}$, which is summable by the second half of (2) at $\rho = |\xi|$. So the defining sum exists at every such point, and $H(\xi) = \sum_m T_{n+m}(\xi)$ there. Second, $H(0) = A_n$ by definition.

Step 1: continuity at a point $x \neq 0$ of $\mathbb{B}$. Put $\delta = |x|/2$ and $\rho = (|x|+1)/2$, so that

$$0 < \delta < |x| < \rho < 1,$$

and let

$$U = \big\{z : \delta < |z| < \rho\big\},$$

an open annulus, being the intersection of the two open sets $\{|z| > \delta\}$ and $\{|z| < \rho\}$; it contains x. Every $z \in U$ is nonzero, since $|z| > \delta > 0$.

On U each summand T_{n+m} is continuous, by part (a) of `Thm2TermSolvesEq18` as recalled above, and is bounded by a constant independent of z:

$$\|T_{n+m}(z)\| \leq 2\,|A_{n+m}|\,|z|^{\lambda_{n+m}} \leq 2\,|A_{n+m}|\,\rho^{\lambda_{n+m}},$$

the first inequality by (2) and the second because $|z| < \rho$ and $\lambda_{n+m} \geq 0$, so that $t \mapsto t^{\lambda_{n+m}}$ is nondecreasing on $[0, \infty)$. The bounding series $\sum_m 2|A_{n+m}|\rho^{\lambda_{n+m}}$ converges by (2), since $0 < \rho < 1$. Hence $\sum_m T_{n+m}$ converges uniformly on U and its sum

$$G(z) = \sum_{m=0}^{\infty} T_{n+m}(z)$$

is continuous on U, by the Weierstrass criterion for a uniformly convergent series of continuous functions.

On U we have $H = G$, because every point of U is nonzero. Therefore H is continuous on U, and U is an open neighbourhood of x, so H is continuous at x; in particular continuous at x within $\mathbb{B}$.

Step 2: the tail estimate near the origin. Let $\xi \neq 0$ with $|\xi| < 1$ and suppose moreover $|\xi|^b \leq \frac{1}{2}$. Write $q = |\xi|^b$, so $0 < q \leq \frac{1}{2}$. Splitting off the $m = 0$ term of the (absolutely convergent) defining series and using $T_n(\xi) = A_n$,

$$H(\xi) - A_n = \sum_{m=1}^{\infty} T_{n+m}(\xi). \tag{3}$$

For $m \geq 1$, combining (2), the second half of (1) and $|A_{n+m}| \leq CA$,

$$\|T_{n+m}(\xi)\| \leq 2\,|A_{n+m}|\,|\xi|^{\lambda_{n+m}} \leq 2\,CA\,q^m.$$

Since $m \geq 1$ and $0 < 2q \leq 1$ we have $(2q)^m \leq 2q$, i.e. $q^m \leq 2q \cdot 2^{-m}$, whence

$$\|T_{n+m}(\xi)\| \leq 4\,CA\,q \cdot \Big(\frac{1}{2}\Big)^m \qquad (m \geq 1). \tag{4}$$

The right-hand side of (4) is summable in m, being a constant times a geometric series of ratio $\frac{1}{2}$, so by (3), the triangle inequality for absolutely convergent series and (4),

$$\|H(\xi) - A_n\| \leq \sum_{m=1}^{\infty} 4\,CA\,q\Big(\frac{1}{2}\Big)^m \leq 4\,CA\,q \cdot \frac{1}{1 - \frac{1}{2}} = 8\,CA\,|\xi|^b. \tag{5}$$

(The estimate (5) is stated for $\xi \neq 0$; at $\xi = 0$ its left-hand side is 0 by $H(0) = A_n$, so it holds there as well.)

Step 3: continuity at the origin. Since $b > 0$, the function $t \mapsto t^b$ is continuous at $t = 0$ with value $0^b = 0$, and $\xi \mapsto |\xi|$ is continuous with $|0| = 0$; composing, $|\xi|^b \to 0$ as $\xi \to 0$, hence also as $\xi \to 0$ within $\mathbb{B}$. Consequently $8\,CA\,|\xi|^b \to 0$ along the same filter, and the condition $|\xi|^b \leq \frac{1}{2}$ holds eventually along it.

So, eventually as $\xi \to 0$ within $\mathbb{B}$, the hypotheses of Step 2 are met and $\|H(\xi) - A_n\| \leq 8\,CA\,|\xi|^b$ by (5); the dominating quantity tends to 0; therefore $H(\xi) \to A_n = H(0)$ as $\xi \to 0$ within $\mathbb{B}$. That is continuity of H at 0 within $\mathbb{B}$.

Step 4: conclusion. Let $x \in \mathbb{B}$. If $x = 0$, H is continuous at x within $\mathbb{B}$ by Step 3. If $x \neq 0$, it is continuous at x by Step 1, hence continuous at x within $\mathbb{B}$. As x was arbitrary, H is continuous on $\mathbb{B}$. □

2.44. `Thm2SeriesW12Convergence`.

Lemma 2.44. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, write $\mathbb{B} = \{\xi : |\xi| < 1\}$ and $d\sigma$ for plane Lebesgue measure, and let $A \colon \mathbb{Z} \to \mathbb{C}$ satisfy*

(C) $A_{-k} = \epsilon_k \overline{A_k}$ *for every integer $k \geq n$, where $\epsilon_k = \epsilon_k(n,p)$ is the coefficient of* `EpsCoef`*;*
(S) $\sum_{m=0}^{\infty} (n+m)\,|A_{n+m}|^2 < \infty$, *that is* $\sum_{k \geq n} k|A_k|^2 < \infty$.

Write $\lambda_k = \lambda_{|k|}(n,p)$ for the exponent of `LambdaCoef`*, let T_k be the k-th summand of (19) in the sense of* `Thm2Term`*, and put*

$$S_N(\xi) = \sum_{k=n}^{n+N} T_k(\xi) \qquad (N \geq 0).$$

Then there is a function $H \colon \mathbb{C} \to \mathbb{C}$ with:

(a) *H continuous on $\mathbb{B}$;*
(b) *H of class C^∞ on $\mathbb{B}-\{0\}$;*
(c) *$H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`;*
(d) *$|H_\xi|^2+|H_{\bar\xi}|^2$ integrable on $\mathbb{B}$, where H_ξ, $H_{\bar\xi}$ are the Wirtinger derivatives of `DZ` and `DZbar`;*
(e) *$a_k(r)=A_k\,r^{\lambda_k}$ for every $k\in\mathbb{Z}$ with $|k|\ge n$ and every $r\in(0,1)$, where a_k is the Fourier coefficient of `FourierCoefOfH`;*
(f) *$a_k(r)=0$ for every $k\in\mathbb{Z}$ with $|k|<n$ and every $r\in(0,1)$;*
(g) *the series (19) converges to H in $W^{1,2}(\mathbb{B})$:*

$$\iint_{\mathbb{B}}|H-S_N|^2\,d\sigma+\iint_{\mathbb{B}}\big|H_\xi-(S_N)_\xi\big|^2\,d\sigma+\iint_{\mathbb{B}}\big|H_{\bar\xi}-(S_N)_{\bar\xi}\big|^2\,d\sigma\ \longrightarrow\ 0\qquad(N\to\infty).$$

Clause (g) is the paper's assertion "The series converges in $W^{1,2}(\mathbb{B})$", made by both halves of Theorem 2 at paper.tex:205 and paper.tex:210; nothing else in the development supplies it, the Fourier expansion giving only convergence in L^2 on each circle separately. Clauses (a)–(f) are the remaining properties of the sum that Theorem 2's two halves consume. No restriction on p is imposed.

Proof. Throughout, k and j denote integers, λ_k depends on k only through $|k|$, so $\lambda_{-k}=\lambda_k$, and $\epsilon_k\in\mathbb{R}$ by `EpsCoef`. Set

$$E=\sum_{k\ge n}k|A_k|^2<\infty$$

by (S), and for $N\ge 0$ let $R_N=\sum_{k>n+N}k|A_k|^2$, so $R_N\to 0$ as $N\to\infty$, being the tail of a convergent series of nonnegative terms. Since $k\ge n\ge 1$ we have $|A_k|^2\le k|A_k|^2$, so also $\sum_{k\ge n}|A_k|^2\le E$.

Step 0: the constants, and the basic estimates on λ_k and ϵ_k. By part (a) of `GammaRatioMinAtOne`, $\gamma_n=\lambda_{n+1}(n,p)>0$. Put

$$\beta=\frac{\gamma_n}{n+1}>0,\qquad c_1=2+\sqrt{p-1},\qquad c_2=\max\Big\{2\pi,\ \frac{\pi c_1^2(n+1)}{2\gamma_n}\Big\}.$$

By `LambdaLinearLowerBound`,

$$\lambda_k\ge\beta k\qquad(k\ge n+1),\tag{1}$$

and by parts (23), (24), (25) of `LambdaEpsEstimates`,

$$\lambda_n=0,\quad \epsilon_n=0,\quad |\epsilon_k|<1\ (k\ge n+1),\quad \lambda_k\le k\sqrt{p-1}\ (k\ge n+1).\tag{2}$$

In particular, for every $k\ge n$,

$$0\le\lambda_k\le k\sqrt{p-1},\qquad |\epsilon_k|\le 1,\qquad \lambda_k+k+n\le c_1k,\tag{3}$$

the last because $\lambda_k\le k\sqrt{p-1}$ and $n\le k$; at $k=n$ all three hold by (2), and $\lambda_n=0\le n\sqrt{p-1}$.

Step 1: the polar form of each summand. By part (b) of `Thm2TermSolvesEq18`, for every $k\ge n$, $r>0$ and $\theta\in\mathbb{R}$,

$$T_k\big(re^{i\theta}\big)=r^{\lambda_k}\Phi_k(\theta),\qquad \Phi_k(\theta)=A_ke^{i(k-n)\theta}+\epsilon_k\overline{A_k}e^{i(-k-n)\theta},\tag{4}$$

and by part (a) of the same node T_k is C^∞ at every $\xi\ne 0$. From (4) and (3),

$$|T_k(\xi)|\le 2|A_k|\,|\xi|^{\lambda_k},\qquad \big|\Phi_k^{(b)}(\theta)\big|\le 2|A_k|(k+n)^b\quad(b\ge 0),\tag{5}$$

the second because differentiating (4)'s profile b times multiplies each of its two terms by $\big(i(k-n)\big)^b$, resp. $\big(i(-k-n)\big)^b$, whose moduli are at most $(k+n)^b$.

Step 2: a summability estimate, used repeatedly. We claim that for every integer $m\ge 0$ and every $\rho\in(0,1)$,

$$\sum_{k\ge n}\big(1+\lambda_k+k+n\big)^m|A_k|\,\rho^{\lambda_k}<\infty.\tag{6}$$

Indeed, by (3), $1+\lambda_k+k+n \le (1+c_1)k$ for $k \ge n \ge 1$. By (1), $\rho^{\lambda_k} \le \rho^{\beta k} = q^k$ for $k \ge n+1$, where $q = \rho^\beta \in (0,1)$. Hence, by the Cauchy–Schwarz inequality applied to the sequences $\big(\sqrt{k}|A_k|\big)$ and $\big((1+c_1)^m k^m q^k/\sqrt{k}\big)$ over $k \ge n+1$,

$$\sum_{k\ge n+1} \big(1+\lambda_k+k+n\big)^m |A_k|\rho^{\lambda_k} \le (1+c_1)^m \Big(\sum_{k\ge n+1} k|A_k|^2\Big)^{1/2} \Big(\sum_{k\ge n+1} k^{2m-1}q^{2k}\Big)^{1/2},$$

and the last series converges because $0<q<1$, a geometric factor dominating any fixed power of k. The single remaining term $k=n$ is finite. This proves (6).

The linear rate β of (1), the uniform coefficient bound $|A_k| \le C_A$ that the energy hypothesis supplies, the geometric domination $\rho^{\lambda_{n+m}} \le (\rho^\beta)^m$ and the resulting convergence of the weighted tails at every polynomial weight are collected as `Thm2TermGeometricSummability`, which also records the constant value of T_n off the origin and the bound $|T_{n+m}(\xi)| \le 2|A_{n+m}||\xi|^{\lambda_{n+m}}$.

Step 3: locally uniform convergence of the series and all its derivatives off the origin. Fix four radii

$$0 < \delta' < \delta < \rho < \rho' < 1$$

and let $K = \{\xi : \delta \le |\xi| \le \rho\}$, a compact subset of the open annulus $\Omega' = \{\xi : \delta' < |\xi| < \rho'\}$, itself a subset of $\mathbb{B}-\{0\}$. We claim that for every multi-index ν of order $m = |\nu|$ in the Cartesian variables x, y there is a constant $C = C(m, \delta', \rho')$, independent of k, with

$$\sup_{\xi\in K} \big|\partial^\nu T_k(\xi)\big| \le C\big(1+\lambda_k+k+n\big)^m \cdot 2|A_k|\,(\rho')^{\lambda_k} \qquad (k \ge n). \tag{7}$$

An explicit form of the k-dependence in (7), together with the smoothness of each $\partial^\nu T_k$ off the origin, is `Thm2TermWordDerivBound`; that node proves both by the monomial calculus of the class $E(s,j,l)(\xi) = |\xi|^s \xi^j \bar\xi^{\,l}$ instead of by sectors, and either route serves what follows.

To see this, cover K by finitely many open sectors, each contained in Ω', on each of which a single branch of the argument is defined. On such a sector the map $\Psi(t,\theta) = e^t e^{i\theta}$ is a C^∞ diffeomorphism from an open subset of $(\log\delta', \log\rho') \times \mathbb{R}$ onto the sector, with inverse $\xi \mapsto (\log|\xi|, \arg\xi)$; all derivatives of Ψ and of Ψ^{-1} of order at most m are bounded there by a constant depending only on m, δ' and ρ', because $|\xi| \ge \delta' > 0$ on Ω'. By (4), $T_k \circ \Psi(t,\theta) = e^{\lambda_k t}\Phi_k(\theta)$, so for $a, b \ge 0$ and $e^t < \rho'$,

$$\big|\partial_t^a \partial_\theta^b \big(T_k\circ\Psi\big)(t,\theta)\big| = \lambda_k^a e^{\lambda_k t}\big|\Phi_k^{(b)}(\theta)\big| \le \lambda_k^a (\rho')^{\lambda_k}\cdot 2|A_k|(k+n)^b \le 2|A_k|(\rho')^{\lambda_k}\big(1+\lambda_k+k+n\big)^{a+b},$$

using (5), $\lambda_k \ge 0$ from (3), and $e^{\lambda_k t} \le (\rho')^{\lambda_k}$. The chain rule expresses $\partial^\nu T_k$ on each sector as a finite sum of terms, each a product of derivatives of Ψ^{-1} of order at most m — all bounded as just said — with a derivative $\partial_t^a\partial_\theta^b(T_k\circ\Psi)$ for some $a+b \le m$. Taking the maximum over the finitely many sectors gives (7).

By (6) applied with $m = |\nu|$ and ρ' in place of ρ, which is legitimate since $\rho' \in (0,1)$, the right-hand sides of (7) are summable over $k \ge n$. Hence for every ν the series $\sum_{k\ge n}\partial^\nu T_k$ converges uniformly on K. Each T_k is C^∞ on Ω' by Step 1. The conclusion of this paragraph — that the sum of the series is C^∞ on $\mathbb{B}-\{0\}$ — is `Thm2SeriesSmoothOffZero`, stated there for the series itself and transferred to H here by congruence, since H agrees with the sum off the origin. By the standard theorem on differentiation of uniformly convergent series — if a series of C^∞ functions on an open set converges uniformly on compact subsets together with each of its derived series, the sum is C^∞ and its derivatives are the sums of the derived series — the function

$$H(\xi) = \sum_{k\ge n} T_k(\xi) \tag{8}$$

is well defined and C^∞ on the interior of K, with $\partial^\nu H = \sum_k \partial^\nu T_k$ there, and $S_N \to H$ uniformly on K, as do $\partial^\nu S_N \to \partial^\nu H$ for every ν. Every point of $\mathbb{B}-\{0\}$ lies in the interior of such a K for a suitable choice of the four radii, so (8) defines a function C^∞ on $\mathbb{B}-\{0\}$, which is (b), and

$$S_N \to H, \quad (S_N)_\xi \to H_\xi, \quad (S_N)_{\bar\xi} \to H_{\bar\xi} \quad \text{uniformly on compact subsets of } \mathbb{B}-\{0\}, \tag{9}$$

since by DZ and DZbar the operators $\partial/\partial\xi$ and $\partial/\partial\bar{\xi}$ are fixed complex linear combinations of $\partial/\partial x$ and $\partial/\partial y$. We complete the definition of H by setting

$$H(0) = A_n, \qquad H(\xi) = 0 \quad \text{for } |\xi| \geq 1.$$

Step 4: proof of (a). By (4) at $k = n$ and (2), $T_n(re^{i\theta}) = r^0\big(A_n e^0 + 0\big) = A_n$, so T_n is the constant A_n on $\mathbb{C} - \{0\}$. Hence for $0 < |\xi| < 1$, by (8) and (5),

$$\big|H(\xi) - A_n\big| \leq \sum_{k\geq n+1} |T_k(\xi)| \leq 2 \sum_{k\geq n+1} |A_k|\,|\xi|^{\lambda_k} \leq 2 \sum_{k\geq n+1} |A_k|\,q^k, \qquad q = |\xi|^\beta \in (0,1),$$

using (1). By Cauchy–Schwarz and $\sum_{k\geq 1} q^{2k} = q^2/(1-q^2)$,

$$\sum_{k\geq n+1} |A_k| q^k \leq \Big(\sum_{k\geq n+1} k|A_k|^2\Big)^{1/2}\Big(\sum_{k\geq n+1} \frac{q^{2k}}{k}\Big)^{1/2} \leq E^{1/2}\Big(\frac{q^2}{1-q^2}\Big)^{1/2}.$$

As $\xi \to 0$ we have $q = |\xi|^\beta \to 0$, so the right-hand side tends to 0; hence $H(\xi) \to A_n = H(0)$ as $\xi \to 0$. Together with the continuity of H on $\mathbb{B} - \{0\}$ from (b), H is continuous on $\mathbb{B}$, which is (a).

This whole step, together with the continuity off the origin that Step 3 supplies, is Thm2SeriesContinuousOnD stated there for the sum of the series itself. It also records what this computation shows about the value at the origin: since T_n is the constant A_n and every later summand carries a positive power of $|\xi|$, the sum tends to A_n, so A_n is the only completing value for which H is continuous at 0.

Step 5: the polar form of the two Wirtinger derivatives of T_k. The Cartesian closed form of these same two derivatives, obtained instead by differentiating the product $(A_k\xi^k + \epsilon_k\overline{A_k}\bar{\xi}^k)|\xi|^{\lambda_k+n-k}\xi^{-n}$ factor by factor, is Thm2TermWirtingerDerivs; it holds for every $k \geq 0$ with no constraint on n, p or k, and either form serves what follows. In polar coordinates, let $\xi = re^{i\theta}$ with $r > 0$. By (4), the radial and angular derivatives of T_k at ξ, in the sense of HRad and HAng, are

$$(T_k)_r = \lambda_k r^{\lambda_k - 1}\Phi_k(\theta), \qquad (T_k)_\theta = r^{\lambda_k}\Phi_k'(\theta).$$

By WirtingerPolarForm, applied at $\xi \neq 0$ where T_k is differentiable by Step 1, and with $e^{i\theta} = \xi/|\xi|$,

$$(T_k)_\xi = \tfrac{1}{2}\Big((T_k)_r - \frac{i}{r}(T_k)_\theta\Big)e^{-i\theta} = \tfrac{1}{2}r^{\lambda_k-1}\big(\lambda_k\Phi_k(\theta) - i\Phi_k'(\theta)\big)e^{-i\theta}, \tag{10}$$

$$(T_k)_{\bar\xi} = \tfrac{1}{2}\Big((T_k)_r + \frac{i}{r}(T_k)_\theta\Big)e^{i\theta} = \tfrac{1}{2}r^{\lambda_k-1}\big(\lambda_k\Phi_k(\theta) + i\Phi_k'(\theta)\big)e^{i\theta}. \tag{11}$$

Substituting $\Phi_k'(\theta) = i(k-n)A_k e^{i(k-n)\theta} + i(-k-n)\epsilon_k\overline{A_k}e^{i(-k-n)\theta}$ and multiplying out the factors $e^{\mp i\theta}$,

$$(T_k)_\xi = \tfrac{1}{2}r^{\lambda_k-1}\Big(\big[\lambda_k + k - n\big]A_k e^{i(k-n-1)\theta} + \big[\lambda_k - k - n\big]\epsilon_k\overline{A_k}e^{i(-k-n-1)\theta}\Big), \tag{12}$$

$$(T_k)_{\bar\xi} = \tfrac{1}{2}r^{\lambda_k-1}\Big(\big[\lambda_k - k + n\big]A_k e^{i(k-n+1)\theta} + \big[\lambda_k + k + n\big]\epsilon_k\overline{A_k}e^{i(-k-n+1)\theta}\Big). \tag{13}$$

Identities (12) and (13) — the polar form of the two Wirtinger derivatives of a single summand of (19), which is the summand-level instance of the transformation formulas (4) — are Thm2TermPolarWirtingerDerivs, proved there by substituting $\xi = re^{i\theta}$ into the Cartesian closed form and using $|re^{i\theta}| = r$ and $\overline{e^{i\theta}} = e^{-i\theta}$ to collect the modulus power and the two angular frequencies. All four bracketed numbers have modulus at most $\lambda_k + k + n \leq c_1 k$ by (3). The resulting pointwise bound, in the form actually used below — namely $|(T_{n+m})_\xi|, |(T_{n+m})_{\bar\xi}| \leq (4 + \sqrt{p-1})(n+m)|A_{n+m}|\,|\xi|^{\lambda_{n+m}-1}$ for every $m \geq 0$ and every $\xi \neq 0$, with a constant independent of m — is Thm2TermDerivBound. At $k = n$ both derivatives vanish identically, since there $\lambda_n = 0$ and $\epsilon_n = 0$ give $\lambda_n + k - n = 0$ and $\epsilon_n = 0$ in (12), and $\lambda_n - k + n = 0$ and $\epsilon_n = 0$ in (13) — as it must be, T_n being constant.

Step 6: the three $L^2(\mathbb{B})$ bounds. Let $k \geq n$. In polar coordinates $d\sigma = r\,dr\,d\theta$; the identity $\int_{\mathbb{B}} |g|^2 d\sigma = \int_0^1 r \int_0^{2\pi} |g(re^{i\theta})|^2 d\theta\,dr$, for an arbitrary measurable g and with both sides read in $[0,\infty]$, is DiscEnergyPolarCoord, and it is used below at each of T_k, $(T_k)_\xi$, $(T_k)_{\bar\xi}$ and the partial

sums. The two exponentials in Φ_k have distinct frequencies $k-n$ and $-k-n$, their difference being $2k \neq 0$, so they are orthogonal on $(0, 2\pi)$; the finite-sum form of that orthogonality, for two frequency families each injective on the index set and never colliding with one another, is `TwoFamilyTrigParseval`, and it is what the partial-sum computation below uses. At a single k it gives

$$\int_0^{2\pi} |\Phi_k(\theta)|^2 d\theta = 2\pi\big(|A_k|^2 + \epsilon_k^2 |A_k|^2\big) \leq 4\pi |A_k|^2$$

by (3). The whole of this step – the passage to polar coordinates, the angular Parseval evaluation and the radial integral $\int_0^1 r^{2\lambda_k - 1} dr = 1/(2\lambda_k)$ – is packaged for a finite sum of polar-separated summands as `PolarSeparatedSumEnergy`, which is what the partial-sum estimate below applies. Hence, by (4),

$$\iint_{\mathbb{B}} |T_k|^2 d\sigma = \int_0^1 r^{2\lambda_k+1} \Big(\int_0^{2\pi} |\Phi_k|^2 d\theta \Big) dr \leq 4\pi |A_k|^2 \int_0^1 r^{2\lambda_k+1} dr = \frac{4\pi |A_k|^2}{2\lambda_k + 2} \leq 2\pi |A_k|^2,$$

the integral being finite because $2\lambda_k + 1 \geq 1 > -1$ by (3). Since $k \geq 1$, $|A_k|^2 \leq k|A_k|^2$, so

$$\iint_{\mathbb{B}} |T_k|^2 d\sigma \leq 2\pi\, k |A_k|^2 \leq c_2\, k |A_k|^2. \tag{14}$$

For the derivatives, take $k \geq n+1$ first, so that $\lambda_k > 0$ by (2). The two exponentials in (12) have distinct frequencies, their difference being $2k \neq 0$, hence are orthogonal, and $|e^{i(\cdot)}| = 1$; so by (12), (3),

$$\int_0^{2\pi} \big|(T_k)_\xi\big|^2 d\theta = \tfrac{1}{4} r^{2\lambda_k - 2} \cdot 2\pi \Big(\big[\lambda_k + k - n\big]^2 |A_k|^2 + \big[\lambda_k - k - n\big]^2 \epsilon_k^2 |A_k|^2 \Big) \leq \frac{\pi}{2} r^{2\lambda_k - 2} c_1^2 k^2 \cdot 2 |A_k|^2.$$

Multiplying by r and integrating, and using $\int_0^1 r^{2\lambda_k - 1} dr = 1/(2\lambda_k)$, finite because $\lambda_k > 0$,

$$\iint_{\mathbb{B}} \big|(T_k)_\xi\big|^2 d\sigma \leq \pi c_1^2 k^2 |A_k|^2 \cdot \frac{1}{2\lambda_k} \leq \frac{\pi c_1^2 (n+1)}{2\gamma_n}\, k |A_k|^2 \leq c_2\, k |A_k|^2,$$

where the middle step used $\lambda_k \geq \beta k = \gamma_n k/(n+1)$ from (1), so that $k^2/\lambda_k \leq k(n+1)/\gamma_n$. The identical computation applies to (13), the four bracketed numbers obeying the same bound. For $k = n$ both derivatives vanish by Step 5, so the bound holds trivially there. Therefore

$$\iint_{\mathbb{B}} \big|(T_k)_\xi\big|^2 d\sigma \leq c_2 k |A_k|^2, \qquad \iint_{\mathbb{B}} \big|(T_k)_{\bar\xi}\big|^2 d\sigma \leq c_2 k |A_k|^2 \qquad (k \geq n). \tag{15}$$

Step 7: orthogonality, and the Cauchy property. Let $k \neq k'$, both $\geq n$. The frequency set of Φ_k is $\{k-n, -k-n\}$. These sets are disjoint for $k \neq k'$: $k - n = k' - n$ and $-k - n = -k' - n$ both force $k = k'$, while $k - n = -k' - n$ forces $k = -k'$, impossible since $k, k' \geq n \geq 1$. Hence, by (4),

$$\iint_{\mathbb{B}} T_k \overline{T_{k'}}\, d\sigma = \int_0^1 r^{\lambda_k + \lambda_{k'} + 1} \Big(\int_0^{2\pi} \Phi_k \overline{\Phi_{k'}}\, d\theta \Big) dr = 0,$$

the inner integral vanishing term by term. The same argument applies to the derivatives: by (12) the frequency set of the angular profile of $(T_k)_\xi$ is $\{k-n-1, -k-n-1\}$, and $k-n-1 = -k'-n-1$ forces $k = -k'$, again impossible; by (13) the frequency set of $(T_k)_{\bar\xi}$ is $\{k-n+1, -k-n+1\}$, and $k - n + 1 = -k' - n + 1$ forces $k = -k'$. So $\{T_k\}_{k \geq n}$, $\{(T_k)_\xi\}_{k \geq n}$ and $\{(T_k)_{\bar\xi}\}_{k \geq n}$ are three mutually orthogonal families in $L^2(\mathbb{B})$.

Now let $M > N \geq 0$. Then $S_M - S_N = \sum_{k=n+N+1}^{n+M} T_k$, and on $\mathbb{B} - \{0\}$ the Wirtinger derivatives of a finite sum are the sums of the derivatives, by linearity of the Fréchet derivative and `DZ`, `DZbar`; the omitted point 0 is a Lebesgue null set and does not affect any integral. By orthogonality (Pythagoras) and (14), (15),

$$\iint_{\mathbb{B}} |S_M - S_N|^2 d\sigma = \sum_{k=n+N+1}^{n+M} \iint_{\mathbb{B}} |T_k|^2 d\sigma \leq c_2 R_N,$$

and likewise for $(S_M - S_N)_\xi$ and $(S_M - S_N)_{\bar\xi}$. Since $R_N \to 0$, the three sequences (S_N), $\big((S_N)_\xi\big)$, $\big((S_N)_{\bar\xi}\big)$ are Cauchy in $L^2(\mathbb{B})$. By completeness of $L^2(\mathbb{B})$ they converge in $L^2(\mathbb{B})$ to functions $G_0, G_1, G_2 \in L^2(\mathbb{B})$.

Step 8: identification of the limits, and proof of (d) and (g). An $L^2(\mathbb{B})$-convergent sequence has a subsequence converging almost everywhere. Applying this to each of the three sequences and comparing with (9), which gives pointwise convergence on $\mathbb{B} - \{0\}$, we get

$$G_0 = H, \qquad G_1 = H_\xi, \qquad G_2 = H_{\bar\xi} \qquad \text{almost everywhere on } \mathbb{B} - \{0\},$$

hence almost everywhere on $\mathbb{B}$, since $\{0\}$ is null. In particular H_ξ and $H_{\bar\xi}$ agree a.e. with elements of $L^2(\mathbb{B})$, so $|H_\xi|^2 + |H_{\bar\xi}|^2$ is integrable on $\mathbb{B}$, which is (d). And the three integrals in (g) are exactly $\|H - S_N\|_{L^2}^2 + \|H_\xi - (S_N)_\xi\|_{L^2}^2 + \|H_{\bar\xi} - (S_N)_{\bar\xi}\|_{L^2}^2$, each of which tends to 0 by Step 7; this is (g).

Step 9: proof of (c). By `DZ` and `DZbar`, for any function f real-differentiable at a point,

$$\frac{\partial f}{\partial x} = f_\xi + f_{\bar\xi}, \qquad \frac{\partial f}{\partial y} = i\big(f_\xi - f_{\bar\xi}\big),$$

since $f_\xi + f_{\bar\xi} = \frac{1}{2}(\partial_x - i\partial_y)f + \frac{1}{2}(\partial_x + i\partial_y)f = \partial_x f$ and $f_\xi - f_{\bar\xi} = -i\,\partial_y f$. Put

$$u_0 = G_1 + G_2, \qquad u_1 = i\big(G_1 - G_2\big),$$

both in $L^2(\mathbb{B})$, so that $\partial_x S_N \to u_0$ and $\partial_y S_N \to u_1$ in $L^2(\mathbb{B})$ by Step 7.

Let φ be a test function for $\mathbb{B}$ in the sense of `IsTestFn` and let $j \in \{0, 1\}$, with $\partial_0 = \partial/\partial x$ and $\partial_1 = \partial/\partial y$, matching the indexing of `IsWeakPartialOn`. Fix N. For $0 < \varepsilon < 1$ the function S_N is C^∞ on a neighbourhood of $\overline{\mathbb{B}} - \mathbb{B}_\varepsilon$ intersected with $\operatorname{supp}\varphi$, so the divergence theorem on $\mathbb{B} - \mathbb{B}_\varepsilon$ gives

$$\iint_{\mathbb{B}-\mathbb{B}_\varepsilon} S_N\,\partial_j\varphi\,d\sigma = -\iint_{\mathbb{B}-\mathbb{B}_\varepsilon} (\partial_j S_N)\varphi\,d\sigma + \oint_{|\xi|=\varepsilon} S_N\varphi\,\nu_j\,ds,$$

with ν the outward unit normal. By (5) and (3), $\lambda_k \geq 0$, so $|S_N(\xi)| \leq 2\sum_{k=n}^{n+N}|A_k| =: M_N$ for $0 < |\xi| < 1$; hence the boundary term is at most $2\pi\varepsilon M_N \sup|\varphi|$, which tends to 0 as $\varepsilon \to 0$. Moreover S_N and $\partial_j S_N$ lie in $L^2(\mathbb{B}) \subseteq L^1(\mathbb{B})$, $\mathbb{B}$ having finite measure, so the two area integrals converge to the corresponding integrals over $\mathbb{B}$. Letting $\varepsilon \to 0$,

$$\iint_{\mathbb{B}} S_N\,\partial_j\varphi\,d\sigma = -\iint_{\mathbb{B}} (\partial_j S_N)\varphi\,d\sigma.$$

Now let $N \to \infty$. Since φ and $\partial_j\varphi$ are bounded with compact support, they lie in $L^2(\mathbb{B})$, so by the Cauchy–Schwarz inequality both sides converge, giving

$$\iint_{\mathbb{B}} H\,\partial_j\varphi\,d\sigma = -\iint_{\mathbb{B}} u_j\varphi\,d\sigma,$$

where on the left we used $S_N \to H$ in $L^2(\mathbb{B})$ from Step 8. As φ was arbitrary, u_j is the distributional j-th partial derivative of H on $\mathbb{B}$ in the sense of `IsWeakPartialOn`. Finally define $D[\,] = H$, $D[j] = u_j$ for $j \in \{0, 1\}$ and $D(w) = 0$ for words w of length at least 2. Then $D[\,] = H$; every $D(w)$ with $|w| \leq 1$ lies in $L^2(\mathbb{B})$, hence in $L^2_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemLpLocOn`, because the indicator of a compact subset of $\mathbb{B}$ times an $L^2(\mathbb{B})$ function is in L^2; and the only words of length < 1 are the empty word, for which the weak-derivative requirement is what we just proved. So $H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, which is (c).

Step 10: proof of (e) and (f). Fix $r \in (0, 1)$ and $j \in \mathbb{Z}$. Choose radii as in Step 3 with $\delta < r < \rho$; the circle $|\xi| = r$ is contained in the corresponding K, so by Step 3 the convergence $S_N \to H$ is uniform on it. Uniform convergence permits passing to the limit under the integral in `FourierCoefOfH`, so

$$a_j(r) = \frac{1}{2\pi}\int_0^{2\pi} H\big(re^{i\theta}\big)e^{i(n-j)\theta}d\theta = \lim_{N\to\infty}\sum_{k=n}^{n+N} r^{\lambda_k}\cdot\frac{1}{2\pi}\int_0^{2\pi}\Phi_k(\theta)e^{i(n-j)\theta}d\theta,$$

using (4). Now, for any integer ℓ, $\frac{1}{2\pi}\int_0^{2\pi} e^{i\ell\theta}d\theta$ equals 1 if $\ell = 0$ and 0 otherwise; applying this to $\ell = k - j$ and $\ell = -k - j$ gives

$$\frac{1}{2\pi}\int_0^{2\pi} \Phi_k(\theta)e^{i(n-j)\theta}d\theta = \begin{cases} A_k & \text{if } k = j, \\ \epsilon_k\overline{A_k} & \text{if } k = -j, \\ 0 & \text{otherwise,} \end{cases}$$

the two cases being mutually exclusive because $k = j = -k$ would force $k = 0$, while $k \geq n \geq 1$. We read off the three regimes.

Both halves of this computation — the polar form of one summand tested against $e^{i(n-j)\theta}$, and the term-by-term integration that leaves only the indices $k = j$ and $k = -j$ — are `Thm2SeriesCircleFourierCoef`, stated there for the series itself; the interchange of sum and integral is justified there by the geometric domination of `Thm2TermGeometricSummability`, which bounds $\|T_k(re^{i\theta})\|$ uniformly in θ by a summable sequence, in place of the uniform convergence used above.

If $|j| < n$: no $k \geq n$ satisfies $k = j$ or $k = -j$, since $|j| < n \leq k$. So every term vanishes and $a_j(r) = 0$. This is (f).

If $j \geq n$: the only contributing index is $k = j$ (the alternative $k = -j$ would need $-j \geq n \geq 1$, i.e. $j \leq -1$). Hence $a_j(r) = A_j r^{\lambda_j}$.

If $j \leq -n$: the only contributing index is $k = -j \geq n$, through the second case, so $a_j(r) = \epsilon_{-j}\overline{A_{-j}}\, r^{\lambda_{-j}}$. By hypothesis (C) applied at the index $-j \geq n$, we have $A_j = A_{-(-j)} = \epsilon_{-j}\overline{A_{-j}}$; and $\lambda_{-j} = \lambda_j$ because λ depends only on the absolute value of its index. Hence $a_j(r) = A_j r^{\lambda_j}$.

The last two regimes together are exactly (e), and the proof is complete. □

2.45. `SeriesTermwiseDeriv`.

Lemma 2.45. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, write $\mathbb{B} = \{\xi : |\xi| < 1\}$, let $\lambda_k = \lambda_{|k|}(n,p)$ be the exponent of `LambdaCoef`, $\epsilon_k = \epsilon_k(n,p)$ the coefficient of `EpsCoef`, $\gamma_n = \lambda_{n+1}(n,p)$ the exponent of `GammaExp`, a_k the Fourier coefficient of `FourierCoefOfH`, T_k the k-th summand of (19) in the sense of `Thm2Term`, and D^w the iterated partial derivative of `PartialWord`. Let $H\colon \mathbb{C} \to \mathbb{C}$ and $A\colon \mathbb{Z} \to \mathbb{C}$ satisfy*

- (H1) *H is continuous on $\mathbb{B}$, lies in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, and is differentiable at almost every point of $\mathbb{B}$;*
- (H2) *H solves (18) on $\mathbb{B}$ in the sense of `SolvesEq18`;*
- (H3) *$a_k(r) = A_k\, r^{\lambda_k}$ for every $k \in \mathbb{Z}$ with $|k| \geq n$ and every $r \in (0,1)$;*
- (H4) *$A_{-k} = \epsilon_k\overline{A_k}$ for every integer $k \geq n$;*
- (H5) *$\sum_{m\geq 0}(n+m)|A_{n+m}|^2 < \infty$, that is $\sum_{k\geq n} k|A_k|^2 < \infty$.*

Then:

(a) *for every ξ with $0 < |\xi| < 1$ the series (19) converges and*

$$H(\xi) = \sum_{m=0}^{\infty} T_{n+m}(\xi);$$

(b) *for every $\rho \in (0,1)$ and every integer $j \geq n$ there is $C \geq 0$ with*

$$\Big|H(\xi) - \sum_{k=n}^{j} T_k(\xi)\Big| \leq C\,|\xi|^{\lambda_{j+1}} \qquad \text{for } 0 < |\xi| \leq \rho;$$

(c) *for every $\rho \in (0,1)$ and every integer $j \geq n$ there is $C \geq 0$ with*

$$\Big|H_r(\xi) - \sum_{k=n}^{j} \big(T_k\big)_r(\xi)\Big| \leq C\,|\xi|^{\lambda_{j+1}-1} \qquad \text{for } 0 < |\xi| \leq \rho,$$

where $(\cdot)_r$ is the radial derivative of `HRad`;

(d) *if moreover* $A_n = 0$, *then for every integer* $m \geq 0$ *and every* $\rho \in (0,1)$ *there is* $C_m \geq 0$ *with*

$$\sum_{|w|=m} \left|D^w H(\xi)\right| \leq C_m\, |\xi|^{\gamma_n - m} \qquad \text{for } 0 < |\xi| \leq \rho,$$

the sum running over all 2^m *words* w *of length* m *in* $\{0,1\}$;

(e) H *is of class* C^∞ *on* $\mathbb{B} - \{0\}$.

Clause (e) is where the smoothness of H *away from the origin is* produced, *and it is the paper's own step: paper.tex:387–388 notes that* $\lambda_k \to \infty$ *and concludes "therefore, we can perform any finite number of term-by-term differentiations in formula (19) for* $0 < |\xi| < 1$*". No smoothness is assumed in (H1) — only almost-everywhere differentiability, which is what item (C2) of the paper's note grants for the inverse of a quasiconformal homeomorphism and which (H1) needs because the hypotheses of the Section-4 nodes are stated with it. Clause (d) is estimate (43) of Corollary 1; the words of length* m *realize exactly the operators* D^ν *with* $|\nu| = m$, *as* `PartialWord` *records, so the sum over words is the paper's* $\sum_{|\nu|=m}$ *up to the multiplicities with which each* D^ν *occurs, which are at most* 2^m *and are absorbed into* C_m. *Clause (a) is the pointwise form of (19); clauses (b) and (c) are the tail estimates with which the lower bounds in (41) and (42) are obtained.*

Proof. Throughout, k and j denote integers, $\lambda_{-k} = \lambda_k$, and $\epsilon_k \in \mathbb{R}$ by `EpsCoef`. Write

$$E = \sum_{k \geq n} k|A_k|^2 < \infty$$

by (H5); since $k \geq n \geq 1$ we have $|A_k|^2 \leq k|A_k|^2$, so also $\sum_{k\geq n} |A_k|^2 \leq E$.

Step 0: the exponents. By part (a) of `GammaRatioMinAtOne`, $\gamma_n > 0$. Put $\beta = \gamma_n/(n+1) > 0$. By `LambdaLinearLowerBound`,

$$\lambda_k \geq \beta k \qquad (k \geq n+1), \tag{1}$$

and by parts (23), (24), (25) of `LambdaEpsEstimates`, $\lambda_n = 0$, $\epsilon_n = 0$, $|\epsilon_k| < 1$ and $\lambda_k \leq k\sqrt{p-1}$ for $k \geq n+1$. Hence, for every $k \geq n$,

$$0 \leq \lambda_k \leq k\sqrt{p-1}, \qquad |\epsilon_k| \leq 1, \tag{2}$$

the case $k = n$ holding because $\lambda_n = 0$ and $\epsilon_n = 0$.

Also λ_k is *strictly increasing* in $k \geq 0$: by `LambdaCoef`, $2\lambda_k + np = \sqrt{4k^2(p-1) + n^2(p-2)^2}$, the radicand is strictly increasing in $k \geq 0$ because $p - 1 > 0$ and $(k+1)^2 > k^2$, and the square root is strictly increasing on $[0, \infty)$. Therefore

$$\lambda_k < \lambda_{k+1} \quad (k \geq 0), \qquad \text{so} \qquad \lambda_k \geq \lambda_{j+1} \ \text{ for } k \geq j+1. \tag{3}$$

Step 1: the summands. By part (b) of `Thm2TermSolvesEq18`, for every $k \geq n$, every $r > 0$ and every θ,

$$T_k\big(re^{i\theta}\big) = r^{\lambda_k}\Phi_k(\theta), \qquad \Phi_k(\theta) = A_k e^{i(k-n)\theta} + \epsilon_k \overline{A_k} e^{i(-k-n)\theta}, \tag{4}$$

and by its part (a) each T_k is C^∞ at every point $\xi \neq 0$. From (4) and (2),

$$\left|T_k(\xi)\right| \leq 2|A_k|\,|\xi|^{\lambda_k}, \qquad \left|\Phi_k^{(b)}(\theta)\right| \leq 2|A_k|(k+n)^b \quad (b \geq 0), \tag{5}$$

the second because each differentiation multiplies the two terms of Φ_k by $i(k-n)$ and $i(-k-n)$ respectively, of modulus at most $k+n$.

Step 2: the master summability estimate. We claim that for every integer $N \geq 0$ and every $\rho \in (0,1)$,

$$\Sigma_N(\rho) := \sum_{k \geq n} (1+k)^N |A_k|\, \rho^{\lambda_k} < \infty. \tag{6}$$

Indeed, by (1), for $k \geq n+1$ we have $\lambda_k \geq \beta k$ and hence $\rho^{\lambda_k} \leq \rho^{\beta k} = q^k$, where

$$q = \rho^\beta \in (0,1)$$

because $0 < \rho < 1$ and $\beta > 0$. By the Cauchy–Schwarz inequality applied to the sequences $\big(k^{1/2}|A_k|\big)_{k\ge n+1}$ and $\big((1+k)^N q^k k^{-1/2}\big)_{k\ge n+1}$,

$$\sum_{k\ge n+1}(1+k)^N|A_k|\rho^{\lambda_k} \le \Big(\sum_{k\ge n+1} k|A_k|^2\Big)^{1/2}\Big(\sum_{k\ge n+1}\frac{(1+k)^{2N}q^{2k}}{k}\Big)^{1/2} \le E^{1/2}\Big(\sum_{k\ge 1}(1+k)^{2N}q^{2k}\Big)^{1/2},$$

and the last series converges because $0 < q^2 < 1$ and a geometric series dominates any fixed power of k (the standard fact that $\sum_{k\ge1} k^M x^k < \infty$ for $0 \le x < 1$ and every M). Adding the single term $k = n$, which is finite, gives (6). This is the paper's Cauchy–Schwarz split at paper.tex:391, and it is here that the *linear* lower bound (1) is used: the constant bound $\lambda_k \ge (p-1)/(np)$ of (25) would leave ρ^{λ_k} bounded below by a positive constant and (6) would fail.

The linear rate β of (1), the uniform coefficient bound $|A_k| \le C_A$ that the energy hypothesis supplies, the geometric domination $\rho^{\lambda_{n+m}} \le (\rho^\beta)^m$ and the resulting convergence of the weighted tails at every polynomial weight are collected as `Thm2TermGeometricSummability`, which also records the constant value of T_n off the origin and the bound $|T_{n+m}(\xi)| \le 2|A_{n+m}||\xi|^{\lambda_{n+m}}$.

Step 3: homogeneity of the summands and their derivatives. Fix $k \ge n$ and a word w of length m. By (4), for $u > 0$ and $\xi \ne 0$, writing $\xi = re^{i\theta}$,

$$T_k(u\xi) = (ur)^{\lambda_k}\Phi_k(\theta) = u^{\lambda_k}T_k(\xi), \tag{7}$$

so T_k is homogeneous of degree λ_k. Next, for any function F that is C^∞ on $\mathbb{C} - \{0\}$ and any $u > 0$, an induction on w using `PartialWord` and the chain rule for the $\mathbb{R}$-linear map $\xi \mapsto u\xi$, whose Fréchet derivative is that same map, gives

$$D^w\big(\xi \mapsto F(u\xi)\big)(\xi) = u^{|w|}\big(D^wF\big)(u\xi)$$

— the base case w empty being trivial, and the inductive step being $\partial_j\big(u^{|w'|}(D^{w'}F)\circ(u\cdot)\big)(\xi) = u^{|w'|}\cdot u\cdot\partial_j(D^{w'}F)(u\xi)$, by $\mathbb{R}$-linearity of the derivative in its argument. Applying D^w in ξ to both sides of (7) therefore yields

$$u^m\big(D^wT_k\big)(u\xi) = u^{\lambda_k}\big(D^wT_k\big)(\xi),$$

that is

$$\big(D^wT_k\big)(u\xi) = u^{\lambda_k-m}\big(D^wT_k\big)(\xi) \qquad (u>0,\ \xi\ne0). \tag{8}$$

Taking $u = 1/|\xi|$ and rearranging,

$$\big|D^wT_k(\xi)\big| = |\xi|^{\lambda_k-m}\,\big|D^wT_k\big(\xi/|\xi|\big)\big| \qquad (\xi\ne0). \tag{9}$$

Step 4: the bound on the unit circle. We claim that for every $m \ge 0$ there is a constant $C_0(m)$, depending only on m, with

$$\sup_{|\eta|=1}\big|D^wT_k(\eta)\big| \le C_0(m)\big(1+\lambda_k+k+n\big)^m\cdot 2|A_k| \tag{10}$$

for every word w of length m and every $k \ge n$.

Cover the unit circle by finitely many open sectors, each contained in the annulus $\Omega_0 = \{\frac12 < |\eta| < 2\}$, on each of which a single branch of the argument is defined. On such a sector the map $\Psi(t,\theta) = e^te^{i\theta}$ is a C^∞ diffeomorphism from an open subset of $(\log\frac12, \log 2)\times\mathbb{R}$ onto the sector, with inverse $\eta \mapsto (\log|\eta|, \arg\eta)$; all derivatives of Ψ and of Ψ^{-1} of order at most m are bounded on the relevant compact sets by a constant depending only on m, since $|\eta| \ge \frac12$ on Ω_0. By (4), $T_k\circ\Psi(t,\theta) = e^{\lambda_k t}\Phi_k(\theta)$, so at the points of the unit circle, where $t = 0$ and $e^{\lambda_k t} = 1$,

$$\big|\partial_t^a\partial_\theta^b\big(T_k\circ\Psi\big)(0,\theta)\big| = \lambda_k^a\big|\Phi_k^{(b)}(\theta)\big| \le \lambda_k^a\cdot 2|A_k|(k+n)^b \le 2|A_k|\big(1+\lambda_k+k+n\big)^{a+b}$$

by (5). The chain rule expresses D^wT_k on each sector as a finite sum of terms, each a product of derivatives of Ψ^{-1} of order at most m, all bounded as just said, with a derivative $\partial_t^a\partial_\theta^b(T_k\circ\Psi)$ for some $a + b \le m$. Taking the maximum over the finitely many sectors gives (10).

Combining (9), (10) and (2), and using $n \le k$ and $1 \le k$, so that $1 + \lambda_k + k + n \le \big(3 + \sqrt{p-1}\big)(1+k)$, we obtain a constant $C_1(m)$ depending only on m, n, p with

$$\big|D^w T_k(\xi)\big| \le C_1(m)(1+k)^m |A_k|\,|\xi|^{\lambda_k - m} \qquad (\xi \neq 0,\ |w| = m,\ k \ge n). \tag{11}$$

An explicit form of (11), with the constant $C_1(m)$ named and with the accompanying fact that each $D^w T_k$ is C^∞ off the origin, is `Thm2TermWordDerivBound`; that node proves both by the monomial calculus of the class $E(s,j,l)(\xi) = |\xi|^s \xi^j \bar{\xi}^l$ instead of by sectors, and either route serves what follows.

Step 5: proof of (a). Fix $r \in (0,1)$. By (5) and (6) with $N = 0$ and $\rho = r$, the series $\sum_{k\ge n} T_k(re^{i\theta}) = \sum_{k\ge n} r^{\lambda_k}\Phi_k(\theta)$ converges absolutely, and uniformly in θ, since its terms are bounded in modulus by $2|A_k|r^{\lambda_k}$, whose sum is $\Sigma_0(r) < \infty$. Let $F_r(\theta)$ be its sum; F_r is continuous and 2π-periodic, being a uniform limit of such functions.

We compute the Fourier coefficients of F_r. Uniform convergence permits term-by-term integration, so for $j \in \mathbb{Z}$

$$\frac{1}{2\pi}\int_0^{2\pi} F_r(\theta) e^{i(n-j)\theta} d\theta = \sum_{k\ge n} r^{\lambda_k} \cdot \frac{1}{2\pi}\int_0^{2\pi} \Phi_k(\theta) e^{i(n-j)\theta} d\theta.$$

Since $\frac{1}{2\pi}\int_0^{2\pi} e^{i\ell\theta} d\theta$ is 1 for $\ell = 0$ and 0 otherwise, applying this with $\ell = k - j$ and $\ell = -k - j$ gives

$$\frac{1}{2\pi}\int_0^{2\pi} \Phi_k(\theta) e^{i(n-j)\theta} d\theta = \begin{cases} A_k & k = j, \\ \epsilon_k \overline{A_k} & k = -j, \\ 0 & \text{otherwise}, \end{cases}$$

the first two cases being mutually exclusive because $k = j = -k$ would force $k = 0$, while $k \ge n \ge 1$. Reading off the three regimes: if $|j| < n$, no $k \ge n$ satisfies $k = j$ or $k = -j$, so the j-th coefficient of F_r is 0; if $j \ge n$ it is $A_j r^{\lambda_j}$; and if $j \le -n$ it is $\epsilon_{-j}\overline{A_{-j}} r^{\lambda_{-j}}$, which by (H4) at the index $-j \ge n$ and $\lambda_{-j} = \lambda_j$ equals $A_j r^{\lambda_j}$. So

$$\text{the } j\text{-th Fourier coefficient of } F_r \text{ is } A_j r^{\lambda_j}\ (|j| \ge n) \quad \text{and} \quad 0\ (|j| < n). \tag{12}$$

Now the Fourier coefficients of $\theta \mapsto H(re^{i\theta})$. By `FourierCoefOfH` the j-th one is $a_j(r)$. For $|j| \ge n$, (H3) gives $a_j(r) = A_j r^{\lambda_j}$. For $|j| < n$: the hypotheses of `FourierCoefOde` are met — $n \ge 1$, $1 < p < \infty$, H continuous on $\mathbb{B}$, in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ and differentiable at almost every point of $\mathbb{B}$, all three by (H1), and H a solution of (18) by (H2) — so its second clause gives, for this j and every $r \in (0,1)$,

$$r\big(r\,a_j'\big)'(r) + pn\,r\,a_j'(r) + (n^2 - j^2)(p-1)\,a_j(r) = 0,$$

the Euler equation (35); and a_j is continuous on $[0,1)$ by part (a) of `FourierCoefRegularity` and C^∞ on $(0,1)$ by the third clause of `FourierCoefOde`. These are exactly the hypotheses of `FourierCoefForm36` at the index j, whose first case ($|j| < n$) gives $a_j \equiv 0$ on $[0,1)$.

So $\theta \mapsto H(re^{i\theta})$ and F_r are continuous 2π-periodic functions with the same Fourier coefficients, hence are equal, by the uniqueness theorem for Fourier coefficients of continuous periodic functions. Given ξ with $0 < |\xi| < 1$, write $\xi = re^{i\theta}$ with $r = |\xi| \in (0,1)$; then $H(\xi) = F_r(\theta) = \sum_{k\ge n} T_k(\xi)$. The convergence is *absolute*: by (5) and (6) with $N = 0$ and $\rho = r$, $\sum_{k\ge n} |T_k(\xi)| \le \sum_{k\ge n} 2|A_k| r^{\lambda_k} = \Sigma_0(r) < \infty$. An absolutely convergent series of complex numbers is unconditionally summable, with the same sum, so the family $\big(T_{n+m}(\xi)\big)_{m\ge 0}$ is summable with sum $H(\xi)$; reindexing $k = n + m$, this is (a) in the unconditional form in which it is stated.

Step 6: term-by-term differentiation. Let $0 < \delta \le \rho < 1$ and let w be a word of length m. For $\delta \le |\xi| \le \rho$ we have, by (11) and (2),

$$\big|D^w T_k(\xi)\big| \le C_1(m)(1+k)^m |A_k|\,|\xi|^{\lambda_k}|\xi|^{-m} \le C_1(m)\delta^{-m}(1+k)^m |A_k|\,\rho^{\lambda_k},$$

using $|\xi|^{\lambda_k} \le \rho^{\lambda_k}$ (valid since $\lambda_k \ge 0$ and $|\xi| \le \rho$) and $|\xi|^{-m} \le \delta^{-m}$. By (6) with $N = m$ these bounds are summable over $k \ge n$. Hence, for every word w, the series $\sum_{k\ge n} D^w T_k$ converges

uniformly on each closed annulus $\{\delta \le |\xi| \le \rho\} \subseteq \mathbb{B} - \{0\}$; each T_k is C^∞ there by Step 1; and by Step 5 the series with w empty converges to H there. By the standard theorem on differentiation of uniformly convergent series — if a series of C^∞ functions on an open set converges uniformly on compact subsets together with each of its derived series, then the sum is C^∞ and its derivatives are the sums of the derived series — we conclude

$$D^w H(\xi) = \sum_{k \ge n} D^w T_k(\xi) \qquad (0 < |\xi| < 1), \tag{13}$$

for every word w, since every point of $\mathbb{B} - \{0\}$ lies in the interior of such an annulus for a suitable choice of δ and ρ. The same theorem gives that the sum is of class C^∞ on the open set $\mathbb{B} - \{0\}$, and by Step 5 that sum is H there; smoothness on an open set being a local property, and every point of $\mathbb{B} - \{0\}$ lying in the interior of such an annulus, we conclude

$$H \text{ is of class } C^\infty \text{ on } \mathbb{B} - \{0\}, \tag{13$'$}$$

which is (e). Note that this is the only route to the derivatives $D^w H$ appearing in (c) and (d): no smoothness of H is assumed.

Step 7: proof of (b). Let $\rho \in (0,1)$ and $j \ge n$. For $0 < |\xi| \le \rho$, by (a) and (5),

$$\Big| H(\xi) - \sum_{k=n}^{j} T_k(\xi) \Big| = \Big| \sum_{k \ge j+1} T_k(\xi) \Big| \le \sum_{k \ge j+1} 2|A_k|\,|\xi|^{\lambda_k}.$$

For $k \ge j+1$ we have $\lambda_k - \lambda_{j+1} \ge 0$ by (3), and $|\xi| \le \rho$, so $|\xi|^{\lambda_k} = |\xi|^{\lambda_{j+1}} |\xi|^{\lambda_k - \lambda_{j+1}} \le |\xi|^{\lambda_{j+1}} \rho^{\lambda_k - \lambda_{j+1}}$. Hence

$$\Big| H(\xi) - \sum_{k=n}^{j} T_k(\xi) \Big| \le |\xi|^{\lambda_{j+1}} \cdot 2\rho^{-\lambda_{j+1}} \sum_{k \ge j+1} |A_k| \rho^{\lambda_k} = C\,|\xi|^{\lambda_{j+1}}$$

with $C = 2\rho^{-\lambda_{j+1}} \sum_{k \ge j+1} |A_k| \rho^{\lambda_k}$, which is finite by (6) with $N = 0$ and nonnegative. This is (b).

Step 8: proof of (c). First the radial derivative of a summand. By `HRad`, $f_r(\xi)$ is the derivative of f at ξ in the direction $\xi/|\xi|$, which at $\xi = re^{i\theta}$ with $r > 0$ is $\partial\big(f(re^{i\theta})\big)/\partial r$. By (4) the function $r \mapsto T_k(re^{i\theta}) = r^{\lambda_k} \Phi_k(\theta)$ is differentiable on $(0,\infty)$ with derivative $\lambda_k r^{\lambda_k - 1} \Phi_k(\theta)$, so

$$\big(T_k\big)_r\big(re^{i\theta}\big) = \lambda_k\, r^{\lambda_k - 1} \Phi_k(\theta), \qquad \big|\big(T_k\big)_r(\xi)\big| \le 2\lambda_k |A_k|\,|\xi|^{\lambda_k - 1}, \tag{14}$$

the bound by (5). This is the expansion of H_r displayed at paper.tex:396.

Next, $H_r = \sum_{k \ge n} (T_k)_r$ on $\mathbb{B} - \{0\}$. Indeed, writing $\xi = |\xi| e^{i\theta}$ we have $\xi/|\xi| = \cos\theta + i \sin\theta$, so by $\mathbb{R}$-linearity of the Fréchet derivative and `HRad`,

$$f_r(\xi) = \cos\theta \cdot D^{[0]} f(\xi) + \sin\theta \cdot D^{[1]} f(\xi)$$

for every f differentiable at ξ, the words [0] and [1] giving the derivatives in the directions 1 and i. Applying (13) with $w = [0]$ and $w = [1]$ and forming this fixed linear combination gives

$$H_r(\xi) = \sum_{k \ge n} \big(T_k\big)_r(\xi) \qquad (0 < |\xi| < 1). \tag{15}$$

The series converges absolutely: on a closed annulus $\{\delta \le |\xi| \le \rho\} \subseteq \mathbb{B} - \{0\}$ the bound (14) together with $\lambda_k \le \sqrt{p-1}\,(1+k)$ from (2) and $|\xi|^{\lambda_k - 1} \le \delta^{-1} \rho^{\lambda_k}$ gives majorants $2\sqrt{p-1}\,\delta^{-1}(1+k)|A_k|\rho^{\lambda_k}$, summable by (6) with $N = 1$.

Now let $\rho \in (0,1)$ and $j \ge n$. For $0 < |\xi| \le \rho$, by (15) and (14),

$$\Big| H_r(\xi) - \sum_{k=n}^{j} \big(T_k\big)_r(\xi) \Big| \le \sum_{k \ge j+1} 2\lambda_k |A_k|\,|\xi|^{\lambda_k - 1}.$$

As in Step 7, $|\xi|^{\lambda_k-1} = |\xi|^{\lambda_{j+1}-1}|\xi|^{\lambda_k-\lambda_{j+1}} \le |\xi|^{\lambda_{j+1}-1}\rho^{\lambda_k-\lambda_{j+1}}$, using $\lambda_k - \lambda_{j+1} \ge 0$ from (3). And $\lambda_k \le k\sqrt{p-1} \le \sqrt{p-1}\,(1+k)$ by (2). Hence

$$\Big|H_r(\xi) - \sum_{k=n}^{j}\big(T_k\big)_r(\xi)\Big| \le C\,|\xi|^{\lambda_{j+1}-1}, \qquad C = 2\sqrt{p-1}\,\rho^{-\lambda_{j+1}}\sum_{k\ge j+1}(1+k)|A_k|\rho^{\lambda_k},$$

finite by (6) with $N = 1$ and nonnegative. This is (c).

Step 9: proof of (d). Assume $A_n = 0$. Then T_n vanishes identically: by `Thm2Term` its two terms carry the factors A_n and $\overline{A_n}$, both zero. So in (13) the sum may be taken over $k \ge n+1$.

Let $m \ge 0$, let $\rho \in (0,1)$ and let w be a word of length m. For $k \ge n+1$ we have $\lambda_k \ge \lambda_{n+1} = \gamma_n$ by (3), so for $0 < |\xi| \le \rho$,

$$|\xi|^{\lambda_k-m} = |\xi|^{\gamma_n-m}\,|\xi|^{\lambda_k-\gamma_n} \le |\xi|^{\gamma_n-m}\rho^{\lambda_k-\gamma_n}.$$

Hence by (13) (summed over $k \ge n+1$) and (11),

$$\big|D^w H(\xi)\big| \le \sum_{k\ge n+1}\big|D^w T_k(\xi)\big| \le |\xi|^{\gamma_n-m}\,C_1(m)\rho^{-\gamma_n}\sum_{k\ge n+1}(1+k)^m|A_k|\rho^{\lambda_k} = C_w\,|\xi|^{\gamma_n-m},$$

where $C_w = C_1(m)\rho^{-\gamma_n}\Sigma_m(\rho)$ is finite by (6) with $N = m$ and nonnegative; note C_w depends only on m, n, p, ρ and A, not on w. Summing over the 2^m words of length m,

$$\sum_{|w|=m}\big|D^w H(\xi)\big| \le 2^m C_w\,|\xi|^{\gamma_n-m} \qquad (0 < |\xi| \le \rho),$$

which is (d) with $C_m = 2^m C_w$. This is the paper's chain of estimates at paper.tex:388–391, the final Cauchy–Schwarz factor $\big[\sum k^{2m-1}\rho^{2\lambda_k-2\lambda_{n+1}}\big]^{1/2}$ there being exactly the second factor produced by (6). □

2.46. `WindingCalculus`.

Lemma 2.46. *Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and let $q\colon \mathbb{R} \to \mathbb{T}$ be the quotient map. Let $\gamma\colon [0,1] \to \mathbb{C}$ be continuous with $\gamma(t) \ne 0$ for every t, and let $\operatorname{ind}(\gamma)$ be its index about 0 in the sense of `WindingNumber`. Then:*

(a) (Existence of a continuous branch.) *There is a continuous $\Theta\colon [0,1] \to \mathbb{R}$ with*

$$\Theta(0) = \arg\gamma(0), \qquad q\big(\Theta(t)\big) = q\big(\arg\gamma(t)\big) \quad \text{for all } t,$$

and consequently

$$\gamma(t) = |\gamma(t)|\,e^{i\Theta(t)} \qquad \text{for all } t.$$

(b) (Characterisation of the index.) *Conversely, for* every *continuous $\Theta\colon [0,1] \to \mathbb{R}$ satisfying $\Theta(0) = \arg\gamma(0)$ and $q(\Theta(t)) = q(\arg\gamma(t))$ for all t,*

$$2\pi\operatorname{ind}(\gamma) = \Theta(1) - \Theta(0).$$

(c) (Integrality.) *If γ is a loop, i.e. $\gamma(0) = \gamma(1)$, then $\operatorname{ind}(\gamma)$ is an integer.*

(d) (Model circles.) *If $w \ne 0$, $k \in \mathbb{Z}$ and $\gamma(t) = w\,e^{2\pi i k t}$ for all $t \in [0,1]$, then $\operatorname{ind}(\gamma) = k$.*

Proof. We first recall the data through which `WindingNumber` defines the index. The map q is a covering map. Since $\gamma(t) \ne 0$ for every t and the reduction $z \mapsto q(\arg z)$ of the principal argument is continuous at every nonzero z, the composite

$$\alpha\colon [0,1] \to \mathbb{T}, \qquad \alpha(t) = q\big(\arg\gamma(t)\big),$$

is continuous. By the path-lifting property of the covering q there is a continuous $\Theta_\gamma\colon [0,1] \to \mathbb{R}$, unique with these two properties, such that

$$q\big(\Theta_\gamma(t)\big) = \alpha(t) = q\big(\arg\gamma(t)\big) \quad \text{for all } t, \qquad \Theta_\gamma(0) = \arg\gamma(0), \tag{1}$$

and `WindingNumber` defines

$$\operatorname{ind}(\gamma) = \frac{\Theta_\gamma(1) - \arg\gamma(0)}{2\pi} = \frac{\Theta_\gamma(1) - \Theta_\gamma(0)}{2\pi}. \tag{2}$$

Uniqueness in (1) is the standard uniqueness of lifts along a covering map: two lifts of α that agree at one point of the connected space $[0,1]$ agree everywhere.

We shall use twice the elementary description of the kernel of q: for $x, y \in \mathbb{R}$,

$$q(x) = q(y) \iff x - y = 2\pi j \quad \text{for some } j \in \mathbb{Z}, \tag{3}$$

and that q is a homomorphism of additive groups, so $q(x+y) = q(x) + q(y)$.

Proof of (a). Take $\Theta = \Theta_\gamma$. The two displayed properties are (1). For the polar representation, fix t. By (1) and (3) there is $j \in \mathbb{Z}$ with $\Theta_\gamma(t) - \arg\gamma(t) = 2\pi j$, whence

$$e^{i\Theta_\gamma(t)} = e^{i\arg\gamma(t)} e^{2\pi i j} = e^{i\arg\gamma(t)},$$

because $e^{2\pi i j} = 1$ for integer j. Since $\gamma(t) \neq 0$, the polar decomposition of a nonzero complex number gives $\gamma(t) = |\gamma(t)| e^{i\arg\gamma(t)} = |\gamma(t)| e^{i\Theta_\gamma(t)}$.

Proof of (b). Let Θ be continuous with $\Theta(0) = \arg\gamma(0)$ and $q(\Theta(t)) = q(\arg\gamma(t))$ for all t. Then Θ and Θ_γ are both lifts of α along q agreeing at $t = 0$, so by the uniqueness recorded in (1), $\Theta = \Theta_\gamma$. Now (2) gives $2\pi \operatorname{ind}(\gamma) = \Theta_\gamma(1) - \Theta_\gamma(0) = \Theta(1) - \Theta(0)$.

Proof of (c). Suppose $\gamma(0) = \gamma(1)$. Then $\arg\gamma(1) = \arg\gamma(0)$, so by (1) applied at $t = 1$ and at $t = 0$,

$$q\big(\Theta_\gamma(1)\big) = q\big(\arg\gamma(1)\big) = q\big(\arg\gamma(0)\big) = q\big(\Theta_\gamma(0)\big).$$

By (3) there is $k \in \mathbb{Z}$ with $\Theta_\gamma(1) - \Theta_\gamma(0) = 2\pi k$, and by (2), $2\pi \operatorname{ind}(\gamma) = 2\pi k$. Since $2\pi \neq 0$, $\operatorname{ind}(\gamma) = k$.

Proof of (d). Let $w \neq 0$, $k \in \mathbb{Z}$ and suppose $\gamma(t) = w\, e^{2\pi i k t}$ for all $t \in [0,1]$. Taking $t = 0$ gives $\gamma(0) = w$. Define

$$\Theta(t) = \arg w + 2\pi k t,$$

which is continuous on $[0,1]$ and satisfies $\Theta(0) = \arg w = \arg\gamma(0)$. We check the lifting property. Fix t. Both w and $e^{2\pi i k t}$ are nonzero, so the additivity of the argument modulo 2π gives

$$q\big(\arg\gamma(t)\big) = q\big(\arg w\big) + q\big(\arg e^{2\pi i k t}\big),$$

and since $\arg e^{i\vartheta} \equiv \vartheta \pmod{2\pi}$ for every real ϑ, the second term is $q(2\pi k t)$. As q is additive,

$$q\big(\arg\gamma(t)\big) = q(\arg w) + q(2\pi k t) = q\big(\arg w + 2\pi k t\big) = q\big(\Theta(t)\big).$$

So Θ satisfies both hypotheses of (b), and therefore

$$2\pi \operatorname{ind}(\gamma) = \Theta(1) - \Theta(0) = \big(\arg w + 2\pi k\big) - \arg w = 2\pi k.$$

Dividing by $2\pi \neq 0$ gives $\operatorname{ind}(\gamma) = k$. □

2.47. `WindingAdditivity`.

Lemma 2.47. *Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, let $q\colon \mathbb{R} \to \mathbb{T}$ be the quotient map, and let* ind *denote the index about* 0 *of* `WindingNumber`.

(a) (Pointwise products.) *Let $\alpha, \beta, \sigma\colon [0,1] \to \mathbb{C}$ be continuous and nowhere zero with $\sigma(t) = \alpha(t)\beta(t)$ for every t. Then*

$$\operatorname{ind}(\sigma) = \operatorname{ind}(\alpha) + \operatorname{ind}(\beta).$$

(b) (d-fold traversal.) *Let $\psi\colon \mathbb{R} \to \mathbb{C}$ be continuous with $\psi(x) \neq 0$ for all x and $\psi(x+1) = \psi(x)$ for all x, let $d \in \mathbb{Z}$, and let $\gamma_1, \gamma_d\colon [0,1] \to \mathbb{C}$ be continuous and nowhere zero with*

$$\gamma_1(t) = \psi(t), \qquad \gamma_d(t) = \psi(dt) \qquad \text{for all } t \in [0,1].$$

Then $\operatorname{ind}(\gamma_d) = d \operatorname{ind}(\gamma_1)$.

Proof. Throughout we use that q is an additive homomorphism, that

$$q(x) = q(y) \iff x - y \in 2\pi\mathbb{Z}, \tag{0}$$

and the characterisation of the index in part (b) of `WindingCalculus`: if $\Theta\colon [0,1] \to \mathbb{R}$ is continuous with $\Theta(0) = \arg\gamma(0)$ and $q(\Theta(t)) = q(\arg\gamma(t))$ for all t, then $2\pi \operatorname{ind}(\gamma) = \Theta(1) - \Theta(0)$.

Proof of (a). By part (a) of `WindingCalculus` choose continuous $\Theta_\alpha, \Theta_\beta\colon [0,1] \to \mathbb{R}$ with

$$\Theta_\alpha(0) = \arg\alpha(0), \quad q\big(\Theta_\alpha(t)\big) = q\big(\arg\alpha(t)\big), \qquad \Theta_\beta(0) = \arg\beta(0), \quad q\big(\Theta_\beta(t)\big) = q\big(\arg\beta(t)\big).$$

Put

$$c = \arg\sigma(0) - \Theta_\alpha(0) - \Theta_\beta(0), \qquad \Theta(t) = \Theta_\alpha(t) + \Theta_\beta(t) + c.$$

Then Θ is continuous and $\Theta(0) = \arg\sigma(0)$ by construction.

We check that Θ lifts the argument of σ. First, $q(c) = 0$: indeed $\sigma(0) = \alpha(0)\beta(0)$ with $\alpha(0), \beta(0) \neq 0$, so by the additivity of the argument modulo 2π,

$$q\big(\arg\sigma(0)\big) = q\big(\arg\alpha(0)\big) + q\big(\arg\beta(0)\big) = q\big(\Theta_\alpha(0)\big) + q\big(\Theta_\beta(0)\big),$$

and therefore $q(c) = q(\arg\sigma(0)) - q(\Theta_\alpha(0)) - q(\Theta_\beta(0)) = 0$. Now for every t, using additivity of q, the same argument-additivity applied to $\sigma(t) = \alpha(t)\beta(t)$ with $\alpha(t), \beta(t) \neq 0$, and $q(c) = 0$,

$$q\big(\Theta(t)\big) = q\big(\Theta_\alpha(t)\big) + q\big(\Theta_\beta(t)\big) + q(c) = q\big(\arg\alpha(t)\big) + q\big(\arg\beta(t)\big) = q\big(\arg\sigma(t)\big).$$

So Θ satisfies both hypotheses of `WindingCalculus`(b) for σ, and likewise Θ_α, Θ_β do for α, β. Hence

$$2\pi\,\mathrm{ind}(\sigma) = \Theta(1) - \Theta(0) = \big(\Theta_\alpha(1) - \Theta_\alpha(0)\big) + \big(\Theta_\beta(1) - \Theta_\beta(0)\big) = 2\pi\,\mathrm{ind}(\alpha) + 2\pi\,\mathrm{ind}(\beta),$$

the constant c cancelling in the difference. Dividing by $2\pi \neq 0$ gives (a).

Proof of (b). Note first that γ_1 is a loop: $\gamma_1(0) = \psi(0)$ and $\gamma_1(1) = \psi(1) = \psi(0+1) = \psi(0)$. So by part (c) of `WindingCalculus` there is $c \in \mathbb{Z}$ with $\mathrm{ind}(\gamma_1) = c$.

By part (a) of `WindingCalculus` choose a continuous $\Theta\colon [0,1] \to \mathbb{R}$ with $\Theta(0) = \arg\gamma_1(0) = \arg\psi(0)$ and $q(\Theta(t)) = q(\arg\gamma_1(t)) = q(\arg\psi(t))$ for all $t \in [0,1]$. By part (b) of `WindingCalculus`,

$$\Theta(1) - \Theta(0) = 2\pi\,\mathrm{ind}(\gamma_1) = 2\pi c. \tag{1}$$

A branch on all of $\mathbb{R}$. Define $\Phi\colon \mathbb{R} \to \mathbb{R}$ by

$$\Phi(x) = \Theta(x - m) + 2\pi c\, m, \qquad \text{where } m = \lfloor x \rfloor,$$

so that $x - m \in [0,1)$. We record four properties.

(i) Φ *is continuous.* On each interval $[m, m+1)$ with $m \in \mathbb{Z}$ the formula reads $\Phi(x) = \Theta(x-m) + 2\pi cm$, which is continuous there. It remains to check that the left-hand limit at an integer $m+1$ equals the value there. As $x \uparrow m+1$ we have $\lfloor x \rfloor = m$ and $x - m \uparrow 1$, so $\Phi(x) \to \Theta(1) + 2\pi cm$, while $\Phi(m+1) = \Theta(0) + 2\pi c(m+1) = \Theta(0) + 2\pi c + 2\pi cm$, and these agree by (1). Hence Φ is continuous on $\mathbb{R}$.

(ii) Φ *lifts the argument of* ψ. Let $x \in \mathbb{R}$ and $m = \lfloor x \rfloor$. Iterating $\psi(y+1) = \psi(y)$ exactly $|m|$ times gives $\psi(x) = \psi(x-m)$. Since $c \in \mathbb{Z}$ and $m \in \mathbb{Z}$ we have $q(2\pi cm) = 0$ by (0), so

$$q\big(\Phi(x)\big) = q\big(\Theta(x-m)\big) + q(2\pi cm) = q\big(\arg\psi(x-m)\big) = q\big(\arg\psi(x)\big).$$

(iii) $\Phi(0) = \Theta(0) = \arg\psi(0)$, since $\lfloor 0 \rfloor = 0$.

(iv) $\Phi(m) = \Phi(0) + 2\pi cm$ *for every* $m \in \mathbb{Z}$, directly from the formula: $\lfloor m \rfloor = m$, so $\Phi(m) = \Theta(0) + 2\pi cm$.

Conclusion. Define $\Theta_d\colon [0,1] \to \mathbb{R}$ by $\Theta_d(t) = \Phi(dt)$. It is continuous by (i). By (iii), $\Theta_d(0) = \Phi(0) = \arg\psi(0) = \arg\gamma_d(0)$, using $\gamma_d(0) = \psi(d \cdot 0) = \psi(0)$. By (ii), for every $t \in [0,1]$,

$$q\big(\Theta_d(t)\big) = q\big(\arg\psi(dt)\big) = q\big(\arg\gamma_d(t)\big).$$

So Θ_d satisfies both hypotheses of `WindingCalculus`(b) for γ_d, and therefore, using (iv) with $m = d$,

$$2\pi\,\mathrm{ind}(\gamma_d) = \Theta_d(1) - \Theta_d(0) = \Phi(d) - \Phi(0) = 2\pi c\, d.$$

Dividing by $2\pi \neq 0$ gives $\mathrm{ind}(\gamma_d) = d\,c = d\,\mathrm{ind}(\gamma_1)$. □

2.48. `WindingHomotopyInvariance`.

Lemma 2.48. *Let $\gamma_0, \gamma_1\colon [0,1] \to \mathbb{C}$ be continuous and nowhere zero, and suppose they are homotopic relative to $\{0,1\}$ through $\mathbb{C} - \{0\}$: that is, writing $P = \{z \in \mathbb{C} : z \neq 0\}$ with its subspace topology, there are continuous maps $g_0, g_1\colon [0,1] \to P$ with $g_j(t) = \gamma_j(t)$ for all t, and a homotopy from g_0 to g_1 inside P which is stationary at $t = 0$ and at $t = 1$. Then*

$$\mathrm{ind}(\gamma_0) = \mathrm{ind}(\gamma_1),$$

with ind *the index about* 0 *of* ***WindingNumber***.

Proof. Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and let $q\colon \mathbb{R} \to \mathbb{T}$ be the quotient map, a covering map. Recall from `WindingNumber` that for a continuous nowhere-zero $\gamma\colon [0,1] \to \mathbb{C}$ the index is

$$\operatorname{ind}(\gamma) = \frac{\Theta_\gamma(1) - \arg\gamma(0)}{2\pi}, \tag{1}$$

where Θ_γ is the lift along q of the continuous map $\alpha_\gamma\colon [0,1] \to \mathbb{T}$, $\alpha_\gamma(t) = q(\arg\gamma(t))$, normalised by $\Theta_\gamma(0) = \arg\gamma(0)$.

Step 1: a continuous map $P \to \mathbb{T}$. Define

$$u\colon P \to \mathbb{T}, \qquad u(z) = q(\arg z).$$

This is continuous: the reduction $z \mapsto q(\arg z)$ of the principal argument is continuous at every nonzero complex number, and every point of P is such a point. By construction, for $j \in \{0,1\}$ and every t,

$$\big(u \circ g_j\big)(t) = q\big(\arg g_j(t)\big) = q\big(\arg \gamma_j(t)\big) = \alpha_{\gamma_j}(t), \tag{2}$$

using $g_j(t) = \gamma_j(t)$.

Step 2: the endpoints agree. A homotopy relative to $\{0,1\}$ is stationary at 0 and at 1, so its two ends agree there: $g_0(0) = g_1(0)$ and $g_0(1) = g_1(1)$. Hence

$$\gamma_0(0) = \gamma_1(0), \qquad \gamma_0(1) = \gamma_1(1), \tag{3}$$

and in particular $\arg\gamma_0(0) = \arg\gamma_1(0)$. Call this common value e, so that both lifts Θ_{γ_0} and Θ_{γ_1} are normalised by the same initial value e.

Step 3: transport the homotopy. Post-composition with a continuous map carries a homotopy relative to a set to a homotopy relative to the same set: applying u to the given homotopy from g_0 to g_1 produces a homotopy from $u \circ g_0$ to $u \circ g_1$ relative to $\{0,1\}$, inside $\mathbb{T}$. By (2) this says

$$\alpha_{\gamma_0} \text{ and } \alpha_{\gamma_1} \text{ are homotopic relative to } \{0,1\}.$$

Step 4: invariance of the lifted endpoint. For a covering map, two paths in the base that are homotopic relative to $\{0,1\}$ have lifts with the same terminal value, once the lifts are started at the same point: this is the homotopy invariance of the endpoint of the lift. Applying it to q, to the two paths α_{γ_0} and α_{γ_1} of Step 3, and to the common initial value e of Step 2, we get

$$\Theta_{\gamma_0}(1) = \Theta_{\gamma_1}(1).$$

Step 5: conclusion. By (1), (3) and Step 4,

$$\operatorname{ind}(\gamma_0) = \frac{\Theta_{\gamma_0}(1) - \arg\gamma_0(0)}{2\pi} = \frac{\Theta_{\gamma_1}(1) - \arg\gamma_1(0)}{2\pi} = \operatorname{ind}(\gamma_1).$$

□

2.49. `PunctDiscLoopClass`.

Lemma 2.49. *Let $\delta > 0$ and put*

$$D = \{z \in \mathbb{C} : z \neq 0 \text{ and } |z| < \delta\},$$

with its subspace topology. Let $\gamma_0, \gamma_1\colon [0,1] \to \mathbb{C}$ be continuous with, for $j = 0, 1$ and all t,

$$\gamma_j(t) \neq 0, \qquad |\gamma_j(t)| < \delta, \qquad \gamma_j(0) = \gamma_j(1),$$

and suppose further that

$$\gamma_0(0) = \gamma_1(0) \qquad \text{and} \qquad \operatorname{ind}(\gamma_0) = \operatorname{ind}(\gamma_1),$$

with ind *the index about* 0 *of* `WindingNumber`*. Then there are continuous maps $g_0, g_1\colon [0,1] \to D$ with $g_j(t) = \gamma_j(t)$ for all t, and a homotopy from g_0 to g_1 inside D which is stationary at $t = 0$ and at $t = 1$; that is, g_0 and g_1 are homotopic relative to $\{0,1\}$ through D.*

Proof. Write $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and let $q\colon \mathbb{R} \to \mathbb{T}$ be the quotient map.

Step 1: polar data for the two loops. By part (a) of `WindingCalculus`, applied to γ_0 and to γ_1, there are continuous $\Theta_0, \Theta_1\colon [0,1] \to \mathbb{R}$ with, for $j = 0, 1$,

$$\Theta_j(0) = \arg \gamma_j(0), \qquad q\big(\Theta_j(t)\big) = q\big(\arg \gamma_j(t)\big), \qquad \gamma_j(t) = |\gamma_j(t)|\, e^{i\Theta_j(t)}. \tag{1}$$

By part (b) of `WindingCalculus`,

$$\Theta_j(1) - \Theta_j(0) = 2\pi \operatorname{ind}(\gamma_j) \qquad (j = 0, 1). \tag{2}$$

Put $m_j(t) = |\gamma_j(t)|$, a continuous function of t with

$$0 < m_j(t) < \delta \qquad \text{for all } t. \tag{3}$$

Step 2: the polar data agree at both ends. At $t = 0$: by hypothesis $\gamma_0(0) = \gamma_1(0)$, so

$$m_0(0) = m_1(0) \qquad \text{and} \qquad \Theta_0(0) = \arg \gamma_0(0) = \arg \gamma_1(0) = \Theta_1(0), \tag{4}$$

the second by the first identity of (1). At $t = 1$: since each γ_j is a loop, $\gamma_0(1) = \gamma_0(0) = \gamma_1(0) = \gamma_1(1)$, so $m_0(1) = m_1(1)$; and by (2), (4) and the hypothesis $\operatorname{ind}(\gamma_0) = \operatorname{ind}(\gamma_1)$,

$$\Theta_0(1) = \Theta_0(0) + 2\pi \operatorname{ind}(\gamma_0) = \Theta_1(0) + 2\pi \operatorname{ind}(\gamma_1) = \Theta_1(1). \tag{5}$$

This is the only place where the equality of the two indices is used.

Step 3: the homotopy. For $s, t \in [0,1]$ put

$$F(s,t) = \Big((1-s)m_0(t) + s\, m_1(t)\Big)\, \exp\Big(i\big((1-s)\Theta_0(t) + s\,\Theta_1(t)\big)\Big).$$

We verify the required properties.

(3a) *F is continuous on $[0,1] \times [0,1]$.* The functions m_j and Θ_j are continuous, and F is obtained from them and from the coordinate s by sums, products, real-to-complex inclusion and the complex exponential, all continuous.

(3b) *The modulus is the interpolated modulus, and it is positive.* Put $R(s,t) = (1-s)m_0(t) + s\, m_1(t)$. Since $0 \le s \le 1$ and $m_0(t), m_1(t) > 0$ by (3), we have $R(s,t) > 0$: if $s < 1$ then $(1-s)m_0(t) > 0$ and $s\, m_1(t) \ge 0$, while if $s = 1$ then $R(1,t) = m_1(t) > 0$. As $|e^{i\vartheta}| = 1$ for real ϑ and $R(s,t)$ is a positive real,

$$|F(s,t)| = R(s,t) > 0.$$

In particular $F(s,t) \neq 0$.

(3c) *The modulus stays below δ.* A convex combination is at most the larger of the two numbers combined, so by (3),

$$|F(s,t)| = R(s,t) \le \max\big(m_0(t), m_1(t)\big) < \delta.$$

Together with (3b), $F(s,t) \in D$ for all s, t.

(3d) *The two ends in s.* Taking $s = 0$ and $s = 1$ and using the polar representation in (1),

$$F(0,t) = m_0(t) e^{i\Theta_0(t)} = \gamma_0(t), \qquad F(1,t) = m_1(t) e^{i\Theta_1(t)} = \gamma_1(t).$$

(3e) *Stationarity at $t = 0$ and $t = 1$.* Let $t \in \{0, 1\}$. By Step 2, $m_0(t) = m_1(t)$ and $\Theta_0(t) = \Theta_1(t)$. Hence for every s,

$$(1-s)m_0(t) + s\, m_1(t) = m_0(t), \qquad (1-s)\Theta_0(t) + s\,\Theta_1(t) = \Theta_0(t),$$

so $F(s,t) = m_0(t) e^{i\Theta_0(t)} = \gamma_0(t)$ does not depend on s.

Step 4: conclusion. Define $g_j\colon [0,1] \to D$ by $g_j(t) = \gamma_j(t)$; this is legitimate because $\gamma_j(t) \neq 0$ and $|\gamma_j(t)| < \delta$ by hypothesis, and g_j is continuous as a map into the subspace D because γ_j is continuous into $\mathbb{C}$. By (3b) and (3c) the map F takes values in D, so by (3a) it is a continuous map $[0,1] \times [0,1] \to D$; by (3d) it equals g_0 at $s = 0$ and g_1 at $s = 1$; and by (3e) it is stationary at $t = 0$ and at $t = 1$. Therefore F is a homotopy from g_0 to g_1 inside D, relative to $\{0, 1\}$. $\square$

2.50. `HomeoCircleIndex`.

Theorem 2.50. *Let $V \subset \mathbb{C}$ be open with $0 \in V$, and let $H, \chi \colon \mathbb{C} \to \mathbb{C}$ satisfy*

(i) *H is continuous on V and injective on V;*
(ii) *$H(V)$ is open;*
(iii) *$\chi(H(z)) = z$ for every $z \in V$, and χ is continuous on $H(V)$;*
(iv) *$H(0) = 0$.*

Then there is $\rho > 0$ such that for every r with $0 < r \le \rho$ and every continuous $\gamma \colon [0,1] \to \mathbb{C}$ which is nowhere zero and satisfies

$$\gamma(t) = H\big(re^{2\pi i t}\big) \qquad \text{for all } t \in [0,1],$$

one has

$$\operatorname{ind}(\gamma) = 1 \quad \text{or} \quad \operatorname{ind}(\gamma) = -1,$$

where ind *is the index about* 0 *of* `WindingNumber`.

Proof. For $\delta > 0$ write $B_\delta = \{w \in \mathbb{C} : |w| < \delta\}$.

Step 0: χ is a two-sided inverse on $H(V)$, and the two non-vanishing facts. Let $w \in H(V)$, say $w = H(z)$ with $z \in V$. By (iii), $\chi(w) = \chi(H(z)) = z$, hence

$$H\big(\chi(w)\big) = H(z) = w \qquad \text{for every } w \in H(V). \tag{1}$$

Consequently, if $w \in H(V)$ and $\chi(w) = 0$, then by (1) and (iv) $w = H(\chi(w)) = H(0) = 0$; that is,

$$\chi(w) \neq 0 \qquad \text{for every } w \in H(V) \text{ with } w \neq 0. \tag{2}$$

Also, if $\xi \in V$ and $\xi \neq 0$, then $H(\xi) \neq H(0) = 0$ by injectivity on V (both ξ and 0 lie in V); that is,

$$H(\xi) \neq 0 \qquad \text{for every } \xi \in V \text{ with } \xi \neq 0. \tag{3}$$

Step 1: choice of δ and ρ. By (iv), $0 = H(0) \in H(V)$, and $H(V)$ is open by (ii), so there is $\delta > 0$ with

$$B_\delta \subseteq H(V). \tag{4}$$

Put

$$W = \{z \in V : H(z) \in B_\delta\}.$$

Then W is open: H is continuous on V and B_δ is open, so $\{z \in V : H(z) \in B_\delta\}$ is open in the subspace V, and V is open in $\mathbb{C}$. Moreover $0 \in W$, since $0 \in V$ and $H(0) = 0 \in B_\delta$. As W is open and contains 0, there is $\rho > 0$ with

$$\{\xi \in \mathbb{C} : |\xi| \le \rho\} \subseteq W \tag{5}$$

(choose $\rho' > 0$ with the open disc of radius ρ' inside W and take $\rho = \rho'/2$). We claim this ρ has the required property.

Fix r with $0 < r \le \rho$, and let γ be continuous, nowhere zero, with $\gamma(t) = H(re^{2\pi i t})$ for all $t \in [0,1]$.

Step 2: γ is a loop in the punctured disc $B_\delta - \{0\}$. For every t we have $|re^{2\pi i t}| = r \le \rho$, so $re^{2\pi i t} \in W$ by (5); in particular $re^{2\pi i t} \in V$ and $H(re^{2\pi i t}) \in B_\delta$, i.e.

$$|\gamma(t)| < \delta \qquad \text{for all } t. \tag{6}$$

Also $\gamma(0) = H(r)$ and $\gamma(1) = H(re^{2\pi i}) = H(r)$, so

$$\gamma(0) = \gamma(1) = H(r), \tag{7}$$

and $H(r) \neq 0$ by (3), since $r \in W \subseteq V$ and $r \neq 0$.

Step 3: the index of γ is an integer, and the model loop. By (7) γ is a loop, so by part (c) of `WindingCalculus` there is $d \in \mathbb{Z}$ with

$$\operatorname{ind}(\gamma) = d. \tag{8}$$

Define $\rho_d, \rho_1 \colon [0,1] \to \mathbb{C}$ by

$$\rho_d(t) = H(r)\, e^{2\pi i d t}, \qquad \rho_1(t) = H(r)\, e^{2\pi i t}.$$

Both are continuous. Since $|e^{i\vartheta}| = 1$ for real ϑ,

$$|\rho_d(t)| = |\rho_1(t)| = |H(r)|,$$

which is nonzero and, by (6) at $t = 0$ together with (7), smaller than δ. So ρ_d and ρ_1 are nowhere-zero paths with values in $B_\delta - \{0\}$, and they are loops with

$$\rho_d(0) = \rho_d(1) = \rho_1(0) = \rho_1(1) = H(r) = \gamma(0). \tag{9}$$

By part (d) of `WindingCalculus`, applied with $w = H(r) \neq 0$ and $k = d$ respectively $k = 1$,

$$\operatorname{ind}(\rho_d) = d, \qquad \operatorname{ind}(\rho_1) = 1. \tag{10}$$

Step 4: γ and ρ_d are homotopic in the punctured disc. The loops γ and ρ_d are continuous, nowhere zero, of modulus $< \delta$ by (6) and Step 3, they are loops by (7) and (9), they share the basepoint $\gamma(0) = \rho_d(0)$ by (9), and they have the same index d by (8) and (10). So `PunctDiscLoopClass` applies with this δ and yields continuous maps $g, h\colon [0,1] \to D$, where $D = \{z \in \mathbb{C} : z \neq 0,\ |z| < \delta\}$, with $g(t) = \gamma(t)$ and $h(t) = \rho_d(t)$ for all t, and a homotopy from g to h inside D relative to $\{0,1\}$.

Step 5: transport the homotopy through χ. By (4), $D \subseteq B_\delta \subseteq H(V)$, and by (2) χ does not vanish on D (every $w \in D$ is a nonzero element of $H(V)$). Hence

$$c\colon D \to P = \{z \in \mathbb{C} : z \neq 0\}, \qquad c(w) = \chi(w),$$

is well defined, and it is continuous because χ is continuous on $H(V) \supseteq D$ by (iii). Post-composition with a continuous map carries a homotopy relative to a set to a homotopy relative to the same set, so $c \circ g$ and $c \circ h$ are homotopic relative to $\{0,1\}$ inside P.

Step 6: the two transported loops. Define $\sigma_0, \sigma_d, \sigma_1\colon [0,1] \to \mathbb{C}$ by

$$\sigma_0(t) = \chi\big(\gamma(t)\big), \qquad \sigma_d(t) = \chi\big(\rho_d(t)\big), \qquad \sigma_1(t) = \chi\big(\rho_1(t)\big).$$

All three are continuous (composites of continuous maps, χ being continuous on $H(V)$, which contains the values of γ, ρ_d and ρ_1 by Step 2, Step 3 and (4)), and all three are nowhere zero by (2). Moreover, for every t,

$$\sigma_0(t) = \chi\big(H(re^{2\pi it})\big) = re^{2\pi it}, \tag{11}$$

by (iii) applied at $z = re^{2\pi it} \in W \subseteq V$.

Now $c \circ g$ and $c \circ h$ are maps into P with $(c\circ g)(t) = \chi(\gamma(t)) = \sigma_0(t)$ and $(c\circ h)(t) = \chi(\rho_d(t)) = \sigma_d(t)$. Applying `WindingHomotopyInvariance` to the pair σ_0, σ_d, with the maps $c \circ g$, $c \circ h$ into P and the homotopy of Step 5, gives

$$\operatorname{ind}(\sigma_0) = \operatorname{ind}(\sigma_d). \tag{12}$$

By (11) and part (d) of `WindingCalculus` with $w = r \neq 0$ and $k = 1$,

$$\operatorname{ind}(\sigma_0) = 1. \tag{13}$$

Step 7: the divisibility. Define $\psi\colon \mathbb{R} \to \mathbb{C}$ by

$$\psi(x) = \chi\big(H(r)e^{2\pi ix}\big).$$

Then:

- ψ is continuous. Indeed $x \mapsto H(r)e^{2\pi ix}$ is continuous and takes values of modulus $|H(r)|$, which lies in $(0, \delta)$ by Step 3; so its values lie in $D \subseteq H(V)$, where χ is continuous.
- $\psi(x) \neq 0$ for every x, by (2), since $H(r)e^{2\pi ix}$ is a nonzero element of $H(V)$.
- $\psi(x+1) = \psi(x)$ for every x, because $e^{2\pi i(x+1)} = e^{2\pi ix}e^{2\pi i} = e^{2\pi ix}$.

Furthermore, for $t \in [0,1]$,

$$\sigma_1(t) = \chi\big(H(r)e^{2\pi it}\big) = \psi(t), \qquad \sigma_d(t) = \chi\big(H(r)e^{2\pi idt}\big) = \psi(dt).$$

So part (b) of `WindingAdditivity`, applied to this ψ, to d, and to the paths σ_1 and σ_d, gives

$$\operatorname{ind}(\sigma_d) = d \operatorname{ind}(\sigma_1). \tag{14}$$

Step 8: conclusion. By (9), $\rho_1(0) = \rho_1(1)$, so $\sigma_1(0) = \chi(\rho_1(0)) = \chi(\rho_1(1)) = \sigma_1(1)$ and σ_1 is a loop; by part (c) of `WindingCalculus` there is $e \in \mathbb{Z}$ with $\operatorname{ind}(\sigma_1) = e$. Combining (13), (12) and (14),

$$1 = \operatorname{ind}(\sigma_0) = \operatorname{ind}(\sigma_d) = d\,e.$$

Thus d and e are integers with $de = 1$, which forces $d = e = 1$ or $d = e = -1$. In either case, by (8), $\operatorname{ind}(\gamma) = d \in \{1, -1\}$, which is the assertion. □

2.51. `LeadingCoefNonzero`.

Lemma 2.51. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, let $K \geq 1$, write $\mathbb{B} = \{\xi : |\xi| < 1\}$, let $\lambda_k = \lambda_{|k|}(n,p)$ be the exponent of `LambdaCoef`, $\epsilon_k = \epsilon_k(n,p)$ the coefficient of `EpsCoef` and a_k the Fourier coefficient of `FourierCoefOfH`. Let $H\colon \mathbb{C} \to \mathbb{C}$ and $A\colon \mathbb{Z} \to \mathbb{C}$ satisfy*

(H1) *H is continuous on $\mathbb{B}$ and differentiable at almost every point of $\mathbb{B}$;*
(H2) *H solves (18) on $\mathbb{B}$ in the sense of `SolvesEq18`;*
(H3) *H is K-quasiconformal on $\mathbb{B}$ in the sense of `QuasiconformalOn`, and $H(0) = 0$;*
(H4) *$a_k(r) = A_k\, r^{\lambda_k}$ for every $k \in \mathbb{Z}$ with $|k| \geq n$ and every $r \in (0,1)$;*
(H5) *$A_{-k} = \epsilon_k \overline{A_k}$ for every integer $k \geq n$;*
(H6) *$\sum_{m \geq 0} (n+m)|A_{n+m}|^2 < \infty$.*

Then

$$A_n = 0 \qquad \text{and} \qquad A_{n+1} \neq 0.$$

The second assertion is the paper's step at paper.tex:382, "Applying argument principle, we find that $A_{n+1} \neq 0$"; the first is the paper's remark at paper.tex:377 that under the hypotheses of Corollary 1 the expansion (19) has $A_n = 0$.

Proof. Write T_k for the k-th summand of (19) in the sense of `Thm2Term`, $\gamma_n = \lambda_{n+1}$, and $I = [0,1]$.

Step 0: the exponents. By part (a) of `GammaRatioMinAtOne`, $\gamma_n > 0$. By parts (23) and (24) of `LambdaEpsEstimates`,

$$\lambda_n = 0, \qquad \epsilon_n = 0, \qquad |\epsilon_k| < 1 \quad (k \geq n+1). \tag{1}$$

Moreover λ_k is strictly increasing in $k \geq 0$: by `LambdaCoef`, $2\lambda_k + np = \sqrt{4k^2(p-1) + n^2(p-2)^2}$, the radicand is strictly increasing in $k \geq 0$ because $p - 1 > 0$ and $(k+1)^2 > k^2$, and the square root is strictly increasing on $[0, \infty)$; so

$$\lambda_k < \lambda_{k+1} \qquad (k \geq 0). \tag{2}$$

The hypotheses (H1)–(H5) of `SeriesTermwiseDeriv` hold here. Its (H1) asks that H be continuous on $\mathbb{B}$, lie in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ and be differentiable at almost every point of $\mathbb{B}$: the first and third are our (H1), and the second is contained in our (H3), since by `QuasiconformalOn` and `QuasiregularOn` a K-quasiconformal map on $\mathbb{B}$ lies in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$. Its (H2) is our (H2), its (H3) is our (H4), its (H4) is our (H5) and its (H5) is our (H6). We may therefore invoke it; write (a), (b) for its first two conclusions.

Step 1: $A_n = 0$. Apply (b) with $\rho = \frac{1}{2}$ and $j = n$: there is $C \geq 0$ with

$$\Big|H(\xi) - \sum_{k=n}^{n} T_k(\xi)\Big| = \big|H(\xi) - T_n(\xi)\big| \leq C|\xi|^{\lambda_{n+1}} = C|\xi|^{\gamma_n} \qquad \text{for } 0 < |\xi| \leq \tfrac{1}{2}.$$

By part (b) of `Thm2TermSolvesEq18` at $k = n$, for $r > 0$,

$$T_n\big(re^{i\theta}\big) = r^{\lambda_n}\big(A_n e^{i(n-n)\theta} + \epsilon_n \overline{A_n} e^{i(-n-n)\theta}\big) = A_n,$$

using $\lambda_n = 0$ and $\epsilon_n = 0$ from (1); so $T_n(\xi) = A_n$ for every $\xi \neq 0$. Hence

$$\big|H(\xi) - A_n\big| \leq C|\xi|^{\gamma_n} \qquad (0 < |\xi| \leq \tfrac{1}{2}),$$

and since $\gamma_n > 0$ we have $|\xi|^{\gamma_n} \to 0$ as $\xi \to 0$; therefore $H(\xi) \to A_n$ as $\xi \to 0$ through $\mathbb{B} - \{0\}$. On the other hand H is continuous on $\mathbb{B}$ by (H1) and $0 \in \mathbb{B}$, so $H(\xi) \to H(0) = 0$ as $\xi \to 0$. A limit along a filter is unique, and $\xi \to 0$ through $\mathbb{B} - \{0\}$ is a nontrivial filter since every punctured neighbourhood of 0 in $\mathbb{B}$ is nonempty; so $A_n = 0$.

Step 2: a least nonvanishing index. Suppose $A_k = 0$ for every integer $k \geq n+1$. With $A_n = 0$ from Step 1 this gives $A_k = 0$ for every $k \geq n$, hence by `Thm2Term` every T_k with $k \geq n$ vanishes identically, both of its terms carrying a factor A_k or $\overline{A_k}$. By (a) then $H(\xi) = 0$ for every ξ with $0 < |\xi| < 1$, and $H(0) = 0$ by (H3); so $H \equiv 0$ on $\mathbb{B}$. But H is injective on $\mathbb{B}$ by `QuasiconformalOn`, while 0 and $\frac{1}{2}$ are two distinct points of $\mathbb{B}$ with $H(0) = H(\frac{1}{2}) = 0$ — a contradiction.

Hence $\{k \in \mathbb{Z} : k > n,\ A_k \neq 0\}$ is a nonempty set of integers bounded below, so it has a least element m, and $m \geq n+1$. By minimality together with Step 1,

$$A_k = 0 \ \text{ for } n \leq k \leq m-1, \qquad A_m \neq 0. \tag{3}$$

Consequently $T_k \equiv 0$ for $n \leq k \leq m-1$, so $\sum_{k=n}^{m} T_k = T_m$.

Step 3: the leading profile and its lower bound. By part (b) of `Thm2TermSolvesEq18` at $k = m$, for $r > 0$ and $\theta \in \mathbb{R}$,

$$T_m\big(re^{i\theta}\big) = r^{\lambda_m}\Phi(\theta), \qquad \Phi(\theta) = A_m e^{i(m-n)\theta} + \epsilon_m \overline{A_m} e^{i(-m-n)\theta}. \tag{4}$$

Since $A_m \neq 0$ we may put $u = \overline{A_m}/A_m$, so that $|u| = 1$ and

$$\Phi(\theta) = A_m e^{i(m-n)\theta}\big(1 + \epsilon_m\, u\, e^{-2im\theta}\big), \tag{5}$$

because $A_m\, \epsilon_m u\, e^{i(m-n)\theta} e^{-2im\theta} = \epsilon_m \overline{A_m} e^{i(-m-n)\theta}$. Since $m \geq n+1$, (1) gives $|\epsilon_m| < 1$, so for every θ

$$\operatorname{Re}\big(1 + \epsilon_m u e^{-2im\theta}\big) \ \geq\ 1 - \big|\epsilon_m u e^{-2im\theta}\big| = 1 - |\epsilon_m| > 0, \tag{6}$$

using $|u| = 1$ and $\epsilon_m \in \mathbb{R}$.

Step 4: the tail. Apply (b) with $\rho = \frac{1}{2}$ and $j = m$, which is legitimate since $m \geq n$: there is $C_1 \geq 0$ with

$$\big|H(\xi) - T_m(\xi)\big| \leq C_1 |\xi|^{\lambda_{m+1}} \qquad \text{for } 0 < |\xi| \leq \tfrac{1}{2}, \tag{7}$$

using $\sum_{k=n}^{m} T_k = T_m$ from Step 2.

Step 5: H does not vanish off the origin, and the index is ± 1. If $\xi \in \mathbb{B}$ and $H(\xi) = 0$, then $H(\xi) = H(0)$ and injectivity of H on $\mathbb{B}$ (`QuasiconformalOn`) forces $\xi = 0$. So

$$H(\xi) \neq 0 \qquad \text{for every } \xi \in \mathbb{B} - \{0\}. \tag{8}$$

The set $\mathbb{B}$ is open and contains 0. By `QuasiconformalOn`, H is continuous and injective on $\mathbb{B}$, the image $H(\mathbb{B})$ is open, and there is χ with $\chi(H(z)) = z$ for every $z \in \mathbb{B}$ and χ continuous on $H(\mathbb{B})$; and $H(0) = 0$ by (H3). These are exactly the hypotheses of `HomeoCircleIndex` with $V = \mathbb{B}$, which therefore provides $\rho_0 > 0$ such that

$$\begin{gathered}\text{for every } r \text{ with } 0 < r \leq \rho_0 \text{ and every continuous } \gamma\colon I \to \mathbb{C} \text{ with} \\ \gamma(t) = H\big(re^{2\pi it}\big) \text{ for all } t \text{ and } \gamma(t) \neq 0 \text{ for all } t, \quad \operatorname{ind}\gamma = 1 \text{ or } \operatorname{ind}\gamma = -1,\end{gathered} \tag{9}$$

where ind is the winding number of `WindingNumber`.

Step 6: the index computed from the leading term. Since $\lambda_{m+1} > \lambda_m$ by (2), the number $\lambda_{m+1} - \lambda_m$ is positive, so $r \mapsto r^{\lambda_{m+1}-\lambda_m}$ tends to 0 as $r \to 0^+$. As $1 - |\epsilon_m| > 0$ by (1), we may choose ρ_1 with

$$0 < \rho_1 \leq \min\big\{\rho_0, \tfrac{1}{2}\big\}, \qquad \frac{C_1}{|A_m|}\, \rho_1^{\lambda_{m+1}-\lambda_m} < 1 - |\epsilon_m|. \tag{10}$$

Fix r with $0 < r \leq \rho_1$ and define three maps on I:

$$\gamma(t) = H\big(re^{2\pi it}\big), \qquad \alpha(t) = A_m r^{\lambda_m} e^{2\pi i(m-n)t}, \qquad \beta(t) = \frac{\gamma(t)}{\alpha(t)}.$$

Then γ is continuous, because $t \mapsto re^{2\pi it}$ is continuous with values in $\mathbb{B}$ (its modulus is $r < 1$) and H is continuous on $\mathbb{B}$; and $\gamma(t) \neq 0$ for every t by (8). The map α is continuous and never 0, since $A_m \neq 0$ and $r > 0$. Hence β is continuous and never 0, and $\gamma = \alpha \cdot \beta$ pointwise.

By part (d) of `WindingCalculus`, applied with $w = A_m r^{\lambda_m} \neq 0$ and $k = m - n \in \mathbb{Z}$, we get $\operatorname{ind}\alpha = m - n$. By the first clause of `WindingAdditivity` applied to $\gamma = \alpha \cdot \beta$,

$$\operatorname{ind}\gamma = \operatorname{ind}\alpha + \operatorname{ind}\beta = (m-n) + \operatorname{ind}\beta. \tag{11}$$

We now show $\operatorname{Re}\beta(t) > 0$ for every t. Put $\xi = re^{2\pi it}$, so $|\xi| = r$. By (4) and (5) with $\theta = 2\pi t$,

$$T_m(\xi) = r^{\lambda_m}\Phi(2\pi t) = A_m r^{\lambda_m} e^{2\pi i(m-n)t}\big(1 + \epsilon_m u e^{-4\pi i m t}\big) = \alpha(t)\big(1 + \epsilon_m u e^{-4\pi i m t}\big),$$

hence

$$\beta(t) = \big(1 + \epsilon_m u e^{-4\pi i m t}\big) + \frac{H(\xi) - T_m(\xi)}{\alpha(t)}.$$

Since $|\alpha(t)| = |A_m| r^{\lambda_m}$ and $0 < r \le \rho_1 \le \frac{1}{2}$, (7) gives

$$\left|\frac{H(\xi) - T_m(\xi)}{\alpha(t)}\right| \le \frac{C_1 r^{\lambda_{m+1}}}{|A_m| r^{\lambda_m}} = \frac{C_1}{|A_m|} r^{\lambda_{m+1}-\lambda_m} \le \frac{C_1}{|A_m|}\rho_1^{\lambda_{m+1}-\lambda_m} < 1 - |\epsilon_m|,$$

the middle inequality because $\lambda_{m+1} - \lambda_m > 0$ and $r \le \rho_1$, and the last by (10). Combining with (6),

$$\operatorname{Re}\beta(t) \;\ge\; \big(1 - |\epsilon_m|\big) - \frac{C_1}{|A_m|}\rho_1^{\lambda_{m+1}-\lambda_m} \;>\; 0 \qquad \text{for every } t \in I. \tag{12}$$

Step 7: $\operatorname{ind}\beta = 0$. First, β is a loop: $\gamma(0) = \gamma(1)$ because $e^0 = e^{2\pi i} = 1$, and $\alpha(0) = \alpha(1)$ because $m - n$ is an integer; hence $\beta(0) = \beta(1)$.

By part (a) of `WindingCalculus` applied to β there is a continuous $\Theta\colon I \to \mathbb{R}$ with

$$\Theta(0) = \arg\beta(0), \qquad \Theta(t) \equiv \arg\beta(t) \pmod{2\pi}, \qquad \beta(t) = |\beta(t)|\, e^{i\Theta(t)} \quad \text{for every } t,$$

the first two of these being exactly the two hypotheses that part (b) of `WindingCalculus` requires of a branch, so (b) is applicable to this Θ below. Taking real parts, $\operatorname{Re}\beta(t) = |\beta(t)|\cos\Theta(t)$, and $|\beta(t)| > 0$; so by (12), $\cos\Theta(t) > 0$ for every t. The set $\{x \in \mathbb{R} : \cos x > 0\}$ is the disjoint union of the open intervals $\big(-\frac{\pi}{2} + 2\pi j,\ \frac{\pi}{2} + 2\pi j\big)$, $j \in \mathbb{Z}$. The image $\Theta(I)$ is connected, being the image of an interval under a continuous map, so it lies in exactly one of them. Also $\cos\Theta(0) = \cos(\arg\beta(0)) > 0$ and $\arg\beta(0) \in (-\pi, \pi]$, which forces $\arg\beta(0) \in (-\frac{\pi}{2}, \frac{\pi}{2})$; so the interval containing $\Theta(I)$ is the one with $j = 0$:

$$\Theta(t) \in \big(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\big) \qquad \text{for every } t \in I.$$

By part (b) of `WindingCalculus`, applied to this Θ,

$$2\pi \operatorname{ind}\beta = \Theta(1) - \Theta(0) \in (-\pi, \pi),$$

so $|\operatorname{ind}\beta| < \frac{1}{2}$. Since β is a loop, part (c) of `WindingCalculus` gives that $\operatorname{ind}\beta$ is an integer. The only integer of modulus less than $\frac{1}{2}$ is 0, so $\operatorname{ind}\beta = 0$.

Step 8: conclusion. Take $r = \rho_1$. By (11) and Step 7, $\operatorname{ind}\gamma = m - n$. By (10) we have $0 < r \le \rho_0$, and γ is a continuous non-vanishing map with $\gamma(t) = H(re^{2\pi it})$, so (9) gives $\operatorname{ind}\gamma = 1$ or $\operatorname{ind}\gamma = -1$; that is, $m - n = 1$ or $m - n = -1$. Since $m \ge n + 1$ we have $m - n \ge 1 > 0$, so $m - n \ne -1$ and therefore $m - n = 1$, i.e. $m = n + 1$. By (3), $A_{n+1} = A_m \ne 0$. Together with Step 1 this is the assertion. □

2.52. `PClassMulClosed`.

Lemma 2.52. *Let $H\colon \mathbb{C} \to \mathbb{C}$, let $s, k, t, l \ge 0$ be integers, and let $P \in \mathbb{P}(s, k)$ and $Q \in \mathbb{P}(t, l)$ relative to H in the sense of `PClass`. Then the pointwise product PQ belongs to $\mathbb{P}(s + t, k + l)$ relative to H.*

Proof. By `PClass` applied to P, choose an integer $r \ge 0$, complex numbers $c_1, \dots, c_r$ and lists $M_1, \dots, M_r$ of derivative symbols with

$$s(M_\alpha) = s, \qquad k(M_\alpha) = k \qquad (\alpha = 1, \dots, r), \tag{1}$$

$$P(\xi) = \sum_{\alpha=1}^{r} c_\alpha H^{M_\alpha}(\xi) \qquad \text{for every } \xi \in \mathbb{C}, \tag{2}$$

and by `PClass` applied to Q, choose an integer $q \ge 0$, complex numbers $b_1, \dots, b_q$ and lists $N_1, \dots, N_q$ with

$$s(N_\beta) = t, \qquad k(N_\beta) = l \qquad (\beta = 1, \dots, q), \tag{3}$$

$$Q(\xi) = \sum_{\beta=1}^{q} b_\beta H^{N_\beta}(\xi) \qquad \text{for every } \xi \in \mathbb{C}. \tag{4}$$

Consider the rq pairs (α, β) with $1 \le \alpha \le r$, $1 \le \beta \le q$. To each we attach the complex number $c_\alpha b_\beta$ and the list

$$M_\alpha * N_\beta$$

obtained by concatenating M_α and N_β, i.e. by listing the entries of M_α followed by the entries of N_β.

Step 1: every attached list has type $(s+t, k+l)$. By `MonomialValue`, $k(\,\cdot\,)$ is the length of a list and $s(\,\cdot\,)$ is the sum of the orders of its entries. Concatenation lists the entries of the two lists one after the other, so its length is the sum of the two lengths and the sum of the orders of its entries is the sum of the two sums:

$$k(M_\alpha * N_\beta) = k(M_\alpha) + k(N_\beta) = k + l, \qquad s(M_\alpha * N_\beta) = s(M_\alpha) + s(N_\beta) = s + t,$$

using (1) and (3).

Step 2: the monomial of a concatenation is the product of the monomials. Again by `MonomialValue`, $H^M(\xi)$ is the product of the symbol values $H^d(\xi)$ over the entries d of M. For a concatenation the list of entries is the concatenation of the two lists of entries, so the product splits as the product of the two partial products:

$$H^{M_\alpha * N_\beta}(\xi) = H^{M_\alpha}(\xi)\, H^{N_\beta}(\xi) \qquad \text{for every } \xi \in \mathbb{C}. \tag{5}$$

Step 3: the linear combination. Let $\xi \in \mathbb{C}$. Using (5) and then the distributive law for finite sums of complex numbers,

$$\sum_{\alpha=1}^{r}\sum_{\beta=1}^{q} c_\alpha b_\beta\, H^{M_\alpha * N_\beta}(\xi) = \sum_{\alpha=1}^{r}\sum_{\beta=1}^{q} \big(c_\alpha H^{M_\alpha}(\xi)\big)\big(b_\beta H^{N_\beta}(\xi)\big) = \Big(\sum_{\alpha=1}^{r} c_\alpha H^{M_\alpha}(\xi)\Big)\Big(\sum_{\beta=1}^{q} b_\beta H^{N_\beta}(\xi)\Big),$$

and by (2) and (4) the right-hand side is $P(\xi)Q(\xi)$.

The finite family of coefficients $c_\alpha b_\beta$ together with the lists $M_\alpha * N_\beta$, which all have type $(s+t, k+l)$ by Step 1, therefore presents PQ as a finite $\mathbb{C}$-linear combination of monomials of type $(s+t, k+l)$, valid at every point of $\mathbb{C}$. By `PClass` this says exactly that $PQ \in \mathbb{P}(s+t, k+l)$ relative to H. □

2.53. `WirtingerIterSwap`.

Lemma 2.53. *Let $\Omega \subset \mathbb{C}$ be open, let $H\colon \mathbb{C} \to \mathbb{C}$ be C^∞ on Ω, and let $i, j \ge 0$ be integers. Then*

(a) $\dfrac{\partial^{i+j} H}{\partial \xi^i \partial \bar{\xi}^j}$ *is C^∞ on Ω, and*

(b) $\dfrac{\partial}{\partial \bar{\xi}} \dfrac{\partial^{i+j} H}{\partial \xi^i \partial \bar{\xi}^j}(\xi) = \dfrac{\partial^{i+j+1} H}{\partial \xi^i \partial \bar{\xi}^{j+1}}(\xi)$ *for every $\xi \in \Omega$,*

where the iterated derivatives are those of `WirtingerIter` and $\partial/\partial\bar{\xi}$ is as in `DZbar`.

Proof. Throughout, ∂_x and ∂_y denote the two real first partial derivatives, taken through the real Fréchet derivative as in `DZ`; thus for any g and any point, $\partial_x g = Dg[1]$ and $\partial_y g = Dg[i]$, and by `DZ` and `DZbar`

$$\frac{\partial g}{\partial \xi} = \tfrac{1}{2}\big(\partial_x g - i\,\partial_y g\big), \qquad \frac{\partial g}{\partial \bar{\xi}} = \tfrac{1}{2}\big(\partial_x g + i\,\partial_y g\big). \tag{0}$$

Step 1: both Wirtinger operators preserve $C^\infty(\Omega)$. Let g be C^∞ on the open set Ω. Then g is in particular real-differentiable at each point of Ω, and each of its first partial derivatives $\partial_x g$, $\partial_y g$ is again C^∞ on Ω (the partial derivatives of a C^∞ function on an open set are C^∞ there). By (0), $\partial g/\partial\xi$ and $\partial g/\partial\bar{\xi}$ are fixed $\mathbb{C}$-linear combinations of $\partial_x g$ and $\partial_y g$, hence both are C^∞ on Ω.

Step 2: proof of (a). By `WirtingerIter`, $\partial^{i+j} H/\partial\xi^i\partial\bar{\xi}^j$ is obtained from H by applying j times the operator $\partial/\partial\bar{\xi}$ and then i times the operator $\partial/\partial\xi$. Each of these $i+j$ operators

carries $C^\infty(\Omega)$ into $C^\infty(\Omega)$ by Step 1, and $H \in C^\infty(\Omega)$ by hypothesis. Induction on the number of operators applied gives $\partial^{i+j}H/\partial\xi^i\partial\bar\xi^j \in C^\infty(\Omega)$.

Step 3: the two operators commute on $C^\infty(\Omega)$. Let g be C^∞ on Ω and let $\xi \in \Omega$. Using (0) twice, and writing $\partial_u\partial_v g$ for $\partial_u(\partial_v g)$,

$$\begin{aligned}\frac{\partial}{\partial\xi}\Big(\frac{\partial g}{\partial\bar\xi}\Big) &= \tfrac12\big(\partial_x - i\,\partial_y\big)\Big[\tfrac12\big(\partial_x g + i\,\partial_y g\big)\Big]\\ &= \tfrac14\Big(\partial_x\partial_x g + \partial_y\partial_y g + i\big(\partial_x\partial_y g - \partial_y\partial_x g\big)\Big),\end{aligned}$$

and symmetrically

$$\frac{\partial}{\partial\bar\xi}\Big(\frac{\partial g}{\partial\xi}\Big) = \tfrac14\Big(\partial_x\partial_x g + \partial_y\partial_y g - i\big(\partial_x\partial_y g - \partial_y\partial_x g\big)\Big),$$

both at every point of Ω; in each case the step is the expansion of the product of the two first-order operators, legitimate because the first derivatives of g are themselves differentiable on Ω by Step 1. Since g is C^∞ on the open set Ω, its second derivative is symmetric at every point of Ω (Schwarz's theorem on mixed partials, valid for C^2 functions on an open set), so $\partial_x\partial_y g = \partial_y\partial_x g$ on Ω. The two displayed expressions therefore coincide:

$$\frac{\partial}{\partial\xi}\Big(\frac{\partial g}{\partial\bar\xi}\Big)(\xi) = \frac{\partial}{\partial\bar\xi}\Big(\frac{\partial g}{\partial\xi}\Big)(\xi) \qquad \text{for every } \xi \in \Omega. \tag{1}$$

Step 4: proof of (b), by induction on i with j fixed.

Base case $i = 0$. By `WirtingerIter`, $\partial^j H/\partial\xi^0\partial\bar\xi^j$ is $(\partial/\partial\bar\xi)^j H$, so applying $\partial/\partial\bar\xi$ to it gives $(\partial/\partial\bar\xi)^{j+1}H$, which is $\partial^{j+1}H/\partial\xi^0\partial\bar\xi^{j+1}$. This is an identity of functions on all of $\mathbb{C}$ and needs no hypothesis.

Inductive step. Assume

$$\frac{\partial}{\partial\bar\xi}\Big(\frac{\partial^{i+j}H}{\partial\xi^i\partial\bar\xi^j}\Big)(\zeta) = \frac{\partial^{i+j+1}H}{\partial\xi^i\partial\bar\xi^{j+1}}(\zeta) \qquad \text{for every } \zeta \in \Omega. \tag{2}$$

Put $g = \partial^{i+j}H/\partial\xi^i\partial\bar\xi^j$, which is C^∞ on Ω by Step 2. By `WirtingerIter`,

$$\frac{\partial^{i+1+j}H}{\partial\xi^{i+1}\partial\bar\xi^j} = \frac{\partial g}{\partial\xi}, \qquad \frac{\partial^{i+j+1}H}{\partial\xi^i\partial\bar\xi^{j+1}} \text{ is a } C^\infty(\Omega) \text{ function by Step 2.}$$

Let $\xi \in \Omega$. Then

$$\frac{\partial}{\partial\bar\xi}\Big(\frac{\partial^{i+1+j}H}{\partial\xi^{i+1}\partial\bar\xi^j}\Big)(\xi) = \frac{\partial}{\partial\bar\xi}\Big(\frac{\partial g}{\partial\xi}\Big)(\xi) = \frac{\partial}{\partial\xi}\Big(\frac{\partial g}{\partial\bar\xi}\Big)(\xi),$$

by (1) applied to g. By the induction hypothesis (2), the functions $\partial g/\partial\bar\xi$ and $\partial^{i+j+1}H/\partial\xi^i\partial\bar\xi^{j+1}$ agree at every point of Ω. Since Ω is open and $\xi \in \Omega$, two functions agreeing on Ω agree on a neighbourhood of ξ, hence have the same real Fréchet derivative at ξ and therefore, by (0), the same $\partial/\partial\xi$ at ξ. Consequently

$$\frac{\partial}{\partial\xi}\Big(\frac{\partial g}{\partial\bar\xi}\Big)(\xi) = \frac{\partial}{\partial\xi}\Big(\frac{\partial^{i+j+1}H}{\partial\xi^i\partial\bar\xi^{j+1}}\Big)(\xi) = \frac{\partial^{i+1+j+1}H}{\partial\xi^{i+1}\partial\bar\xi^{j+1}}(\xi),$$

the last equality again by `WirtingerIter`. Combining the two displays gives (b) for $i+1$, and the induction is complete. □

2.54. `PClassDerivClosed`.

Lemma 2.54. *Let $\Omega \subset \mathbb{C}$ be open, let $H\colon \mathbb{C} \to \mathbb{C}$ be C^∞ on Ω, let $s, k \geq 0$ be integers, and let $P \in \mathbb{P}(s,k)$ relative to H in the sense of `PClass`. Then there exist functions $Q_1, Q_2 \in \mathbb{P}(s+1,k)$ relative to H such that*

$$\frac{\partial P}{\partial\xi}(\xi) = Q_1(\xi) \quad \text{and} \quad \frac{\partial P}{\partial\bar\xi}(\xi) = Q_2(\xi) \qquad \text{for every } \xi \in \Omega,$$

with $\partial/\partial\xi$ as in `DZ` and $\partial/\partial\bar\xi$ as in `DZbar`.

Proof. By `PClass` fix an integer $r \geq 0$, complex numbers $c_1, \dots, c_r$ and lists $M_1, \dots, M_r$ of derivative symbols with

$$s(M_\alpha) = s, \qquad k(M_\alpha) = k \qquad (\alpha = 1, \dots, r), \tag{1}$$

$$P(\xi) = \sum_{\alpha=1}^{r} c_\alpha H^{M_\alpha}(\xi) \qquad \text{for every } \xi \in \mathbb{C}. \tag{2}$$

Step 1: each symbol value is C^∞ on Ω. Let $d = (i, j, \beta)$ be a derivative symbol, and put

$$g_d = \frac{\partial^{i+j} H}{\partial \xi^i \partial \bar{\xi}^j},$$

the iterated derivative of `WirtingerIter`. By part (a) of `WirtingerIterSwap`, g_d is C^∞ on Ω. By `SymbolValue`, H^d is either g_d or $\overline{g_d}$; complex conjugation is an $\mathbb{R}$-linear isometry of $\mathbb{C}$, so composing a C^∞ function on Ω with it again gives a C^∞ function on Ω. Hence H^d is C^∞ on Ω, and in particular g_d and H^d are real-differentiable at every point of Ω.

Step 2: differentiating one symbol value raises its order by one. Let $d = (i, j, \beta)$ be a derivative symbol and define two triples

$$d' = \begin{cases} (i+1,\, j,\, \beta) & \text{if } \beta \text{ is false,} \\ (i,\, j+1,\, \beta) & \text{if } \beta \text{ is true,} \end{cases} \qquad d'' = \begin{cases} (i,\, j+1,\, \beta) & \text{if } \beta \text{ is false,} \\ (i+1,\, j,\, \beta) & \text{if } \beta \text{ is true.} \end{cases}$$

In each of the four cases the two indices are non-negative with positive sum, so d' and d'' are derivative symbols in the sense of `DerivSymbol`, and

$$|d'| = |d| + 1, \qquad |d''| = |d| + 1. \tag{3}$$

We claim that for every $\xi \in \Omega$

$$\frac{\partial H^d}{\partial \xi}(\xi) = H^{d'}(\xi), \qquad \frac{\partial H^d}{\partial \bar{\xi}}(\xi) = H^{d''}(\xi). \tag{4}$$

Suppose first that β is false, so $H^d = g_d$ by `SymbolValue`. By `WirtingerIter`, $g_d = (\partial/\partial\xi)^i (\partial/\partial\bar{\xi})^j H$, so applying $\partial/\partial\xi$ once gives $(\partial/\partial\xi)^{i+1}(\partial/\partial\bar{\xi})^j H$, which is $\partial^{i+1+j} H / \partial\xi^{i+1}\partial\bar{\xi}^j$ and hence, by `SymbolValue`, equals $H^{(i+1,j,\mathrm{false})} = H^{d'}$; this is an identity of functions on all of $\mathbb{C}$. For the second identity, part (b) of `WirtingerIterSwap` gives, for $\xi \in \Omega$,

$$\frac{\partial g_d}{\partial \bar{\xi}}(\xi) = \frac{\partial^{i+j+1} H}{\partial \xi^i \partial \bar{\xi}^{j+1}}(\xi) = H^{(i,j+1,\mathrm{false})}(\xi) = H^{d''}(\xi),$$

the middle equality by `SymbolValue`.

Suppose now that β is true, so $H^d = \overline{g_d}$ by `SymbolValue`. By Step 1, g_d is real-differentiable at every $\xi \in \Omega$, so `WirtingerConjRule` applies to g_d at each such ξ and gives

$$\frac{\partial \overline{g_d}}{\partial \xi}(\xi) = \overline{\frac{\partial g_d}{\partial \bar{\xi}}(\xi)}, \qquad \frac{\partial \overline{g_d}}{\partial \bar{\xi}}(\xi) = \overline{\frac{\partial g_d}{\partial \xi}(\xi)}.$$

For the first, part (b) of `WirtingerIterSwap` gives $\dfrac{\partial g_d}{\partial \bar{\xi}}(\xi) = \dfrac{\partial^{i+j+1} H}{\partial \xi^i \partial \bar{\xi}^{j+1}}(\xi)$, whence by `SymbolValue`

$$\frac{\partial H^d}{\partial \xi}(\xi) = \overline{\frac{\partial^{i+j+1} H}{\partial \xi^i \partial \bar{\xi}^{j+1}}(\xi)} = H^{(i,j+1,\mathrm{true})}(\xi) = H^{d'}(\xi).$$

For the second, $\partial g_d / \partial \xi = \partial^{i+1+j} H / \partial \xi^{i+1} \partial \bar{\xi}^j$ on all of $\mathbb{C}$ by `WirtingerIter`, as in the case β false, whence by `SymbolValue`

$$\frac{\partial H^d}{\partial \bar{\xi}}(\xi) = \overline{\frac{\partial^{i+1+j} H}{\partial \xi^{i+1} \partial \bar{\xi}^j}(\xi)} = H^{(i+1,j,\mathrm{true})}(\xi) = H^{d''}(\xi).$$

This proves (4) in all cases.

Step 3: Leibniz on a monomial. Let $M = (d_1, \dots, d_k)$ be a list of derivative symbols. For $1 \leq t \leq k$ let

$$M[t] = (d_1, \dots, d_{t-1},\, d_t',\, d_{t+1}, \dots, d_k), \qquad M\langle t \rangle = (d_1, \dots, d_{t-1},\, d_t'',\, d_{t+1}, \dots, d_k)$$

be the lists obtained from M by replacing its t-th entry d_t by d_t', respectively by d_t'', in the notation of Step 2. Then

$$k(M[t]) = k(M), \qquad k(M\langle t\rangle) = k(M), \qquad s(M[t]) = s(M\langle t\rangle) = s(M) + 1. \tag{5}$$

Indeed, by `MonomialValue` $k(\,\cdot\,)$ is the length of the list, and replacing one entry does not change the length; and $s(\,\cdot\,)$ is the sum of the orders of the entries, of which exactly one summand changes, from $|d_t|$ to $|d_t'| = |d_t| + 1$ respectively $|d_t''| = |d_t| + 1$, by (3).

We claim that for every $\xi \in \Omega$

$$\frac{\partial H^M}{\partial \xi}(\xi) = \sum_{t=1}^{k} H^{M[t]}(\xi), \qquad \frac{\partial H^M}{\partial \bar{\xi}}(\xi) = \sum_{t=1}^{k} H^{M\langle t\rangle}(\xi), \tag{6}$$

and prove both by induction on the length k of M, the two arguments being word for word the same: the first uses the first identity of (4), the first identity of `WirtingerProductRule` and the operator $\partial/\partial\xi$ of `DZ`, and the second uses the second identity of (4), the second identity of `WirtingerProductRule` and the operator $\partial/\partial\bar{\xi}$ of `DZbar`. We write out the first.

If $k = 0$, then H^M is the constant function 1 by `MonomialValue`. A constant function is real-differentiable with zero derivative at every point, so by `DZ` $\partial H^M/\partial\xi = 0$, which is the empty sum on the right of (6).

Let $k \geq 1$ and write $M = (d_1) * M''$ with $M'' = (d_2, \ldots, d_k)$, where $*$ denotes concatenation. By `MonomialValue`, $H^M = H^{d_1} \cdot H^{M''}$ pointwise. By Step 1 every symbol value is real-differentiable at every point of Ω; hence so is H^{d_1}, and so is the finite pointwise product $H^{M''}$. Therefore `WirtingerProductRule` applies at each $\xi \in \Omega$ and gives

$$\frac{\partial H^M}{\partial \xi}(\xi) = \frac{\partial H^{d_1}}{\partial \xi}(\xi)\, H^{M''}(\xi) + H^{d_1}(\xi)\, \frac{\partial H^{M''}}{\partial \xi}(\xi).$$

By the first identity of (4), the first term equals $H^{d_1'}(\xi) H^{M''}(\xi)$, which by `MonomialValue` is $H^{M[1]}(\xi)$, because $M[1] = (d_1') * M''$ and the monomial of a concatenation is the product of the two monomials. By the induction hypothesis applied to M'', whose length is $k - 1$,

$$H^{d_1}(\xi)\, \frac{\partial H^{M''}}{\partial \xi}(\xi) = H^{d_1}(\xi) \sum_{t=2}^{k} H^{\,M''[t-1]}(\xi) = \sum_{t=2}^{k} H^{M[t]}(\xi),$$

again by `MonomialValue`, since $M[t] = (d_1) * M''[t-1]$ for $2 \leq t \leq k$. Adding the two contributions gives the first identity of (6).

Step 4: assembling Q_1 and Q_2. Define, for $\xi \in \mathbb{C}$,

$$Q_1(\xi) = \sum_{\alpha=1}^{r} \sum_{t=1}^{k} c_\alpha\, H^{M_\alpha[t]}(\xi), \qquad Q_2(\xi) = \sum_{\alpha=1}^{r} \sum_{t=1}^{k} c_\alpha\, H^{M_\alpha\langle t\rangle}(\xi).$$

By (1) and (5), each of the rk lists $M_\alpha[t]$ and each of the rk lists $M_\alpha\langle t\rangle$ has $s(\,\cdot\,) = s + 1$ and $k(\,\cdot\,) = k$, i.e. type $(s+1, k)$. Hence the coefficients c_α together with these lists present Q_1, respectively Q_2, as a finite $\mathbb{C}$-linear combination of monomials of type $(s+1, k)$ at every point of $\mathbb{C}$, so $Q_1, Q_2 \in \mathbb{P}(s+1, k)$ relative to H by `PClass`.

Finally let $\xi \in \Omega$. By (2) the function P agrees at every point of $\mathbb{C}$ with $\zeta \mapsto \sum_\alpha c_\alpha H^{M_\alpha}(\zeta)$, so the two have the same real Fréchet derivative at ξ and hence, by `DZ` and `DZbar`, the same $\partial/\partial\xi$ and $\partial/\partial\bar{\xi}$ there. Each H^{M_α} is real-differentiable at ξ, being a finite product of functions differentiable there by Step 1. The real Fréchet derivative is additive on functions differentiable at ξ and satisfies $D(cf)(\xi) = c\, Df(\xi)$ for a constant $c \in \mathbb{C}$; since by `DZ` and `DZbar` each of $\partial/\partial\xi$ and $\partial/\partial\bar{\xi}$ at ξ is a fixed $\mathbb{C}$-linear expression in the values of that derivative at the two vectors 1 and i, both operators are likewise additive at ξ and commute with multiplication by constants. Therefore, using (6) applied to each M_α, which is legitimate because $k(M_\alpha) = k$ by (1),

$$\frac{\partial P}{\partial \xi}(\xi) = \sum_{\alpha=1}^{r} c_\alpha \frac{\partial H^{M_\alpha}}{\partial \xi}(\xi) = \sum_{\alpha=1}^{r} c_\alpha \sum_{t=1}^{k} H^{M_\alpha[t]}(\xi) = Q_1(\xi),$$

and in the same way

$$\frac{\partial P}{\partial \bar{\xi}}(\xi) = \sum_{\alpha=1}^{r} c_\alpha \sum_{t=1}^{k} H^{M_\alpha\langle t\rangle}(\xi) = Q_2(\xi).$$

□

2.55. `PClassConjClosed`.

Lemma 2.55. *Let $H\colon \mathbb{C} \to \mathbb{C}$, let $s, k \geq 0$ be integers and let $P \in \mathbb{P}(s,k)$ relative to H in the sense of `PClass`. Then the pointwise complex conjugate $\bar{P}$, defined by $\bar{P}(\xi) = \overline{P(\xi)}$, also belongs to $\mathbb{P}(s,k)$ relative to H.*

Proof. By `PClass` fix an integer $r \geq 0$, complex numbers $c_1, \dots, c_r$ and lists $M_1, \dots, M_r$ of derivative symbols with

$$s(M_\alpha) = s, \qquad k(M_\alpha) = k \qquad (\alpha = 1, \dots, r), \tag{1}$$

$$P(\xi) = \sum_{\alpha=1}^{r} c_\alpha H^{M_\alpha}(\xi) \qquad \text{for every } \xi \in \mathbb{C}. \tag{2}$$

Step 1: flipping one symbol. For a derivative symbol $d = (i, j, \beta)$ in the sense of `DerivSymbol` put

$$d^\flat = (i,\ j,\ \neg\beta),$$

the symbol with the same two indices and the opposite Boolean. Then $d^\flat$ is again a derivative symbol, since its indices are the same non-negative integers and therefore still satisfy $i + j > 0$, and its order is unchanged:

$$|d^\flat| = i + j = |d|. \tag{3}$$

Write $g_d = \partial^{i+j} H / \partial\xi^i \partial\bar{\xi}^j$ for the iterated derivative of `WirtingerIter`, which depends only on i and j and so is the same for d and $d^\flat$. By `SymbolValue`, $H^d = g_d$ and $H^{d^\flat} = \overline{g_d}$ when β is false, while $H^d = \overline{g_d}$ and $H^{d^\flat} = g_d$ when β is true. Since complex conjugation is an involution, $\overline{\overline{g_d}} = g_d$, and in both cases

$$\overline{H^d(\xi)} = H^{d^\flat}(\xi) \qquad \text{for every } \xi \in \mathbb{C}. \tag{4}$$

Step 2: flipping a monomial. For a list $M = (d_1, \dots, d_k)$ of derivative symbols put $M^\flat = (d_1^\flat, \dots, d_k^\flat)$. By `MonomialValue`, $k(\,\cdot\,)$ is the length of a list and $s(\,\cdot\,)$ is the sum of the orders of its entries; flipping every entry changes neither the length nor, by (3), any of the orders, so

$$k(M^\flat) = k(M), \qquad s(M^\flat) = s(M). \tag{5}$$

Moreover H^M is by `MonomialValue` the pointwise product of the symbol values over the entries of M, and complex conjugation is multiplicative, so by (4)

$$\overline{H^M(\xi)} = \overline{\prod_{t=1}^{k} H^{d_t}(\xi)} = \prod_{t=1}^{k} \overline{H^{d_t}(\xi)} = \prod_{t=1}^{k} H^{d_t^\flat}(\xi) = H^{M^\flat}(\xi) \tag{6}$$

for every $\xi \in \mathbb{C}$. For the empty list both sides of (6) are the constant 1, since $\bar{1} = 1$.

Step 3: conclusion. Complex conjugation is additive and satisfies $\overline{cw} = \bar{c}\,\bar{w}$. Hence, for every $\xi \in \mathbb{C}$, by (2) and (6),

$$\overline{P(\xi)} = \overline{\sum_{\alpha=1}^{r} c_\alpha H^{M_\alpha}(\xi)} = \sum_{\alpha=1}^{r} \overline{c_\alpha}\ \overline{H^{M_\alpha}(\xi)} = \sum_{\alpha=1}^{r} \overline{c_\alpha}\, H^{M_\alpha^\flat}(\xi).$$

By (1) and (5) each list $M_\alpha^\flat$ satisfies $s(M_\alpha^\flat) = s$ and $k(M_\alpha^\flat) = k$, i.e. has type (s,k). The complex numbers $\overline{c_\alpha}$ together with the lists $M_\alpha^\flat$ therefore present $\bar{P}$ as a finite $\mathbb{C}$-linear combination of monomials of type (s,k), valid at every point of $\mathbb{C}$. By `PClass` this says exactly that $\bar{P} \in \mathbb{P}(s,k)$ relative to H. □

2.56. `PClassSmooth`.

Lemma 2.56. *Let $\Omega \subset \mathbb{C}$ be open, let $H\colon \mathbb{C} \to \mathbb{C}$ be C^∞ on Ω, let $s, k \geq 0$ be integers and let $P \in \mathbb{P}(s,k)$ relative to H in the sense of `PClass`. Then P is C^∞ on Ω.*

Proof. By `PClass` fix an integer $r \geq 0$, complex numbers $c_1, \dots, c_r$ and lists $M_1, \dots, M_r$ of derivative symbols, each of type (s,k), with

$$P(\xi) = \sum_{\alpha=1}^{r} c_\alpha H^{M_\alpha}(\xi) \qquad \text{for every } \xi \in \mathbb{C}. \tag{1}$$

Step 1: every symbol value is C^∞ on Ω. Let $d = (i, j, \beta)$ be a derivative symbol and put $g_d = \partial^{i+j} H / \partial \xi^i \partial \bar{\xi}^j$, the iterated derivative of `WirtingerIter`. By part (a) of `WirtingerIterSwap`, g_d is C^∞ on Ω. By `SymbolValue`, H^d is either g_d or its pointwise complex conjugate $\overline{g_d}$. Complex conjugation is an $\mathbb{R}$-linear isometry of $\mathbb{C}$, so composing a C^∞ function on Ω with it again yields a C^∞ function on Ω. Hence H^d is C^∞ on Ω in either case.

Step 2: every monomial is C^∞ on Ω. By `MonomialValue`, H^{M_α} is the pointwise product of the symbol values H^d over the finitely many entries d of M_α, and is the constant function 1 when M_α is empty. A finite pointwise product of C^∞ functions on Ω is C^∞ on Ω, and constants are C^∞; so H^{M_α} is C^∞ on Ω for each $\alpha = 1, \dots, r$ by Step 1.

Step 3: conclusion. By (1), P agrees at every point of $\mathbb{C}$, hence in particular at every point of Ω, with the function $\xi \mapsto \sum_\alpha c_\alpha H^{M_\alpha}(\xi)$. A finite $\mathbb{C}$-linear combination of C^∞ functions on Ω is C^∞ on Ω, so that function is C^∞ on Ω by Step 2, and therefore so is P. □

2.57. `JacInPClass`.

Lemma 2.57. *Let $\Omega \subset \mathbb{C}$ be open and let $H\colon \mathbb{C} \to \mathbb{C}$ be C^∞ on Ω. Write*

$$H_\xi = \frac{\partial H}{\partial \xi}, \quad H_{\bar\xi} = \frac{\partial H}{\partial \bar\xi}, \quad H_{\xi\xi} = \frac{\partial^2 H}{\partial \xi^2}, \quad H_{\xi\bar\xi} = \frac{\partial^2 H}{\partial \xi \partial \bar\xi}, \quad H_{\bar\xi\bar\xi} = \frac{\partial^2 H}{\partial \bar\xi^2}$$

for the iterated derivatives of `WirtingerIter`, and let $J = |H_\xi|^2 - |H_{\bar\xi}|^2$ be the Jacobian of `Jac`, regarded as a complex-valued function. Then, relative to H and in the sense of `PClass`,

(a) *$J \in \mathbb{P}(2,2)$;*

(b) *the function*

$$R = H_{\xi\xi}\overline{H_\xi} + \overline{H_{\xi\bar\xi}} H_\xi - H_{\xi\bar\xi}\overline{H_{\bar\xi}} - \overline{H_{\bar\xi\bar\xi}} H_{\bar\xi}$$

belongs to $\mathbb{P}(3,2)$; and

(c) *$\dfrac{\partial J}{\partial \xi}(\xi) = R(\xi)$ for every $\xi \in \Omega$.*

Proof. By `WirtingerIter`, H_ξ is the iterated derivative with indices $(1,0)$ and $H_{\bar\xi}$ the one with indices $(0,1)$, while $H_{\xi\xi}$, $H_{\xi\bar\xi}$, $H_{\bar\xi\bar\xi}$ are those with indices $(2,0)$, $(1,1)$, $(0,2)$ respectively.

Step 1: J as a combination of symbol values. For any $w \in \mathbb{C}$ one has $|w|^2 = w\bar w$ as complex numbers. Hence, by `Jac`, for every $\xi \in \mathbb{C}$

$$J(\xi) = |H_\xi(\xi)|^2 - |H_{\bar\xi}(\xi)|^2 = H_\xi(\xi)\overline{H_\xi(\xi)} - H_{\bar\xi}(\xi)\overline{H_{\bar\xi}(\xi)}. \tag{1}$$

This is the expression $J = H_\xi \overline{H_\xi} - H_{\bar\xi}\overline{H_{\bar\xi}}$ displayed in Section 5 of the paper.

Step 2: proof of (a). Consider the derivative symbols

$$e_1 = (1, 0, \text{false}), \quad e_1^c = (1, 0, \text{true}), \quad e_2 = (0, 1, \text{false}), \quad e_2^c = (0, 1, \text{true}),$$

each of order 1, and the two lists

$$M = (e_1, e_1^c), \qquad N = (e_2, e_2^c).$$

By `SymbolValue`, $H^{e_1} = H_\xi$, $H^{e_1^c} = \overline{H_\xi}$, $H^{e_2} = H_{\bar\xi}$, $H^{e_2^c} = \overline{H_{\bar\xi}}$, and by `MonomialValue`

$$H^M = H_\xi \overline{H_\xi}, \qquad H^N = H_{\bar\xi}\overline{H_{\bar\xi}}.$$

Each of M and N has length 2, and the orders of its two entries are 1 and 1, so $s(M) = s(N) = 2$ and $k(M) = k(N) = 2$: both lists have type $(2,2)$. By (1), $J = 1 \cdot H^M + (-1) \cdot H^N$ at every

point of $\mathbb{C}$. The coefficients $1, -1$ together with the lists M, N therefore present J as a finite $\mathbb{C}$-linear combination of monomials of type $(2,2)$, so $J \in \mathbb{P}(2,2)$ by `PClass`.

Step 3: proof of (b). Consider the four lists

$$A_1 = \big((2,0,\text{false}),(1,0,\text{true})\big), \qquad A_2 = \big((1,1,\text{true}),(1,0,\text{false})\big),$$

$$A_3 = \big((1,1,\text{false}),(0,1,\text{true})\big), \qquad A_4 = \big((0,2,\text{true}),(0,1,\text{false})\big),$$

with coefficients $1, 1, -1, -1$ respectively. Each A_μ has length 2, and the orders of its two entries are 2 and 1; hence $s(A_\mu) = 3$ and $k(A_\mu) = 2$, so each has type $(3,2)$. By `SymbolValue` and `MonomialValue`,

$$H^{A_1} = H_{\xi\xi}\overline{H_\xi}, \quad H^{A_2} = \overline{H_{\xi\bar\xi}}H_\xi, \quad H^{A_3} = H_{\xi\bar\xi}\overline{H_{\bar\xi}}, \quad H^{A_4} = \overline{H_{\bar\xi\bar\xi}}H_{\bar\xi},$$

so $R = 1\cdot H^{A_1} + 1\cdot H^{A_2} + (-1)\cdot H^{A_3} + (-1)\cdot H^{A_4}$ at every point of $\mathbb{C}$. By `PClass`, $R \in \mathbb{P}(3,2)$.

Step 4: differentiability of the four first-order factors. By part (a) of `WirtingerIterSwap`, each of H_ξ and $H_{\bar\xi}$ is C^∞ on Ω; in particular each is real-differentiable at every point of Ω. Complex conjugation is an $\mathbb{R}$-linear isometry of $\mathbb{C}$, so $\overline{H_\xi}$ and $\overline{H_{\bar\xi}}$ are real-differentiable at every point of Ω as well.

Step 5: the two products. Let $\xi \in \Omega$. By Step 4 and `WirtingerProductRule`,

$$\frac{\partial\big(H_\xi\overline{H_\xi}\big)}{\partial\xi}(\xi) = \frac{\partial H_\xi}{\partial\xi}(\xi)\,\overline{H_\xi(\xi)} + H_\xi(\xi)\,\frac{\partial\overline{H_\xi}}{\partial\xi}(\xi).$$

Here $\partial H_\xi/\partial\xi = H_{\xi\xi}$ on all of $\mathbb{C}$, directly by `WirtingerIter`, since applying $\partial/\partial\xi$ to the iterated derivative with indices $(1,0)$ gives the one with indices $(2,0)$. For the second term, `WirtingerConjRule` applies to H_ξ at ξ by Step 4 and gives $\partial\overline{H_\xi}/\partial\xi(\xi) = \overline{\partial H_\xi/\partial\bar\xi(\xi)}$, while part (b) of `WirtingerIterSwap` with indices $i = 1$, $j = 0$ gives $\partial H_\xi/\partial\bar\xi(\xi) = H_{\xi\bar\xi}(\xi)$. Hence

$$\frac{\partial\big(H_\xi\overline{H_\xi}\big)}{\partial\xi}(\xi) = H_{\xi\xi}(\xi)\overline{H_\xi(\xi)} + H_\xi(\xi)\overline{H_{\xi\bar\xi}(\xi)}. \tag{2}$$

In the same way, by Step 4 and `WirtingerProductRule`,

$$\frac{\partial\big(H_{\bar\xi}\overline{H_{\bar\xi}}\big)}{\partial\xi}(\xi) = \frac{\partial H_{\bar\xi}}{\partial\xi}(\xi)\,\overline{H_{\bar\xi}(\xi)} + H_{\bar\xi}(\xi)\,\frac{\partial\overline{H_{\bar\xi}}}{\partial\xi}(\xi).$$

Here $\partial H_{\bar\xi}/\partial\xi = H_{\xi\bar\xi}$ on all of $\mathbb{C}$ by `WirtingerIter`, since applying $\partial/\partial\xi$ to the iterated derivative with indices $(0,1)$ gives the one with indices $(1,1)$; and by `WirtingerConjRule` together with part (b) of `WirtingerIterSwap` with indices $i = 0$, $j = 1$,

$$\frac{\partial\overline{H_{\bar\xi}}}{\partial\xi}(\xi) = \overline{\frac{\partial H_{\bar\xi}}{\partial\bar\xi}(\xi)} = \overline{H_{\bar\xi\bar\xi}(\xi)}.$$

Hence

$$\frac{\partial\big(H_{\bar\xi}\overline{H_{\bar\xi}}\big)}{\partial\xi}(\xi) = H_{\xi\bar\xi}(\xi)\overline{H_{\bar\xi}(\xi)} + H_{\bar\xi}(\xi)\overline{H_{\bar\xi\bar\xi}(\xi)}. \tag{3}$$

Step 6: proof of (c). Let $\xi \in \Omega$. By (1), the function J agrees at every point of $\mathbb{C}$ with $H_\xi\overline{H_\xi} - H_{\bar\xi}\overline{H_{\bar\xi}}$, so the two have the same real Fréchet derivative at ξ and hence, by `DZ`, the same $\partial/\partial\xi$ at ξ. Both products are real-differentiable at ξ by Step 4, and by `DZ` the operator $\partial/\partial\xi$ at ξ is a fixed $\mathbb{C}$-linear expression in the values of the real Fréchet derivative at the vectors 1 and i; since that derivative is additive on functions differentiable at ξ, so is $\partial/\partial\xi$. Therefore, by (2) and (3),

$$\frac{\partial J}{\partial\xi}(\xi) = \frac{\partial\big(H_\xi\overline{H_\xi}\big)}{\partial\xi}(\xi) - \frac{\partial\big(H_{\bar\xi}\overline{H_{\bar\xi}}\big)}{\partial\xi}(\xi) = H_{\xi\xi}(\xi)\overline{H_\xi(\xi)} + \overline{H_{\xi\bar\xi}(\xi)}H_\xi(\xi) - H_{\xi\bar\xi}(\xi)\overline{H_{\bar\xi}(\xi)} - \overline{H_{\bar\xi\bar\xi}(\xi)}H_{\bar\xi}(\xi),$$

which is $R(\xi)$. This is exactly the formula for J_ξ displayed as property (ii) in Section 5 of the paper. □

2.58. `PClassGrowthBound`.

Lemma 2.58. *Let $H\colon \mathbb{C}\to\mathbb{C}$, let $\Omega\subseteq\mathbb{C}$ be open with H of class C^∞ on Ω and $\xi\neq 0$ for every $\xi\in\Omega$, let $g\in\mathbb{R}$, and let C_m, $m=0,1,2,\dots$, be reals such that*

$$\sum_{|w|=m}\left|D^w H(\xi)\right|\le C_m|\xi|^{g-m}\qquad \text{for every } m\ge 0 \text{ and every } \xi\in\Omega,$$

the sum running over the 2^m words w of length m in $\{0,1\}$ and D^w being the iterated partial derivative of `PartialWord`. Let $s,k\ge 0$ be integers and let $P\in\mathbb{P}(s,k)$ relative to H in the sense of `PClass`. Then there is $M\ge 0$ with

$$\left|P(\xi)\right|\le M\,|\xi|^{kg-s}\qquad \text{for every } \xi\in\Omega.$$

This is the growth estimate the paper records at paper.tex:488 for a monomial $Q=H^{s_1}\cdots H^{s_k}$ of type (s,k), namely $|Q(\xi)|\le M|\xi|^{g-s_1}\cdots|\xi|^{g-s_k}=M|\xi|^{kg-s}$, extended to all of $\mathbb{P}(s,k)$ by summing over the finite linear combination that membership in $\mathbb{P}(s,k)$ carries. Applied with $g=\gamma_n$, the hypothesis is estimate (43) of Corollary 1 and the conclusion at $P_\nu\in\mathbb{P}(4l-3,3l-2)$ is the paper's $|P_\nu(\xi)|\le M|\xi|^{(3l-2)\gamma_n-4l+3}$ of paper.tex:492.

Proof. If $\Omega=\emptyset$ the assertion holds with $M=0$, the conclusion being vacuous. So assume $\Omega\neq\emptyset$ and fix $\xi_0\in\Omega$.

Step 1: $C_m\ge 0$ for every m. The left-hand side of the hypothesis at ξ_0 is a sum of moduli, hence ≥ 0, so $0\le C_m|\xi_0|^{g-m}$. By assumption $\xi_0\neq 0$, so $|\xi_0|>0$ and $|\xi_0|^{g-m}>0$; dividing, $C_m\ge 0$.

Step 2: every Wirtinger derivative of order q is controlled by the real ones. We claim: if G is C^∞ on Ω and $T_1,\dots,T_q$ is any list with each $T_a\in\{\partial/\partial\xi,\partial/\partial\bar\xi\}$, then on Ω

$$T_1T_2\cdots T_q\,G=2^{-q}\sum_{|w|=q}\varepsilon_w\,D^wG,\tag{1}$$

for some complex numbers ε_w with $|\varepsilon_w|\le 1$, the sum being over the 2^q words of length q.

We induct on q. For $q=0$ the composite is $G=D^{[]}G$, so (1) holds with the single coefficient 1. For the step, let $T_1\cdots T_qG=T_1\big(T_2\cdots T_qG\big)$ and write $G'=T_2\cdots T_qG$. By `DZ` and `DZbar`,

$$\frac{\partial G'}{\partial\xi}=\tfrac12\big(D^{[0]}G'-i\,D^{[1]}G'\big),\qquad \frac{\partial G'}{\partial\bar\xi}=\tfrac12\big(D^{[0]}G'+i\,D^{[1]}G'\big),$$

the two real partials being the values of the Fréchet derivative on 1 and i, i.e. $D^{[0]}$ and $D^{[1]}$ in the sense of `PartialWord`. By the inductive hypothesis $G'=2^{-(q-1)}\sum_{|w'|=q-1}\varepsilon_{w'}D^{w'}G$ as functions on the open set Ω, where each $D^{w'}G$ is C^∞ on Ω because G is; so ∂_t may be applied term by term, and by the recursion $D^{t::w'}=\partial_tD^{w'}$ of `PartialWord`,

$$D^{[t]}G'=2^{-(q-1)}\sum_{|w'|=q-1}\varepsilon_{w'}D^{t::w'}G\qquad(t=0,1).$$

Substituting into the two displays above expresses T_1G' as $2^{-q}\sum_{|w|=q}\varepsilon_wD^wG$ with each ε_w equal to $\pm\varepsilon_{w'}$ or $\pm i\varepsilon_{w'}$ for the corresponding w', hence of modulus at most 1. This proves (1).

Now let d be a derivative symbol in the sense of `DerivSymbol`, with orders i,j and total order $q=i+j$; note $q\ge 1$, since `DerivSymbol` requires $0<i+j$. By `WirtingerIter` the iterated Wirtinger derivative $\partial^qH/\partial\xi^i\partial\bar\xi^j$ is the composite of j copies of $\partial/\partial\bar\xi$ followed by i copies of $\partial/\partial\xi$ applied to H, which is a composite as in (1) with $G=H$; and by `SymbolValue` the value of d on H is that iterated derivative or its complex conjugate, which has the same modulus. Hence, for $\xi\in\Omega$, by (1) and $|\varepsilon_w|\le 1$,

$$\left|H^d(\xi)\right|\le 2^{-q}\sum_{|w|=q}\left|D^wH(\xi)\right|\le\sum_{|w|=q}\left|D^wH(\xi)\right|\le C_q|\xi|^{g-q},$$

writing H^d for the value of d on H, using $2^{-q}\le 1$ and all terms nonnegative, and applying the hypothesis with $m=q$. So

$$\left|H^d(\xi)\right|\le C_{q_d}\,|\xi|^{g-q_d}\qquad(\xi\in\Omega),\tag{2}$$

where $q_d \geq 1$ is the order of d.

Step 3: the monomial bound. Let $\mathcal{M} = [d_1, \dots, d_k]$ be a list of derivative symbols with $\sum_{a=1}^{k} q_{d_a} = s$, i.e. a monomial of type (s,k), and let $\xi \in \Omega$. By `MonomialValue` its value is the product $\prod_{a=1}^{k} H^{d_a}(\xi)$, so by (2), all factors and all bounds being nonnegative by Step 1,

$$\big|Q_{\mathcal{M}}(\xi)\big| = \prod_{a=1}^{k} \big|H^{d_a}(\xi)\big| \leq \prod_{a=1}^{k} C_{q_{d_a}} |\xi|^{g-q_{d_a}} = \Big(\prod_{a=1}^{k} C_{q_{d_a}}\Big)\, |\xi|^{\sum_a (g-q_{d_a})} = \Big(\prod_{a=1}^{k} C_{q_{d_a}}\Big) |\xi|^{kg-s},$$

where the powers of the positive real $|\xi|$ were combined by $|\xi|^{\alpha}|\xi|^{\beta} = |\xi|^{\alpha+\beta}$, legitimate because $|\xi| > 0$, and $\sum_a (g - q_{d_a}) = kg - \sum_a q_{d_a} = kg - s$. This is exactly the paper's display at paper.tex:488.

Step 4: the linear combination. By `PClass` there is a finite list L of pairs $(c, \mathcal{M})$, with every $\mathcal{M}$ occurring in L of type (s,k) — that is, of length k and with orders summing to s — such that

$$P(\xi) = \sum_{(c,\mathcal{M})\in L} c\, Q_{\mathcal{M}}(\xi) \qquad \text{for every } \xi.$$

Hence, by the triangle inequality and Step 3, for $\xi \in \Omega$,

$$\big|P(\xi)\big| \leq \sum_{(c,\mathcal{M})\in L} |c|\, \big|Q_{\mathcal{M}}(\xi)\big| \leq \Big(\sum_{(c,\mathcal{M})\in L} |c| \prod_{a=1}^{k} C_{q_{d_a(\mathcal{M})}} \Big) |\xi|^{kg-s} = M\, |\xi|^{kg-s},$$

where M denotes the bracketed sum. It is a finite sum of products of nonnegative numbers by Step 1, hence $M \geq 0$. Note that the exponent is the same for every term because `PClass` requires every monomial in the presentation to have the *same* type (s,k). This is the assertion. □

2.59. `ChiHigherDerivBound`.

Lemma 2.59. *Let $n \geq 1$ be an integer, let $1 < p < \infty$ and let $K \geq 1$. Let $V, U \subseteq \mathbb{C}$ be open with $0 \in V$, and let $H, \chi \colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *H is continuous on V and differentiable at almost every point of V;*
(H2) *H is K-quasiconformal on V in the sense of `QuasiconformalOn`, and $H(0) = 0$;*
(H3) *H solves equation (18) on V in the sense of `SolvesEq18`;*
(H4) *χ is C^∞ on U, $H(\xi) \in U$ for every $\xi \in V - \{0\}$, and $\chi(H(\xi)) = \xi$ for every $\xi \in V$.*

Write γ_n for the exponent of `GammaExp` and D^w for the iterated partial derivative of `PartialWord`. Then there is $\delta > 0$ with $\{z : |z| < \delta\} \subseteq H(V)$ and there are constants $B_l \geq 0$, $l = 0, 1, 2, \dots$, such that

$$\big|D^w \chi(z)\big| \leq B_l\, |z|^{1/\gamma_n - l} \qquad \text{for every word } w \text{ of length } l \text{ and every } 0 < |z| < \delta.$$

This is the estimate the paper reaches at paper.tex:506 in the proof of Corollary 2, with the case $l = 0$ included as the paper notes at paper.tex:502 ("This also holds true for $l = 0$ since $\chi(z) = \xi$"). No smoothness of H is assumed: `Cor1` produces it, on the ball of its own radius ρ, and since the conclusion here is existential in δ the proof simply works inside that ball.

Proof. Write $\gamma = \gamma_n$ and $J(\xi) = \operatorname{Jac} H(\xi)$ in the sense of `Jac`.

Step 0: $\gamma > 0$. By part (a) of `GammaRatioMinAtOne`, using $n \geq 1$ and $1 < p$, we have $\gamma = \gamma_n > 0$.

Step 1: Corollary 1. The hypotheses of `Cor1` hold with this n, p, K, V and H: V is open with $0 \in V$; H is continuous on V by (H1); H is differentiable at almost every point of V by (H1); H is K-quasiconformal on V by (H2); H solves (18) on V by (H3); and $H(0) = 0$ by (H2). It therefore provides $\rho > 0$, constants $0 < c < C$ and a family C_m with

$$c|\xi|^{\gamma} \leq |H(\xi)| \leq C|\xi|^{\gamma} \qquad (|\xi| \leq \rho), \tag{41}$$

$$c|\xi|^{2\gamma-2} \leq J(\xi) \leq C|\xi|^{2\gamma-2} \qquad (0 < |\xi| \leq \rho), \tag{42}$$

$$\sum_{|w|=m} \big|D^w H(\xi)\big| \leq C_m |\xi|^{\gamma-m} \qquad (m \geq 0,\ 0 < |\xi| \leq \rho), \tag{43}$$

and, as its first conclusion,

$$H \text{ is of class } C^\infty \text{ on } \{\xi : |\xi| < \rho\} - \{0\}. \tag{43′}$$

This is the only source of the derivatives $D^w H$ that (43) speaks of, and it is why no smoothness hypothesis is needed: the radius is ρ, and every estimate below is made inside the ball of radius ρ.

Step 2: the set Ω. Put

$$\Omega = \big\{\xi \in V : 0 < |\xi| \text{ and } |\xi| < \rho\big\} = V \cap \{\xi : |\xi| < \rho\} \cap \big(\mathbb{C} - \{0\}\big).$$

This is open, being an intersection of three open sets. Every $\xi \in \Omega$ satisfies $\xi \neq 0$ and $|\xi| < \rho$, so $\Omega \subseteq \{\xi : |\xi| < \rho\} - \{0\}$ and therefore H is C^∞ on Ω by (43′); a function of class C^∞ on a set is of class C^∞ on every subset of it. Since $|\xi| < \rho$ implies $|\xi| \le \rho$, the estimates (41), (42), (43) all hold at every $\xi \in \Omega$.

Step 3: $J \neq 0$ *on* Ω *— this is where (42)-lower enters.* For $\xi \in \Omega$, by the lower half of (42),

$$J(\xi) \ge c|\xi|^{2\gamma-2} > 0, \tag{1}$$

because $c > 0$ and $|\xi| > 0$. In particular $J(\xi) \neq 0$ on Ω.

Step 4: formula (49). Let $l \ge 1$ and let w be a word of length l. The hypotheses of `Lemma1` hold with the open sets Ω and U, the maps χ and H, the smoothness of H on Ω from Step 2 and of χ on U from (H4), the inclusion $H(\xi) \in U$ for $\xi \in \Omega$ — by (H4), since $\Omega \subseteq V - \{0\}$ — the relation $\chi(H(\xi)) = \xi$ for $\xi \in \Omega$ from (H4), since $\Omega \subseteq V$, and $J \neq 0$ on Ω from Step 3. It therefore provides P_w with $P_w \in \mathbb{P}(4l-3, 3l-2)$ relative to H in the sense of `PClass` and

$$D^w \chi\big(H(\xi)\big) = J(\xi)^{1-2l} P_w(\xi) \qquad (\xi \in \Omega). \tag{49}$$

Note that for $l \ge 1$ the two indices are genuine positive integers, $4l - 3 \ge 1$ and $3l - 2 \ge 1$, so no truncation occurs and they equal $4l-3$ and $3l-2$ as real numbers.

Step 5: the bound on P_w. The hypotheses of `PClassGrowthBound` hold with H, the open set Ω, the smoothness of H on Ω, the fact that every $\xi \in \Omega$ is nonzero, the exponent $g = \gamma$, the family C_m, and the estimate (43) on Ω from Step 2. Applied to $P_w \in \mathbb{P}(4l-3, 3l-2)$ it gives $M_w \ge 0$ with

$$\big|P_w(\xi)\big| \le M_w |\xi|^{(3l-2)\gamma-(4l-3)} \qquad (\xi \in \Omega), \tag{2}$$

which is the paper's display at paper.tex:492.

Step 6: the J^{1-2l} *factor.* Let $\xi \in \Omega$ and $l \ge 1$. Since $l \ge 1$ we have $1 - 2l \le -1 < 0$. The function $t \mapsto t^{1-2l}$ is strictly decreasing on $(0, \infty)$ because its exponent is negative; and $0 < c|\xi|^{2\gamma-2} \le J(\xi)$ by (1). Therefore

$$\big|J(\xi)^{1-2l}\big| = J(\xi)^{1-2l} \le \big(c|\xi|^{2\gamma-2}\big)^{1-2l} = c^{1-2l}\,|\xi|^{(2\gamma-2)(1-2l)}, \tag{3}$$

the first equality because $J(\xi) > 0$. This is the step the paper writes at paper.tex:496; note that although it is displayed as an upper bound on J^{1-2l}, what it consumes is the *lower* bound of (42), precisely because $1 - 2l < 0$.

Step 7: the exponents collapse. Combining (49), (2) and (3), for $\xi \in \Omega$ and $l \ge 1$,

$$\big|D^w \chi(H(\xi))\big| \le c^{1-2l} M_w\,|\xi|^{(2\gamma-2)(1-2l)}\,|\xi|^{(3l-2)\gamma-(4l-3)} = c^{1-2l} M_w\,|\xi|^{(1-2l)(2\gamma-2)+(3l-2)\gamma-4l+3},$$

the powers of the positive real $|\xi|$ combining additively. The exponent is

$$(1-2l)(2\gamma-2) + (3l-2)\gamma - 4l + 3 = \big(2\gamma - 2 - 4l\gamma + 4l\big) + \big(3l\gamma - 2\gamma - 4l + 3\big) = 1 - l\gamma,$$

every other term cancelling. Hence

$$\big|D^w \chi(H(\xi))\big| \le c^{1-2l} M_w\,|\xi|^{1-l\gamma} \qquad (\xi \in \Omega,\ |w| = l \ge 1), \tag{4}$$

which is the paper's display at paper.tex:500.

Step 8: the case $l = 0$. The only word of length 0 is the empty word, and $D^{[]}\chi = \chi$ by `PartialWord`. For $\xi \in \Omega$ we have $\chi(H(\xi)) = \xi$ by (H4), so

$$\big|D^{[]}\chi(H(\xi))\big| = |\xi| = |\xi|^{1-0\cdot\gamma}, \tag{5}$$

so (4) holds for $l = 0$ as well, with the constant 1 in place of $c^{1-2l}M_w$. This is the paper's remark at paper.tex:502; it must be made separately, because (49) is stated only for $l \geq 1$ and the indices $4l-3$, $3l-2$ are not meaningful at $l = 0$.

Step 9: change of variable — this is where both halves of (41) enter. Let $\xi \in \Omega$ and put $z = H(\xi)$. By (41), since $|\xi| \leq \rho$,

$$c|\xi|^\gamma \leq |z| \leq C|\xi|^\gamma. \tag{6}$$

In particular $|z| \geq c|\xi|^\gamma > 0$, so $z \neq 0$. Since $\gamma > 0$, the map $t \mapsto t^{1/\gamma}$ is increasing on $[0, \infty)$, so (6) gives

$$\Big(\frac{|z|}{C}\Big)^{1/\gamma} \leq |\xi| \leq \Big(\frac{|z|}{c}\Big)^{1/\gamma}. \tag{7}$$

Fix $l \geq 0$ and put $e = 1 - l\gamma$. We bound $|\xi|^e$ by a multiple of $|z|^{e/\gamma}$, distinguishing the sign of e, which can be either.

If $e \geq 0$ then $t \mapsto t^e$ is nondecreasing, so by the upper bound in (7),

$$|\xi|^e \leq \Big(\big(|z|/c\big)^{1/\gamma}\Big)^e = c^{-e/\gamma}|z|^{e/\gamma};$$

this uses the *lower* half of (41), which is what bounds $|\xi|$ above.

If $e < 0$ then $t \mapsto t^e$ is decreasing on $(0, \infty)$, so by the lower bound in (7), whose left side is positive since $|z| > 0$,

$$|\xi|^e \leq \Big(\big(|z|/C\big)^{1/\gamma}\Big)^e = C^{-e/\gamma}|z|^{e/\gamma};$$

this uses the *upper* half of (41), which is what bounds $|\xi|$ below.

In either case, with $\kappa_l = \max\{c^{-e/\gamma}, C^{-e/\gamma}\} > 0$,

$$|\xi|^{1-l\gamma} \leq \kappa_l\, |z|^{1/\gamma - l}, \tag{8}$$

because $e/\gamma = (1 - l\gamma)/\gamma = 1/\gamma - l$, the division being legitimate since $\gamma \neq 0$ by Step 0.

Step 10: the radius δ. Put $V_2 = \{\xi \in V : |\xi| < \rho\}$, an open set containing 0, since $0 \in V$ and $|0| = 0 < \rho$. By `QuasiconformalOn` applied to H on V there is ψ with $\psi(H(\xi)) = \xi$ for every $\xi \in V$ and ψ continuous on $H(V)$, and $H(V)$ is open. We claim

$$H(V_2) = \psi^{-1}(V_2) \cap H(V). \tag{9}$$

If $w = H(\xi)$ with $\xi \in V_2 \subseteq V$, then $w \in H(V)$ and $\psi(w) = \xi \in V_2$. Conversely if $w \in H(V)$ with $\psi(w) \in V_2$, write $w = H(\xi)$ with $\xi \in V$; then $\psi(w) = \xi$, so $\xi \in V_2$ and $w \in H(V_2)$. The right side of (9) is open in the subspace $H(V)$ by continuity of ψ there, that is, of the form $G \cap H(V)$ with $G \subseteq \mathbb{C}$ open; and $H(V)$ is open in $\mathbb{C}$. So $H(V_2)$ is open in $\mathbb{C}$, and it contains $H(0) = 0$ by (H2). Hence there is $\delta > 0$ with

$$\{z : |z| < \delta\} \subseteq H(V_2) \subseteq H(V), \tag{10}$$

which in particular is the asserted inclusion $\{z : |z| < \delta\} \subseteq H(V)$.

Moreover, every z with $0 < |z| < \delta$ is $H(\xi)$ for some $\xi \in \Omega$: by (10), $z = H(\xi)$ with $\xi \in V_2$, i.e. $\xi \in V$ and $|\xi| < \rho$; and $\xi \neq 0$, since $\xi = 0$ would give $z = H(0) = 0$, contradicting $|z| > 0$. So $\xi \in \Omega$.

Step 11: the constants. For each $l \geq 0$ there are only 2^l words of length l, so

$$M_l^* = \max\big\{M_w : |w| = l\big\}$$

is a maximum over a finite nonempty set, and $M_l^* \geq 0$; for $l = 0$ put $M_0^* = 1$. Define

$$B_l = \kappa_l \cdot \max\big\{1,\ c^{1-2l}M_l^*\big\} \quad (l \geq 1), \qquad B_0 = \kappa_0.$$

Then $B_l \geq 0$ for every l, since $\kappa_l > 0$ and the other factor is positive.

Let $l \geq 0$, let w be a word of length l, and let z satisfy $0 < |z| < \delta$. By Step 10 there is $\xi \in \Omega$ with $H(\xi) = z$. If $l \geq 1$ then by (4) and (8),

$$\big|D^w\chi(z)\big| = \big|D^w\chi(H(\xi))\big| \leq c^{1-2l}M_w\, |\xi|^{1-l\gamma} \leq c^{1-2l}M_l^*\, \kappa_l\, |z|^{1/\gamma - l} \leq B_l\, |z|^{1/\gamma - l},$$

all factors being nonnegative. If $l = 0$ then by (5) and (8),

$$\left|D^w\chi(z)\right| = |\xi| = |\xi|^{1-0\cdot\gamma} \le \kappa_0|z|^{1/\gamma-0} = B_0|z|^{1/\gamma-0}.$$

This is the assertion, with this δ and these B_l. □

2.60. `ChiSmoothOffZero`.

Lemma 2.60. *Let $m \ge 1$ be an integer, let $U \subseteq \mathbb{C}$ be open and let $f, \chi\colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *χ is continuous on U;*
(H2) *$f(z) = [\chi(z)]^m$ for every $z \in U$;*
(H3) *f is C^∞ on U;*
(H4) *$f(z) \ne 0$ for every $z \in U$.*

Then χ is C^∞ on U.

This is the smoothness of the quasiconformal factor of the factorization (15) away from the singular point. The paper differentiates χ at paper.tex:170–174, where it passes from (12) to (16), and Lemma 1 differentiates it arbitrarily often, whereas the factorization theorem quoted in item (C2) of the note for the formalization supplies only that χ is a quasiconformal homeomorphism of a neighborhood of the singular point, hence only that χ is continuous. The statement here is what closes that gap: applied on a punctured neighborhood of the singular point, on which (C3) makes f smooth and discreteness of the zero set makes f nonvanishing, it gives $\chi \in C^\infty$ there. The hypotheses are exactly what that application provides, and neither (H1) nor (H4) can be dropped: for $m = 2$, $U = \mathbb{C}$, $f(z) = z\bar z = |z|^2$ and $\chi(z) = |z|$ all of (H1), (H2), (H3) hold while χ fails to be differentiable at $z = 0$, the one point where (H4) fails; and without (H1) an arbitrary pointwise choice of m-th roots of $f(z)$ satisfies (H2) and is in general not even measurable.

Proof. Smoothness on an open set is a local property: if every point of the open set U has an open neighborhood contained in U on which χ is C^∞, then χ is C^∞ on U. So fix $z_1 \in U$; we produce such a neighborhood.

Step 1: a holomorphic m-th root on a disc about $f(z_1)$. Put $w_1 = f(z_1)$, so $w_1 \ne 0$ by (H4), and let

$$D = \{w \in \mathbb{C} : |w - w_1| < |w_1|\},$$

an open disc with $w_1 \in D$. For $w \in D$ we have $|w/w_1 - 1| = |w - w_1|/|w_1| < 1$, and $|\zeta - 1| < 1$ forces $\operatorname{Re}\zeta > 0$: writing $\zeta = s + it$, the inequality $(s-1)^2 + t^2 < 1$ gives $s^2 + t^2 < 2s$, hence $s > 0$. Thus w/w_1 lies in the open right half-plane for every $w \in D$; in particular $0 \notin D$. The principal logarithm Log is holomorphic on the right half-plane and satisfies $e^{\operatorname{Log}\zeta} = \zeta$ there. Choose $\ell_1 \in \mathbb{C}$ with $e^{\ell_1} = w_1$, which is possible because $w_1 \ne 0$, and define

$$\varphi(w) = \exp\Big(\frac{\ell_1 + \operatorname{Log}(w/w_1)}{m}\Big), \qquad w \in D.$$

Then φ is holomorphic on D, being a composition of the holomorphic maps $w \mapsto w/w_1$, Log, an affine map and exp, it does not vanish, being a value of exp, and, using $m \ge 1$,

$$\varphi(w)^m = \exp\big(\ell_1 + \operatorname{Log}(w/w_1)\big) = e^{\ell_1}\,\frac{w}{w_1} = w \qquad (w \in D). \tag{1}$$

Step 2: a ball on which f takes values in D. By (H3) the function f is continuous on U. Since U is open there is $r_0 > 0$ with $\{z : |z - z_1| < r_0\} \subseteq U$, and on that ball continuity of f relative to U is continuity of f at z_1; as D is an open neighborhood of $f(z_1) = w_1$ there is therefore $r \in (0, r_0]$ with

$$B := \{z : |z - z_1| < r\} \subseteq U \qquad \text{and} \qquad f(z) \in D \text{ for every } z \in B.$$

The set B is open and connected, being an open ball, and $z_1 \in B$.

Step 3: a C^∞ nonvanishing m-th root of f on B. Put $\psi = \varphi \circ f$ on B. A holomorphic function on an open subset of $\mathbb{C}$ is C^∞ as a map of the two real variables, and f is C^∞ on $B \subseteq U$ by (H3) with $f(B) \subseteq D$, so ψ is C^∞ on B. Moreover ψ does not vanish on B, φ not vanishing on D, and by (1)

$$\psi(z)^m = \varphi\big(f(z)\big)^m = f(z) \qquad (z \in B). \tag{2}$$

Step 4: the quotient χ/ψ is constant on B. Define $q = \chi/\psi$ on B, which is legitimate since ψ has no zero there. For $z \in B$, (H2) and (2) give $\chi(z)^m = f(z) = \psi(z)^m$, hence

$$q(z)^m = \frac{\chi(z)^m}{\psi(z)^m} = 1 \qquad (z \in B), \tag{3}$$

so q takes its values in the set $\mu_m = \{\omega \in \mathbb{C} : \omega^m = 1\}$. That set is finite with at least one element: it is the zero set in $\mathbb{C}$ of the polynomial $X^m - 1$, which is not the zero polynomial because $m \geq 1$, so it has at most m elements, and $1 \in \mu_m$. The function q is continuous on B: χ is continuous on $U \supseteq B$ by (H1), and ψ is continuous and nonvanishing on B by Step 3.

We claim a continuous function on a connected set with values in a finite subset $S \subseteq \mathbb{C}$ is constant. Let $\rho > 0$ be smaller than $|\omega - \omega'|$ for all distinct $\omega, \omega' \in S$ (any $\rho > 0$ if S has at most one element). For $\omega \in S$ the set $q^{-1}(\{\omega\}) \cap B$ equals $q^{-1}\big(\{w : |w - \omega| < \rho\}\big) \cap B$, because a value of q lies in S and the only point of S within ρ of ω is ω itself; the latter set is open in B by continuity of q. It is also closed in B, being the preimage of the closed set $\{\omega\}$. So each of the finitely many sets $q^{-1}(\{\omega\}) \cap B$, $\omega \in S$, is a clopen subset of the connected set B and hence is empty or all of B; they cover B, which is nonempty, so one of them equals B and q is constant on B. Hence there is ω_0 with $\omega_0^m = 1$ and

$$\chi(z) = \omega_0\, \psi(z) \qquad (z \in B). \tag{4}$$

Step 5: conclusion. By (4) the function χ agrees on B with the constant multiple $\omega_0\psi$ of the C^∞ function ψ, so χ is C^∞ on the open set B, which contains z_1 and is contained in U. Since $z_1 \in U$ was arbitrary, the locality of smoothness recorded at the start gives that χ is C^∞ on U. □

2.61. `HodographEq18`.

Lemma 2.61. *Let $n \geq 1$ be an integer, let $1 < p < \infty$, let $W, V \subseteq \mathbb{C}$ be open, let $z_0 \in W$, and let $f, \chi, H \colon \mathbb{C} \to \mathbb{C}$ satisfy*

- (H1) *$f(z) = [\chi(z)]^n$ for every $z \in W$;*
- (H2) *$\chi(z_0) = 0$ and χ is injective on W;*
- (H3) *χ is C^∞ on $W - \{z_0\}$;*
- (H3′) *H is differentiable at almost every point of V, and $J(\xi) = |H_\xi(\xi)|^2 - |H_{\bar\xi}(\xi)|^2 \neq 0$ for almost every $\xi \in V$, with J as in `Jac`;*
- (H3″) *$H \in W^{1,2}_{\mathrm{loc}}(V)$ in the sense of `MemWLocOn`;*
- (H4) *$H(\xi) \in W$ and $\chi(H(\xi)) = \xi$ for every $\xi \in V$, and $H(0) = z_0$;*
- (H5) *f solves equation (12) on W in the sense of `SolvesEq12`.*

Then H solves equation (18) on V in the sense of `SolvesEq18`.

This is the hodograph step of Section 3 of the paper: (H1) is the factorization (15), and the conclusion is the paper's assertion at paper.tex:179–181 that "from (16) we see that H satisfies the linear equation (18)". It is what connects the f-side of the development, where (12) is the hypothesis, to the H-side, where (18) is.

The hypotheses on H are exactly what (17) and the weak form of (18) consume: differentiability almost everywhere, $J \neq 0$ almost everywhere, and membership in $W^{1,2}_{\mathrm{loc}}(V)$. All three are supplied by item (C2) of the paper's note for the inverse of a quasiconformal homeomorphism, whose clause reads "the inverse H of a quasiconformal homeomorphism belongs to $W^{1,2}_{\mathrm{loc}}$, has Jacobian $J > 0$ almost everywhere, and satisfies (17)". Hypothesis (H3″) is what makes the conclusion statable: by `SolvesEq18` solving (18) is an identity between the weak Wirtinger derivatives of H, and those exist exactly because H lies in a local Sobolev space. No smoothness of H is assumed, and none may be: the paper's arrow runs $H \to \chi$ (paper.tex:416), and H's regularity off the origin is produced only later, from (18) itself, so assuming it here would invert the development.

Proof. Write $c' = \frac{1}{p} - \frac{1}{2}$ and $c = \frac{1}{2} - \frac{1}{p} = -c'$.

Step 1: χ and f do not vanish on $W - \{z_0\}$. Let $z \in W$ with $z \neq z_0$. If $\chi(z) = 0$ then $\chi(z) = \chi(z_0)$ with $z, z_0 \in W$ by (H2), so injectivity gives $z = z_0$, a contradiction. Hence

$\chi(z) \neq 0$, and by (H1) and $n \geq 1$,

$$f(z) = [\chi(z)]^n \neq 0 \qquad (z \in W - \{z_0\}). \tag{1}$$

Step 2: (12) holds at every *point of* $W - \{z_0\}$. The set $W - \{z_0\}$ is open, W being open. On it f agrees with $z \mapsto \chi(z)^n$ by (H1), and χ is C^∞ there by (H3), so f is C^∞ on $W - \{z_0\}$. Consequently $z \mapsto \mathrm{d}f(z)$ is continuous on $W - \{z_0\}$, hence so are f_z and $f_{\bar z}$, these being the fixed linear combinations $\frac{1}{2}(\mathrm{d}f[1] \mp i\,\mathrm{d}f[i])$ of `DZ` and `DZbar`; and f is continuous and nonvanishing there by (1). Therefore both sides of

$$f_{\bar z}(z) = c'\Big[\frac{\overline{f(z)}}{f(z)} f_z(z) + \frac{f(z)}{f(z)}\overline{f_z(z)}\Big] \tag{12}$$

are continuous functions of z on $W - \{z_0\}$.

By (H5) and `SolvesEq12` there are weak Wirtinger derivatives a_f, b_f of f on W, in the sense of `IsWeakWirtingerOn`, and a Lebesgue null set Z_0 such that

$$b_f(z) = c'\Big[\frac{\overline{f(z)}}{f(z)} a_f(z) + \frac{f(z)}{f(z)}\overline{a_f(z)}\Big] \tag{12$'$}$$

at every $z \in W - Z_0$ with $f(z) \neq 0$. We pass from a_f, b_f to the pointwise derivatives $f_z, f_{\bar z}$. By `IsWeakWirtingerOn` there are f_x, f_y with $f, f_x, f_y \in L^1_{\mathrm{loc}}(W)$, with f_x and f_y the distributional ∂_0- and ∂_1-derivatives of f on W, and with $a_f = \frac{1}{2}(f_x - i f_y)$, $b_f = \frac{1}{2}(f_x + i f_y)$ at every point. These are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set W, the function f and the pair $D_0 = f_x$, $D_1 = f_y$; and its (H3) holds, because f is C^∞ on $W - \{z_0\}$ as just shown and hence differentiable at every point of W except possibly the single point z_0, which is a null set. That lemma therefore gives a null set Z_1 with $f_x = \partial_0 f$ and $f_y = \partial_1 f$ on $W - Z_1$, where $\partial_j = D^{[j]}$ is the classical partial derivative of `PartialWord`, that is $\partial_0 f(z) = \mathrm{d}f(z)[1]$ and $\partial_1 f(z) = \mathrm{d}f(z)[i]$. Since f_z and $f_{\bar z}$ are by `DZ` and `DZbar` the same two combinations $\frac{1}{2}\big(\mathrm{d}f(z)[1] \mp i\,\mathrm{d}f(z)[i]\big)$ of those numbers,

$$a_f(z) = f_z(z), \qquad b_f(z) = f_{\bar z}(z) \qquad (z \in W - Z_1). \tag{12$''$}$$

Put $Z = Z_0 \cup Z_1$, a null set. Substituting (12″) into (12′), the identity (12) holds at every $z \in W - Z$ with $f(z) \neq 0$; by (1) the last condition is automatic on $W - \{z_0\}$. So (12) holds on $\big(W - \{z_0\}\big) - Z$. A null set has empty interior, so its complement meets every nonempty open subset of $W - \{z_0\}$; that is, $\big(W - \{z_0\}\big) - Z$ is dense in $W - \{z_0\}$. Two continuous functions on an open set that agree on a dense subset of it agree on all of it. Hence (12) holds at every $z \in W - \{z_0\}$.

Step 3: equation (16) at every $z \in W - \{z_0\}$. Fix $z \in W - \{z_0\}$. On the open set $W - \{z_0\}$ the map χ is C^∞ and nonvanishing by Step 1, so `WirtingerPowerRule` applies to it there with the integer exponent n and gives

$$\partial_z\big(\chi^n\big)(z) = n\,\chi(z)^{n-1}\chi_z(z), \qquad \partial_{\bar z}\big(\chi^n\big)(z) = n\,\chi(z)^{n-1}\chi_{\bar z}(z). \tag{2}$$

Since f agrees with χ^n on the open set $W - \{z_0\}$, the two sides have the same Wirtinger derivatives at z, so $f_z(z)$ and $f_{\bar z}(z)$ are given by the right-hand sides of (2). Also $\overline{f(z)} = \overline{\chi(z)}^{\,n}$ and $\overline{f_z(z)} = n\,\overline{\chi(z)}^{\,n-1}\overline{\chi_z(z)}$. Substituting all of this into (12),

$$n\chi^{n-1}\chi_{\bar z} = c'\Big[\frac{\bar\chi^{\,n}}{\chi^n}\, n\chi^{n-1}\chi_z + \frac{\chi^n}{\bar\chi^{\,n}}\, n\bar\chi^{\,n-1}\overline{\chi_z}\Big],$$

all arguments being z. Dividing by $n\chi(z)^{n-1} \neq 0$, which is legitimate by Step 1 and $n \geq 1$, and simplifying $\dfrac{\chi^n}{\bar\chi^{\,n}} \cdot \dfrac{\bar\chi^{\,n-1}}{\chi^{n-1}} = \dfrac{\chi}{\bar\chi}$, we obtain

$$\chi_{\bar z}(z) = c'\Big[\frac{\overline{\chi(z)}^{\,n}}{\chi(z)^n}\chi_z(z) + \frac{\chi(z)}{\chi(z)}\,\overline{\chi_z(z)}\Big] \qquad (z \in W - \{z_0\}), \tag{16}$$

which is equation (16) of the paper.

Step 4: $H(\xi) \in W - \{z_0\}$, and the data (17) needs. Let $\xi \in V$ with $\xi \neq 0$. If $H(\xi) = z_0$ then $\xi = \chi(H(\xi)) = \chi(z_0) = 0$ by (H4) and (H2), a contradiction; so, using $H(\xi) \in W$ from (H4),

$$H(\xi) \in W - \{z_0\} \qquad (\xi \in V - \{0\}). \tag{3}$$

In particular χ is C^∞ at $H(\xi)$ by (H3), the set $W - \{z_0\}$ being open, so χ is differentiable at $H(\xi)$. No corresponding pointwise statement about H is available, and none is claimed: (H3′) controls H only almost everywhere, so the next paragraph has to discard a null set.

By (H3′), with J as in `Jac`, there are null sets N_1 and N_2 such that H is differentiable at every $\xi \in V - N_1$ and $J(\xi) \neq 0$ for every $\xi \in V - N_2$. Put $N = N_1 \cup N_2$, again null. Combining these two facts with the differentiability of χ at $H(\xi)$ just obtained, which holds at every $\xi \in V$ with $\xi \neq 0$, we conclude that for every $\xi \in V - N$ with $\xi \neq 0$,

$$H \text{ is differentiable at } \xi, \qquad \chi \text{ is differentiable at } H(\xi), \qquad J(\xi) \neq 0. \tag{4}$$

Step 5: equation (18) off a null set. Fix $\xi \in V - N$ with $\xi \neq 0$ and write $z = H(\xi)$, $J = J(\xi)$. The hypotheses of `ChiFirstDerivs` hold with the open set V, the pair (χ, H), the point $\xi \in V$, the relation $\chi(H(\zeta)) = \zeta$ for $\zeta \in V$ from (H4), differentiability of H at ξ and of χ at $H(\xi)$ from (4), and $J \neq 0$ from (4). It gives

$$\chi_z(z) = \frac{\overline{H_\xi(\xi)}}{J}, \qquad \chi_{\bar z}(z) = -\frac{H_{\bar\xi}(\xi)}{J}. \tag{17}$$

Note $\chi(z) = \chi(H(\xi)) = \xi$ by (H4), and J is real, so $\overline{\chi_z(z)} = \overline{\overline{H_\xi(\xi)}/J} = H_\xi(\xi)/J$. Substituting (17) and $\chi(z) = \xi$ into (16), which applies because $z \in W - \{z_0\}$ by (3),

$$-\frac{H_{\bar\xi}(\xi)}{J} = c'\Big[\frac{\bar\xi^{\,n}}{\xi^n} \cdot \frac{\overline{H_\xi(\xi)}}{J} + \frac{\xi}{\bar\xi} \cdot \frac{H_\xi(\xi)}{J}\Big].$$

Multiplying by $-J \neq 0$ and using $-c' = c = \frac{1}{2} - \frac{1}{p}$,

$$H_{\bar\xi}(\xi) = c\Big[\frac{\xi}{\bar\xi} H_\xi(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{H_\xi(\xi)}\Big],$$

which is the pointwise form of equation (18) at ξ.

Step 6: the weak form of (18), and conclusion. Step 5 establishes the pointwise identity (18), between $H_{\bar\xi}$ and H_ξ, at every $\xi \in V - N$ with $\xi \neq 0$. It remains to read it for the weak Wirtinger derivatives, which is what `SolvesEq18` asks for.

By (H3″) and `MemWLocOn` there is a family (H_w), indexed by the words w in $\{0,1\}$, with $H_() = H$, with $H_w \in L^2_{\mathrm{loc}}(V)$ in the sense of `MemLpLocOn` for every w with $|w| \leq 1$, and with $H_{(j)}$ the distributional ∂_j-derivative of H on V in the sense of `IsWeakPartialOn` for $j = 0, 1$. Put $H_x = H_{(0)}$, $H_y = H_{(1)}$.

Local square integrability implies local integrability: if $u \in L^2_{\mathrm{loc}}(V)$ and $\kappa \subseteq V$ is compact, then $\mathbf{1}_\kappa u$ is measurable and, by the Cauchy–Schwarz inequality,

$$\int_\kappa |u|\, d\sigma \leq \sigma(\kappa)^{1/2}\Big(\int_\kappa |u|^2\, d\sigma\Big)^{1/2} < \infty,$$

$\sigma(\kappa)$ being finite because κ is bounded; so $u \in L^1_{\mathrm{loc}}(V)$ in the sense of `MemLpLocOn`. Applying this to H, H_x and H_y gives $H, H_x, H_y \in L^1_{\mathrm{loc}}(V)$.

Define $a = \frac{1}{2}(H_x - iH_y)$ and $b = \frac{1}{2}(H_x + iH_y)$ at every point of $\mathbb{C}$. Then a and b are weak Wirtinger derivatives of H on V in the sense of `IsWeakWirtingerOn`: its clause (a) is the local integrability just proved, its clause (b) is the distributional-derivative property of H_x, H_y, and its clause (c) is the definition of a and b.

Those same two clauses are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set V, the function H and the pair $D_0 = H_x$, $D_1 = H_y$, and its (H3) is the first half of (H3′). So there is a null set N_3 with $H_x = \partial_0 H$ and $H_y = \partial_1 H$ on $V - N_3$, with $\partial_j = D^{[j]}$ the classical partial derivative of `PartialWord`; as in Step 2, H_ξ and $H_{\bar\xi}$ are by `DZ` and `DZbar` the same two combinations of $\mathrm{d}H(\xi)[1] = \partial_0 H(\xi)$ and $\mathrm{d}H(\xi)[i] = \partial_1 H(\xi)$, whence

$$a(\xi) = H_\xi(\xi), \qquad b(\xi) = H_{\bar\xi}(\xi) \qquad (\xi \in V - N_3). \tag{5}$$

Combining (5) with Step 5, for every $\xi \in V - (N \cup N_3)$ with $\xi \neq 0$,

$$b(\xi) = c\Big[\frac{\xi}{\bar{\xi}} a(\xi) + \frac{\bar{\xi}^n}{\xi^n}\overline{a(\xi)}\Big].$$

The set $N \cup N_3$ is null, so this holds at almost every $\xi \in V$ with $\xi \neq 0$. With the pair a, b exhibited above, this is precisely the assertion that H solves (18) on V in the sense of `SolvesEq18`. □

2.62. `DExpEqGammaOneInv`.

Lemma 2.62. *Let $1 < p < \infty$. Then $\gamma_1 = \gamma_1(p) > 0$ and*

$$\gamma_1(p) \cdot d(p) = 1,$$

where γ_1 is as in `GammaExp` and d is the exponent (50) of `DExp`. Equivalently, $d(p) = 1/\gamma_1(p)$.

Proof. Fix $p > 1$, so $p - 1 > 0$.

Step 1 (closed form for γ_1). By `GammaExp` and `LambdaCoef` with $n = 1$, $k = 2$,

$$2\gamma_1 = 2\lambda_2(1,p) = -p + \sqrt{4 \cdot 2^2(p-1) + 1^2 \cdot (p-2)^2} = -p + \sqrt{16(p-1) + (p-2)^2},$$

and

$$16(p-1) + (p-2)^2 = 16p - 16 + p^2 - 4p + 4 = p^2 + 12p - 12.$$

Write $S = \sqrt{p^2 + 12p - 12}$, so that

$$2\gamma_1 = -p + S.$$

Step 2 ($S > p > 0$, and $\gamma_1 > 0$). We have

$$\big(p^2 + 12p - 12\big) - p^2 = 12(p-1) > 0,$$

so $p^2 + 12p - 12 > p^2 \geq 0$. In particular the radicand is positive, so $S > 0$ and $S^2 = p^2 + 12p - 12$; and since the square root is strictly increasing on $[0, \infty)$ and $p > 0$,

$$S = \sqrt{p^2 + 12p - 12} > \sqrt{p^2} = p.$$

Hence $2\gamma_1 = -p + S > 0$, i.e. $\gamma_1 > 0$.

Step 3 (closed form for d). Since $p - 1 > 0$,

$$1 + \frac{14}{p-1} + \frac{1}{(p-1)^2} = \frac{(p-1)^2 + 14(p-1) + 1}{(p-1)^2} = \frac{p^2 - 2p + 1 + 14p - 14 + 1}{(p-1)^2} = \frac{p^2 + 12p - 12}{(p-1)^2}.$$

Taking square roots, and using $p - 1 > 0$ together with $\sqrt{X/(p-1)^2} = \sqrt{X}/(p-1)$ for $X \geq 0$,

$$\sqrt{1 + \frac{14}{p-1} + \frac{1}{(p-1)^2}} = \frac{S}{p-1}.$$

Also $1 + \dfrac{1}{p-1} = \dfrac{p}{p-1}$. Hence, by `DExp`,

$$6\, d(p) = 1 + \frac{1}{p-1} + \frac{S}{p-1} = \frac{p+S}{p-1}.$$

Step 4 (the product). Multiplying the formulas of Steps 1 and 3,

$$2\gamma_1 \cdot 6\, d(p) = (S-p) \cdot \frac{p+S}{p-1} = \frac{S^2 - p^2}{p-1} = \frac{\big(p^2 + 12p - 12\big) - p^2}{p-1} = \frac{12(p-1)}{p-1} = 12,$$

using $S^2 = p^2 + 12p - 12$ from Step 2 and $p - 1 \neq 0$. Thus $12\,\gamma_1 d(p) = 12$, and dividing by 12 gives $\gamma_1 d(p) = 1$. □

2.63. `Eq12ImpliesPHarmonic`.

Lemma 2.63. *Let $1 < p < \infty$, let $\Omega \subseteq \mathbb{C}$ be open, let $S \subseteq \Omega$, let $f\colon \mathbb{C} \to \mathbb{C}$ and let $v\colon \mathbb{C} \to \mathbb{R}$. Assume*

(H1) *$S \cap K$ is finite for every compact $K \subseteq \Omega$;*
(H2) *for $z \in \Omega$, $f(z) = 0$ if and only if $z \in S$;*
(H3) *f is continuous on Ω;*
(H4) *f is C^∞ on $\Omega - S$;*
(H5) *f solves equation (12) on Ω in the sense of `SolvesEq12`;*
(H6) *v is continuous on Ω;*
(H7) *v is C^∞ on $\Omega - S$;*
(H8) *$\partial v/\partial z = f$ on $\Omega - S$, with $\partial/\partial z$ as in `DZ`;*
(H9) *v and f belong to $L^p_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`;*
(H10) *both first-order partial derivatives $\partial_j w$, $j \in \{0, 1\}$, of $w = |f|^{p-2}f$ in the sense of `PartialWord` belong to $L^1_{\mathrm{loc}}(\Omega)$.*

Then v is p-harmonic on Ω in the sense of `PHarmonicOn`.

This is the converse of the quoted result (C1), and the paper's note lists it among the facts that must be proved: "the inverse of (54) is the complex gradient of a p-harmonic function on $\mathbb{C}$". The potential is taken as a hypothesis in the paper's own form $\partial v/\partial z = f$ of (55); constructing a potential is a separate matter, which in Section 7 uses the homogeneity of the extremal inverse.

Proof. Write $c' = \frac{1}{p} - \frac{1}{2} = \frac{2-p}{2p}$, and let $\varphi_v(z)$ denote $v(z)$ regarded as a complex number, so that (H8) reads $\partial_z \varphi_v = f$ on $\Omega - S$. Set

$$w(z) = |f(z)|^{p-2} f(z) \qquad (z \in \mathbb{C}),$$

with the convention that the product is 0 when $f(z) = 0$, which it is, the second factor vanishing.

Step 0: $\Omega - S$ is open, and S is a null set. Let $z \in \Omega - S$ and choose $\rho > 0$ with $\overline{B}(z, \rho) \subseteq \Omega$. By (H1) the set $S \cap \overline{B}(z, \rho)$ is finite and does not contain z, so it has a positive distance δ to z; then $B(z, \min(\rho, \delta))$ misses S. Hence $\Omega - S$ is open. Moreover Ω is a countable union of compact subsets of Ω (for instance the closed balls $\overline{B}(q, r) \subseteq \Omega$ with q of rational coordinates and r rational), so by (H1) the set S is countable and therefore Lebesgue null.

Step 1: (12) holds at every point of $\Omega - S$. By (H2), f does not vanish on $\Omega - S$, and by (H4) it is C^∞ there; by Step 0 that set is open. Hence $z \mapsto \mathrm{d}f(z)$ is continuous on $\Omega - S$, so f_z and $f_{\bar z}$, being the fixed combinations $\frac{1}{2}(\mathrm{d}f[1] \mp i\,\mathrm{d}f[i])$ of `DZ` and `DZbar`, are continuous there, and so are $\overline{f}/f$ and $f/\overline{f}$. Both sides of

$$f_{\bar z} = c'\Big[\frac{\overline{f}}{f} f_z + \frac{f}{\overline{f}}\overline{f_z}\Big] \tag{12}$$

are therefore continuous on $\Omega - S$.

By (H5) and `SolvesEq12` there are weak Wirtinger derivatives a_f, b_f of f on Ω, in the sense of `IsWeakWirtingerOn`, and a null set Z_0 such that

$$b_f(z) = c'\Big[\frac{\overline{f(z)}}{f(z)} a_f(z) + \frac{f(z)}{\overline{f(z)}}\overline{a_f(z)}\Big]$$

at every $z \in \Omega - Z_0$ with $f(z) \neq 0$. We replace a_f, b_f by f_z, $f_{\bar z}$. By `IsWeakWirtingerOn` there are f_x, f_y with $f, f_x, f_y \in L^1_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`, with f_x and f_y the distributional ∂_0- and ∂_1-derivatives of f on Ω in the sense of `IsWeakPartialOn`, and with $a_f = \frac{1}{2}(f_x - i f_y)$, $b_f = \frac{1}{2}(f_x + i f_y)$ at every point. These are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set Ω, the function f and the pair $D_0 = f_x$, $D_1 = f_y$; and its (H3) holds, because by (H4) and Step 0 f is differentiable at every point of the open set $\Omega - S$, while S is null by Step 0. So there is a null set Z_1 with $f_x = \partial_0 f$ and $f_y = \partial_1 f$ on $\Omega - Z_1$, where $\partial_j = D^{[j]}$ is the classical partial derivative of `PartialWord`, that is $\partial_0 f(z) = \mathrm{d}f(z)[1]$ and $\partial_1 f(z) = \mathrm{d}f(z)[i]$. Since f_z and $f_{\bar z}$ are by `DZ` and `DZbar` the same two combinations $\frac{1}{2}\big(\mathrm{d}f(z)[1] \mp i\,\mathrm{d}f(z)[i]\big)$, we get $a_f = f_z$ and $b_f = f_{\bar z}$ on $\Omega - Z_1$.

Put $Z = Z_0 \cup Z_1$, a null set. By the last two paragraphs, (12) holds at every $z \notin Z$ with $z \in \Omega$ and $f(z) \neq 0$; on $\Omega - S$ the last condition is automatic by (H2). Thus (12) holds on $(\Omega - S) - Z$, which is dense in $\Omega - S$ because a null set has empty interior, and two continuous functions on an open set agreeing on a dense subset agree everywhere on it. Hence (12) holds at every point of $\Omega - S$.

Step 2: $f_{\bar z}$ is real on $\Omega - S$. Put $a = (\overline{f}/f) f_z$. Then

$$\frac{f}{\overline{f}}\overline{f_z} = \overline{\Big(\frac{\overline{f}}{f} f_z\Big)} = \bar a,$$

so the bracket in (12) is $a + \bar a$, which is real; and c' is real. By Step 1, $f_{\bar z} = c'(a + \bar a)$ at each point of $\Omega - S$, so

$$\overline{f_{\bar z}} = f_{\bar z} \qquad \text{on } \Omega - S. \tag{1}$$

Step 3: w is continuous on Ω and C^∞ on $\Omega - S$. On $\Omega - S$ the function f is C^∞ and non-vanishing, and $\zeta \mapsto |\zeta|^{p-2}$ is C^∞ on $\mathbb{C} - \{0\}$ by the first clause of `WirtingerModulusPowerRule`; hence $|f|^{p-2}$, and with it w, is C^∞ on $\Omega - S$. For continuity on Ω it remains to check the points of S. Where $f \neq 0$ we have $|w| = |f|^{p-2}|f| = |f|^{p-1}$, and $w = 0$ where $f = 0$; so $|w(z)| = |f(z)|^{p-1}$ for every z. If $z_0 \in S$ then $f(z_0) = 0$ by (H2), and f is continuous at z_0 by (H3), so $|w(z)| = |f(z)|^{p-1} \to 0 = |w(z_0)|$ as $z \to z_0$ in Ω, using $p - 1 > 0$. Hence w is continuous on Ω.

Step 4: the formula for $w_{\bar z}$ on $\Omega - S$. Fix $z \in \Omega - S$ and let $F(\zeta) = |\zeta|^{p-2}$. By `WirtingerModulusPowerRule` at the exponent $\mu = p - 2$ and the point $f(z) \neq 0$, F is C^∞, in particular differentiable, at $f(z)$, with

$$F_\zeta(f(z)) = \frac{p-2}{2} \cdot \frac{|f(z)|^{p-2}}{f(z)}, \qquad F_{\bar\zeta}(f(z)) = \frac{p-2}{2} \cdot \frac{|f(z)|^{p-2}}{\overline{f(z)}}.$$

Since f is differentiable at z, `WirtingerChainRule` applies to the pair (F, f) at z and gives

$$\partial_{\bar z}\big(|f|^{p-2}\big) = F_\zeta(f) f_{\bar z} + F_{\bar\zeta}(f)\overline{f_z} = \frac{p-2}{2}|f|^{p-2}\Big[\frac{f_{\bar z}}{f} + \frac{\overline{f_z}}{\overline{f}}\Big],$$

all arguments being z. Both $|f|^{p-2}$ and f are differentiable at z, so `WirtingerProductRule` gives

$$w_{\bar z} = \partial_{\bar z}\big(|f|^{p-2}\big)\, f + |f|^{p-2} f_{\bar z} = |f|^{p-2}\Big[\frac{p-2}{2}\Big(f_{\bar z} + \frac{f}{\overline{f}}\overline{f_z}\Big) + f_{\bar z}\Big] = |f|^{p-2}\Big[\frac{p}{2} f_{\bar z} + \frac{p-2}{2} \cdot \frac{f}{\overline{f}}\overline{f_z}\Big].$$

Multiplying the bracket by $|f|^2/|f|^2$ and using $|f|^2 = f\overline{f}$, so that $(f/\overline{f})|f|^2 = f^2$, we obtain

$$w_{\bar z} = |f|^{p-4} X, \qquad X = \frac{p}{2}|f|^2 f_{\bar z} + \frac{p-2}{2} f^2 \overline{f_z}, \tag{2}$$

at every point of $\Omega - S$.

Step 5: $\operatorname{Re} w_{\bar z} = 0$ on $\Omega - S$. Fix $z \in \Omega - S$. Using (1) and the reality of $|f|^2$,

$$\bar X = \frac{p}{2}|f|^2 f_{\bar z} + \frac{p-2}{2}\overline{f}^2 f_z,$$

so

$$X + \bar X = p|f|^2 f_{\bar z} + \frac{p-2}{2}\big(f^2\overline{f_z} + \overline{f}^2 f_z\big).$$

By Step 1 and $c' = (2-p)/(2p)$,

$$p|f|^2 f_{\bar z} = p\, f\overline{f} \cdot \frac{2-p}{2p}\Big[\frac{\overline{f}}{f} f_z + \frac{f}{\overline{f}}\overline{f_z}\Big] = \frac{2-p}{2}\big(\overline{f}^2 f_z + f^2\overline{f_z}\big),$$

the two cancellations being legitimate because $f(z) \neq 0$. Hence

$$X + \bar X = \frac{2-p}{2}\big(\overline{f}^2 f_z + f^2\overline{f_z}\big) + \frac{p-2}{2}\big(f^2\overline{f_z} + \overline{f}^2 f_z\big) = 0,$$

the two coefficients being exact negatives. Since $X + \bar{X} = 2\operatorname{Re} X$ we get $\operatorname{Re} X = 0$, and by (2), $|f|^{p-4}$ being real,

$$\operatorname{Re} w_{\bar{z}} = 0 \qquad \text{on } \Omega - S. \tag{3}$$

Step 6: the gradient field. Define

$$g(z) = 2\overline{f(z)} \qquad (z \in \mathbb{C}).$$

On $\Omega - S$ we have, by (H8) and DZ, $\frac{1}{2}\big(\partial_x\varphi_v - i\,\partial_y\varphi_v\big) = f$; the function v is real-valued, so $\partial_x\varphi_v$ and $\partial_y\varphi_v$ are real, and comparing real and imaginary parts,

$$\partial_x\varphi_v = 2\operatorname{Re} f = \operatorname{Re} g, \qquad \partial_y\varphi_v = -2\operatorname{Im} f = \operatorname{Im} g \qquad \text{on } \Omega - S. \tag{4}$$

Next, at every $z \in \mathbb{C}$,

$$|g|^{p-2}\operatorname{Re} g = 2^{p-1}\operatorname{Re} w, \qquad |g|^{p-2}\operatorname{Im} g = -2^{p-1}\operatorname{Im} w. \tag{5}$$

Indeed, if $f(z) \neq 0$ then $|g| = 2|f| \neq 0$ and $|g|^{p-2}\operatorname{Re} g = 2^{p-2}|f|^{p-2}\cdot 2\operatorname{Re} f = 2^{p-1}\operatorname{Re}\big(|f|^{p-2}f\big)$, and similarly for the imaginary part; and if $f(z) = 0$ then $g(z) = 0$, so both sides of each identity vanish, the left ones because the factors $\operatorname{Re} g$ and $\operatorname{Im} g$ do.

Step 7: the weak gradient of v on all of Ω. We apply WeakPartialAcrossPoint on Ω with the exceptional set S, which satisfies its local finiteness hypothesis by (H1), and with growth exponent $\beta = 0$. Take first $j = 0$, the function φ_v and the candidate derivative $\operatorname{Re} g$. Its hypotheses hold: φ_v is C^1 on $\Omega - S$ by (H7); its classical x-partial there is $\operatorname{Re} g$ by (4); φ_v is continuous on Ω by (H6) and $\operatorname{Re} g$ is continuous on Ω by (H3), so both are bounded on every compact subset of Ω and hence lie in $L^1_{\mathrm{loc}}(\Omega)$; and for $z_0 \in S$, continuity of φ_v at z_0 gives $M, \rho > 0$ with $|\varphi_v(z)| \le M = M|z - z_0|^{-0}$ for $0 < |z - z_0| < \rho$. The conclusion is that $\operatorname{Re} g$ is the distributional x-partial of φ_v on Ω in the sense of IsWeakPartialOn. Taking $j = 1$ with the candidate $\operatorname{Im} g$ gives in the same way that $\operatorname{Im} g$ is the distributional y-partial of φ_v on Ω.

Step 8: the weak form of the p-harmonic equation. Let $W_1 = \operatorname{Re} w$ and $W_2 = \operatorname{Im} w$, regarded as complex-valued functions. By Step 3 they are continuous on Ω and C^∞ on $\Omega - S$. Apply WeakPartialAcrossPoint on Ω with exceptional set S, $\beta = 0$, first to $j = 0$, the function W_1 and its classical x-partial $\partial_x W_1$ on $\Omega - S$, and then to $j = 1$, the function W_2 and its classical y-partial $\partial_y W_2$. The local integrability of W_1, W_2 follows from their continuity on Ω; the local integrability of $\partial_x W_1$ and $\partial_y W_2$ follows from (H10): taking real or imaginary parts is a continuous $\mathbb{R}$-linear operation and so commutes with differentiation, so on $\Omega - S$ we have $\partial_x W_1 = \operatorname{Re}(\partial_0 w)$ and $\partial_y W_2 = \operatorname{Im}(\partial_1 w)$, dominated in modulus by $|\partial_0 w|$ and $|\partial_1 w|$ respectively, and S is null by Step 0, so the domination off a null set suffices for integrability on compact subsets of Ω; and the growth hypothesis holds with $\beta = 0$ because W_1, W_2 are continuous, hence bounded near each point of S. We obtain

$$\int W_1\,\partial_x\phi = -\int (\partial_x W_1)\phi, \qquad \int W_2\,\partial_y\phi = -\int (\partial_y W_2)\phi \tag{6}$$

for every test function ϕ for Ω in the sense of IsTestFn.

On $\Omega - S$ we compute, from DZbar,

$$\operatorname{Re} w_{\bar{z}} = \operatorname{Re}\Big(\tfrac{1}{2}(\partial_x w + i\,\partial_y w)\Big) = \tfrac{1}{2}\big(\partial_x \operatorname{Re} w - \partial_y \operatorname{Im} w\big) = \tfrac{1}{2}\big(\partial_x W_1 - \partial_y W_2\big),$$

so by (3),

$$\partial_x W_1 - \partial_y W_2 = 0 \qquad \text{on } \Omega - S. \tag{7}$$

Now let ϕ be a real-valued test function for Ω; regarded as complex-valued it is a test function in the sense of IsTestFn. Using (5), then (6), then (7) together with the fact that S is null by Step 0 and that ϕ vanishes off a compact subset of Ω,

$$\int |g|^{p-2}\big(\operatorname{Re} g\cdot\partial_x\phi + \operatorname{Im} g\cdot\partial_y\phi\big) = 2^{p-1}\int \big(W_1\partial_x\phi - W_2\partial_y\phi\big) = -2^{p-1}\int \big(\partial_x W_1 - \partial_y W_2\big)\phi = 0.$$

Finally, since $\partial_x\phi$ and $\partial_y\phi$ are real, at every z

$$\operatorname{Re}\Big(g(z)\,\overline{\partial_x\phi(z) + i\,\partial_y\phi(z)}\Big) = \operatorname{Re} g(z)\,\partial_x\phi(z) + \operatorname{Im} g(z)\,\partial_y\phi(z).$$

Hence

$$\int |g|^{p-2}\,\mathrm{Re}\Big(g\,\overline{\partial_x\phi + i\,\partial_y\phi}\Big) = 0$$

for every real test function ϕ for Ω, which is the equation clause of `PHarmonicOn`.

Step 9: conclusion. Take $g = 2\overline{f}$ as the witness required by `PHarmonicOn`. Then $v \in L^p_{\mathrm{loc}}(\Omega)$ by (H9); and $g \in L^p_{\mathrm{loc}}(\Omega)$ because $|g| = 2|f|$ and $f \in L^p_{\mathrm{loc}}(\Omega)$ by (H9). Step 7 supplies the two weak partial derivatives of v, namely $\mathrm{Re}\,g$ in the x direction and $\mathrm{Im}\,g$ in the y direction, and Step 8 supplies the equation. Hence v is p-harmonic on Ω in the sense of `PHarmonicOn`. □

2.64. `ExtremalEpsBound`.

Lemma 2.64. *Let $1 < p < \infty$, let $\epsilon_k(n,p)$ be as in `EpsCoef` and let $d = d(p)$ be as in `DExp`. Then*

(a) $\epsilon_2(1,p) = \dfrac{1-d}{1+3d}$,

(b) $\big|\epsilon_2(1,p)\big| < \dfrac{1}{3}$, *and*

(c) $\epsilon_2(1,p) = 0$ *if and only if* $p = 2$.

Thus the number ϵ of equation (54) of the paper, which Section 7 introduces as $\epsilon = \epsilon_2 = \frac{\lambda_2-1}{\lambda_2+3} = \frac{1-d}{1+3d}$, satisfies $|\epsilon| < \frac{1}{3}$, and $\epsilon \neq 0$ exactly when $p \neq 2$.

Proof. Fix $p > 1$ and write $\gamma = \gamma_1(p)$. By `DExpEqGammaOneInv`,

$$\gamma > 0 \qquad \text{and} \qquad \gamma\, d = 1.$$

In particular $d = 1/\gamma$.

Step 0 (what $\epsilon_2(1,p)$ is). By `EpsCoef` with $n = 1$, $k = 2$, and since $\lambda_2(1,p) = \gamma_1(p) = \gamma$ by `GammaExp`,

$$\epsilon_2(1,p) = \frac{\lambda_2(1,p) + 1 - 2}{\lambda_2(1,p) + 1 + 2} = \frac{\gamma - 1}{\gamma + 3}.$$

Since $\gamma > 0$ we have $\gamma + 3 > 3 > 0$, so this is a well-defined real number.

Step 1 (a closed form for γ, and $\gamma < 3$). By `GammaExp` and `LambdaCoef` with $n = 1$, $k = 2$,

$$2\gamma = -p + \sqrt{4 \cdot 2^2(p-1) + (p-2)^2} = -p + \sqrt{p^2 + 12p - 12},$$

because $16(p-1) + (p-2)^2 = p^2 + 12p - 12$. Since $p > 1$ we have $p^2 + 12p - 12 > p^2 \geq 0$, so the radicand is positive; write $S = \sqrt{p^2 + 12p - 12} > 0$, so $S^2 = p^2 + 12p - 12$ and $2\gamma = S - p$.

We claim $\gamma < 3$, i.e. $S < p + 6$. Both $S \geq 0$ and $p + 6 > 0$, so it suffices to compare squares:

$$(p+6)^2 - S^2 = \big(p^2 + 12p + 36\big) - \big(p^2 + 12p - 12\big) = 48 > 0.$$

Hence $S < p + 6$, so $2\gamma = S - p < 6$ and $\gamma < 3$.

Step 2 (proof of (a)). Since $\gamma > 0$,

$$1 - d = 1 - \frac{1}{\gamma} = \frac{\gamma - 1}{\gamma}, \qquad 1 + 3d = 1 + \frac{3}{\gamma} = \frac{\gamma + 3}{\gamma},$$

and $\dfrac{\gamma+3}{\gamma} > 0$, so $1 + 3d \neq 0$ and

$$\frac{1-d}{1+3d} = \frac{(\gamma-1)/\gamma}{(\gamma+3)/\gamma} = \frac{\gamma-1}{\gamma+3} = \epsilon_2(1,p)$$

by Step 0.

Step 3 (proof of (b)). By Step 0, $|\epsilon_2(1,p)| = \dfrac{|\gamma - 1|}{\gamma + 3}$ with $\gamma + 3 > 0$, so the assertion $|\epsilon_2(1,p)| < \frac{1}{3}$ is equivalent to $3|\gamma - 1| < \gamma + 3$. We distinguish two cases.

If $\gamma \geq 1$, then $|\gamma - 1| = \gamma - 1$ and

$$3(\gamma - 1) < \gamma + 3 \iff 2\gamma < 6 \iff \gamma < 3,$$

which holds by Step 1.

If $\gamma < 1$, then $|\gamma - 1| = 1 - \gamma$ and

$$3(1-\gamma) < \gamma + 3 \iff 0 < 4\gamma \iff \gamma > 0,$$

which holds since $\gamma > 0$.

In both cases $3|\gamma - 1| < \gamma + 3$, hence $|\epsilon_2(1,p)| < \frac{1}{3}$.

Step 4 (proof of (c)). Since $\gamma + 3 > 0$, Step 0 gives

$$\epsilon_2(1,p) = 0 \iff \gamma - 1 = 0 \iff \gamma = 1.$$

By Step 1, $2\gamma = S - p$ with $S = \sqrt{p^2 + 12p - 12} \geq 0$, so $\gamma = 1 \iff S = p + 2$.

If $p = 2$, then $p^2 + 12p - 12 = 4 + 24 - 12 = 16$, so $S = 4 = p + 2$ and $\gamma = 1$. Conversely, if $S = p+2$, then squaring (both sides are nonnegative, as $p+2 > 0$) gives $p^2+12p-12 = p^2+4p+4$, i.e. $8p = 16$, i.e. $p = 2$.

Hence $\epsilon_2(1,p) = 0$ if and only if $p = 2$. □

2.65. `ExtremalPolarForm`.

Lemma 2.65. *Let $1 < p < \infty$, let $d = d(p)$ and $\epsilon = \epsilon_2(1,p)$ be as in `ExtremalH`, and let H be the extremal function of `ExtremalH`. Then*

(a) *for every $r > 0$ and every $\theta \in \mathbb{R}$,*

$$H\big(re^{i\theta}\big) = r^{1/d}\Big(e^{i\theta} + \epsilon\, e^{-3i\theta}\Big);$$

(b) *for every $t > 0$ and every $\xi \in \mathbb{C}$,*

$$t\, H(\xi) = H\big(t^d \xi\big).$$

Proof. Throughout, $1 < p < \infty$ is fixed. We first record that $d = d(p) > 0$. Indeed, by `DExp`,

$$6\,d = 1 + \frac{1}{p-1} + \sqrt{1 + \frac{14}{p-1} + \frac{1}{(p-1)^2}},$$

and since $p - 1 > 0$ the first two summands are positive and the square root is nonnegative, so $6d \geq 1 + \frac{1}{p-1} > 0$ and hence $d > 0$. In particular $1/d$ is defined and positive, and $d \neq 0$.

Proof of (a). Fix $r > 0$ and $\theta \in \mathbb{R}$, and put $\xi = re^{i\theta}$.

Step 1: $|\xi| = r$ and $\xi \neq 0$. Since $|e^{i\theta}| = 1$ and $r > 0$,

$$|\xi| = |r|\,\big|e^{i\theta}\big| = r.$$

As $r > 0$ this gives $|\xi| \neq 0$, hence $\xi \neq 0$.

Step 2: the first quotient. By Step 1 and $r \neq 0$,

$$\frac{\xi}{|\xi|} = \frac{re^{i\theta}}{r} = e^{i\theta}.$$

Step 3: the second quotient. We have $\xi^3 = r^3 e^{3i\theta}$ and, by Step 1, $|\xi|^3 = r^3$. Both r^3 and $e^{3i\theta}$ are nonzero, so

$$\frac{|\xi|^3}{\xi^3} = \frac{r^3}{r^3 e^{3i\theta}} = \frac{1}{e^{3i\theta}} = e^{-3i\theta},$$

the last equality because $e^{-3i\theta}e^{3i\theta} = e^0 = 1$.

Step 4: the modulus factor. By Step 1, $|\xi|^{1/d} = r^{1/d}$.

Step 5. Substituting Steps 2, 3 and 4 into the formula of `ExtremalH`,

$$H(\xi) = \big(e^{i\theta} + \epsilon\, e^{-3i\theta}\big)\, r^{1/d} = r^{1/d}\big(e^{i\theta} + \epsilon\, e^{-3i\theta}\big),$$

which is (a).

Proof of (b). Fix $t > 0$ and $\xi \in \mathbb{C}$, and put $\zeta = t^d \xi$. Note $t^d > 0$, since $t > 0$.

Case $\xi = 0$. Then $\zeta = 0$ as well. By `ExtremalH` and the convention $w/0 = 0$, the first factor of $H(0)$ is $0/0 + \epsilon \cdot (0/0) = 0$, so $H(0) = 0$. Hence both sides of (b) are 0: the left is $t \cdot 0 = 0$ and the right is $H(0) = 0$.

Case $\xi \neq 0$. Then $|\xi| > 0$ and, since $t^d > 0$,

$$\zeta \neq 0, \qquad |\zeta| = \big|t^d\big|\,|\xi| = t^d|\xi| > 0.$$

We compare the three ingredients of `ExtremalH` at ζ and at ξ.

First, since $t^d \neq 0$,

$$\frac{\zeta}{|\zeta|} = \frac{t^d \xi}{t^d |\xi|} = \frac{\xi}{|\xi|}.$$

Second, $\zeta^3 = t^{3d}\xi^3$ and $|\zeta|^3 = t^{3d}|\xi|^3$, and both t^{3d} and ξ^3 are nonzero, so

$$\frac{|\zeta|^3}{\zeta^3} = \frac{t^{3d}|\xi|^3}{t^{3d}\xi^3} = \frac{|\xi|^3}{\xi^3}.$$

Hence the first factor of `ExtremalH` takes the same value at ζ as at ξ.

Third, using $t^d > 0$, $|\xi| > 0$ and the rules for real powers of positive numbers,

$$|\zeta|^{1/d} = \big(t^d|\xi|\big)^{1/d} = \big(t^d\big)^{1/d}|\xi|^{1/d} = t^{d\cdot(1/d)}|\xi|^{1/d} = t\,|\xi|^{1/d},$$

the last step because $d \cdot (1/d) = 1$, which uses $d \neq 0$.

Combining the three computations with the formula of `ExtremalH`,

$$H(\zeta) = \Big(\frac{\xi}{|\xi|} + \epsilon\frac{|\xi|^3}{\xi^3}\Big)\, t\,|\xi|^{1/d} = t\Big(\frac{\xi}{|\xi|} + \epsilon\frac{|\xi|^3}{\xi^3}\Big)|\xi|^{1/d} = t\,H(\xi),$$

which is (b). □

2.66. `ExtremalHomogeneous`.

Lemma 2.66. *Let $1 < p < \infty$, let H be the extremal function of `ExtremalH`, let $d = d(p)$ be the exponent of `DExp`, and let $f\colon \mathbb{C} \to \mathbb{C}$ be a two-sided inverse of H, that is*

$$f\big(H(\xi)\big) = \xi \quad \text{and} \quad H\big(f(z)\big) = z \qquad (\xi, z \in \mathbb{C}),$$

which is the relation (55) of the paper. Assume also that f is of class C^∞ on $\mathbb{C} - \{0\}$. Then

(57) *$f(tz) = t^d f(z)$ for every $z \in \mathbb{C}$ and every real $t \geq 0$;*

(H) *for every word w of length l, every real $t > 0$ and every $z \neq 0$,*

$$\big(D^w f\big)(tz) = t^{\,d-l}\,\big(D^w f\big)(z),$$

where D^w is the iterated partial derivative of `PartialWord`: each partial derivative of f of order l is homogeneous of degree $d - l$.

Clause (57) is the paper's (57) at paper.tex:597, and clause (H) is the sentence that follows it, "Hence, partial derivatives $D^\nu f$ are homogeneous functions of degree $d - |\nu|$". The paper's displays (58) and (59) are the two cases $l = k - 1$ and $l = k + 1$ of (H): since $d = k - 1 + \alpha$, the exponents are $d - (k-1) = \alpha$ and $d - (k+1) = \alpha - 2$, and (59) is the latter rearranged.

Proof. Write $d = d(p)$.

Step 0: $d > 0$. By `DExpEqGammaOneInv`, $\gamma_1(p) > 0$ and $\gamma_1(p)\, d = 1$; hence $d = 1/\gamma_1(p) > 0$.

Step 1: proof of (57) for $t > 0$. Let $t > 0$ and $z \in \mathbb{C}$, and put $\xi = f(z)$, so that $H(\xi) = z$ by the second inverse relation. By the second clause of `ExtremalPolarForm`, applied with this $t > 0$ and this ξ,

$$t\,H(\xi) = H\big(t^d \xi\big).$$

The left-hand side is tz. Applying f to both sides and using the first inverse relation at the point $t^d\xi$,

$$f(tz) = f\big(H(t^d\xi)\big) = t^d\xi = t^d f(z).$$

Step 2: proof of (57) at $t = 0$. By `ExtremalH`, $H(0) = 0$; so the first inverse relation at $\xi = 0$ gives $f(0) = f(H(0)) = 0$. Hence $f(0 \cdot z) = f(0) = 0$, while $0^d f(z) = 0$ because $0^d = 0$ for $d > 0$ by Step 0. So (57) holds at $t = 0$ as well, and therefore for every $t \geq 0$.

Step 3: how D^w behaves under a dilation. Fix $t > 0$. We claim that for every word w and every function F that is C^∞ on $\mathbb{C} - \{0\}$,

$$D^w\big(\zeta \mapsto F(t\zeta)\big)(z) = t^{\,|w|}\big(D^w F\big)(tz) \qquad (z \neq 0). \tag{1}$$

We induct on w. If w is empty, D^w is the identity by `PartialWord` and both sides are $F(tz) = t^0 F(tz)$. For the step, write $w = j\colon w'$, so that $D^w = \partial_j \circ D^{w'}$ by `PartialWord`, where ∂_0 and ∂_1 are the derivatives in the directions 1 and i. The dilation $\zeta \mapsto t\zeta$ is an $\mathbb{R}$-linear homeomorphism

of $\mathbb{C}$ carrying $\mathbb{C}-\{0\}$ onto itself, with Fréchet derivative at every point equal to itself. By the inductive hypothesis the two functions

$$D^{w'}\big(\zeta \mapsto F(t\zeta)\big) \qquad \text{and} \qquad t^{\,|w'|}\big(D^{w'}F\big)(t\,\cdot)$$

agree at every point of the open set $\mathbb{C}-\{0\}$, and both are C^∞ there, F being C^∞ on $\mathbb{C}-\{0\}$ and the dilation mapping that set into itself. Differentiating at $z\neq 0$ in the direction v ($v=1$ if $j=0$, $v=i$ if $j=1$) and using the chain rule together with $\mathbb{R}$-linearity of the derivative in its argument,

$$\partial_j\Big(t^{\,|w'|}\big(D^{w'}F\big)(t\,\cdot)\Big)(z) = t^{\,|w'|}\,\mathrm{d}\big(D^{w'}F\big)(tz)[tv] = t^{\,|w'|+1}\,\partial_j\big(D^{w'}F\big)(tz),$$

which is (1) for w, since $|w|=|w'|+1$.

Step 4: proof of (H). Fix $t>0$ and a word w of length l. By Step 1 the two functions $\zeta\mapsto f(t\zeta)$ and $\zeta\mapsto t^d f(\zeta)$ agree at every point of $\mathbb{C}$, in particular on the open set $\mathbb{C}-\{0\}$, where both are C^∞ by hypothesis and by Step 3's observation about the dilation. Apply D^w to both and evaluate at $z\neq 0$. On the left, (1) with $F=f$ gives $t^{\,l}\big(D^w f\big)(tz)$. On the right, t^d is a constant scalar and D^w is built from Fréchet derivatives, which are linear in the function, so $D^w\big(t^d f\big)=t^d D^w f$ and the value is $t^d\big(D^w f\big)(z)$. Therefore

$$t^{\,l}\big(D^w f\big)(tz) = t^d\big(D^w f\big)(z),$$

and dividing by $t^{\,l}>0$, together with $t^d/t^{\,l}=t^{\,d-l}$ for $t>0$,

$$\big(D^w f\big)(tz) = t^{\,d-l}\big(D^w f\big)(z),$$

which is (H).

Step 5: the two displays of the paper. Let k and α be as in (9), so that $d=k-1+\alpha$. Taking $l=k-1$ in (H) gives $\big(D^w f\big)(tz)=t^\alpha\big(D^w f\big)(z)$, which is (58); and taking $l=k+1$ gives $\big(D^w f\big)(tz)=t^{\alpha-2}\big(D^w f\big)(z)$, that is $\big(D^w f\big)(z)=t^{\,2-\alpha}\big(D^w f\big)(tz)$, which is (59). □

2.67. `ExtremalInjective`.

Lemma 2.67. *Let $1<p<\infty$ and let H be the extremal function of `ExtremalH`. Then H is continuous on $\mathbb{C}$, is a bijection of $\mathbb{C}$ onto $\mathbb{C}$, and its inverse is continuous; that is, H is a homeomorphism of the plane onto itself.*

This is the paper's assertion at paper.tex:586 that "in this case H is a quasiconformal homeomorphism on the whole plane", in the form that Section 7 actually uses: the inverse f of (55) is defined and continuous on all of $\mathbb{C}$. It rests on the strict *inequality $|\epsilon|<1/3$ of `ExtremalEpsBound`, and needs no degree-theoretic input.*

Proof. Write $d=d(p)$ for the exponent of `DExp`, $\epsilon=\epsilon_2(1,p)$ for the coefficient of `EpsCoef` as used in `ExtremalH`, $a=1/d$, and

$$G(\theta)=e^{i\theta}+\epsilon e^{-3i\theta}, \qquad w(\theta)=\epsilon e^{-4i\theta}.$$

Note $\epsilon\in\mathbb{R}$, so $\bar\epsilon=\epsilon$.

Step 0: the two constants. By `DExpEqGammaOneInv`, $\gamma_1(p)>0$ and $\gamma_1(p)\,d=1$, so

$$d=1/\gamma_1(p)>0, \qquad a=1/d=\gamma_1(p)>0. \tag{1}$$

By part (b) of `ExtremalEpsBound`,

$$|\epsilon|<\tfrac13, \qquad \text{hence} \qquad 1-3|\epsilon|>0 \ \text{ and } \ 1-|\epsilon|>\tfrac23>0. \tag{2}$$

Step 1: polar form, and two-sided control of $|H|$. By the first clause of `ExtremalPolarForm`, for every $r>0$ and every θ,

$$H\big(re^{i\theta}\big)=r^a\,G(\theta). \tag{3}$$

Also

$$G(\theta)=e^{i\theta}\big(1+w(\theta)\big), \tag{4}$$

since $e^{i\theta}\cdot\epsilon e^{-4i\theta}=\epsilon e^{-3i\theta}$. As $|e^{-4i\theta}|=1$ we have $|w(\theta)|=|\epsilon|$ for every θ, so by the triangle inequality and (2),

$$0<1-|\epsilon|\le\big|1+w(\theta)\big|\le 1+|\epsilon|,\qquad\text{and}\qquad |G(\theta)|=\big|1+w(\theta)\big|, \tag{5}$$

the last equality from (4) and $|e^{i\theta}|=1$. Combining with (3),

$$\big(1-|\epsilon|\big)r^a\le\big|H(re^{i\theta})\big|\le\big(1+|\epsilon|\big)r^a\qquad(r>0). \tag{6}$$

In particular $H(\xi)\neq 0$ for every $\xi\neq 0$. And $H(0)=0$, as `ExtremalH` records.

Step 2: continuity of H. On $\mathbb{C}-\{0\}$ the formula of `ExtremalH` is a product of continuous functions — there $|\xi|>0$ and $\xi\neq 0$, so both quotients are continuous, and $\xi\mapsto|\xi|^a$ is continuous — so H is continuous there. At 0: by (6) applied with $r=|\xi|$, for $\xi\neq 0$,

$$\big|H(\xi)-H(0)\big|=\big|H(\xi)\big|\le\big(1+|\epsilon|\big)|\xi|^a,$$

and $|\xi|^a\to 0$ as $\xi\to 0$ because $a>0$ by (1). So H is continuous at 0 as well, hence on $\mathbb{C}$.

Step 3: a strictly increasing argument for G. By (5), $\operatorname{Re}\big(1+w(\theta)\big)\ge 1-|w(\theta)|=1-|\epsilon|>0$, so $1+w(\theta)$ lies in the open right half plane for every θ. On that half plane the principal argument Arg is C^∞, so

$$\Theta(\theta)=\theta+\operatorname{Arg}\big(1+w(\theta)\big)$$

defines a C^∞ function $\Theta\colon\mathbb{R}\to\mathbb{R}$, and by (4) and (5),

$$G(\theta)=\big|G(\theta)\big|\,e^{i\Theta(\theta)},\qquad\big|G(\theta)\big|>0. \tag{7}$$

We compute Θ'. From $w'(\theta)=-4i\,w(\theta)$ and the formula $\frac{d}{d\theta}\operatorname{Arg}z(\theta)=\operatorname{Im}\big(z'(\theta)/z(\theta)\big)$, valid for a differentiable curve z in the right half plane,

$$\frac{d}{d\theta}\operatorname{Arg}\big(1+w\big)=\operatorname{Im}\Big(\frac{-4iw}{1+w}\Big)=-4\operatorname{Re}\Big(\frac{w}{1+w}\Big),$$

so

$$\Theta'(\theta)=1-4\operatorname{Re}\Big(\frac{w}{1+w}\Big)=\operatorname{Re}\Big(\frac{1+w-4w}{1+w}\Big)=\operatorname{Re}\Big(\frac{1-3w}{1+w}\Big). \tag{8}$$

Now, multiplying numerator and denominator by $\overline{1+w}$ and using $\epsilon\in\mathbb{R}$,

$$\operatorname{Re}\Big(\frac{1-3w}{1+w}\Big)=\frac{\operatorname{Re}\big((1-3w)(1+\bar w)\big)}{|1+w|^2}=\frac{1+\operatorname{Re}(\bar w)-3\operatorname{Re}(w)-3|w|^2}{|1+w|^2}=\frac{1-2\operatorname{Re}(w)-3|w|^2}{|1+w|^2},$$

since $\operatorname{Re}(\bar w)=\operatorname{Re}(w)$. Using $\operatorname{Re}(w)\le|w|=|\epsilon|$, the factorization $1-2t-3t^2=(1-3t)(1+t)$, and (5),

$$\Theta'(\theta)\ \ge\ \frac{1-2|\epsilon|-3|\epsilon|^2}{|1+w|^2}=\frac{\big(1-3|\epsilon|\big)\big(1+|\epsilon|\big)}{|1+w|^2}\ \ge\ \frac{\big(1-3|\epsilon|\big)\big(1+|\epsilon|\big)}{\big(1+|\epsilon|\big)^2}=\frac{1-3|\epsilon|}{1+|\epsilon|}=:\mu>0, \tag{9}$$

by (2). So Θ is strictly increasing, with $\Theta(\theta_2)-\Theta(\theta_1)\ge\mu(\theta_2-\theta_1)$ for $\theta_1\le\theta_2$, by the mean value theorem.

Step 4: Θ is a bijection of $\mathbb{R}$ with $\Theta(\theta+2\pi)=\Theta(\theta)+2\pi$. Since $w(\theta+2\pi)=\epsilon e^{-4i\theta}e^{-8\pi i}=w(\theta)$, the function $\operatorname{Arg}(1+w)$ is 2π-periodic, so

$$\Theta(\theta+2\pi)=\Theta(\theta)+2\pi. \tag{10}$$

By Step 3, $\Theta(\theta)\ge\Theta(0)+\mu\theta$ for $\theta\ge 0$ and $\Theta(\theta)\le\Theta(0)+\mu\theta$ for $\theta\le 0$, so $\Theta(\theta)\to\pm\infty$ as $\theta\to\pm\infty$; being continuous, Θ is onto $\mathbb{R}$ by the intermediate value theorem, and it is injective by strict monotonicity. So $\Theta\colon\mathbb{R}\to\mathbb{R}$ is a strictly increasing continuous bijection.

Step 5: H is injective. Since $H(0)=0$ and $H(\xi)\neq 0$ for $\xi\neq 0$ by Step 1, it suffices to show that H is injective on $\mathbb{C}-\{0\}$. Let $\xi_1,\xi_2\neq 0$ with $H(\xi_1)=H(\xi_2)=:z$, and write $\xi_j=r_je^{i\theta_j}$ with $r_j=|\xi_j|>0$. By (3) and (7),

$$z=r_j^a\big|G(\theta_j)\big|\,e^{i\Theta(\theta_j)}\qquad(j=1,2),$$

with $r_j^a|G(\theta_j)|>0$. Comparing moduli and then unit vectors,

$$r_1^a\big|G(\theta_1)\big|=r_2^a\big|G(\theta_2)\big|,\qquad e^{i\Theta(\theta_1)}=e^{i\Theta(\theta_2)}. \tag{11}$$

The second equality gives $\Theta(\theta_1) - \Theta(\theta_2) = 2\pi k$ for some $k \in \mathbb{Z}$. By (10), $\Theta(\theta_2 + 2\pi k) = \Theta(\theta_2) + 2\pi k = \Theta(\theta_1)$, so $\theta_1 = \theta_2 + 2\pi k$ by injectivity of Θ (Step 4). Hence $e^{i\theta_1} = e^{i\theta_2}$, and G, being a combination of $e^{i\theta}$ and $e^{-3i\theta}$, is 2π-periodic, so $G(\theta_1) = G(\theta_2)$. Dividing the first equality of (11) by $|G(\theta_1)| > 0$ gives $r_1^a = r_2^a$, and since $a > 0$ the map $t \mapsto t^a$ is strictly increasing on $(0, \infty)$, hence injective; so $r_1 = r_2$. Therefore $\xi_1 = r_1 e^{i\theta_1} = r_2 e^{i\theta_2} = \xi_2$.

Step 6: H is surjective. We have $H(0) = 0$. Let $z \neq 0$ and write $z = |z|e^{i\varphi}$ with $\varphi \in \mathbb{R}$. By Step 4 there is θ with $\Theta(\theta) = \varphi$, and then by (7) $G(\theta) = |G(\theta)|e^{i\varphi}$ with $|G(\theta)| > 0$. Put

$$s = \frac{|z|}{|G(\theta)|} > 0, \qquad r = s^d > 0.$$

Then $r^a = s^{da} = s^1 = s$, using $da = 1$ from (1) and $s > 0$. Hence by (3)

$$H\big(re^{i\theta}\big) = r^a G(\theta) = s\,\big|G(\theta)\big|e^{i\varphi} = |z|e^{i\varphi} = z.$$

So H is onto $\mathbb{C}$, and with Step 5 it is a bijection of $\mathbb{C}$.

Step 7: H is a closed map, hence a homeomorphism. Let $C \subseteq \mathbb{C}$ be closed and let y lie in the closure of $H(C)$; choose $\xi_k \in C$ with $H(\xi_k) \to y$. The sequence $\big(|H(\xi_k)|\big)$ is convergent, hence bounded by some R, so by (6), which also holds trivially at $\xi_k = 0$,

$$\big(1 - |\epsilon|\big)|\xi_k|^a \leq \big|H(\xi_k)\big| \leq R,$$

whence $|\xi_k|^a \leq R/(1-|\epsilon|)$ by (2). Since $a > 0$, the map $t \mapsto t^a$ is an increasing bijection of $[0, \infty)$ onto itself, so $|\xi_k| \leq \big(R/(1-|\epsilon|)\big)^{1/a}$: the sequence (ξ_k) is bounded. By the Bolzano–Weierstrass theorem a subsequence converges, say $\xi_{k_i} \to \xi$, and $\xi \in C$ because C is closed. By Step 2, $H(\xi) = \lim_i H(\xi_{k_i}) = y$, so $y \in H(C)$. Hence $H(C)$ is closed.

Let g be the inverse of the bijection H, so that $g(H(\xi)) = \xi$ for every ξ and $H(g(z)) = z$ for every z. For closed $C \subseteq \mathbb{C}$ we have $g^{-1}(C) = H(C)$, which is closed by the previous paragraph; as preimages of closed sets under g are closed, g is continuous. Together with Step 2 and Steps 5–6 this is the assertion. □

2.68. `InverseSolvesEq12`.

Lemma 2.68. *Let $1 < p < \infty$, let $V \subseteq \mathbb{C}$ be open, let $H, f \colon \mathbb{C} \to \mathbb{C}$, and assume*

(H1) *$H(V)$ is open;*
(H2) *H is C^1 on $V - \{0\}$;*
(H3) *H solves equation (18) at $n = 1$ on V in the sense of `SolvesEq18`;*
(H4) *$f(H(\xi)) = \xi$ for every $\xi \in V$;*
(H5) *f is differentiable at almost every point of $H(V)$, and $J_f(z) = |f_z(z)|^2 - |f_{\bar z}(z)|^2 \neq 0$ for almost every $z \in H(V)$, with J_f as in `Jac`;*
(H6) *$f \in W^{1,1}_{\mathrm{loc}}(H(V))$ in the sense of `MemWLocOm`.*

Then f solves equation (12) on $H(V)$ in the sense of `SolvesEq12`.

This is the converse of the hodograph step of Section 3, which consumes (12) and produces (18), whereas Section 7 of the paper starts from the explicit solution (54) of (18) and must conclude that its inverse satisfies (12). The two coefficient brackets match only at $n = 1$, which is the only case Section 7 uses.

Hypothesis (H6) is what the conclusion needs, and nothing more: by `SolvesEq12`, solving (12) is an identity between the weak Wirtinger derivatives of f, and membership in $W^{1,1}_{\mathrm{loc}}$ is exactly the data those consist of. For the f of Section 7 it follows from the explicit homogeneity of the inverse of (54), and in general from the quoted result (C1), which places the complex gradient of a p-harmonic function in $W^{1,2}_{\mathrm{loc}} \subseteq W^{1,1}_{\mathrm{loc}}$.

Proof. Throughout put

$$c = \tfrac{1}{2} - \tfrac{1}{p}, \qquad c' = \tfrac{1}{p} - \tfrac{1}{2} = -c.$$

Step 1: the inverse relations on $H(V)$. Let $z \in H(V)$. Then $z = H(\xi)$ for some $\xi \in V$, and (H4) gives $f(z) = f(H(\xi)) = \xi$. Hence

$$f(z) \in V \quad \text{and} \quad H(f(z)) = H(\xi) = z \qquad (z \in H(V)). \tag{1}$$

Step 2: (18) holds at every *point of* $V-\{0\}$. The set $V-\{0\}$ is open, V being open. By (H2) the map $z\mapsto \mathrm{d}H(z)$ is continuous on $V-\{0\}$, hence so are $H_\xi=\frac12(\mathrm{d}H[1]-i\,\mathrm{d}H[i])$ and $H_{\bar\xi}=\frac12(\mathrm{d}H[1]+i\,\mathrm{d}H[i])$ of `DZ` and `DZbar`; and $\xi\mapsto\xi/\bar\xi$ and $\xi\mapsto\bar\xi/\xi$ are continuous on $V-\{0\}$. Therefore both sides of

$$H_{\bar\xi}(\xi)=c\Big[\frac{\xi}{\bar\xi}H_\xi(\xi)+\frac{\bar\xi}{\xi}\overline{H_\xi(\xi)}\Big] \tag{18}$$

are continuous functions of ξ on $V-\{0\}$.

By (H3) and `SolvesEq18` at $n=1$ there are weak Wirtinger derivatives a_H, b_H of H on V, in the sense of `IsWeakWirtingerOn`, and a Lebesgue null set Z_0 with

$$b_H(\xi)=c\Big[\frac{\xi}{\bar\xi}a_H(\xi)+\frac{\bar\xi}{\xi}\overline{a_H(\xi)}\Big]\qquad(\xi\in V-Z_0,\ \xi\neq0). \tag{18$'$}$$

We first replace a_H, b_H by the pointwise derivatives. By `IsWeakWirtingerOn` there are H_x, H_y with $H, H_x, H_y\in L^1_{\mathrm{loc}}(V)$, with H_x and H_y the distributional ∂_0- and ∂_1-derivatives of H on V, and with $a_H=\frac12(H_x-iH_y)$, $b_H=\frac12(H_x+iH_y)$ at every point. These are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set V, the function H and the pair $D_0=H_x$, $D_1=H_y$; its (H3) holds because H is C^1 on $V-\{0\}$ by (H2), hence differentiable at every point of V except possibly the single point 0, which is a null set. So there is a null set Z_1 with $H_x=\partial_0H$ and $H_y=\partial_1H$ on $V-Z_1$, where $\partial_j=D^{[j]}$ is the classical partial derivative of `PartialWord`, that is $\partial_0H(\xi)=\mathrm{d}H(\xi)[1]$ and $\partial_1H(\xi)=\mathrm{d}H(\xi)[i]$. Since H_ξ and $H_{\bar\xi}$ are by `DZ` and `DZbar` the same two combinations $\frac12\big(\mathrm{d}H(\xi)[1]\mp i\,\mathrm{d}H(\xi)[i]\big)$,

$$a_H(\xi)=H_\xi(\xi),\qquad b_H(\xi)=H_{\bar\xi}(\xi)\qquad(\xi\in V-Z_1). \tag{18$''$}$$

Put $Z=Z_0\cup Z_1$, a null set. Substituting (18″) into (18′), the identity (18) holds at every $\xi\in V-Z$ with $\xi\neq0$. Thus (18) holds on $\big(V-\{0\}\big)-Z$. A null set has empty interior, so its complement meets every nonempty open subset of $V-\{0\}$; that is, $\big(V-\{0\}\big)-Z$ is dense in $V-\{0\}$. Two continuous functions on an open set which agree on a dense subset of it agree on all of it. Hence (18) holds at every $\xi\in V-\{0\}$.

Step 3: the exceptional null set in the z *variable.* By (H5) there are null sets N_1, N_2 such that f is differentiable at every $z\in H(V)-N_1$ and $J_f(z)\neq0$ for every $z\in H(V)-N_2$. Put $N=N_1\cup N_2$, again null.

Step 4: the inverse-derivative formulas at a good point. Fix $z\in H(V)-N$ with $f(z)\neq0$, and put $\xi=f(z)$ and $J=J_f(z)$. By (1), $\xi\in V$, and $\xi\neq0$ by assumption, so $\xi\in V-\{0\}$. Since $V-\{0\}$ is open and H is C^1 there by (H2), H is differentiable at $\xi=f(z)$. Also f is differentiable at z and $J\neq0$, by the choice of z.

Apply `ChiFirstDerivs` with the open set $U=H(V)$, which is open by (H1), with the pair of mutually inverse maps taken in the order (H,f) — that is, H in the role of the map differentiated at the image point and f in the role of the map differentiated at z — with the point $z\in U$, with the relation $H(f(\zeta))=\zeta$ for $\zeta\in U$ supplied by (1), with differentiability of f at z and of H at $f(z)$, and with $J\neq0$. Its hypotheses constrain the pair only through that relation, so it applies in this order, and it gives

$$H_\xi(\xi)=\frac{\overline{f_z(z)}}{J},\qquad H_{\bar\xi}(\xi)=-\frac{f_{\bar z}(z)}{J}. \tag{2}$$

Since J is real by `Jac`, conjugating the first identity of (2) gives

$$\overline{H_\xi(\xi)}=\frac{f_z(z)}{J}. \tag{3}$$

Step 5: (12) at z. By Step 2, (18) holds at ξ. Substituting (2) and (3) into (18),

$$-\frac{f_{\bar z}(z)}{J}=c\Big[\frac{\xi}{\bar\xi}\cdot\frac{\overline{f_z(z)}}{J}+\frac{\bar\xi}{\xi}\cdot\frac{f_z(z)}{J}\Big].$$

Multiplying by $-J \neq 0$ and using $-c = c'$,

$$f_{\bar z}(z) = c'\Big[\frac{\bar\xi}{\xi} f_z(z) + \frac{\xi}{\bar\xi}\overline{f_z(z)}\Big].$$

Finally $\xi = f(z)$, so $\bar\xi/\xi = \overline{f(z)}/f(z)$ and $\xi/\bar\xi = f(z)/\overline{f(z)}$, and the display reads

$$f_{\bar z}(z) = c'\Big[\frac{\overline{f(z)}}{f(z)} f_z(z) + \frac{f(z)}{\overline{f(z)}}\overline{f_z(z)}\Big], \tag{12}$$

which is the pointwise form of equation (12) at z.

Step 6: the weak form of (12), and conclusion. Step 5 establishes the pointwise identity (12), between $f_{\bar z}$ and f_z, at every $z \in H(V) - N$ with $f(z) \neq 0$. It remains to read it for the weak Wirtinger derivatives of f, which is what `SolvesEq12` asks for. Write $U = H(V)$, open by (H1).

By (H6) and `MemWLocOn` there is a family (f_w), indexed by the words w in $\{0,1\}$, with $f_{()} = f$, with $f_w \in L^1_{\mathrm{loc}}(U)$ in the sense of `MemLpLocOn` for every w with $|w| \leq 1$, and with $f_{(j)}$ the distributional ∂_j-derivative of f on U in the sense of `IsWeakPartialOn` for $j = 0, 1$. Put $f_x = f_{(0)}$, $f_y = f_{(1)}$, and define $a = \frac{1}{2}(f_x - i f_y)$, $b = \frac{1}{2}(f_x + i f_y)$ at every point of $\mathbb{C}$. Then a and b are weak Wirtinger derivatives of f on U in the sense of `IsWeakWirtingerOn`: its clause (a) is the local integrability of f, f_x, f_y, its clause (b) the distributional-derivative property of f_x, f_y, and its clause (c) the definition of a, b.

Those two clauses are also hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for the open set U, the function f and the pair $D_0 = f_x$, $D_1 = f_y$, and its (H3) is the first half of (H5). So there is a null set N_3 with $f_x = \partial_0 f$ and $f_y = \partial_1 f$ on $U - N_3$, whence, exactly as in Step 2 and by `DZ` and `DZbar`,

$$a(z) = f_z(z), \qquad b(z) = f_{\bar z}(z) \qquad (z \in U - N_3). \tag{4}$$

Combining (4) with Step 5, for every $z \in U - (N \cup N_3)$ with $f(z) \neq 0$,

$$b(z) = c'\Big[\frac{\overline{f(z)}}{f(z)} a(z) + \frac{f(z)}{\overline{f(z)}}\overline{a(z)}\Big].$$

The set $N \cup N_3$ is null, so this holds at almost every $z \in U$ with $f(z) \neq 0$; with the pair a, b exhibited above, this is exactly the assertion that f solves (12) on $H(V)$ in the sense of `SolvesEq12`. □

2.69. `ExtremalInverseGradient`.

Theorem 2.69. *Let $1 < p < \infty$. Then there are maps $f\colon \mathbb{C} \to \mathbb{C}$ and $v\colon \mathbb{C} \to \mathbb{R}$ with the following properties, where $H = H_p$ is the extremal function (54) of `ExtremalH` and $d = d(p)$ is the exponent of `DExp`:*

(a) *f is continuous on $\mathbb{C}$, $f(H(\xi)) = \xi$ for every $\xi \in \mathbb{C}$ and $H(f(z)) = z$ for every $z \in \mathbb{C}$;*
(b) *f is C^∞ on $\mathbb{C} - \{0\}$;*
(c) *$f(tz) = t^d f(z)$ for every $t \geq 0$ and every z, and $D^\nu f(tz) = t^{d-|\nu|} D^\nu f(z)$ for every $t > 0$, every $z \neq 0$ and every iterated partial derivative D^ν of `PartialWord`;*
(d) *v is continuous on $\mathbb{C}$, is C^∞ on $\mathbb{C} - \{0\}$, and is p-harmonic on $\mathbb{C}$ in the sense of `PHarmonicOn`;*
(e) *$\partial v/\partial z = f$ at every $z \neq 0$, with $\partial/\partial z$ as in `DZ`.*

This is (55) of the paper together with the assertion of paper.tex:592 that it defines a p-harmonic function of complex gradient f; the paper's note lists that assertion among the facts which must be proved. Every regularity statement above is a conclusion and none is assumed: the paper's arrow runs from the equation to the regularity.

Proof. Write $\mu = 1/d$, $\varepsilon = \epsilon_2(1,p)$ for the coefficient of `EpsCoef` appearing in `ExtremalH`, $\lambda_2 = \lambda_2(1,p)$ for the exponent of `LambdaCoef`, and $c = \frac{1}{2} - \frac{1}{p}$, so that (18) at $n = 1$ reads

$$H_{\bar\xi}(\xi) = c\Big[\frac{\xi}{\bar\xi} H_\xi(\xi) + \frac{\bar\xi}{\xi}\overline{H_\xi(\xi)}\Big]. \tag{18}$$

Step 1: $d > 0$ and $\lambda_2 = \mu$. By `GammaExp`, $\gamma_1 = \lambda_2(1,p)$. By `DExpEqGammaOneInv`, $\gamma_1 > 0$ and $\gamma_1\, d = 1$. Hence $\lambda_2 > 0$, $d = 1/\lambda_2 > 0$ and $\lambda_2 = 1/d = \mu$.

Step 2: the inverse, and its zero. By `ExtremalInjective`, H is continuous and bijective and there is a continuous f with $f(H(\xi)) = \xi$ and $H(f(z)) = z$ for all $\xi, z \in \mathbb{C}$. Fix this f; this is (a).

The second clause of `ExtremalPolarForm` gives $t\,H(\xi) = H(t^d\xi)$ for $t > 0$; at $\xi = 0$ and $t = 2$ this reads $2H(0) = H(0)$, so

$$H(0) = 0, \qquad \text{and hence} \qquad f(z) = 0 \iff z = 0. \tag{1}$$

Indeed $f(0) = f(H(0)) = 0$, and if $f(z) = 0$ then $z = H(f(z)) = H(0) = 0$.

Step 3: H is the $k = 2$ summand of (19) at $n = 1$. Let $A\colon \mathbb{Z} \to \mathbb{C}$ be $A(2) = 1$ and $A(j) = 0$ for $j \neq 2$. By `Thm2Term` at $n = 1$, $k = 2$,

$$T(\xi) := \big(A(2)\xi^2 + \varepsilon\,\overline{A(2)}\,\bar\xi^{\,2}\big)|\xi|^{\lambda_2+1-2}\xi^{-1} = \big(\xi^2 + \varepsilon\bar\xi^{\,2}\big)|\xi|^{\lambda_2-1}\xi^{-1},$$

ε being real. If $\xi \neq 0$ then $\bar\xi = |\xi|^2/\xi$, so $\bar\xi^{\,2} = |\xi|^4/\xi^2$ and

$$T(\xi) = \Big(\xi^2 + \varepsilon\frac{|\xi|^4}{\xi^2}\Big)\frac{|\xi|^{\lambda_2-1}}{\xi} = \Big(\xi + \varepsilon\frac{|\xi|^4}{\xi^3}\Big)|\xi|^{\lambda_2-1} = \Big(\frac{\xi}{|\xi|} + \varepsilon\frac{|\xi|^3}{\xi^3}\Big)|\xi|^{\lambda_2} = H(\xi),$$

by `ExtremalH` and $\lambda_2 = \mu = 1/d$ from Step 1. At $\xi = 0$ both sides vanish: $T(0) = 0$ because its first factor does, and $H(0) = 0$ by (1). Hence $H = T$ as functions on $\mathbb{C}$.

By `Thm2TermSolvesEq18` at $n = 1$, $k = 2$, whose hypotheses $1 \le 1$, $1 \le 2$ and $1 < p$ hold, T is C^∞ at every $\xi \neq 0$ and solves (18) at $n = 1$ on $\{|\xi| < 1\}$ in the sense of `SolvesEq18`. Therefore

$$H \text{ is } C^\infty \text{ on } \mathbb{C} - \{0\}, \tag{2}$$

the set $\mathbb{C} - \{0\}$ being open, and H solves (18) on $\{|\xi| < 1\}$.

Step 3′: weak and classical Wirtinger derivatives agree. We record once, in general form, the passage between the two readings of the Wirtinger operators, and use it twice below. *Let $\Omega \subseteq \mathbb{C}$ be open, let $u\colon \mathbb{C} \to \mathbb{C}$ be differentiable at almost every point of Ω, and let a, b be weak Wirtinger derivatives of u on Ω in the sense of `IsWeakWirtingerOn`. Then $a = u_z$ and $b = u_{\bar z}$ almost everywhere on Ω, with $u_z, u_{\bar z}$ the pointwise operators of `DZ` and `DZbar`.* Indeed, by `IsWeakWirtingerOn` there are u_x, u_y with $u, u_x, u_y \in L^1_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`, with u_x and u_y the distributional ∂_0- and ∂_1-derivatives of u on Ω in the sense of `IsWeakPartialOn`, and with $a = \frac12(u_x - iu_y)$, $b = \frac12(u_x + iu_y)$ at every point. The first two are hypotheses (H1) and (H2) of `WeakPartialEqFDerivAE` for Ω, u and the pair $D_0 = u_x$, $D_1 = u_y$, and the almost-everywhere differentiability is its (H3); so $u_x = \partial_0 u$ and $u_y = \partial_1 u$ almost everywhere on Ω, where $\partial_j = D^{[j]}$ is the classical partial derivative of `PartialWord`, that is $\partial_0 u = \mathrm{d}u[1]$ and $\partial_1 u = \mathrm{d}u[i]$; and $u_z, u_{\bar z}$ are by `DZ` and `DZbar` the same two combinations $\frac12(\mathrm{d}u[1] \mp i\,\mathrm{d}u[i])$. Call this fact (W). We note for later use the two identities it rests on, valid at every point by `DZ` and `DZbar`:

$$\partial_0 u = u_z + u_{\bar z}, \qquad \partial_1 u = i\,(u_z - u_{\bar z}). \tag{2′}$$

Step 4: (18) at every point of the punctured unit disc. By (2) the map $\xi \mapsto \mathrm{d}H(\xi)$ is continuous on $\mathbb{C} - \{0\}$, so H_ξ and $H_{\bar\xi}$ of `DZ` and `DZbar`, being the fixed combinations $\frac12(\mathrm{d}H[1] \mp i\,\mathrm{d}H[i])$, are continuous there, and so are $\xi \mapsto \xi/\bar\xi$ and $\xi \mapsto \bar\xi/\xi$. Hence both sides of (18) are continuous on the open set $\{0 < |\xi| < 1\}$. By Step 3 and `SolvesEq18` there are weak Wirtinger derivatives a_H, b_H of H on $\mathbb{B} = \{|\xi| < 1\}$ and a null set Z_0 such that the identity (18) holds for the pair a_H, b_H at every $\xi \in \mathbb{B} - Z_0$ with $\xi \neq 0$; and by (W), applied to $u = H$ on the open set $\mathbb{B}$ — legitimate because by (2) H is differentiable at every point of $\mathbb{B}$ except possibly 0, a null set — there is a null set Z_1 with $a_H = H_\xi$ and $b_H = H_{\bar\xi}$ on $\mathbb{B} - Z_1$. So, with $Z = Z_0 \cup Z_1$, a null set, (18) holds for $H_\xi, H_{\bar\xi}$ at every $\xi \notin Z$ with $|\xi| < 1$ and $\xi \neq 0$. A null set has empty interior, so $\{0 < |\xi| < 1\} - Z$ is dense in $\{0 < |\xi| < 1\}$, and two continuous functions on an open set agreeing on a dense subset agree everywhere on it. Hence (18) holds at every ξ with $0 < |\xi| < 1$.

Step 5: (18) at every $\xi \neq 0$. Let $\lambda > 0$ and put $t = \lambda^\mu > 0$, so that $t^d = \lambda^{\mu d} = \lambda$; the second clause of `ExtremalPolarForm` then gives

$$H(\lambda\xi) = \lambda^\mu H(\xi) \qquad (\xi \in \mathbb{C},\ \lambda > 0). \tag{3}$$

Let $m(\xi) = \lambda\xi$. Then $\mathrm{d}m(\xi)[1] = \lambda$ and $\mathrm{d}m(\xi)[i] = \lambda i$, so by DZ and DZbar,

$$\partial_\xi m = \tfrac{1}{2}(\lambda - i\cdot\lambda i) = \lambda, \qquad \partial_{\bar\xi} m = \tfrac{1}{2}(\lambda + i\cdot\lambda i) = 0.$$

Fix $\xi_0 \neq 0$ and choose λ with $0 < \lambda < 1/|\xi_0|$, so that $0 < |\lambda\xi_0| < 1$. By (2), H is differentiable at $\lambda\xi_0 = m(\xi_0)$, and m is differentiable at ξ_0, so WirtingerChainRule applies to the pair (H, m) at ξ_0 and gives

$$\partial_\xi(H\circ m)(\xi_0) = \lambda\, H_\xi(\lambda\xi_0), \qquad \partial_{\bar\xi}(H\circ m)(\xi_0) = \lambda\, H_{\bar\xi}(\lambda\xi_0),$$

using $\partial_\xi m = \lambda$, $\partial_{\bar\xi} m = 0$ and $\overline{\lambda} = \lambda$. On the other hand $H\circ m = \lambda^\mu H$ as functions by (3), and ∂_ξ, $\partial_{\bar\xi}$ are $\mathbb{R}$-linear in the function, so $\partial_\xi(H\circ m)(\xi_0) = \lambda^\mu H_\xi(\xi_0)$ and likewise for $\partial_{\bar\xi}$. Comparing, and dividing by $\lambda \neq 0$,

$$H_\xi(\lambda\xi_0) = \lambda^{\mu-1} H_\xi(\xi_0), \qquad H_{\bar\xi}(\lambda\xi_0) = \lambda^{\mu-1} H_{\bar\xi}(\xi_0). \tag{4}$$

Since $\lambda > 0$ is real, $\lambda\xi_0/\overline{\lambda\xi_0} = \xi_0/\bar\xi_0$ and $\overline{\lambda\xi_0}/(\lambda\xi_0) = \bar\xi_0/\xi_0$. Substituting (4) into (18) at the point $\lambda\xi_0$, which is legitimate by Step 4, and dividing by $\lambda^{\mu-1} \neq 0$, we obtain (18) at ξ_0. Hence

$$\text{(18) holds at every } \xi \neq 0, \tag{5}$$

where (18) is read for the pointwise derivatives H_ξ, $H_{\bar\xi}$.

Step 5′: H solves (18) on $\mathbb{C}$ in the sense of SolvesEq18. We must produce weak Wirtinger derivatives of H on $\mathbb{C}$ and check (18) for them almost everywhere.

First a growth bound near the origin. Let $0 < |\xi| < 1$ and put $\xi_0 = \xi/|\xi|$ and $\lambda = |\xi|$, so that $|\xi_0| = 1$, $\lambda > 0$, $\lambda < 1 = 1/|\xi_0|$ and $\lambda\xi_0 = \xi$; these are exactly the conditions under which (4) was established, so $H_\xi(\xi) = \lambda^{\mu-1} H_\xi(\xi_0)$ and $H_{\bar\xi}(\xi) = \lambda^{\mu-1} H_{\bar\xi}(\xi_0)$. By (2) the functions H_ξ and $H_{\bar\xi}$ are continuous on $\mathbb{C} - \{0\}$, hence bounded on the compact unit circle, say by C_0. Therefore

$$|H_\xi(\xi)| \le C_0|\xi|^{\mu-1}, \qquad |H_{\bar\xi}(\xi)| \le C_0|\xi|^{\mu-1} \qquad (0 < |\xi| < 1),$$

whence by (2′), for $j = 0, 1$,

$$|\partial_j H(\xi)| \le 2C_0\,|\xi|^{\mu-1} \qquad (0 < |\xi| < 1). \tag{5′}$$

Next, H and $\partial_0 H, \partial_1 H$ lie in $L^1_{\mathrm{loc}}(\mathbb{C})$ in the sense of MemLpLocOn. For H this holds because H is continuous on $\mathbb{C}$ by Step 2, hence bounded on every compact set, which has finite measure. For $\partial_j H$: it is continuous on $\mathbb{C}-\{0\}$ by (2), hence measurable off the null set $\{0\}$ and so measurable. Let $\kappa \subseteq \mathbb{C}$ be compact and split $\kappa = \kappa_1 \cup \kappa_2$ with $\kappa_1 = \kappa \cap \{|\xi| \le \tfrac{1}{2}\}$ and $\kappa_2 = \kappa \cap \{|\xi| \ge \tfrac{1}{2}\}$. On κ_1, by (5′) and polar coordinates,

$$\int_{\kappa_1} |\partial_j H|\, d\sigma \le 2C_0 \int_{\{|\xi|<1/2\}} |\xi|^{\mu-1} d\sigma(\xi) = 4\pi C_0 \int_0^{1/2} r^{\mu-1}\, r\, dr = 4\pi C_0 \int_0^{1/2} r^\mu\, dr < \infty,$$

because $\mu > 0 > -1$ by Step 1. The set κ_2 is closed and bounded, hence compact, and contained in $\mathbb{C} - \{0\}$, where $\partial_j H$ is continuous; so $\partial_j H$ is bounded on κ_2, which has finite measure, and $\int_{\kappa_2} |\partial_j H|\, d\sigma < \infty$. Adding the two bounds gives $\int_\kappa |\partial_j H|\, d\sigma < \infty$.

Now fix $j \in \{0, 1\}$ and apply WeakPartialAcrossPoint with $\Omega = \mathbb{C}$, $S = \{0\}$, this j, $g = H$, $v = \partial_j H$ and $\beta = 0$. Its hypotheses hold: $\mathbb{C}$ is open and $S \subseteq \mathbb{C}$ with $S \cap \kappa$ finite for every compact κ, S being a singleton; $0 \le \beta = 0 < 1$; g is C^1 on $\mathbb{C} - S$ by (2), a function C^∞ on an open set being C^1 there; $D^{[j]} g = v$ on $\mathbb{C} - S$ by the definition of v; $g, v \in L^1_{\mathrm{loc}}(\mathbb{C})$ by the previous paragraph; and the growth hypothesis holds at the only point 0 of S with $\rho = 1$ and the constant $M = \sup_{|\xi|\le 1} |H(\xi)| < \infty$, finite because H is continuous on the compact set $\{|\xi| \le 1\}$, since $\|H(\xi)\| \le M = M|\xi - 0|^{-0}$ for $0 < |\xi| < 1$. Therefore $\partial_j H$ is the distributional ∂_j-derivative of H on $\mathbb{C}$ in the sense of IsWeakPartialOn.

Consequently the three clauses of IsWeakWirtingerOn hold for H on $\mathbb{C}$ with $u_x = \partial_0 H$, $u_y = \partial_1 H$ and the pair $a = H_\xi$, $b = H_{\bar\xi}$: clause (a) is the local integrability just proved, clause (b) is the last paragraph, and clause (c) is (2′) solved for H_ξ and $H_{\bar\xi}$, that is $H_\xi = \tfrac{1}{2}(\partial_0 H - i\partial_1 H)$ and $H_{\bar\xi} = \tfrac{1}{2}(\partial_0 H + i\partial_1 H)$, which hold at every point by DZ and DZbar. So H_ξ and $H_{\bar\xi}$ are themselves weak Wirtinger derivatives of H on $\mathbb{C}$, and by (5) the identity (18) holds for them

at every $\xi \neq 0$, hence at almost every $\xi \in \mathbb{C}$ with $\xi \neq 0$. This is exactly the assertion that H solves (18) at $n = 1$ on $\mathbb{C}$ in the sense of `SolvesEq18`; record it as

($5''$) H solves (18) at $n = 1$ on $\mathbb{C}$ in the sense of `SolvesEq18`.

Step 6: $H_\xi \neq 0$ off the origin. Let $\xi \neq 0$. By `ExtremalH` and $|\xi| > 0$,

$$H(\xi) = \xi\,|\xi|^{\mu-1} + \varepsilon\,|\xi|^{\mu+3}\xi^{-3} =: T_1(\xi) + T_2(\xi).$$

By `DZ` applied to the identity map, $\partial_\xi \xi = 1$ and $\partial_{\bar\xi}\xi = 0$. By `WirtingerModulusPowerRule`, $\partial_\xi(|\xi|^m) = \frac{m}{2}|\xi|^m/\xi$ for every real m, and these functions are C^∞ at ξ. By `WirtingerPowerRule` applied to the identity map on the open set $\mathbb{C} - \{0\}$, where it is C^∞ and non-vanishing, with the integer exponent -3, $\partial_\xi(\xi^{-3}) = -3\xi^{-4}$. Hence, by `WirtingerProductRule`,

$$\partial_\xi T_1 = |\xi|^{\mu-1} + \xi \cdot \frac{\mu-1}{2}\frac{|\xi|^{\mu-1}}{\xi} = \frac{\mu+1}{2}|\xi|^{\mu-1},$$

$$\partial_\xi T_2 = \varepsilon\Big[\frac{\mu+3}{2}\frac{|\xi|^{\mu+3}}{\xi}\xi^{-3} - 3|\xi|^{\mu+3}\xi^{-4}\Big] = \varepsilon\frac{\mu-3}{2}|\xi|^{\mu+3}\xi^{-4} = \varepsilon\frac{\mu-3}{2}|\xi|^{\mu-1}\frac{|\xi|^4}{\xi^4},$$

where the last step uses $|\xi|^{\mu+3} = |\xi|^{\mu-1}|\xi|^4$. Since $\big||\xi|^4/\xi^4\big| = 1$,

$$|H_\xi(\xi)| \;\geq\; |\xi|^{\mu-1} \cdot \frac{(\mu+1) - |\varepsilon|\,|\mu-3|}{2}. \tag{6}$$

We check that the last numerator is positive. By `ExtremalEpsBound`, $\varepsilon = (1-d)/(1+3d)$, and $d > 0$ by Step 1, so $1 + 3d > 0$ and $|\varepsilon| = |1-d|/(1+3d)$. With $\mu = 1/d$,

$$\mu + 1 = \frac{1+d}{d}, \qquad |\mu - 3| = \frac{|1-3d|}{d},$$

so $(\mu+1) - |\varepsilon||\mu-3| > 0$ is equivalent, after multiplying by the positive number $d(1+3d)$, to

$$(1+d)(1+3d) > |1-d|\,|1-3d| = |(1-d)(1-3d)|,$$

that is, to $1+4d+3d^2 > |1-4d+3d^2|$. This holds because $(1+4d+3d^2)-(1-4d+3d^2) = 8d > 0$ and $(1+4d+3d^2)+(1-4d+3d^2) = 2+6d^2 > 0$. Hence the numerator in (6) is positive and

$$H_\xi(\xi) \neq 0 \qquad (\xi \neq 0). \tag{7}$$

Step 7: $\det \mathrm{d}H = J_H$, and $J_H > 0$ off the origin. Let $F\colon \mathbb{C} \to \mathbb{C}$ be differentiable at w and put $L = \mathrm{d}F(w)$, $a = F_z(w)$, $b = F_{\bar z}(w)$. By `DZ` and `DZbar`, $a = \frac12(L[1] - iL[i])$ and $b = \frac12(L[1] + iL[i])$, so $L[1] = a + b$ and $L[i] = i(a-b)$. In the basis $\{1, i\}$ the matrix of L has columns $(\mathrm{Re}(a+b), \mathrm{Im}(a+b))$ and $(-\mathrm{Im}(a-b), \mathrm{Re}(a-b))$, so

$$\det L = \mathrm{Re}(a+b)\,\mathrm{Re}(a-b) + \mathrm{Im}(a+b)\,\mathrm{Im}(a-b) = \mathrm{Re}\big((a+b)\overline{(a-b)}\big) = |a|^2 - |b|^2,$$

because $b\bar a - a\bar b = b\bar a - \overline{b\bar a}$ is purely imaginary. By `Jac` this says

$$\det \mathrm{d}F(w) = J_F(w) = |F_z(w)|^2 - |F_{\bar z}(w)|^2. \tag{8}$$

Now let $\xi \neq 0$. Taking moduli in (5) and using $|\xi/\bar\xi| = |\bar\xi/\xi| = 1$ and $|\overline{H_\xi}| = |H_\xi|$,

$$|H_{\bar\xi}(\xi)| \leq 2|c|\,|H_\xi(\xi)| = \frac{|p-2|}{p}|H_\xi(\xi)|.$$

For $p > 1$ we have $|p-2| < p$, since $p - 2 < p$ and $-(p-2) = 2 - p < p$ is $1 < p$; so $|p-2|/p < 1$ and, by (7) and `Jac`,

$$J_H(\xi) = |H_\xi(\xi)|^2 - |H_{\bar\xi}(\xi)|^2 \;\geq\; \Big(1 - \frac{(p-2)^2}{p^2}\Big)|H_\xi(\xi)|^2 > 0 \qquad (\xi \neq 0). \tag{9}$$

Step 8: f is C^∞ off the origin, with $J_f \neq 0$ there. Fix $z_0 \neq 0$ and put $\xi_0 = f(z_0)$. By (1), $\xi_0 \neq 0$, and $H(\xi_0) = H(f(z_0)) = z_0$ by Step 2. By (2), H is C^∞ on a neighbourhood of ξ_0, and by (8) and (9) its differential $\mathrm{d}H(\xi_0)$ has non-zero determinant, hence is invertible. By the inverse function theorem there are open sets $N \ni \xi_0$ and $M \ni z_0$ such that H maps N bijectively onto M and the inverse $h\colon M \to N$ of this restriction is C^∞.

Since f is continuous and $f(z_0) = \xi_0 \in N$, there is an open M' with $z_0 \in M' \subseteq M$ and $f(M') \subseteq N$. For $z \in M'$ we have $H(f(z)) = z$ with $f(z) \in N$ and $z \in M$, so $f(z) = h(H(f(z))) = h(z)$.

Thus $f = h$ on M', and f is C^∞ on a neighbourhood of z_0. As $z_0 \neq 0$ was arbitrary, this proves (b).

Moreover, differentiating $H \circ f = \mathrm{id}$ on M' at z_0 gives $\mathrm{d}H(\xi_0) \circ \mathrm{d}f(z_0) = \mathrm{id}$, so $\mathrm{d}f(z_0)$ is invertible and, by (8) applied to f,

$$J_f(z) = \det \mathrm{d}f(z) \neq 0 \qquad (z \neq 0). \tag{10}$$

Step 9: homogeneity. By (b), f is C^∞ on $\{0\}^c = \mathbb{C} - \{0\}$, so `ExtremalHomogeneous` applies with the two inverse relations of Step 2, and yields exactly the two clauses of (c). Nothing in its hypotheses refers to equation (12), so this step may be taken before Step 11.

Step 10: $f \in W^{1,1}_{\mathrm{loc}}(\mathbb{C})$. By the second clause of (c) at $|\nu| = 1$, that is for the one-letter words $w = [j]$,

$$\partial_j f(tz) = t^{\,d-1} \partial_j f(z) \qquad (t > 0,\ z \neq 0,\ j = 0, 1),$$

$\partial_j = D^{[j]}$ being the classical partial derivative of `PartialWord`. By (b) the functions $\partial_0 f, \partial_1 f$ are continuous on $\mathbb{C} - \{0\}$, hence bounded on the compact unit circle, say by C_1; taking $t = |z|$ and the point $z/|z|$ of that circle,

$$|\partial_j f(z)| \leq C_1 |z|^{\,d-1} \qquad (z \neq 0,\ j = 0, 1). \tag{10'}$$

Hence $\partial_j f \in L^1_{\mathrm{loc}}(\mathbb{C})$ in the sense of `MemLpLocOn`: it is continuous off the null set $\{0\}$ and so measurable, and for compact $\kappa \subseteq B(0, R)$, by (10') and polar coordinates,

$$\int_\kappa |\partial_j f|\, d\sigma \leq C_1 \int_{B(0,R)} |z|^{\,d-1} d\sigma(z) = 2\pi C_1 \int_0^R r^{\,d-1} r\, dr = 2\pi C_1 \int_0^R r^{\,d}\, dr < \infty,$$

because $d > 0 > -1$ by Step 1. Also $f \in L^1_{\mathrm{loc}}(\mathbb{C})$, being continuous on $\mathbb{C}$ by (a) and hence bounded on each compact set, which has finite measure.

Fix $j \in \{0, 1\}$ and apply `WeakPartialAcrossPoint` with $\Omega = \mathbb{C}$, $S = \{0\}$, this j, $g = f$, $v = \partial_j f$ and $\beta = 0$: $\mathbb{C}$ is open, S is a singleton contained in it so $S \cap \kappa$ is finite for every compact κ, $0 \leq \beta = 0 < 1$, g is C^1 on $\mathbb{C} - S$ by (b), a function C^∞ on an open set being C^1 there, $D^{[j]} g = v$ there by the definition of v, g and v are in $L^1_{\mathrm{loc}}(\mathbb{C})$ by the last paragraph, and the growth hypothesis holds at 0 with $\rho = 1$ and $M = \sup_{|z| \leq 1} |f(z)| < \infty$, finite by continuity of f on the compact disc $\{|z| \leq 1\}$. So $\partial_j f$ is the distributional ∂_j-derivative of f on $\mathbb{C}$ in the sense of `IsWeakPartialOn`.

Therefore the family D with $D_{()} = f$, $D_{(j)} = \partial_j f$ for $j = 0, 1$ and $D_w = 0$ for $|w| \geq 2$ witnesses the three clauses of `MemWLocOn` for $k = 1$, $s = 1$ on $\mathbb{C}$: clause (a) is $D_{()} = f$; clause (b) is the local integrability of f and of $\partial_0 f, \partial_1 f$ just proved, the functions D_w with $|w| \geq 2$ being irrelevant since $|w| \leq k = 1$; and clause (c), which concerns only the empty word, is the previous paragraph. Hence

$$f \in W^{1,1}_{\mathrm{loc}}(\mathbb{C}) \quad \text{in the sense of } \texttt{MemWLocOn}. \tag{10''}$$

Step 11: f solves (12) on $\mathbb{C}$. We apply `InverseSolvesEq12` with $V = \mathbb{C}$ and the pair (H, f). Its hypotheses hold: $\mathbb{C}$ is open; $H(\mathbb{C}) = \mathbb{C}$ is open, H being surjective by Step 2; H is C^1 on $\mathbb{C} - \{0\}$ by (2); H solves (18) at $n = 1$ on $\mathbb{C}$ in the sense of `SolvesEq18` by (5''); $f(H(\xi)) = \xi$ for every ξ by Step 2; by (b) and (10), f is differentiable with $J_f \neq 0$ at every $z \neq 0$, hence at almost every z, the origin being a null set; and its hypothesis (H6), $f \in W^{1,1}_{\mathrm{loc}}(H(\mathbb{C})) = W^{1,1}_{\mathrm{loc}}(\mathbb{C})$, is (10''), the image $H(\mathbb{C})$ being $\mathbb{C}$ by Step 2. The conclusion is that f solves (12) on $\mathbb{C}$ in the sense of `SolvesEq12`: there are weak Wirtinger derivatives a_f, b_f of f on $\mathbb{C}$ and a null set off which

$$b_f = \Big(\frac{1}{p} - \frac{1}{2}\Big)\Big[\frac{\overline{f}}{f} a_f + \frac{f}{\overline{f}} \overline{a_f}\Big]$$

at the points where $f \neq 0$. By (W) of Step 3', applied to $u = f$ on the open set $\Omega = \mathbb{C}$ — legitimate because f is differentiable at every point of $\mathbb{C} - \{0\}$ by (b) and $\{0\}$ is null — we have $a_f = f_z$ and $b_f = f_{\bar z}$ almost everywhere, so, the union of two null sets being null,

$$f_{\bar z} = \Big(\frac{1}{p} - \frac{1}{2}\Big)\Big[\frac{\overline{f}}{f} f_z + \frac{f}{\overline{f}} \overline{f_z}\Big] \tag{11}$$

almost everywhere where $f \neq 0$, the derivatives now being the pointwise ones of `DZ` and `DZbar`.

Step 12: (12) holds at every $z \neq 0$, and $f_{\bar z}$ is real there. By (1), f does not vanish on the open set $\mathbb{C} - \{0\}$, and by (b) it is C^∞ there, so, as in Step 4, both sides of (11) are continuous on $\mathbb{C} - \{0\}$ while (11) holds off a null set; the complement of a null set is dense, so (11) holds at every $z \neq 0$. Writing $a = (\overline{f}/f) f_z$ we have $(f/\overline{f})\overline{f_z} = \bar a$, so the bracket in (11) is $a + \bar a$, which is real, and $1/p - 1/2$ is real. Hence

$$\overline{f_{\bar z}(z)} = f_{\bar z}(z) \qquad (z \neq 0). \tag{12}$$

Step 13: Euler's identity. Let F be differentiable at w and $L = \mathrm{d}F(w)$, $a = F_z(w)$, $b = F_{\bar z}(w)$. By the computation in Step 7, $L[1] = a + b$ and $L[i] = i(a - b)$, so for $h = h_1 + ih_2$ with h_1, h_2 real,

$$L[h] = h_1(a + b) + h_2\, i(a - b) = a(h_1 + ih_2) + b(h_1 - ih_2) = a\,h + b\,\bar h.$$

Fix $z \neq 0$ and put $\psi(t) = f(tz)$ for $t > 0$. By (b) and the chain rule, ψ is differentiable and $\psi'(1) = \mathrm{d}f(z)[z] = z f_z(z) + \bar z f_{\bar z}(z)$. By the first clause of (c), $\psi(t) = t^d f(z)$ for $t > 0$, so $\psi'(1) = d\, f(z)$. Hence

$$z f_z(z) + \bar z f_{\bar z}(z) = d\, f(z) \qquad (z \neq 0). \tag{13}$$

Step 14: the potential. By Step 1, $d + 1 > 0$. Define

$$v(z) = \frac{2}{d+1} \operatorname{Re}\big(z f(z)\big) \qquad (z \in \mathbb{C}),$$

a real-valued function, continuous on $\mathbb{C}$ because f is. Let φ_v denote v regarded as complex-valued, so that

$$\varphi_v(z) = \frac{1}{d+1}\Big(z f(z) + \overline{z f(z)}\Big).$$

Since f is C^∞ on $\mathbb{C} - \{0\}$ and conjugation is a real-linear isometry, φ_v is C^∞ on $\mathbb{C} - \{0\}$. Fix $z \neq 0$. By $\partial_z z = 1$, $\partial_{\bar z} z = 0$ and `WirtingerProductRule`,

$$\partial_z\big(z f(z)\big) = f(z) + z f_z(z), \qquad \partial_{\bar z}\big(z f(z)\big) = z f_{\bar z}(z),$$

and by `WirtingerConjRule` and (12),

$$\partial_z \overline{\big(z f(z)\big)} = \overline{\partial_{\bar z}\big(z f(z)\big)} = \overline{z f_{\bar z}(z)} = \bar z\, f_{\bar z}(z).$$

Adding and using (13),

$$\partial_z \varphi_v(z) = \frac{1}{d+1}\big(f(z) + z f_z(z) + \bar z f_{\bar z}(z)\big) = \frac{1}{d+1}\big(f(z) + d\, f(z)\big) = f(z) \qquad (z \neq 0), \tag{14}$$

which is (e).

Step 15: local integrability of the derivatives of $|f|^{p-2}f$. Put $w = |f|^{p-2} f$ and $\kappa = (p-1)d > 0$. By (1) and (b), f is C^∞ and non-vanishing on $\mathbb{C} - \{0\}$, and $\zeta \mapsto |\zeta|^{p-2}$ is C^∞ on $\mathbb{C} - \{0\}$ by `WirtingerModulusPowerRule`, so w is C^∞ on $\mathbb{C} - \{0\}$. For $t > 0$ and $z \neq 0$, the first clause of (c) gives $f(tz) = t^d f(z)$ with $t^d > 0$, so

$$w(tz) = \big|t^d f(z)\big|^{p-2} t^d f(z) = t^{(p-2)d} t^d w(z) = t^\kappa w(z).$$

Fix $t > 0$ and $j \in \{0, 1\}$. Differentiating the identity $w(tz) = t^\kappa w(z)$ in z on the open set $\mathbb{C} - \{0\}$, the left side has j-th partial $t\,(\partial_j w)(tz)$ by the chain rule and the right side has $t^\kappa \partial_j w(z)$, so

$$(\partial_j w)(tz) = t^{\kappa - 1} \partial_j w(z) \qquad (z \neq 0,\ t > 0).$$

The function $\partial_j w$ is continuous on $\mathbb{C} - \{0\}$, hence bounded on the unit circle, say by C. For $z \neq 0$, applying the last display with $t = |z|$ and the point $z/|z|$ of the unit circle,

$$|\partial_j w(z)| = |z|^{\kappa - 1} \big|\partial_j w(z/|z|)\big| \leq C |z|^{\kappa - 1}.$$

Let $K \subseteq \mathbb{C}$ be compact and choose R with $K \subseteq B(0, R)$. In polar coordinates

$$\int_{B(0,R)} |z|^{\kappa - 1}\, \mathrm{d}\sigma(z) = 2\pi \int_0^R r^{\kappa - 1}\, r\, \mathrm{d}r = 2\pi \int_0^R r^\kappa\, \mathrm{d}r < \infty,$$

because $\kappa > 0 > -1$. Hence $\partial_j w$, which is continuous off the null set $\{0\}$ and so measurable, is integrable on K; that is, $\partial_j w \in L^1_{\mathrm{loc}}(\mathbb{C})$ in the sense of `MemLpLocOn`.

Step 16: v is p-harmonic. We apply `Eq12ImpliesPHarmonic` with $\Omega = \mathbb{C}$, $S = \{0\}$, and the pair (f, v). Its hypotheses hold: $S \cap K \subseteq \{0\}$ is finite for every compact K; $f(z) = 0$ exactly for $z \in S$ by (1); f is continuous by (a) and C^∞ on $\mathbb{C} - \{0\}$ by (b); f solves (12) on $\mathbb{C}$ by Step 11; v is continuous on $\mathbb{C}$ and C^∞ on $\mathbb{C} - \{0\}$ by Step 14; $\partial v/\partial z = f$ on $\mathbb{C} - \{0\}$ by (14); v and f are continuous, hence bounded on compact sets, hence in $L^p_{\mathrm{loc}}(\mathbb{C})$ in the sense of `MemLpLocOn`; and the two first-order partial derivatives of $|f|^{p-2}f$ are in $L^1_{\mathrm{loc}}(\mathbb{C})$ by Step 15. The conclusion is that v is p-harmonic on $\mathbb{C}$ in the sense of `PHarmonicOn`, which together with the continuity of v on $\mathbb{C}$ and its smoothness on $\mathbb{C} - \{0\}$, both established in Step 14, is (d).

Steps 2, 8, 9, 14 and 16 establish (a), (b), (c), (e) and (d) respectively, so the pair (f, v) has all the stated properties. □

2.70. `FDerivAcrossPoint`.

Lemma 2.70. *Let $\Omega \subseteq \mathbb{C}$ be open, let $z_0 \in \Omega$, let $g\colon \mathbb{C} \to \mathbb{C}$ and let $L\colon \mathbb{C} \to \mathbb{C}$ be a continuous $\mathbb{R}$-linear map. Assume*

(H1) *g is continuous on Ω;*
(H2) *g is (real) differentiable at every point of $\Omega - \{z_0\}$;*
(H3) *$\mathrm{d}g(z) \to L$ as $z \to z_0$ within $\Omega - \{z_0\}$.*

Then g is differentiable at z_0 with $\mathrm{d}g(z_0) = L$, and $z \mapsto \mathrm{d}g(z)$ is continuous at z_0 within Ω.

This is the step Section 6 of the paper passes over. It is what upgrades C^1 to C^k across the singular point: by (52), for $|\nu| = m+1 \le k$ the exponent $d-m = k-1+\alpha-m$ is at least $\alpha > 0$, so the derivative in question tends to 0 at the singular point, and this lemma turns that into genuine differentiability there. It is also the continuity at the singular point that the hypothesis "F continuous on U" of Lemma 2 of the paper (paper.tex:538–539) silently presumes.

Proof. Write $h(z) = g(z) - g(z_0) - L(z - z_0)$ for $z \in \Omega$. By (H1), h is continuous on Ω with $h(z_0) = 0$; by (H2) and linearity of L, h is differentiable at every $z \in \Omega - \{z_0\}$, with

$$\mathrm{d}h(z) = \mathrm{d}g(z) - L \qquad (z \in \Omega - \{z_0\}). \tag{1}$$

Since Ω is open and $z_0 \in \Omega$, fix $r > 0$ with $B = \{z : |z - z_0| < r\} \subseteq \Omega$.

Step 1: the radial mean value estimate. We claim: if $0 < \varepsilon$, $0 < \delta \le r$, and

$$\big\|\mathrm{d}g(w) - L\big\| \le \varepsilon \qquad \text{for every } w \text{ with } 0 < |w - z_0| < \delta, \tag{2}$$

then

$$\big\|h(z)\big\| \le \varepsilon\,|z - z_0| \qquad \text{for } |z - z_0| < \delta. \tag{3}$$

Let z with $|z - z_0| < \delta$. If $z = z_0$ then $h(z) = 0$ and (3) is clear, so assume $z \ne z_0$. For $t \in (0, 1]$ put $z_t = z_0 + t(z - z_0)$, and let $S_t = [z_t, z]$ be the closed segment joining z_t to z. Every point of S_t has the form $z_0 + s(z - z_0)$ with $s \in [t, 1]$, so its distance to z_0 is $s|z - z_0| \in (0, \delta)$; hence $S_t \subseteq B - \{z_0\} \subseteq \Omega - \{z_0\}$ and, by (2) and (1), $\|\mathrm{d}h\| \le \varepsilon$ at every point of S_t. The segment S_t is convex and h is differentiable at each of its points, so the mean value inequality on a convex set — if $\|\mathrm{d}h\| \le C$ on a convex set S then $\|h(y) - h(x)\| \le C|y - x|$ for $x, y \in S$ — gives

$$\big\|h(z) - h(z_t)\big\| \le \varepsilon\,|z - z_t| = \varepsilon(1 - t)|z - z_0| \le \varepsilon\,|z - z_0|.$$

Now let $t \to 0^+$. Then $z_t \to z_0$, and h is continuous at z_0 with $h(z_0) = 0$, so $h(z_t) \to 0$; the left-hand side therefore tends to $\|h(z)\|$, and (3) follows.

Step 2: g is differentiable at z_0 with derivative L. Let $\varepsilon > 0$. By (H3) there is $\delta_0 > 0$ such that (2) holds with $\delta = \min\{\delta_0, r\}$: indeed (H3) says precisely that for every $\varepsilon > 0$ the set of $w \in \Omega - \{z_0\}$ with $\|\mathrm{d}g(w) - L\| \le \varepsilon$ contains the intersection of $\Omega - \{z_0\}$ with a punctured ball about z_0, and shrinking that ball into $B \subseteq \Omega$ makes every w with $0 < |w - z_0| < \delta$ lie in $\Omega - \{z_0\}$. By Step 1,

$$\big\|g(z) - g(z_0) - L(z - z_0)\big\| \le \varepsilon\,|z - z_0| \qquad \text{for } |z - z_0| < \delta.$$

As $\varepsilon > 0$ was arbitrary, $g(z) - g(z_0) - L(z - z_0) = o(|z - z_0|)$ as $z \to z_0$, which is exactly the statement that g is differentiable at z_0 with $\mathrm{d}g(z_0) = L$.

Step 3: continuity of the derivative at z_0 within Ω. By Step 2, $\mathrm{d}g(z_0) = L$. Let $\varepsilon > 0$ and take δ as in Step 2. For $z \in \Omega$ with $|z - z_0| < \delta$ there are two cases: if $z \neq z_0$ then $\|\mathrm{d}g(z) - L\| \leq \varepsilon$ by (2); and if $z = z_0$ then $\mathrm{d}g(z) - L = L - L = 0$. In both cases $\|\mathrm{d}g(z) - \mathrm{d}g(z_0)\| \leq \varepsilon$. Hence $\mathrm{d}g(z) \to \mathrm{d}g(z_0)$ as $z \to z_0$ within Ω, i.e. $z \mapsto \mathrm{d}g(z)$ is continuous at z_0 within Ω. □

2.71. `HomogPolyModFour.`

Lemma 2.71. *Let $\epsilon \in \mathbb{C}$ with $\epsilon \neq 0$, let $d \geq 1$ be an integer, and let $A_{mn} \in \mathbb{C}$ for $m, n \geq 0$. Then the identity*

$$\xi = \sum_{m+n=d} A_{mn}\big(\xi + \epsilon\xi^{-3}\big)^m\big(\xi^{-1} + \epsilon\xi^3\big)^n \tag{60}$$

cannot hold for every $\xi \in \mathbb{C}$ with $\xi \neq 0$. Here the sum runs over the $d + 1$ pairs (m, n) with $m + n = d$, $m, n \geq 0$, and powers are taken with the convention $w^0 = 1$ for every $w \in \mathbb{C}$, including $w = 0$.

This is the contradiction with which Section 7 of the paper closes (paper.tex:636–642), and with it the optimality half of Theorem 1. The only input is $\epsilon \neq 0$, which for the extremal function (54) is exactly the condition $p \neq 2$.

Proof. Suppose (60) holds for every $\xi \neq 0$.

Step 1: the two sets of fourth roots. Since $\epsilon \neq 0$ we have $-\epsilon \neq 0$ and $-1/\epsilon \neq 0$. Every nonzero complex number has a fourth root, so fix $\zeta \neq 0$ with $\zeta^4 = -\epsilon$ and $\eta \neq 0$ with $\eta^4 = -1/\epsilon$. Note that $(i\zeta)^4 = i^4\zeta^4 = \zeta^4 = -\epsilon$ and likewise $(i\eta)^4 = -1/\epsilon$, and that $i\zeta \neq 0$, $i\eta \neq 0$.

Step 2: the first evaluation. Let $\xi \neq 0$ with $\xi^4 = -\epsilon$. Then

$$\xi + \epsilon\xi^{-3} = \xi^{-3}\big(\xi^4 + \epsilon\big) = \xi^{-3}\big(-\epsilon + \epsilon\big) = 0.$$

Hence in (60) each term with $m \geq 1$ carries the factor $0^m = 0$ and vanishes, while the term with $m = 0$, $n = d$ carries $0^0 = 1$. So (60) reduces to

$$\xi = A_{0d}\big(\xi^{-1} + \epsilon\xi^3\big)^d.$$

Moreover

$$\xi^{-1} + \epsilon\xi^3 = \xi^{-1}\big(1 + \epsilon\xi^4\big) = \xi^{-1}\big(1 + \epsilon(-\epsilon)\big) = \xi^{-1}\big(1 - \epsilon^2\big),$$

so $\xi = A_{0d}\,\xi^{-d}\big(1 - \epsilon^2\big)^d$. Multiplying by ξ^d, which is legitimate since $\xi \neq 0$,

$$\xi^{d+1} = A_{0d}\big(1 - \epsilon^2\big)^d \qquad \text{for every } \xi \neq 0 \text{ with } \xi^4 = -\epsilon. \tag{1}$$

The right-hand side of (1) does not depend on which such ξ is taken.

Step 3: 4 divides $d + 1$. Apply (1) to $\xi = \zeta$ and to $\xi = i\zeta$, both legitimate by Step 1. The two right-hand sides are equal, so

$$\zeta^{d+1} = (i\zeta)^{d+1} = i^{d+1}\zeta^{d+1}.$$

Since $\zeta \neq 0$ we have $\zeta^{d+1} \neq 0$, and dividing by it gives $i^{d+1} = 1$. The element i has order 4 in the multiplicative group of $\mathbb{C}$ — its powers cycle through $i, -1, -i, 1$ — so $i^N = 1$ if and only if $4 \mid N$. Hence

$$4 \mid d + 1. \tag{2}$$

Step 4: the second evaluation. Let $\xi \neq 0$ with $\xi^4 = -1/\epsilon$. Then

$$\xi^{-1} + \epsilon\xi^3 = \xi^{-1}\big(1 + \epsilon\xi^4\big) = \xi^{-1}\Big(1 + \epsilon \cdot \Big(-\frac{1}{\epsilon}\Big)\Big) = 0,$$

using $\epsilon \neq 0$. Hence in (60) each term with $n = d - m \geq 1$, that is with $m \leq d - 1$, carries the factor $0^n = 0$ and vanishes, while the term with $m = d$, $n = 0$ carries $0^0 = 1$. So (60) reduces to

$$\xi = A_{d0}\big(\xi + \epsilon\xi^{-3}\big)^d,$$

and

$$\xi + \epsilon\xi^{-3} = \xi^{-3}\big(\xi^4 + \epsilon\big) = \xi^{-3}\Big(\epsilon - \frac{1}{\epsilon}\Big),$$

so $\xi = A_{d0}\,\xi^{-3d}\big(\epsilon - 1/\epsilon\big)^d$. Multiplying by ξ^{3d},

$$\xi^{3d+1} = A_{d0}\Big(\epsilon - \frac{1}{\epsilon}\Big)^d \qquad \text{for every } \xi \neq 0 \text{ with } \xi^4 = -1/\epsilon, \tag{3}$$

again with a right-hand side independent of the choice of root.

Step 5: 4 divides $3d+1$. Exactly as in Step 3, applying (3) to $\xi = \eta$ and $\xi = i\eta$ gives $\eta^{3d+1} = i^{3d+1}\eta^{3d+1}$ with $\eta^{3d+1} \neq 0$, hence $i^{3d+1} = 1$ and

$$4 \mid 3d+1. \tag{4}$$

Step 6: the contradiction. By (2) and (4), 4 divides the sum

$$(d+1) + (3d+1) = 4d+2.$$

But $4 \mid 4d$, so $4 \mid 4d+2$ would give $4 \mid 2$, which is false. This contradiction shows that (60) cannot hold for every $\xi \neq 0$, which is the assertion. □

2.72. `KAlphaEqDExpAddOne`.

Lemma 2.72. *Let $1 < p < \infty$, let $k+\alpha$ denote the right-hand side of equation (9) of the paper, as defined in `KAlphaExp`, and let $d = d(p)$ be the exponent (50), as defined in `DExp`. Then*

$$k + \alpha = d + 1, \qquad \textit{equivalently} \qquad d = k - 1 + \alpha.$$

This is the identity the paper records at paper.tex:530, "Let the integer k and the exponent $\alpha \in (0,1]$ be determined by (9), so $d = k-1+\alpha$". Both halves of Theorem 1 use it: the regularity half converts Corollary 2's exponent $d-m$ into $k-1+\alpha-m$, and the optimality half reads the cases (58), (59) off the same identity.

Proof. By `KAlphaExp` and `DExp`, writing

$$S = \sqrt{1 + \frac{14}{p-1} + \frac{1}{(p-1)^2}},$$

we have

$$k + \alpha = \frac{1}{6}\Big(7 + \frac{1}{p-1} + S\Big), \qquad d = \frac{1}{6}\Big(1 + \frac{1}{p-1} + S\Big).$$

The two expressions have the same second and third summands, so

$$(k+\alpha) - d = \frac{1}{6}\big(7-1\big) = \frac{6}{6} = 1,$$

that is $k + \alpha = d + 1$. Subtracting 1 from both sides and rearranging gives $d = k - 1 + \alpha$. Note that no property of p beyond the definitions is used: the identity is a consequence of the fact that (9) and (50) differ only in their leading constant. □

2.73. `KAlphaWellDefined`.

Lemma 2.73. *Let $1 < p < \infty$ and let $\kappa(p)$ be the right-hand side of equation (9), as in `KAlphaExp`. Then there is exactly one pair (k, α) consisting of an integer $k \geq 1$ and a real number α with $0 < \alpha \leq 1$ such that*

$$k + \alpha = \kappa(p).$$

Proof. Fix $p > 1$ and write $v = \kappa(p)$ and $q = \dfrac{1}{p-1}$, so that $q > 0$ and, by `KAlphaExp`,

$$6v = 7 + q + \sqrt{1 + 14q + q^2}.$$

Step 1 ($v > \frac{4}{3}$, in particular $v > 1$). Since $q > 0$ we have $14q > 0$ and $q^2 \geq 0$, so $1+14q+q^2 > 1$. The square root is strictly increasing on $[0, \infty)$ and $\sqrt{1} = 1$, so $\sqrt{1 + 14q + q^2} > 1$. Hence

$$6v = 7 + q + \sqrt{1 + 14q + q^2} > 7 + 0 + 1 = 8,$$

so $v > \frac{8}{6} = \frac{4}{3} > 1$.

Step 2 (existence). Let $m = \lceil v \rceil$ be the least integer $\geq v$, and set

$$k = m - 1, \qquad \alpha = v - k.$$

Since $v > 1$, the integer 1 is not $\geq v$, so $m \geq 2$ and therefore $k = m - 1 \geq 1$ is an integer ≥ 1. By definition of the ceiling we have $v \leq m$, and also $m - 1 < v$, for otherwise $m - 1$ would be an integer $\geq v$ smaller than m. Thus

$$k < v \leq k + 1,$$

and hence $\alpha = v - k$ satisfies $0 < \alpha \leq 1$. Finally $k + \alpha = k + (v - k) = v$. So the pair (k, α) has all the required properties.

Step 3 (uniqueness). Suppose (k, α) and (k', α') both consist of an integer ≥ 1 and a real number in $(0, 1]$, and that

$$k + \alpha = v = k' + \alpha'.$$

Then $k - k' = \alpha' - \alpha$. From $0 < \alpha \leq 1$ and $0 < \alpha' \leq 1$ we get

$$-1 \leq -\alpha < \alpha' - \alpha < \alpha' \leq 1,$$

so $\alpha' - \alpha$ lies in the open interval $(-1, 1)$. Hence the integer $k - k'$ lies in $(-1, 1)$, which forces $k - k' = 0$, i.e. $k = k'$. Then $\alpha = v - k = v - k' = \alpha'$. So the two pairs coincide. □

2.74. `NormalizedFactorization`.

Lemma 2.74. *Let $1 < p < \infty$, let $K \in \mathbb{R}$, let $\Omega \subseteq \mathbb{C}$ be open and connected, and let $f \colon \mathbb{C} \to \mathbb{C}$ be K-quasiregular on Ω in the sense of `QuasiregularOn`. Let $g \colon \mathbb{C} \to \mathbb{C}$ satisfy*

(i) *$g = f$ almost everywhere on Ω;*
(ii) *g is continuous on Ω;*
(iii) *g is real-differentiable at almost every point of Ω;*
(iv) *g solves equation (12) of the paper on Ω in the sense of `SolvesEq12`.*

Assume $g(z^*) \neq 0$ for some $z^* \in \Omega$, and let $z_0 \in \Omega$ satisfy $g(z_0) = 0$. Assume the bundles (C2) and (C3) of `QCPackage` and `SmoothOffZeros`. Then there are an integer $m \geq 1$, a constant $K' \geq 1$, an open set V' with $z_0 \in V' \subseteq \Omega$ and maps $\chi, H \colon \mathbb{C} \to \mathbb{C}$ such that

(a) *χ is K'-quasiconformal on V' in the sense of `QuasiconformalOn`, $\chi(z_0) = 0$, and $g(z) = [\chi(z)]^m$ for every $z \in V'$;*
(b) *χ is C^∞ on $V' - \{z_0\}$;*
(c) *$H(\chi(z)) = z$ for every $z \in V'$, H is continuous on $\chi(V')$, $H(0) = z_0$, and for every $\xi \in \chi(V')$ one has $H(\xi) \in V'$ and $\chi(H(\xi)) = \xi$.*

This is the normalised form of the factorization (15) that Sections 3 to 5 use. Clause (a) is the factorization theorem quoted in (C2), shrunk so that z_0 is the only zero of g in V'; (b) is `ChiSmoothOffZero` applied on the punctured neighborhood, where (C3) makes g smooth and the discreteness of the zero set makes it nonvanishing; and (c) is the inverse furnished by the quasiconformality of χ, with the two inversion identities and $H(0) = z_0$.

The factorization, the equation and the singular point are all read for the continuous representative g of f, not for f itself, and they must be. Quasiregularity as `QuasiregularOn` defines it constrains f only through $W^{1,2}_{\mathrm{loc}}$ membership and an almost-everywhere inequality, both unchanged by altering f on a null set; so no hypothesis on f alone can give a pointwise identity $f = \chi^m$ on an open set, and the set $f^{-1}(0)$ is not determined modulo null sets, whereas $g^{-1}(0)$ is. This is the same reading of the quoted (C2) and (C3) that `QCPackage` and `SmoothOffZeros` carry — the paper's own at paper.tex:156, where the singular set is written $f^{-1}(0) = \{z \in \Omega : \nabla u(z) = 0\}$ — and a single representative is threaded through all of them.

No smoothness of H is claimed, and none is available here: the paper's arrow runs $H \to \chi$ (paper.tex:416), and H's regularity off the origin is produced downstream, by term-by-term differentiation of the expansion (19) as at paper.tex:387–388. Claiming it here would invert the development, and it is what forced the illegal fourth bundle in earlier cycles. Nothing else about H is claimed either. Its $W^{1,2}_{\mathrm{loc}}$ membership, almost everywhere differentiability and $J_H > 0$ almost everywhere are one application of (C2)'s inverse clause away; and its quasiconformality — which the paper asserts flatly at paper.tex:179 but which (C2) does not quote, so that paper.tex:113 requires it to be proved — is assembled from those, from (18) and from the homeomorphism clauses of χ in the node that applies Corollary 1, which already carries the bundle (C2) and the χ data.

Proof. Step 1: the factorization and the isolated zero. The clause `qr_factorization` of `QCPackage`, applied with K, Ω, the pair f, g, the point z_0, the openness and connectedness of Ω, the quasiregularity of f, the hypotheses (i) and (ii) on g, the point z^* with $g(z^*) \neq 0$ and $g(z_0) = 0$, provides an integer $m \geq 1$, a constant K', an open set V_0 with $z_0 \in V_0 \subseteq \Omega$ and a map χ with

$$\chi \text{ is } K'\text{-quasiconformal on } V_0, \qquad \chi(z_0) = 0, \qquad g(z) = [\chi(z)]^m \ (z \in V_0). \tag{1}$$

By `QuasiconformalOn` the first clause of (1) contains K'-quasiregularity of χ on V_0, whose first clause is $1 \leq K'$ by `QuasiregularOn`; so

$$1 \leq m, \qquad 1 \leq K'. \tag{2}$$

The clause `qr_discrete_zeros` of `QCPackage`, applied with the same K, Ω, the pair f, g, the openness and connectedness of Ω, the quasiregularity of f, the hypotheses (i) and (ii) on g and the point z^*, read at the point $z_0 \in \Omega$, provides $r_0 > 0$ with

$$g(z) \neq 0 \qquad \text{for every } z \in \Omega \text{ with } z \neq z_0 \text{ and } |z - z_0| < r_0. \tag{3}$$

Step 2: the neighborhood V'. Put

$$V' = V_0 \cap \{z : |z - z_0| < r_0\}.$$

It is open, being an intersection of two open sets; $z_0 \in V'$, since $z_0 \in V_0$ and $|z_0 - z_0| = 0 < r_0$; and $V' \subseteq V_0 \subseteq \Omega$. By (3) and $V' \subseteq \Omega$,

$$g(z) \neq 0 \qquad (z \in V' - \{z_0\}), \tag{4}$$

and by (1), restricted to $V' \subseteq V_0$,

$$\chi(z_0) = 0, \qquad g(z) = [\chi(z)]^m \quad (z \in V'). \tag{5}$$

Step 3: χ is K'-quasiconformal on V'. By `QuasiconformalOn` applied to V_0 we have: χ is K'-quasiregular on V_0, injective on V_0, continuous on V_0, the image $\chi(V_0)$ is open, and there is ψ with $\psi(\chi(z)) = z$ for $z \in V_0$ and ψ continuous on $\chi(V_0)$. We verify the same five clauses for V'.

(3a) K'-quasiregularity on V'. By `QuasiregularOn` this asks for $1 \leq K'$, which is (2), for $\chi \in W^{1,2}_{\mathrm{loc}}(V')$ in the sense of `MemWLocOn`, and for a pair of weak Wirtinger derivatives of χ on V' satisfying the distortion inequality (11) of the paper almost everywhere on V'.

For the last clause, let a, b be weak Wirtinger derivatives of χ on V_0, with underlying family χ_x, χ_y, such that $|b| \leq \kappa_0 |a|$ almost everywhere on V_0, where $\kappa_0 = (K' - 1)/(K' + 1)$; these exist by `QuasiregularOn` applied on V_0. The same a, b are weak Wirtinger derivatives of χ on V': the local integrability clause (a) of `IsWeakWirtingerOn` restricts, because every compact subset of V' is a compact subset of V_0 and `MemLpLocOn` quantifies over those; the distributional-derivative clause (b) restricts, because by `IsTestFn` the only clause of "φ is a test function for a set" which mentions the set is $\operatorname{tsupp} \varphi \subseteq V' \subseteq V_0$, so every test function for V' is one for V_0, while the integral identity of `IsWeakPartialOn` does not mention the set; and clause (c) is a pointwise identity, independent of the set. The inequality also restricts, since $V' \subseteq V_0$ and an almost-everywhere implication with a smaller antecedent set is weaker.

For the Sobolev clause, take the family D witnessing $\chi \in W^{1,2}_{\mathrm{loc}}(V_0)$: each $D(w)$ with $|w| \leq 1$ lies in $L^2_{\mathrm{loc}}(V')$ in the sense of `MemLpLocOn`, because every compact subset of V' is a compact subset of V_0; and for $|w| < 1$ and $j \in \{0, 1\}$ the identity of `IsWeakPartialOn` holds on V' because every test function for V' is a test function for V_0 — by `IsTestFn` the only clause involving the set is $\operatorname{tsupp} \varphi \subseteq V' \subseteq V_0$ — and the identity itself does not mention the set. So D witnesses $\chi \in W^{1,2}_{\mathrm{loc}}(V')$.

(3b) injectivity and continuity on V'. Both restrict from V_0 along $V' \subseteq V_0$.

(3c) $\chi(V')$ is open. We claim

$$\chi(V') = \chi(V_0) \cap \psi^{-1}(V'). \tag{6}$$

If $\xi = \chi(z)$ with $z \in V'$, then $z \in V_0$, so $\xi \in \chi(V_0)$ and $\psi(\xi) = \psi(\chi(z)) = z \in V'$. Conversely, let $\xi \in \chi(V_0)$ with $\psi(\xi) \in V'$ and write $\xi = \chi(y)$ with $y \in V_0$; then $\psi(\xi) = \psi(\chi(y)) = y$, so $y \in V'$ and $\xi = \chi(y) \in \chi(V')$. This proves (6). Now ψ is continuous on $\chi(V_0)$ and V' is open, so the

right-hand side of (6) is open in the subspace $\chi(V_0)$; and $\chi(V_0)$ is open in $\mathbb{C}$. A subset of $\mathbb{C}$ open in an open subspace is open, so $\chi(V')$ is open.

(3d) the inverse clause. The same ψ satisfies $\psi(\chi(z)) = z$ for $z \in V' \subseteq V_0$ and is continuous on $\chi(V') \subseteq \chi(V_0)$.

With (3a)–(3d) and (5), χ is K'-quasiconformal on V' with $\chi(z_0) = 0$ and $g = \chi^m$ on V', which is assertion (a).

Step 4: g is C^∞ on $V' - \{z_0\}$. By `QuasiregularOn` the quasiregularity of f on Ω contains $f \in W^{1,2}_{\mathrm{loc}}(\Omega)$. The clause `smooth_off_zeros` of `SmoothOffZeros`, applied with p, Ω, the pair f, g, the openness of Ω, $1 < p$, that membership and the four hypotheses (i)–(iv) on g, gives that g is C^∞ on $\Omega - \{z : g(z) = 0\}$. By (4) every $z \in V' - \{z_0\}$ lies in Ω with $g(z) \neq 0$, so

$$V' - \{z_0\} \subseteq \Omega - \{z : g(z) = 0\}, \tag{7}$$

and a function of class C^∞ on a set is of class C^∞ on every subset of it; hence g is C^∞ on $V' - \{z_0\}$.

Step 5: χ is C^∞ on $V' - \{z_0\}$. The set $U = V' - \{z_0\}$ is open, V' being open and $\{z_0\}$ closed. The hypotheses of `ChiSmoothOffZero` hold with this m, this U, and g, χ in the roles of its two maps: $m \geq 1$ is (2); (H1), continuity of χ on U, is the restriction of (3b); (H2), $g = \chi^m$ on U, is (5) restricted to $U \subseteq V'$; (H3), g of class C^∞ on U, is Step 4; and (H4), $g \neq 0$ on U, is (4). Therefore

$$\chi \text{ is } C^\infty \text{ on } U = V' - \{z_0\}, \tag{8}$$

which is assertion (b).

Step 6: the inverse H. Put $H = \psi$, the map of Step 3. By (3d),

$$H(\chi(z)) = z \quad (z \in V'), \qquad H \text{ is continuous on } \chi(V'). \tag{9}$$

Three consequences. First, taking $z = z_0 \in V'$ in (9) and using $\chi(z_0) = 0$ from (5),

$$H(0) = H\big(\chi(z_0)\big) = z_0. \tag{10}$$

Second, let $\xi \in \chi(V')$ and write $\xi = \chi(z)$ with $z \in V'$; then $H(\xi) = z$ by (9), so

$$H(\xi) \in V' \qquad \text{and} \qquad \chi\big(H(\xi)\big) = \chi(z) = \xi \qquad \big(\xi \in \chi(V')\big). \tag{11}$$

With (9), (10) and (11), assertion (c) is proved.

Step 7: conclusion. The data m, K', V', χ and H satisfy $m \geq 1$ and $K' \geq 1$ by (2); V' is open with $z_0 \in V' \subseteq \Omega$ by Step 2; assertion (a) holds by Step 3 with (5); (b) by (8); and (c) by (9), (10) and (11). □

2.75. `VanishingDerivsIsPolynomial`.

Lemma 2.75. *Let $U \subseteq \mathbb{C}$ be open, non-empty and connected, let $m \geq 0$ be an integer, and let $g\colon \mathbb{C} \to \mathbb{C}$ be C^∞ on U. Assume that*

$$D^\nu g(z) = 0 \qquad \text{for every } z \in U \text{ and every } \nu \text{ with } |\nu| = m + 1,$$

where D^ν runs over the iterated real partial derivatives $\partial_{j_1} \cdots \partial_{j_{m+1}}$ of `PartialWord`, $\partial_0 = \partial/\partial x$ and $\partial_1 = \partial/\partial y$. Then there are coefficients $a_{ij} \in \mathbb{C}$, $i, j \geq 0$, with $a_{ij} = 0$ whenever $i + j > m$, such that

$$g(z) = \sum_{i=0}^{m} \sum_{j=0}^{m} a_{ij}\, z^i \bar{z}^{\,j} \qquad (z \in U).$$

This is the step Section 7 of the paper states as "in each case, we conclude that f must be a homogeneous polynomial". Only the polynomial character is proved here; the homogeneity, which turns the degree bound into an exact degree, comes from the homogeneity (57) of the extremal inverse and is applied separately.

Proof. Throughout, $e_0 = 1$ and $e_1 = i$ are the two coordinate directions, so that $\partial_j h = \mathrm{d}h[e_j]$, and for $h \in \mathbb{C}$ we write $h^{(0)} = \operatorname{Re} h$ and $h^{(1)} = \operatorname{Im} h$, so that $h = h^{(0)} e_0 + h^{(1)} e_1$.

Step 1: the directional-derivative expansion. Let $\Omega \subseteq U$ be open, let $h \in \mathbb{C}$, let $w \in \Omega$, and suppose the segment $\{w + th : 0 \le t \le 1\}$ lies in Ω. Put $\psi(t) = g(w + th)$. We claim that for every $s \ge 0$ and every t with $w + th \in \Omega$,

$$\psi^{(s)}(t) = \sum_{|\nu|=s} c_\nu(h)\, D^\nu g(w + th), \tag{1}$$

where the sum is over the 2^s words $\nu = (j_1, \dots, j_s)$ of length s in $\{0, 1\}$ and $c_\nu(h) = h^{(j_1)} \cdots h^{(j_s)}$. Indeed, for $s = 0$ this is the definition of ψ. Assume (1) for s. Each $D^\nu g$ is C^∞ on Ω, and by the chain rule, for any C^1 function F on Ω,

$$\frac{\mathrm{d}}{\mathrm{d}t} F(w+th) = \mathrm{d}F(w+th)[h] = h^{(0)}\, \mathrm{d}F(w+th)[e_0] + h^{(1)}\, \mathrm{d}F(w+th)[e_1] = \sum_{j\in\{0,1\}} h^{(j)} (\partial_j F)(w+th),$$

using the $\mathbb{R}$-linearity of $\mathrm{d}F(w + th)$ in the direction. Applying this to each $F = D^\nu g$ in (1) and using $\partial_j D^\nu = D^{(j,\nu)}$ — which is how `PartialWord` composes, the word being extended on the left — gives (1) for $s + 1$. This proves the claim.

Step 2: all partials of order at least $m + 1$ vanish on U. We show by induction on $s \ge m + 1$ that $D^\nu g = 0$ on U for every $|\nu| = s$. For $s = m + 1$ this is the hypothesis. Assume it for s and let $|\nu| = s + 1$. Write $\nu = (j, \nu')$ with $|\nu'| = s$, so that $D^\nu g = \partial_j (D^{\nu'} g)$. By the inductive hypothesis $D^{\nu'} g$ vanishes identically on the open set U, so its real Fréchet derivative vanishes at every point of U, and therefore $D^\nu g = 0$ on U.

Step 3: g is a polynomial on each ball contained in U. Let $z_0 \in U$ and let $r > 0$ be such that the open ball $B = B(z_0, r)$ is contained in U. Fix $z \in B$ and put $h = z - z_0$ and $\psi(t) = g(z_0 + th)$; the segment $\{z_0 + th : 0 \le t \le 1\}$ lies in the convex set $B \subseteq U$. By Step 1, ψ is C^∞ on a neighbourhood of $[0, 1]$, and by (1) with $s = m + 1$ together with the hypothesis,

$$\psi^{(m+1)}(t) = \sum_{|\nu|=m+1} c_\nu(h)\, D^\nu g(z_0 + th) = 0 \qquad (0 \le t \le 1).$$

Taylor's formula with integral remainder for the one-variable C^{m+1} function ψ on $[0, 1]$ therefore gives

$$g(z) = \psi(1) = \sum_{s=0}^{m} \frac{\psi^{(s)}(0)}{s!} + \int_0^1 \frac{(1-t)^m}{m!} \psi^{(m+1)}(t)\, \mathrm{d}t = \sum_{s=0}^{m} \frac{1}{s!} \sum_{|\nu|=s} c_\nu(z - z_0)\, D^\nu g(z_0),$$

where the second equality uses (1) at $t = 0$. Each $c_\nu(z - z_0)$ is a product of s factors, each equal to $\mathrm{Re}(z - z_0)$ or $\mathrm{Im}(z - z_0)$. Writing

$$\mathrm{Re}(z - z_0) = \tfrac{1}{2}\big((z - z_0) + \overline{(z - z_0)}\big), \qquad \mathrm{Im}(z - z_0) = \tfrac{1}{2i}\big((z - z_0) - \overline{(z - z_0)}\big),$$

and expanding, and then expanding the powers of $z - z_0$ and $\bar{z} - \bar{z}_0$ by the binomial theorem, we obtain coefficients $a_{ij}(z_0) \in \mathbb{C}$, vanishing unless $i + j \le m$, with

$$g(z) = \sum_{i+j\le m} a_{ij}(z_0)\, z^i \bar{z}^j \qquad (z \in B(z_0, r)). \tag{2}$$

Step 4: rigidity. We claim that if coefficients b_{ij}, $i + j \le m$, satisfy $\sum_{i+j\le m} b_{ij} z^i \bar{z}^j = 0$ for every z in some nonempty open set $O \subseteq \mathbb{C}$, then every $b_{ij} = 0$.

Let $\mathcal{P}_m$ denote the complex vector space of functions $\mathbb{R}^2 \to \mathbb{C}$ of the form $\sum_{a+b\le m} \gamma_{ab} x^a y^b$. Its dimension is the number N of pairs (a, b) with $a + b \le m$, because the monomials $x^a y^b$ are linearly independent as functions (a nonzero polynomial in two real variables is not the zero function). Under $z = x + iy$ each $z^i \bar{z}^j$ with $i + j \le m$ lies in $\mathcal{P}_m$, and conversely each $x^a y^b$ with $a + b \le m$ is, by the two displays of Step 3, a linear combination of the $z^i \bar{z}^j$ with $i + j \le m$; so the N functions $z^i \bar{z}^j$, $i + j \le m$, span the N-dimensional space $\mathcal{P}_m$ and hence form a basis of it.

Now $Q = \sum_{i+j\le m} b_{ij} z^i \bar{z}^j$ is an element of $\mathcal{P}_m$ vanishing on the nonempty open set O. Writing $Q = \sum_{a+b\le m} \gamma_{ab} x^a y^b$ and fixing a point $(x_0, y_0) \in O$, all partial derivatives $\partial_x^a \partial_y^b Q$ vanish at (x_0, y_0), since Q vanishes on a neighbourhood of it; and those derivatives at (x_0, y_0) determine all the γ_{ab} (by downward induction on $a + b$, the derivative of order (a, b) at (x_0, y_0) equals

$a!\,b!\,\gamma_{ab}$ plus a combination of $\gamma_{a'b'}$ with $a'+b' > a+b$). Hence every $\gamma_{ab} = 0$, so $Q = 0$ in $\mathcal{P}_m$, and since the $z^i\bar{z}^j$ form a basis, every $b_{ij} = 0$. This proves the claim.

Step 5: the coefficients do not depend on the base point. For $z_0 \in U$ choose $r(z_0) > 0$ with $B(z_0, r(z_0)) \subseteq U$ and let $\Phi(z_0) = \big(a_{ij}(z_0)\big)_{i+j\le m}$ be the coefficient family of (2); it is well defined, because by Step 4 two families representing g on the same ball coincide. We claim Φ is locally constant. Let $z_0 \in U$ and $z_1 \in B(z_0, r(z_0))$. The set $B(z_0, r(z_0)) \cap B(z_1, r(z_1))$ is open and contains z_1, hence is nonempty, and on it both representations (2) hold, so

$$\sum_{i+j\le m} \big(a_{ij}(z_0) - a_{ij}(z_1)\big) z^i \bar{z}^j = 0$$

there. By Step 4, $a_{ij}(z_0) = a_{ij}(z_1)$ for all i, j. Thus Φ is constant on $B(z_0, r(z_0))$, and so locally constant on U.

The set U is connected and nonempty, so a locally constant function on it is constant: $\Phi \equiv (a_{ij})$ for a single family a_{ij}, $i + j \le m$. Extend it by $a_{ij} = 0$ for $i + j > m$. Finally, for $z \in U$ apply (2) with $z_0 = z$, which is legitimate because $z \in B(z, r(z))$:

$$g(z) = \sum_{i+j\le m} a_{ij}\, z^i \bar{z}^j = \sum_{i=0}^{m}\sum_{j=0}^{m} a_{ij}\, z^i \bar{z}^j,$$

the second equality because the added terms have $i + j > m$ and so vanishing coefficients. This is the assertion. □

2.76. `Section7Heart`.

Theorem 2.76. *Let $1 < p < \infty$ with $p \ne 2$, let $k \ge 1$ be an integer and $0 < \alpha \le 1$ with $k + \alpha$ equal to the right-hand side of (9), as in `KAlphaExp`, and put $q = \frac{2}{2-\alpha}$. Let $U \subseteq \mathbb{C}$ be open with $0 \in U$, let $H = H_p$ be the extremal function (54) of `ExtremalH`, let $d = d(p)$ be the exponent of `DExp`, and let $f\colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *f is C^∞ on $\mathbb{C} - \{0\}$;*
(H2) *$f(H(\xi)) = \xi$ for every $\xi \in \mathbb{C}$, and $H(f(z)) = z$ for every $z \in \mathbb{C}$;*
(H3) *$f(tz) = t^d f(z)$ for every $t \ge 0$ and every $z \in \mathbb{C}$;*
(H4) *$D^\nu f(tz) = t^{d-|\nu|} D^\nu f(z)$ for every $t > 0$, every $z \ne 0$ and every iterated partial derivative D^ν of `PartialWord`.*

Then

$$f \notin C^{k-1+\alpha}_{\mathrm{loc}}(U) \quad \text{and} \quad f \notin W^{k+1,q}_{\mathrm{loc}}(U),$$

in the senses of `MemHolderSmallOn` and `MemWLocOn`; here $k \ge 1$, so $k - 1$ is the ordinary predecessor.

This is (56) of the paper, the heart of the optimality half of Theorem 1. Two features of the statement are forced. The two non-memberships are asserted on an arbitrary open U containing the singular point, not on $\mathbb{C}$: a failure of the little-o condition, or of the integrability, is witnessed by a compact subset of the domain, and a witness inside $\mathbb{C}$ need not meet a prescribed subdomain, so a statement about $\mathbb{C}$ would not restrict to one about U. And the Sobolev exponent is the endpoint $q = 2/(2-\alpha)$ and nothing smaller: $(2-\alpha)q = 2$ is what makes the Jacobian factor cancel in the scaling identity (59), and for smaller q the assertion is false, which is why the positive half (10) of Theorem 1 carries a strict inequality.

The properties of f are taken as hypotheses, as they are in the other nodes of the Theorem 1 chain; they are properties of the inverse of (54), and are discharged where this theorem is applied.

Proof. Throughout, $e_0 = 1$ and $e_1 = i$, and for $h \in \mathbb{C}$ we write $h^{(0)} = \operatorname{Re} h$, $h^{(1)} = \operatorname{Im} h$, so that $h = h^{(0)}e_0 + h^{(1)}e_1$. A word $\nu = (j_1, \dots, j_m)$ in $\{0, 1\}$ has length $|\nu| = m$ and $D^\nu = \partial_{j_1} \cdots \partial_{j_m}$ as in `PartialWord`, so that $D^{(j,\nu)} = \partial_j D^\nu$. Finally $c_\nu(h) = h^{(j_1)} \cdots h^{(j_m)}$.

Step 0: elementary consequences of the hypotheses.
(0a) $d = k - 1 + \alpha$. By `KAlphaEqDExpAddOne` the right-hand side of (9) equals $d + 1$, and by hypothesis it equals $k + \alpha$; hence $d = k + \alpha - 1$.

(0b) $d > 0$, since $k \geq 1$ and $\alpha > 0$ give $d \geq \alpha > 0$.
(0c) $f(0) = 0$. Apply (H3) with $t = 0$ and any z: $f(0) = 0^d f(z) = 0$, because $d \neq 0$ by (0b).
(0d) f is not identically zero: $f(H(1)) = 1$ by (H2).
(0e) $\alpha < 1$ if and only if $d < k$, and $\alpha = 1$ if and only if $d = k$; both by (0a).
(0f) $q \geq 1$ and $(2 - \alpha)q = 2$. Indeed $0 < \alpha \leq 1$ gives $1 \leq 2 - \alpha < 2$, so $q = 2/(2 - \alpha)$ satisfies $q > 1$ and $(2 - \alpha)q = 2$.
(0g) If $\alpha < 1$ then $d - j \neq 0$ for every integer $0 \leq j \leq k$. By (0a), $k - 1 < d < k$; so for $j \leq k - 1$ we have $d - j \geq d - (k - 1) = \alpha > 0$, and $d - k = \alpha - 1 < 0$.

Step 1: three standard facts, recorded for use below.
(1a) *Decomposition of a differential.* Let F be differentiable at w, $L = \mathrm{d}F(w)$, $a = F_z(w)$, $b = F_{\bar{z}}(w)$. By DZ and DZbar, $a = \frac{1}{2}(L[1] - iL[i])$ and $b = \frac{1}{2}(L[1] + iL[i])$, so $L[1] = a + b$ and $L[i] = i(a - b)$, and for $h = h^{(0)} + ih^{(1)}$,

$$L[h] = h^{(0)}(a + b) + h^{(1)}i(a - b) = a\,h + b\,\bar{h}.$$

(1b) *Derivatives along a ray.* Let Ω be open, let g be C^∞ on Ω, let $z_0 \in \mathbb{C}$ and let I be an interval of positive reals with $tz_0 \in \Omega$ for $t \in I$. Put $\psi(t) = g(tz_0)$. Then ψ is C^∞ on I and

$$\psi^{(m)}(t) = \sum_{|\nu|=m} c_\nu(z_0)\, D^\nu g(tz_0) \qquad (t \in I,\ m \geq 0). \tag{1}$$

For $m = 0$ this is the definition. If it holds for m, then each $D^\nu g$ is C^∞ on Ω and, for any C^1 function G on Ω,

$$\frac{\mathrm{d}}{\mathrm{d}t}G(tz_0) = \mathrm{d}G(tz_0)[z_0] = \sum_{j\in\{0,1\}} z_0^{(j)}\,(\partial_j G)(tz_0)$$

by the chain rule and the $\mathbb{R}$-linearity of $\mathrm{d}G(tz_0)$ in the direction. Applying this to each $G = D^\nu g$ and using $\partial_j D^\nu = D^{(j,\nu)}$ gives (1) for $m + 1$.
(1c) *Partial derivatives and iterated Fréchet derivatives.* Let Ω be open and g of class C^m on Ω. Since Ω is open, the iterated Fréchet derivative of g within Ω coincides on Ω with the iterated Fréchet derivative of g, which we write $\mathrm{d}^m g(x)$; it is the symmetric m-linear map characterized by the recursion $\mathrm{d}^{m+1}g(x)[v_0, v_1, \ldots, v_m] = \big(\mathrm{d}(\mathrm{d}^m g)(x)[v_0]\big)[v_1, \ldots, v_m]$, with $\mathrm{d}^0 g(x) = g(x)$. Comparing this recursion with $D^{(j,\nu)} = \partial_j D^\nu$ gives, for every word $\nu = (j_1, \ldots, j_m)$,

$$D^\nu g(x) = \mathrm{d}^m g(x)[e_{j_1}, \ldots, e_{j_m}] \qquad (x \in \Omega), \tag{2}$$

and consequently, $\|e_j\| = 1$,

$$\big|D^\nu g(x) - D^\nu g(y)\big| \leq \big\|\mathrm{d}^m g(x) - \mathrm{d}^m g(y)\big\| \qquad (x, y \in \Omega). \tag{3}$$

Also, g of class C^m on Ω makes each $D^\nu g$ with $|\nu| = m$ continuous on Ω, and each $D^\nu g$ with $|\nu| = m - 1$ of class C^1 on Ω.

Step 2: the radial vanishing lemma. Let $m \geq 0$, let $R > 0$, let $|z_0| = 1$, and suppose

$$D^\nu f(tz_0) = 0 \qquad \text{for every } |\nu| = m \text{ and every } 0 < t < R.$$

Suppose moreover that $d - j \neq 0$ for every integer $0 \leq j \leq m - 1$. Then $f(z_0) = 0$.

Indeed, put $\psi(t) = f(tz_0)$ for $t \in (0, R)$. By (H1) and (1b) with $\Omega = \mathbb{C} - \{0\}$, which contains tz_0 for $0 < t < R$, the function ψ is C^∞ on $(0, R)$ and, by (1) and the hypothesis,

$$\psi^{(m)}(t) = \sum_{|\nu|=m} c_\nu(z_0) D^\nu f(tz_0) = 0 \qquad (0 < t < R).$$

On the other hand (H3) gives $\psi(t) = t^d f(z_0)$ for $t > 0$, whence $\psi^{(m)}(t) = d(d - 1)\cdots(d - m + 1)\, t^{d-m} f(z_0)$, the product being empty and equal to 1 when $m = 0$. Each factor is non-zero by assumption and $t^{d-m} > 0$, so $f(z_0) = 0$.

Step 3: f cannot vanish on the whole unit circle. Suppose $f(z_0) = 0$ for every $|z_0| = 1$. For $z \neq 0$, (H3) with $t = |z| > 0$ and the point $z/|z|$ gives $f(z) = |z|^d f(z/|z|) = 0$, and $f(0) = 0$ by (0c). So $f \equiv 0$, contradicting (0d). Hence there is z_0 with $|z_0| = 1$ and $f(z_0) \neq 0$.

Step 4: the $\alpha = 1$ endgame. Assume $\alpha = 1$, let $R > 0$ with $\overline{B}(0, R) \subseteq U$, and assume

$$D^\nu f(z) = 0 \qquad \text{for every } |\nu| = k+1 \text{ and every } 0 < |z| < R. \tag{4}$$

We derive a contradiction.

(4a) The set $B = \{z : 0 < |z| < R\}$ is open, non-empty and connected: it is the image of the connected set $(0, R) \times \mathbb{R}$ under the continuous map $(r, \theta) \mapsto re^{i\theta}$.

(4b) By (H1), f is C^∞ on $B \subseteq \mathbb{C} - \{0\}$, and by (4) every partial derivative of order $k+1$ of f vanishes on B. Hence `VanishingDerivsIsPolynomial` applies on B with $m = k$ and yields coefficients a_{ij}, vanishing unless $i + j \le k$, with

$$f(z) = \sum_{i+j \le k} a_{ij} z^i \bar{z}^j \qquad (z \in B). \tag{5}$$

(4c) *f agrees on B with the top-degree part.* Put $Q(z) = \sum_{i+j=k} a_{ij} z^i \bar{z}^j$. By (0e), $d = k$. Let $z \in B$ and $0 < s < 1$; then $sz \in B$, and (H3) gives $f(sz) = s^k f(z)$. Substituting (5) on both sides,

$$\sum_{n=0}^{k} s^n \sum_{i+j=n} a_{ij} z^i \bar{z}^j = s^k \sum_{i+j \le k} a_{ij} z^i \bar{z}^j \qquad (0 < s < 1).$$

Both sides are polynomials in s of degree at most k which agree at the infinitely many points of $(0, 1)$, so they agree coefficientwise. Comparing the coefficients of s^k and using (5),

$$Q(z) = \sum_{i+j=k} a_{ij} z^i \bar{z}^j = \sum_{i+j \le k} a_{ij} z^i \bar{z}^j = f(z) \qquad (z \in B). \tag{6}$$

(4d) *$f = Q$ on $\mathbb{C} - \{0\}$.* Note $Q(tz) = t^k Q(z)$ for $t > 0$, each monomial having $i + j = k$ and t being real. Let $z \neq 0$ and put $t = R/(2|z|) > 0$, so that $0 < |tz| = R/2 < R$ and $tz \in B$. By (H3), $f(tz) = t^k f(z)$, so by (6)

$$f(z) = t^{-k} f(tz) = t^{-k} Q(tz) = t^{-k} t^k Q(z) = Q(z).$$

(4e) *The identity on the unit circle.* If $\xi \neq 0$ then $H(\xi) \neq 0$: otherwise (H2) and (0c) give $\xi = f(H(\xi)) = f(0) = 0$. Let $|\xi| = 1$. Then $H(\xi) \neq 0$, so by (H2) and (4d),

$$\xi = f(H(\xi)) = Q(H(\xi)) = \sum_{m=0}^{k} a_{m,k-m} H(\xi)^m \overline{H(\xi)}^{k-m}. \tag{7}$$

Write $\xi = e^{i\theta}$, which is possible for every ξ of modulus 1. The first clause of `ExtremalPolarForm` at $r = 1$ gives

$$H(\xi) = 1^{1/d}\big(e^{i\theta} + \varepsilon e^{-3i\theta}\big) = \xi + \varepsilon \xi^{-3},$$

where $\varepsilon = \epsilon_2(1, p)$ is the real coefficient of `EpsCoef`, and $e^{-3i\theta} = \xi^{-3}$. Since $|\xi| = 1$ we have $\bar{\xi} = \xi^{-1}$, so $\overline{\xi^{-3}} = \xi^3$ and, ε being real,

$$\overline{H(\xi)} = \xi^{-1} + \varepsilon \xi^3.$$

Substituting into (7),

$$\xi = \sum_{m=0}^{k} a_{m,k-m} \big(\xi + \varepsilon \xi^{-3}\big)^m \big(\xi^{-1} + \varepsilon \xi^3\big)^{k-m} \qquad (|\xi| = 1). \tag{8}$$

(4f) *Continuation of (8) to all $\xi \neq 0$.* For $\xi \neq 0$,

$$\big(\xi + \varepsilon \xi^{-3}\big)^m = \xi^{-3m} \big(\xi^4 + \varepsilon\big)^m, \qquad \big(\xi^{-1} + \varepsilon \xi^3\big)^{k-m} = \xi^{-(k-m)} \big(1 + \varepsilon \xi^4\big)^{k-m},$$

so multiplying the m-th summand of (8) by ξ^{3k} produces $a_{m,k-m} \xi^{3k-3m-(k-m)} (\xi^4 + \varepsilon)^m (1 + \varepsilon \xi^4)^{k-m}$ with $3k - 3m - (k - m) = 2(k - m) \ge 0$. Hence, setting

$$P_1(X) = X^{3k+1}, \qquad P_2(X) = \sum_{m=0}^{k} a_{m,k-m} X^{2(k-m)} \big(X^4 + \varepsilon\big)^m \big(1 + \varepsilon X^4\big)^{k-m},$$

both P_1 and P_2 are polynomials, and multiplying (8) by ξ^{3k} gives $P_1(\xi) = P_2(\xi)$ for every ξ with $|\xi| = 1$. The unit circle is infinite, $\theta \mapsto e^{i\theta}$ being injective on $[0, 2\pi)$, so the polynomial $P_1 - P_2$ has infinitely many roots and is therefore the zero polynomial. Hence $P_1(\xi) = P_2(\xi)$ for every $\xi \in \mathbb{C}$, and dividing by $\xi^{3k} \neq 0$ and undoing the two displays above,

$$\xi = \sum_{m=0}^{k} a_{m,k-m}\big(\xi + \varepsilon\xi^{-3}\big)^m\big(\xi^{-1} + \varepsilon\xi^{3}\big)^{k-m} \qquad (\xi \neq 0). \tag{9}$$

(4g) *The contradiction.* By `ExtremalEpsBound`, $\varepsilon = 0$ if and only if $p = 2$; since $p \neq 2$ we have $\varepsilon \neq 0$. Also $k \geq 1$. So `HomogPolyModFour` applies with the coefficient ε, the degree k and the coefficient family a_{ij}, whose hypothesis is exactly (9), and it yields a contradiction. This completes Step 4.

Step 5: $f \notin C^{k-1+\alpha}_{\mathrm{loc}}(U)$. Suppose to the contrary that $f \in C^{k-1+\alpha}_{\mathrm{loc}}(U)$ in the sense of `MemHolderSmallOn` at the index $k-1$. Since $0 \in U$ and U is open, fix $\rho > 0$ with $K = \overline{B}(0, \rho) \subseteq U$; K is compact. By `MemHolderSmallOn` one of two cases holds.

Case A: $\alpha < 1$. Then f is of class C^{k-1} on U and, for every compact $K' \subseteq U$ and every $\delta > 0$, there is $\rho' > 0$ with

$$\big\|\mathrm{d}^{k-1}f(x) - \mathrm{d}^{k-1}f(y)\big\| \leq \delta|x-y|^{\alpha} \qquad (x, y \in K',\ |x-y| < \rho'), \tag{10}$$

where we have used (1c) to replace the derivative within U by the derivative at the point, U being open.

Fix a word ν with $|\nu| = k-1$.

First, $D^\nu f(0) = 0$. By (1c), $D^\nu f$ is continuous on U. Pick any z with $0 < |z| \leq \rho$; then $tz \in K \subseteq U$ for $0 < t \leq 1$, and (H4) gives

$$D^\nu f(tz) = t^{\,d-(k-1)} D^\nu f(z) = t^\alpha D^\nu f(z) \longrightarrow 0 \qquad (t \to 0^+),$$

using (0a) and $\alpha > 0$. Since $tz \to 0$ and $D^\nu f$ is continuous at $0 \in U$, we get $D^\nu f(0) = 0$.

Next, let $\delta > 0$ and let $\rho' > 0$ be as in (10) for $K' = K$. Fix z with $0 < |z| \leq \rho$ and let $0 < t \leq 1$ satisfy $t|z| < \rho'$. Then $x = tz$ and $y = 0$ lie in K and $|x - y| = t|z| < \rho'$, so by (10) and (3),

$$\big|D^\nu f(tz)\big| = \big|D^\nu f(tz) - D^\nu f(0)\big| \leq \delta\,(t|z|)^\alpha.$$

Combining with $D^\nu f(tz) = t^\alpha D^\nu f(z)$ and dividing by $t^\alpha > 0$,

$$\big|D^\nu f(z)\big| \leq \delta|z|^\alpha.$$

As $\delta > 0$ was arbitrary, $D^\nu f(z) = 0$. Thus

$$D^\nu f(z) = 0 \qquad \text{for every } |\nu| = k-1 \text{ and } 0 < |z| \leq \rho. \tag{11}$$

Now let $|z_0| = 1$. For $0 < t < \rho$ we have $0 < |tz_0| = t \leq \rho$, so (11) gives $D^\nu f(tz_0) = 0$ for every $|\nu| = k-1$. By (0g) the numbers $d - j$ are non-zero for $0 \leq j \leq k-2$, so Step 2 with $m = k-1$ and $R = \rho$ gives $f(z_0) = 0$. This holds for every z_0 on the unit circle, contradicting Step 3.

Case B: $\alpha = 1$. Then, by `MemHolderSmallOn` at the index $k-1$ and $\alpha = 1$, the function f is of class $C^{(k-1)+1} = C^k$ on U, using $k \geq 1$.

Fix a word ν with $|\nu| = k-1$. By (1c), $D^\nu f$ is of class C^1 on U; in particular it is continuous on U and differentiable at 0. Exactly as in Case A — using (H4) with $d-(k-1) = \alpha = 1$ — we get $D^\nu f(0) = 0$. Put $a = \partial_z(D^\nu f)(0)$ and $b = \partial_{\bar z}(D^\nu f)(0)$. By differentiability at 0, $D^\nu f(0) = 0$ and (1a),

$$D^\nu f(\zeta) = a\zeta + b\bar\zeta + o(|\zeta|) \qquad (\zeta \to 0). \tag{12}$$

Fix $z \neq 0$. For all small $t > 0$ we have $tz \in U$, and (H4) with $|\nu| = k-1$ and $d-(k-1) = 1$ gives $D^\nu f(tz) = t\,D^\nu f(z)$. Hence by (12) with $\zeta = tz$,

$$t\,D^\nu f(z) = a(tz) + b\overline{(tz)} + o(t) = t\big(az + b\bar z\big) + o(t),$$

and dividing by t and letting $t \to 0^+$,

$$D^\nu f(z) = a_\nu z + b_\nu \bar z \qquad (z \neq 0), \tag{13}$$

where we now write a_ν, b_ν for the constants a, b attached to ν.

Let ν' be any word with $|\nu'| = k+1$. Since $k \geq 1$, $|\nu'| \geq 2$, so we may write $\nu' = (j_1, j_2, \nu)$ with $|\nu| = k-1$, and then $D^{\nu'} f = \partial_{j_1}\partial_{j_2}(D^\nu f)$. On the open set $\mathbb{C} - \{0\}$ the function $D^\nu f$ agrees by (13) with the $\mathbb{R}$-linear map $z \mapsto a_\nu z + b_\nu \bar{z}$, whose first partial derivatives are the constants $a_\nu e_j + b_\nu \overline{e_j}$; hence $\partial_{j_2}(D^\nu f)$ is constant on $\mathbb{C} - \{0\}$ and $D^{\nu'} f = \partial_{j_1}\partial_{j_2}(D^\nu f) = 0$ there.

Finally choose $R > 0$ with $\overline{B}(0, R) \subseteq U$. Then (4) holds, and Step 4 gives a contradiction.

In both cases we have reached a contradiction, so $f \notin C^{k-1+\alpha}_{\mathrm{loc}}(U)$.

Step 6: $f \notin W^{k+1,q}_{\mathrm{loc}}(U)$. Suppose to the contrary that $f \in W^{k+1,q}_{\mathrm{loc}}(U)$ in the sense of `MemWLocOn`. Let $(D_w)_w$ be the corresponding family, indexed by words, so that $D_{()} = f$, each D_w with $|w| \leq k+1$ lies in $L^q_{\mathrm{loc}}(U)$ in the sense of `MemLpLocOn`, and $D_{w \frown (j)}$ is the distributional j-th partial derivative of D_w on U in the sense of `IsWeakPartialOn` whenever $|w| < k+1$, where $w \frown (j)$ denotes w with the letter j appended on the right. Put $\Omega' = U - \{0\}$, an open set on which f is C^∞ by (H1).

(6a) *Identification of the weak derivatives.* We claim that for every word w with $|w| \leq k+1$,

$$D_w = D^{\overleftarrow{w}} f \quad \text{almost everywhere on } \Omega', \tag{14}$$

where $\overleftarrow{w}$ is w reversed. We induct on $|w|$. For $|w| = 0$ both sides are f. Assume (14) for all words of length $s < k+1$ and let $|w| = s$, $j \in \{0, 1\}$; we prove it for $w \frown (j)$, whose reverse is $(j, \overleftarrow{w})$.

Put $u = D^{\overleftarrow{w}} f$, which is C^∞ on Ω'. Every test function for Ω' is a test function for U, its support being contained in $\Omega' \subseteq U$; so, restricting the integration-by-parts identity of `IsWeakPartialOn` to such test functions, $D_{w \frown (j)}$ is the distributional j-th partial of D_w on Ω'. Both integrals in that identity are unchanged if D_w is replaced by a function agreeing with it almost everywhere on Ω', the integrands being supported in Ω'; so by the inductive hypothesis $D_{w \frown (j)}$ is also the distributional j-th partial of u on Ω', for each of the two letters j.

We may now apply `WeakPartialEqFDerivAE` on the open set Ω' to the function u and to the pair of candidate derivatives $j \mapsto D_{w \frown (j)}$. Its hypotheses hold: u is continuous on Ω', hence bounded on compact subsets of Ω' and therefore in $L^1_{\mathrm{loc}}(\Omega')$; each $D_{w \frown (j)}$ lies in $L^q_{\mathrm{loc}}(U)$ with $|w \frown (j)| = s+1 \leq k+1$, hence in $L^1_{\mathrm{loc}}(\Omega')$, because compact subsets of Ω' are compact subsets of U of finite measure and $q \geq 1$ by (0f), so Hölder's inequality bounds the L^1 norm on such a set by the L^q norm; the two integration-by-parts identities were just established; and u, being C^∞ on Ω', is differentiable at every point of Ω'. The conclusion is that, almost everywhere on Ω',

$$D_{w \frown (j)} = \partial_j u = \partial_j D^{\overleftarrow{w}} f = D^{(j, \overleftarrow{w})} f = D^{\overleftarrow{w \frown (j)}} f,$$

which is (14) for $w \frown (j)$. This proves the claim.

(6b) *Finiteness of the integral.* Fix $R > 0$ with $\overline{B}(0, R) \subseteq U$ and fix a word ν with $|\nu| = k+1$; put $F = D^\nu f$. By (6a) applied to $w = \overleftarrow{\nu}$, we have $F = D_w$ almost everywhere on Ω', hence almost everywhere on $B(0, R)$, the origin being a null set. Since $\overline{B}(0, R)$ is a compact subset of U and $D_w \in L^q_{\mathrm{loc}}(U)$,

$$\int_{B(0,R)} |F|^q \, \mathrm{d}\sigma = \int_{B(0,R)} |D_w|^q \, \mathrm{d}\sigma < \infty. \tag{15}$$

Note also that F is continuous on the open set $B = \{0 < |z| < R\}$ by (H1), hence measurable.

(6c) *The scaling identity (59) and its integrated form.* By (H4) with $|\nu| = k+1$ and (0a), $d - (k+1) = \alpha - 2$, so

$$F(z) = t^{2-\alpha} F(tz) \qquad (z \neq 0,\ t > 0). \tag{16}$$

Let $0 < t \leq 1$. Using (16), then $(2-\alpha)q = 2$ from (0f), then the change of variables $\zeta = tz$, whose Jacobian is t^2,

$$\int_{B(0,R)} |F(z)|^q \, \mathrm{d}\sigma(z) = t^{(2-\alpha)q} \int_{B(0,R)} |F(tz)|^q \, \mathrm{d}\sigma(z) = t^2 \, t^{-2} \int_{B(0,tR)} |F(\zeta)|^q \, \mathrm{d}\sigma(\zeta) = \int_{B(0,tR)} |F|^q \, \mathrm{d}\sigma,$$

the first equality holding because (16) is valid at every $z \neq 0$, that is almost everywhere. This is the paper's identity $\iint_{|z|<R} |D^\nu f|^q = \iint_{|z|<tR} |D^\nu f|^q$, and it is exactly here that the endpoint exponent is used: for $q < 2/(2-\alpha)$ the two powers of t do not cancel.

(6d) *F vanishes.* By (15) the function $|F|^q$ is integrable on $B(0,R)$, and the sets $B(0,tR)$ decrease to $\{0\}$ as $t \to 0^+$, a null set; so by dominated convergence $\int_{B(0,tR)} |F|^q \to 0$. By (6c) the left-hand side of the display is independent of t, so

$$\int_{B(0,R)} |F|^q \, \mathrm{d}\sigma = 0,$$

whence $F = 0$ almost everywhere on $B(0,R)$. Since F is continuous on the open set $B = \{0 < |z| < R\}$ and the complement of a null set is dense, $F = 0$ at every point of B. As ν with $|\nu| = k+1$ was arbitrary,

$$D^\nu f(z) = 0 \qquad \text{for every } |\nu| = k+1 \text{ and every } 0 < |z| < R. \tag{17}$$

(6e) *The contradiction, in two branches.* If $\alpha < 1$, let $|z_0| = 1$; for $0 < t < R$ we have $0 < |tz_0| = t < R$, so (17) gives $D^\nu f(tz_0) = 0$ for every $|\nu| = k+1$, and by (0g) the numbers $d - j$ are non-zero for $0 \le j \le k$, so Step 2 with $m = k+1$ and this R gives $f(z_0) = 0$. This holds for every z_0 on the unit circle, contradicting Step 3. If $\alpha = 1$, then (17) is exactly (4) and Step 4 gives the contradiction directly.

Hence $f \notin W^{k+1,q}_{\mathrm{loc}}(U)$, and with Step 5 the theorem is proved. □

2.77. `Thm1DerivBounds`.

Lemma 2.77. *Let $n \ge 1$ be an integer, let $1 < p < \infty$, let $K' \ge 1$, and assume the bundle (C2) of `QCPackage`. Let $W \subseteq \mathbb{C}$ be open, let $z_0 \in W$, let $u\colon \mathbb{C} \to \mathbb{R}$ and $f, \chi, H\colon \mathbb{C} \to \mathbb{C}$, and assume*

- (H1) *f solves (12) on W in the sense of `SolvesEq12`, and $f(z_0) = 0$;*
- (H2) *χ is K'-quasiconformal on W in the sense of `QuasiconformalOn`, $\chi(z_0) = 0$, and $f = \chi^n$ on W;*
- (H3) *χ is C^∞ on $W - \{z_0\}$;*
- (H4) *$H(\chi(z)) = z$ for every $z \in W$ and H is continuous on $\chi(W)$;*
- (H5) *$\partial_0 u = f + \bar{f}$ and $\partial_1 u = i(f - \bar{f})$ on W, where ∂_0, ∂_1 are the partial derivatives of `PartialWord` along the words $[0]$, $[1]$; that is, f is the complex gradient of u on W.*

Write $d = d(p)$ for the exponent of `DExp`. Then there are δ with $0 < \delta \le 1$ and constants B_m, $m = 0, 1, 2, \ldots$, such that for $0 < |z - z_0| < \delta$

$$\sum_{|w|=m} \left|D^w f(z)\right| \le B_m |z - z_0|^{d-m}, \tag{51}$$

$$\sum_{|w|=m+1} \left|D^w u(z)\right| \le B_m |z - z_0|^{d-m}, \tag{52}$$

the sums running over all words w of the indicated length in $\{0,1\}$ and D^w being the iterated partial derivative of `PartialWord`.

These are the estimates (51) and (52) of the paper, which Section 6 obtains from Corollary 2 in the two sentences at paper.tex:515 and paper.tex:520: the reduction to a singular point at the origin, and the observation that the smallest of the numbers n/γ_n occurs at $n = 1$ and equals d. The restriction $\delta \le 1$ is not cosmetic; see Step 4.

Proof. Write $d = d(p)$, $\gamma_n = \gamma_n(p)$ for the exponent of `GammaExp`, and $\tau(\zeta) = \zeta + z_0$ for the translation carrying 0 to z_0.

Step 0: $d > 0$. By `DExpEqGammaOneInv`, $\gamma_1 > 0$ and $\gamma_1 d = 1$, so $d = 1/\gamma_1 > 0$.

Step 1: how the iterated partials behave under a translation. We claim that for every word w and every $F\colon \mathbb{C} \to \mathbb{C}$,

$$D^w\big(F \circ \tau\big) = \big(D^w F\big) \circ \tau. \tag{1}$$

Induct on w. If w is empty, both sides are $F \circ \tau$ by `PartialWord`. For $w = j : w'$ we have $D^w = \partial_j \circ D^{w'}$ by `PartialWord`, and by the inductive hypothesis $D^{w'}(F \circ \tau) = (D^{w'}F) \circ \tau$

as functions on $\mathbb{C}$. The translation τ is affine with Fréchet derivative the identity at every point, and so is $\tau^{-1}(\zeta) = \zeta - z_0$. Hence $G \circ \tau$ is real-differentiable at ζ *if and only if* G is real-differentiable at $\tau\zeta$: in one direction compose with τ and use the chain rule, in the other write $G = (G \circ \tau) \circ \tau^{-1}$ and compose with τ^{-1}. We claim that for every G and every ζ,

$$\partial_j\big(G \circ \tau\big)(\zeta) = \mathrm{d}G(\tau\zeta)\big[\mathrm{d}\tau(\zeta)[e_j]\big] = \mathrm{d}G(\tau\zeta)[e_j] = \big(\partial_j G\big)(\tau\zeta),$$

where $e_0 = 1$ and $e_1 = i$. This needs the two cases of the equivalence just proved, because the identification of $\mathrm{d}G(\tau\zeta)$ with a genuine derivative is not automatic: by `PartialWord` the symbol $\mathrm{d}G(\tau\zeta)$ denotes the zero map wherever G fails to be real-differentiable. If G is real-differentiable at $\tau\zeta$, then $G\circ\tau$ is real-differentiable at ζ and the first equality is the chain rule, while the second uses $\mathrm{d}\tau(\zeta) = \mathrm{id}$. If G is not real-differentiable at $\tau\zeta$, then $G \circ \tau$ is not real-differentiable at ζ, so $\mathrm{d}G(\tau\zeta)$ and $\mathrm{d}(G\circ\tau)(\zeta)$ are both the zero map by that convention and all four expressions in the display are 0. So the display holds for every G and every ζ, with no differentiability hypothesis, which is what the induction needs: (1) is asserted for arbitrary F, and Steps 2(b), 2(d) and 3 apply it to test functions, to χ and to members of a `MemWLocOn` family, none of which is known to be differentiable at the point in question. Applying it with $G = D^{w'}F$ gives (1) for w.

Step 2: the translated data. Put

$$\widetilde{W} = \tau^{-1}(W), \qquad \tilde{f} = f \circ \tau, \qquad \tilde{\chi} = \chi \circ \tau, \qquad \widetilde{H} = H - z_0, \qquad \tilde{u} = u \circ \tau.$$

We check that these satisfy every hypothesis of `Cor2` with the same n, p, K' and (C2).

(a) $\widetilde{W}$ is open and $0 \in \widetilde{W}$. τ is a homeomorphism of $\mathbb{C}$, so $\widetilde{W}$ is open; and $\tau(0) = z_0 \in W$.

(b) $\tilde{f}$ solves (12) on $\widetilde{W}$. We first record how the two clauses that make up the weak data behave under τ; this is used again in (d). *Let $u, u' \colon \mathbb{C} \to \mathbb{C}$ and $j \in \{0,1\}$. (i) If $u \in L^s_{\mathrm{loc}}(W)$ in the sense of `MemLpLocOn`, $s \in \{1,2\}$, then $u \circ \tau \in L^s_{\mathrm{loc}}(\widetilde{W})$. (ii) If u' is the distributional ∂_j-derivative of u on W in the sense of `IsWeakPartialOn`, then $u' \circ \tau$ is the distributional ∂_j-derivative of $u \circ \tau$ on $\widetilde{W}$.* For (i): τ carries every compact $\kappa \subseteq \widetilde{W}$ onto a compact subset $\tau(\kappa)$ of W, is a homeomorphism, and preserves Lebesgue measure, so $u \circ \tau$ is measurable and $\int_\kappa |u\circ\tau|^s d\sigma = \int_{\tau(\kappa)} |u|^s d\sigma < \infty$. For (ii): let φ be a test function for $\widetilde{W}$ in the sense of `IsTestFn` and put $\tilde{\varphi} = \varphi \circ \tau^{-1}$. Then $\tilde{\varphi}$ is C^∞ and $\operatorname{tsupp}\tilde{\varphi} = \tau(\operatorname{tsupp}\varphi)$ is compact and contained in $\tau(\widetilde{W}) = W$, so $\tilde{\varphi}$ is a test function for W; and the argument of Step 1 applies verbatim to the translation τ^{-1}, giving $\partial_j\tilde{\varphi} = (\partial_j\varphi) \circ \tau^{-1}$. Hence, by translation invariance of σ and the hypothesis on u, u' applied to $\tilde{\varphi}$,

$$\int (u \circ \tau)\, \partial_j\varphi \, d\sigma = \int u \,\big((\partial_j\varphi) \circ \tau^{-1}\big) d\sigma = \int u\, \partial_j\tilde{\varphi}\, d\sigma = -\int u'\, \tilde{\varphi}\, d\sigma = -\int (u' \circ \tau)\, \varphi \, d\sigma,$$

which is the required identity for φ. Call this fact (T).

By (H1) and `SolvesEq12` there are weak Wirtinger derivatives a, b of f on W, in the sense of `IsWeakWirtingerOn`, with underlying family f_x, f_y — so that $f, f_x, f_y \in L^1_{\mathrm{loc}}(W)$, f_x and f_y are the distributional ∂_0- and ∂_1-derivatives of f on W, and $a = \frac{1}{2}(f_x - i f_y)$, $b = \frac{1}{2}(f_x + i f_y)$ at every point — and a null set Z such that

$$b(z) = \left(\tfrac{1}{p} - \tfrac{1}{2}\right)\left[\frac{\overline{f(z)}}{f(z)}a(z) + \frac{f(z)}{\overline{f(z)}}\overline{a(z)}\right]$$

at every $z \in W - Z$ with $f(z) \neq 0$. By (T), the functions $\tilde{f} = f \circ \tau$, $f_x \circ \tau$ and $f_y \circ \tau$ lie in $L^1_{\mathrm{loc}}(\widetilde{W})$ and $f_x\circ\tau$, $f_y\circ\tau$ are the distributional ∂_0- and ∂_1-derivatives of $\tilde{f}$ on $\widetilde{W}$; and composing the two identities for a and b with τ gives $a\circ\tau = \frac{1}{2}(f_x\circ\tau - i\, f_y\circ\tau)$ and $b\circ\tau = \frac{1}{2}(f_x\circ\tau + i\, f_y\circ\tau)$ at every point. So $a \circ \tau$ and $b \circ \tau$ are weak Wirtinger derivatives of $\tilde{f}$ on $\widetilde{W}$.

Let now $\zeta \notin \tau^{-1}(Z)$ with $\zeta \in \widetilde{W}$ and $\tilde{f}(\zeta) \neq 0$; then $z = \tau\zeta \in W$, $z \notin Z$ and $f(z) = \tilde{f}(\zeta) \neq 0$, so the displayed identity holds at z, and it is verbatim the same identity for the pair $a \circ \tau$, $b \circ \tau$ and the function $\tilde{f}$ at ζ. Finally $\tau^{-1}(Z)$ is null, Lebesgue measure being translation invariant. So $\tilde{f}$ solves (12) on $\widetilde{W}$ in the sense of `SolvesEq12`.

(c) $\tilde{f}(0) = f(z_0) = 0$, $\tilde{\chi}(0) = \chi(z_0) = 0$, and $\tilde{f} = \tilde{\chi}^n$ on $\widetilde{W}$ — the last because $\tilde{f}(\zeta) = f(\tau\zeta) = \chi(\tau\zeta)^n = \tilde{\chi}(\zeta)^n$ for $\zeta \in \widetilde{W}$, using (H2).

(d) $\tilde\chi$ is K'-quasiconformal on $\widetilde{W}$. We verify the five clauses of `QuasiconformalOn`. For K'-quasiregularity (`QuasiregularOn`): $1 \le K'$ is a hypothesis; the distortion clause transports exactly as the equation did in (b), namely `QuasiregularOn` applied to χ on W provides weak Wirtinger derivatives a_χ, b_χ of χ on W, with underlying family χ_x, χ_y, and a null set Z_1 with $|b_\chi| \le \kappa_0 |a_\chi|$ on $W - Z_1$, where $\kappa_0 = (K'-1)/(K'+1)$; by (T) and the same composition with τ as in (b), $a_\chi \circ \tau$ and $b_\chi \circ \tau$ are weak Wirtinger derivatives of $\tilde\chi$ on $\widetilde{W}$, and for $\zeta \in \widetilde{W} - \tau^{-1}(Z_1)$ the inequality $|b_\chi(\tau\zeta)| \le \kappa_0 |a_\chi(\tau\zeta)|$ is exactly the required inequality for the transported pair at ζ, the set $\tau^{-1}(Z_1)$ being null; and $\tilde\chi \in W^{1,2}_{\mathrm{loc}}(\widetilde{W})$ in the sense of `MemWLocOn` because, given a coherent family D for χ on W, the family $\widetilde{D}(w) = D(w) \circ \tau$ is coherent for $\tilde\chi$ on $\widetilde{W}$: each $\widetilde{D}(w)$ lies in $L^2_{\mathrm{loc}}(\widetilde{W})$ in the sense of `MemLpLocOn`, since τ carries compact subsets of $\widetilde{W}$ to compact subsets of W and preserves Lebesgue measure; and the integration-by-parts identity of `IsWeakPartialOn` transports because $\varphi \mapsto \varphi \circ \tau^{-1}$ is a bijection from the test functions for $\widetilde{W}$ onto those for W in the sense of `IsTestFn`, and the two integrals agree by translation invariance together with (1). Clause (a) of `MemWLocOn` also holds for the transported family: $\widetilde{D}(()) = D(()) \circ \tau = \chi \circ \tau = \tilde\chi$, since D satisfies that clause for χ. Injectivity and continuity of $\tilde\chi$ on $\widetilde{W}$ are immediate from those of χ on W. The image is unchanged, $\tilde\chi(\widetilde{W}) = \chi(W)$, so it is open. And if ψ is the continuous inverse supplied for χ by `QuasiconformalOn`, then $\tilde\psi = \psi - z_0$ satisfies $\tilde\psi(\tilde\chi(\zeta)) = \psi(\chi(\tau\zeta)) - z_0 = \tau\zeta - z_0 = \zeta$ for $\zeta \in \widetilde{W}$, and is continuous on $\tilde\chi(\widetilde{W}) = \chi(W)$.

(e) $\tilde\chi$ is C^∞ on $\widetilde{W} - \{0\}$. Indeed $\tau(\widetilde{W} - \{0\}) = W - \{z_0\}$, where χ is C^∞ by (H3), and τ is C^∞.

(f) The inverse data. For $\zeta \in \widetilde{W}$, $\widetilde{H}(\tilde\chi(\zeta)) = H(\chi(\tau\zeta)) - z_0 = \tau\zeta - z_0 = \zeta$ by (H4). Also $\tilde\chi(\widetilde{W}) = \chi(W)$, on which H is continuous, so $\widetilde{H} = H - z_0$ is continuous there. That is all `Cor2` asks of the inverse: it assumes no smoothness of it, deriving everything else about $\widetilde{H}$ from the bundle (C2) and from the equation.

Step 3: Corollary 2. By Step 2 the hypotheses of `Cor2` hold for (n, p, K'), the bundle (C2), $\widetilde{W}$, and $\tilde f, \tilde\chi, \widetilde{H}$. It therefore provides $\delta_0 > 0$ and constants B^0_m with

$$\sum_{|w|=m} \big|D^w \tilde f(\zeta)\big| \le B^0_m |\zeta|^{\,n/\gamma_n - m} \qquad (0 < |\zeta| < \delta_0). \tag{48}$$

By (1), $D^w \tilde f(\zeta) = \big(D^w f\big)(\zeta + z_0)$, so substituting $z = \zeta + z_0$,

$$\sum_{|w|=m} \big|D^w f(z)\big| \le B^0_m |z - z_0|^{\,n/\gamma_n - m} \qquad (0 < |z - z_0| < \delta_0). \tag{48$'$}$$

Each $B^0_m \ge 0$: the left side of (48′) is a sum of moduli, hence nonnegative, and $|z - z_0|^{\,n/\gamma_n - m} > 0$ for any admissible z, of which there is at least one since $\delta_0 > 0$.

Step 4: eliminating n, and why $\delta \le 1$. By part (b) of `GammaRatioMinAtOne`, $1/\gamma_1 \le n/\gamma_n$ for every integer $n \ge 1$; and $d = 1/\gamma_1$ by Step 0. Hence

$$d \le \frac{n}{\gamma_n}, \qquad \text{so} \qquad \frac{n}{\gamma_n} - m \;\ge\; d - m. \tag{3}$$

Since W is open and $z_0 \in W$, choose $\delta_W > 0$ with $\{z : |z - z_0| < \delta_W\} \subseteq W$, and put

$$\delta = \min\{\delta_0,\ \delta_W,\ 1\} > 0, \qquad \text{so} \qquad \delta \le 1.$$

Let $0 < |z - z_0| < \delta$ and set $t = |z - z_0|$, so $0 < t < 1$ and $z \in W$. For real $a \ge b$ one has $t^a = t^b \cdot t^{a-b} \le t^b$, because $0 < t < 1$ and $a - b \ge 0$ give $t^{a-b} \le 1$. Applying this with $a = n/\gamma_n - m$ and $b = d - m$, which is legitimate by (3), and using $B^0_m \ge 0$, (48′) gives

$$\sum_{|w|=m} \big|D^w f(z)\big| \le B^0_m |z - z_0|^{\,d - m} \qquad (0 < |z - z_0| < \delta). \tag{51$'$}$$

This is where $\delta \le 1$ is needed: for $t > 1$ the inequality $t^a \le t^b$ reverses, and since n/γ_n may be strictly larger than d the estimate would fail.

Step 5: from (51) to (52). Let $m \ge 0$ and $0 < |z - z_0| < \delta$, so $z \in W$ by Step 4. (Throughout this step u, which is real-valued, is read as a map $\mathbb{C} \to \mathbb{C}$ through the inclusion $\mathbb{R} \subseteq \mathbb{C}$, since ∂_0,

∂_1 and D^w of `PartialWord` are defined for maps $\mathbb{C} \to \mathbb{C}$; this is the reading under which (H5) and (52) are stated, and the inclusion is $\mathbb{R}$-linear, so it commutes with every D^w and changes none of the moduli below.) Every word w of length $m+1$ is uniquely $w = w'{+}{+}[j]$ with $|w'| = m$ and $j \in \{0,1\}$, and this is a bijection between the words of length $m+1$ and the pairs (w', j).

By `PartialWord` the operator attached to $w = (j_1, \dots, j_m)$ is $D^w = \partial_{j_1}\partial_{j_2}\cdots\partial_{j_m}$, so the recursion it supplies peels the *first* letter, $D^{j::w'} = \partial_j \circ D^{w'}$, which is the form Step 1 used. The form needed here peels the last letter,

$$D^{w'{+}{+}[j]}G = D^{w'}\big(D^{[j]}G\big), \tag{4}$$

and follows from the first-letter form by induction on w'. If w' is empty then $w'{+}{+}[j] = [j]$ and both sides are $D^{[j]}G$, since $D^{()}G = G$. If $w' = i :: w''$ then $w'{+}{+}[j] = i :: (w''{+}{+}[j])$, so by the first-letter form and the inductive hypothesis applied to w'',

$$D^{w'{+}{+}[j]}G = \partial_i\big(D^{w''{+}{+}[j]}G\big) = \partial_i\big(D^{w''}\big(D^{[j]}G\big)\big) = D^{i::w''}\big(D^{[j]}G\big) = D^{w'}\big(D^{[j]}G\big),$$

which is (4) for w'. By (H5) the functions $D^{[0]}u$ and $f + \bar{f}$ agree on the open set W, as do $D^{[1]}u$ and $i(f - \bar{f})$; iterated partial derivatives are local operators, so $D^{w'}$ of either side agrees at points of W.

It remains to apply $D^{w'}$ to the sums $f + \bar{f}$ and $i(f - \bar{f})$, and for that we must first record *where* $D^{w'}$ is additive. It is not additive unconditionally: by the convention of `PartialWord` a derivative is taken to be 0 wherever the function fails to be real-differentiable, so if G is real-differentiable at a point and G' is not, then $G + G'$ is not either and $\partial_j(G + G') = 0$ there while $\partial_j G + \partial_j G'$ need not vanish. What is true, and all that is used, is that the Fréchet derivative is additive at a point at which both summands are differentiable, so ∂_j and hence $D^{w'}$ are additive on any open set on which both summands are C^∞. Such a set is at hand. By (H2) $f = \chi^n$ on W, and by (H3) χ is C^∞ on the open set $W - \{z_0\}$; since $W - \{z_0\}$ is open and $\zeta \mapsto \zeta^n$ is C^∞ on $\mathbb{C}$, the function f is C^∞ on $W - \{z_0\}$, and so is $\bar{f}$, conjugation being $\mathbb{R}$-linear. Consequently every iterated partial derivative of f and of $\bar{f}$ is again C^∞ on $W - \{z_0\}$, so each of the m applications of ∂_0 or ∂_1 making up $D^{w'}$ acts on a map that is C^∞ there. Finally the point z lies in $W - \{z_0\}$: $z \in W$ by Step 4, and $|z - z_0| > 0$ gives $z \neq z_0$. So f and $\bar{f}$ are C^∞ on a neighbourhood of z, and $D^{w'}(f + \bar{f}) = D^{w'}f + D^{w'}\bar{f}$ and $D^{w'}\big(i(f - \bar{f})\big) = i\big(D^{w'}f - D^{w'}\bar{f}\big)$ hold at z. The subtraction is the additivity just justified applied to f and $-\bar{f}$, and the scalar i passes through unconditionally: for a nonzero constant $c \in \mathbb{C}$ the map $\zeta \mapsto c\zeta$ is an $\mathbb{R}$-linear homeomorphism of $\mathbb{C}$, so G is real-differentiable at a point exactly when cG is, and then $\mathrm{d}(cG) = c\,\mathrm{d}G$ by the chain rule, while at a point of non-differentiability both $D^{w'}(cG)$ and $c\,D^{w'}G$ are 0 by the convention of `PartialWord`.

Commuting with conjugation, by contrast, needs no hypothesis at all: $\zeta \mapsto \bar{\zeta}$ is an $\mathbb{R}$-linear isometric involution of $\mathbb{C}$, so a map is real-differentiable at a point exactly when its conjugate is, and then their derivatives are conjugate; while at a point of non-differentiability both $D^{w'}(\bar{G})$ and $\overline{D^{w'}G}$ are 0 by the same convention. Hence $D^{w'}(\bar{G}) = \overline{D^{w'}G}$ at every point and for every G, and in particular $D^{w'}\bar{f}(z) = \overline{D^{w'}f(z)}$. Hence at z

$$D^{w'{+}{+}[0]}u(z) = D^{w'}f(z) + \overline{D^{w'}f(z)}, \qquad D^{w'{+}{+}[1]}u(z) = i\Big(D^{w'}f(z) - \overline{D^{w'}f(z)}\Big),$$

and both have modulus at most $2\big|D^{w'}f(z)\big|$, by the triangle inequality and $|\bar{\zeta}| = |\zeta|$, $|i| = 1$. Summing over the bijection,

$$\sum_{|w|=m+1}\big|D^w u(z)\big| = \sum_{|w'|=m}\Big(\big|D^{w'{+}{+}[0]}u(z)\big| + \big|D^{w'{+}{+}[1]}u(z)\big|\Big) \le 4\sum_{|w'|=m}\big|D^{w'}f(z)\big| \le 4B^0_m|z-z_0|^{d-m},$$

the last step by (51′).

Step 6: conclusion. Put $B_m = 4B^0_m \ge 0$. Then (52) is Step 5, and (51) follows from (51′) because $B^0_m \le 4B^0_m$ and $|z - z_0|^{d-m} \ge 0$. Together with $0 < \delta \le 1$ from Step 4 this is the assertion. □

2.78. `Thm1SmoothAcrossZeros`.

Lemma 2.78. *Let $\Omega \subseteq \mathbb{C}$ be open, let $Z \subseteq \Omega$ meet every compact subset of Ω in a finite set, let $k \geq 1$ be an integer and $\alpha > 0$, and let $w\colon \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *w is of class C^1 on Ω;*

(H2) *w is of class C^∞ on $\Omega - Z$;*

(H3) *for every $z_0 \in Z$ there are $\delta > 0$ and constants B_j with*

$$\sum_{|v|=j} \left|D^v w(z)\right| \leq B_j \, |z - z_0|^{k+\alpha-j} \qquad (1 \leq j \leq k,\ 0 < |z - z_0| < \delta),$$

the sum running over the words v of length j in $\{0,1\}$ and D^v being the iterated partial derivative of `PartialWord`.

Then w is of class C^k on Ω.

This is the step Section 6 of the paper passes over. Under the hypotheses of Theorem 1 the exponent in (52) at order $j = m + 1 \leq k$ is $d - m = k - 1 + \alpha - m = k + \alpha - j \geq \alpha > 0$, so the derivatives of order at most k tend to 0 at a singular point; this lemma is what turns that into differentiability there, and it also supplies the continuity at the singular point which the hypothesis "F continuous on U" of Lemma 2 of the paper (paper.tex:538–539) silently presumes.

The hypothesis on Z is the paper's own. At paper.tex:156 the complex gradient f is asserted to be continuous and, in the same sentence, its zero set $f^{-1}(0)$ to be discrete; a discrete subset that is closed in Ω — which $f^{-1}(0)$ is, by that continuity — meets every compact subset of Ω in a finite set. So local finiteness is what the paper supplies, and it is also what the argument below needs: mere isolation of each point of Z within Z would leave $\Omega - Z$ possibly non-open, since such a Z may accumulate at a point of Ω outside Z, and steps (b) and (d) both read $\Omega - Z$ as an open set.

Proof. Step 0: two consequences of the hypothesis on Z. Both are used below, and both come from the local finiteness of Z. Fix the following construction, available at any $\zeta \in \Omega$: since Ω is open there is $\rho > 0$ with $\kappa_\zeta = \{z : |z - \zeta| \leq \rho\} \subseteq \Omega$, and κ_ζ is compact, so $Z \cap \kappa_\zeta$ is finite by hypothesis.

(0a) $\Omega - Z$ is open. Let $z \in \Omega - Z$ and form κ_z and ρ as above. The set $Z \cap \kappa_z$ is finite and does not contain z, so each of its finitely many elements ζ has $|\zeta - z| > 0$; let ε be the least of those distances if $Z \cap \kappa_z \neq \emptyset$, and $\varepsilon = \rho$ otherwise, and put $s = \min\{\varepsilon, \rho\} > 0$. If $|z' - z| < s$ then $z' \in \kappa_z \subseteq \Omega$, and $z' \notin Z$: otherwise $z' \in Z \cap \kappa_z$ with $|z' - z| < s \leq \varepsilon \leq |z' - z|$, a contradiction. So $\{z' : |z' - z| < s\} \subseteq \Omega - Z$, and z is interior to $\Omega - Z$.

(0b) Each point of Z is isolated in Z. Let $z_0 \in Z$ and form κ_{z_0} and ρ as above. The set $Z \cap \kappa_{z_0}$ is finite, so its elements other than z_0 are finitely many and each has $|\zeta - z_0| > 0$; let ε be the least of those distances if there are any, and $\varepsilon = \rho$ otherwise, and put $r_0 = \min\{\varepsilon, \rho\} > 0$. If $z \in Z$ with $|z - z_0| < r_0$ then $z \in \kappa_{z_0}$, so $z \in Z \cap \kappa_{z_0}$, and $z \neq z_0$ would give $\varepsilon \leq |z - z_0| < r_0 \leq \varepsilon$; hence $z = z_0$. That is, $\{z : |z - z_0| < r_0\} \cap Z = \{z_0\}$.

We prove by induction on i that w is of class C^i on Ω for every i with $1 \leq i \leq k$; the case $i = k$ is the assertion.

Base case $i = 1$. This is (H1).

Inductive step. Let $1 \leq i < k$ and assume w is of class C^i on Ω. We show that w is of class C^{i+1} on Ω. It suffices to show that for every word v' of length i the function

$$g = D^{v'} w$$

is of class C^1 on Ω: a function of class C^i on an open set all of whose iterated partial derivatives of order i are of class C^1 there is of class C^{i+1} there, since the derivative of order $i + 1$ is assembled from the first derivatives of the derivatives of order i, and these are then continuous.

Fix such a v'. We check that g is differentiable at every point of Ω and that $\mathrm{d}g$ is continuous on Ω.

(a) g is continuous on Ω. w is C^i on Ω by the inductive hypothesis and $|v'| = i$, so $g = D^{v'} w$ is continuous on Ω: the iterated derivative of order i is continuous there, and g is its value on the fixed tuple of basis vectors prescribed by v', by `PartialWord`.

(b) g is C^∞, in particular C^1, on $\Omega - Z$. The set $\Omega - Z$ is open by (0a), and by (H2) w is C^∞ on it; hence so is each of its iterated partial derivatives, in particular g. This is where the openness of $\Omega - Z$ is needed: on a non-open set the smoothness of (H2) would be smoothness within the set and would not give differentiability at a point of Ω.

(c) At each point of Z. Let $z_0 \in Z$ and take $\delta > 0$ and B_j as in (H3) for this z_0. Choose $r > 0$ with

$$r < \delta, \qquad \{z : |z - z_0| < r\} \subseteq \Omega, \qquad \{z : |z - z_0| < r\} \cap Z = \{z_0\},$$

which is possible: the first two by $\delta > 0$ and openness of Ω at $z_0 \in Z \subseteq \Omega$, and the third by (0b), after shrinking r below the r_0 it provides. Put $\Omega' = \{z : |z - z_0| < r\}$, an open set containing z_0 with $\Omega' \subseteq \Omega$ and $\Omega' - \{z_0\} \subseteq \Omega - Z$.

By (a), g is continuous on Ω'. By (b), g is differentiable at every point of $\Omega' - \{z_0\}$. For the derivative limit, let $z \in \Omega' - \{z_0\}$. The two first partial derivatives of g at z are $D^{[0]{+}{+}v'}w(z)$ and $D^{[1]{+}{+}v'}w(z)$: by the recursion of `PartialWord` one has $D^{\iota::v'} = \partial_\iota \circ D^{v'}$, so prefixing a letter to the word is applying that partial derivative last. And an $\mathbb{R}$-linear map $\mathbb{C} \to \mathbb{C}$ has operator norm at most the sum of the moduli of its values on 1 and i. Hence

$$\big\|\mathrm{d}g(z)\big\| \le \big|D^{[0]{+}{+}v'}w(z)\big| + \big|D^{[1]{+}{+}v'}w(z)\big| \le \sum_{|v|=i+1} \big|D^{v}w(z)\big| \le B_{i+1}\,|z - z_0|^{\,k+\alpha-(i+1)},$$

the middle inequality because the two words $[0]{+}{+}v'$ and $[1]{+}{+}v'$ are among the words of length $i+1$ and all terms of the sum are nonnegative, and the last by (H3) applied with $j = i+1$, which is legitimate since $1 \le i+1 \le k$ and $0 < |z - z_0| < r < \delta$.

Now $i + 1 \le k$ gives $k + \alpha - (i+1) \ge \alpha > 0$, so $|z - z_0|^{\,k+\alpha-(i+1)} \to 0$ as $z \to z_0$. Also $B_{i+1} \ge 0$, since the left-hand side of (H3) is a sum of moduli and the power of $|z - z_0|$ is positive for at least one admissible z. Therefore

$$\mathrm{d}g(z) \;\longrightarrow\; 0 \qquad \text{as } z \to z_0 \text{ within } \Omega' - \{z_0\}.$$

The hypotheses of `FDerivAcrossPoint` now hold with the open set Ω', the point $z_0 \in \Omega'$, the function g, and $L = 0$: g is continuous on Ω', differentiable at every point of $\Omega' - \{z_0\}$, and its derivative tends to 0 at z_0 within $\Omega' - \{z_0\}$. It therefore gives that g is differentiable at z_0 with $\mathrm{d}g(z_0) = 0$, and that $\mathrm{d}g$ is continuous at z_0 within Ω'. Since Ω' is an open neighbourhood of z_0, continuity within Ω' at z_0 is continuity at z_0, hence also continuity at z_0 within Ω.

(d) Conclusion for g. By (b), g is differentiable at every point of $\Omega - Z$ and $\mathrm{d}g$ is continuous there; by (c), g is differentiable at every point of Z and $\mathrm{d}g$ is continuous at each such point within Ω. As $\Omega = (\Omega - Z) \cup Z$, the function g is differentiable everywhere on Ω and $\mathrm{d}g$ is continuous on Ω; that is, g is of class C^1 on Ω.

Since v' was an arbitrary word of length i, the remark opening the inductive step gives that w is of class C^{i+1} on Ω, completing the induction and the proof. □

2.79. `WeakGradContinuousRep`.

Lemma 2.79. *Let $\Omega \subseteq \mathbb{C}$ be open, let $u\colon \mathbb{C} \to \mathbb{R}$ and let $F_0, F_1\colon \mathbb{C} \to \mathbb{C}$. Assume*

(H1) *$u \in L^1_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`;*
(H2) *F_0 and F_1 are continuous on Ω;*
(H3) *for $j = 0, 1$, F_j is the distributional j-th partial derivative of u on Ω, in the sense of `IsWeakPartialOn`.*

Then there is $w\colon \mathbb{C} \to \mathbb{R}$ with

(a) *$w(z) = u(z)$ for almost every z with $z \in \Omega$;*
(b) *w of class C^1 on Ω;*
(c) *$\partial_j w = F_j$ on Ω for $j = 0, 1$, where ∂_0 and ∂_1 are the partial derivatives of `PartialWord` along the words $[0]$ and $[1]$, and where the real number $\partial_j w(z)$ is read as a complex number. Clause (c) in particular asserts that F_0 and F_1 are real-valued on Ω; that is a consequence of (H3), proved as Step 0 below, and not an extra assumption.*

This is the first of the three steps Section 6 of the paper states in one line. It is what makes the conclusion of Theorem 1 a true statement about u: a p-harmonic function is constrained only through integrals, so it may be altered on a null set without ceasing to be p-harmonic, while membership in a Hölder class is not invariant under such an alteration; the regularity assertion must therefore be made of a representative. For a p-harmonic u the hypothesis (H2) holds because the complex gradient f is continuous, so that $\nabla u = (2\operatorname{Re} f, -2\operatorname{Im} f)$ is continuous on all of Ω. The instance $F_0 = F_1 = 0$ is the constant-u branch.

Proof. Write $d\sigma$ for plane Lebesgue measure, $B(z,r) = \{\zeta : |\zeta - z| < r\}$ and $\bar{B}(z,r) = \{\zeta : |\zeta - z| \le r\}$.

Step 0: F_0 and F_1 are real-valued on Ω. Let $j \in \{0,1\}$ and let φ be a test function for Ω in the sense of `IsTestFn` which is real-valued. Then $\partial_j\varphi$ is real-valued, so $\int u\,\partial_j\varphi\,d\sigma$ is a real number; by (H3) it equals $-\int F_j\varphi\,d\sigma$, whence $\int(\operatorname{Im} F_j)\varphi\,d\sigma = 0$. Now $\operatorname{Im} F_j$ is continuous on Ω by (H2); if it were nonzero at some $z \in \Omega$, say positive, it would be positive on a ball $B(z,r) \subseteq \Omega$, and taking for φ a nonnegative real test function supported in that ball and positive at z — a standard smooth bump — would make $\int(\operatorname{Im} F_j)\varphi\,d\sigma > 0$, a contradiction; the negative case is symmetric. Hence

$$F_j(z) \in \mathbb{R} \qquad (z \in \Omega,\ j = 0,1). \tag{0}$$

Step 1: the local set-up. Fix $z_* \in \Omega$. Since Ω is open there is $\rho > 0$ with $\bar{B}(z_*, 4\rho) \subseteq \Omega$. Put

$$U = B(z_*, 2\rho), \qquad K = \bar{B}(z_*, 2\rho),$$

so K is compact and convex, $K \subseteq B(z_*, 3\rho)$, and $\bar{B}(z_*, 3\rho) \subseteq \Omega$. Let

$$\hat{u} = \mathbf{1}_{\bar{B}(z_*,4\rho)}\,u, \qquad \hat{F}_j = \mathbf{1}_{\bar{B}(z_*,4\rho)}\,F_j.$$

By (H1) and compactness of $\bar{B}(z_*, 4\rho) \subseteq \Omega$ the function $\hat{u}$ is integrable on $\mathbb{C}$, hence locally integrable; and $\hat{F}_j$ is bounded and measurable, hence also locally integrable, F_j being continuous on the compact $\bar{B}(z_*, 4\rho)$.

For $0 < \varepsilon < \rho$ let $\rho_\varepsilon \colon \mathbb{C} \to \mathbb{R}$ be a smooth mollifier: $\rho_\varepsilon \ge 0$, smooth, supported in $\bar{B}(0,\varepsilon)$, with $\int \rho_\varepsilon\,d\sigma = 1$ (the normalized smooth bump functions). Put

$$u_\varepsilon = \rho_\varepsilon * \hat{u}, \qquad G_{j,\varepsilon} = \rho_\varepsilon * \hat{F}_j.$$

Both are C^∞ on $\mathbb{C}$, being convolutions of a smooth compactly supported function with a locally integrable one, and u_ε is real-valued because $\hat{u}$ and ρ_ε are. Note that for $x \in B(z_*, 3\rho)$ and $\varepsilon < \rho$ the convolution integrals see only points of $\bar{B}(x,\varepsilon) \subseteq \bar{B}(z_*, 4\rho)$, where $\hat{u} = u$ and $\hat{F}_j = F_j$.

Step 2: $\partial_j u_\varepsilon = G_{j,\varepsilon}$ on $B(z_*, 3\rho)$. Let $x \in B(z_*, 3\rho)$ and put $\varphi_x(z) = \rho_\varepsilon(x - z)$. Then φ_x is smooth with support contained in $\bar{B}(x,\varepsilon) \subseteq \bar{B}(z_*, 4\rho) \subseteq \Omega$, since $|x - z_*| < 3\rho$ and $\varepsilon < \rho$; so φ_x is a test function for Ω in the sense of `IsTestFn`.

Differentiating the convolution under the integral sign — legitimate because ρ_ε is C^∞ with compact support and $\hat{u}$ is integrable — gives

$$\partial_j u_\varepsilon(x) = \int \big(\partial_j \rho_\varepsilon\big)(x - z)\,\hat{u}(z)\,d\sigma(z).$$

On the other hand $\partial_{z_j}\varphi_x(z) = -(\partial_j\rho_\varepsilon)(x - z)$ by the chain rule, so, using that φ_x and $\partial_j\varphi_x$ are supported where $\hat{u} = u$ and $\hat{F}_j = F_j$, and then (H3),

$$\partial_j u_\varepsilon(x) = -\int u\,\partial_j\varphi_x\,d\sigma = \int F_j\,\varphi_x\,d\sigma = \int \hat{F}_j(z)\,\rho_\varepsilon(x - z)\,d\sigma(z) = G_{j,\varepsilon}(x).$$

Thus

$$\partial_j u_\varepsilon = G_{j,\varepsilon} \quad \text{on } B(z_*, 3\rho), \qquad j = 0,1,\ 0 < \varepsilon < \rho. \tag{1}$$

Step 3: $G_{j,\varepsilon} \to F_j$ uniformly on K. The function F_j is continuous on the compact set $\bar{B}(z_*, 3\rho)$, hence uniformly continuous there; let ω be a modulus of continuity, so $|F_j(a) - F_j(b)| \le$

$\omega(|a-b|)$ for $a, b \in \bar{B}(z_*, 3\rho)$ with $\omega(t) \to 0$ as $t \to 0^+$. For $x \in K$ and $0 < \varepsilon < \rho$, using $\int \rho_\varepsilon = 1$, $\rho_\varepsilon \geq 0$ and $\operatorname{supp} \rho_\varepsilon \subseteq \bar{B}(0, \varepsilon)$,

$$\big|G_{j,\varepsilon}(x) - F_j(x)\big| = \Big|\int \rho_\varepsilon(y)\big(F_j(x-y) - F_j(x)\big)d\sigma(y)\Big| \leq \sup_{|y|\leq\varepsilon} \big|F_j(x-y) - F_j(x)\big| \leq \omega(\varepsilon),$$

the points $x - y$ lying in $\bar{B}(z_*, 3\rho)$ because $|x - z_*| \leq 2\rho$ and $|y| \leq \varepsilon < \rho$. So

$$\sup_K \big|G_{j,\varepsilon} - F_j\big| \leq \omega(\varepsilon) \;\longrightarrow\; 0 \qquad (\varepsilon \to 0^+). \tag{2}$$

Step 4: $u_\varepsilon \to u$ almost everywhere on U. The mollifications of a locally integrable function converge to it almost everywhere as the radius tends to 0. Applied to $\hat{u}$ this gives a null set N with

$$u_\varepsilon(x) \to \hat{u}(x) = u(x) \qquad (\varepsilon \to 0^+), \quad x \in U - N, \tag{3}$$

the identification $\hat{u} = u$ holding on $U \subseteq \bar{B}(z_*, 4\rho)$.

Step 5: the local representative. Fix a sequence $\varepsilon_n = \rho/(n+2) \downarrow 0$ and choose a base point $z_1 \in U - N$; this is possible because U has positive measure and N is null. Let $h = u_{\varepsilon_n} - u_{\varepsilon_m}$. By (1), h is C^∞ on $B(z_*, 3\rho)$ with $\partial_j h = G_{j,\varepsilon_n} - G_{j,\varepsilon_m}$ there, so by (2), writing $\sigma_{n,m} = 2\big(\omega(\varepsilon_n) + \omega(\varepsilon_m)\big)$,

$$\sup_K \big\|\mathrm{d}h\big\| \leq \sup_K \big|\partial_0 h\big| + \sup_K \big|\partial_1 h\big| \leq \sigma_{n,m},$$

since an $\mathbb{R}$-linear map $\mathbb{C} \to \mathbb{C}$ has operator norm at most the sum of the moduli of its values on 1 and i. The set K is convex and h is differentiable at each of its points, so the mean value inequality gives, for $x \in K$,

$$\big|h(x) - h(z_1)\big| \leq \sigma_{n,m}\,|x - z_1| \leq 4\rho\,\sigma_{n,m}.$$

Moreover $h(z_1) = u_{\varepsilon_n}(z_1) - u_{\varepsilon_m}(z_1) \to 0$ as $n, m \to \infty$ by (3), since $z_1 \in U - N$. As $\sigma_{n,m} \to 0$ as $n, m \to \infty$, the sequence (u_{ε_n}) is uniformly Cauchy on K, hence converges uniformly on K to a continuous real-valued function w_*.

Three consequences. First, by (3), for $x \in U - N$ we have $u_{\varepsilon_n}(x) \to u(x)$ and $u_{\varepsilon_n}(x) \to w_*(x)$, so

$$w_* = u \qquad \text{almost everywhere on } U. \tag{4}$$

Second, let $L(x)$ denote the $\mathbb{R}$-linear map $\mathbb{C} \to \mathbb{C}$ determined by $L(x)[1] = F_0(x)$ and $L(x)[i] = F_1(x)$, which is real-valued by (0). By (1) each u_{ε_n} is differentiable at each point of the open set U with derivative the map sending $1, i$ to $G_{0,\varepsilon_n}, G_{1,\varepsilon_n}$; by (2) those derivatives converge to L uniformly on $K \supseteq U$, hence locally uniformly on U; and $u_{\varepsilon_n} \to w_*$ pointwise on U. By the theorem that a locally uniform limit of derivatives is the derivative of the limit, w_* is differentiable at every $x \in U$ with $\mathrm{d}w_*(x) = L(x)$, that is

$$\partial_j w_* = F_j \qquad \text{on } U,\ j = 0, 1. \tag{5}$$

Third, by (5) and (H2) the derivative $\mathrm{d}w_*$ is continuous on U, so

$$w_* \text{ is of class } C^1 \text{ on } U. \tag{6}$$

Step 6: a canonical global definition. Define

$$w(z) = \lim_{r \to 0^+} \frac{1}{\pi r^2} \int_{B(z,r)} u\, d\sigma \quad (z \in \Omega), \qquad w(z) = 0 \quad (z \notin \Omega),$$

and let us check that the limit exists for every $z \in \Omega$ and that w agrees with every local representative.

Let $z_* \in \Omega$ and let U, w_* be as in Steps 1–5 for this z_*. Let $z \in U$. For $r > 0$ small enough that $B(z, r) \subseteq U$, (4) gives

$$\frac{1}{\pi r^2} \int_{B(z,r)} u\, d\sigma = \frac{1}{\pi r^2} \int_{B(z,r)} w_*\, d\sigma,$$

the two integrands agreeing almost everywhere on $B(z,r) \subseteq U$; and since w_* is continuous at z by (6), the averages of w_* over $B(z,r)$ tend to $w_*(z)$ as $r \to 0^+$. Hence the limit defining $w(z)$ exists and

$$w = w_* \qquad \text{on } U. \tag{7}$$

In particular w is real-valued, since w_* is.

Step 7: conclusion. Each $z_* \in \Omega$ has a neighbourhood $U(z_*) = B(z_*, 2\rho(z_*))$ as in Step 1, and by (7), (6), (5),

$$w = w_*^{(z_*)} \text{ on } U(z_*), \qquad w \in C^1\big(U(z_*)\big), \qquad \partial_j w = F_j \text{ on } U(z_*).$$

Since Ω is open and is covered by these neighbourhoods, w is of class C^1 on a neighbourhood of each of its points in Ω, hence C^1 on Ω, which is (b); and $\partial_j w = F_j$ at every point of Ω, which is (c).

For (a): Ω is an open subset of $\mathbb{C}$, hence second countable, so the cover $\{U(z_*)\}_{z_* \in \Omega}$ has a countable subcover $U_1, U_2, \dots$ of Ω. By (4) and (7), for each i there is a null set N_i with $w = u$ on $U_i - N_i$. The set $N = \bigcup_i N_i$ is null, being a countable union of null sets, and every $z \in \Omega - N$ lies in some U_i and hence satisfies $w(z) = u(z)$. So $w = u$ almost everywhere on Ω, which is (a). □

3. Main results

3.1. `Thm2Expansion` (paper, Theorem 2, expansion).

Theorem 3.1. *Let $n \geq 1$ be an integer and let $1 < p < \infty$. Write $\mathbb{B} = \{\xi : |\xi| < 1\}$ and $d\sigma$ for plane Lebesgue measure. Then there is a constant $C = C(n,p) > 0$ with the following property.*

Let $H \colon \mathbb{C} \to \mathbb{C}$ satisfy

- (H1) *$H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, H continuous on $\mathbb{B}$, and H differentiable at almost every point of $\mathbb{B}$;*
- (H2) *H solves equation (18) on $\mathbb{B}$ in the sense of `SolvesEq18`;*
- (H3) *$|H_\xi|^2 + |H_{\bar\xi}|^2$ integrable on $\mathbb{B}$, where H_ξ, $H_{\bar\xi}$ are the Wirtinger derivatives of `DZ` and `DZbar`.*

Then there are complex numbers A_k, $k \in \mathbb{Z}$, such that, with $\lambda_k = \lambda_{|k|}(n,p)$ the exponent of `LambdaCoef`, $\epsilon_k = \epsilon_k(n,p)$ the coefficient of `EpsCoef`, a_k the Fourier coefficient of `FourierCoefOfH` and T_k the k-th summand of (19) in the sense of `Thm2Term`:

- (39) *$A_{-k} = \epsilon_k \overline{A_k}$ for every integer $k \geq n$;*
- (b) *$a_k(r) = A_k\, r^{\lambda_k}$ for every $k \in \mathbb{Z}$ with $|k| \geq n$ and every $r \in (0,1)$;*
- (22) *the series $\sum_{m \geq 0} (n+1+m)|A_{n+1+m}|^2$ converges, and*

$$\sum_{m=0}^{\infty} (n+1+m)\,|A_{n+1+m}|^2 \leq C \iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big) d\sigma,$$

that is $\sum_{k=n+1}^{\infty} k|A_k|^2 < \infty$ and $\sum_{k=n+1}^{\infty} k|A_k|^2 \leq C \iint_{\mathbb{B}} (|H_\xi|^2 + |H_{\bar\xi}|^2) d\sigma$;

- (19) *the series (19) converges to H in $W^{1,2}(\mathbb{B})$:*

$$\iint_{\mathbb{B}} \Big|H - \sum_{k=n}^{n+N} T_k\Big|^2 d\sigma + \iint_{\mathbb{B}} \Big|H_\xi - \Big(\sum_{k=n}^{n+N} T_k\Big)_\xi\Big|^2 d\sigma + \iint_{\mathbb{B}} \Big|H_{\bar\xi} - \Big(\sum_{k=n}^{n+N} T_k\Big)_{\bar\xi}\Big|^2 d\sigma \longrightarrow 0$$

as $N \to \infty$.

This is the forward half of Theorem 2 of the paper (paper.tex:187–205). Its hypothesis surface is now the printed one, $H \in W^{1,2}(\mathbb{B})$ solving (18), together with two clauses that cost nothing: continuity of H on $\mathbb{B}$, which the inverse of a quasiconformal homeomorphism has because it is a homeomorphism, and differentiability almost everywhere, which item (C2) of the paper's note supplies for that same inverse and which is needed because (18) and the polar derivatives are written with the Fréchet derivative. In particular the C^∞ half of the normalisation licensed at paper.tex:183, "$H \in C^\infty(\mathbb{B} - \{0\})$", is no longer assumed: it is a consequence *of the expansion:*

it is produced downstream by term-by-term differentiation of (19), exactly as at paper.tex:387–388, and so cannot be assumed here. In accordance with the statement of Theorem 2 there is no quasiregularity hypothesis and no restriction $p \neq 2$: the paper's exclusion of $p = 2$ at paper.tex:230 comes after *the Remark that computes the case $p = 2$ explicitly, so it governs Sections 3–6 and not the theorem.*

Proof. Step 0: the constant. Put

$$K = \max\Big\{p - 1,\ \frac{1}{p-1}\Big\}.$$

Since $p > 1$ we have $p - 1 > 0$; if $p \geq 2$ then $K \geq p - 1 \geq 1$, and if $p < 2$ then $0 < p - 1 < 1$ so $K \geq 1/(p-1) > 1$. Thus $K \geq 1$, and `EnergyIdentity` applies with this n, p and K; let $C > 0$ be the constant it provides. It depends only on n, p and K, and K depends only on p, so $C = C(n, p)$. Now fix H satisfying (H1)–(H3).

Step 1: H is K-quasiregular on $\mathbb{B}$. By `Eq18ImpliesQuasiregular`, applied with $\Omega = \mathbb{B}$, using $H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ from (H1) and (H2),

(1) H is K-quasiregular on $\mathbb{B}$ in the sense of `QuasiregularOn`.

Step 2: the coefficients. Let $j \in \mathbb{Z}$. By part (a) of `FourierCoefRegularity`, applied with the three clauses of (H1), the function a_j is continuous on $[0, 1)$; and by the third clause of `FourierCoefOde`, applied with $n \geq 1$, $1 < p < \infty$, (H1) and (H2), it is C^∞ on $(0, 1)$. By the second clause of `FourierCoefOde`, applied with the same data, a_j satisfies

$$r\big(r\,a_j'\big)'(r) + pn\,r\,a_j'(r) + (n^2 - j^2)(p-1)\,a_j(r) = 0 \qquad \text{for every } r \in (0,1),$$

which is the Euler equation (35). These are exactly the hypotheses of `FourierCoefForm36` at the index j, which imposes no restriction on p. Its two cases give:

(2) $$|j| < n \implies a_j(r) = 0 \ \text{ for every } r \in [0,1),$$

(3) $$|j| \geq n \implies \exists\, A_j \in \mathbb{C} :\ a_j(r) = A_j\, r^{\lambda_{|j|}} \ \text{ for every } r \in [0,1).$$

Choose, for each j with $|j| \geq n$, such a constant A_j, and put $A_j = 0$ for $|j| < n$. This defines $A\colon \mathbb{Z} \to \mathbb{C}$, and since $(0, 1) \subseteq [0, 1)$ and $\lambda_j = \lambda_{|j|}$, (3) is exactly assertion (b).

Step 3: proof of (39). The hypotheses of `CoefCompatAndSeries` are met: $n \geq 1$, $1 < p < \infty$, H continuous on $\mathbb{B}$, in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ and differentiable at almost every point of $\mathbb{B}$, all three by (H1), H a solution of (18) by (H2), and the coefficient hypothesis is (b) of Step 2. Its second and third conclusions give

(4) $$A_{-n} = 0, \qquad A_{-k} = \epsilon_k \overline{A_k} \quad \text{for every integer } k > n.$$

It remains to treat $k = n$. By part (23) of `LambdaEpsEstimates`, $\epsilon_n(n, p) = 0$, so $\epsilon_n \overline{A_n} = 0 = A_{-n}$ by the first part of (4). Hence

(5) $$A_{-k} = \epsilon_k \overline{A_k} \qquad \text{for every integer } k \geq n,$$

which is (39).

Step 4: proof of (22). Apply `EnergyIdentity` with this H and this A. Its hypotheses hold: (H0) there is our (H2); (H1) there is the continuity and the almost-everywhere differentiability of our (H1); its quasiregularity hypothesis is (1); its integrability hypothesis is our (H3); its hypothesis (H4) is assertion (b) of Step 2; and its hypothesis (H5), namely $A_{-k} = \epsilon_k \overline{A_k}$ for $k > n$, is contained in (5). Its clause (22a) is precisely the pair of assertions

(6)
$$\sum_{m=0}^{\infty} (n + 1 + m)|A_{n+1+m}|^2 \text{ converges}, \qquad \sum_{m=0}^{\infty} (n + 1 + m)|A_{n+1+m}|^2 \leq C \iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big)\, d\sigma,$$

which is (22); the reindexing $k = n + 1 + m$ turns the left side into $\sum_{k \geq n+1} k|A_k|^2$.

Step 5: summability of $\sum_{k\ge n} k|A_k|^2$. By the first part of (6) the series $\sum_{m\ge 0}(n+1+m)|A_{n+1+m}|^2$ converges. Adding the single further term $n|A_n|^2$,

$$\sum_{m=0}^{\infty}(n+m)|A_{n+m}|^2 = n|A_n|^2 + \sum_{m=0}^{\infty}(n+1+m)|A_{n+1+m}|^2 < \infty, \tag{7}$$

so the series of nonnegative terms $\sum_m (n+m)|A_{n+m}|^2$ converges, which is hypothesis (S) of `Thm2SeriesW12Convergence`.

Step 6: the series and its sum H'. By (5) and (7) the two hypotheses (C) and (S) of `Thm2SeriesW12Convergence` hold for this A. That node therefore provides a function H' which is continuous on $\mathbb{B}$ and C^∞ on $\mathbb{B}-\{0\}$, whose Fourier coefficients satisfy

$$a_j(H')(r) = A_j\, r^{\lambda_j} \quad (|j|\ge n), \qquad a_j(H')(r) = 0 \quad (|j|<n), \qquad r\in(0,1), \tag{8}$$

and for which the series (19) converges to H' in $W^{1,2}(\mathbb{B})$ in the sense of its clause (g).

Step 7: $H = H'$ *on* $\mathbb{B}$. Fix $r\in(0,1)$ and consider the 2π-periodic continuous function

$$g_r(\theta) = H\big(re^{i\theta}\big) - H'\big(re^{i\theta}\big),$$

continuous because both H and H' are continuous on $\mathbb{B}$ and $\theta\mapsto re^{i\theta}$ is continuous with values in $\mathbb{B}$. By `FourierCoefOfH`, the j-th Fourier coefficient of g_r in the relevant normalisation is $a_j(H)(r) - a_j(H')(r)$ for every $j\in\mathbb{Z}$. For $|j|\ge n$ this is $A_j r^{\lambda_j} - A_j r^{\lambda_j} = 0$ by Step 2 (b) and (8); for $|j|<n$ it is $0-0=0$ by (2) and (8). So all Fourier coefficients of the continuous 2π-periodic function g_r vanish, and therefore $g_r\equiv 0$, by the uniqueness theorem for Fourier coefficients of continuous periodic functions. As $r\in(0,1)$ was arbitrary and every ξ with $0<|\xi|<1$ is of the form $re^{i\theta}$ with $r=|\xi|\in(0,1)$, we get $H=H'$ on $\mathbb{B}-\{0\}$. At the origin, both H and H' are continuous on $\mathbb{B}$, so

$$H(0) = \lim_{\xi\to 0} H(\xi) = \lim_{\xi\to 0} H'(\xi) = H'(0),$$

the limits being along $\mathbb{B}-\{0\}$. Hence $H=H'$ on $\mathbb{B}$.

Step 8: proof of (19). Since $H=H'$ on the open set $\mathbb{B}$, for every $\xi\in\mathbb{B}$ the two functions agree on a neighbourhood of ξ, so their real Fréchet derivatives agree there and hence, by `DZ` and `DZbar`, $H_\xi = H'_\xi$ and $H_{\bar\xi} = H'_{\bar\xi}$ on $\mathbb{B}-\{0\}$, where both are differentiable. Therefore the three integrands of clause (g) of Step 6 coincide with those of the assertion (19) almost everywhere on $\mathbb{B}$ — everywhere except possibly at the single point 0, which is a Lebesgue null set — so the three integrals are equal for every N, and the convergence to 0 asserted in (19) is clause (g).

Steps 3, 2, 4 and 8 give (39), (b), (22) — both its convergence assertion and its inequality — and (19) respectively, with the constant C of Step 0, which completes the proof. □

3.2. `Cor1` (paper, Corollary 1).

Corollary 3.2. *Let $n\ge 1$ be an integer, let $1<p<\infty$ and let $K\ge 1$. Let $V\subseteq\mathbb{C}$ be open with $0\in V$ and let $H\colon\mathbb{C}\to\mathbb{C}$ be continuous on V, differentiable at almost every point of V, K-quasiconformal on V in the sense of `QuasiconformalOn`, a solution of equation (18) on V in the sense of `SolvesEq18`, and such that $H(0)=0$. Write $\gamma_n=\gamma_n(p)$ for the exponent of `GammaExp`, $J=|H_\xi|^2-|H_{\bar\xi}|^2$ for the Jacobian of `Jac`, and D^w for the iterated partial derivative of `PartialWord`.*

Then there is $\rho>0$ and there are constants $0<c<C$ and C_m, $m=0,1,2,\dots$, such that H is of class C^∞ on $\{\xi : 0<|\xi|<\rho\}$ and

$$c|\xi|^{\gamma_n} \le |H(\xi)| \le C|\xi|^{\gamma_n} \qquad \text{for } 0\le|\xi|\le\rho, \tag{41}$$

$$c|\xi|^{2\gamma_n-2} \le J(\xi) \le C|\xi|^{2\gamma_n-2} \qquad \text{for } 0<|\xi|\le\rho, \tag{42}$$

and

$$\sum_{|w|=m}\big|D^w H(\xi)\big| \le C_m|\xi|^{\gamma_n-m} \qquad \text{for } 0<|\xi|\le\rho, \tag{43}$$

the sum running over all 2^m words w of length m in $\{0,1\}$, and moreover

$$\gamma_n = \lambda_{n+1}(n,p) = \frac{n}{2}\Big\{-p + \sqrt{4\big(1+\tfrac{1}{n}\big)^2(p-1) + (p-2)^2}\Big\}. \tag{44}$$

This is Corollary 1 of the paper (paper.tex:360–374). The printed corollary assumes only that H is a local solution of (18), quasiconformal in a neighborhood of the origin, with $H(0) = 0$; no smoothness. Yet (43) bounds $\sum_{|\nu|=m} |D^\nu H|$ for every m, which presupposes that H is C^∞ off the origin. That smoothness is therefore a conclusion *here, and it is the paper's own step: at paper.tex:387–388 it notes $\lambda_k \to \infty$ and performs any finite number of term-by-term differentiations of the expansion (19). The two clauses retained beyond the printed hypotheses — continuity on V, which a quasiconformal homeomorphism has, and differentiability almost everywhere, which item (C2) of the paper's note supplies for the inverse of one — are what make (18), J and the D^ν meaningful, every one of them being written with the Fréchet derivative. The words of length m realize exactly the operators D^ν with $|\nu| = m$, as* `PartialWord` *records, so the sum in (43) is the paper's $\sum_{|\nu|=m}$ up to the multiplicities with which each D^ν occurs, at most 2^m, absorbed into C_m. The single radius ρ and the single pair $c < C$ shared by (41), (42) and (43) are as the paper states them.*

Proof. Write $\lambda_k = \lambda_{|k|}(n,p)$ for the exponent of `LambdaCoef` and $\epsilon_k = \epsilon_k(n,p)$ for the coefficient of `EpsCoef`, so that $\gamma_n = \lambda_{n+1}$ by `GammaExp`. By part (a) of `GammaRatioMinAtOne`, $\gamma_n > 0$. By (2) of Step 0 below, $\lambda_{n+2} > \lambda_{n+1} = \gamma_n$.

Step 0: monotonicity of the exponents, and (44). By `LambdaCoef`, $2\lambda_k + np = \sqrt{4k^2(p-1) + n^2(p-2)^2}$; the radicand is strictly increasing in $k \geq 0$ because $p - 1 > 0$ and $(k+1)^2 > k^2$, and the square root is strictly increasing on $[0,\infty)$, so

$$\lambda_k < \lambda_{k+1} \qquad (k \geq 0). \tag{2}$$

For (44): by `GammaExp` and `LambdaCoef`,

$$\gamma_n = \lambda_{n+1}(n,p) = \tfrac{1}{2}\Big(-np + \sqrt{4(n+1)^2(p-1) + n^2(p-2)^2}\Big).$$

Since $n \geq 1$ we have $n > 0$ and $n\big(1+\frac{1}{n}\big) = n+1$, so $n^2\big(1+\frac{1}{n}\big)^2 = (n+1)^2$ and therefore, using $n\sqrt{x} = \sqrt{n^2 x}$ for $n > 0$ and $x \geq 0$,

$$\frac{n}{2}\Big\{-p + \sqrt{4\big(1+\tfrac{1}{n}\big)^2(p-1) + (p-2)^2}\Big\} = \tfrac{1}{2}\Big(-np + \sqrt{4n^2\big(1+\tfrac{1}{n}\big)^2(p-1) + n^2(p-2)^2}\Big) = \gamma_n.$$

This is (44); note it does not involve H.

Step 1: rescaling. Since V is open and $0 \in V$, there is $\varepsilon > 0$ with $\{z : |z| < \varepsilon\} \subseteq V$. Put $t = \varepsilon/2 > 0$; then $\{z : |z| \leq t\} \subseteq \{z : |z| < \varepsilon\} \subseteq V$. Define

$$G(\zeta) = H(t\zeta).$$

The hypotheses of `Eq18Rescale` hold with this t, V, K and H: its hypotheses are exactly the present ones, with $\{z : |z| \leq t\} \subseteq V$ as just arranged. So that node gives: G is continuous on $\mathbb{B} = \{\zeta : |\zeta| < 1\}$; G is differentiable at almost every point of $\mathbb{B}$; G solves (18) on $\mathbb{B}$; $G \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$; $|G_\zeta|^2 + |G_{\bar\zeta}|^2$ is integrable on $\mathbb{B}$; G is K-quasiconformal on $\mathbb{B}$; and the conditional clause

(3) if G is C^∞ on $\mathbb{B} - \{0\}$, then H is C^∞ on $\{z : 0 < |z| < t\}$ and $\big(D^w H\big)(t\zeta) = t^{-|w|} D^w G(\zeta)$

for every word w and every ζ with $0 < |\zeta| < 1$. Also $G(0) = H(0) = 0$. The hypothesis of (3) is discharged in Step 5 below, where the smoothness of G is produced; nothing before that point uses it.

Step 2: the coefficients. The hypotheses of `Thm2Expansion` on the unit disk — membership in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$, continuity on $\mathbb{B}$, differentiability at almost every point of $\mathbb{B}$, equation (18), and integrability of $|G_\zeta|^2 + |G_{\bar\zeta}|^2$ — are among the conclusions of Step 1. Applying it to G yields $A\colon \mathbb{Z} \to \mathbb{C}$ with

$$A_{-k} = \epsilon_k \overline{A_k} \qquad (k \geq n), \tag{4}$$

$$a_k(G)(r) = A_k\, r^{\lambda_k} \qquad (|k| \geq n,\ r \in (0,1)), \tag{5}$$

where $a_k(G)$ is the Fourier coefficient of FourierCoefOfH for G, and with $\sum_{m\ge0}(n+1+m)|A_{n+1+m}|^2$ convergent. Adding the single further term $n|A_n|^2$,

$$\sum_{m\ge0}(n+m)|A_{n+m}|^2 = n|A_n|^2 + \sum_{m\ge0}(n+1+m)|A_{n+1+m}|^2 < \infty. \tag{6}$$

Step 3: $A_n = 0$ and $A_{n+1} \neq 0$. The hypotheses (H1)–(H6) of LeadingCoefNonzero hold for G and A: (H1), (H2) and the quasiconformality in (H3) are from Step 1, $G(0) = 0$ is from Step 1, (H4) is (5), (H5) is (4), and (H6) is (6). Hence

$$A_n = 0, \qquad A_{n+1} \neq 0. \tag{7}$$

By Thm2Term, $A_n = 0$ forces $T_n \equiv 0$, where T_k denotes the k-th summand of (19) for G and A, both of its terms carrying a factor A_n or $\overline{A_n}$. Consequently

$$\sum_{k=n}^{n+1} T_k = T_{n+1}. \tag{8}$$

Step 4: the leading profile. By part (b) of Thm2TermSolvesEq18 at $k = n+1$, for $r > 0$ and $\theta \in \mathbb{R}$,

$$T_{n+1}\big(re^{i\theta}\big) = r^{\gamma_n}\Phi(\theta), \qquad \Phi(\theta) = A_{n+1}e^{i\theta} + \epsilon_{n+1}\overline{A_{n+1}}e^{i(-2n-1)\theta},$$

since $\lambda_{n+1} = \gamma_n$ and $(n+1) - n = 1$, $-(n+1) - n = -2n-1$. As $A_{n+1} \neq 0$ by (7) we may set $u = \overline{A_{n+1}}/A_{n+1}$, so $|u| = 1$ and

$$\Phi(\theta) = A_{n+1}e^{i\theta}\big(1 + \epsilon_{n+1}u\, e^{-2i(n+1)\theta}\big),$$

because $A_{n+1}\epsilon_{n+1}u\, e^{i\theta}e^{-2i(n+1)\theta} = \epsilon_{n+1}\overline{A_{n+1}}e^{i(-2n-1)\theta}$. By part (24) of LambdaEpsEstimates, $|\epsilon_{n+1}| < 1$, so for every θ, ϵ_{n+1} being real and $|u| = 1$,

$$\big|1 + \epsilon_{n+1}u\, e^{-2i(n+1)\theta}\big| \ge 1 - |\epsilon_{n+1}| > 0.$$

Hence, writing

$$a = |A_{n+1}|\big(1 - |\epsilon_{n+1}|\big) > 0, \tag{9}$$

we have $|\Phi(\theta)| \ge a$ for every θ and therefore

$$\big|T_{n+1}(\zeta)\big| \ge a\,|\zeta|^{\gamma_n} \qquad (\zeta \neq 0). \tag{10}$$

Also, by HRad the radial derivative $f_r(\zeta)$ is the derivative of f at ζ in the direction $\zeta/|\zeta|$, which at $\zeta = re^{i\theta}$ with $r > 0$ is $\partial\big(f(re^{i\theta})\big)/\partial r$; so differentiating $r \mapsto r^{\gamma_n}\Phi(\theta)$ gives

$$\big(T_{n+1}\big)_r\big(re^{i\theta}\big) = \gamma_n\, r^{\gamma_n - 1}\Phi(\theta), \qquad \big|\big(T_{n+1}\big)_r(\zeta)\big| \ge \gamma_n\, a\,|\zeta|^{\gamma_n - 1} \quad (\zeta \neq 0). \tag{11}$$

Step 5: the series estimates for G, and the smoothness. The hypotheses (H1)–(H5) of SeriesTermwiseDeriv hold for G and A, by Step 1, (5), (4) and (6); in particular its (H1) asks that G be continuous on $\mathbb{B}$, lie in $W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ and be differentiable at almost every point of $\mathbb{B}$, and Step 1 records all three. Write (a′), (b′), (c′), (d′), (e′) for its five conclusions.

By (e′), G is of class C^∞ on $\mathbb{B} - \{0\}$. This discharges the hypothesis of the conditional (3) of Step 1, which therefore gives

$$H \text{ is } C^\infty \text{ on } \{z : 0 < |z| < t\}, \qquad \big(D^w H\big)(t\zeta) = t^{-|w|}D^w G(\zeta) \quad (0 < |\zeta| < 1), \tag{11′}$$

for every word w. This is the only route to the derivatives $D^w H$ that (43) speaks of: no smoothness of H was assumed.

Upper bound in (41) for G. By (d′), which applies because $A_n = 0$ by (7), with $m = 0$ and $\rho = \frac12$, there is $C_2 \ge 0$ with $\sum_{|w|=0}|D^w G(\zeta)| \le C_2|\zeta|^{\gamma_n}$ for $0 < |\zeta| \le \frac12$. The only word of length 0 is the empty word, and $D^{[]}G = G$ by PartialWord, so

$$|G(\zeta)| \le C_2|\zeta|^{\gamma_n} \qquad (0 < |\zeta| \le \tfrac12), \tag{12}$$

and this also holds at $\zeta = 0$, both sides being 0 there because $G(0) = 0$ and $0^{\gamma_n} = 0$ for $\gamma_n > 0$.

Lower bound in (41) for G. By (b′) with $\rho = \frac{1}{2}$ and $j = n+1$, which is legitimate as $n+1 \geq n$, there is $C_3 \geq 0$ with

$$\Big|G(\zeta) - \sum_{k=n}^{n+1} T_k(\zeta)\Big| \leq C_3|\zeta|^{\lambda_{n+2}} \qquad (0 < |\zeta| \leq \tfrac{1}{2}),$$

hence, by (8) and (10) and the triangle inequality,

$$|G(\zeta)| \;\geq\; \big|T_{n+1}(\zeta)\big| - C_3|\zeta|^{\lambda_{n+2}} \;\geq\; |\zeta|^{\gamma_n}\Big(a - C_3|\zeta|^{\lambda_{n+2}-\gamma_n}\Big),$$

where we used $|\zeta|^{\lambda_{n+2}} = |\zeta|^{\gamma_n}|\zeta|^{\lambda_{n+2}-\gamma_n}$. Since $\lambda_{n+2} - \gamma_n > 0$ by (2) and $a > 0$ by (9), and since $s \mapsto s^{\lambda_{n+2}-\gamma_n}$ is continuous at 0 with value 0, we may choose ρ_1 with

$$0 < \rho_1 \leq \tfrac{1}{2}, \qquad C_3\,\rho_1^{\lambda_{n+2}-\gamma_n} \leq \tfrac{a}{2}, \tag{13}$$

and then, $s \mapsto s^{\lambda_{n+2}-\gamma_n}$ being nondecreasing,

$$|G(\zeta)| \geq \tfrac{a}{2}\,|\zeta|^{\gamma_n} \qquad (0 < |\zeta| \leq \rho_1), \tag{14}$$

which also holds at $\zeta = 0$, both sides being 0.

Estimate (43) for G, and the upper bound in (42) for G. By (d′), for each $m \geq 0$ with $\rho = \frac{1}{2}$ there is $C'_m \geq 0$ with

$$\sum_{|w|=m} \big|D^w G(\zeta)\big| \leq C'_m|\zeta|^{\gamma_n - m} \qquad (0 < |\zeta| \leq \tfrac{1}{2}). \tag{15}$$

Take $m = 1$. The two words of length 1 are [0] and [1], and by `PartialWord` $D^{[0]}G = \partial_x G$, $D^{[1]}G = \partial_y G$, these being the values of the real Fréchet derivative on 1 and i. By `DZ` and `DZbar`, $G_\zeta = \frac{1}{2}(\partial_x G - i\partial_y G)$ and $G_{\bar\zeta} = \frac{1}{2}(\partial_x G + i\partial_y G)$, so by the triangle inequality and (15),

$$\big|G_\zeta(\zeta)\big| \leq \tfrac{1}{2}\big(|\partial_x G(\zeta)| + |\partial_y G(\zeta)|\big) \leq \tfrac{1}{2}C'_1|\zeta|^{\gamma_n - 1} \qquad (0 < |\zeta| \leq \tfrac{1}{2}).$$

Hence, by `Jac` and $|G_{\bar\zeta}|^2 \geq 0$,

$$J_G(\zeta) = \big|G_\zeta(\zeta)\big|^2 - \big|G_{\bar\zeta}(\zeta)\big|^2 \leq \big|G_\zeta(\zeta)\big|^2 \leq \tfrac{1}{4}(C'_1)^2|\zeta|^{2\gamma_n - 2} \qquad (0 < |\zeta| \leq \tfrac{1}{2}), \tag{16}$$

using $\big(|\zeta|^{\gamma_n - 1}\big)^2 = |\zeta|^{2\gamma_n - 2}$.

Lower bound in (42) for G. By (c′) with $\rho = \frac{1}{2}$ and $j = n+1$ there is $C_4 \geq 0$ with

$$\Big|G_r(\zeta) - \sum_{k=n}^{n+1} (T_k)_r(\zeta)\Big| \leq C_4|\zeta|^{\lambda_{n+2}-1} \qquad (0 < |\zeta| \leq \tfrac{1}{2}).$$

Since $T_n \equiv 0$ we have $(T_n)_r \equiv 0$, so the sum is $(T_{n+1})_r$, and by (11) and the triangle inequality

$$\big|G_r(\zeta)\big| \;\geq\; \gamma_n a|\zeta|^{\gamma_n - 1} - C_4|\zeta|^{\lambda_{n+2}-1} \;=\; |\zeta|^{\gamma_n - 1}\Big(\gamma_n a - C_4|\zeta|^{\lambda_{n+2}-\gamma_n}\Big).$$

As $\gamma_n a > 0$, choose ρ_2 with

$$0 < \rho_2 \leq \rho_1, \qquad C_4\,\rho_2^{\lambda_{n+2}-\gamma_n} \leq \tfrac{\gamma_n a}{2}, \tag{17}$$

so that, putting $b = \gamma_n a/2 > 0$,

$$\big|G_r(\zeta)\big| \geq b\,|\zeta|^{\gamma_n - 1} \qquad (0 < |\zeta| \leq \rho_2). \tag{18}$$

Now we pass from $|G_r|$ to J_G by (26). The hypotheses of `QCDistortionPolar` hold with $\Omega = \mathbb{B}$, which is open, with this K, with G K-quasiregular on $\mathbb{B}$ — this is part of K-quasiconformality, by `QuasiconformalOn` — and with G differentiable at almost every point of $\mathbb{B}$, both from Step 1. Its conclusion (26) gives

$$K^{-1}\big|G_r(\zeta)\big|^2 \leq J_G(\zeta) \qquad \text{for almost every } \zeta \in \mathbb{B} \text{ with } \zeta \neq 0. \tag{19}$$

We upgrade (19) to *every* such ζ. Both sides are continuous on the open set $\mathbb{B} - \{0\}$: there G is C^∞ by (e′) of Step 5, so $\zeta \mapsto \mathrm{d}G(\zeta)$ is continuous; $G_r(\zeta) = \mathrm{d}G(\zeta)[\zeta/|\zeta|]$ by `HRad`, with $\zeta \mapsto \zeta/|\zeta|$ continuous and $|\zeta| > 0$; and $G_\zeta, G_{\bar\zeta}$ are the fixed linear combinations $\frac{1}{2}(\mathrm{d}G[1] \mp i\,\mathrm{d}G[i])$ of `DZ` and `DZbar`, so J_G is continuous by `Jac`. The set where (19) holds has full measure in $\mathbb{B} - \{0\}$, so

its complement contains no nonempty open set and it is therefore dense in $\mathbb{B}-\{0\}$. A non-strict inequality between continuous functions that holds on a dense subset of an open set holds on all of it. Hence (19) holds at every $\zeta \in \mathbb{B}$ with $\zeta \neq 0$, and with (18),

$$J_G(\zeta) \geq K^{-1}b^2|\zeta|^{2\gamma_n-2} \qquad (0<|\zeta|\leq \rho_2). \tag{20}$$

Step 6: transfer back to H. Put

$$\rho = t\rho_2 > 0.$$

Let ξ with $|\xi| \leq \rho$ and put $\zeta = \xi/t$, so that $|\zeta| = |\xi|/t \leq \rho_2 \leq \frac{1}{2} < 1$ and $t\zeta = \xi$, whence $H(\xi) = H(t\zeta) = G(\zeta)$ by the definition of G. Note $|\zeta|^e = t^{-e}|\xi|^e$ for every real e, since $t > 0$.

(41). By (12) and (14),

$$\tfrac{a}{2}t^{-\gamma_n}|\xi|^{\gamma_n} \;\leq\; |H(\xi)| \;\leq\; C_2 t^{-\gamma_n}|\xi|^{\gamma_n} \qquad (|\xi|\leq\rho),$$

including $\xi = 0$, where all three quantities vanish.

(43). Let $m \geq 0$ and $0 < |\xi| \leq \rho$, so $0 < |\zeta| < 1$. By (11′), $\big(D^wH\big)(\xi) = t^{-m}D^wG(\zeta)$ for every word w of length m, so $|D^wH(\xi)| = t^{-m}|D^wG(\zeta)|$ and, by (15),

$$\sum_{|w|=m}\big|D^wH(\xi)\big| = t^{-m}\sum_{|w|=m}\big|D^wG(\zeta)\big| \leq t^{-m}C'_m|\zeta|^{\gamma_n-m} = C'_m\, t^{-\gamma_n}\,|\xi|^{\gamma_n-m},$$

the last step because $t^{-m}|\zeta|^{\gamma_n-m} = t^{-m}t^{-(\gamma_n-m)}|\xi|^{\gamma_n-m} = t^{-\gamma_n}|\xi|^{\gamma_n-m}$.

(42). By (11′) with $w=[0]$ and $w=[1]$ we get $\partial_x G(\zeta) = t\,\partial_x H(\xi)$ and $\partial_y G(\zeta) = t\,\partial_y H(\xi)$, so by the linearity in `DZ` and `DZbar`,

$$G_\zeta(\zeta) = t\,H_\xi(\xi), \qquad G_{\bar\zeta}(\zeta) = t\,H_{\bar\xi}(\xi),$$

and hence, by `Jac`, $J_G(\zeta) = t^2 J_H(\xi)$, that is $J_H(\xi) = t^{-2}J_G(\zeta)$. With (16) and (20), and $t^{-2}|\zeta|^{2\gamma_n-2} = t^{-2}t^{-(2\gamma_n-2)}|\xi|^{2\gamma_n-2} = t^{-2\gamma_n}|\xi|^{2\gamma_n-2}$,

$$K^{-1}b^2t^{-2\gamma_n}|\xi|^{2\gamma_n-2} \;\leq\; J_H(\xi) \;\leq\; \tfrac{1}{4}(C'_1)^2t^{-2\gamma_n}|\xi|^{2\gamma_n-2} \qquad (0<|\xi|\leq\rho).$$

Step 7: the constants. Put

$$c = \min\Big\{\tfrac{a}{2}t^{-\gamma_n},\; K^{-1}b^2t^{-2\gamma_n}\Big\}, \qquad C_m = C'_m\,t^{-\gamma_n} \quad (m\geq 0),$$

$$C = \max\Big\{C_2t^{-\gamma_n},\; \tfrac{1}{4}(C'_1)^2t^{-2\gamma_n},\; c+1\Big\}.$$

Then $c > 0$, being the minimum of two positive numbers ($a>0$, $b>0$, $K^{-1}>0$, $t^{-\gamma_n}>0$), and $c < c+1 \leq C$. Replacing the two lower constants by the smaller c preserves the lower bounds, and replacing the two upper constants by the larger C preserves the upper bounds, because the powers $|\xi|^{\gamma_n}$ and $|\xi|^{2\gamma_n-2}$ are nonnegative. So (41) and (42) hold with c, C and this ρ, and (43) holds with C_m, while (44) was proved in Step 0.

Finally the smoothness assertion: $\rho = t\rho_2 \leq t/2 < t$ by (13) and the choice of $\rho_2 \leq \frac{1}{2}$, so $\{\xi : 0<|\xi|<\rho\} \subseteq \{z : 0<|z|<t\}$, on which H is C^∞ by (11′); and a function of class C^∞ on a set is of class C^∞ on every subset of it. This completes the proof. □

3.3. `Lemma1` (paper, Lemma 1).

Lemma 3.3. *Let Ω and U be open subsets of $\mathbb{C}$ and let $\chi, H : \mathbb{C}\to\mathbb{C}$ be such that*

(a) *H is C^∞ on Ω and χ is C^∞ on U;*
(b) *$H(\xi)\in U$ and $\chi(H(\xi)) = \xi$ for every $\xi\in\Omega$;*
(c) *$J(\xi)\neq 0$ for every $\xi\in\Omega$, where $J = |H_\xi|^2 - |H_{\bar\xi}|^2$ is the Jacobian of `Jac`.*

Let $m\geq 1$ be an integer and let w be a word of length m in the two real partial derivatives $\partial/\partial x$ and $\partial/\partial y$, so that D^w is a partial derivative operator of order m in the sense of `PartialWord`. Then there exists

$$P = P_w \in \mathbb{P}(4m-3,\, 3m-2)$$

relative to H, in the sense of `PClass`, such that

$$D^w\chi(z) = J(\xi)^{1-2m}P_w(\xi) \qquad \text{for every } \xi\in\Omega,\ \text{where } z = H(\xi).$$

Proof. Throughout, Ω, U, χ, H are fixed as in the hypotheses, and all class memberships are relative to H. Write

$$H_\xi = \frac{\partial H}{\partial \xi}, \qquad H_{\bar\xi} = \frac{\partial H}{\partial \bar\xi}$$

for the iterated derivatives of `WirtingerIter` with indices $(1,0)$ and $(0,1)$, and let $\mathbf{J}$ denote the Jacobian J of `Jac` regarded as a complex-valued function, $\mathbf{J}(\xi) = J(\xi) \in \mathbb{R} \subset \mathbb{C}$. By hypothesis (c), $\mathbf{J}(\xi) \neq 0$ for $\xi \in \Omega$, so integer powers of $\mathbf{J}(\xi)$ are defined there.

Step 0. Five preliminary facts.

(0a) $\mathbb{P}(s,k)$ is a $\mathbb{C}$-linear space. If $P, Q \in \mathbb{P}(s,k)$ and $c \in \mathbb{C}$ then $P + Q \in \mathbb{P}(s,k)$ and $cP \in \mathbb{P}(s,k)$. This is immediate from `PClass`: if $(c_\alpha, M_\alpha)_{\alpha \le r}$ presents P and $(b_\beta, N_\beta)_{\beta \le q}$ presents Q, with all lists of type (s,k), then the concatenated family presents $P+Q$ and the family $(c\,c_\alpha, M_\alpha)_{\alpha \le r}$ presents cP, and in both cases every list in the new family is one of the old ones and so still has type (s,k).

(0b) The four first-order factors lie in $\mathbb{P}(1,1)$. Consider the derivative symbols of `DerivSymbol`

$$e = (1,0,\text{false}), \quad e^c = (1,0,\text{true}), \quad e' = (0,1,\text{false}), \quad e'^c = (0,1,\text{true}),$$

each of order 1. By `SymbolValue` their values on H are H_ξ, $\overline{H_\xi}$, $H_{\bar\xi}$, $\overline{H_{\bar\xi}}$ respectively. Each of the one-entry lists (e), (e^c), (e'), (e'^c) has length 1 and total order 1, i.e. type $(1,1)$, and by `MonomialValue` its monomial is the corresponding symbol value. Taking the single coefficient 1 in `PClass`,

$$H_\xi,\ \overline{H_\xi},\ H_{\bar\xi},\ \overline{H_{\bar\xi}} \ \in\ \mathbb{P}(1,1).$$

(0c) $\mathbf{J} \in \mathbb{P}(2,2)$, $\mathbf{J}$ is C^∞ on Ω, and both $\partial\mathbf{J}/\partial\xi$ and $\partial\mathbf{J}/\partial\bar\xi$ agree on Ω with members of $\mathbb{P}(3,2)$. The membership $\mathbf{J} \in \mathbb{P}(2,2)$ is part (a) of `JacInPClass`. Hence $\mathbf{J}$ is C^∞ on Ω by `PClassSmooth`, using hypothesis (a) and that Ω is open. Part (b) of `JacInPClass` provides a function $R \in \mathbb{P}(3,2)$, namely

$$R = H_{\xi\xi}\overline{H_\xi} + \overline{H_{\xi\bar\xi}}H_\xi - H_{\xi\bar\xi}\overline{H_{\bar\xi}} - \overline{H_{\bar\xi\bar\xi}}H_{\bar\xi},$$

and part (c) states

$$\frac{\partial \mathbf{J}}{\partial \xi}(\xi) = R(\xi) \qquad \text{for every } \xi \in \Omega. \tag{1}$$

For the other operator we use that J is real-valued. By `Jac`, $J(\xi) = \|H_\xi(\xi)\|^2 - \|H_{\bar\xi}(\xi)\|^2$ is a difference of squared norms, hence a real number, so the complex number $\mathbf{J}(\xi)$ satisfies $\overline{\mathbf{J}(\xi)} = \mathbf{J}(\xi)$ for every ξ; that is, the function $\overline{\mathbf{J}}$ equals $\mathbf{J}$. Since $\mathbf{J}$ is C^∞ on Ω it is real-differentiable at each $\xi \in \Omega$, so `WirtingerConjRule` applies to $\mathbf{J}$ at ξ and gives

$$\frac{\partial \overline{\mathbf{J}}}{\partial \bar\xi}(\xi) = \overline{\frac{\partial \mathbf{J}}{\partial \xi}(\xi)}.$$

The left-hand side is $\partial\mathbf{J}/\partial\bar\xi(\xi)$, because $\overline{\mathbf{J}}$ and $\mathbf{J}$ are the same function. Combining with (1),

$$\frac{\partial \mathbf{J}}{\partial \bar\xi}(\xi) = \overline{R(\xi)} \qquad \text{for every } \xi \in \Omega, \tag{2}$$

and $\overline{R} \in \mathbb{P}(3,2)$ by `PClassConjClosed`.

(0d) For every word v, $D^v\chi$ is C^∞ on U. By hypothesis (a), χ is C^∞ on the open set U. By `PartialWord`, $D^v\chi$ is obtained from χ by applying $|v|$ of the two real first-order operators $\partial/\partial x$, $\partial/\partial y$, and each of these carries $C^\infty(U)$ into $C^\infty(U)$, because the first partial derivatives of a C^∞ function on an open set are again C^∞ there. Induction on $|v|$ gives the claim. In particular $D^v\chi$ is real-differentiable at every point of U, hence at $H(\xi)$ for every $\xi \in \Omega$ by hypothesis (b).

(0e) The two real partials in terms of the Wirtinger operators. Let F be real-differentiable at a point ζ and write $F_x(\zeta) = DF(\zeta)[1]$, $F_y(\zeta) = DF(\zeta)[i]$ for its two real first partials. By `DZ` and `DZbar`,

$$\frac{\partial F}{\partial z}(\zeta) + \frac{\partial F}{\partial \bar z}(\zeta) = \tfrac{1}{2}\big(F_x - iF_y\big)(\zeta) + \tfrac{1}{2}\big(F_x + iF_y\big)(\zeta) = F_x(\zeta),$$

$$\frac{\partial F}{\partial z}(\zeta) - \frac{\partial F}{\partial \bar z}(\zeta) = -i\,F_y(\zeta), \qquad \text{hence} \qquad i\Big(\frac{\partial F}{\partial z}(\zeta) - \frac{\partial F}{\partial \bar z}(\zeta)\Big) = i\big(-i\,F_y(\zeta)\big) = F_y(\zeta).$$

By `PartialWord`, for a word v and a direction $j \in \{0,1\}$ the word $(j)*v$ obtained by prepending j satisfies $D^{(j)*v}F = \partial_j\big(D^vF\big)$ with $\partial_0 = \partial/\partial x$ and $\partial_1 = \partial/\partial y$. Applying the two displays to $F = D^v\chi$, which is real-differentiable on U by (0d), gives at every point of U

$$D^{(0)*v}\chi = \big(D^v\chi\big)_z + \big(D^v\chi\big)_{\bar z}, \qquad D^{(1)*v}\chi = i\Big(\big(D^v\chi\big)_z - \big(D^v\chi\big)_{\bar z}\Big). \tag{3}$$

Step 1. The base case $m = 1$.

Let $w = (j)$ be a word of length 1 and let $\xi \in \Omega$. We apply `ChiFirstDerivs` with its open set taken to be Ω: it contains ξ; the inverse relation $\chi(H(\zeta)) = \zeta$ holds for every $\zeta \in \Omega$ by hypothesis (b); H is real-differentiable at ξ by hypothesis (a) and openness of Ω; χ is real-differentiable at $H(\xi)$ by hypotheses (a), (b) and openness of U; and $J(\xi) \neq 0$ by hypothesis (c). It yields

$$\frac{\partial\chi}{\partial z}\big(H(\xi)\big) = \frac{\overline{H_\xi(\xi)}}{\mathbf{J}(\xi)}, \qquad \frac{\partial\chi}{\partial\bar z}\big(H(\xi)\big) = -\frac{H_{\bar\xi}(\xi)}{\mathbf{J}(\xi)}.$$

Taking v to be the empty word in (3), for which $D^v\chi = \chi$ by `PartialWord`, and evaluating at the point $H(\xi) \in U$,

$$D^{(0)}\chi\big(H(\xi)\big) = \mathbf{J}(\xi)^{-1}\Big(\overline{H_\xi(\xi)} - H_{\bar\xi}(\xi)\Big), \qquad D^{(1)}\chi\big(H(\xi)\big) = \mathbf{J}(\xi)^{-1}\Big(i\,\overline{H_\xi(\xi)} + i\,H_{\bar\xi}(\xi)\Big),$$

the second because $i\big(\overline{H_\xi}/\mathbf{J} - (-H_{\bar\xi}/\mathbf{J})\big) = \mathbf{J}^{-1}\big(i\overline{H_\xi} + iH_{\bar\xi}\big)$. Put

$$P_{(0)} = \overline{H_\xi} - H_{\bar\xi}, \qquad P_{(1)} = i\,\overline{H_\xi} + i\,H_{\bar\xi}.$$

By (0b) and (0a) both belong to $\mathbb{P}(1,1)$, and $\mathbb{P}(1,1) = \mathbb{P}(4\cdot 1 - 3,\, 3\cdot 1 - 2)$. Since $\mathbf{J}(\xi)^{-1} = \mathbf{J}(\xi)^{1-2\cdot 1}$, the conclusion holds for $m = 1$ with $P = P_w$.

Step 2. The inductive step.

Let $m \geq 1$ and assume the conclusion for every word of length m: for each such word v there is $P_v \in \mathbb{P}(4m-3, 3m-2)$ with

$$D^v\chi\big(H(\xi)\big) = \mathbf{J}(\xi)^{1-2m}P_v(\xi) \qquad \text{for every } \xi \in \Omega. \tag{IH}$$

Let w' be a word of length $m+1$ and write $w' = (j)*v$ with $|v| = m$.

We claim it suffices to produce $Q_1, Q_2 \in \mathbb{P}(4m+1, 3m+1)$ with

$$\big(D^v\chi\big)_z\big(H(\xi)\big) = \mathbf{J}(\xi)^{-1-2m}Q_1(\xi), \qquad \big(D^v\chi\big)_{\bar z}\big(H(\xi)\big) = \mathbf{J}(\xi)^{-1-2m}Q_2(\xi) \tag{4}$$

for every $\xi \in \Omega$. Indeed, granting (4), equation (3) evaluated at $H(\xi)$ gives

$$D^{w'}\chi\big(H(\xi)\big) = \begin{cases} \mathbf{J}(\xi)^{-1-2m}\big(Q_1(\xi) + Q_2(\xi)\big) & \text{if } j = 0,\\ \mathbf{J}(\xi)^{-1-2m}\big(iQ_1(\xi) - iQ_2(\xi)\big) & \text{if } j = 1,\end{cases}$$

and in either case the second factor is a $\mathbb{C}$-linear combination of Q_1 and Q_2, hence lies in $\mathbb{P}(4m+1, 3m+1)$ by (0a). Since $m \geq 1$,

$$-1-2m = 1-2(m+1), \qquad 4m+1 = 4(m+1)-3, \qquad 3m+1 = 3(m+1)-2,$$

so this is exactly the conclusion for the word w' of length $m+1$.

Step 2a: the chain-rule system. Let $\xi \in \Omega$ be arbitrary; it is fixed only for Steps 2a–2c, and the functions Q_1, Q_2 constructed in Step 2d will not depend on it. Abbreviate

$$a = \big(D^v\chi\big)_z\big(H(\xi)\big), \quad b = \big(D^v\chi\big)_{\bar z}\big(H(\xi)\big), \quad A = H_\xi(\xi), \quad B = H_{\bar\xi}(\xi).$$

By (0d), $D^v\chi$ is real-differentiable at $H(\xi)$, and by hypothesis (a) with Ω open, H is real-differentiable at ξ. Hence `WirtingerChainRule` applies with $F = D^v\chi$ and $G = H$ and gives

$$\frac{\partial\big(D^v\chi\circ H\big)}{\partial\xi}(\xi) = a\,A + b\,\overline{B}, \qquad \frac{\partial\big(D^v\chi\circ H\big)}{\partial\bar\xi}(\xi) = a\,B + b\,\overline{A}. \tag{5}$$

This is the pair of equations displayed in the proof of Lemma 1 in the paper.

Step 2b: computing the left-hand sides. By (IH) the function $\zeta \mapsto D^v\chi(H(\zeta))$ agrees at every point of Ω with $\zeta \mapsto \mathbf{J}(\zeta)^{1-2m}P_v(\zeta)$. Since Ω is open and the real Fréchet derivative at ξ depends only on the restriction of a function to a neighbourhood of ξ, the two functions have the same $\partial/\partial\xi$ and $\partial/\partial\bar\xi$ at ξ, by `DZ` and `DZbar`.

Both factors are C^∞ on Ω: P_v by `PClassSmooth` applied to $P_v \in \mathbb{P}(4m-3, 3m-2)$, and $\zeta \mapsto \mathbf{J}(\zeta)^{1-2m}$ because $\mathbf{J}$ is C^∞ on Ω by (0c) and non-vanishing there by hypothesis (c), so that a negative integer power of it is a finite product of copies of the C^∞ function $1/\mathbf{J}$. In particular both are real-differentiable at ξ. By `WirtingerProductRule` and then `WirtingerPowerRule` applied to $g = \mathbf{J}$ with exponent $a = 1-2m$ (legitimate by (0c) and hypothesis (c)),

$$\frac{\partial\big(\mathbf{J}^{1-2m}P_v\big)}{\partial\xi}(\xi) = (1-2m)\,\mathbf{J}(\xi)^{-2m}\frac{\partial\mathbf{J}}{\partial\xi}(\xi)\,P_v(\xi) + \mathbf{J}(\xi)^{1-2m}\frac{\partial P_v}{\partial\xi}(\xi),$$

using $(1-2m)-1 = -2m$. Since $\mathbf{J}(\xi) \neq 0$ we may write $\mathbf{J}(\xi)^{1-2m} = \mathbf{J}(\xi)^{-2m}\mathbf{J}(\xi)$ and factor out $\mathbf{J}(\xi)^{-2m}$. Writing L_1 and L_2 for the two left-hand sides of (5), we obtain

$$L_1 = \mathbf{J}(\xi)^{-2m}\Big[\mathbf{J}(\xi)\frac{\partial P_v}{\partial\xi}(\xi) + (1-2m)P_v(\xi)\frac{\partial\mathbf{J}}{\partial\xi}(\xi)\Big], \tag{6}$$

$$L_2 = \mathbf{J}(\xi)^{-2m}\Big[\mathbf{J}(\xi)\frac{\partial P_v}{\partial\bar\xi}(\xi) + (1-2m)P_v(\xi)\frac{\partial\mathbf{J}}{\partial\bar\xi}(\xi)\Big], \tag{7}$$

(7) by the identical computation with $\partial/\partial\bar\xi$ in place of $\partial/\partial\xi$, using the second identity of `WirtingerProductRule` and the second identity of `WirtingerPowerRule`. Equations (5)–(7) are the system

$$\begin{cases} a\,H_\xi + b\,\overline{H_{\bar\xi}} = \mathbf{J}^{-2m}\big[\mathbf{J}P_\xi + (1-2m)P\mathbf{J}_\xi\big] \\ a\,H_{\bar\xi} + b\,\overline{H_\xi} = \mathbf{J}^{-2m}\big[\mathbf{J}P_{\bar\xi} + (1-2m)P\mathbf{J}_{\bar\xi}\big] \end{cases}$$

of the paper, evaluated at ξ, with $P = P_v$.

Step 2c: solving the system. First the determinant. For $\omega \in \mathbb{C}$ one has $|\omega|^2 = \omega\bar\omega$ as complex numbers, so by `Jac`

$$\mathbf{J}(\xi) = A\overline{A} - B\overline{B}. \tag{8}$$

Multiply the first equation of (5) by $\overline{A}$ and the second by $\overline{B}$ and subtract. The terms in b cancel, because $\overline{B}\,\overline{A} = \overline{A}\,\overline{B}$, leaving

$$a\big(A\overline{A} - B\overline{B}\big) = L_1\overline{A} - L_2\overline{B}, \qquad \text{i.e.} \qquad a\,\mathbf{J}(\xi) = L_1\overline{A} - L_2\overline{B}$$

by (8). Since $\mathbf{J}(\xi) \neq 0$,

$$a = \mathbf{J}(\xi)^{-1}\big(L_1\overline{A} - L_2\overline{B}\big). \tag{9}$$

Multiply the first equation of (5) by B and the second by A and subtract. The terms in a cancel, because $AB = BA$, leaving

$$b\big(\overline{B}B - \overline{A}A\big) = L_1B - L_2A, \qquad \text{i.e.} \qquad -\,b\,\mathbf{J}(\xi) = L_1B - L_2A,$$

again by (8), so

$$b = \mathbf{J}(\xi)^{-1}\big(L_2A - L_1B\big). \tag{10}$$

Substituting (6) and (7) into (9) and (10), and using $\mathbf{J}(\xi)^{-1}\mathbf{J}(\xi)^{-2m} = \mathbf{J}(\xi)^{-1-2m}$ (valid since $\mathbf{J}(\xi) \neq 0$),

$$a = \mathbf{J}(\xi)^{-1-2m}\Big[\mathbf{J}P_\xi\overline{A} - \mathbf{J}P_{\bar\xi}\overline{B} + (1-2m)P\mathbf{J}_\xi\overline{A} - (1-2m)P\mathbf{J}_{\bar\xi}\overline{B}\Big](\xi), \tag{11}$$

$$b = \mathbf{J}(\xi)^{-1-2m}\Big[\mathbf{J}P_{\bar\xi}A - \mathbf{J}P_\xi B + (1-2m)P\mathbf{J}_{\bar\xi}A - (1-2m)P\mathbf{J}_\xi B\Big](\xi), \tag{12}$$

where $P = P_v$. These are exactly the two expressions the paper displays for $(D^\nu\chi)_z$ and $(D^\nu\chi)_{\bar z}$. The single extra factor $\mathbf{J}(\xi)^{-1}$, which turns $\mathbf{J}^{-2m}$ into $\mathbf{J}^{-1-2m}$, is precisely the division by the determinant of the system, and the determinant is $\mathbf{J}(\xi)$ by (8).

Step 2d: every bracket term lies in $\mathbb{P}(4m+1, 3m+1)$. Collect the memberships, all relative to H. Each item below produces a function on $\mathbb{C}$ that does not depend on the point ξ fixed in Step 2a: the existential quantifiers of `PClassDerivClosed` and the class members of `JacInPClass` stand outside the "for every $\xi \in \Omega$" of those statements. The list is:

- $\mathbf{J} \in \mathbb{P}(2, 2)$, by (0c);

- by `PClassDerivClosed` applied to $P_v \in \mathbb{P}(4m-3, 3m-2)$ (its hypotheses hold: Ω is open and H is C^∞ on Ω), there are
$$Q,\, Q' \in \mathbb{P}\big((4m-3)+1,\, 3m-2\big) = \mathbb{P}(4m-2,\, 3m-2)$$
with $\partial P_v/\partial\xi = Q$ and $\partial P_v/\partial\bar\xi = Q'$ at every point of Ω;
- $R,\ \overline{R} \in \mathbb{P}(3,2)$ with $\partial\mathbf{J}/\partial\xi = R$ and $\partial\mathbf{J}/\partial\bar\xi = \overline{R}$ on Ω, by (0c);
- $P_v \in \mathbb{P}(4m-3, 3m-2)$, by (IH);
- $H_\xi,\ \overline{H_\xi},\ H_{\bar\xi},\ \overline{H_{\bar\xi}} \in \mathbb{P}(1,1)$, by (0b).

Now `PClassMulClosed` closes $\mathbb{P}$ under pointwise products, adding both indices; applying it twice gives the class of a product of three factors. The eight products occurring in the brackets of (11) and (12) have only two shapes, and both land in $\mathbb{P}(4m+1, 3m+1)$:

$$\mathbf{J}\cdot Q\cdot\overline{H_\xi},\quad \mathbf{J}\cdot Q'\cdot\overline{H_{\bar\xi}},\quad \mathbf{J}\cdot Q'\cdot H_\xi,\quad \mathbf{J}\cdot Q\cdot H_{\bar\xi}:\quad \big(2+(4m-2)+1,\ 2+(3m-2)+1\big) = (4m+1,\ 3m+$$
$$P_v\cdot R\cdot\overline{H_\xi},\quad P_v\cdot\overline{R}\cdot\overline{H_{\bar\xi}},\quad P_v\cdot\overline{R}\cdot H_\xi,\quad P_v\cdot R\cdot H_{\bar\xi}:\quad \big((4m-3)+3+1,\ (3m-2)+2+1\big) = (4m+1,\ 3m+$$

Both index computations use $m \ge 1$, so that $4m-3$ and $3m-2$ are the ordinary differences: $2+(4m-2)+1 = 4m+1$ and $2+(3m-2)+1 = 3m+1$, while $(4m-3)+3+1 = 4m+1$ and $(3m-2)+2+1 = 3m+1$.

Define

$$Q_1 = \mathbf{J}\cdot Q\cdot\overline{H_\xi} - \mathbf{J}\cdot Q'\cdot\overline{H_{\bar\xi}} + (1-2m)\,P_v\cdot R\cdot\overline{H_\xi} - (1-2m)\,P_v\cdot\overline{R}\cdot\overline{H_{\bar\xi}},$$
$$Q_2 = \mathbf{J}\cdot Q'\cdot H_\xi - \mathbf{J}\cdot Q\cdot H_{\bar\xi} + (1-2m)\,P_v\cdot\overline{R}\cdot H_\xi - (1-2m)\,P_v\cdot R\cdot H_{\bar\xi},$$

as pointwise products and sums of functions on $\mathbb{C}$. Each of the eight summands lies in $\mathbb{P}(4m+1, 3m+1)$ by the computation just made, and the scalars ±1 and $\pm(1-2m)$ are complex numbers, so $Q_1, Q_2 \in \mathbb{P}(4m+1, 3m+1)$ by (0a).

Finally let $\xi \in \Omega$. The bracket in (11) and the value $Q_1(\xi)$ differ only in that the bracket contains $\partial P_v/\partial\xi(\xi)$, $\partial P_v/\partial\bar\xi(\xi)$, $\partial\mathbf{J}/\partial\xi(\xi)$, $\partial\mathbf{J}/\partial\bar\xi(\xi)$ where $Q_1(\xi)$ contains $Q(\xi)$, $Q'(\xi)$, $R(\xi)$, $\overline{R(\xi)}$, and these pairs agree at every point of Ω by the four identifications just listed; also $\overline{A} = \overline{H_\xi(\xi)}$ and $\overline{B} = \overline{H_{\bar\xi}(\xi)}$ by definition of A and B. Hence (11) reads $a = \mathbf{J}(\xi)^{-1-2m}Q_1(\xi)$, and in the same way (12) reads $b = \mathbf{J}(\xi)^{-1-2m}Q_2(\xi)$. Since Q_1 and Q_2 were defined without reference to ξ, and $\xi \in \Omega$ was arbitrary, these two identities hold for every $\xi \in \Omega$. This is (4), and the inductive step is complete.

Step 3. Conclusion. By Step 1 the conclusion holds for every word of length 1, and by Step 2 its validity for all words of length $m \ge 1$ implies its validity for all words of length $m+1$. By induction it holds for every $m \ge 1$ and every word w of length m, which is the assertion of the lemma. □

3.4. `Cor2` (paper, Corollary 2).

Corollary 3.4. *Let $n \ge 1$ be an integer, let $1 < p < \infty$ and let $K' \ge 1$. Assume the bundle (C2) of `QCPackage`. Let $W \subseteq \mathbb{C}$ be open with $0 \in W$ and let $f, \chi, H : \mathbb{C} \to \mathbb{C}$ satisfy*

(H1) *f solves equation (12) on W in the sense of `SolvesEq12`, and $f(0) = 0$;*
(H2) *χ is K'-quasiconformal on W in the sense of `QuasiconformalOn`, $\chi(0) = 0$, and $f(z) = [\chi(z)]^n$ for every $z \in W$;*
(H3) *χ is C^∞ on $W - \{0\}$;*
(H4) *$H(\chi(z)) = z$ for every $z \in W$, and H is continuous on $\chi(W)$.*

Write γ_n for the exponent of `GammaExp` and D^w for the iterated partial derivative of `PartialWord`. Then there is $\delta > 0$ and there are constants B_m, $m = 0, 1, 2, \dots$, such that

$$\sum_{|w|=m} \big|D^w f(z)\big| \le B_m |z|^{n/\gamma_n - m} \qquad \text{for } 0 < |z| < \delta, \tag{48}$$

the sum running over all 2^m words w of length m in $\{0,1\}$.

This is Corollary 2 of the paper (paper.tex:473–479). Hypothesis (H1) is the paper's "f is a local solution to the system (12) which has the form (15), $f(0) = 0$", and (H2) is the factorization (15) itself, whose existence and whose quasiconformality of χ are part of the quoted bundle (C2);

(H3) and (H4) are Section 3's normalisation of the hodograph pair. The words of length m realize exactly the operators D^ν with $|\nu| = m$, as `PartialWord` *records, so the sum is the paper's $\sum_{|\nu|=m}$ up to the multiplicities with which each D^ν occurs, at most 2^m, absorbed into B_m. Equation (18) for H is not assumed; it is a consequence of (H1) and (H2).*

Proof. Write $\gamma = \gamma_n$ and $V = \chi(W)$. Note first that $f(0) = 0$ is also a consequence of (H2): $f(0) = \chi(0)^n = 0$ since $\chi(0) = 0$ and $n \geq 1$. It is recorded in (H1) because the paper states it, and is not used separately below.

Step 0: $\gamma > 0$. By part (a) of `GammaRatioMinAtOne`, with $n \geq 1$ and $1 < p$, we have $\gamma > 0$.

Step 1: V is an open neighbourhood of 0. By `QuasiconformalOn` applied to χ on W, the set $V = \chi(W)$ is open, χ is injective and continuous on W. Since $0 \in W$ and $\chi(0) = 0$, we have $0 = \chi(0) \in V$.

Step 2: H is a two-sided inverse of χ between V and W. Let $\xi \in V$, say $\xi = \chi(z)$ with $z \in W$. By (H4), $H(\xi) = H(\chi(z)) = z$, so

$$H(\xi) \in W \quad \text{and} \quad \chi(H(\xi)) = \chi(z) = \xi \qquad (\xi \in V). \tag{1}$$

Taking $z = 0$ in (H4) and using $\chi(0) = 0$ gives

$$H(0) = H(\chi(0)) = 0. \tag{2}$$

Also $H(V) = W$: the inclusion $H(V) \subseteq W$ is (1), and conversely each $z \in W$ is $H(\chi(z))$ with $\chi(z) \in V$.

Finally, H does not vanish on $V - \{0\}$: if $\xi \in V$ and $H(\xi) = 0$ then $\xi = \chi(H(\xi)) = \chi(0) = 0$ by (1). Hence

$$H(\xi) \in W - \{0\} \qquad (\xi \in V - \{0\}). \tag{3}$$

Step 3: the inverse clauses of (C2), and (18) on V. The clause `qc_inverse` of `QCPackage`, applied with K', the open set W, the map χ and the map H — its hypotheses being that W is open, that χ is K'-quasiconformal on W, which is (H2), and that $H(\chi(z)) = z$ for every $z \in W$, which is (H4) — gives

$$H \in W^{1,2}_{\mathrm{loc}}(V), \qquad H \text{ differentiable a.e. on } V, \qquad J_H > 0 \text{ a.e. on } V, \tag{3$'$}$$

where $V = \chi(W)$ and J_H is the Jacobian of `Jac`; in particular $J_H \neq 0$ almost everywhere on V, a positive number being nonzero.

Now the hypotheses of `HodographEq18` hold with this n, p, the open sets W and V, the point $z_0 = 0 \in W$, the maps f, χ, H, the factorization of (H2), $\chi(0) = 0$, injectivity of χ on W from Step 1, the smoothness of χ on $W - \{0\}$ from (H3), the second and third clauses of (3′) in place of any smoothness of H — these are its (H3′) — its (H3″), namely $H \in W^{1,2}_{\mathrm{loc}}(V)$ in the sense of `MemWLocOn`, which is the first clause of (3′), the two relations (1), $H(0) = 0$ from (2), and equation (12) from (H1). Therefore

$$H \text{ solves (18) on } V \text{ in the sense of } \texttt{SolvesEq18}. \tag{18}$$

Step 4: $H \in W^{1,2}_{\mathrm{loc}}(V)$. This is the first clause of (3′):

$$H \in W^{1,2}_{\mathrm{loc}}\big(\chi(W)\big) = W^{1,2}_{\mathrm{loc}}(V) \tag{4}$$

in the sense of `MemWLocOn`.

Step 5: H is K-quasiconformal on V with $K = \max\{p - 1, 1/(p-1)\}$. Since $p > 1$ we have $p - 1 > 0$; if $p \geq 2$ then $K \geq p - 1 \geq 1$, and if $p < 2$ then $0 < p - 1 < 1$ so $K \geq 1/(p-1) > 1$. Thus $K \geq 1$. We verify the five clauses of `QuasiconformalOn` for H on V.

Quasiregularity. By `Eq18ImpliesQuasiregular`, applied with p, n, the set V and the map H, using (4) and (18), H is K-quasiregular on V with exactly this K.

Injectivity. If $\xi_1, \xi_2 \in V$ and $H(\xi_1) = H(\xi_2)$ then by (1) $\xi_1 = \chi(H(\xi_1)) = \chi(H(\xi_2)) = \xi_2$.

Continuity on $V = \chi(W)$ is part of (H4).

Openness of the image. $H(V) = W$ by Step 2, which is open.

The inverse. Take $\psi = \chi$: by (1), $\chi(H(\xi)) = \xi$ for every $\xi \in V$, and χ is continuous on $H(V) = W$ by Step 1.

Step 6: the derivatives of χ. The hypotheses of `ChiHigherDerivBound` hold with this n, p, K, the open sets V and $U = W - \{0\}$ — the latter open since W is — the point $0 \in V$ from Step 1, the maps H and χ, continuity of H on V from (H4) and its differentiability almost everywhere on V from (3′), K-quasiconformality from Step 5, equation (18) from Step 3, $H(0) = 0$ from (2), smoothness of χ on $U = W - \{0\}$ from (H3), the inclusion $H(\xi) \in U$ for $\xi \in V - \{0\}$ from (3), and $\chi(H(\xi)) = \xi$ on V from (1). It therefore provides $\delta > 0$ with

$$\{z : |z| < \delta\} \subseteq H(V) = W \tag{5}$$

and constants $B'_l \geq 0$ with

$$\left|D^w \chi(z)\right| \leq B'_l \, |z|^{1/\gamma - l} \qquad (|w| = l,\ 0 < |z| < \delta). \tag{6}$$

Step 7: the Leibniz rule for words. We record the two-factor Leibniz rule in the form needed. Let u, v be C^∞ on an open set G and let w be a word of length m, with letters indexed by $\{1, \dots, m\}$. For $S \subseteq \{1, \dots, m\}$ let w_S denote the subword of w on the positions in S, in their original order. Then on G

$$D^w(uv) = \sum_{S \subseteq \{1,\dots,m\}} D^{w_S} u \cdot D^{w_{S^c}} v. \tag{7}$$

This is proved by induction on m. For $m = 0$ both sides are uv, the only subset being $\emptyset$. For the step, write $w = j\colon w'$ with w' of length $m - 1$. By `PartialWord`, $D^w = \partial_j \circ D^{w'}$, so by the inductive hypothesis and the ordinary product rule for ∂_j,

$$D^w(uv) = \partial_j\Big(\sum_{S'} D^{w'_{S'}} u \cdot D^{w'_{S'^c}} v\Big) = \sum_{S'}\Big(D^{j::w'_{S'}} u \cdot D^{w'_{S'^c}} v + D^{w'_{S'}} u \cdot D^{j::w'_{S'^c}} v\Big),$$

again using $D^{j::} = \partial_j \circ D^{\cdot}$. Subsets S of the letters of w correspond bijectively to pairs consisting of a subset S' of the letters of w' together with the decision whether the new letter lies in S; the two summands above are exactly the two cases, since the new letter in S gives $w_S = j\colon w'_{S'}$ and $w_{S^c} = w'_{S'^c}$, and the new letter in S^c gives the other pair. This is (7).

Step 8: the derivatives of χ^N. We claim that for every integer $N \geq 1$ and every $m \geq 0$ there is $A_{N,m} \geq 0$ with

$$\left|D^w\big(\chi^N\big)(z)\right| \leq A_{N,m} \, |z|^{N/\gamma - m} \qquad (|w| = m,\ 0 < |z| < \delta). \tag{8}$$

Note first that χ is C^∞ on the open set $\{z : 0 < |z| < \delta\}$: that set is contained in $W - \{0\}$ by (5), where χ is C^∞ by (H3). Hence so is every power χ^N, and all the derivatives below exist.

For $N = 1$, (8) is (6) with $A_{1,m} = B'_m$. For the inductive step, apply (7) on $G = \{z : 0 < |z| < \delta\}$ with $u = \chi$ and $v = \chi^N$, so that $uv = \chi^{N+1}$, and estimate each term by (6) and (8): for $|w| = m$ and $0 < |z| < \delta$,

$$\left|D^w\big(\chi^{N+1}\big)(z)\right| \leq \sum_S \left|D^{w_S}\chi(z)\right| \left|D^{w_{S^c}}(\chi^N)(z)\right| \leq \sum_S B'_{|S|} \, |z|^{1/\gamma - |S|} \cdot A_{N,m-|S|} \, |z|^{N/\gamma - (m-|S|)},$$

where $|w_S| = |S|$ and $|w_{S^c}| = m - |S|$. The powers of the positive real $|z|$ combine additively, and

$$\Big(\frac{1}{\gamma} - |S|\Big) + \Big(\frac{N}{\gamma} - m + |S|\Big) = \frac{N+1}{\gamma} - m,$$

so the right-hand side equals $A_{N+1,m}|z|^{(N+1)/\gamma - m}$ with

$$A_{N+1,m} = \sum_{S \subseteq \{1,\dots,m\}} B'_{|S|} \, A_{N,m-|S|} = \sum_{j=0}^{m} \binom{m}{j} B'_j \, A_{N,m-j} \ \geq 0,$$

a finite sum of products of nonnegative numbers. This proves (8) for $N+1$, and hence for every $N \geq 1$ by induction.

Step 9: conclusion. Let $m \geq 0$, let w be a word of length m, and let $0 < |z| < \delta$. By (5), $z \in W$, and W is open, so f agrees with χ^n on a neighbourhood of z by (H2); hence $D^w f(z) = D^w(\chi^n)(z)$. By (8) with $N = n$, which is legitimate since $n \geq 1$,

$$\left|D^w f(z)\right| \leq A_{n,m} \, |z|^{n/\gamma - m}.$$

Summing over the 2^m words of length m,

$$\sum_{|w|=m} \left|D^w f(z)\right| \le 2^m A_{n,m}\,|z|^{n/\gamma-m} \qquad (0<|z|<\delta),$$

which is (48) with $B_m = 2^m A_{n,m} \ge 0$, with the δ of Step 6. This is the paper's closing step at paper.tex:510. □

3.5. `Lemma2` (paper, Lemma 2).

Lemma 3.5. *Let U be a convex open subset of $\mathbb{R}^n$ containing the origin, let $M \ge 0$ and $\alpha \in (0,1]$, and let $F\colon \mathbb{R}^n \to \mathbb{C}$ be a function of class $C(U) \cap C^1(U-\{0\})$ such that*

(i) *$|F(x)| \le M|x|^\alpha$ for $x \in U$,*
(ii) *$\|DF(x)\| \le M|x|^{\alpha-1}$ for $x \in U-\{0\}$,*

where $DF(x)$ denotes the real Fréchet derivative of F at x, a real-linear map $\mathbb{R}^n \to \mathbb{C}$, and $\|DF(x)\|$ its operator norm. Then

$$|F(x)-F(y)| \le 5M|x-y|^\alpha \qquad \text{for all } x,y \in U.$$

Proof. Let $x, y \in U$.

The degenerate case $x = y$. Then $|F(x)-F(y)| = 0$, and since $\alpha > 0$ we have $|x-y|^\alpha = 0^\alpha = 0$, so the right-hand side is 0 as well and the inequality holds. Assume from now on that $x \ne y$, so that $|x-y| > 0$.

Exactly one of the following two cases occurs.

Case 1: $|y| \le 2|x-y|$. By the triangle inequality in $\mathbb{C}$,

$$|F(x)-F(y)| \le |F(x)| + |F(y)|.$$

Now apply the unweighted two-term power-mean inequality $(z_1+z_2)^p \le 2^{p-1}(z_1^p+z_2^p)$, valid for $z_1, z_2 \ge 0$ and $p \ge 1$ (mathlib's `NNReal.rpow_add_le_mul_rpow_add_rpow`), at the exponent $p = 1/\alpha$, which is ≥ 1 because $0 < \alpha \le 1$, and with $z_1 = |F(x)| \ge 0$, $z_2 = |F(y)| \ge 0$. Raising the result to the power α, which is monotone since $\alpha > 0$, and using $\left(t^{1/\alpha}\right)^\alpha = t$ for $t \ge 0$ on the left and $\left(2^{1/\alpha-1}\right)^\alpha = 2^{1-\alpha}$ on the right, gives

$$|F(x)| + |F(y)| \le 2^{1-\alpha}\Big(|F(x)|^{1/\alpha} + |F(y)|^{1/\alpha}\Big)^\alpha.$$

By hypothesis (i), $|F(x)| \le M|x|^\alpha$ and $|F(y)| \le M|y|^\alpha$, all four quantities being nonnegative. Since $1/\alpha > 0$, the map $t \mapsto t^{1/\alpha}$ is nondecreasing on $[0,\infty)$, so

$$|F(x)|^{1/\alpha} \le \left(M|x|^\alpha\right)^{1/\alpha} = M^{1/\alpha}\left(|x|^\alpha\right)^{1/\alpha} = M^{1/\alpha}|x|,$$

and likewise $|F(y)|^{1/\alpha} \le M^{1/\alpha}|y|$, using $(t^\alpha)^{1/\alpha} = t$ for $t \ge 0$ and the multiplicativity of real powers on nonnegative bases. Since $\alpha > 0$, the map $t \mapsto t^\alpha$ is nondecreasing on $[0,\infty)$, whence

$$2^{1-\alpha}\Big(|F(x)|^{1/\alpha} + |F(y)|^{1/\alpha}\Big)^\alpha \le 2^{1-\alpha}\Big(M^{1/\alpha}\big(|x|+|y|\big)\Big)^\alpha = 2^{1-\alpha}M\big(|x|+|y|\big)^\alpha,$$

where we used $\left(M^{1/\alpha}\right)^\alpha = M$ (valid since $M \ge 0$).

We now bound $|x|+|y|$. By the triangle inequality $|x| \le |x-y|+|y|$, so

$$|x|+|y| \le |x-y|+2|y| \le |x-y|+4|x-y| = 5|x-y|,$$

the second inequality being the case hypothesis $|y| \le 2|x-y|$. Hence, again by monotonicity of $t \mapsto t^\alpha$ on $[0,\infty)$ and $M \ge 0$,

$$|F(x)-F(y)| \le 2^{1-\alpha}M\big(5|x-y|\big)^\alpha = 2^{1-\alpha}5^\alpha M|x-y|^\alpha.$$

Finally

$$2^{1-\alpha}5^\alpha = 2\cdot\left(\frac{5}{2}\right)^\alpha \le 2\cdot\frac{5}{2} = 5,$$

because $\frac{5}{2} > 1$ and $0 < \alpha \le 1$, so that $t \mapsto (\frac{5}{2})^t$ is nondecreasing. Since $M|x-y|^\alpha \ge 0$, we conclude

$$|F(x)-F(y)| \le 5M|x-y|^\alpha.$$

Case 2: $|y| > 2|x-y|$. For $t \in [0,1]$ put

$$z(t) = tx + (1-t)y.$$

Since U is convex and $x, y \in U$, we have $z(t) \in U$ for every $t \in [0,1]$. Moreover $z(t) = y + t(x-y)$, so for $t \in [0,1]$,

$$|z(t)| \geq |y| - t\,|x-y| \geq |y| - |x-y| > 2|x-y| - |x-y| = |x-y| > 0,$$

using $t \leq 1$ and the case hypothesis. In particular $z(t) \neq 0$, so

$$z(t) \in U - \{0\} \qquad \text{for every } t \in [0,1].$$

Since U is open and $\{0\}$ is closed, $U - \{0\}$ is open, and F is of class C^1 on it. The segment $\{z(t) : t \in [0,1]\}$ is contained in $U - \{0\}$, so the map $t \mapsto F(z(t))$ is of class C^1 on $[0,1]$, with derivative

$$\frac{d}{dt} F\big(z(t)\big) = DF\big(z(t)\big)[x-y]$$

by the chain rule, $t \mapsto z(t)$ being affine with derivative the constant vector $x - y$. By the fundamental theorem of calculus, applied to the real and imaginary parts of this C^1 function on $[0,1]$,

$$F(x) - F(y) = F\big(z(1)\big) - F\big(z(0)\big) = \int_0^1 DF\big(z(t)\big)[x-y]\,dt.$$

We estimate the integrand. For each $t \in [0,1]$, by the definition of the operator norm,

$$\big|DF\big(z(t)\big)[x-y]\big| \leq \big\|DF\big(z(t)\big)\big\|\,|x-y|,$$

and by hypothesis (ii), applicable since $z(t) \in U - \{0\}$,

$$\big\|DF\big(z(t)\big)\big\| \leq M\,|z(t)|^{\alpha-1}.$$

Now $\alpha - 1 \leq 0$, so $r \mapsto r^{\alpha-1}$ is nonincreasing on $(0,\infty)$; since $|z(t)| \geq |x-y| > 0$, this gives

$$|z(t)|^{\alpha-1} \leq |x-y|^{\alpha-1}.$$

As $M \geq 0$, combining the last three displays yields, for every $t \in [0,1]$,

$$\big|DF\big(z(t)\big)[x-y]\big| \leq M\,|x-y|^{\alpha-1}\,|x-y| = M\,|x-y|^{\alpha},$$

the last step using $|x-y| > 0$ and $|x-y|^{\alpha-1}|x-y| = |x-y|^{(\alpha-1)+1} = |x-y|^{\alpha}$.

Therefore

$$|F(x)-F(y)| = \Big|\int_0^1 DF\big(z(t)\big)[x-y]\,dt\Big| \leq \int_0^1 \big|DF\big(z(t)\big)[x-y]\big|\,dt \leq \int_0^1 M|x-y|^{\alpha}\,dt = M|x-y|^{\alpha}.$$

Since $M|x-y|^{\alpha} \geq 0$, we have $M|x-y|^{\alpha} \leq 5M|x-y|^{\alpha}$, so

$$|F(x) - F(y)| \leq 5M|x-y|^{\alpha}$$

in Case 2 as well.

As Cases 1 and 2 are exhaustive, the estimate (53) holds for all $x, y \in U$. □

3.6. `Thm1Optimality` (paper, Theorem 1, optimality).

Theorem 3.6. *Let $1 < p < \infty$ with $p \neq 2$, let $k \geq 1$ be an integer and $0 < \alpha \leq 1$ with $k + \alpha$ equal to the right-hand side of (9), as in `KAlphaExp`, and let $\Omega \subseteq \mathbb{C}$ be open, non-empty and connected. Then there is a function $v\colon \mathbb{C} \to \mathbb{R}$ such that*

(a) *v is p-harmonic on Ω in the sense of `PHarmonicOn`;*
(b) *v is continuous on Ω;*
(c) *$v \notin C^{k+\alpha}_{\mathrm{loc}}(\Omega)$ in the sense of `MemHolderSmallOn`;*
(d) *$v \notin W^{k+2,q}_{\mathrm{loc}}(\Omega)$ in the sense of `MemWLocOn`, where $q = \frac{2}{2-\alpha}$.*

This is the optimality clause of Theorem 1 of the paper: "for each $1 < p < \infty$, $p \neq 2$, there is a p-harmonic function $v \in W^{1,p}_{\mathrm{loc}}(\Omega)$ which is not in the class $C^{k+\alpha}_{\mathrm{loc}}(\Omega) \cup W^{k+2,\frac{2}{2-\alpha}}_{\mathrm{loc}}(\Omega)$"; the membership $v \in W^{1,p}_{\mathrm{loc}}(\Omega)$ is part of `PHarmonicOn`*. Together with the regularity half it covers the target: neither half covers it alone.*

Two features of the statement deserve comment. The conclusion concerns v, whose two indices exceed by one those of (56), which concerns the complex gradient $f = \partial v/\partial z$; the exponent q does not move. And the continuity (b) is asserted deliberately. Being p-harmonic is a property of v seen only through integrals, so it is unchanged when v is altered on a set of measure zero, whereas membership in a Hölder class is not; without (b) the non-membership (c) could be arranged by altering any function on a null set, and the statement would be vacuous. It is not a strengthening the paper does not justify: the paper's own witness, the potential of the continuous function f of (55), is continuous. Connectedness of Ω carries its non-emptiness, without which (c) and (d) are false, every function lying in every class on the empty set; and it is the paper's hypothesis on the domain.

Proof. One remark on well-posedness before the argument proper. The hypothesis above presents the pair (k, α) as given, whereas Theorem 1 of the paper says at paper.tex:87–90 that "the integer $k \geq 1$ and the exponent $\alpha \in (0, 1]$ are determined by the equation (9)". By `KAlphaWellDefined` there is, for the given p, exactly one pair consisting of an integer $k \geq 1$ and a real number α with $0 < \alpha \leq 1$ such that $k + \alpha$ equals the right-hand side of (9); so the hypothesis on (k, α) is satisfiable for every $1 < p < \infty$, and in particular for every $p \neq 2$, and determines the pair from p, exactly as the paper asserts, and the statement proved below is therefore not vacuous for any such p. This is a well-posedness remark and not a step in the argument: nothing in the rest of this proof invokes that lemma, and in particular the bundle fed to `Section7Heart` in Step 3 below uses only the hypothesis on (k, α) as stated, for whichever pair satisfies it.

Step 1: the extremal data. By `ExtremalInverseGradient` there are $f : \mathbb{C} \to \mathbb{C}$ and $v_0 : \mathbb{C} \to \mathbb{R}$ such that: f is continuous on $\mathbb{C}$ and satisfies $f(H(\xi)) = \xi$ and $H(f(z)) = z$ for all ξ, z, where $H = H_p$ is the extremal function of `ExtremalH`; f is C^∞ on $\mathbb{C} - \{0\}$; f satisfies the two homogeneity identities of degree $d = d(p)$ and $d - |\nu|$ of `DExp` and `PartialWord`; v_0 is continuous on $\mathbb{C}$, is C^∞ on $\mathbb{C} - \{0\}$ and is p-harmonic on $\mathbb{C}$ in the sense of `PHarmonicOn`; and

$$\frac{\partial v_0}{\partial z}(z) = f(z) \qquad (z \neq 0), \tag{1}$$

with $\partial/\partial z$ as in `DZ`.

Step 2: translation of the singular point into Ω. Since Ω is connected it is non-empty; fix $z_0 \in \Omega$ and put

$$U = \{z - z_0 : z \in \Omega\}, \qquad v(z) = v_0(z - z_0) \quad (z \in \mathbb{C}).$$

Then U is open, being the image of the open set Ω under the homeomorphism $z \mapsto z - z_0$, and $0 = z_0 - z_0 \in U$. Write $\tau(z) = z - z_0$, so that $v = v_0 \circ \tau$, $U = \tau(\Omega)$ and $\Omega = \tau^{-1}(U)$.

Step 3: (56) on U. The hypotheses of `Section7Heart` hold with this U and this f: $1 < p < \infty$ and $p \neq 2$; $k \geq 1$, $0 < \alpha \leq 1$ and $k + \alpha$ is the right-hand side of (9); U is open with $0 \in U$; and the four properties (H1)–(H4) of f — smoothness off the origin, the two inverse relations, and the two homogeneity identities — are precisely those listed in Step 1. Therefore

$$f \notin C^{k-1+\alpha}_{\mathrm{loc}}(U) \quad \text{and} \quad f \notin W^{k+1,q}_{\mathrm{loc}}(U). \tag{2}$$

Step 4: two remarks on norms and on translation.
(4a) Let Ω' be open, let g be of class C^m on Ω' and let $\mathrm{d}^m g$ be its m-th iterated Fréchet derivative, which on the open set Ω' agrees with the derivative within Ω'. As in `MemHolderSmallOn` the little-o condition is expressed through $\mathrm{d}^m g$, whereas the partial derivatives $D^\nu g$ of `PartialWord` with $|\nu| = m$ are the values $\mathrm{d}^m g(x)[e_{j_1}, \ldots, e_{j_m}]$ on the basis vectors $e_0 = 1$, $e_1 = i$. Expanding each argument of an m-linear map in that basis and using $|u^{(j)}| \leq \|u\|$ gives, for $x, y \in \Omega'$,

$$\left|D^\nu g(x) - D^\nu g(y)\right| \leq \left\|\mathrm{d}^m g(x) - \mathrm{d}^m g(y)\right\| \leq \sum_{|\nu'|=m} \left|D^{\nu'} g(x) - D^{\nu'} g(y)\right|, \tag{3}$$

the last sum having 2^m terms.

(4b) Translation covariance. Lebesgue measure on $\mathbb{C}$ is translation invariant, τ is a C^∞ diffeomorphism with $\mathrm{d}\tau = \mathrm{id}$, τ carries compact subsets of Ω to compact subsets of U and preserves distances, and $\phi \mapsto \phi \circ \tau^{-1}$ is a bijection between the test functions for Ω of `IsTestFn` and those for U. Consequently each of the notions used below is covariant under τ: for $u\colon \mathbb{C} \to \mathbb{C}$,

- $u \in L^s_{\mathrm{loc}}(\Omega)$ if and only if $u \circ \tau^{-1} \in L^s_{\mathrm{loc}}(U)$, by `MemLpLocOn` and translation invariance of the measure;
- w is the distributional j-th partial of u on Ω in the sense of `IsWeakPartialOn` if and only if $w \circ \tau^{-1}$ is that of $u \circ \tau^{-1}$ on U, by the change of variables $\zeta = \tau(z)$ in both integrals, which also converts ϕ into $\phi \circ \tau^{-1}$;
- $u \in C^{m+\alpha}_{\mathrm{loc}}(\Omega)$ if and only if $u \circ \tau^{-1} \in C^{m+\alpha}_{\mathrm{loc}}(U)$, by `MemHolderSmallOn`: the C^m clause is preserved by composition with a diffeomorphism whose differential is the identity, which also gives $\mathrm{d}^m(u \circ \tau^{-1})(\zeta) = \mathrm{d}^m u(\tau^{-1}(\zeta))$, and the little-o clause then transfers because τ^{-1} carries compact subsets of U to compact subsets of Ω and preserves distances;
- $u \in W^{m,s}_{\mathrm{loc}}(\Omega)$ if and only if $u \circ \tau^{-1} \in W^{m,s}_{\mathrm{loc}}(U)$, by `MemWLocOn`: the family (D_w) for u on Ω transforms into the family $(D_w \circ \tau^{-1})$ for $u \circ \tau^{-1}$ on U by the two previous items;
- for $h\colon \mathbb{C} \to \mathbb{C}$, the integral identity of clause (c) of `PHarmonicOn` holds with witness h on Ω if and only if it holds with witness $h \circ \tau^{-1}$ on U. Written out, with $\nabla\phi = \phi_x + i\phi_y$ and the integrand that `PHarmonicOn` records for clause (c), the two assertions are

$$\int_{\mathbb{C}} |h|^{p-2} \operatorname{Re}\big(h\,\overline{\nabla\phi}\big)\, d\sigma = 0 \quad \text{for every real test function } \phi \text{ for } \Omega$$

  and the same with h replaced by $h \circ \tau^{-1}$ and ϕ ranging over the real test functions for U. Take the direction $U \to \Omega$, which is the one Step 5 uses, and assume the second. Let ϕ be a real test function for Ω and put $\psi = \phi \circ \tau^{-1}$, a real test function for U by the bijection recorded above. Substituting $\zeta = \tau(z)$,

$$\int_{\mathbb{C}} |h(z)|^{p-2} \operatorname{Re}\big(h(z)\,\overline{\nabla\phi(z)}\big)\, d\sigma(z) = \int_{\mathbb{C}} \big|(h \circ \tau^{-1})(\zeta)\big|^{p-2} \operatorname{Re}\big((h \circ \tau^{-1})(\zeta)\,\overline{\nabla\psi(\zeta)}\big)\, d\sigma(\zeta),$$

  by three facts, each already recorded in the preamble to this list: the measure is unchanged because Lebesgue measure on $\mathbb{C}$ is translation invariant; the witness is carried along because $h(z) = (h \circ \tau^{-1})(\tau z)$ by definition; and the gradient is carried along because $\mathrm{d}\tau = \mathrm{id}$ gives $\nabla\psi(\zeta) = \nabla\phi(\tau^{-1}\zeta) = \nabla\phi(z)$. The right-hand integral vanishes by the assumption on U, so the left-hand one does too; and since $\phi \mapsto \phi \circ \tau^{-1}$ is a *bijection* onto the real test functions for U, the quantifier transfers as well. The converse direction is the same computation run through τ^{-1}, which is again a translation. The convention of `PHarmonicOn` by which $|h|^{p-2}$ is read as 0 where h vanishes causes no difficulty here: the integrand is a fixed function of the two values $h(z)$ and $\nabla\phi(z)$, so the substitution transports it along with them.

Finally $v \circ \tau^{-1} = v_0$ and $\tau(\Omega) = U$.

Step 5: (a) and (b). By Step 1, v_0 is p-harmonic on $\mathbb{C}$. Restricting to Ω is legitimate: by `PHarmonicOn` with the witness g_0, the two local L^p memberships and the two integration-by-parts identities and the integral identity are required against test functions for the ambient set, and every test function for a subset of $\mathbb{C}$ is one for $\mathbb{C}$, while compact subsets of a subset are compact subsets of $\mathbb{C}$; so v_0 is p-harmonic on U with the same witness. Now transfer that back to Ω. By `PHarmonicOn` what has to be checked for $v = v_0 \circ \tau$ on Ω with witness $g = g_0 \circ \tau$ is exactly its three clauses, and each is an instance of (4b) read in the direction $U \to \Omega$ — that is, right to left in the stated equivalence, with the function on the Ω side taken to be $v_0 \circ \tau$ or $g_0 \circ \tau$, so that composing it with τ^{-1} returns the function already known on U. Clause (a), that v and g lie in $L^p_{\mathrm{loc}}(\Omega)$, is the first item applied once to each of the two functions. Clause (b), that $\operatorname{Re} g$ and $\operatorname{Im} g$ are the distributional x- and y-partials of v on Ω, is the second item applied once for each of the two letters, using $\operatorname{Re}(g_0 \circ \tau) = (\operatorname{Re} g_0) \circ \tau$ and likewise for the imaginary part. Clause (c), the integral identity, is the fifth item with $h = g_0 \circ \tau$. All three therefore hold, so v is p-harmonic on Ω, which is (a). And v is continuous on $\mathbb{C}$, being the composition of the continuous v_0 with τ; in particular it is continuous on Ω, which is (b).

Step 6: (c). Suppose $v \in C^{k+\alpha}_{\mathrm{loc}}(\Omega)$. By the third item of (4b), $v_0 = v \circ \tau^{-1} \in C^{k+\alpha}_{\mathrm{loc}}(U)$. We show that then $f \in C^{k-1+\alpha}_{\mathrm{loc}}(U)$, contradicting the first half of (2).

By `MemHolderSmallOn` there are two cases, and in both v_0 is at least of class C^k on U: if $\alpha = 1$ the clause is $v_0 \in C^{k+1}(U)$, and if $\alpha < 1$ it is $v_0 \in C^k(U)$ together with the little-o condition on $\mathrm{d}^k v_0$. Put

$$w_0 = \frac{\partial v_0}{\partial z} = \tfrac{1}{2}\big(\partial_x v_0 - i\,\partial_y v_0\big),$$

which by `DZ` is the stated combination of first partial derivatives, and which is therefore of class C^{k-1} on U, with

$$D^\nu w_0 = \tfrac{1}{2}\big(D^{\nu\frown(0)} v_0 - i\, D^{\nu\frown(1)} v_0\big) \qquad (|\nu| \le k-1) \tag{4}$$

at every point of U, where $\nu \frown (j)$ is ν with the letter j appended on the right, a word of length $|\nu|+1$. Two things make (4) legitimate, and both need the set named. First, $D^{\nu\frown(j)} = D^\nu \circ \partial_j$: by `PartialWord` the operator attached to a word is the composite $\partial_{j_1}\partial_{j_2}\cdots\partial_{j_m}$ in which the last letter is applied first, so appending j on the right applies ∂_j first. Second, the step is linearity of D^ν, which is *not* available unconditionally: by the convention of `PartialWord` a derivative is taken to be 0 wherever the function fails to be real-differentiable, so these operators are additive only where both summands are differentiable. Here the instance is located. The set is the open set U, on which v_0 is of class C^k as just established; for $|\nu| \le k-1$ each of the at most k applications of ∂_0 or ∂_1 occurring on either side of (4) acts on a function of class C^k on U, so at every point of U all the derivatives involved are genuine and the convention never intervenes. Since U is open, that is all (4) asserts.

First, $w_0 = f$ on U. Both are continuous on U: w_0 because $v_0 \in C^k(U)$ with $k \ge 1$, and f by Step 1. They agree on $U - \{0\}$ by (1). The singleton $\{0\}$ has empty interior, so $U - \{0\}$ is dense in U, and two continuous functions on U agreeing on a dense subset agree on U.

Now the index shift. If $\alpha = 1$, then $v_0 \in C^{k+1}(U)$, so $w_0 \in C^k(U)$ by (4), and by `MemHolderSmallOn` at the index $k-1$ and $\alpha = 1$ — whose clause is $C^{(k-1)+1} = C^k$, using $k \ge 1$ — we get $w_0 \in C^{k-1+\alpha}_{\mathrm{loc}}(U)$. If $\alpha < 1$, then $w_0 \in C^{k-1}(U)$, and we must verify the little-o clause for w_0 at the index $k-1$. Let $K \subseteq U$ be compact and $\delta > 0$. Apply the little-o clause for v_0 at the index k with the same K and with $\delta/2^{k-1} > 0$, obtaining $\rho > 0$ such that $\|\mathrm{d}^k v_0(x) - \mathrm{d}^k v_0(y)\| \le 2^{-(k-1)}\delta|x-y|^\alpha$ for $x, y \in K$ with $|x-y| < \rho$. For such x, y and for $|\nu| = k-1$, (4) and the first inequality of (3) applied to v_0 at the order k give

$$\big|D^\nu w_0(x) - D^\nu w_0(y)\big| \le \tfrac{1}{2}\sum_{j\in\{0,1\}} \big|D^{\nu\frown(j)} v_0(x) - D^{\nu\frown(j)} v_0(y)\big| \le \big\|\mathrm{d}^k v_0(x) - \mathrm{d}^k v_0(y)\big\| \le \frac{\delta|x-y|^\alpha}{2^{k-1}},$$

and then the second inequality of (3) applied to w_0 at the order $k-1$, whose sum has 2^{k-1} terms, gives

$$\big\|\mathrm{d}^{k-1} w_0(x) - \mathrm{d}^{k-1} w_0(y)\big\| \le 2^{k-1} \cdot \frac{\delta|x-y|^\alpha}{2^{k-1}} = \delta|x-y|^\alpha.$$

This is the little-o clause for w_0, so again $w_0 \in C^{k-1+\alpha}_{\mathrm{loc}}(U)$.

Since $w_0 = f$ on U, and since both clauses of `MemHolderSmallOn` involve the function only through its restriction to U — the smoothness being smoothness on U and the derivatives being taken within U — we conclude $f \in C^{k-1+\alpha}_{\mathrm{loc}}(U)$, contradicting (2). Hence (c) holds.

Step 7: (d). Suppose $v \in W^{k+2,q}_{\mathrm{loc}}(\Omega)$. By the fourth item of (4b), $v_0 \in W^{k+2,q}_{\mathrm{loc}}(U)$. We show that then $f \in W^{k+1,q}_{\mathrm{loc}}(U)$, contradicting the second half of (2).

Let $(D_w)_w$ be a family as in `MemWLocOn` for v_0 on U at the order $k+2$: $D_{()} = v_0$, each D_w with $|w| \le k+2$ lies in $L^q_{\mathrm{loc}}(U)$, and $D_{w\frown(j)}$ is the distributional j-th partial of D_w on U whenever $|w| < k+2$. Note that `IsWeakPartialOn` is an identity between integrals which is linear in the pair of functions involved: if w_1 is the distributional j-th partial of u_1 and w_2 that of u_2, then $c_1 w_1 + c_2 w_2$ is that of $c_1 u_1 + c_2 u_2$; and it is unchanged if either function is replaced by one agreeing with it almost everywhere.

Define

$$E_{()} = f, \qquad E_w = \tfrac{1}{2}\big(D_{(0)\frown w} - i\, D_{(1)\frown w}\big) \quad \text{for } 1 \le |w| \le k+1,$$

where $(j) \frown w$ is the word w with the letter j prepended, of length $|w| + 1 \le k + 2$.
(7a) $E_{()}$ *agrees almost everywhere with* $\frac{1}{2}(D_{(0)} - iD_{(1)})$. On the open set $U - \{0\}$ the function v_0 is C^∞ by Step 1. Restricting the integration-by-parts identities to test functions for $U - \{0\}$, each $D_{(j)}$ is the distributional j-th partial of v_0 on $U - \{0\}$. Apply `WeakPartialEqFDerivAE` on $U - \{0\}$ to v_0 and the pair $j \mapsto D_{(j)}$: its hypotheses hold, since v_0 is continuous on U and hence in $L^1_{\mathrm{loc}}(U - \{0\})$; each $D_{(j)}$ lies in $L^q_{\mathrm{loc}}(U)$ with $q \ge 1$, hence in $L^1_{\mathrm{loc}}(U - \{0\})$ by Hölder's inequality on compact sets, which have finite measure; the two identities were just noted; and v_0 is differentiable at every point of $U - \{0\}$. The conclusion is that $D_{(j)} = \partial_j v_0$ almost everywhere on $U - \{0\}$ for each j, so by `DZ` and (1),

$$\tfrac{1}{2}\big(D_{(0)} - iD_{(1)}\big) = \tfrac{1}{2}\big(\partial_x v_0 - i\,\partial_y v_0\big) = \frac{\partial v_0}{\partial z} = f = E_{()}$$

almost everywhere on $U - \{0\}$, hence almost everywhere on U, the origin being a null set.
(7b) (E_w) *is a family as required.* We have $E_{()} = f$. Each E_w with $1 \le |w| \le k + 1$ is a linear combination of two members of $L^q_{\mathrm{loc}}(U)$, hence lies in $L^q_{\mathrm{loc}}(U)$; and $E_{()} = f$ is continuous on U, hence bounded on compact subsets of U and so in $L^q_{\mathrm{loc}}(U)$ as well. Let $|w| \le k$ and $j \in \{0, 1\}$; we check that $E_{w\frown(j)}$ is the distributional j-th partial of E_w on U. Since $|(j') \frown w| = |w| + 1 \le k + 1 < k + 2$ for each letter j', the family (D_w) gives that $D_{(j')\frown w\frown(j)}$ is the distributional j-th partial of $D_{(j')\frown w}$ on U; by linearity,

$$E_{w\frown(j)} = \tfrac{1}{2}\big(D_{(0)\frown w\frown(j)} - i\,D_{(1)\frown w\frown(j)}\big)$$

is the distributional j-th partial of $\frac{1}{2}(D_{(0)\frown w} - iD_{(1)\frown w})$ on U. If $|w| \ge 1$ the latter is E_w and we are done; if $w = ()$ the latter agrees with $E_{()} = f$ almost everywhere by (7a), and the integral identity is insensitive to that, so $E_{(j)}$ is the distributional j-th partial of $E_{()}$ as well.

Hence $f \in W^{k+1,q}_{\mathrm{loc}}(U)$ in the sense of `MemWLocOn`, contradicting (2). This proves (d), and with Steps 5 and 6 the theorem. □

3.7. `Thm1Regularity` (paper, Theorem 1, regularity).

Theorem 3.7. *Let $1 < p < \infty$, let $k \ge 1$ be an integer and $\alpha \in (0, 1]$ with $k + \alpha$ equal to the right-hand side of (9), as given by `KAlphaExp`. Assume the three quoted bundles (C1), (C2), (C3) of `CGradPackage`, `QCPackage` and `SmoothOffZeros`. Let $\Omega \subseteq \mathbb{C}$ be a plane domain, that is open and connected, and let $u\colon \mathbb{C} \to \mathbb{R}$ be p-harmonic on Ω in the sense of `PHarmonicOn`. Then there is $\tilde{u}\colon \mathbb{C} \to \mathbb{R}$ with*

(a) $\tilde{u}(z) = u(z)$ *for almost every z with $z \in \Omega$;*
(b) $\tilde{u} \in C^{k,\alpha}_{\mathrm{loc}}(\Omega)$ *in the sense of `MemHolderLocOn`;*
(c) $\tilde{u} \in W^{k+2,q}_{\mathrm{loc}}(\Omega)$ *in the sense of `MemWLocOn` for every q with $1 \le q < 2/(2 - \alpha)$.*

This is the assertion (8) of Theorem 1 of the paper, with the integrability range (10). It covers every $1 < p < \infty$: Section 6 opens at paper.tex:515 with $p \ne 2$, but Theorem 1 claims (8) for all p, and only its optimality clause at paper.tex:95 excludes $p = 2$.

Two points about the shape. First, the conclusion is asserted of a representative $\tilde{u}$ rather than of u, and it must be: p-harmonicity constrains u only through integrals against test functions and through membership in L^p_{loc}, all unaffected by changing u on a null set, whereas $C^{k,\alpha}_{\mathrm{loc}}$ requires C^k-smoothness and so is destroyed by changing a function at a single point. Since $W^{k+2,q}_{\mathrm{loc}}$ is itself invariant under changes on null sets, (c) holds for u as well as for $\tilde{u}$. For the same reason the quoted bundles (C2) and (C3) are read of a representative of the complex gradient: quasiregularity and $W^{1,2}_{\mathrm{loc}}$ membership see that gradient only almost everywhere, so its continuity, the discreteness of its zero set and its smoothness off that set are statements about its continuous representative, as the paper's own $f^{-1}(0) = \{z : \nabla u(z) = 0\}$ at paper.tex:156 already indicates. The proof below introduces that representative once, in Step 2, and runs on it throughout. Second, Section 3's normalisation of the factorization (15) — that χ and its inverse may be taken C^∞ off the singular point — is not *a hypothesis here: it is derived from the three bundles by `NormalizedFactorization`, which are exactly the ones the note at paper.tex:103 enumerates. Nothing beyond them is assumed: in particular the smoothness of the hodograph inverse off the*

origin, which Section 3 takes as a standing assumption at paper.tex:183 and which paper.tex:115 licenses for Theorem 2 only, is not a hypothesis anywhere on this chain — it is produced from the expansion (19) by `Cor1`*, as the paper does at paper.tex:387–388.*

Proof. One remark on well-posedness before the argument proper. The hypothesis above presents the pair (k,α) as given, whereas Theorem 1 of the paper says at paper.tex:87–90 that "the integer $k \geq 1$ and the exponent $\alpha \in (0,1]$ are determined by the equation (9)". By `KAlphaWellDefined` there is, for the given p, exactly one pair consisting of an integer $k \geq 1$ and a real number α with $0 < \alpha \leq 1$ such that $k+\alpha$ equals the right-hand side of (9); so the hypothesis on (k,α) is satisfiable for every $1 < p < \infty$ and determines the pair from p, exactly as the paper asserts, and the statement proved below is therefore not vacuous for any p. This is a well-posedness remark and not a step in the argument: nothing in the rest of this proof invokes that lemma, every step below being valid for whichever pair satisfies the hypothesis.

Write $d = d(p)$ for the exponent of `DExp` and $d\sigma$ for plane Lebesgue measure. By `KAlphaEqDExpAddOne` and the hypothesis on (k,α),

$$d+1 = (\text{right-hand side of (9)}) = k+\alpha, \qquad \text{so} \qquad d = k-1+\alpha. \tag{1}$$

In particular $d - m = k + \alpha - (m+1)$ for every m, which is the bookkeeping used below.

Step 1: the complex gradient. By `PHarmonicOn` there is $g\colon \mathbb{C} \to \mathbb{C}$ with $u \in L^p_{\mathrm{loc}}(\Omega)$ and $g \in L^p_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`, with

$$F_0 := \operatorname{Re} g \text{ the weak } \partial_0\text{-derivative of } u, \qquad F_1 := \operatorname{Im} g \text{ the weak } \partial_1\text{-derivative of } u \tag{2}$$

on Ω in the sense of `IsWeakPartialOn`, and with the weak p-harmonic equation. Here F_0 and F_1 are real-valued, and are read as complex-valued functions wherever the ambient notions require it, as `PHarmonicOn` does. Put

$$f = \tfrac{1}{2}\bar{g}.$$

The clause `complex_gradient` of `CGradPackage`, applied with p, Ω, u, g, the openness of Ω, $1 < p$, the p-harmonicity of u and the two clauses (2), gives

$$f \in W^{1,2}_{\mathrm{loc}}(\Omega), \qquad f \text{ solves (12) on } \Omega, \qquad f \text{ is } K\text{-quasiregular on } \Omega, \tag{3}$$

with $K = \max\{p-1, 1/(p-1)\}$, in the senses of `MemWLocOn`, `SolvesEq12` and `QuasiregularOn`. Note $K \geq 1$: if $p \geq 2$ then $p-1 \geq 1$, and if $p < 2$ then $0 < p-1 < 1$ so $1/(p-1) > 1$.

Step 2: the continuous representative of the gradient. First we record the relation between g and f: as $f = \frac{1}{2}\bar{g}$,

$$F_0 = \operatorname{Re} g = f + \bar{f}, \qquad F_1 = \operatorname{Im} g = i\big(f - \bar{f}\big), \tag{4}$$

the first because $f + \bar{f} = \frac{1}{2}(\bar{g} + g) = \operatorname{Re} g$ and the second because $i(f - \bar{f}) = \frac{i}{2}(\bar{g} - g) = \frac{i}{2}(-2i \operatorname{Im} g) = \operatorname{Im} g$.

The clause `qr_continuous` of `QCPackage`, applied with K, Ω, f, the openness of Ω and the quasiregularity in (3), provides $\hat{f}\colon \mathbb{C} \to \mathbb{C}$ with

$$\hat{f} = f \text{ almost everywhere on } \Omega, \qquad \hat{f} \text{ is continuous on } \Omega. \tag{4$'$}$$

It does not give continuity of f itself and cannot: by `QuasiregularOn` quasiregularity constrains f only through $W^{1,2}_{\mathrm{loc}}$ membership, which `MemWLocOn` expresses by integral identities and indicator L^2-conditions, and through an almost-everywhere inequality between the weak Wirtinger derivatives of f, which by `IsWeakWirtingerOn` are themselves determined by the class of f up to null sets; none of this is affected by altering f on a null set. Put

$$\hat{F}_0 = \hat{f} + \overline{\hat{f}}, \qquad \hat{F}_1 = i\big(\hat{f} - \overline{\hat{f}}\big),$$

which are real-valued and continuous on Ω by the second clause of (4′), and which by (4) and the first clause of (4′) satisfy $\hat{F}_j = F_j$ almost everywhere on Ω. By `IsWeakPartialOn` the assertion that a function is the weak ∂_j-derivative of u on Ω is the identity $\int u\,\partial_j\varphi = -\int F_j\varphi$ over test functions for Ω, and the right-hand integral is unchanged when F_j is altered on a null set; so by (2),

$$\hat{F}_0 \text{ and } \hat{F}_1 \text{ are the weak } \partial_0\text{- and } \partial_1\text{-derivatives of } u \text{ on } \Omega. \tag{4$''$}$$

Finally, the clause `complex_gradient_rep` of `CGradPackage`, applied with p, Ω, u, g and $\hat{f}$, the openness of Ω, $1 < p$, the p-harmonicity of u, the two clauses (2), and the two clauses of $(4')$ — note that $\hat{f} = f = \frac{1}{2}\bar{g}$ almost everywhere on Ω is the first of them — gives

$$(4''') \qquad \hat{f} \text{ is differentiable at almost every point of } \Omega, \qquad \hat{f} \text{ solves (12) on } \Omega$$

in the sense of `SolvesEq12`. Of these two clauses it is the first, the almost-everywhere differentiability of $\hat{f}$, that is genuinely new: the second could equally be read off from the second clause of (3), because by `SolvesEq12` and `IsWeakWirtingerOn` solving (12) is a property of the almost-everywhere class of its argument — the local integrability and integration-by-parts clauses see the argument only through integrals, and its pointwise values enter (12) only inside an "almost every z" quantifier — and $\hat{f} = f$ almost everywhere on Ω by $(4')$. Nothing in (3), by contrast, gives differentiability of any representative at any point, and that is what the smoothness bundle (C3) consumes.

Step 3: u is locally integrable. Let $\kappa \subseteq \Omega$ be compact. By Step 1 the indicator of κ times u lies in L^p, and κ has finite measure, so by Hölder's inequality with exponents $p \geq 1$ and $p/(p-1)$ it lies in L^1. Hence $u \in L^1_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`.

Step 4: the representative. The hypotheses of `WeakGradContinuousRep` hold for Ω, u and $\hat{F}_0, \hat{F}_1$: local integrability of u is Step 3, continuity of $\hat{F}_0, \hat{F}_1$ on Ω is Step 2, and the weak-derivative clauses are $(4'')$. It therefore provides $\tilde{u}\colon \mathbb{C} \to \mathbb{R}$ with

$$(5) \qquad \tilde{u} = u \text{ a.e. on } \Omega, \qquad \tilde{u} \in C^1(\Omega), \qquad \partial_j \tilde{u} = \hat{F}_j \text{ on } \Omega \ (j = 0, 1),$$

where ∂_0, ∂_1 are the partial derivatives of `PartialWord` along [0], [1]. The first clause of (5) is assertion (a). Combining the third with the definition of $\hat{F}_j$,

$$(6) \qquad \partial_0 \tilde{u} = \hat{f} + \overline{\hat{f}}, \qquad \partial_1 \tilde{u} = i\big(\hat{f} - \overline{\hat{f}}\big) \qquad \text{on } \Omega.$$

Throughout the rest of the proof we write $w = \tilde{u}$, regarded as a complex-valued function with real values where the ambient notions require it.

Step 5: the degenerate case $\hat{f} \equiv 0$. Suppose $\hat{f}(z) = 0$ for every $z \in \Omega$. Then $\hat{F}_0 = \hat{F}_1 = 0$ on Ω by their definitions, and by (5) both first partial derivatives of w vanish at every point of Ω. An $\mathbb{R}$-linear map $\mathbb{C} \to \mathbb{C}$ is determined by its values on 1 and i, so $\mathrm{d}w = 0$ on Ω; and w is C^1 on Ω by (5). A C^1 function with vanishing derivative on a connected open set is constant, so $w \equiv c$ on Ω for some real c.

Then (b) and (c) hold. For (b): w agrees with a constant on the open set Ω, hence is C^∞ there, so in particular $\mathrm{ContDiff}^k$ on Ω; and the iterated derivative of order $k \geq 1$ of a locally constant function vanishes identically on Ω, so for any compact $\kappa \subseteq \Omega$ the Hölder condition of `MemHolderLocOn` holds with constant 0. For (c): let $1 \leq q < 2/(2-\alpha)$, and take the family D with $D[\,] = w$ and $D(v) = 0$ for every nonempty word v. Each $D(v)$ lies in $L^q_{\mathrm{loc}}(\Omega)$, being either a constant or 0 and compact sets having finite measure; and for every word v with $|v| < k+2$ and every j, the identity of `IsWeakPartialOn` holds: for $v = [\,]$ it reads $\int w\, \partial_j \varphi = -\int 0 \cdot \varphi = 0$, and $\int w\, \partial_j \varphi = c \int \partial_j \varphi = 0$ because the integral of a partial derivative of a test function vanishes, as in Step 2 of `WeakPartialAcrossPoint`; and for v nonempty both sides are 0. So $w \in W^{k+2,q}_{\mathrm{loc}}(\Omega)$ in the sense of `MemWLocOn`. This settles the theorem in this case, and we assume from now on that $\hat{f}(z^*) \neq 0$ for some $z^* \in \Omega$.

Step 6: the singular set. Put

$$Z = \{z \in \Omega : \hat{f}(z) = 0\}.$$

The clause `qr_discrete_zeros` of `QCPackage`, applied with K, Ω, the pair $f, \hat{f}$, the openness and connectedness of Ω, the quasiregularity in (3), the two clauses of $(4')$ and the point z^*, gives for each $z_0 \in \Omega$ an $r > 0$ with $\hat{f}(z) \neq 0$ for every $z \in \Omega$ with $z \neq z_0$ and $\mathrm{dist}(z, z_0) < r$. Read at $z_0 \in Z$ this says

$$(7) \qquad \forall z_0 \in Z,\ \exists r > 0,\ \forall z \in \Omega,\ z \neq z_0,\ \mathrm{dist}(z, z_0) < r \ \Rightarrow\ z \notin Z.$$

Moreover Z meets every compact $\kappa \subseteq \Omega$ in a finite set. Indeed, Z is closed in Ω, because $\hat{f}$ is continuous on Ω by $(4')$; if $Z \cap \kappa$ were infinite it would have an accumulation point $\zeta \in \kappa \subseteq \Omega$,

and $\zeta \in Z$ by closedness, contradicting (7) at ζ. So

$$Z \cap \kappa \text{ is finite for every compact } \kappa \subseteq \Omega. \tag{8}$$

Step 7: smoothness off the singular set. The clause `smooth_off_zeros` of `SmoothOffZeros`, applied with p, Ω, the pair $f, \hat{f}$, the openness of Ω, $1 < p$, the first clause of (3), the two clauses of (4′) and the two clauses of (4‴), gives that $\hat{f}$ is C^∞ on $\Omega - \{z : \hat{f}(z) = 0\} = \Omega - Z$. By (6) the two first partial derivatives of w are the $\mathbb{R}$-linear combinations $\hat{f} + \overline{\hat{f}}$ and $i(\hat{f} - \overline{\hat{f}})$ of $\hat{f}$ and $\overline{\hat{f}}$, hence are C^∞ on $\Omega - Z$; and w is C^1 on Ω by (5). A C^1 function on an open set whose first partial derivatives are C^∞ there is itself C^∞ there, so

$$w \text{ is } C^\infty \text{ on } \Omega - Z. \tag{9}$$

Step 8: the estimates (51), (52) at each singular point. Fix $\zeta \in Z$, so $\zeta \in \Omega$ and $\hat{f}(\zeta) = 0$. The hypotheses of `NormalizedFactorization` hold with p, K, the bundles (C2), (C3), the domain Ω, the pair $f, \hat{f}$ and the point ζ: the openness and connectedness of Ω are hypotheses of the present theorem, the quasiregularity of f is the third clause of (3), its hypotheses (i) and (ii) on $\hat{f}$ are the two clauses of (4′) and its (iii) and (iv) are the two clauses of (4‴), the point z^* has $\hat{f}(z^*) \neq 0$ by the case assumption of Step 5, and $\hat{f}(\zeta) = 0$. It provides $m \geq 1$, $K' \geq 1$, an open V' with $\zeta \in V' \subseteq \Omega$, and χ, H with: χ K'-quasiconformal on V', $\chi(\zeta) = 0$, $\hat{f} = \chi^m$ on V', χ of class C^∞ on $V' - \{\zeta\}$, $H(\chi(z)) = z$ on V', H continuous on $\chi(V')$, $H(0) = \zeta$, and $H(\xi) \in V'$ with $\chi(H(\xi)) = \xi$ for $\xi \in \chi(V')$. No smoothness of H is provided and none is needed: `Thm1DerivBounds` asks only for its continuity.

The hypotheses of `Thm1DerivBounds` now hold with m, p, K', the bundle (C2), the point ζ, the open set V', the function w in place of u, and $\hat{f}, \chi, H$ in place of its f, χ, H: (H1) is the restriction to V' of the second clause of (4‴) — an almost-everywhere statement restricts to a subset — together with $\hat{f}(\zeta) = 0$; (H2), (H3), (H4) are the data just produced; and (H5) is (6) restricted to $V' \subseteq \Omega$. It therefore provides δ_ζ with $0 < \delta_\zeta \leq 1$ and constants B_m^ζ with

$$\sum_{|v|=m} \left|D^v \hat{f}(z)\right| \leq B_m^\zeta |z - \zeta|^{d-m}, \tag{51$_\zeta$}$$

$$\sum_{|v|=m+1} \left|D^v w(z)\right| \leq B_m^\zeta |z - \zeta|^{d-m}, \tag{52$_\zeta$}$$

for $0 < |z - \zeta| < \delta_\zeta$ and every $m \geq 0$. Using (1), (52$_\zeta$) reads in terms of the order $j = m + 1$:

$$\sum_{|v|=j} \left|D^v w(z)\right| \leq B_{j-1}^\zeta |z - \zeta|^{k+\alpha-j} \qquad (j \geq 1,\ 0 < |z - \zeta| < \delta_\zeta). \tag{10}$$

Each $B_{j-1}^\zeta \geq 0$, the left side being a sum of moduli and the power positive for some admissible z. Shrinking δ_ζ we may and do also assume

$$\{z : |z - \zeta| < \delta_\zeta\} \subseteq \Omega \quad \text{and} \quad \{z : |z - \zeta| < \delta_\zeta\} \cap Z = \{\zeta\}, \tag{11}$$

the first by openness of Ω and the second by (7).

Step 9: w is of class C^k on Ω. The hypotheses of `Thm1SmoothAcrossZeros` hold with Ω, Z, k, α and w: $Z \subseteq \Omega$; the hypothesis that Z meets every compact subset of Ω in a finite set is (8); (H1) is the second clause of (5); (H2) is (9); and (H3) is (10), with $\delta = \delta_\zeta$ and $B_j = B_{j-1}^\zeta$. Hence

$$w \text{ is of class } C^k \text{ on } \Omega. \tag{12}$$

This is the first clause of `MemHolderLocOn`.

Step 10: from the derivatives of order k to the Hölder condition. Write $T(x) =$ the iterated derivative of order k of w within Ω at x, the object appearing in `MemHolderLocOn`. We first record a comparison with the D^v.

(10a) The comparison. For $x, y \in \Omega$,

$$\big\|T(x) - T(y)\big\| \le \sum_{|v|=k} \big|D^v w(x) - D^v w(y)\big|. \tag{13}$$

Indeed $T(x) - T(y)$ is a continuous $\mathbb{R}$-multilinear map of k arguments on $\mathbb{C}$, and its norm is the supremum of its values on k-tuples of unit vectors. Writing a unit vector as $a \cdot 1 + b \cdot i$ with $a^2 + b^2 = 1$, so $|a|, |b| \le 1$, and expanding each of the k arguments by multilinearity, the value on such a tuple is a sum over the 2^k words v of length k of a coefficient of modulus at most 1 times the value of $T(x) - T(y)$ on the corresponding tuple of basis vectors. That value is the iterated directional derivative of w along those basis vectors, taken at x and at y, which by `PartialWord` is $D^v w(x) - D^v w(y)$ for the word v of the corresponding indices — the order in which the indices are read is a convention, and immaterial here because the sum in (13) is over all words of length k. Taking absolute values and summing gives (13).

(10b) The three cases. Let $\kappa \subseteq \Omega$ be compact; we must produce $C \ge 0$ with $\|T(x) - T(y)\| \le C|x-y|^\alpha$ for all $x, y \in \kappa$. Since Ω is open and κ compact, fix $r_0 > 0$ with $\kappa_{3r_0} := \{x : \operatorname{dist}(x, \kappa) \le 3r_0\} \subseteq \Omega$; this set is compact. By (8) the set $Z \cap \kappa_{3r_0}$ is finite. Choose $r \in (0, r_0]$ with

$$r \le 1, \qquad 4r < \delta_\zeta \ \text{ for every } \zeta \in Z \cap \kappa_{3r_0}, \qquad 4r < \min|\zeta - \zeta'|$$

over distinct $\zeta, \zeta' \in Z \cap \kappa_{3r_0}$ (the last condition omitted if there are fewer than two such points). Let $x, y \in \kappa$.

Case 1: $|x - y| \ge r$. By (12) the map T is continuous on Ω, hence bounded on the compact κ, say by M_T. Then

$$\|T(x) - T(y)\| \le 2M_T = 2M_T r^{-\alpha} r^\alpha \le 2M_T r^{-\alpha} |x - y|^\alpha.$$

Case 2: $|x - y| < r$ *and* $|x - \zeta| \le 2r$ *for some* $\zeta \in Z$. Such a ζ lies in κ_{3r_0}, since $\operatorname{dist}(\zeta, \kappa) \le 2r \le 3r_0$. Also $|y - \zeta| \le |y - x| + |x - \zeta| < 3r$. Fix a word v of length k and put

$$U = \{\eta : |\eta| < 4r\}, \qquad F(\eta) = D^v w(\eta + \zeta) \quad (\eta \in U).$$

We verify the hypotheses of `Lemma2` for F on U, in dimension $n = 2$. The plane $\mathbb{C}$ with its modulus is carried onto $\mathbb{R}^2$ with its Euclidean norm by the $\mathbb{R}$-linear isometry $\eta \mapsto (\operatorname{Re}\eta, \operatorname{Im}\eta)$, under which convexity, openness, membership of the origin, continuity, membership of C^1, norms of vectors and operator norms of Fréchet derivatives all correspond; so it is equivalent, and we argue in $\mathbb{C}$.

U is open and convex and contains 0. By (11) and $4r < \delta_\zeta$ the translate $\zeta + U$ is contained in Ω and meets Z only at ζ. Hence F is continuous on U, by (12); and F is C^1 on $U - \{0\}$, by (9). For hypothesis (i), (10) at $j = k$ gives for $0 < |\eta| < 4r < \delta_\zeta$

$$|F(\eta)| \le \sum_{|v'|=k} \big|D^{v'} w(\eta + \zeta)\big| \le B^\zeta_{k-1} |\eta|^{\,k+\alpha-k} = B^\zeta_{k-1} |\eta|^\alpha,$$

and at $\eta = 0$ the same bound holds because the right-hand side tends to 0 as $\eta \to 0$ while F is continuous at 0, forcing $F(0) = 0$. For hypothesis (ii), let $0 < |\eta| < 4r$. As in Step 9, the operator norm of the Fréchet derivative of F at η is at most the sum of the moduli of its two first partial derivatives, which are $D^{[0]+\!+v} w$ and $D^{[1]+\!+v} w$ at $\eta + \zeta$ — prefixing a letter applies that partial derivative last, by the recursion of `PartialWord` — and both words have length $k + 1$, so by (10) at $j = k + 1$

$$\|\mathrm{d}F(\eta)\| \le \sum_{|v'|=k+1} \big|D^{v'} w(\eta + \zeta)\big| \le B^\zeta_k |\eta|^{\,k+\alpha-(k+1)} = B^\zeta_k |\eta|^{\alpha-1}.$$

So `Lemma2` applies with $M = \max\{B^\zeta_{k-1}, B^\zeta_k\} \ge 0$ and this $\alpha \in (0, 1]$, and yields

$$\big|D^v w(x) - D^v w(y)\big| = \big|F(x - \zeta) - F(y - \zeta)\big| \le 5M|x - y|^\alpha,$$

the two arguments lying in U because $|x - \zeta| \le 2r < 4r$ and $|y - \zeta| < 3r < 4r$. Summing over the 2^k words v and using (13),

$$\|T(x) - T(y)\| \le 2^k \cdot 5M_* |x - y|^\alpha, \qquad M_* = \max_{\zeta \in Z \cap \kappa_{3r_0}} \max\{B^\zeta_{k-1}, B^\zeta_k\},$$

a maximum over a finite set by (8).

Case 3: $|x-y|<r$ and $|x-\zeta|>2r$ for every $\zeta\in Z$. Then the ball $\{\xi : |\xi-x|<r\}$ is contained in $\Omega-Z$: it lies in $\kappa_{3r_0}\subseteq\Omega$, and any ξ in it has $|\xi-\zeta|\geq|x-\zeta|-|x-\xi|>2r-r=r>0$ for every $\zeta\in Z$. Put

$$S=\big\{\xi\in\kappa_{3r_0} : \operatorname{dist}(\xi,Z)\geq r\big\},$$

a compact subset of $\Omega-Z$, and let L be the supremum over S and over the 2^k words v of length k of $\|\mathrm{d}\big(D^v w\big)\|$, finite because each $D^v w$ is C^1 on the open $\Omega-Z\supseteq S$ by (9), so its derivative is continuous on the compact S. Every point of the segment from x to y lies in $\{\xi : |\xi-x|<r\}$, hence in S; the segment is convex, so the mean value inequality gives, for each word v of length k,

$$\big|D^v w(x)-D^v w(y)\big|\leq L|x-y|=L|x-y|^{\alpha}|x-y|^{1-\alpha}\leq L\,r^{1-\alpha}|x-y|^{\alpha},$$

using $|x-y|<r$ and $1-\alpha\geq 0$. Summing over v and using (13), $\|T(x)-T(y)\|\leq 2^k L r^{1-\alpha}|x-y|^{\alpha}$.

Taking for C the maximum of the three constants $2M_T r^{-\alpha}$, $2^k\cdot 5M_*$ and $2^k L r^{1-\alpha}$ — and noting that the three cases exhaust the possibilities — gives the Hölder condition on κ. With (12) this proves (b).

Step 11: the Sobolev clause. Let q satisfy $1\leq q<2/(2-\alpha)$. Define a family of functions indexed by words by

$$D(v)=D^{\,\mathrm{rev}(v)}w,$$

where $\mathrm{rev}(v)$ is v read backwards. This convention is chosen so that the coherence required by `MemWLocOn` is a tautology at the level of functions: $D([\,])=w$, and for every v and j, by the recursion of `PartialWord`,

$$D\big(v\mathbin{+\!\!+}[j]\big)=D^{\,j::\mathrm{rev}(v)}w=\partial_j\big(D^{\,\mathrm{rev}(v)}w\big)=\partial_j\big(D(v)\big), \tag{14}$$

so no commutation of mixed partial derivatives is needed anywhere.

(11a) Local L^q bounds. Let v be a word with $|v|\leq k+2$ and let $\kappa\subseteq\Omega$ be compact. If $|v|\leq k$ then $D(v)$ is continuous on Ω by (12), hence bounded on κ, hence in L^q there, κ having finite measure. Suppose $|v|=j\in\{k+1,k+2\}$. By (8) the set $Z\cap\kappa$ is finite; enlarge κ to the compact $\kappa'=\{x : \operatorname{dist}(x,\kappa)\leq\varrho\}\subseteq\Omega$ for some $\varrho>0$, and for each $\zeta\in Z\cap\kappa'$ pick $\varrho_\zeta\in(0,\min\{\varrho,\delta_\zeta,1\})$. Off the union of the balls $\{|z-\zeta|<\varrho_\zeta\}$ the function $D(v)$ is continuous on a compact subset of $\Omega-Z$ by (9), hence bounded there, so contributes a finite amount to $\int_\kappa\|D(v)\|^q$. On each such ball, (10) gives $\|D(v)(z)\|\leq B^\zeta_{j-1}|z-\zeta|^{\,k+\alpha-j}$, and since $|z-\zeta|<1$ and $k+\alpha-j\geq\alpha-2$ we get $|z-\zeta|^{\,k+\alpha-j}\leq|z-\zeta|^{\alpha-2}$; hence

$$\int_{\{|z-\zeta|<\varrho_\zeta\}}\|D(v)\|^q\,d\sigma\leq\big(B^\zeta_{j-1}\big)^q\int_{\{|z-\zeta|<\varrho_\zeta\}}|z-\zeta|^{(\alpha-2)q}\,d\sigma<\infty$$

by `PowerIntegrableNearZero`, applied with this α, the radius ϱ_ζ and this q, after the translation $z\mapsto z-\zeta$, which preserves Lebesgue measure. Summing the finitely many contributions, $D(v)\in L^q(\kappa)$. As κ was arbitrary, $D(v)\in L^q_{\mathrm{loc}}(\Omega)$ in the sense of `MemLpLocOn`. (The case $q=1$ of this computation will be used in (11b).)

(11b) The weak-derivative clauses. Let v be a word with $|v|<k+2$ and let $\iota\in\{0,1\}$; we apply `WeakPartialAcrossPoint` with Ω, the exceptional set $S=Z$, the index ι, the functions $G=D(v)$ and $V=D(v\mathbin{+\!\!+}[\iota])$, and the exponent $\beta=1-\alpha$, which lies in $[0,1)$ because $0<\alpha\leq 1$. Its hypotheses hold: $Z\subseteq\Omega$; local finiteness is (8); G is C^1 on $\Omega-Z$ by (9); $\partial_\iota G=V$ on $\Omega-Z$ — indeed everywhere — by (14); G and V lie in $L^1_{\mathrm{loc}}(\Omega)$ by (11a) with $q=1$, since $|v|\leq k+1$ and $|v\mathbin{+\!\!+}[\iota]|\leq k+2$; and for the growth bound, let $\zeta\in Z$ and $0<|z-\zeta|<\min\{\delta_\zeta,1\}$, so that by (10) at the order $|v|$ — or trivially if $|v|=0$, $G=w$ being continuous hence bounded near ζ —

$$\|G(z)\|\leq B^\zeta_{|v|-1}\,|z-\zeta|^{\,k+\alpha-|v|}\leq B^\zeta_{|v|-1}\,|z-\zeta|^{\alpha-1}=B^\zeta_{|v|-1}\,|z-\zeta|^{-\beta},$$

using $k+\alpha-|v|\geq k+\alpha-(k+1)=\alpha-1$ and $|z-\zeta|<1$. Hence `WeakPartialAcrossPoint` gives that $D(v\mathbin{+\!\!+}[\iota])$ is the distributional ι-th partial derivative of $D(v)$ on Ω, in the sense of `IsWeakPartialOn`.

By (11a), (11b) and $D([\,]) = w$, the family D witnesses $w \in W^{k+2,q}_{\mathrm{loc}}(\Omega)$ in the sense of `MemWLocOn`. As q was arbitrary in the range $1 \le q < 2/(2-\alpha)$, this is (c), and with (a) from Step 4 and (b) from Step 10 the proof is complete. □

3.8. `Thm2Converse` (paper, Theorem 2, converse).

Theorem 3.8. *Let $n \ge 1$ be an integer and let $1 < p < \infty$. Write $\mathbb{B} = \{\xi : |\xi| < 1\}$ and $d\sigma$ for plane Lebesgue measure, let $\lambda_k = \lambda_{|k|}(n,p)$ be the exponent of `LambdaCoef`, $\epsilon_k = \epsilon_k(n,p)$ the coefficient of `EpsCoef`, and T_k the k-th summand of (19) in the sense of `Thm2Term`. Then there is a constant $C = C(n,p) > 0$ with the following property.*

Let $A\colon \mathbb{Z} \to \mathbb{C}$ satisfy

$$\sum_{m=0}^{\infty} (n+m)\,|A_{n+m}|^2 < \infty, \qquad \text{that is} \qquad \sum_{k=n}^{\infty} k|A_k|^2 < \infty.$$

Then there is a function $H\colon \mathbb{C} \to \mathbb{C}$ with

(a) *$H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemWLocOn`, H continuous on $\mathbb{B}$, and H of class C^∞ on $\mathbb{B} - \{0\}$;*

(b) *H a solution of equation (18) on $\mathbb{B}$ in the sense of `SolvesEq18`;*

(c) *the series (19) converging to H in $W^{1,2}(\mathbb{B})$:*

$$\iint_{\mathbb{B}} \Big|H - \sum_{k=n}^{n+N} T_k\Big|^2 d\sigma + \iint_{\mathbb{B}} \Big|H_\xi - \Big(\sum_{k=n}^{n+N} T_k\Big)_\xi\Big|^2 d\sigma + \iint_{\mathbb{B}} \Big|H_{\bar\xi} - \Big(\sum_{k=n}^{n+N} T_k\Big)_{\bar\xi}\Big|^2 d\sigma \longrightarrow 0$$

as $N \to \infty$, where H_ξ, $H_{\bar\xi}$ are the Wirtinger derivatives of `DZ` and `DZbar`;

(d) *the energy bound*

$$\iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big)d\sigma \le C \sum_{m=0}^{\infty} (n+1+m)\,|A_{n+1+m}|^2 = C \sum_{k=n+1}^{\infty} k|A_k|^2.$$

This is the converse half of Theorem 2 of the paper (paper.tex:207–212). Note the paper's own asymmetry, reproduced here: the hypothesis sums from $k = n$ while the bound (d) sums from $k = n+1$. It is correct, because $\lambda_n = 0$ makes the $k = n$ summand the constant A_n, which contributes nothing to the energy. As in the forward half there is no quasiregularity hypothesis and no restriction $p \ne 2$.

Proof. Step 0: the constant. Put

$$K = \max\Big\{p - 1,\ \frac{1}{p-1}\Big\}.$$

Since $p > 1$ we have $p - 1 > 0$; if $p \ge 2$ then $K \ge p - 1 \ge 1$, and if $p < 2$ then $0 < p - 1 < 1$, so $K \ge 1/(p-1) > 1$. Thus $K \ge 1$ and `EnergyIdentity` applies with this n, p and K; let $C > 0$ be the constant it provides. It depends only on n, p and K, and K on p alone, so $C = C(n,p)$. Now fix A with the stated summability.

Step 1: normalising the coefficient family. Define $\widetilde{A}\colon \mathbb{Z} \to \mathbb{C}$ by

$$\widetilde{A}_j = \begin{cases} A_j & \text{if } j \ge n, \\ \epsilon_{-j}\overline{A_{-j}} & \text{if } j \le -n, \\ 0 & \text{if } |j| < n. \end{cases}$$

The three cases are mutually exclusive and exhaust $\mathbb{Z}$: $j \ge n$ and $j \le -n$ cannot both hold, since $n \ge 1$. So $\widetilde{A}$ is well defined. Three properties follow at once.

First, $\widetilde{A}_j = A_j$ for every $j \ge n$, by the first case. Second, for every integer $k \ge n$,

$$\widetilde{A}_{-k} = \epsilon_k \overline{A_k} = \epsilon_k \overline{\widetilde{A}_k}, \tag{1}$$

by the second case applied at $j = -k \le -n$ and then the first case at k. Third, for every $k \ge n$,

$$T_k = \text{the } k\text{-th summand of (19) for } \widetilde{A} = \text{the } k\text{-th summand of (19) for } A, \tag{2}$$

because by Thm2Term the k-th summand depends on the coefficient family only through its value at the single index k, namely

$$\big(A_k\xi^k + \epsilon_k\overline{A_k}\bar{\xi}^k\big)|\xi|^{\lambda_k+n-k}\xi^{-n},$$

and $\widetilde{A}_k = A_k$ there. In particular the partial sums $\sum_{k=n}^{n+N} T_k$ formed from $\widetilde{A}$ and from A are the same function, so clauses (c) below may be read for either family.

Finally, the summability hypothesis of the theorem reads, for $\widetilde{A}$,

$$\sum_{m=0}^{\infty}(n+m)|\widetilde{A}_{n+m}|^2 = \sum_{m=0}^{\infty}(n+m)|A_{n+m}|^2 < \infty, \tag{3}$$

since $n+m \geq n$ for every $m \geq 0$.

Step 2: the sum of the series. By (1) and (3) the hypotheses (C) and (S) of Thm2SeriesW12Convergence hold for $\widetilde{A}$. That node provides H with:

- H continuous on $\mathbb{B}$ and C^∞ on $\mathbb{B} - \{0\}$ (its (a), (b));
- $H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ (its (c));
- $|H_\xi|^2 + |H_{\bar\xi}|^2$ integrable on $\mathbb{B}$ (its (d));
- $a_j(r) = \widetilde{A}_j\, r^{\lambda_j}$ for $|j| \geq n$ and $r \in (0,1)$ (its (e));
- the $W^{1,2}(\mathbb{B})$ convergence of $\sum_{k=n}^{n+N} T_k$ to H (its (g)).

The first three bullets give assertion (a), and the last, read through (2), gives assertion (c).

Step 3: proof of (b). Write $S_N = \sum_{k=n}^{n+N} T_k$.

We shall use twice the following passage between the two readings of the Wirtinger operators. *Let $\Omega \subseteq \mathbb{C}$ be open, let $u\colon \mathbb{C} \to \mathbb{C}$ be differentiable at almost every point of Ω, and let a, b be weak Wirtinger derivatives of u on Ω in the sense of IsWeakWirtingerOn. Then $a = u_z$ and $b = u_{\bar z}$ almost everywhere on Ω, with $u_z, u_{\bar z}$ the pointwise operators of DZ and DZbar.* Indeed, by IsWeakWirtingerOn there are u_x, u_y with $u, u_x, u_y \in L^1_{\mathrm{loc}}(\Omega)$ in the sense of MemLpLocOn, with u_x, u_y the distributional ∂_0- and ∂_1-derivatives of u on Ω in the sense of IsWeakPartialOn, and with $a = \frac{1}{2}(u_x - iu_y)$, $b = \frac{1}{2}(u_x + iu_y)$ at every point; the first two are hypotheses (H1) and (H2) of WeakPartialEqFDerivAE for Ω, u and the pair $D_0 = u_x$, $D_1 = u_y$, and the almost-everywhere differentiability is its (H3), so $u_x = \partial_0 u$ and $u_y = \partial_1 u$ almost everywhere on Ω, $\partial_j = D^{[j]}$ being the classical partial derivative of PartialWord, that is $\partial_0 u = \mathrm{d}u[1]$, $\partial_1 u = \mathrm{d}u[i]$; and $u_z, u_{\bar z}$ are by DZ and DZbar the same two combinations $\frac{1}{2}(\mathrm{d}u[1] \mp i\,\mathrm{d}u[i])$. Call this fact (W).

By part (a) of Thm2TermSolvesEq18, each T_k with $k \geq n$ is C^∞ at every $\xi \neq 0$, hence differentiable at every point of $\mathbb{B}$ except possibly 0, which is a null set. By its part (c) each T_k solves (18) on $\mathbb{B}$ in the sense of SolvesEq18: there are weak Wirtinger derivatives a_k, b_k of T_k on $\mathbb{B}$ and a null set Z_k' off which the identity holds for a_k, b_k at the points of $\mathbb{B} - \{0\}$. By (W) applied to $u = T_k$ on $\Omega = \mathbb{B}$, which is open, there is a null set Z_k'' with $a_k = (T_k)_\xi$ and $b_k = (T_k)_{\bar\xi}$ on $\mathbb{B} - Z_k''$. So, with $Z_k = Z_k' \cup Z_k''$, a null set, for $\xi \notin Z_k$ with $\xi \in \mathbb{B}$, $\xi \neq 0$,

$$(T_k)_{\bar\xi}(\xi) = c\Big[\frac{\xi}{\bar\xi}(T_k)_\xi(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{(T_k)_\xi(\xi)}\Big], \qquad c = \tfrac{1}{2} - \tfrac{1}{p}. \tag{4}$$

Let $Z = \bigcup_{k\geq n} Z_k$, a countable union of null sets, hence null. Fix $\xi \in \mathbb{B}$ with $\xi \neq 0$ and $\xi \notin Z$. For each N, all the summands of S_N are differentiable at ξ, so by linearity of the Fréchet derivative and DZ, DZbar, $(S_N)_\xi(\xi) = \sum_{k=n}^{n+N}(T_k)_\xi(\xi)$ and likewise for $\partial/\partial\bar\xi$. The map

$$w \;\longmapsto\; c\Big[\frac{\xi}{\bar\xi}w + \frac{\bar\xi^{\,n}}{\xi^n}\overline{w}\Big]$$

is additive, because $w \mapsto \overline{w}$ is, and c is real. Summing (4) over $k = n, \dots, n+N$ therefore gives

$$(S_N)_{\bar\xi}(\xi) = c\Big[\frac{\xi}{\bar\xi}(S_N)_\xi(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{(S_N)_\xi(\xi)}\Big] \qquad \text{for every } N \geq 0. \tag{5}$$

By assertion (c) of Step 2, $(S_N)_\xi \to H_\xi$ and $(S_N)_{\bar\xi} \to H_{\bar\xi}$ in $L^2(\mathbb{B})$. An $L^2(\mathbb{B})$-convergent sequence has a subsequence converging almost everywhere, so there are a null set Z' and a

subsequence $N_1 < N_2 < \cdots$ with

$$(S_{N_i})_\xi(\xi) \to H_\xi(\xi), \qquad (S_{N_i})_{\bar\xi}(\xi) \to H_{\bar\xi}(\xi) \qquad \text{for every } \xi \notin Z'.$$

The map displayed above is continuous in w, being the sum of a $\mathbb{C}$-linear and a conjugate-linear term with fixed coefficients. Hence, letting $i \to \infty$ in (5) along the subsequence, for every $\xi \in \mathbb{B}$ with $\xi \neq 0$ and $\xi \notin Z \cup Z'$,

$$H_{\bar\xi}(\xi) = c\Big[\frac{\xi}{\bar\xi} H_\xi(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{H_\xi(\xi)}\Big], \tag{5′}$$

an identity between the pointwise Wirtinger derivatives of H, valid off the null set $Z \cup Z'$.

It remains to read (5′) for the weak Wirtinger derivatives, which is what `SolvesEq18` asks for. By Step 2, $H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$, so by `MemWLocOn` there is a family (H_w) indexed by the words in $\{0,1\}$ with $H_{()} = H$, with $H_w \in L^2_{\mathrm{loc}}(\mathbb{B})$ in the sense of `MemLpLocOn` for $|w| \le 1$, and with $H_{(j)}$ the distributional ∂_j-derivative of H on $\mathbb{B}$ in the sense of `IsWeakPartialOn`. Put $H_x = H_{(0)}$, $H_y = H_{(1)}$ and $a = \frac12(H_x - iH_y)$, $b = \frac12(H_x + iH_y)$. Each of H, H_x, H_y lies in $L^1_{\mathrm{loc}}(\mathbb{B})$: for compact $\kappa \subseteq \mathbb{B}$ the Cauchy–Schwarz inequality gives $\int_\kappa |g|\,d\sigma \le \sigma(\kappa)^{1/2}\big(\int_\kappa |g|^2 d\sigma\big)^{1/2} < \infty$, $\sigma(\kappa)$ being finite because κ is bounded. So the three clauses of `IsWeakWirtingerOn` hold and a, b are weak Wirtinger derivatives of H on $\mathbb{B}$.

By Step 2, H is C^∞ on $\mathbb{B} - \{0\}$, hence differentiable at every point of $\mathbb{B}$ except possibly 0; so (W) applies to $u = H$ on the open set $\Omega = \mathbb{B}$ and gives a null set Z'' with $a = H_\xi$ and $b = H_{\bar\xi}$ on $\mathbb{B} - Z''$. Combining with (5′), for every $\xi \in \mathbb{B}$ with $\xi \neq 0$ and $\xi \notin Z \cup Z' \cup Z''$,

$$b(\xi) = c\Big[\frac{\xi}{\bar\xi} a(\xi) + \frac{\bar\xi^{\,n}}{\xi^n}\overline{a(\xi)}\Big].$$

The set $Z \cup Z' \cup Z''$ is null, so with the pair a, b this says precisely that H solves (18) on $\mathbb{B}$ in the sense of `SolvesEq18`, which is (b).

Step 4: proof of (d). By `Eq18ImpliesQuasiregular`, applied with $\Omega = \mathbb{B}$, using $H \in W^{1,2}_{\mathrm{loc}}(\mathbb{B})$ from Step 2 and (b) from Step 3,

$$H \text{ is } K\text{-quasiregular on } \mathbb{B}. \tag{6}$$

Now apply `EnergyIdentity` with this H and the family $\widetilde{A}$. Its hypotheses hold: (H0) is (b); (H1) is the continuity and C^∞-smoothness from Step 2; its quasiregularity hypothesis is (6); its integrability hypothesis is the third bullet of Step 2; its hypothesis (H4) is the fourth bullet of Step 2; and its hypothesis (H5), namely $\widetilde{A}_{-k} = \epsilon_k \overline{\widetilde{A}_k}$ for $k > n$, is contained in (1). Its third conclusion reads

$$\iint_{\mathbb{B}} \big(|H_\xi|^2 + |H_{\bar\xi}|^2\big)\,d\sigma \le C \sum_{m=0}^{\infty} (n+1+m)\,|\widetilde{A}_{n+1+m}|^2.$$

Finally $\widetilde{A}_{n+1+m} = A_{n+1+m}$ for every $m \ge 0$, because $n+1+m \ge n$, so the right-hand side equals $C\sum_{m\ge0}(n+1+m)|A_{n+1+m}|^2$. This is (d), and with Steps 2 and 3 the proof is complete. □